

# Choupo

The Open Source Chemical Process Simulator

# Tutorials Guide

Choupo-dev

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# How to use this guide

Every Choupo tutorial is a complete, runnable case:

```
source etc/bashrc
runCase tutorials/<category>/<group>/<name>
```

This guide gives each tutorial the CONTEXT the case file cannot: the theory block at the head of each group states the physics and the equations you need (condensed – the Theory Guide carries the full derivations), and each entry tells you **what the case does** and **what to expect** before you run it. The per-tutorial text is assembled from the cases' own headers – the case file is always the source of truth, and reading it IS part of the exercise.

Three reading rules:

1. **Run first, read second, re-run third.** The log at verbosity 3 shows every Newton iteration, every K-value, every announcement – the glass box is the point.
2. **Expect the announcements.** Choupo never silently helps: initial guesses, bounds that bind, model fallbacks, unit overrides – all are printed. If a run is quiet about something, it did not happen.
3. **Golden numbers are commitments.** Cases carrying an `expected` file pin their KPIs; the validation AADs quoted in the entries are locked by the regression suite and re-verified on every change.

# Learning paths

**The process-simulation path.** Start with `flash01_benzene_toluene` (the Rachford–Rice loop, visible), then `flash` non-ideal cases (activity models bend the K-values), `distillation` (stages couple; Wang–Henke vs simultaneous), one `flowsheets` recycle case (tears and convergence), and close with a `gibbs` flame (equilibrium without mechanism). Then the batch side: an `ignition` case for stiffness – run the `_rk4` twin and watch the explicit method die – and a `ctrl` loop for feedback.

**The electrolytes path.** `pitzer01_nacl` ($\gamma_{\pm}$, $\varphi$, $a_w$ of one salt), then `pitzer_gamma_hamer_wu` (the catalogue vs NBS data, zero refit), `pitzer_nacl_sp77_hot` (temperature, to 200°C), speciation (`scaling` cases: SI, pH, Davies-vs-Pitzer forking an engineering answer), the eNRTL series (`enrtl_mixed_nacl_ethanol_esteso` mixed-solvent; `enrtl_multisalt_nacl_kcl` two salts, two formulations, an experimental verdict), and the process payoff: evaporators and crystallisers where all of it sets duties and yields.

**The combustion path.** `gibbs06_h2_flame_radicals` (the radical pool at equilibrium; the $c_p$ extrapolation trap), `ignition01_h2o2` (Ó Conaire kinetics, the stiffness lesson via the `_rk4` twin), `ignition02_h2air_slack` (quantitative: $\tau_{\mathrm{ign}}$ vs shock-tube data), `nox01_zeldovich_postflame` (thermal NO; kinetics meets Gibbs at 98.9 %).

**The props-bench path.** `estimate_acetone` (CREATE: groups → constants, error shown), a `scan` case (SEE), `vle01_consistency` (TEST the data), a `fit` case (Levenberg–Marquardt with identifiability), then the promote-and-use loop into a steady case. The Props Guide is this path's companion.

# Category I: Steady-state flowsheets (choupoSolve)

Solved as one nonlinear system $F(\mathbf{x}) = \mathbf{0}$: units in sequence, recycles as tears, everything announced. The electrodialysis stacks sit here as the `electrodialysis` sub-class since 2026-09-20 (a top-level folder is a discipline, not a binary; the retired `electrochem/` folder held these seven steady cases).

## absorption

**The theory you need.** Gas absorption transfers a solute from gas to liquid across stages; Henry's law ($p_i = H_i x_i$) anchors the equilibrium at dilution. Expect the L/G ratio to control the operating line's slope, and a minimum solvent rate at the pinch with the equilibrium line. *Full derivation: Theory Guide, the absorption section.*

## absorber01_NH3_water

tutorials/steady/absorption/absorber01_NH3_water




**What it does:** NH3 absorption from air into water (Kremser, 6 stages)

**The story:** The textbook gas-absorption case: scrubbing ammonia out of an air stream with water, in a counter-current packed/plate column, solved by the Kremser group method (the built-in 'absorber').

Ammonia is very soluble in water (Henry's constant H ≈ 0.99 bar at 298 K → K ≈ 1 at atmospheric pressure), so a modest water rate and a handful of stages recover most of it — unlike CO2/water (H ≈ 1640 bar), where water is a poor solvent.

Feed gas: 100 kmol/h, 10 % NH3 / 90 % N2 (N2 = inert carrier) Solvent: 150 kmol/h pure water (L/V = 1.5) Column: 6 theoretical stages, 298.15 K, 1 atm

The model is NON-ISOTHERMAL: ammonia dissolution is strongly exothermic (ΔH ≈ -34 kJ/mol), so the heat of absorption heats the column. The energy balance is solved coupled with the mass balance.

Expected: liquid heats up ≈ +17 K (top 309 K, internal bulge ~318 K, bottom 315 K — the classic absorber "temperature bulge") recovery_NH3 ≈ 0.675 (only ~68 %, NOT the 97 % an isothermal Kremser would predict — because T rises, H rises, K rises, A falls: the exotherm self-limits the absorption. Real NH3 / amine absorbers are cooled for this reason.) N2 stays in the gas (K_N2 » 1 → no absorption) NH3 mass balance closes (in = clean-gas NH3 + rich-liquid NH3)

Design question ("how many stages for 90 %?") is a DesignSpec on '$stages', NOT a mode of this unit — the column has the stages it has; the recovery is what comes out.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | cleanGas | F | 0.0259019084984 |
| stream | gasFeed | F | 0.0277777777778 |
| stream | richLiquid | F | 0.043542535946 |
| stream | solvent | F | 0.0416666666667 |
| kpi | scrubber | A_N2 | 0.0027498297201 |
| kpi | scrubber | A_NH3 | 1.53522727273 |
| kpi | scrubber | F_cleanGas | 0.0259019084984 |
| kpi | scrubber | F_richLiquid | 0.043542535946 |
| kpi | scrubber | K_N2 | 545.488322072 |
| kpi | scrubber | K_NH3 | 0.977054034049 |
| kpi | scrubber | L_in | 0.0416666666667 |
| kpi | scrubber | L_over_V | 1.5 |
| kpi | scrubber | T_bottom | 315.417418114 |
| kpi | scrubber | T_top | 308.88795701 |
| *... and 14 more reference values in the case's `expected`.* | | | |

## absorption01_CO2_water

tutorials/steady/absorption/absorption01_CO2_water




**What it does:** Single-stage CO2 absorption in water (Henry's law)

**The story:** single-stage CO2 absorption in water, modelled via the v0.28 Henry's-law infrastructure.

Feed: a 5100 kmol/h mixed stream of CO2 / N2 / water already at 298.15 K and 5 bar, with mole fractions matching what you'd obtain by mixing 100 kmol/h of a 10/90 CO2/N2 sour-gas with 5000 kmol/h of liquid water (L/G ~ 50:1 mole). An IsothermalFlash then partitions the species between vapour and liquid.

Why a pre-mixed feed and not Mixer + Flash? The Mixer carries a liquid-enthalpy balance that needs Cp_liq for every component, and CO2 is a permanent gas at 298 K with no liquid-Cp data (Tc = 304 K). Pre-mixing externally keeps the tutorial focused on the Henry's-law K-value selector — the point of the case.

Physics: H_CO2(298 K) = 1.64e8 Pa = 1640 bar (Henry's law) K_CO2 = H / P = 1.64e8 / 5e5 = 328 K_N2 » 1 (no Henry entry, no Antoine close to 298 K either — falls back to Raoult and Psat extrapolation) K_water « 1 (most water stays in liquid; Psat(298) ~ 3 kPa)

Observed result: cleanGas F = 66.0 kmol/h y_CO2 = 0.123 y_N2 = 0.871 y_water = 0.0063 richLiquid F = 5034 kmol/h x_CO2 = 3.75e-4 x_N2 = 6.5e-3 x_water = 0.993 V/F = 0.013 K_CO2 (y/x) = 328 ? exactly H_CO2(298)/P = 1.64e8 / 5e5 K_water = 0.0063 ? Psat_water(298)/P = 3.17 kPa / 500 kPa mass balance CO2: 10 in = 8.11 vapour + 1.89 liquid ~ 19 % of feed CO2 absorbed (a single-stage scrubber is far from perfect; an industrial column with 5-10 theoretical stages would recover > 95 %).

Toggling the GUI Units menu to Mass-basis shows the absorption as CO2 mass transfer between the two streams in kg/h, useful for scale-up arithmetic.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | cleanGas | F | 0.0164210205273 |
| stream | feed | F | 1.41666666667 |
| stream | richLiquid | F | 1.40024564614 |
| kpi | scrubber | F_alpha | 1.40024564614 |
| kpi | scrubber | F_beta | 0.0164210205273 |
| kpi | scrubber | F_in | 1.41666666667 |
| kpi | scrubber | K_CO2 | 327.999999984 |
| kpi | scrubber | K_N2 | 110.543208463 |
| kpi | scrubber | K_water | 0.00633349747981 |
| kpi | scrubber | P | 500000 |
| kpi | scrubber | Q | 0 |
| kpi | scrubber | Q_kW | 0 |
| kpi | scrubber | T | 298.15 |
| kpi | scrubber | V_over_F | 0.0115913086075 |

*... and 7 more reference values in the case's `expected`.*

## acetone05_luyben_absorber

tutorials/steady/absorption/acetone05_luyben_absorber




**What it does:** Luyben's acetone absorber: 9 stages, water washing acetone out of the hydrogen offgas – against his own acetone and bottoms

**The story:** The unit acetone04 was measured FOR. Luyben's absorber washes acetone out of the reactor's hydrogen-rich offgas with water: 9 stages at 1.5 atm, 20 kmol/h of water on top, 39.80 kmol/h of gas at the bottom.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a, Figure 1.

WHAT IS INDEPENDENT HERE, stated before the numbers because the last case in this family had to say the opposite. The gas inlet is Luyben's own ("Gas (to absorber)": 39.80 kmol/h, acetone 0.1147, IPA 0.0033) EXCEPT for its hydrogen and water, which were derived from the gas LEAVING using H2's inertness and a balance. So:

NOT independent – the outlet H2 and water (the inlet was built from them) INDEPENDENT – everything about the ACETONE, and the whole bottoms stream. Nothing in 0/ was built from those.

Luyben's answer, for the record, before this case is run:

bottoms 20.05 kmol/h acetone 0.1020 → 2.045 kmol/h absorbed offgas 39.76 kmol/h acetone 0.0634 → 2.521 kmol/h lost recovery 2.045 / 4.565 = 44.8 %

His own table closes on itself to 0.02 % in both acetone and water, which is two checks on a hand transcription and the reason those numbers are trusted as targets.

WHAT acetone04 PREDICTS FOR THIS UNIT. Measured there, on the same thermodynamics this case runs: gamma_acetone = 10.27 at 1 mol % in water and 318 K, giving K = 6.84. A model that holds acetone too weakly in the water will absorb LESS of it than Luyben does. That is the prediction; the run below is the test.

The thermodynamics is PREDICTIVE UNIFAC with H2 authorised outside the activity model – the same declared package as acetone03, for the same reasons, recorded in constant/thermoPhysPropDict.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.00574366573439 |
| stream | gasIn | F | 0.0110555555556 |
| stream | offgas | F | 0.0108674453767 |
| stream | washWater | F | 0.00555555555556 |
| kpi | absorber | A_H2 | 0.00044153375233 |
| kpi | absorber | A_acetone | 0.0857422914441 |
| kpi | absorber | A_isopropanol | 4.51803004485 |
| kpi | absorber | A_water | 8.0125706439 |
| kpi | absorber | F_cleanGas | 0.0108674453767 |
| kpi | absorber | F_richLiquid | 0.00574366573439 |
| kpi | absorber | K_H2 | 1138.10679288 |
| kpi | absorber | K_acetone | 5.86073166871 |
| kpi | absorber | K_isopropanol | 0.11122382052 |
| kpi | absorber | K_water | 0.0627155235376 |
| *... and 20 more reference values in the case's `expected`.* | | | |

## extract01_ethanol_water_benzene

tutorials/steady/absorption/extract01_ethanol_water_benzene




**What it does:** Counter-current LL extraction: recover ethanol from water with benzene

**The story:** Counter-current LL extraction column

Recover ETHANOL (the solute) from a dilute aqueous feed (WATER = the carrier) using BENZENE as the extracting SOLVENT. This is the classic liquid-liquid extraction problem: a multistage counter-current liquid-liquid cascade.

Topology (the 'extractor' unit op): the feed (water + ethanol) enters at stage 1; fresh solvent (benzene) enters at stage N; the two liquid phases pass counter-currently — the raffinate (water-rich, ethanol-depleted) flows 1 → N, the extract (benzene-rich, ethanol-enriched) flows N → 1; EXTRACT leaves stage 1, RAFFINATE leaves stage N.

Each theoretical stage is ONE liquid-liquid equilibrium, solved by the SHARED Gibbs-energy-minimisation LL flash (IsothermalFlash, phaseSet LL) — robust against the K = 1 saddle that traps Newton-on-residual methods in LLE (CLAUDE.md sec.9). The cascade is torn by successive substitution (mirroring the distillation bubble-point loop).

Thermo: predictive UNIFAC for both liquid phases (water / ethanol / benzene groups inline), the same model used by the ternary LLE scan tutorial — no fitted pairs needed; self-DESCRIBING (inline manifest), UNIFAC records resolve from data/standards/ at runtime.

KPIs to watch: recovery_ethanol (fraction of ethanol leaving in the extract), distribution_ethanol (z_E / z_R), F_extract / F_raffinate, massClosure (must be ~0).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | extract | F | 0.0359317152957 |
| stream | feed | F | 0.0277777777778 |
| stream | raffinate | F | 0.0251793952707 |
| stream | solvent | F | 0.0333333333333 |
| kpi | extractor01 | F_extract | 0.0359317152957 |
| kpi | extractor01 | F_feed | 0.0277777777778 |
| kpi | extractor01 | F_raffinate | 0.0251793952707 |
| kpi | extractor01 | F_solvent | 0.0333333333333 |
| kpi | extractor01 | P | 101325 |
| kpi | extractor01 | T | 298.15 |
| kpi | extractor01 | converged | 1 |
| kpi | extractor01 | distribution_benzene | 295.380144934 |
| kpi | extractor01 | distribution_ethanol | 0.555183850838 |
| kpi | extractor01 | distribution_water | 0.00705390428172 |

*... and 17 more reference values in the case's `expected`.*

## extract02_declared_interior

tutorials/steady/absorption/extract02_declared_interior




**What it does:** An extraction cascade that starts from the interior declared in 0/

**The story:** the interior the case declares

PHYSICALLY THIS IS ‘extract01_ethanol_water_benzene‘: the same feed, the same benzene solvent, the same five theoretical stages, the same UNIFAC. The answer is the same answer. What differs is where the cascade STARTS.

A state directory is a RESTARTABLE SNAPSHOT. ‘0/‘ carries one file per stream – the boundary – and, under ‘internalStates/‘, one file per unit: what the equipment holds between them. This case declares

0/internalStates/extractor01

a ‘stageProfile‘ block carrying the extract and raffinate leaving each of the five stages with their compositions, in exactly the grammar the run writes to ‘converged/internalStates/extractor01‘. THE SAME OBJECT TRAVELS BOTH WAYS: run the case, copy the file from ‘converged/‘ into ‘0/‘, and the next run starts where the last one finished.

A DECLARED PROFILE THAT DOES NOT SATISFY THE BALANCES IS A SEED, NOT AN ANSWER. Nothing checks this file against the stage equilibria – the sweeps do, and they move off any profile that is wrong. What IS checked is that the declaration describes THIS cascade: the stage count, the component set, every value finite, no negative flow. A mismatch REFUSES by name.

Both routes announce, on every run:

[seed] interior read from 0/ (stageProfile): 5 stages, extract F ... [seed] interior seeded by the unit: the fresh solvent as every stage’s extract, the feed as every stage’s raffinate – declare a stageProfile block in 0/internalStates/<SECTOR>/<unit> to own it

TRY IT: read the ‘iterations‘ KPI here, then delete ‘0/internalStates/‘ and run again. Both runs reach the same cascade; only one had to search for it. ‘bin/choupo-init0‘ materialises the unit’s OWN seed (the fresh solvent and the raw feed on every stage) – never a better guess chosen for you.

Record: docs/design/a-state-directory-is-a-restartable-snapshot.md

*Physically this is extract01_ethanol_water_benzene: recover ethanol from a dilute aqueous feed with benzene, five theoretical stages, counter-current, UNIFAC for both liquid phases, each stage split by the shared Gibbs-minimisation LL flash. The answer is the same answer — ethanol recovery to the extract 44.2 %, distribution 0.555. What differs is where the cascade starts.*

*∗Same to the digits above, not to every digit:∗ both runs stop inside the same 1e-4 mass-closure target, and they stop at slightly different points inside it (recovery 0.44157 here against 0.44176 unseeded). That is what a restart from an answer looks like — it is not obliged to take the same path back — which is why this case carries its own golden rather than sharing extract01’s.*

*In OpenFOAM a time directory holds one file per field, and that file carries boundaryField ∗and∗ internalField — the conditions at the edges of the domain and the values inside it, together, so the directory is a complete snapshot you can restart from.*

*This case declares 0/internalStates/extractor01 — a stageProfile block with the extract and raffinate leaving each of the five stages and their compositions, in exactly the grammar the run writes to converged/internalStates/extractor01. The same object travels both ways.*

*The extractor’s unknowns are its stage profile. The sweep starts by putting something on every stage and grinding the counter-current balance down; until this slice that something was written in code — ∗"every stage’s extract = the fresh solvent, every stage’s raffinate = the feed"∗ — honest, built from the streams the case declares, and undeclared, unseen, unowned. That is the one thing the 2026-05-30 rule forbids: every solver aid is explicit in a dict, and the solver announces what it does to converge.*

*Nothing checks this file against the stage equilibria. The sweeps do, and they move off any profile that is wrong. Edit a stage's extract flow by half and the cascade still converges to the same recovery — it simply has further to travel.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | extract | F | 0.0359317155144 |
| stream | extract | T | 298.15 |
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 298.15 |
| stream | raffinate | F | 0.0251793960966 |
| stream | raffinate | T | 298.15 |
| stream | solvent | F | 0.0333333333333 |
| stream | solvent | T | 298.15 |
| kpi | extractor01 | F_extract | 0.0359317155144 |
| kpi | extractor01 | F_feed | 0.0277777777778 |
| kpi | extractor01 | F_raffinate | 0.0251793960966 |
| kpi | extractor01 | F_solvent | 0.0333333333333 |
| kpi | extractor01 | P | 101325 |
| kpi | extractor01 | T | 298.15 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## extraction01_waterButanol

tutorials/steady/absorption/extraction01_waterButanol




**What it does:** Water/n-butanol LL flash at 298 K (experimental)

**The story:** Water + 1-butanol at 25 °C. This system has a well-known liquid-liquid miscibility gap below 125.5 °C (upper critical solution temperature). A 70/30 water/nButanol feed (inside the gap) → splits into a water-rich (aqueous) phase ~98 mol% water and a butanol-rich (organic) phase ~59 mol% nButanol / 41 mol% water.

Demonstrates: 'phases (...)' syntax with TWO liquid phases 'phaseSet LL' in the IsothermalFlash Michelsen TPD stability test to find a robust initial guess

*Water + 1-butanol at 25 °C and 1 atm. The pair has a well-known liquid-liquid miscibility gap below its upper critical solution temperature (125.5 °C), so a feed whose composition lies inside the gap spontaneously splits into two liquid phases. This case feeds a 70/30 water/nButanol mixture (inside the gap) to an isothermal decanter and lets the flash find the two-liquid split.*

*A single feed splits into two liquids, both at 298.15 K, vf = 0:*

*The split is found by Michelsen TPD stability testing (to seed a robust initial guess) followed by direct Gibbs-energy minimisation — the LL path the engine uses precisely because a Newton/successive-substitution flash on a symmetric $\gamma$-model would collapse to the trivial K = 1 saddle.*

*- phases (...) syntax with TWO liquid phases; - phaseSet LL in the IsothermalFlash; - the Michelsen TPD stability test as the initial-guess generator.*

*- The LL flash reuses the VL solver's printout, so the two liquids are shown under Vapor flow V / Liquid flow L headings (and x/y/K labels). Read them as phase-$\alpha$ (organic) and phase-$\beta$ (aqueous), not as a vapour and a liquid. - The decanter is isothermal and adiabatic in reality (an enthalpy hand-check gives H_in ≈ H_out), so the physically correct duty is ≈ 0. The KPI Q and any utility/cost the run reports for this LL split are a known artifact of the shared flash duty path — do not cost this unit from them.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | aqueous | F | 0.0140642365954 |
| stream | feed | F | 0.0277777777778 |
| stream | organic | F | 0.0137135411824 |
| kpi | decanter01 | F_alpha | 0.0140642365954 |
| kpi | decanter01 | F_beta | 0.0137135411824 |
| kpi | decanter01 | F_in | 0.0277777777778 |
| kpi | decanter01 | P | 101325 |
| kpi | decanter01 | T | 298.15 |
| kpi | decanter01 | betaFraction | 0.493687482565 |
| kpi | decanter01 | phaseSet | 1 |
| stream | aqueous | T | 298.15 |
| stream | feed | T | 298.15 |
| stream | organic | T | 298.15 |
| boundary | global | H_feeds_kW | -8280.54358316 |
| *... and 2 more reference values in the case's expected.* | | | |

### stripper01_NH3_water

tutorials/steady/absorption/stripper01_NH3_water




**What it does:** NH3 stripping from water with air (Kremser, 6 stages, warm)

**The story:** The inverse of absorber01: instead of scrubbing NH3 OUT of a gas, here we strip NH3 OUT of water with a lean gas (air ~ N2). Counter-current Kremser; the rich liquid enters the top, the stripping gas the bottom.

Run warm (330 K): NH3 is less soluble hot (K rises), so the stripping factor S = K V/L exceeds 1 and the strip is efficient. But desorption is endothermic, so the column cools – which lowers K and self-limits the stripping (the mirror image of the absorber's exotherm).

Liquid: 100 kmol/h, 5 % NH3 in water, 330 K Gas: 100 kmol/h N2 (V/L = 1), 330 K Column: 6 theoretical stages

Expected: S_NH3 > 1 → most of the NH3 stripped into the gas; stripped liquid nearly NH3-free; rich gas carries the NH3; mass closes 100 %.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | liquidFeed | F | 0.0277777777778 |
| stream | richGas | F | 0.029140357434 |
| stream | strippedLiquid | F | 0.0264151981216 |
| stream | strippingGas | F | 0.0277777777778 |
| kpi | stripper | F_richGas | 0.029140357434 |
| kpi | stripper | F_strippedLiquid | 0.0264151981216 |
| kpi | stripper | K_NH3 | 3.62862883208 |
| kpi | stripper | K_water | 0.170562530402 |
| kpi | stripper | L_in | 0.0277777777778 |
| kpi | stripper | S_NH3 | 3.62862883208 |
| kpi | stripper | S_water | 0.170562530402 |
| kpi | stripper | T_bottom | 313.524974667 |
| kpi | stripper | T_top | 315.973930965 |
| kpi | stripper | V_in | 0.0277777777778 |

*... and 14 more reference values in the case's `expected`.*

## crystallisation

**The theory you need.** Crystallisation removes solute when the solution crosses saturation: $m > m_{\mathrm{sat}}(T)$, with $K_{\mathrm{sp}}(T)$ from $\Delta G_{\mathrm{diss}}$ and its $T$-dependence from $\Delta H_{\mathrm{diss}}$ (van 't Hoff). The heat of crystallisation is ION-DERIVED – $\Delta H_{\mathrm{cryst}} = -(\Delta H^{\infty}_{\mathrm{soln}} + \bar{L}_2(m_{\mathrm{sat}}))$ – never a stored constant (a solid's formation enthalpy is $\sum \nu_i h^{\mathrm{aq}}_{f,i} - \Delta H_{\mathrm{soln}}$, derived, single-sourced). Antisolvent cases add the mixed-solvent electrolyte model (eNRTL): ethanol drops the dielectric constant, $\gamma_{\pm}$ rises, salt crashes out. Expect supersaturation as the driving force and yields set by the solubility curve, not the kinetics (these cases are equilibrium-staged). *Full derivation: Theory Guide, the crystallisation + electrolyte-enthalpy chapters.*

## crystalliser01_sugar

tutorials/steady/crystallisation/crystalliser01_sugar

**What it does:** Cooling crystalliser: sugar solution 80 → 20 C, yield by solubility

**The story:** / flowsheetDict (LEAF node)

A hot sugar solution (z_sucrose 0.15, ~3.35 kg/kg water, below saturation at 80 C where c_sat ~ 3.8) is cooled to 20 C (c_sat ~ 2.0 kg/kg). The excess crystallises. Magma = crystals (s[]) + saturated mother liquor.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | magma | F | 0.0277777777778 |
| kpi | cryst | Q_removed | 271677.114808 |
| kpi | cryst | T_op | 293.15 |
| kpi | cryst | c_sat | 2.026 |
| kpi | cryst | crystal_flow | 0.564469958333 |
| kpi | cryst | crystal_mol | 0.00164906487154 |
| kpi | cryst | liquorFlow | 0.0261287129062 |
| kpi | cryst | solute_dissolved | 0.861767541667 |
| kpi | cryst | solute_in | 1.4262375 |
| kpi | cryst | yield | 0.395775569169 |
| stream | feed | T | 353.15 |
| stream | magma | T | 293.15 |
| kpi | cryst | Q_kW | -271.677114808 |

*... and 7 more reference values in the case's `expected`.*

## crystalliser02_msmpr

tutorials/steady/crystallisation/crystalliser02_msmpr

**What it does:** MSMPR sugar crystalliser: population balance by method of moments

**The story:** / flowsheetDict (LEAF node)

Continuous MSMPR crystalliser. A hot near-saturated sugar solution is fed to a cooled, well-mixed vessel of working volume V; the magma (crystals + mother liquor) is withdrawn at the same volumetric rate, so the residence time is tau = V / Q. 'model MSMPR' selects the population-balance path (method of moments); the kinetics are referenced from constant/crystallisation.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | magma | F | 0.0277777777778 |
| kpi | cryst | CV | 1 |
| kpi | cryst | L_dominant | 0.000223952804481 |
| kpi | cryst | L_meanMass | 0.000298603739308 |
| kpi | cryst | L_meanNumber | 7.46509348269e-05 |
| kpi | cryst | Q_removed | 269525.057151 |
| kpi | cryst | T_op | 293.15 |
| kpi | cryst | c_sat | 2.026 |
| kpi | cryst | crystal_flow | 0.396190602513 |
| kpi | cryst | crystal_mol | 0.00115744690287 |
| kpi | cryst | growthRate | 1.95272330048e-08 |
| kpi | cryst | liquorFlow | 0.0266203308749 |
| kpi | cryst | magmaDensity | 302.920529908 |

*... and 21 more reference values in the case's `expected`.*

## crystalliser03_pbe_size_independent

`tutorials/steady/crystallisation/crystalliser03_pbe_size_independent`

**What it does:** Crystalliser FVM-PBE: sanity vs MSMPR (size-independent growth)

**The story:** / flowsheetDict (LEAF node)

Same physical setup as crystalliser02_msmpr (hot sugar syrup, 5 $m^3$ vessel at 20 °C, k_g 1e-7, k_b 1e9) – only the solver method changes: 'model FVM' discretises the population balance n(L) on Nbins evenly-spaced size bins, marching the steady PBE forward in L with first-order upwind. With sizeFactor 0 (the default), G(L) is constant and the FVM result must match the analytical MSMPR exponential.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | magma | F | 0.0277777777778 |
| kpi | cryst | CV | 0.999540291345 |
| kpi | cryst | L_dominant | 0.000224609375 |
| kpi | cryst | L_meanMass | 0.000298633437171 |
| kpi | cryst | L_meanNumber | 7.48298114726e-05 |
| kpi | cryst | Q_removed | 269576.1605 |
| kpi | cryst | T_op | 293.15 |
| kpi | cryst | bins | 256 |
| kpi | cryst | c_sat | 2.026 |
| kpi | cryst | crystal_flow | 0.400186610054 |
| kpi | cryst | crystal_mol | 0.00116912099742 |
| kpi | cryst | growthRate_L0 | 1.90635340048e-08 |
| kpi | cryst | liquorFlow | 0.0266086567804 |

*... and 22 more reference values in the case's `expected`.*

### crystalliser04_size_dependent_growth

tutorials/steady/crystallisation/crystalliser04_size_dependent_growth

**What it does:** Crystalliser FVM-PBE with size-dependent growth (sizeFactor > 0)

**The story:** / flowsheetDict (LEAF node)

Same hot-sugar feed and 5 $m^3$ vessel at 20 °C as crystalliser03, only 'sizeFactor' is non-zero — the size-dependent growth law that breaks the method-of-moments closure but is handled natively by the FVM.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | magma | F | 0.0277777777778 |
| kpi | cryst | CV | 1.03562508094 |
| kpi | cryst | L_dominant | 0.000232421875 |
| kpi | cryst | L_meanMass | 0.000329374475484 |
| kpi | cryst | L_meanNumber | 7.45169890057e-05 |
| kpi | cryst | Q_removed | 269660.031771 |
| kpi | cryst | T_op | 293.15 |
| kpi | cryst | bins | 256 |
| kpi | cryst | c_sat | 2.026 |
| kpi | cryst | crystal_flow | 0.406744893212 |
| kpi | cryst | crystal_mol | 0.00118828062534 |
| kpi | cryst | growthRate_L0 | 1.83025070562e-08 |
| kpi | cryst | liquorFlow | 0.0265894971524 |

*... and 22 more reference values in the case's `expected`.*

## crystalliser05_nacl_pitzer

`tutorials/steady/crystallisation/crystalliser05_nacl_pitzer`

**What it does:** NaCl crystalliser: saturation from the PITZER ion activity (Ksp = (gamma_pm∗m_sat)^2). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

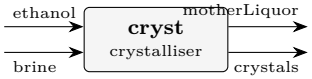
**The story:** NaCl crystalliser whose SATURATION comes from the Pitzer ion activity (Ksp = (gamma_pm ∗ m_sat)^2), via the 'electrolyte {}' block – NaCl carries no solubility{} curve, so the electrolyte model is what makes this possible. The feed is a SUPERSATURATED brine (~7.6 mol/kg, as it would leave an evaporator); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess. Components live case-local in constant/components/; other curated records resolve from data/standards/ at runtime.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | magma | F | 0.0277777777778 |
| kpi | cryst | Ksp_activity | 38.4846213082 |
| kpi | cryst | Q_removed | 1251.26049958 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | c_sat | 0.35905536 |
| kpi | cryst | crystal_flow | 0.036683987968 |
| kpi | cryst | crystal_mol | 0.000627720533333 |
| kpi | cryst | gamma_pm_sat | 1.00970010562 |
| kpi | cryst | liquorFlow | 0.0271500572444 |
| kpi | cryst | m_sat | 6.144 |
| kpi | cryst | solute_dissolved | 0.158116012032 |
| kpi | cryst | solute_in | 0.1948 |
| kpi | cryst | yield | 0.18831616 |
| *... and 7 more reference values in the case's `expected`.* | | | |

## crystalliser06_nacl_antisolvent

tutorials/steady/crystallisation/crystalliser06_nacl_antisolvent




**What it does:** Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises gamma_pm, and DROPS the saturation molality m_sat – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product Ksp anchored in pure water, solved with the mixed-solvent gamma_pm.

**The story:** ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity gamma_pm (eNRTL mixed-solvent + Born), so the saturation molality m_sat COLLAPSES (6.14 → ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product Ksp = (gamma_pm ∗ m_sat)^2 anchored in pure water, solved with the mixed-solvent gamma_pm at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine (x_water 0.900, x_NaCl 0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol → salt-free x_ethanol ~ 0.40. Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

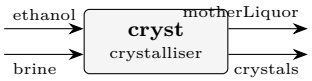
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brine | F | 0.0277777777778 |
| stream | crystals | F | 0.00202035108762 |
| stream | ethanol | F | 0.0166666666667 |
| stream | motherLiquor | F | 0.0424240933568 |
| kpi | cryst | Ksp_activity | 34.8439846653 |
| kpi | cryst | Q_removed | 6559.87377671 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | antisolvent_xfrac | 0.399920015997 |
| kpi | cryst | c_sat | 0.0517444790334 |
| kpi | cryst | cakeLiquidFlow | 0.000329661969914 |
| kpi | cryst | cakeMoisture | 0.1 |
| kpi | cryst | cakeWetness_pct | 9.09090909091 |
| kpi | cryst | crystal_flow | 0.0988038720389 |
| kpi | cryst | crystal_mol | 0.00169068911771 |

*... and 17 more reference values in the case's `expected`.*

## crystalliser07_nacl_antisolvent_cooled

tutorials/steady/crystallisation/crystalliser07_nacl_antisolvent_cooled




**What it does:** Antisolvent + COOLING crystallisation of NaCl: the same drowning-out brine + ethanol as crystalliser06, but the crystalliser is COOLED to 5 C. The eNRTL is now temperature-dependent (tau(T)=tau_25∗298.15/T, A_DH(T) via eps_w(T)/rho_w(T), Born at T) and Ksp(T) shifts by van't Hoff with NaCl's enthalpy of solution – so cooling adds to the antisolvent's effect (yield 61% → 70%). Exercises the T-dependent electrolyte path.

**The story:** brine [cryst @ 5 C] crystals (wet NaCl cake) ethanol motherLiquor (saturated supernatant)

Identical drowning-out feed to crystalliser06 (saturated brine + ethanol, salt-free x_ethanol ~ 0.40), but the crystalliser is COOLED to 5 C. Two levers now stack: ANTISOLVENT – ethanol lowers the dielectric → mixed-solvent gamma_pm(T). COOLING – the electrolyte model is temperature-dependent: tau(T) = tau_25 ∗ 298.15/T (the Delta_g/RT form, like the molecular NRTL), the Debye-Huckel A_DH(T) follows eps_w(T)/rho_w(T), the Born term is at T, and the saturation product Ksp(T) shifts by van't Hoff with NaCl's enthalpy of solution (dissolutionEnthalpy in NaCl.dat). Result: m_sat ~ 0.885 → 0.681 mol/kg, yield 61% → 70% vs the isothermal 06. (For NaCl alone cooling is weak – solubility is nearly T-flat; the antisolvent carries most of the effect. A salt with a large dHsol, e.g. KNO3, would gain far more from cooling.)

Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brine | F | 0.0277777777778 |
| stream | crystals | F | 0.00231987381407 |
| stream | ethanol | F | 0.0166666666667 |
| stream | motherLiquor | F | 0.0421245706304 |
| kpi | cryst | Ksp_activity | 31.135192385 |
| kpi | cryst | Q_removed | 82721.44242 |
| kpi | cryst | T_op | 278.15 |
| kpi | cryst | antisolvent_xfrac | 0.399920015997 |
| kpi | cryst | c_sat | 0.0398065230351 |
| kpi | cryst | cakeLiquidFlow | 0.000380305320939 |
| kpi | cryst | cakeMoisture | 0.1 |
| kpi | cryst | cakeWetness_pct | 9.09090909091 |
| kpi | cryst | crystal_flow | 0.113348382739 |
| kpi | cryst | crystal_mol | 0.00193956849313 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## crystalliser08_nacl_antisolvent_msmpr

tutorials/steady/crystallisation/crystalliser08_nacl_antisolvent_msmpr




**What it does:** Continuous MSMPR antisolvent crystallisation of NaCl: a saturated brine + ethanol fed to a well-mixed vessel (residence time tau = V/Q). The eNRTL antisolvent supersaturation S = c/c_sat(eNRTL) drives nucleation + growth; the method of moments closes the population balance and returns the CRYSTAL-SIZE DISTRIBUTION n(L) (mean size, CV) – what the equilibrium model cannot give.

**The story:** CONTINUOUS MSMPR + ANTISOLVENT

brine [cryst MSMPR] magma (crystals WITH a size distribution n(L) ethanol + saturated mother liquor)

Same drowning-out feed as crystalliser06 (saturated brine + ethanol), but the vessel is a CONTINUOUS, well-mixed MSMPR crystalliser of working volume V; the magma leaves at the same volumetric rate, so the residence time is tau = V/Q.

Where the equilibrium model stops at the YIELD, the population balance goes on to the CRYSTAL-SIZE DISTRIBUTION: the antisolvent supersaturation S = c / c_sat(eNRTL, T, x_ethanol) drives growth G = k_g (S-1)^g and nucleation B0 = k_b (S-1)^b; the method of moments gives n(L) = n0 exp(-L/Gtau), the mean sizes (L43 = 4 Gtau), the CV, and the magma PSD – plotted in the GUI.

The supersaturation here comes from the eNRTL ion-activity Ksp (the antisolvent lowers m_sat), NOT a solubility curve – the MSMPR now reads the electrolyte saturation, the same helper the equilibrium model uses. Components + kinetics live under constant/; the eNRTL ion/tau pair resolves from data/standards/ at runtime (self-DESCRIBING inline manifest).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brine | F | 0.0277777777778 |
| stream | ethanol | F | 0.0166666666667 |
| stream | magma | F | 0.0444444444444 |
| kpi | cryst | CV | 1 |
| kpi | cryst | L_dominant | 2.02534396647e-05 |
| kpi | cryst | L_meanMass | 2.70045862196e-05 |
| kpi | cryst | L_meanNumber | 6.75114655489e-06 |
| kpi | cryst | Q_kW | -6.1566687571 |
| kpi | cryst | Q_removed | 6156.6687571 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | c_sat | 0.0517444790334 |
| kpi | cryst | crystal_flow | 0.092730856228 |
| kpi | cryst | crystal_mol | 0.00158677029822 |
| kpi | cryst | growthRate | 1.92664299147e-09 |

*... and 20 more reference values in the case's `expected`.*

## crystalliser09_KHT_KCl_series

tutorials/steady/crystallisation/crystalliser09_KHT_KCl_series




**What it does:** Two crystallisers in series separating KCl from potassium bitartrate (KHT, cream of tartar) by manipulating T and adding ethanol. CRYST1 COOLS the hot feed to 5 C → KHT (sparingly soluble, steep solubility curve) crystallises while KCl stays dissolved. CRYST2 doses ETHANOL into the KCl-rich mother liquor → KCl drops out by drowning-out (eNRTL mixed-solvent). Two salts, two saturation routes (curve vs ion-activity), one package – a stress test of the property system.

**The story:** SELECTIVE crystallisation of two salts

feed (hot: KCl + KHT) [cryst1: COOL to 5 C] KHT_cake (cream of tartar)

ethanol [cryst2: ANTISOLVENT] KCl_cake

finalLiquor (EtOH+water+salt)

[recovery: DISTILLATION] recoveredEthanol (recycle-grade)

stillage (water + residual salt)

Two salts with very different solubilities are separated by two DIFFERENT levers, each crystalliser targeting one salt ('solute ...'):

CRYST1 cools the hot feed. Potassium bitartrate (KHT) is sparingly soluble with a STEEP solubility curve, so cooling 50 → 5 C precipitates most of it; KCl, far more soluble and only mildly T-sensitive, stays dissolved. KHT's saturation is its measured solubility CURVE c_sat(T) – no ion parameters.

CRYST2 doses ethanol into the KCl-rich mother liquor. KCl drops out by drowning-out: the eNRTL mixed-solvent term raises gamma_pm as the dielectric falls, collapsing m_sat. KCl's saturation is the ion-activity Ksp.

The SAME package carries both routes; 'useElec' is decided per solute, so KHT uses its curve while KCl uses the eNRTL. Self-DESCRIBING (inline manifest); curated records resolve from data/standards/ at runtime.

Basis (per kg water): 55.5 water, 3.0 KCl (below the 5 C saturation ~3.9), and 0.078 KHT (saturated at 50 C).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | KCl_cake | F | 0.00138951804787 |
| stream | KHT_cake | F | 5.38404641059e-05 |
| stream | feed | F | 0.0277777777792 |
| stream | finalLiquor | F | 0.0541131192255 |
| stream | freshEthanol | F | 0.00416666666667 |
| stream | liquor1 | F | 0.0277239373151 |
| stream | recoveredEthanol | F | 0.0236111111111 |
| stream | stillage | F | 0.0305020081144 |
| kpi | cryst1 | Q_kW | -93.1162690166 |
| kpi | cryst1 | Q_removed | 93116.2690166 |
| kpi | cryst1 | T_op | 278.15 |
| kpi | cryst1 | c_sat | 0.0034013435 |
| kpi | cryst1 | cakeLiquidFlow | 2.54671494088e-05 |
| kpi | cryst1 | cakeMoisture | 0.1 |

*... and 65 more reference values in the case's `expected`.*

## crystalliser10_naoh_antisolvent

tutorials/steady/crystallisation/crystalliser10_naoh_antisolvent




**What it does:** NaOH antisolvent crystallisation (water + ethanol, eNRTL drowning-out) – the acceptance-case core

**The story:** H2O + NaOH + ethanol, every effect that IS calibrated active, every one that is NOT loudly announced: ACTIVE: drowning-out saturation (eNRTL mixed-solvent; Na-OH taus anchored to the Parker-slotted Pitzer surface), BPE-grade water activity, differential-Ksp lineage, wet-cake split. ANNOUNCED OMISSIONS (honest): heat of dilution / crystallisation heat use the dHcryst constant (the eNRTL kernel is NOT calorimetrically fitted – per-kernel flag); water-ethanol heat of mixing gated until the H_E refit; the solid is ANHYDROUS NaOH until the NaOH.H2O hydrate forms entry lands.

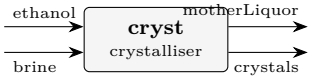
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | NaOH_cake | F | 0.00183771747926 |
| stream | antisolvent | F | 0.00833333333333 |
| stream | liquor | F | 0.0138888888889 |
| stream | motherLiquor | F | 0.020384504743 |
| kpi | cryst | Ksp_activity | 7365.8380048 |
| kpi | cryst | Q_kW | 72.7276039114 |
| kpi | cryst | Q_removed | -72727.6039114 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | antisolvent_xfrac | 0.461538461538 |
| kpi | cryst | c_sat | 0.181200024556 |
| kpi | cryst | cakeLiquidFlow | 0.000203756483719 |
| kpi | cryst | cakeMoisture | 0.1 |
| kpi | cryst | cakeWetness_pct | 9.09090909091 |
| kpi | cryst | crystal_flow | 0.0653535379386 |

*... and 17 more reference values in the case's `expected`.*

## crystalliser11_kcl_antisolvent_ethanol

tutorials/steady/crystallisation/crystalliser11_kcl_antisolvent_ethanol




**What it does:** Antisolvent (drowning-out) crystallisation of KCl: a SATURATED aqueous KCl brine (4.76 mol/kg) is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises gamma_pm, and DROPS the saturation molality – so KCl precipitates with NO cooling and NO evaporation. Saturation is the SAME ion-activity product Ksp anchored in pure water, solved with the mixed-solvent gamma_pm at the feed's salt-free ethanol fraction. KCl-water-ethanol experimental reference: Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

**The story:** ANTISOLVENT (drowning-out) of KCl

[cryst] crystals (solid KCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual KCl)

The KCl counterpart of crystalliser06 (NaCl). KCl is the second salt the catalogue's KCl.dat names as "recovered by ANTISOLVENT (drowning-out with ethanol)"; this case realises it.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity gamma_pm (eNRTL mixed-solvent + Born), so the KCl saturation molality m_sat COLLAPSES from its aqueous 4.76 mol/kg and KCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product Ksp = (gamma_pm ∗ m_sat)^2 anchored in pure water, solved with the mixed-solvent gamma_pm at the feed's salt-free ethanol fraction.

Basis: 100 kmol/h saturated aqueous brine (x_water 0.9210, x_KCl 0.0790; 4.76 mol KCl / kg water) + 60.0 kmol/h ethanol → salt-free x_ethanol ~ 0.39. Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

Experimental KCl-water-ethanol reference (the drowning-out anchor): Lopes, Farelo & Ferra, J. Solution Chem. 28 (1999) 117.

*The KCl counterpart of crystalliser06 (NaCl): a saturated aqueous KCl brine is mixed with ethanol, whose low dielectric collapses the KCl solubility, so KCl crystallises with no cooling and no evaporation. KCl is the salt the catalogue's KCl.dat names as ∗"recovered by ANTISOLVENT (drowning-out with ethanol)"∗ — this case realises it.*

*Ethanol's relative permittivity (24.3) is far below water's (78.5). Dosing it lowers the mixed-solvent dielectric, which raises the mean ionic activity coefficient $\gamma\pm$ (eNRTL mixed-solvent + Born term), so the saturation molality collapses. The saturation is the same ion-activity product Ksp = $(\gamma\pm \cdot m_sat)^2$ anchored in pure water, re-solved with the mixed-solvent $\gamma\pm$ at the feed's salt-free ethanol fraction — the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).*

*100 kmol/h saturated brine (x_water 0.921, x_KCl 0.079 = 4.76 mol/kg) + 60 kmol/h ethanol → salt-free x_ethanol ≈ 0.39:*

*The KCl–water–ethanol activity behaviour modelled here is measured in A. Lopes, F. Farelo & M. I. A. Ferra, "Activity Coefficients of Potassium Chloride in Water–Ethanol Mixtures", ∗J. Solution Chem.∗ 28 (1999) 117 — mean activity coefficients of KCl by emf in 5–20 % w/w ethanol, 25–45 °C. That $\gamma\pm$ rise with ethanol content is exactly the driver of the drowning-out captured above.*

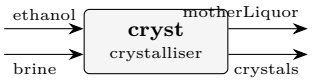
*- The eNRTL $\tau$ for K–Cl is the illustrative 1:1-alkali-chloride proxy (= Chen & Evans water–NaCl), so the numbers are qualitative. The Farelo data is the path to a quantitative refit of the KCl–water–ethanol interaction (the paper reports Pitzer $\beta^0/\beta^1/C?$ per solvent composition). - The mass / yield is the reliable output. The crystalliser's heat term carries the shared electrolyte-enthalpy caveat (a dissolved salt's formation datum lives in its ions, so the stream enthalpy falls back to the sensible datum and is announced) — identical to crystalliser06.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brine | F | 0.0277777777778 |
| stream | crystals | F | 0.0021050886953 |
| stream | ethanol | F | 0.0166666666667 |
| stream | motherLiquor | F | 0.0423393557491 |
| kpi | cryst | Ksp_activity | 7.69319557877 |
| kpi | cryst | Q_kW | -28.9281906957 |
| kpi | cryst | Q_removed | 28928.1906957 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | antisolvent_xfrac | 0.394477317554 |
| kpi | cryst | c_sat | 0.0311001943033 |
| kpi | cryst | cakeLiquidFlow | 0.000423217143227 |
| kpi | cryst | cakeMoisture | 0.1 |
| kpi | cryst | cakeWetness_pct | 9.09090909091 |
| kpi | cryst | crystal_flow | 0.125385206079 |
| *… and 17 more reference values in the case's `expected`.* | | | |

## crystalliser12_nacl_pitzer_antisolvent

tutorials/steady/crystallisation/crystalliser12_nacl_pitzer_antisolvent




**What it does:** Pitzer + antisolvent: the calorimetric L_phi fit is AQUEOUS, so a mixed solvent drops it – and the duty basis says so

**The story:** crystalliser06_nacl_antisolvent – ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity gamma_pm (eNRTL mixed-solvent + Born), so the saturation molality m_sat COLLAPSES (6.14 → ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product Ksp = (gamma_pm ∗ m_sat)^2 anchored in pure water, solved with the mixed-solvent gamma_pm at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine (x_water 0.900, x_NaCl 0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol → salt-free x_ethanol ~ 0.40. Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brine | F | 0.0277777777778 |
| stream | crystals | F | 0 |
| stream | ethanol | F | 0.0166666666667 |
| stream | motherLiquor | F | 0.0444444444444 |
| kpi | cryst | Ksp_activity | 38.4846213082 |
| kpi | cryst | Q_kW | -0 |
| kpi | cryst | Q_removed | 0 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | antisolvent_xfrac | 0.399920015997 |
| kpi | cryst | c_sat | 0.35905536 |
| kpi | cryst | cakeLiquidFlow | 0 |
| kpi | cryst | cakeMoisture | 0.1 |
| kpi | cryst | cakeWetness_pct | 0 |
| kpi | cryst | crystal_flow | 0 |

*... and 16 more reference values in the case's `expected`.*

### overlay02_nacl_calorimetric

tutorials/steady/crystallisation/overlay02_nacl_calorimetric

**What it does:** NaCl crystalliser: saturation from the PITZER ion activity (Ksp = (gamma_pm∗m_sat)^2). Feed is a SUPERSATURATED brine (post-evaporation, ~7.6 mol/kg); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess.

**The story:** crystalliser05_nacl_pitzer (LEAF node)

NaCl crystalliser whose SATURATION comes from the Pitzer ion activity (Ksp = (gamma_pm ∗ m_sat)^2), via the 'electrolyte {}' block – NaCl carries no solubility{} curve, so the electrolyte model is what makes this possible. The feed is a SUPERSATURATED brine (~7.6 mol/kg, as it would leave an evaporator); the unit drops it to saturation (6.144 mol/kg) and crystallises the excess. Components live case-local in constant/components/; other curated records resolve from data/standards/ at runtime.

*A NaCl crystalliser (cloned from crystalliser05) whose case-local constant/components/NaCl.dat declares overlayOf NaCl; and recalibrates ONLY solidPhases.halite.calorimetric.dissolutionEnthalpy (3880 → 5880 J/mol).*

*The ThermoPackageBuilder deep-merges this delta over the standard record, so the crystalliser duty reflects it: Q_removed = 3690.997 W here vs 2435.556 in crystalliser05 (ratio 1.515 = 5880/3880). Every other KPI (Ksp_activity, m_sat, c_sat, crystal_mol, gamma) is byte-identical to crystalliser05 – logK25, dH, the dissolution reaction and the crystal block all still come from the standard.*

*This is the second half of the roadmap Phase A acceptance test: the SINGLE unified reader applies the overlay at ALL three component-loading sites (SpeciationSolver, Database, ThermoPackageBuilder), not just the SI path.*

**What to expect (golden-locked):**

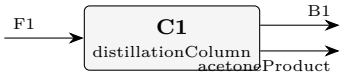
| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | magma | F | 0.0277777777778 |
| kpi | cryst | Ksp_activity | 38.4846213082 |
| kpi | cryst | Q_kW | -2.50670156624 |
| kpi | cryst | Q_removed | 2506.70156624 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | c_sat | 0.35905536 |
| kpi | cryst | crystal_flow | 0.036683987968 |
| kpi | cryst | crystal_mol | 0.000627720533333 |
| kpi | cryst | gamma_pm_sat | 1.00970010562 |
| kpi | cryst | liquorFlow | 0.0271500572444 |
| kpi | cryst | m_sat | 6.144 |
| kpi | cryst | solute_dissolved | 0.158116012032 |
| kpi | cryst | solute_in | 0.1948 |
| *... and 7 more reference values in the case's `expected`.* | | | |

## distillation

**The theory you need.** A column is $N$ flash stages coupled by counter-current flows. Each stage obeys MESH equations: Material balance, Equilibrium ($y_i = K_i x_i$), Summation ($\sum y = \sum x = 1$), and Heat balance. Choupo ships two solution strategies, selectable per column: Wang–Henke (bubble-point, solving the tridiagonal material balances stage by stage; robust for near-ideal systems, slow near azeotropes) and `simultaneous` (one Newton on ALL the MESH residuals; ∼5–6 iterations even through azeotropes). Expect: reflux and boilup set the split; a minimum reflux below which no stage count reaches the spec; and azeotropes as fixed points no column can cross ($x_D$ pinned at the azeotropic composition). *Full derivation: Theory Guide, the distillation chapter (McCabe–Thiele + MESH).*

### acetone06_luyben_column_C1

tutorials/steady/distillation/acetone06_luyben_column_C1




**What it does:** Luyben's acetone column C1 on predictive UNIFAC: the test of a prediction written before the column was built

**The story:** THE COLUMN THAT TESTS A PREDICTION WRITTEN BEFORE IT EXISTED.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. Column C1 of his Figure 1: 66 stages, feed on stage 54, reflux ratio 2.78, distillate 32.25 kmol/h, 1 atm. It takes the combined flash liquid plus absorber bottoms (his F1) and is asked to make the ACETONE PRODUCT at 99.9 mol %.

THE PREDICTION, quoted from the sibling props case rather than reinvented. 'props/compare/acetone04_acetone_water_vle' measured the acetone/water VLE this column rests on BEFORE any column was built, and found that predictive UNIFAC puts a minimum-boiling azeotrope at x_acetone ~ 0.978 where Luyben's UNIQUAC has only a pinch. It is a tiny feature – 0.044 K deep, |y - x| never above 7.7e-4 – and acetone04 recorded, in advance and in writing:

"A column C1 built on this thermodynamics should be expected to asymptote near 97.9 % and fail his spec – and if it had been built before this was measured, the column would have been blamed for a defect that lives in the activity model."

So THIS case has a pass/fail criterion that is not its own golden: either the distillate stalls near the predicted ceiling, or the prediction was wrong. Both outcomes are results. Neither is a reason to touch the thermodynamics.

THE FEED IS LUYBEN'S OWN F1, not this project's upstream answer. acetone03 computes a flash liquid and acetone05 an absorber bottoms, and their sum is what physically feeds C1 – but wiring them in would test the column and two upstream units at once, and acetone05 already reports that it recovers a sixth of what the paper does. Feeding the PUBLISHED stream isolates the column: whatever it gets wrong here is the column and its thermodynamics. The composition in 0/F1 is his mole fractions times his 72.85 kmol/h.

WHAT IS NOT LUYBEN'S, and why each choice is the honest one:

UNIQUAC vs UNIFAC. He runs UNIQUAC; Choupo ships no fitted pair for any of these binaries and inventing one would convert unsourced into falsely sourced. Predictive UNIFAC is what is available, and acetone04 measured what it costs on the binary that decides this column.

The rigorous MESH ('model simultaneous'). Wang-Henke is the corpus default and is documented as unstable through an azeotrope, which is precisely the feature under test here. Choosing the method that CAN resolve the feature is not tuning; choosing the one that cannot would have hidden it.

Isopropanol's liquid heat capacity is Rowlinson-Bondi off the same Joback fragmentation as the rest of its record. Its measured accuracy on hydrogen-bonding molecules is 30-70 %, stated in the record itself. It is 5 mol % of the feed and it moves the DUTIES, not the split.

THIS COLUMN IS MODELLED DOWNSTREAM OF LUYBEN'S VENT, so it carries no hydrogen. His C1 has a partial condenser with a vent; Choupo's column has a total condenser and no vapour distillate. Both facts are stated before the reason, because the reason has to be argued rather than asserted:

- his vent is 0.0465 kmol/h at H2 0.2798, i.e. 0.0130 kmol/h of hydrogen, against the 0.0146 kmol/h that F1 carries. The vent removes essentially ALL of it. A column model with no vent and a total condenser therefore represents the column AFTER the vent, which is a statement about WHICH UNIT is being modelled – not an approximation to the physics of the one that is.

- what that costs is quantified rather than hidden: the same vent carries 0.0335 kmol/h of ACETONE (his own figure), 0.10 % of the product. Not modelling the vent therefore FLATTERS this column's acetone recovery by that much, and the comparison below must be read knowing it.

IT WAS RUN THE OTHER WAY FIRST, and the run is the reason this paragraph exists rather than a preference. With H2 in the feed the MESH converged (16 Newton iterations, ||F|| 2.4e-11) to essentially

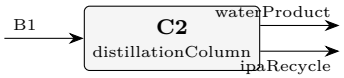
the same distillate – x_acetone 0.9722 vs the shipped run's goldened 0.9731 – and then the energy report REFUSED: hydrogen has no liquid heat capacity, because at 330 K it has no liquid. The refusal is right, and it is the honest form of the modelling error: a total condenser was silently asking a permanent gas to leave as a liquid. See this case's CLAUDE.md.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | B1 | F | 0.0112717083333 |
| stream | F1 | F | 0.0202300416667 |
| stream | acetoneProduct | F | 0.00895833333333 |
| kpi | C1 | B | 0.0112717083333 |
| kpi | C1 | D | 0.00895833333333 |
| kpi | C1 | Q_condenser_kW | -1008.41512564 |
| kpi | C1 | Q_reboiler_kW | 1041.15999682 |
| kpi | C1 | R | 2.78 |
| kpi | C1 | T_bottom | 353.606283382 |
| kpi | C1 | T_top | 329.375698025 |
| kpi | C1 | feedStage | 54 |
| kpi | C1 | nFeeds | 1 |
| kpi | C1 | nSideDraws | 0 |
| kpi | C1 | nStages | 66 |
| *... and 21 more reference values in the case's `expected`.* | | | |

## acetone07_luyben_column_C2

tutorials/steady/distillation/acetone07_luyben_column_C2




**What it does:** Luyben's water column C2: the IPA recycle meets the azeotrope acetone01 measured

**The story:** THE SECOND COLUMN, AND THE SECOND PREDICTION WRITTEN BEFORE IT.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. Column C2 of his Figure 1: 19 stages, feed on stage 16, reflux ratio 0.849, 1 atm. It takes C1's bottoms and makes the plant's WATER PRODUCT at the bottom (34.67 kmol/h, 99.9 mol % water) while sending the unconverted isopropanol back overhead as the RECYCLE (5.88 kmol/h at x_IPA 0.6500).

THE PREDICTION, quoted from acetone01 rather than reinvented, and committed to the repository BEFORE this case was written:

"C2's overhead should cap at roughly x_IPA 0.59 and cannot reach his 0.650. The gap is 0.06 in mole fraction, about 9 % relative, and it is a HARD ceiling rather than a shortfall in stages or reflux."

The reason is one measured number: acetone01 puts the UNIFAC IPA/water azeotrope at x_IPA 0.5901 against Luyben's UNIQUAC value of 0.6732. A column cannot distil past a minimum-boiling azeotrope from below.

AND THE DIRECTION OF THIS ERROR IS THE OPPOSITE OF C1's, which is the part worth carrying away. C1's azeotrope is a UNIFAC INVENTION – Luyben's model has none in acetone/water – so it costs purity the paper does not lose. Here the azeotrope is REAL in both models and UNIFAC merely puts it in the wrong place. One failure is a spurious feature; the other is a misplaced real one. They are not the same defect and should not be reported as one.

THE FEED IS LUYBEN'S PUBLISHED B1, not acetone06's answer – the same choice and the same reason as in C1: it isolates the column. His B1 is 40.56 kmol/h at 370 K, A 0.0001 / IPA 0.0951 / W 0.9048 – entered here as the saturated liquid he means, at this model's 1 atm bubble point (357.46 K), because at 370 K and 1 atm this UNIFAC makes it a vapour (0/B1 says why). The CLOSED flowsheet (acetone08) is where the two columns finally see each other's real streams, and the difference between this case and that one is exactly the propagation the plant exists to show.

THE DISTILLATE RATE IS HIS, 5.88 kmol/h, and that matters here in a way it did not in C1. Fixing the rate means the column must send that much material overhead whatever its composition can be – so if the ceiling binds, it shows up as a WETTER recycle at the same flow, not as a smaller one. That is the honest way to pose it: the plant's IPA inventory has to go somewhere, and the question is what quality it comes back at.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | B1 | F | 0.0112666944444 |
| stream | ipaRecycle | F | 0.00163333333333 |
| stream | waterProduct | F | 0.00963336111111 |
| kpi | C2 | B | 0.00963336111111 |
| kpi | C2 | D | 0.00163333333333 |
| kpi | C2 | Q_condenser_kW | -121.100002882 |
| kpi | C2 | Q_reboiler_kW | 127.189514012 |
| kpi | C2 | R | 0.849 |
| kpi | C2 | T_bottom | 366.17478554 |
| kpi | C2 | T_top | 355.876498071 |
| kpi | C2 | feedStage | 16 |
| kpi | C2 | nFeeds | 1 |
| kpi | C2 | nSideDraws | 0 |
| kpi | C2 | nStages | 19 |

*... and 21 more reference values in the case's `expected`.*

## column01_benzene_toluene

`tutorials/steady/distillation/column01_benzene_toluene`




**What it does:** Benzene/toluene distillation column (Wang-Henke)

**The story:** Benzene/toluene distillation — classical textbook problem with Raoult, no azeotrope, easy convergence. Used to validate the column code.

*The same equimolar benzene/toluene feed (100 kmol/h, now at 370 K) in a rigorous column: 15 stages (reboiler counted), feed on stage 8, R = 2, distillate 50 kmol/h, Raoult VLE, solved by the Wang–Henke bubble-point method. At 370 K and 1 atm the feed resolves to about 70 % vapour (q = 0.30), so the reboiler does far less than the condenser. The golden: x_D = 0.9510 benzene, x_B = 0.9510 toluene, T_top 355.69 K, T_bottom 381.51 K, condenser -1283 kW, reboiler 647 kW.*

*1. The specification is the column, not the split. Compare with shortcut01: there you declared the recoveries and got N; here you declare N, the feed stage, R and D — and the purity is the ∗answer∗. Both cases separate the same feed; only the question differs. 2. Watch it converge. The Log tab prints Pass max balance/F T_top T_bottom for the 41 bubble-point passes. The step and the balance fall by the same factor, about 0.65, every pass: that is what linear convergence looks like. The column stops on the balance, not the step — when every stage balance and the balance over its ports close to 1e-9 of the feed (here 7.4e-10, with a last step of 2.3e-9). A small step alone is not a closed balance: a column that contracts slowly takes small steps while still far from its answer. column02_simultaneous solves the identical column by a Newton on all the MESH equations in a handful of iterations — run both and count. 3. The profile is printed stage by stage. T and x for every tray, the feed tray marked, the reboiler last: 355.69 K and 88.3 % benzene in the liquid on tray 1 (the vapour leaving it, the distillate, is 95.1 %), 381.51 K and 95.1 % toluene in the reboiler. That table is the McCabe–Thiele diagram in numbers. 4. Two duties, two utilities, priced from the catalogue. The reboiler is served by low-pressure steam (0.300 kg/s, 28.31 ?/h at the catalogue price) and the condenser by cooling water (30.89 kg/s, 1.86 ?/h) — the same picker every steady report uses, and the ?/h is a catalogue number, not a market one. 5. Read the caveats. Both Antoine fits are asked slightly outside their declared windows on the way to the answer (benzene at 370 K, toluene at 380.7 K), announced and still returned.*

*Raise refluxRatio to 3: purities climb, both duties climb with them — the trade-off every column design turns on. Then try feedStage 4: the same reflux buys less separation when the feed lands in the wrong place.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | column01 | T_top | 355.689111404 |
| kpi | column01 | T_bottom | 381.514100345 |
| kpi | column01 | x_D_LK | 0.951017315221 |
| kpi | column01 | x_B_HK | 0.951017316705 |
| utility | column01 | heating.steamLP.duty_kW | 679.510516909 |
| utility | column01 | heating.steamLP.T | 381.514100345 |
| utility | column01 | heating.steamLP.kg_s | 0.310846531065 |
| utility | column01 | heating.steamLP.eur_h | 29.3548543305 |
| utility | column01 | cooling.coolingWater.duty_kW | -1315.28783372 |
| utility | column01 | cooling.coolingWater.T | 355.689111404 |
| utility | column01 | cooling.coolingWater.kg_s | 31.4662161178 |
| utility | column01 | cooling.coolingWater.eur_h | 1.89401448056 |
| stream | bottoms | F | 0.0138888888889 |
| stream | bottoms | T | 381.514100345 |
| *... and 19 more reference values in the case's `expected`.* | | | |

## column02_simultaneous

tutorials/steady/distillation/column02_simultaneous




**What it does:** Rigorous distillation by simultaneous MESH Newton (benzene/toluene; validates against column01)

**The story:** The SAME benzene/toluene column as column01, but solved by the rigorous SIMULTANEOUS-correction (MESH Newton) method instead of the sequential bubble-point (Wang-Henke). Selected by 'model simultaneous;'.

Pedagogical point: on an ideal system both methods give the SAME answer — this case validates the MESH solver against column01. But the simultaneous Newton converges in ~6 iterations (quadratic) vs ~100 for the bubble-point, and — the reason it exists — it stays stable on azeotropic systems where the sequential method does not.

Expected (identical to column01): T_top ≈ 354.2 K, T_bottom ≈ 382.9 K, x_D(benzene) ≈ 0.981, x_B(toluene) ≈ 0.981.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | R | 2 |
| kpi | column01 | T_bottom | 382.893527916 |
| kpi | column01 | T_top | 354.210890137 |
| kpi | column01 | feedStage | 8 |
| kpi | column01 | nStages | 15 |
| kpi | column01 | x_B_HK | 0.981245125322 |
| kpi | column01 | x_D_LK | 0.981245125322 |
| utility | column01 | heating.steamLP.duty_kW | 1324.55862478 |
| utility | column01 | heating.steamLP.T | 382.893527916 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## column03_azeotrope_mesh

tutorials/steady/distillation/column03_azeotrope_mesh




**What it does:** Azeotropic distillation (ethanol/water, NRTL) by the rigorous simultaneous MESH method

**The story:** Ethanol/water (NRTL) distillation by the rigorous SIMULTANEOUS (MESH Newton) method. This is the payoff of the v0.48 solver: ethanol/water forms an azeotrope at x_eth ~ 0.894 (1 atm), and the sequential bubble-point (Wang-Henke) is unstable here – on this very case it does NOT converge (200 outer iters). The MESH Newton converges in ~5 iterations to a PHYSICAL distillate (x_eth ~ 0.80, below the azeotrope).

Feed: 100 kmol/h, 30/70 ethanol/water, 1 atm, a SATURATED LIQUID – said once, by the stream: 'phase liquid;' at the feed's own bubble point under this case's NRTL, 354.42 K (the engine's bubbleT on this stream gives 354.4248 K; rounded DOWN so the feed's own equilibrium agrees with the pin). Until 2026-10-05 the stream said T = 363 K and the column said 'feedQuality 1.0': at 363 K this feed is a VAPOUR (V/F = 1 by the engine's own flash), so the case said two different things. The column priced a liquid, the energy report priced the vapour, and the plant boundary carried a 1149.69 kW residual – the latent heat of the feed – that this header used to describe as closing. The column now reads a feed by what it MEANS (DEV.md D1), so the contradiction refused, and the case was corrected to declare the saturated liquid it was designed around. A vapour feed at 363 K is NOT a drop-in alternative: with q = 0 this specification (R = 2.5, D = 25 kmol/h) leaves the stripping section without vapour, which the MESH reports as a non-positive flow. Column: 16 stages, feed at 8, R = 2.5, D = 25 kmol/h, method simultaneous.

Expected: converged; x_D(ethanol) ~ 0.80 (< 0.894 azeotrope – physical); T_top ~ 351.4 K (the azeotropic temperature); mass closes 100%.

NOTE: asking for a distillate ABOVE the azeotrope (x_eth > 0.894) is physically impossible – no method converges, and that is correct. The MESH method does not invent a non-physical "past the azeotrope" answer (which the bubble-point used to do).

*A 100 kmol/h ethanol/water feed, 30/70, into a 16-stage column with the feed on stage 8, R = 2.5, distillate 25 kmol/h, at 1.01325 bar, under the same NRTL parameters compare_activity_etoh_water plotted and flash02_ethanol_water used. Everything about that specification is ordinary. The one line that is not is in the unit's third slot: model simultaneous;.*

*The golden: x_D = 0.8046 ethanol, x_B = 0.8682 water, T_top 351.39 K, T_bottom 357.71 K, reboiler 956.24 kW, condenser -952.97 kW, and the plant-boundary first law closing at a residual of 0 kW. The feed is a saturated liquid at 354.42 K, declared as one in 0/feed (phase liquid; at its own bubble point).*

*1. type, then model, then operation — and here model is the numerical method. distillationColumn accepts WangHenke (the default, and what solved column01) or simultaneous, the rigorous MESH Newton that solves every stage equation at once. The operation block is unchanged in kind from column01: the same four specifications, the same grammar. Only the machine behind them was chosen differently, and this case exists because on this mixture that choice decides whether there is an answer at all. The case's own flowsheetDict header records that the sequential bubble-point method does not converge here. 2. The azeotrope is a ceiling, and the answer sits under it. Ethanol and water form an azeotrope the case names at roughly x_eth 0.894 at 1 atm; the distillate comes out at 0.804616545279 ethanol. A distillate specified above the azeotrope is not a hard convergence problem — it does not exist, and the header is explicit that no method reaches it. A solver that ∗did∗ return such a number would be reporting something physically impossible. 3. The temperature span is the azeotrope showing itself. Sixteen stages buy only 6.32 K from top to bottom here (351.39 K to 357.71 K). column01 separates benzene from toluene over 28.68 K on fifteen stages. Same kind of equipment, same kind of specification; the mixture is what collapsed the driving force, and the duties tell the same story — 956.24 kW of reboiler heat for a 25 kmol/h distillate. 4. The case asks for the profile, and the profile is where a column is read. controlDict declares reports { streamTable massBalance profiles spreadsheet }. The profiles report prints T, x and y stage by stage; read it tray by tray and you can see how much of the column is actually separating and how much of it is barely moving. A column is specified in four numbers and understood in that table.*

*5. A feed's state is said once, by the stream. Until 2026-10-05 this case said two things: 0/feed put the feed at 363 K, and the column said feedQuality 1.0 (a saturated liquid). At 363 K and 1 atm this mixture is all vapour under the case's own NRTL (it boils at 354.42 K, and the engine's own flash at 363 K finds V/F = 1), so the column priced a liquid while the energy report priced the vapour the stream is, and the plant boundary carried a 1149.69 kW residual – the latent heat of the whole feed. The engine now reads a feed by what it MEANS and refuses the contradiction; the case was corrected to declare the saturated liquid it was designed around, and the first law closes. Two things to take from that: a quantity with two homes will one day disagree with itself, and a vapour feed is not a drop-in alternative here – try deleting phase liquid; and setting T 363 K; in 0/feed: with q = 0 this specification leaves the stripping section without vapour, and the MESH says so by name.*

*Change the unit's model to WangHenke and run it. Watch the Log tab: the outer iterations march and max does not fall. That is the whole claim of this case, reproduced in front of you — and it is why "which solver" belongs in the model, not in a footnote. Then open cstr01_first_order, where composition changes for a reason that has nothing to do with phase equilibrium.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0208333333333 |
| stream | distillate | F | 0.00694444444444 |
| stream | feed | F | 0.0277777777778 |
| kpi | column | B | 0.0208333333333 |
| kpi | column | D | 0.00694444444444 |
| kpi | column | R | 2.5 |
| kpi | column | T_bottom | 357.713408954 |
| kpi | column | T_top | 351.388958309 |
| kpi | column | feedStage | 8 |
| kpi | column | nStages | 16 |
| kpi | column | x_B_HK | 0.868205515068 |
| kpi | column | x_D_LK | 0.804616545214 |
| utility | column | heating.steamLP.duty_kW | 1000.15110633 |
| utility | column | heating.steamLP.T | 357.713408954 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## column04_multifeed_sidedraw

tutorials/steady/distillation/column04_multifeed_sidedraw




**What it does:** Simultaneous MESH column: two feeds + a liquid side draw (rigorous MESH)

**The story:** Multiple feeds as FLOWSHEET STREAMS (‘inputs ( feed feed2 )‘) + a liquid side draw. operation.feeds maps each input stream to a stage (information follows the streams → the plant-boundary mass + energy balance closes).

*The rigorous simultaneous MESH column extended to multiple feeds and side draws — the most-used features of a commercial simulator column, on the glass-box Newton solver.*

*A benzene / toluene / o-xylene split (light / intermediate / heavy):*

*- primary feed — 100 kmol/h, benzene/toluene/o-xylene 0.40/0.40/0.20, enters high (stage 5); - second feed — 60 kmol/h, heavier (0.05/0.45/0.50), enters low (stage 11); - liquid side draw — 15 kmol/h pulled from stage 8, between the two feeds.*

*Mass closes (100 + 60 = 42 + 15 + 103 = 160 kmol/h). Q_condenser = -1469 kW, Q_reboiler = +1547 kW. The side draw does exactly what it should — bleed the intermediate component where it is most enriched on the trays.*

*The simultaneous MESH is a bubble-point-reduced Newton (CMO flows). The extension generalises the single fixed feed into a per-stage flow staircase: L[j], V[j] are rebuilt by a CMO march that adds each feed's liquid/vapour parts and subtracts each draw; every stage's material balance carries an F_j z_{F,j} source and (L_j+U_j), (V_j+W_j) outflows. The general balance reduces exactly to the old single-feed equations when there is one feed and no draw — so every existing distillation tutorial is unchanged (verified).*

*outputs ( distillate bottoms <draw1> <draw2> ... ) — side-draw streams follow distillate and bottoms, in stage order.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0286111111111 |
| stream | distillate | F | 0.0116666666667 |
| stream | feed | F | 0.0277777777778 |
| stream | intermediateCut | F | 0.00416666666667 |
| kpi | radfracLite | B | 0.0286111111111 |
| kpi | radfracLite | D | 0.0116666666667 |
| kpi | radfracLite | Q_condenser_kW | -1468.6891022 |
| kpi | radfracLite | Q_reboiler_kW | 1547.3829782 |
| kpi | radfracLite | R | 3 |
| kpi | radfracLite | T_bottom | 395.926996925 |
| kpi | radfracLite | T_top | 354.978489498 |
| kpi | radfracLite | feedStage | 5 |
| kpi | radfracLite | nFeeds | 2 |
| kpi | radfracLite | nSideDraws | 1 |

*... and 21 more reference values in the case's `expected`.*

## column05_reactive_methylacetate

tutorials/steady/distillation/column05_reactive_methylacetate




**What it does:** Reactive distillation: methyl acetate synthesis (validation vs P?pken et al. 2001, run S-1)

**The story:** Both feeds are FLOWSHEET STREAMS (‘inputs ( feed acidFeed )‘) – so the plant-boundary mass + energy balance sees them. ‘operation.feeds‘ only maps each stream to a stage (no inline composition – the credo: information follows the streams).

*The simultaneous MESH column with an equilibrium reaction on the catalytic stages (rigorous column rung 4), compared to a published experiment.*

*Mole-conserving, so the CMO internal-flow profile is untouched. Each catalytic stage carries one extra unknown — the molar extent ?_j — closed by the activity-based equilibrium residual Σ_i ν_i ln(γ_i x_i) = ln K_a(T), with γ_i from the liquid activity model and K_a(T) by van’t Hoff: K_a(T) = K_a(298)·exp(-Δh°_r/R·(1/T - 1/298.15)). Converged by homotopy (solve non-reactive, then switch the reaction on).*

*> T. P?pken, S. Steinigeweg & J. Gmehling, ∗"Synthesis and Hydrolysis of > Methyl Acetate by Reactive Distillation\{}ldots"∗, Ind. Eng. Chem. Res. 40 > (2001) 1566. Run S-1: N = 25, reactive stages 11–19, HOAc fed above / > MeOH below the zone, reflux 2.1, P ≈ 1.017 bar. Measured endpoints, cited, in > constant/experimental/methyl_acetate_RD_profile.csv.*

*The equilibrium constant and thermodynamics are from the companion paper P?pken, G?tze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601: K_a(298) ≈ 38.7, Δh°_r ≈ -5.67 kJ/mol → K_a ≈ 29 in the reactive zone (~65 °C).*

*This is not a tight match, and the reason is physically important: with the correct equilibrium K_a (≈29) the model gives the equilibrium ceiling (96 %), and it correctly brackets the measurement from above. The measured 87.6 % is ∗below∗ the equilibrium because the real column is kinetically limited — the paper’s recommended model is an adsorption-based finite ∗rate∗ (per gram of catalyst), not equilibrium. An earlier run with an (incorrect, too low) K_a = 5.2 gave 90.7 %, a closer ∗number∗ but for the wrong reason; using a wrong constant to fake agreement would be dishonest.*

*So the path to a bit-tight match is the kinetic rate model, not the thermo.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 9.13888888889e-06 |
| stream | distillate | F | 8.36111111111e-06 |
| stream | feed | F | 8.94444444444e-06 |
| kpi | rdColumn | B | 9.13888888889e-06 |
| kpi | rdColumn | D | 8.36111111111e-06 |
| kpi | rdColumn | Q_condenser_kW | -0.803941040437 |
| kpi | rdColumn | Q_reboiler_kW | 0.834961079318 |
| kpi | rdColumn | R | 2.1 |
| kpi | rdColumn | T_bottom | 363.715464674 |
| kpi | rdColumn | T_top | 330.01210931 |
| kpi | rdColumn | conversion | 0.955249509215 |
| kpi | rdColumn | feedStage | 20 |
| kpi | rdColumn | nFeeds | 2 |
| kpi | rdColumn | nSideDraws | 0 |

*. . . and 22 more reference values in the case’s `expected`.*

## column06_fullMESH

tutorials/steady/distillation/column06_fullMESH




**What it does:** Full-MESH (Naphtali-Sandholm) distillation – energy-consistent V_j/L_j, seeded from the CMO solution (benzene/toluene; compare column02)

**The story:** The SAME benzene/toluene column as column02_simultaneous, but solved by the FULL MESH (Naphtali-Sandholm), selected by 'model fullMESH;'. The full MESH promotes the internal flows V_j, L_j to UNKNOWNS and adds a per-stage total-mass + ENERGY balance, so the flows follow the REAL latent heats instead of the constant-molal-overflow (CMO) assumption behind the 'simultaneous' (bubble-point-reduced) method.

Robustness trick: the full MESH is SEEDED from the converged CMO profile — the cheap model bootstraps the rigorous one; ~4 Newton iterations from that seed.

Expected — close to, but NOT equal to, column02 (the gap IS the energy-balance correction the CMO drops): T_top ~ 354.4 K, T_bottom ~ 382.8 K, x_D(benzene) ~ 0.978 (CMO: 0.981). Benzene/toluene latent heats are similar (30.8 vs 33.2 kJ/mol) so the gap is small here; on a wide-boiling column the full MESH earns its cost.

*The same benzene/toluene column as column02_simultaneous, but solved by the full MESH (model fullMESH) instead of the bubble-point/CMO reduction.*

*The simultaneous method is a bubble-point-reduced MESH: it assumes constant molal overflow (CMO — equimolal latent heats), so the internal vapour/liquid flows V_j, L_j are fixed by a mass march and only the compositions x_j and stage temperatures T_j are solved.*

*The full MESH (Naphtali–Sandholm) promotes V_j and L_j to unknowns and adds, per stage, the total-mass balance and the energy balance — so the flows follow the ∗real∗ latent heats instead of assuming them equal. The extra per-stage energy equation is the enthalpy residual the CMO never carried:*

*What stage 1 is follows from the declared condenser type (condenser { type total/partial; }, default total). Under a TOTAL condenser stage 1 is the top tray and carries the same energy balance: the reflux R·D enters it as the condensate of its own vapour, at the bubble point of y_1, and V_1 = (R+1)·D leaves it to a condenser that is not an equilibrium stage and is not counted. Under a PARTIAL condenser stage 1 IS the condenser, and its rows fix V_1 = 0, L_1 = R·D in place of an energy residual. The reboiler fixes L_N = B. Q_cond (the condenser's own envelope) and Q_reb (the reboiler's) are results.*

*h_L and h_V are the enthalpies the column's STREAMS are priced on – the formation datum, each phase at its own composition, the vapour with its equation-of-state departure – so every stage balance, both duties and the stream table are one surface (since 2026-10-26, DEV.md C169; until then the rows used a sensible datum with a Watson latent heat, and a stage that closed on that surface left 3.94 kW open on the streams'). bin/curate/check_column_thermo_consistency.py recomputes every stage.*

*The full-MESH is seeded from the converged CMO profile — the cheap model launches the rigorous one. Here it converges in ~4 Newton iterations from that seed (it would be far harder from a cold start).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column06 | B | 0.0138888888889 |
| kpi | column06 | D | 0.0138888888889 |
| kpi | column06 | Q_condenser_kW | -1308.09838097 |
| kpi | column06 | Q_reboiler_kW | 1325.30888531 |
| kpi | column06 | R | 2 |
| kpi | column06 | T_bottom | 382.762148261 |
| kpi | column06 | T_top | 354.394223856 |
| kpi | column06 | feedStage | 8 |
| kpi | column06 | nFeeds | 1 |
| kpi | column06 | nSideDraws | 0 |
| kpi | column06 | nStages | 15 |
| *. . . and 17 more reference values in the case's `expected`.* | | | |

## column07_naphtaliSandholm

tutorials/steady/distillation/column07_naphtaliSandholm




**What it does:** Full-MESH (Naphtali-Sandholm) validation vs AIChE J. 17 (1971) 148, Problem 2: nC4/iC5/nC5

*Reproduces the ideal-hydrocarbon distillation of Naphtali & Sandholm, "Multicomponent Separation Calculations by Linearization", AIChE Journal 17(1), 148–153 (1971) — DOI [10.1002/aic.690170130](https://doi.org/10.1002/aic.* *— the paper that introduced the very full-MESH method Choupo's model fullMESH implements.*

*n-butane / isopentane / n-pentane, 20 plates, liquid feed ($T = 30\ °F = 272.04$ K, 500 mol of each) on plate 10, reflux ratio 1000, 80 % of the feed recovered in the bottoms ($B = 1200$). A liquid sidestream on plate 1 (the condenser) takes 30 % of the reflux.*

*Equivalent spec used here. The distillate (≈1) and the condenser sidestream (≈299) leave the ∗total∗ condenser with the same composition, so they are combined into ONE top product of 300 with reflux $R = L_1/300 = 996.68/300 = 3.3223$. This gives identical internal flows ($L_1 = 996.68$, $V_2 = 1296.68$), hence the identical column — without needing a condenser side draw.*

*Converges in 3 full-MESH Newton iterations (seeded from the CMO profile).*

*Products Choupo —— top product (300) 99.99 % n-$C_4$ bottoms n-$C_4$ / i-$C_5$ / n-$C_5$ 200 / 500 / 500 B 1200*

*The profile shape — the flat n-butane-rich top, the sharp rise across the feed zone, the pentane-plateau, the reboiler climb — matches stage-by-stage. The ~1–2 K residual is thermo, not method: Choupo uses fitted Antoine vapour pressures, the paper used the K-value tables in Brown's ∗Unit Operations∗ (1950).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.416666666667 |
| kpi | column07 | B | 0.333333333333 |
| kpi | column07 | D | 0.0833333333333 |
| kpi | column07 | Q_condenser_kW | -7934.42286361 |
| kpi | column07 | Q_reboiler_kW | 9226.13269031 |
| kpi | column07 | R | 3.3223 |
| kpi | column07 | T_bottom | 296.58946842 |
| kpi | column07 | T_top | 272.663334648 |
| kpi | column07 | feedStage | 10 |
| kpi | column07 | nFeeds | 1 |
| kpi | column07 | nSideDraws | 0 |
| kpi | column07 | nStages | 20 |
| kpi | column07 | x_B_HK | 0.416665691626 |
| kpi | column07 | x_D_LK | 0.999874145869 |

*. . . and 17 more reference values in the case's `expected`.*

## column08_radfrac_multidraw

tutorials/steady/distillation/column08_radfrac_multidraw




**What it does:** rigorous column: multi-feed + multi-side-draw distillation (C3/C4/C5/C6), full-MESH

*A C3/C4/C5/C6 depropanizer-style column at 16 bar, split four ways by the rigorous full-MESH, exercising the two most-used rigorous column features at once: TWO feeds (a light cut high, a heavy cut low) and TWO liquid side draws.*

*Two saturated-liquid feeds (100 kmol/h each, 290 K / 320 K): a light cut on stage 10, a heavy cut on stage 20. 30 stages, R = 4. Mass closes exactly (200 = 45 + 55 + 55 + 45 kmol/h) and the energy balance closes: the feed-to-product enthalpy gap (1137.7 kW) equals the net duty $Q_{\{}text{reb}}+Q_{\{}text{cond}} = 1942.3 - 804.6$ kW. Converges in a few full-MESH iterations from the CMO seed.*

*This case was first written at 1 atm, and the energy balance would not close — because at 1 atm and 290 K the light hydrocarbons (propane b.p. 231 K, n-butane 272 K) are above their boiling points: the feed streams flash to VAPOUR (vf = 1.0), yet the column was told quality 1.0 (saturated liquid). The latent heat of the unrecognised vapour feed was exactly the missing ~1 MW. At 16 bar the feeds are genuinely liquid (vf = 0, consistent with quality 1.0) and the balance closes — which is also why a real depropanizer runs at elevated pressure: so the propane distillate condenses against cooling water (here 321.6 K). A feed's thermal state must be specified consistently with its actual phase.*

*Both feeds are real flowsheet streams (inputs ( feed1 feed2 ), mapped to stages by operation.feeds), so the plant-boundary mass + energy balance sees them (∗information follows the streams∗). Side draws are outputs in stage order: outputs ( distillate bottoms drawC4 drawC5 ).*

*Self-consistently validated (mass + energy close, four cuts in volatility order) on the full-MESH solver that IS paper-validated against Naphtali & Sandholm (1971) in column07_naphtaliSandholm. No external multi-feed+multi-draw single-column benchmark with published profiles is readily available in the open literature.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0125 |
| stream | distillate | F | 0.0125 |
| stream | drawC4 | F | 0.0152777777778 |
| stream | drawC5 | F | 0.0152777777778 |
| stream | feed1 | F | 0.0277777777778 |
| stream | feed2 | F | 0.0277777777778 |
| kpi | radfrac | B | 0.0125 |
| kpi | radfrac | D | 0.0125 |
| kpi | radfrac | Q_condenser_kW | -912.633625877 |
| kpi | radfrac | Q_reboiler_kW | 1801.00793227 |
| kpi | radfrac | R | 4 |
| kpi | radfrac | T_bottom | 452.977133674 |
| kpi | radfrac | T_top | 322.21378999 |
| kpi | radfrac | feedStage | 10 |

*. . . and 23 more reference values in the case's `expected`.*

## column09_tray_hydraulics

tutorials/steady/distillation/column09_tray_hydraulics




**What it does:** Sieve-tray hydraulics: the column that separates, sized so its trays do not flood

**The story:** column01, asked the question a converged MESH solution never answers: CAN THE TRAYS PASS THE TRAFFIC?

A converged column tells you what separates. It says nothing about whether the vapour rising through the holes will carry the liquid up with it (entrainment flooding), whether the liquid will back up the downcomer until it reaches the tray above (downcomer flooding), or whether it will simply rain through the holes instead of bubbling (weeping). A column that floods separates nothing, and the stage equations never notice.

Add an ‘hydraulics {}‘ block and the column runs a RATING pass on its own converged profile. Two modes, the same equations in opposite directions:

give ‘diameter‘ → RATING. How close to flood is each tray? omit ‘diameter‘ → DESIGN. What diameter keeps every tray at or below ‘floodFraction‘ of its flooding velocity? The widest tray sets the column. That is this case.

Nothing here is guessed. The orifice coefficient C_o is a chart in the literature (Liebson, Kelley & Bullington 1957, against hole-area ratio and plate-thickness/hole-diameter); it is a DECLARED number you own, printed back at you. The weep-point constant K2 is a chart too – leave it out and the column tells you the weep check was not performed rather than inventing it.

Read the printed table. The feed is a saturated liquid, so it joins the liquid traffic and leaves the vapour traffic alone: L doubles across the feed (0.0278 → 0.0556 kmol/s) while V does not move (0.0417 kmol/s). The flow parameter F_LV therefore doubles too (0.0405 on tray 7, 0.0812 on tray 8), which lowers Fair’s capacity parameter and with it the flooding velocity (1.545 → 1.435 m/s). The vapour velocity does NOT fall to compensate – it rises, 1.044 → 1.092 m/s – so the STRIPPING trays are the tight ones here: the flood approach climbs monotonically from 63.7 % on the top tray to 80.0 % on stage 14, and stage 14 is the tray that sets the 1.316 m tower.

(This paragraph used to say the stripping trays were NOT the tight ones, "because the vapour volumetric flow falls faster than the capacity". The run’s own table says otherwise on both counts, and it always did – the vapour velocity rises across the feed and the flood approach never once falls. Corrected 2026-09-07, against the printed table rather than against the argument.)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column09 | B | 0.0138888888889 |
| kpi | column09 | D | 0.0138888888889 |
| kpi | column09 | L_rect | 0.0277777777778 |
| kpi | column09 | L_strip | 0.0361877253979 |
| kpi | column09 | Q_condenser_kW | -1315.28783372 |
| kpi | column09 | Q_reboiler_kW | 679.510516909 |
| kpi | column09 | R | 2 |
| kpi | column09 | T_bottom | 381.514100345 |
| kpi | column09 | T_top | 355.689111404 |
| kpi | column09 | V_rect | 0.0416666666667 |
| kpi | column09 | V_strip | 0.022298836509 |
| *... and 92 more reference values in the case’s `expected`.* | | | |

## column10_flooding

tutorials/steady/distillation/column10_flooding




**What it does:** The same column, built too narrow: every tray floods – and it still converges

**The story:** the SAME column as column09, built 1.10 m wide instead of the 1.316 m that column09 designed for it.

Nothing about the separation changes. The distillate is still 98.12 % benzene, the bottoms still 98.12 % toluene, the MESH equations converge exactly as before, and every number in the stream table is identical. The column is a perfectly good column – on paper.

It floods. The worst tray runs at 114.5 % of its entrainment-flooding velocity: the vapour is fast enough to carry the liquid up to the tray above instead of letting it fall. In a real column the trays would fill, the pressure drop would run away, and the separation printed above would not happen at all.

This is the whole reason hydraulics exists as a separate pass. A converged solution is not a feasible column, and the stage equations have no way of knowing the difference. Widen it (column09), space the trays further apart, or cut the reflux – but do not believe the stream table until a tray can pass the traffic it carries.

Note the downcomer, too: the backup reaches 268.5 mm against a 275 mm limit. Narrow the column a little further and it would flood from below as well, for an entirely different reason.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column10 | B | 0.0138888888889 |
| kpi | column10 | D | 0.0138888888889 |
| kpi | column10 | L_rect | 0.0277777777778 |
| kpi | column10 | L_strip | 0.0361877253979 |
| kpi | column10 | Q_condenser_kW | -1315.28783372 |
| kpi | column10 | Q_reboiler_kW | 679.510516909 |
| kpi | column10 | R | 2 |
| kpi | column10 | T_bottom | 381.514100345 |
| kpi | column10 | T_top | 355.689111404 |
| kpi | column10 | V_rect | 0.0416666666667 |
| kpi | column10 | V_strip | 0.022298836509 |

*. . . and 31 more reference values in the case's `expected`.*

## column11_murphree

tutorials/steady/distillation/column11_murphree




**What it does:** Murphree tray efficiency: 14 REAL trays plus the reboiler (nStages 15, reboiler counted), not ideal stages – and the purity you actually get

**The story:** the same 15-stage benzene/toluene column as column09, with ONE line added: murphreeEfficiency 0.70.

Every column before this one is built from EQUILIBRIUM stages: the vapour leaving a tray is in equilibrium with the liquid leaving it. No real tray does that. The vapour spends a finite time in the froth and comes off somewhere between what entered it and what equilibrium would give. Murphree measured the shortfall on the vapour side:

y_j = E_MV . y*_j + (1 - E_MV) . y_{j+1}, y*_j = K_j x_j

With E_MV = 1 the tray is an ideal stage and nothing changes – every existing case still runs byte for byte. With E_MV < 1, 'nStages' stops counting ideal stages and starts counting REAL TRAYS. Which matters, because the hydraulics pass below sizes those trays: until now we were dimensioning, to the millimetre, a tray we then assumed was perfect.

E_MV is a DECLARED number. It comes from a correlation (O'Connell), a chart, or plant data, and it is yours to own – exactly like the orifice coefficient C_o and the weep constant K2 in the hydraulics block. Choupo will not invent it.

Compare against column09 (same column, E_MV = 1):

distillate benzene 98.12 % (ideal) → 95.05 % (E_MV = 0.70) bottoms toluene 98.12 % (ideal) → 95.05 %

Fourteen real trays do less than fourteen ideal ones. That is the whole content of a tray efficiency, and it is why a column designed on ideal stages and built on real trays comes up short. Sweep E_MV and watch the limit close:

E_MV 0.50 0.70 0.90 0.99 0.999 1.000 x_D 0.9100 0.9505 0.9738 0.9806 0.9812 0.98125 <- the ideal stage

Nothing is switched on at E_MV = 1; the effective K collapses to K and the case reproduces column09 to the last bit.

NOTE: the efficiency is folded into an effective K, with the vapour from the tray below lagged one outer pass – the linear solve keeps its tridiagonal shape. At convergence the lag vanishes and the Murphree relation holds exactly. The MESH 'simultaneous' method REFUSES the keyword rather than quietly returning ideal stages: there the relation would have to enter the residual and Jacobian.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column11 | B | 0.0138888888889 |
| kpi | column11 | D | 0.0138888888889 |
| kpi | column11 | L_rect | 0.0277777777778 |
| kpi | column11 | L_strip | 0.0361877253979 |
| kpi | column11 | Q_condenser_kW | -1320.17731859 |
| kpi | column11 | Q_reboiler_kW | 682.299285003 |
| kpi | column11 | R | 2 |
| kpi | column11 | T_bottom | 380.054233859 |
| kpi | column11 | T_top | 356.362521335 |
| kpi | column11 | V_rect | 0.0416666666667 |
| kpi | column11 | V_strip | 0.022298836509 |
| *. . . and 32 more reference values in the case's `expected`.* | | | |

## column12_stage_is_a_flash

tutorials/steady/distillation/column12_stage_is_a_flash




**What it does:** An equilibrium stage IS a flash: a tray's own two products, recombined and re-flashed, must come back as the same tray

**The story:** AN EQUILIBRIUM STAGE IS A FLASH – the identity witness.

A column's MESH equations and a flash drum's Rachford-Rice are two DIFFERENT pieces of code that claim to compute the same thing: the equilibrium split of a mixture at (T, P). Nothing in the corpus made them say it to each other. This case does.

feed -→ [tower] -→ distillate -→ bottoms -→ stageLiquid (the liquid LEAVING stage 5) -→ stageVapour (the vapour LEAVING stage 5)

stageLiquid + stageVapour -→ [recombine] -→ stageMix stageMix -→ [restage] -→ checkLiquid, checkVapour

THE CLAIM. Stage 5 is an equilibrium stage: by definition the vapour leaving it is in equilibrium with the liquid leaving it. So if the two are put back together and the mixture is allowed to separate again, at the same pressure and with no heat added or removed, NOTHING must happen: the same temperature, the same two compositions, and a vapour fraction equal to the ratio of the two draws.

T(checkLiquid) == T(stageLiquid) == T(stageVapour) z(checkLiquid) == z(stageLiquid) z(checkVapour) == z(stageVapour) F(checkVapour) / (F(checkLiquid) + F(checkVapour)) == W / (U + W)

Nothing above is a number anyone typed. Every one of them is produced by the run, on both sides, and the gate 'check_stage_identity' compares them. That is what makes this a witness rather than a demonstration.

WHY IT IS WORTH A CASE. The column asks the thermodynamic package for K-values per stage; the flash asks it for an equilibrium split. Those are different entry points and, for a package that resolves CHEMISTRY, they are different code. The effective K-value a reacting stage uses ('ThermoPackage::stageK', added 2026-08-04 for the sour-water column) is exactly the sort of definition that can be subtly wrong – wrong normalisation, wrong basis, the reaction counted twice – and still produce a column that converges to a plausible-looking profile. This identity is what catches it. The molecular case here is the CONTROL: it must hold on a system where nobody doubts the answer, before the same construction is trusted on one where they should.

HOW THE RECOMBINATION WORKS. 'recombine' is an ADIABATIC mixer, and since 2026-10-24 its outlet is the state of equal enthalpy RESOLVED at its own equilibrium, priced as the energy report prices it. Handed the two halves of stage 5 it publishes the stage itself – its temperature and V/F = W / (U + W) – one unit early; it does not SEPARATE the phases, 'restage' does. 'restage' is an ADIABATIC flash, so enthalpy is precisely the quantity it reads back, and the identity is therefore an ENERGY statement as well as an equilibrium one: it would fail if the column and the flash disagreed about the enthalpy of either phase. (Before that date the mixer published one dominant-phase label at a fictitious 182.9 K and carried only the enthalpy across; the identity held then too, to the flash's outer tolerance.)

WHY THE DRAWS ARE ON ONE STAGE AND NOT TWO. Both draws come off stage 5. A liquid draw from stage 5 and a vapour draw from stage 6 would also recombine into something – and it would not be an equilibrium mixture, so the flash would move it. The point of the case is that when the two phases genuinely leave the SAME stage, the flash cannot move them at all.

*What it demonstrates: that Choupo's two independent equilibrium routines — the column's MESH equations and the flash drum's Rachford-Rice — give the same answer, by making them say it to each other inside one run.*

*Stage 5 is an equilibrium stage, so the vapour leaving it is in equilibrium with the liquid leaving it. Put them back together, let them separate again at the same pressure with no heat added, and nothing may happen. The run must produce*

*Not one of those numbers is typed into the case. Both sides are computed, by different code, and bin/curate/check_stage_identity.py compares them.*

*Three things at once, and each of them has been wrong in a simulator somewhere:*

*1. The K-value definition. The column asks the package for K-values per stage; the flash asks it for a split. A wrong normalisation or a wrong basis in one of them shifts the profile without ever making the column fail to converge. 2. The enthalpy of each phase. The recombination carries an ∗enthalpy∗ across one arrow and the adiabatic flash reads it back. If the column and the flash priced the vapour differently, the recovered temperature would be wrong even with perfect K-values. 3. That a two-phase inlet is priced as two phases. Building this case is what found adiabaticFlash pricing every inlet as a sub-cooled liquid regardless of its vapour fraction — silently, since no case in the corpus had ever fed it one.*

*Since 2026-10-24 the identity holds to round-off: restage publishes T_drop 0 and H_residual 0, and lands on V/F = 0.5 exactly. Until then it held to about 1e-7 relative – the adiabatic flash stopped its outer Newton at an energy residual of ~2.5e-4 J/mol, because recombine handed it a single-phase stream at a temperature 189 K away from the stage's (see below), and the flash had to walk all the way back. The gate is still set at 1e-6 relative.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.00833333333333 |
| stream | checkLiquid | F | 0.00277777761205 |
| stream | checkVapour | F | 0.00277777794351 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| stream | stageLiquid | F | 0.00277777777778 |
| stream | stageMix | F | 0.00555555555556 |
| stream | stageVapour | F | 0.00277777777778 |
| kpi | recombine | F_out | 0.00555555555556 |
| kpi | recombine | P_out | 101325 |
| kpi | recombine | T_out | 371.883911726 |
| kpi | recombine | n_in | 2 |
| kpi | recombine | vf | 0.5 |
| kpi | restage | F_in | 0.00555555555556 |
| *... and 41 more reference values in the case's `expected`.* | | | |

## column13_sour_water_stage_identity

tutorials/steady/distillation/column13_sour_water_stage_identity




**What it does:** The stage identity under CHEMISTRY: a sour-water tray's own two products, recombined and re-flashed, must come back as the same tray

**The story:** THE STAGE IDENTITY, UNDER CHEMISTRY.

This is column12 ('column12_stage_is_a_flash') with one thing changed: the thermodynamic package resolves an aqueous EQUILIBRIUM. Read that case first – the construction, the mixer's fictitious temperature and the tolerance are all explained there and not repeated here.

feed -→ [tower] -→ distillate, bottoms -→ stageLiquid, stageVapour (stage 2) stageLiquid + stageVapour -→ [recombine] -→ stageMix stageMix -→ [restage] -→ checkLiquid, checkVapour

WHY IT IS A DIFFERENT TEST FROM column12, and not a copy.

In a molecular world a stage's K-values and a flash's split come from the same handful of lines: 'ThermoPackage::stageK' forwards straight to 'Kvec', and the identity, while worth having, is checking arithmetic that was never in doubt.

Here it is not the same code. A package that resolves chemistry has no K-value to hand out: the volatility of NH3 is set by how much of it is NH4+, which depends on the pH, which depends on how much CO2 has left. 'stageK' therefore solves the stage's own equilibrium at its own (T, P, z) and returns an effective APPARENT K – a definition, made by us, which nothing outside this case checks. A wrong normalisation, a basis confusion between apparent components and species, or the speciation counted twice would all produce a column that converges to a plausible profile. This case is what refuses it.

AND THE DEFINITION WAS WRONG ONCE, WHICH IS THE ARGUMENT FOR THE CASE. The first version read K = y/x off the flash. That is right whenever the stage is two-phase, and it hands back a column of ZEROS the moment the trial state is subsaturated – which a column solver's trial states usually are, on the way to the answer. Every bubble-point residual then reads -1, the Jacobian is singular in T, and the MESH stops without moving: "Newton iters: 0". A K-value is an INCIPIENT quantity, defined over a liquid whether or not a vapour happens to be there, so the subsaturated branch now uses the equilibrium partial pressures over the fully speciated liquid, K_i = (p_i/P)/x_i. The two branches agree where they meet, because at a two-phase state the partials sum to P.

WHAT MUST HOLD, all of it produced by the run and none of it authored:

T(checkLiquid) == T(stageLiquid) == T(stageVapour) z(checkLiquid) == z(stageLiquid) (apparent component basis) z(checkVapour) == z(stageVapour) F(checkVapour)/(F(checkLiquid)+F(checkVapour)) == W/(U+W)

and, in addition to column12, the thing only a reacting stage can be asked: the two liquids must carry the SAME SPECIATION. The column's stage liquid and the re-flashed liquid are the same material at the same state, so their ion inventories must agree species by species – not merely their apparent compositions. That is the check that would catch an effective K which happens to reproduce the apparent split while resting on a different chemistry underneath.

THE COST, said out loud. Every stage K-value evaluation is a full reactive flash, and the MESH Jacobian is numerical, so one Newton pass costs of the order of (stages x unknowns) reactive flashes. The column is FOUR stages for that reason and no other: this is an identity witness, not a design study. A student who wants a real sour-water stripper wants the S2 case, which does not exist yet.

WHAT THE COLUMN ACTUALLY DOES, since it is worth reading the profile. It converges in 7 Newton iterations to |F| ~ 8e-10, and the answer is the physics: NH3 climbs from 0.0054 in the reboiler liquid to 0.30 in the overhead vapour while CO2 strips out ahead of it, and T falls from 371.5 K at the bottom to 354.6 K at the top. No gammaPhi column can produce that profile, because in a molecular world ammonia's volatility does not depend on how much carbon dioxide has left the liquid.

*What it demonstrates: a distillation column solved under an aqueous equilibrium, and the proof that the effective K-value it uses per stage is the same equilibrium a flash computes — checked through two independent code paths, on the apparent components ∗and∗ on the ions.*

*Read column12_stage_is_a_flash first. This is the same construction; only the thermodynamic package differs.*

*Four stages, sour water ($NH_3$ + $CO_2$ in water), 1 atm, R = 2, D = 3 kmol/h. It converges in 7 Newton iterations to ?F? ≈ 8e-10, and the profile is the physics:*

*No gammaPhi column can produce that profile, because in a molecular world ammonia's volatility does not depend on how much carbon dioxide has left the liquid. Here it does: the carbon dioxide a liquid still carries is what holds its ammonia as $NH_4$?, so stripping the $CO_2$ sets the ammonia free and the free ammonia then strips. (The usual shorthand for that is "stripping $CO_2$ raises the pH" — which is true of a fixed solution and is ∗not∗ how it reads tray by tray here. See the per-tray table below: it is the carbonate loading that orders cleanly, not the pH.)*

*Both halves of stage 2 are drawn as real streams, recombined, and re-flashed at the same (T, P). Nothing may happen — and nothing does, to about 1e-9 relative:*

*The ion inventories agreeing is the claim that a comparison of apparent compositions alone would miss: an effective K could reproduce the apparent split while resting on a different chemistry underneath.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0259166666667 |
| stream | checkLiquid | F | 0.00041666665957 |
| stream | checkVapour | F | 0.000138888895986 |
| stream | distillate | F | 0.000833333333333 |
| stream | feed | F | 0.0273055555556 |
| stream | stageLiquid | F | 0.000416666666667 |
| stream | stageMix | F | 0.000555555555556 |
| stream | stageVapour | F | 0.000138888888889 |
| kpi | recombine | F_out | 0.000555555555556 |
| kpi | recombine | P_out | 101325 |
| kpi | recombine | T_out | 483.325341357 |
| kpi | recombine | n_in | 2 |
| kpi | recombine | vf | 0 |
| kpi | restage | F_in | 0.000555555555556 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## column14_klemola_c4_splitter

tutorials/steady/distillation/column14_klemola_c4_splitter




**What it does:** Klemola-Ilme industrial C4 splitter: 74 real trays at the measured E_MV = 1.191, products anchored on the plant's own key-component analysis

**The story:** an INDUSTRIAL i-butane/n-butane fractionator, 74 real trays, 2.9 m diameter, run on its own measured specification.

PRIMARY, and the only source used for any number below:

Klemola, K. T.; Ilme, J. K. Distillation Efficiencies of an Industrial-Scale i-Butane/n-Butane Fractionator. Ind. Eng. Chem. Res. 1996, 35 (12), 4579-4586. IE960390R.

The transcription record – Tables 1 and 2 number by number, the efficiency results, and the four things the paper does NOT publish – is docs/design/klemola-c4-splitter-reference-case.md. No number in this case is absent from that record; where the record says the paper is silent, this header DECLARES an assumption and says it is one.

WHY THIS CASE EXISTS. The corpus's other distillation tutorials are pinned on self-recorded goldens; the closest thing to an external anchor (acetone06/07) compares Choupo against another SIMULATION. This is the first column anchored on a measured plant. What it puts under test is the COLUMN MODEL – the MESH solve, the declared tray efficiency, the material balance – not an activity model: the system is nearly ideal, so a disagreement here cannot be blamed on thermodynamics and then forgotten.

1. WHAT IS TRANSCRIBED, PER NUMBER (record Sections 2 and 3, Tables 1 / 2)

number of trays 74 Table 1 → nStages 75 (see 2) feed tray 37 Table 2 → feedStage 37 (see 2) column top pressure 658.6 kPa Table 2 → P (see A2) feed pressure 892.67 kPa Table 2 → used in A1 only feed flow rate 26234 kg/h Table 2 → 0/feed (see 3) distillate flow rate 8115 kg/h Table 2 → distillateRate (see 4) reflux flow rate 92838 kg/h Table 2 → refluxRatio (see 5) bottoms flow rate 18119 kg/h Table 2 → NOT an input (see 6) Murphree tray efficiency 119.1 % text p.4580 → murphreeEfficiency reflux temperature 18.5 C Table 2 → NOT USED (see 7) boiler duty 10.240 MW Table 2 → NOT an input (see 8)

Table 1's tray geometry (2.900 m diameter, 0.600 m spacing, 4.9 m2 active area, 772 valves, 18.82 % free hole area, ...) is transcribed in the record for the reader and is used by NOTHING here: no hydraulics block is declared, because the anchor is the separation, not the rating. Ballast V-1 valve trays are not the sieve tray Choupo's hydraulics pass models, and rating them with a sieve correlation would be a number with no source behind it.

2. STAGE COUNT AND FEED STAGE – and one AMBIGUITY, declared

Choupo's Wang-Henke cascade numbers stage 1 = TOP TRAY and stage N = REBOILER; the total condenser sits outside the stage list (the distillate is y_1, the vapour leaving the top tray, condensed). So 74 real trays plus the reboiler gives

nStages 75 (= 74 trays + 1 reboiler) feedStage 37 (Table 2, taken as counted from the TOP)

ASSUMPTION A0 – the paper prints "feed tray 37" and does not state the direction of its tray numbering; the record does not either, and the primary text was checked. Counted from the bottom instead, the feed would sit on tray 38 of 74 as Choupo numbers them: 37 vs 38 rectifying trays out of 74. Stage 1 = top is taken because it is Choupo's own convention and because the difference is one tray in the middle of a 74-tray column. Anyone who establishes the paper's direction should change this line and say so.

3. THE FEED – wt % to mole flows, arithmetic shown

Table 2 gives the feed as 26234 kg/h at these wt % (the paper's own units and its own rounding to 0.1 wt %). THE COLUMN SUMS TO 100.1, NOT 100 – that is the rounding, and it has to be resolved before anything can be converted. It is resolved by NORMALISING the wt % to unity and applying them to the measured 26234 kg/h, so that the total mass flow – which the paper's own overall balance closes exactly

(18119 + 8115 = 26234) – is preserved. The alternative (applying the raw wt % literally) would invent 26 kg/h of feed.

MW from each component's own data/standards record. kmol/h = kg/h / MW.

component wt% wt% norm kg/h MW kmol/h x

propane 1.5 1.49850 393.1169 44.0960 8.915024 0.0196925 isoButane 29.4 29.37063 7705.0909 58.1222 132.567090 0.2928294 nButane 67.7 67.63237 17742.6753 58.1230 305.260832 0.6742952 1Butene 0.2 0.19980 52.4156 56.1063 0.934219 0.0020636 isoButene 0.2 0.19980 52.4156 56.1063 0.934219 0.0020636 trans2Butene 0.1 0.09990 26.2078 56.1063 0.467110 0.0010318 neopentane 0.1 0.09990 26.2078 72.1488 0.363246 0.0008024 isopentane 0.8 0.79920 209.6623 72.1488 2.905971 0.0064190 nPentane 0.1 0.09990 26.2078 72.1488 0.363246 0.0008024

TOTAL 100.1 100.00000 26234.0000 452.710957 1.0000000

mean feed MW = 26234 / 452.710957 = 57.94867 kg/kmol

Those kmol/h are what 0/feed carries.

ASSUMPTION A1 – THE FEED THERMAL STATE. Table 2 gives the feed PRESSURE (892.67 kPa) and no feed temperature; record Section 5.2 names this gap and names "saturated liquid at the feed pressure" as the natural assumption. That is what is assumed here, and it has a consequence the case must carry rather than hide: a liquid saturated at 892.67 kPa is NOT saturated on a tray at 658.6 kPa – it flashes on the way in. Both steps were computed with THIS CASE'S OWN thermo (the dict in constant/, no other model), by two throwaway runs whose numbers are reproduced here so a reader can repeat them:

bubbleT of the feed composition at 892.67 kPa → T = 334.4218 K adiabaticFlash of that stream to 658.6 kPa → V/F = 0.0929333445 T = 323.5875148 K

feedQuality q = 1 - V/F = 0.9070666555

and 0/feed carries the COLUMN-INLET state (323.5875148 K, 658600 Pa; the V/F is not declared – a mixture's (T, P) fix it, and the engine resolves it), not the line state. The paper's 892.67 kPa therefore appears in this case only as the premise of A1, stated above. The 'feedQuality' key below is a CROSS-CHECK against the feed stream's resolved state, which the column reads (2026-09-12): a key that contradicts it is refused by name, so the two cannot drift apart in silence. (Until 2026-10-03 the stream file also declared 'vaporFraction 0.0929333445' – a third state variable beside T and P, which the reader now refuses as over-specified.)

Sensitivity, MEASURED rather than asserted – the competing assumption was run. Declaring instead "saturated liquid AT THE COLUMN PRESSURE" gives T = 323.2706 K (bubble point at 658.6 kPa) and q = 1.0 exactly, and the whole column moves by:

x_D(i-butane) 0.92571 → 0.92576 (+5e-5) x_B(i-butane) 0.00352 → 0.00350 (-2e-5)

i.e. below the resolution of anything this case compares against. A1 is therefore chosen on FIDELITY TO THE RECORD, not on the answer, and the simpler assumption is available to anyone who prefers one derived number to three.

4. DISTILLATE RATE – a mass spec the engine takes in moles

Table 2 gives 8115 kg/h; 'distillateRate' is a MOLAR flow. The conversion needs a mean MW, and the only one available is the one from the paper's own MEASURED top composition (top wt % of Table 2, which sums to 100.0 exactly):

component wt% kg/h MW kmol/h y

propane 5.3 430.0950 44.0960 9.753606 0.0686760 isoButane 93.5 7587.5250 58.1222 130.544353 0.9191740 nButane 0.2 16.2300 58.1230 0.279235 0.0019661 1Butene 0.4 32.4600 56.1063 0.578545 0.0040736 trans2Butene 0.6 48.6900 56.1063 0.867817 0.0061104 i-butene / neopentane / i-pentane / n-pentane: 0 wt % as printed

TOTAL 100.0 8115.0000 142.023556 1.0000000

mean distillate MW = 8115 / 142.023556 = 57.13841 kg/kmol → distillateRate 142.023556 kmol/h

SAID PLAINLY: this input therefore borrows the measured distillate COMPOSITION, which is one of the quantities the case then predicts. A fully self-consistent conversion would iterate the mean MW against the predicted composition. It is not iterated here, and the size of what is left on the table is MEASURED rather than waved at: the run's own converged/distillate comes out at mean MW 57.2186, i.e. +0.14 % on the 57.13841 used above, so the distillate leaves at 8126.4 kg/h against the Table-2 8115 kg/h. One iteration would close that; it is left open because the residue is smaller than the 0.1 wt % resolution of the composition it came from. If a future edit moves the distillate composition materially, this number must be recomputed with it.

5. REFLUX RATIO – why the MASS ratio IS the MOLAR ratio here

Table 2 gives reflux 92838 kg/h and distillate 8115 kg/h, both MASS. The condenser is TOTAL, so the reflux and the distillate are drawn from one condensed stream and have IDENTICAL composition, hence identical mean MW. L/D in moles = (L_mass/MW)/(D_mass/MW) = L_mass/D_mass exactly – no composition-weighted conversion is needed, and none is done:

R = 92838 / 8115 = 11.440296 (11.4402957...)

A NOTE ON A NUMBER IN CIRCULATION: 11.4441 has been quoted for this ratio. It is not what the division gives. 8115 x 11.4441 = 92859.9 kg/h, 22 kg/h above the Table-2 reflux. This case uses 11.440296.

6. BOTTOMS – an OUTPUT, and it does not land on 18119 kg/h

B = F - D is the engine's mole balance: 452.710957 - 142.023556 = 310.687401 kmol/h. Priced at the PREDICTED bottoms composition that comes out at 18107.6 kg/h, 0.06 % under the Table-2 18119 kg/h. It cannot land on it exactly: F's molar total is fixed by the feed mass and the feed's measured composition, D's by the distillate mass and the distillate's measured composition, and the paper's three compositions do not close per component (record Section 5.4 – the wt % columns sum to 100.1, 100.0 and 100.3). The mass residue is the printing, not the column. 18119 kg/h is therefore transcribed above and used by nothing.

7. REFLUX TEMPERATURE 18.5 C – transcribed, NOT USED, and why

18.5 C is far below the bubble point of the overhead at 658.6 kPa (about 41 C, and the run prints its own top-stage temperature). The plant's condenser therefore SUBCOOLS its reflux. Choupo's Wang-Henke path returns saturated reflux and has no subcooling keyword; declaring one would be inventing a model. The number is transcribed so nobody has to wonder whether it was overlooked, and the consequence is stated: the real column's internal liquid traffic in the top few trays is higher than the constant-molar-overflow value used here, by the amount the subcooled reflux condenses on contact. That is a KNOWN, UNMODELLED difference between this case and the plant, not a tuning knob.

8. BOILER DUTY 10.240 MW – transcribed, and NOT claimed as an anchor

The run reports a reboiler duty of about 8.47 MW against the Table-2 10.240 MW. The case does NOT anchor on that, and the reason is in its own output rather than in an argument.

The Wang-Henke path solves M and E under CONSTANT MOLAR OVERFLOW: the internal traffic is fixed by R, D and q, and NO per-stage energy balance is solved. The duties are then reported by closing the column's overall enthalpy change, and the flowsheet's independent energy audit raises the first-law advisory on this run – 98.42 % closure, a residual of 2.96 kW, 0.017 % of the boundary enthalpy. That is small, and it is NOT evidence that the duty is right: the ledger and the duty are computed from the same three stream states, so agreement between them says the arithmetic is consistent, not that a column with no energy equation found the right vapour traffic. The companion arm column15 makes the point unmissable – the same three streams to within 0.8 kW of enthalpy, and its audit closes at -256.84 % with a 670.7 kW residual. Two runs of one column, one ledger claim each, three orders of magnitude apart. The corpus's other Wang-Henke columns sit further out still (column01/09/11's own energyBalance_byUnit.csv), so this is a property of the solver path and of the reporting around it, and it predates this case by a long way.

Also unmodelled and worth naming beside the duty: the plant's subcooled reflux (7) and the condenser's own subcooling duty are not in this figure.

A duty comparison resting on any of that would be a number pretending to be evidence. 10.240 MW is transcribed; 8.47 MW is reported; neither is an anchor row.

9. THERMO – mirroring the paper, not substituting for it

The paper states its own model: liquid activity by UNIFAC, vapour pressure by Antoine, vapour fugacity by the original Soave-Redlich-Kwong. All three are public in this tree, so constant/thermoPhysPropDict declares exactly that and no comparison here hides a model swap. All nine components carry a UNIFAC group decomposition in data/standards/components/, and the only main-group pair the mixture needs (CH2 <→ C=C, for the three butenes) is in data/standards/parameters/UNIFAC/interactions.dat. There is no approximations{} block and none is needed: every component is priced by the model that is declared.

TWO CAVEATS THE RUN ANNOUNCES FOR ITSELF, repeated here so they are read before the numbers and not after: seven of the nine component records carry 'reviewStatus interim' (imported from CoolProp 7.2.0, not yet checked against primaries). The thermodynamics under this anchor is not itself anchored. propane's Antoine fit is declared valid to 320.7 K and the stripping section runs to about 329 K, so its Psat is EXTRAPOLATED there. Propane is 1.5 wt % of the feed and leaves overhead; the extrapolation is announced by the engine on every run and is not corrected here.

10. METHOD – Wang-Henke, and it is not a preference

'model simultaneous' REFUSES a murphreeEfficiency other than 1: there the Murphree relation would have to enter the residual and the Jacobian, and it does not. The efficiency is implemented on the Wang-Henke path only, folded into an effective K with the vapour from the tray below lagged one outer pass, so this case is Wang-Henke because that is where the measured input is honoured. It costs iterations: the corpus's other columns converge in tens of outer passes, this one in several hundred, because 74 trays of a close-boiling pair is a long cascade. maxOuterIter is raised accordingly.

11. E_MV = 1.191 IS A MEASURED INPUT, AND IT IS ABOVE 1

The paper back-calculates a Murphree TRAY efficiency of 119.1 % (74 real trays behaving as 88 ideal ones; overall column efficiency 118.9 %). Above 100 % is not an error: E_MV compares the vapour leaving a tray with equilibrium against that tray's OUTLET liquid, and on a 2.9 m cross-flow tray the liquid enters rich and leaves lean, so the vapour has met liquid richer than the outlet. Nothing anywhere exceeds equilibrium. The engine ADMITS the value and announces it, and states what it does not check (the plug-flow ceiling, which needs the point efficiency and mV/L – neither is declared).

It is an INPUT. Predicting it needs the rate-based programme Choupo deliberately does not have; the paper spends its own length on that difficulty and its Table 4 shows ten published methods disagreeing across a factor of five. This case reads the measurement; it does not reproduce it.

The companion case column15_klemola_ideal_trays is this column with murphreeEfficiency 1.0 and nothing else changed. The gap between the two is the pedagogical payload: how much of this separation the efficiency carries.

12. HAND ESTIMATE – what the specs imply, before the engine is trusted

Relative volatility of the keys from the same Antoine constants the case uses, at 325 K (mid-column):

Psat(i-C4) = 7.2334 bar, Psat(n-C4) = 5.1636 bar → alpha = 1.4008

FENSKE, on the measured key split (moles from the tables above):

S = (d_iC4/b_iC4)(b_nC4/d_nC4) = (130.544/0.932)(304.898/0.279) = 1.5287e5 N_min = ln S / ln alpha = 11.937 / 0.33698 = 35.4 ideal stages

UNDERWOOD, on the four real components (q = 1 for this estimate):

theta = 1.2483 R_min = 7.626 R / R_min = 11.440296 / 7.626 = 1.500

That last line is the strongest check available on the transcription: the reflux and distillate rates, taken from two separate lines of Table 2 and converted independently, land on exactly 1.5 x R_min – the textbook design point. Two mistranscribed numbers do not do that.

GILLILAND then gives X = 0.3066, Y = 0.3761, N = 57.4 ideal stages – fewer than the 74 trays present, and fewer than the 88 ideal trays the paper's own rigorous fit needs. The disagreement is expected and is not evidence against either: Gilliland is a correlation fitted across many systems and is known to be optimistic at high N/N_min on close-boiling pairs, and 88 is a fitted rigorous simulation, not a measurement. What the estimate establishes is the ORDER: this separation needs tens of stages, not five and not five hundred, and the declared reflux is comfortably above minimum. If the engine returned a near-perfect split on ten trays, or failed to separate at all on 74, the hand estimate would have caught it.

13. THE ANCHOR – and what a PASS on it does NOT mean

Compared: i-butane and n-butane, in BOTH products, against the Table-2 measurements. Only the keys. Record Section 5.4 forbids a trace-component anchor before this case was built – 0.1 wt % resolution on a 0.2 wt % number is +/-25 % before anything physical happens, and the traces do not close at all (propane +23 %, i-butene -100 %). The keys close to about 1 %, which is what a key-component anchor needs. The band on any anchor row belongs to the table's own rounding, not to the solver's tolerance.

This is an ENDPOINT anchor. The paper publishes no measured tray temperature profile and no measured tray composition profile – its figures are EFFICIENCY profiles from the back-calculation – so nothing here tests what the column model predicts INSIDE. The internals stay unanchored; the natural second case is a different primary (Vogelpohl's total-reflux profiles), never blended into this one.

What a PASS would mean: that the MESH solve plus a declared efficiency reproduces an industrial column's products from its own specification, on public thermodynamics. What it would NOT mean: that the internals are right (nothing here measures them), nor that Choupo could have PREDICTED the efficiency, nor that the trace components are modelled well.

14. THE ANCHOR TABLE – observed, and it does NOT all agree

Predicted product compositions, converted back to wt % from this case's own converged/ streams, beside the Table-2 measurements. The "band" column is the measurement's OWN rounding: the paper prints to 0.1 wt %, so a printed value v carries +/-0.05 wt %.

DISTILLATE (predicted 8126.4 kg/h vs the 8115 kg/h spec – see 4)

component predicted measured band verdict

i-butane 94.033 93.5 +/-0.05 +0.53 wt % HIGH n-butane 0.0015 0.2 +/-0.05 LOW by ~130x

BOTTOMS (predicted 18107.6 kg/h vs the Table-2 18119 kg/h – see 6)

component predicted measured band verdict

i-butane 0.351 0.3 +/-0.05 INSIDE the band n-butane 97.984 98.1 +/-0.05 -0.12 wt %, just out

THREE OF THE FOUR KEY ROWS LAND WHERE A KEY-COMPONENT ANCHOR WANTS THEM, around the 1 % level at which record section 5.4 says the keys close at all. THE FOURTH DOES NOT, and it is not a rounding argument: the model rejects n-butane overhead about two orders of magnitude harder than the plant did. 0.2 wt % is at the printing resolution (+/-25 % before any physics), but 130x is not a resolution effect.

It is recorded, not explained and not tuned. What can be said without guessing: the disagreement is NOT the efficiency, because the control arm column15 at E_MV = 1 gives 0.0144 wt % – still an order of magnitude below the measurement. Under this model 74 IDEAL trays already over-separate this key, which is the opposite direction from the paper's own finding that 74 real trays behave as 88 ideal ones. Candidates a future slice could test, each of which this case declares as an assumption or a caveat above: the single column pressure (A2), the tray-numbering direction (A0), the unmodelled subcooled reflux (7), the absent per-stage energy balance (8), and seven interim component records (thermo dict). Whichever it is, an anchor that disagreed and said so is worth more than one that agreed after something was adjusted to make it.

THE OTHER FIVE COMPONENTS ARE NOT ANCHOR ROWS and are listed only so nobody reads their absence as concealment: propane 4.84 vs 5.3 wt % overhead and 0.00 vs 0.3 in the bottoms; trans-2-butene 0.00 vs 0.6 overhead; i-butene 0.58 vs 0 overhead. Record section 5.4 already showed these do not close in the SOURCE – propane +23 %, i-butene -100 %, trans-2-butene +155 % between the feed and the two products as printed – so a number computed here has nothing to be compared against. Anchoring on them would be measuring the printing.

15. THE TWO ARMS, side by side

column14 (E_MV = 1.191) against column15 (E_MV = 1.0), everything else identical:

quantity E_MV 1.191 E_MV 1.0 measured

distillate i-butane, wt % 94.033 93.975 93.5 distillate n-butane, wt % 0.0015 0.0144 0.2 bottoms i-butane, wt % 0.351 0.378 0.3 bottoms n-butane, wt % 97.984 97.977 98.1 top temperature, K 314.250 314.116 – reboiler temperature, K 329.437 329.435 – outer iterations 349 426 –

WHAT THE EFFICIENCY BUYS, read off that table: essentially nothing on the major components – i-butane overhead moves 0.06 wt %, n-butane in the bottoms 0.007 – and a real factor on the MINOR key in each product: the n-butane carried overhead drops about 10x, and the i-butane lost to the bottoms falls 7 %.

That shape is worth more than a single number. At 74 trays and R = 11.4 this column is far past the point where separating power limits the major components; extra power goes entirely into the impurity that is left, which is exactly where a plant's money is. A student who expects a 19 % efficiency bonus to show up as a 19 % anything will not find it, and finding out why is the lesson.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0863020558333 |
| stream | distillate | F | 0.0394509877778 |
| stream | feed | F | 0.125753043611 |
| kpi | splitter | B | 0.0863020558333 |
| kpi | splitter | D | 0.0394509877778 |
| kpi | splitter | L_rect | 0.45133097767 |
| kpi | splitter | L_strip | 0.565397368633 |
| kpi | splitter | Q_condenser_kW | -8453.15398421 |
| kpi | splitter | Q_reboiler_kW | 8257.34269769 |
| kpi | splitter | R | 11.440296 |
| kpi | splitter | T_bottom | 329.436819763 |
| kpi | splitter | T_top | 314.249966091 |
| kpi | splitter | V_rect | 0.490781965448 |
| kpi | splitter | V_strip | 0.479095312799 |
| *... and 20 more reference values in the case's `expected`.* | | | |

## column15_klemola_ideal_trays

tutorials/steady/distillation/column15_klemola_ideal_trays




**What it does:** The Klemola-Ilme C4 splitter with IDEAL trays: the same column at E_MV = 1, the control arm that says what the measured efficiency is worth

**The story:** the CONTROL ARM of the Klemola-Ilme anchor.

This is column14_klemola_c4_splitter with ONE line changed:

murphreeEfficiency 1.191; → murphreeEfficiency 1.0;

Everything else – the 74 trays, the feed on tray 37, the feed state, the pressure, the distillate rate, the reflux ratio, the thermo dict, the three 0/ files, the convergence controls – is byte-identical to that case, and if the two ever drift apart this case stops being a control. Read column14's flowsheetDict for the provenance of every number; it is the primary document and none of it is repeated here.

Klemola, K. T.; Ilme, J. K. Distillation Efficiencies of an Industrial-Scale i-Butane/n-Butane Fractionator. Ind. Eng. Chem. Res. 1996, 35 (12), 4579-4586.

WHY THIS IS A SEPARATE CASE AND NOT A SECOND ARM INSIDE column14

A Choupo case is ONE pass: one converged/ state, one KPI set, one golden. Two efficiencies could have been swept in a single case through system/outerDict, and that was rejected: a sweep publishes the DRIVER's KPI table, so the two arms' products would live in a sweep artefact instead of in two real converged/ directories, and the anchor – which is an ENDPOINT anchor, a comparison of product compositions – would have nowhere natural to sit. Two cases give two converged states, two goldens, and two independently sealable units of evidence, at the cost of the duplication named above.

WHAT THE GAP BETWEEN THE ARMS MEANS – and what it does NOT

E_MV = 1 makes every one of the 74 trays an EQUILIBRIUM stage: the vapour leaving it is in equilibrium with the liquid leaving it. No real tray does that, and this plant's trays do BETTER than that – the paper back-calculates 119.1 %, because on a 2.9 m cross-flow tray the liquid is not well mixed, so the vapour has been in contact with liquid richer than the outlet.

The difference between this case's products and column14's is therefore the answer to a question a student can otherwise only be told: how much of an industrial separation is carried by the tray efficiency, as against the tray COUNT and the reflux? Run both and subtract. The paper's own version of the same arithmetic is that 74 real trays behave as 88 ideal ones.

WHAT IT IS NOT. This case is not an anchor and must never be compared with the Table-2 measurements as though it were one: it deliberately declares an efficiency the plant does not have. Its role is to be the OTHER END of a one-parameter difference. If it happened to sit closer to the measured products than column14 does, that would be a finding about the model, not a reason to prefer E_MV = 1 – 119.1 % is the measurement.

Note also that E_MV = 1 is the engine's DEFAULT, and the key is written out in full below anyway. An absent key would mean the same thing to the solver and something quite different to a reader: it would look like an oversight in a case whose entire subject is the efficiency.

WHAT THE TWO ARMS ACTUALLY GAVE

quantity E_MV 1.191 E_MV 1.0 measured

distillate i-butane, wt % 94.033 93.975 93.5 distillate n-butane, wt % 0.0015 0.0144 0.2 bottoms i-butane, wt % 0.351 0.378 0.3 bottoms n-butane, wt % 97.984 97.977 98.1 top temperature, K 314.250 314.116 – reboiler temperature, K 329.437 329.435 – outer iterations 349 426 –

The efficiency moves the MAJOR components by almost nothing (i-butane overhead by 0.06 wt %) and the MINOR key in each product by a real factor (n-butane carried overhead drops about 10x; i-butane lost to

the bottoms falls 7 %). At 74 trays and R = 11.4 the column is far past the point where separating power limits the majors, so extra power goes entirely into the impurity that is left. column14's flowsheetDict section 15 argues that at length; this is the same table so that either case can be read alone.

A FINDING THIS ARM IS WHAT ESTABLISHED, and it belongs here because this is the case that establishes it: at E_MV = 1 the model ALREADY rejects n-butane overhead an order of magnitude harder than the plant measured (0.0144 wt % against 0.2). So the anchor's one clear disagreement is not caused by the declared efficiency – turning the efficiency off does not recover the measurement, it only halves the distance in the log. Under this model 74 ideal trays over-separate this key, which runs opposite to the paper's own conclusion that 74 real trays behave as 88 ideal ones. Recorded; not explained here and not tuned away.

ONE MORE OBSERVATION, and it is about the ENGINE rather than the column. These two runs produce product streams whose tabulated enthalpies agree to 0.8 kW, and their flowsheet energy audits disagree wildly: column14 closes at 98.42 % (residual 2.96 kW) and this arm at -256.84 % (residual 670.7 kW). A difference that large between two nearly identical states does not come from the physics of the two arms; it is in how the energy report prices the products, and this pair is a compact reproducer for anyone who goes looking. Neither case claims a duty anchor – see column14's section 8.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0863020558333 |
| stream | distillate | F | 0.0394509877778 |
| stream | feed | F | 0.125753043611 |
| kpi | splitter | B | 0.0863020558333 |
| kpi | splitter | D | 0.0394509877778 |
| kpi | splitter | L_rect | 0.45133097767 |
| kpi | splitter | L_strip | 0.565397368633 |
| kpi | splitter | Q_condenser_kW | -8446.257018 |
| kpi | splitter | Q_reboiler_kW | 8250.44278944 |
| kpi | splitter | R | 11.440296 |
| kpi | splitter | T_bottom | 329.435277335 |
| kpi | splitter | T_top | 314.116313405 |
| kpi | splitter | V_rect | 0.490781965448 |
| kpi | splitter | V_strip | 0.479095312799 |
| *... and 19 more reference values in the case's `expected`.* | | | |

## column16_declared_interior

tutorials/steady/distillation/column16_declared_interior




**What it does:** A column that starts from the interior declared in 0/

**The story:** A COLUMN THAT STARTS FROM THE INTERIOR THE CASE DECLARES.

Physically this is column01 – benzene/toluene, Raoult, 15 stages, feed on 8, R = 2, D = 50 kmol/h – and the answer is the same answer. What differs is where the ITERATION STARTS, and that is the whole subject of the case.

THE OPENFOAM READING OF A STATE DIRECTORY. In OpenFOAM a time directory holds one file per FIELD, and that file carries ‘boundaryField‘ AND ‘internalField‘ together: the conditions at the edges of the domain and the values inside it, in one place, so the directory is a COMPLETE, RESTARTABLE snapshot. Choupo‘s ‘0/‘ used to carry only the boundary half – one file per STREAM – while what each unit held inside it existed nowhere a case could declare it. Since 2026-09-06 a state view carries both:

0/<stream> a stream: one file, flat (the boundary) 0/internalStates/<unit> a unit‘s interior: ONE file per unit, one block per kind

This case declares ‘0/internalStates/column16‘, a ‘stageProfile‘ block with the temperature and the liquid and vapour compositions on all 15 stages, in exactly the grammar the run writes to ‘converged/internalStates/column16‘. The same object travels both ways, which is what makes the snapshot restartable – run the case, copy the file from ‘converged/internalStates/‘ to ‘0/internalStates/‘, and the next run starts where the last one finished. The interiors have their own root because a unit and a stream may share a name: identity is (kind, sector, name), never name alone.

WHY IT MATTERS HERE, and it is not the speed. Until this slice the column seeded its own interior in CODE – "initial T-profile (linear between guesses)" – undeclared, unseen and unowned, which is exactly the silent crutch Choupo forbids for every other solver aid (every aid is explicit in a dict and announces itself when it binds). Now the unit says which of the two routes it took, on every run, both ways:

[seed] interior read from 0/ (stageProfile): 15 stages, T ... K [seed] interior seeded by the unit: linear T between the guesses ...

A DECLARED PROFILE THAT DOES NOT SATISFY THE BALANCES IS A SEED, NOT AN ANSWER. Nothing checks this file against the column equations – the iteration does that, and it moves off any profile that is wrong. Edit a stage temperature by fifty kelvin and the column still converges to the same distillate; it simply takes more outer iterations to get there. What IS checked is that the declaration describes THIS column: the stage count, the component set, and that every number is a number. A mismatch REFUSES by name and names the remedy – a seed silently ignored would be the silence this case exists to end.

WHAT THE STUDENT SHOULD DO WITH IT. Run the case and read the ‘iterations‘ KPI; then delete ‘0/internalStates/‘ and run it again. Both runs reach the same column; only one of them had to search for it. That difference is the entire value of a restartable snapshot, and it is why an interior is STATE and not a report.

*Physically this is column01_benzene_toluene: equimolar benzene/toluene, Raoult VLE, 15 stages (reboiler counted), feed on stage 8, R = 2, distillate 50 kmol/h, Wang–Henke. The answer is the same answer — x_D = 0.9812 benzene, x_B = 0.9812 toluene. What differs is where the iteration starts.*

*In OpenFOAM a time directory holds one file per field, and that file carries boundaryField ∗and∗ internalField: the conditions at the edges of the domain and the values inside it, together, so the directory is a complete snapshot you can restart from.*

*Choupo’s 0/ used to carry only the boundary half — one file per stream — while what each unit held ∗inside∗ it existed nowhere a case could declare. Since 2026-09-06 a state view carries both:*

*This case declares 0/internalStates/column16 — a stageProfile block with T and the liquid and vapour compositions on all fifteen stages, in exactly the grammar the run writes to converged/internalStates/column16. The same object travels both ways. Run the case, copy the file from converged/internalStates/ into 0/internalStates/, and the next run starts where the last one finished. The interiors have their own root because a unit and a stream may share a name — identity is (kind, sector, name), never name alone.*

*Nothing checks this file against the column equations. The iteration does that, and it moves off any profile that is wrong. Edit a stage temperature by fifty kelvin and the column still converges to the same distillate — it simply has further to travel.*

*What ∗is∗ checked is that the declaration describes this column: the stage count against nStages, the component set against the case's, every value finite. A mismatch refuses by name and says how to fix it. A seed silently ignored would be exactly the silence this case exists to end.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | bottoms | T | 381.514100298 |
| stream | distillate | F | 0.0138888888889 |
| stream | distillate | T | 354.247899301 |
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 370 |
| kpi | column16 | B | 0.0138888888889 |
| kpi | column16 | D | 0.0138888888889 |
| kpi | column16 | L_rect | 0.0277777777778 |
| kpi | column16 | L_strip | 0.0361877253979 |
| kpi | column16 | Q_condenser_kW | -1315.28783308 |
| kpi | column16 | Q_reboiler_kW | 679.510517779 |
| kpi | column16 | R | 2 |
| kpi | column16 | T_bottom | 381.514100298 |
| *. . . and 19 more reference values in the case's `expected`.* | | | |

## column17_exchanger_design

tutorials/steady/distillation/column17_exchanger_design




**What it does:** The column's condenser and reboiler designed: Nusselt film condensation and Rohsenow boiling under the critical heat flux

**The story:** column09's column, unchanged, with its CONDENSER and REBOILER designed instead of estimated.

The flowsheet below is column09's to the last key: the same benzene/toluene feed, the same 15 stages, the same reflux, the same tray hydraulics. What changes is system/postDict, and the sheets it writes for the two exchangers. Read column09 first; this case is the next question.

column09 sizes its condenser and reboiler as A = Q/(U LMTD), with U and the LMTD TYPED by the author – a first estimate, not a design. Here both are DESIGNED on the same route a flowsheet exchanger has (a Kern shell-and-tube design: the bundle declared, the tube count solved, U a RESULT of the films, the wall and the fouling), and the process side of each is priced by a PHASE-CHANGE film instead of a single-phase one:

the CONDENSER condenses the overhead on the outside of horizontal tubes, cooling water inside: Nusselt's laminar film, the condensate of one tube falling on the next (h_1 N^-1/4 at the deepest tier); the REBOILER boils the bottoms as a pool around a tube bundle (a kettle), LP steam condensing inside: Rohsenow's nucleate flux, HELD UNDER the Zuber critical heat flux.

NOTHING ABOUT THE PROCESS SIDE IS TYPED. The temperatures are the column's own (the top tray's dew point and the distillate's bubble point; the bottoms' bubble point), and every property a film needs – density, viscosity, conductivity, heat capacity, surface tension, latent heat – comes from constant/thermoPhysPropDict. That is why this case's package declares LIQUID transport models where column09's declares only a surface tension: delete one and the exchanger it prices refuses by name.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | bottoms | T | 381.514100345 |
| stream | distillate | F | 0.0138888888889 |
| stream | distillate | T | 354.247899351 |
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 370 |
| kpi | column17 | B | 0.0138888888889 |
| kpi | column17 | D | 0.0138888888889 |
| kpi | column17 | L_rect | 0.0277777777778 |
| kpi | column17 | L_strip | 0.0361877253979 |
| kpi | column17 | Q_condensate_m3_s | 0.00403798363546 |
| kpi | column17 | Q_condenser_kW | -1315.28783372 |
| kpi | column17 | Q_reboiler_kW | 679.510516909 |
| kpi | column17 | R | 2 |

*... and 185 more reference values in the case's `expected`.*

## column18_partial_condenser

tutorials/steady/distillation/column18_partial_condenser




**What it does:** Benzene/toluene column with a PARTIAL condenser (Wang-Henke)

**The story:** column01's benzene/toluene column under a PARTIAL condenser ('condenser { type partial; }').

WHAT IT DEMONSTRATES. The condenser TYPE is a declared choice, and it decides what stage 1 IS. Here stage 1 is the condenser itself, an equilibrium stage: the reflux is its liquid x_1, and the distillate leaves as a saturated VAPOUR y_1 = K_1 x_1 at T_1 (published 'vf 1', pinned). Compared with column01 (the same column, 'type total' by default): the distillate composition is the SAME – both types write the same material balance on stage 1 (x_D = y_1 = K_1 x_1 under a total condenser is the condensate of that vapour; here it is the vapour itself); the distillate PHASE differs: vapour at T_1 = 355.69 K here, liquid at its bubble point 354.25 K there; the condenser duty is the stage's OWN envelope, Q = -[ V_2 h_V(y_2,T_2) - L_1 h_L(x_1,T_1) - D h_V(y_1,T_1) ], about two thirds of the total condenser's (R+1) D latent duty, because D leaves uncondensed; the reboiler duty rises by the enthalpy the vapour distillate carries away; stage 1 is NOT a tray: the hydraulics pass drops it (V_1 = 0), where under a total condenser it is the top tray and is rated.

THE STAGE COUNT. 'nStages' counts EQUILIBRIUM stages, top to bottom, the reboiler included. A partial condenser is one of them, so the 15 stages here are 1 condenser + 13 trays + 1 reboiler; column01's 15 are 14 trays + 1 reboiler under a total condenser that is not counted. A source that counts its TOTAL condenser as "stage 1" describes nStages - 1 here.

*column01_benzene_toluene's 15-stage benzene/toluene column (feed on stage 8, R = 2, D = 50 kmol/h, Raoult, Wang–Henke) with ONE line added:*

*The condenser TYPE is a declared choice (default total, announced when it is taken), and it decides what stage 1 IS.*

*Stage 1 is the condenser itself, an equilibrium stage. The vapour from stage 2 enters it, part of it is condensed and returned as the reflux — the stage's liquid x_1 — and the rest leaves as the distillate, a saturated VAPOUR y_1 = K_1 x_1 at T_1. Under a total condenser (column01) ALL of the vapour leaving stage 1 is condensed and reflux and distillate leave as one liquid of that vapour's composition, at its bubble point; stage 1 is then the top tray, and the condenser above it is not an equilibrium stage and is not counted.*

*Both types write the same material balance on stage 1, so the two cases share their composition profile and their distillate composition exactly (x_D = 0.9510 benzene, both). What differs:*

*and the reboiler duty closes the column, as always. The plant first law closes at 0 kW in both cases: the split between the two duties moves, the sum does not.*

*nStages counts EQUILIBRIUM stages, top to bottom, the reboiler included. A partial condenser is one of them: the 15 stages here are 1 condenser + 13 trays + 1 reboiler. column01's 15 are 14 trays + 1 reboiler, under a total condenser that is not counted. So a source that counts its TOTAL condenser as "stage 1" (many textbooks and simulators do) describes a column of nStages - 1 here, and one that counts a partial condenser as stage 1 maps one to one.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | bottoms | T | 381.514100345 |
| stream | distillate | F | 0.0138888888889 |
| stream | distillate | T | 355.689111404 |
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 370 |
| kpi | column18 | B | 0.0138888888889 |
| kpi | column18 | D | 0.0138888888889 |
| kpi | column18 | L_rect | 0.0277777777778 |
| kpi | column18 | L_strip | 0.0361877253979 |
| kpi | column18 | Q_condenser_kW | -888.554586266 |
| kpi | column18 | Q_reboiler_kW | 691.20654736 |
| kpi | column18 | R | 2 |
| kpi | column18 | T_bottom | 381.514100345 |
| *... and 18 more reference values in the case's `expected`.* | | | |

## shortcut01_benzene_toluene

tutorials/steady/distillation/shortcut01_benzene_toluene




**What it does:** Shortcut (FUG) design of a benzene/toluene column

**The story:** Preliminary column design by the Fenske-Underwood-Gilliland (FUG) shortcut method (the built-in 'shortcutColumn'). The same benzene/ toluene separation as the rigorous 'column01', but designed in closed form: you specify the key recoveries and the reflux, and FUG returns the minimum stages, the minimum reflux, the actual stage count and the feed location.

Feed: 100 kmol/h, 50/50 benzene/toluene, saturated liquid, 1 atm. Keys: LK = benzene, HK = toluene ($\alpha \approx 2.5$). Spec: 99 % of the benzene to the distillate, 1 % of the toluene. Reflux: R = 1.3 · R_min.

Expected ($\alpha \approx 2.5$): N_min ≈ 10.1 (Fenske, total reflux) R_min ≈ 1.3 (Underwood) N ≈ 22 (Gilliland, at R = 1.3 R_min) distillate ~99 % benzene, bottoms ~99 % toluene; mass closes 100 %.

Compare with the rigorous column01 (Wang-Henke): FUG is a fast design estimate; the rigorous column is the verification. FUG does NOT handle azeotropes — it assumes constant relative volatility.

*An equimolar benzene/toluene feed (100 kmol/h at 365 K, 1 atm) is to be split 99 % / 1 %: 99 % of the benzene overhead, 1 % of the toluene with it. Fenske–Underwood–Gilliland answers in one pass, no stages solved. The golden: N_min = 10.07 (Fenske), R_min = 1.294 (Underwood, θ = 1.427), operating R = 1.682 (declared as 1.3·R_min), N = 21.6 theoretical stages with the feed at 10.8 from the top; α_LK/HK = 2.491 at the feed bubble point; D = B = 50 kmol/h; x_D = x_B = 0.99.*

*1. Three declared numbers, four printed results. recoveryLK 0.99; recoveryHK 0.01; refluxFactor 1.3; — and the engine will not invent the reflux factor for you: a column whose reflux the engine chose is one nobody can defend at a viva, so the key is REQUIRED. 2. The only thermodynamics is one bubble point. α is evaluated once, at the feed's bubble temperature, and held constant — the method's founding assumption, and the run says so in its result block. 3. A shortcut is recorded as a different question. Look at the top of the result: problemDivergence carries a declaredApproximation for this unit — ∗requested: distillation separation; solved: Fenske-Underwood-Gilliland shortcut (constant relative volatility at the feed bubble point; no azeotrope representable)∗. It is not an error and it is not hidden: a stream table cannot show that the column was a shortcut, so the record travels with the answer. 4. Now build the rigorous one. column01_benzene_toluene puts 15 real stages with the feed on 8 at R = 2 — read that README and compare N = 21.6 at 1.3·R_min with what 15 stages at a higher reflux actually deliver.*

*Set refluxFactor 1.05: N climbs steeply toward infinity as R → R_min — Gilliland's curve, in two runs. Then recoveryLK 0.999: N_min jumps, because Fenske is logarithmic in the split.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column | B | 0.0138888888889 |
| kpi | column | D | 0.0138888888889 |
| kpi | column | N_min | 10.067785029 |
| kpi | column | N_theoretical | 21.5880731317 |
| kpi | column | R | 1.68248609289 |
| kpi | column | R_min | 1.29422007145 |
| kpi | column | alpha_LK_HK | 2.49137883567 |
| kpi | column | feed_stage | 10.7940365659 |
| kpi | column | theta | 1.42716041594 |
| kpi | column | x_B_HK | 0.99 |
| kpi | column | x_D_LK | 0.99 |

*... and 7 more reference values in the case's `expected`.*

## stripper01_sour_water

tutorials/steady/distillation/stripper01_sour_water




**What it does:** S2: a sour-water stripper over the chemistry – eight reactive trays, the CO2-strips-first mechanism measured across the whole column.

**The story:** S2 of the sour-water programme: a REAL multi-stage column over the chemistry, not a two-stage identity rig.

column13 proved the machinery: one reactive flash per stage per residual, the apparent components the state, the ions internal to each tray. This case is what the machinery exists FOR – a sour-water stripper: feed near the top, most of the trays below it stripping, the reboiler making the vapour that carries NH3 and CO2 overhead.

THE MECHANISM IS THE LESSON (docs/design/sour-water-stripper-scope.md ?4.3): stripping CO2 RAISES the liquid pH down the column, and the rising pH converts NH4+ back to free NH3, which then strips. The separation and the speciation are the same problem. Eight trays instead of column13's four give the mechanism room to show, and it shows STRICTLY, measured on this converged profile: the carbonate loading falls tray by tray from 0.524 mol/kg (top) to 0.024 (reboiler) – over an order of magnitude – while the free-ammonia fraction rises 0.802 → 0.947 with no exception, and the pH climbs 8.46 → 8.80 below the top tray. (The top tray itself sits ~7 K colder than tray 2 and its pH reads higher for the temperature of its own Kw – comparing pH across that jump compares two water dissociations, which is why the pH claim starts at tray 2.) check_tray_chemistry's ordering claim is tested across this whole span rather than across four points.

WHAT BUILDING IT FOUND, because the first honest run is an instrument (both fixes are the engine's, recorded in the scope doc ?6d): a MESH initialisation ramps T across the column, and an 8-stage ramp visits temperatures a 4-stage ramp never did – including ones above the feed's two-phase band, where the reactive Newton has no interior V/F and stalled. stageK now prices such a trial INCIPIENT over the hypothetical speciated liquid (announced once), the same construction its subsaturated branch always used – continuous K, and the aid shapes the path, never the answer (a converged stage sits at saturation, where the constructions agree); the per-tray chemistry report used to FLASH each converged tray liquid – a liquid pinned by the MESH to its own bubble point, i.e. the flash's degenerate corner – and at reflux 0.5 five trays of eight printed NaN for chemistry the package resolves easily. The report now asks the right question (speciate the LIQUID), and NaN is reserved for a tray whose speciation itself fails.

WHAT THIS CASE DECLARES, said plainly (the column12/13 rule): 'model simultaneous;' – the rigorous MESH, whose per-stage equilibrium is 'ThermoPackage::stageK': for this reactive package, one reactive flash per stage per residual evaluation. Nested, not a species-basis MESH – the Jacobian sees apparent components only. the per-tray chemistry (pH, ionic strength, every molality) rides in reports/unitOperations/tower/profile.csv – engine-owned, the same surface column13 writes.

DATA SCOPE: sealed (bin/choupo-import); records mirrored under constant/ and sha256-verified by constant/propertyManifest.

*Eight reactive trays over a sour water (2.5 mol% $NH_3$, 1 mol% $CO_2$), feed on tray 2, reflux 0.5, distillate 5 of 100 kmol/h. Converges in 9 MESH Newton iterations; the overhead carries 33.4 mol% $NH_3$ at 355 K, the bottoms leave at 99.1 % water. This is S2 of the sour-water programme (docs/design/sour-water-stripper-scope.md ?6d) — the case the effective stage K (ThermoPackage::stageK) was built ∗for∗, after column13 proved the machinery on an identity rig.*

*A molecular column cannot express this separation. The volatility of $NH_3$ is set by a chemistry the column itself moves:*

*1. $CO_2$ strips first — it is the more volatile solute, and the carbonate loading of the liquid falls tray by tray, across an order of magnitude (0.52 → 0.024 mol $CO_2$-species per mol $NH_3$-species). 2. The falling carbonate frees the ammonia. The $CO_2$ a liquid still carries is what holds its ammonia protonated as $NH_4^+$; as it leaves, the pH climbs (8.46 → 8.80 below the top tray) and the free-ammonia fraction rises 0.802 → 0.947 — strictly, with no exception, on every tray. 3. The freed ammonia then strips. The whole cascade is*

*visible in reports/unitOperations/tower/profile.csv: per-tray pH, ionic strength and every species molality, engine-owned.*

*The surge — the mechanism's own signature. Free ammonia is not merely stripped; it is ∗produced∗ by the deprotonation the falling carbonate allows. Just below the feed the production outruns the stripping: m_NH3aq rises from 1.316 (tray 2, the feed tray) to 1.424 (tray 3) and only then falls strictly to the reboiler (0.464). A first instinct would be to exempt this from the "everything is stripped downward" check; instead it is pinned as its own claim (check_tray_chemistry T5), because hiding a mechanism's signature to make a monotonicity check pass would be the check defeating its subject. Sabotage-verified: move the feed to tray 5 and T5 alone fires.*

*The top tray sits ~7 K colder than tray 2, so its pH reads higher for the temperature of its own K_w — comparing pH across that jump compares two water dissociations, which is why the pH claim starts at tray 2.*

*∗ The MESH initial ramp vs the two-phase band. An 8-stage linear T ramp visits temperatures a 4-stage ramp never did — including Tf + 15 K, ∗above∗ the feed's two-phase band, where the reactive Newton has no interior V/F and stalled before the first iteration printed. stageK now catches the typed ReactiveVLE::NonConvergence and prices such a trial ∗incipient∗ over the hypothetical speciated liquid (announced once) — the same K_i = (p_i^eq/P)/x_i its subsaturated branch always used, so the K surface is continuous across the band. A converged stage sits at saturation, where the constructions agree: the aid shapes the path, never the answer. ∗ The per-tray chemistry report flashed a liquid pinned to its own bubble point. The MESH's residual (Σy = 1) places every converged tray liquid exactly at its bubble point — the two-phase flash's degenerate corner — and at reflux 0.5 five trays of eight printed NaN for chemistry the package resolves easily. The report now asks the right question (speciateReactiveAsLiquid); NaN is reserved for a tray whose speciation itself fails.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0263888888889 |
| stream | distillate | F | 0.00138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | tower | B | 0.0263888888889 |
| kpi | tower | D | 0.00138888888889 |
| kpi | tower | Q_condenser_kW | -55.8692544947 |
| kpi | tower | Q_reboiler_kW | 76.3532639151 |
| kpi | tower | R | 0.5 |
| kpi | tower | T_bottom | 370.531146763 |
| kpi | tower | T_top | 355.393273713 |
| kpi | tower | feedStage | 2 |
| kpi | tower | nFeeds | 1 |
| kpi | tower | nSideDraws | 0 |
| kpi | tower | nStages | 8 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## stripper02_sour_water_h2s

tutorials/steady/distillation/stripper02_sour_water_h2s




**What it does:** The three-family sour water: CO2 strips overhead while H2S concentrates downward and the pH FALLS – the selectivity lesson the two-family stripper cannot teach.

**The story:** the THREE-family sour water: NH3 + CO2 + H2S, and the SELECTIVITY lesson the two-family stripper cannot teach.

stripper01 shows the mechanism: CO2 strips, pH rises, ammonia frees. Add the third volatile weak electrolyte and the story INVERTS in the most instructive way, measured on this converged profile:

CO2 still strips overhead, hard: its loading falls from 0.195 to 1e-5 (per mol of ammonia species) between tray 2 and the reboiler; H2S does NOT follow – its loading RISES down the column, 0.18 to 0.51: the H2S/CO2 selectivity of a sour-water stripper at these conditions favours CO2 removal, and the sulfide concentrates in the liquid heading for the bottoms; and the pH FALLS down the column (8.49 → 7.61), the OPPOSITE of stripper01's climb: the departing ammonia and the accumulating sulfide acidify faster than the departing CO2 alkalinises.

A student who has read stripper01 should predict a rising pH here and be wrong – that is the point. The chemistry is the same machinery (three networks at once: ammonia, carbonate, sulfide – the H2S bridge is 'aqueousSpeciation sulfide', HS- + H+, declared on the component), and the per-tray profile carries all three families' molalities.

Same column as stripper01 otherwise: 8 reactive trays, feed on tray 2, reflux 0.5, D = 5 of 100 kmol/h, 'model simultaneous' (one reactive flash per stage per residual evaluation; the Jacobian sees apparent components only).

DATA SCOPE: sealed (bin/choupo-import); records mirrored under constant/ and sha256-verified by constant/propertyManifest.

*The three-family sour water: $NH_3$ (2.5) + $CO_2$ (1.0) + $H_2S$ (0.5 kmol/h) in 96 kmol/h water, over the same 8 reactive trays as stripper01. Three speciation networks run at once on every tray — ammonia, carbonate, sulfide — and the per-tray profile carries all of them.*

*stripper01 teaches: $CO_2$ strips → pH rises → $NH_4$? deprotonates → ammonia strips. Add $H_2S$ and a student should predict the same rising pH. The converged profile says otherwise:*

*$CO_2$ is the more volatile acid gas at these conditions, so the stripper is selective: the carbonate leaves overhead while the sulfide stays with the liquid — and with the ammonia also leaving, the residual $H_2S$ acidifies the bottoms faster than the departing $CO_2$ can alkalinise it. The pH direction is not a law of strippers; it is an outcome of ∗which∗ acid gas wins, and this pair of cases shows both outcomes with the same machinery.*

*∗ reports/unitOperations/tower/profile.csv — per-tray pH, ionic strength and every species molality of all three families (m_H2Saq, m_HS beside the carbonate and ammonia columns). ∗ The distillate: an acid-gas overhead ($NH_3$ + $CO_2$ + a little $H_2S$) — the real design question of a sour-water unit (where each contaminant ends up) is answered per tray, not just at the ends. ∗ A tray whose $H_2S$ has stripped to numerical zero reports the sulfide molalities as 0 exactly — a solved tray with an empty family is a zero, never a NaN (NaN stays reserved for a tray whose speciation itself failed).*

*model simultaneous; — one reactive flash per stage per residual evaluation; the Jacobian sees apparent components only. Sealed (bin/choupo-import).*

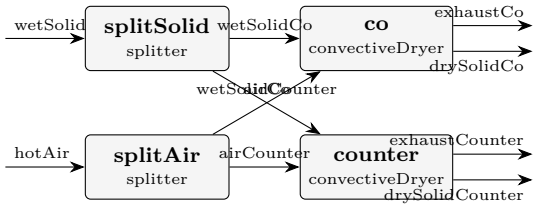
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0263888888889 |
| stream | distillate | F | 0.00138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | tower | B | 0.0263888888889 |
| kpi | tower | D | 0.00138888888889 |
| kpi | tower | Q_condenser_kW | -56.7703732241 |
| kpi | tower | Q_reboiler_kW | 80.8786864139 |
| kpi | tower | R | 0.5 |
| kpi | tower | T_bottom | 372.031545213 |
| kpi | tower | T_top | 356.430868413 |
| kpi | tower | feedStage | 2 |
| kpi | tower | nFeeds | 1 |
| kpi | tower | nSideDraws | 0 |
| kpi | tower | nStages | 8 |
| *. . . and 27 more reference values in the case's `expected`.* | | | |

## drying

## convDryer01_sugar_curve

tutorials/steady/drying/convDryer01_sugar_curve




**What it does:** Continuous convective drying of sugar on its characteristic drying curve: the same feed through a co-current and a counter-current dryer

**The story:** CONTINUOUS CONVECTIVE DRYING ON THE CHARACTERISTIC DRYING CURVE.

solidDryer01's feed – 3423 kg/h dry sucrose at X_in = 0.20 kg/kg, and 800 kmol/h of air at 420 K with 2 mol% water – split in two halves, each sent through the SAME continuous dryer (300 m2 of gas-solid contact, k_Y = 0.05 kg/(m2 s), X_c = 0.12 kg/kg, the characteristic curve f = Phi^1.5), one with the air flowing WITH the solid, one AGAINST it. (src/unitOperations/heatTransfer/ConvectiveDryer.H)

Along the contact area the air COOLS and HUMIDIFIES as it dries the solid, so the wet bulb, the constant rate and the local equilibrium moisture all move – what the batch tray (dryer02) cannot show.

co-current counter-current X out [kg/kg] 0.02342 0.00468 water removed [kg/s] 0.0839 0.0929 air out [K] 359.7 353.2 Y out [kg/kg dry] 0.0393 0.0421 X_c crossed at 6.4 % of A 12.4 % of A floor: X_eq of the air as fed = 0.00110 kg/kg

Counter-current dries five times further on the same area, because the driest solid meets the freshest (hottest, driest) air, where X_eq is lowest and the falling-rate driving force survives; co-current ends where the used-up air meets the driest solid. That is the textbook result, and here it is computed, not asserted.

THE FIRST LAW, on the package's own formation surface: each dryer publishes energyResidual_kW (1e-7 and 5e-10 kW), and the plant energy report closes at 1e-7 kW. The hot air DECLARES 'phase gas;' – without it the stream carries vf = 0, the report prices the air as a liquid and the plant is off by 380 kW; the dryer now refuses that by name.

HYPOTHESES (announced on every run): the solid sits at the LOCAL wet bulb in both periods – it leaves at ~316 K, while a real solid warms toward the air in the falling period (not modelled); Lewis = 1; one k_Y over the whole area; no axial mixing; no re-wetting; adiabatic; bound moisture priced as liquid water. X_c, k_Y and the curve are HYPOTHETICAL teaching values, not measured on this sugar.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | airCo | F | 0.111111111115 |
| stream | airCo | T | 420 |
| stream | airCounter | F | 0.111111111115 |
| stream | airCounter | T | 420 |
| stream | drySolidCo | F | 0.0020068905978 |
| stream | drySolidCo | T | 316.106805335 |
| stream | drySolidCounter | F | 0.00151236672392 |
| stream | drySolidCounter | T | 315.705897235 |
| stream | exhaustCo | F | 0.115770899356 |
| stream | exhaustCo | T | 359.690122884 |
| stream | exhaustCounter | F | 0.11626542323 |
| stream | exhaustCounter | T | 353.240422979 |
| stream | hotAir | F | 0.222222222229 |
| stream | hotAir | T | 420 |
| *... and 30 more reference values in the case's `expected`.* | | | |

## evapDryer01_nacl

tutorials/steady/drying/evapDryer01_nacl




**What it does:** Evaporative dryer: free surface water on NaCl crystals evaporated into dry nitrogen, adiabatic, the binding limit announced

**The story:** FREE WATER, NOT BOUND WATER.

'evaporativeDryer' is the honest counterpart of 'solidDryer': where that one models a HYGROSCOPIC material whose moisture is GAB-sorbed and dries toward an equilibrium isotherm, this one models a NON-sorbing crystal whose moisture is free surface water. There is no isotherm, so there is no equilibrium moisture to stop at – the drying stops at whichever of three real limits binds FIRST, and the unit ANNOUNCES which:

1. the free water runs out; 2. the exhaust reaches saturation (the constant-rate air-capacity limit); 3. the gas cannot supply the latent + sensible heat.

Limits 2 and 3 are coupled – evaporation cools the gas, which lowers its carrying capacity – and the unit bisects on the removed water until the two agree, rather than applying them in sequence and hoping.

Wet salt: 20 kmol/h NaCl crystals (1169 kg/h) carrying 15 kmol/h of free water. Dry nitrogen at 400 K. The gas flow is deliberately generous, so the binding limit here is the FIRST one: the water runs out. Halve the nitrogen and the announcement changes – that is the case to run next.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | drySolid | F | 0.00555555555556 |
| stream | hotGas | F | 0.166805555556 |
| stream | humidExhaust | F | 0.170972222222 |
| stream | wetSolid | F | 0.00972222222222 |
| kpi | dryer | T_out | 360.675490975 |
| kpi | dryer | X_final | 0 |
| kpi | dryer | X_initial | 0.231198665298 |
| kpi | dryer | air_in_kmol_h | 600.5 |
| kpi | dryer | drySolid_flow | 0.324666666667 |
| kpi | dryer | exhaust_humidity | 0.0392946211155 |
| kpi | dryer | moisture_pct_wb | 0 |
| kpi | dryer | water_removed | 0.0750625 |
| stream | drySolid | T | 360.675490975 |
| stream | hotGas | T | 400 |
| *... and 3 more reference values in the case's `expected`.* | | | |

## evapDryer02_energy_limited

tutorials/steady/drying/evapDryer02_energy_limited




**What it does:** The same dryer starved of gas: 40 kmol/h instead of 600, so the gas runs out of heat at its wet bulb and 16.6 wt% of the water stays on the crystal

**The story:** FREE WATER, NOT BOUND WATER.

'evaporativeDryer' is the honest counterpart of 'solidDryer': where that one models a HYGROSCOPIC material whose moisture is GAB-sorbed and dries toward an equilibrium isotherm, this one models a NON-sorbing crystal whose moisture is free surface water. There is no isotherm, so there is no equilibrium moisture to stop at – the drying stops at whichever of three real limits binds FIRST, and the unit ANNOUNCES which:

1. the free water runs out; 2. the exhaust reaches saturation (the constant-rate air-capacity limit); 3. the gas cannot supply the latent + sensible heat.

Limits 2 and 3 are coupled – evaporation cools the gas, which lowers its carrying capacity – and the unit bisects on the removed water until the two agree, rather than applying them in sequence and hoping.

THIS CASE IS THE STARVED ONE. Identical to evapDryer01_nacl except for the gas: 40 kmol/h instead of 600. The announcement changes to [energy-limited]: the solid leaves at the gas's WET-BULB temperature, 311.68 K, this model's floor, having given up 36.75 kg/h of water with the exhaust at a_w 0.886, below its cap, and 16.6 wt% of the water stays on the crystal – a dryer that does not dry, reported as such instead of returning a dry solid it cannot produce.

THE WALL THIS GAS MEETS IS HEAT, AT THE WET BULB – AND THE CASE'S NAME PREDATES THAT (2026-10-05). The wet solid enters at 298.15 K, BELOW the gas's wet bulb, so part of what the gas pays goes to warming the cake up to it, and what is left evaporates less than the exhaust has room for. The name was given the same morning (DEV.md C35 item 12, it was 'evapDryer02_energy_limited'), when the floor was still the solid's FEED temperature: then the exhaust reached its cap at T_out = 310.73 K, BELOW the gas's own wet bulb, and the room wall came first. Vitor's ruling C35 item 7 put the floor at the wet bulb – a wet solid in a gas cools, or warms, toward it – and that answer is no longer allowed. Whether the case is renamed back, or its wet solid fed warm (at 330 K, above the wet bulb, the same gas meets the room wall again: the exhaust at its cap at 314.5 K with 49.0 kg/h carried and 51.6 kg/h payable – the fixture 'check_evaporative_dryer' arm (a') runs), is Vitor's; DEV.md C35 item 7.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | drySolid | F | 0.00915552191059 |
| stream | hotGas | F | 0.01125 |
| stream | humidExhaust | F | 0.0118167003116 |
| stream | wetSolid | F | 0.00972222222222 |
| kpi | dryer | T_out | 311.677565845 |
| kpi | dryer | X_final | 0.199753779936 |
| kpi | dryer | X_initial | 0.231198665298 |
| kpi | dryer | air_in_kmol_h | 40.5 |
| kpi | dryer | drySolid_flow | 0.389520060553 |
| kpi | dryer | exhaust_humidity | 0.886447295542 |
| kpi | dryer | moisture_pct_wb | 16.6495645421 |
| kpi | dryer | water_removed | 0.0102091061141 |
| stream | drySolid | T | 311.677565845 |
| stream | hotGas | T | 400 |

*... and 3 more reference values in the case's `expected`.*

## solidDryer01_sugar

`tutorials/steady/drying/solidDryer01_sugar`

**What it does:** Solid dryer: wet sugar powder dried to sorption equilibrium

**The story:** / flowsheetDict (LEAF node)

Wet sugar dried by a REAL hot-air STREAM (no airTemperature/RH parameters, no phantom duty): the hot air BRINGS the heat (its sensible cooling) and CARRIES AWAY the moisture (a humid exhaust). Adiabatic → the outlet T is a RESULT; the plant energy balance closes with real streams.

Wet solid: 10 kmol/h sucrose (3423 kg/h) + 38 kmol/h bound water (X_in=0.20). Hot air: 800 kmol/h at 420 K, DRY (2 mol% water → RH ~0.4%) – hot dry air, like a real sugar dryer. Its low RH sets a low GAB equilibrium moisture, so the sugar dries to X_eq ~0.001 (~0.1 wt%); the exhaust leaves carrying the evaporated water (838 > 800 kmol/h, the pickup visible in the molar balance).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | drySolid | F | 0.00283583138708 |
| stream | hotAir | F | 0.222222222229 |
| stream | humidExhaust | F | 0.23271974852 |
| stream | wetSolid | F | 0.0133333576778 |
| kpi | solidDryer | T_out | 346.659472289 |
| kpi | solidDryer | X_equilibrium | 0.00109945367934 |
| kpi | solidDryer | X_final | 0.00109945367934 |
| kpi | solidDryer | X_initial | 0.199991235822 |
| kpi | solidDryer | air_in_kmol_h | 800.000000025 |
| kpi | solidDryer | drySolid_flow | 0.951878730237 |
| kpi | solidDryer | moisture_pct_wb | 0.109824620851 |
| kpi | solidDryer | water_activity | 0.00431158404412 |
| kpi | solidDryer | water_removed | 0.189112936127 |
| stream | drySolid | T | 346.659472289 |
| *... and 9 more reference values in the case's `expected`.* | | | |

### solidDryer02_starved_air

`tutorials/steady/drying/solidDryer02_starved_air`

**What it does:** Solid dryer starved of air: the exhaust saturates before the sugar reaches its isotherm

**The story:** / flowsheetDict (LEAF node)

solidDryer01_sugar with its hot air CUT FROM 800 TO 100 kmol/h – the same wet sugar (10 kmol/h sucrose, 38 kmol/h bound water, X_in = 0.20, at 330 K), the same air (420 K, 2 mol% water), one eighth of it.

WHAT IT SHOWS: the third limit of the solid dryer, SATURATION. The air can still PAY for the isotherm's 680.8 kg/h – its sensible heat plus the warm sugar's would cool the exhaust to ~318.6 K – but 100 kmol/h of air at 318.6 K cannot CARRY that water: it saturates (a_w = 1) at about 157 kg/h. The dryer stops there, the powder leaves at X ~0.15 kg/kg instead of the isotherm's X_eq ~0.001, and the run says so (the '[saturation-bound]' result block and a WARNING in the caveat block). Until 2026-10-05 (DEV.md C37 item 3) the unit evaporated all 680.8 kg/h, priced the exhaust at its condensed equilibrium – a fog carrying ~520 kg/h of liquid water – and published it as a vapour, vf = 1.

The outlet temperature barely moves between the two answers: a fog and a saturated gas beside the same water left on the powder hold the same enthalpy. What moves is WHERE the water is – which is the whole point. Witness of check_evaporative_dryer arm (g).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | drySolid | F | 0.0109196509746 |
| stream | drySolid | T | 318.606571264 |
| stream | hotAir | F | 0.0277777777778 |
| stream | hotAir | T | 420 |
| stream | humidExhaust | F | 0.030191484481 |
| stream | humidExhaust | T | 318.606571264 |
| stream | wetSolid | F | 0.0133333576778 |
| stream | wetSolid | T | 330 |
| kpi | solidDryer | T_out | 318.606571264 |
| kpi | solidDryer | T_wetbulb | 315.70439014 |
| kpi | solidDryer | X_equilibrium | 0.00109902098433 |
| kpi | solidDryer | X_final | 0.154259849734 |
| kpi | solidDryer | X_initial | 0.199991235822 |
| kpi | solidDryer | air_in_kmol_h | 100 |

*... and 8 more reference values in the case's `expected`.*

## sprayDryer01_sugar

`tutorials/steady/drying/sprayDryer01_sugar`




**What it does:** Spray dryer: 40 wt% sugar solution → powder (rotary atomiser, Ranz-Marshall)

**The story:** flowsheet (one unit, two feeds, two products).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N2/O2/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.0138888888889 |
| stream | exhaustAir | F | 0.0148043580604 |
| stream | feed | F | 0.000957611111111 |
| stream | powder | F | 4.2141939629e-05 |
| kpi | dryer | Nu | 2.24583682227 |
| kpi | dryer | Pr | 0.746977616919 |
| kpi | dryer | Re | 0.203916784363 |
| kpi | dryer | Sc | 0.6026290984 |
| kpi | dryer | Sh | 2.22885536787 |
| kpi | dryer | T_out | 366.288022769 |
| kpi | dryer | T_wetbulb | 318.053221267 |
| kpi | dryer | d_droplet | 6.04023981348e-05 |
| kpi | dryer | d_droplet_micron | 60.4023981348 |
| kpi | dryer | d_particle | 3.80926671659e-05 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## sprayDryer02_residence_sweep

`tutorials/steady/drying/sprayDryer02_residence_sweep`




**What it does:** Spray dryer: chamber-height sweep → the drying curve (residence = L/v; constant + falling rate → equilibrium)

**The story:** sprayDryer01's sugar dryer with the drying-chamber HEIGHT swept by system/outerDict (0.01 → 0.3 m, 40 pts): residence time = L / v_particle traces the DRYING CURVE – constant-rate then falling-rate, down to the equilibrium moisture.

The base flowsheet is sprayDryer01_sugar's, unchanged: one unit, two feeds, two products.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N2/O2/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.0138888888889 |
| stream | exhaustAir | F | 0.0148043580604 |
| stream | feed | F | 0.000957611111111 |
| stream | powder | F | 4.21419396495e-05 |
| kpi | dryer | Nu | 2.23180639411 |
| kpi | dryer | Pr | 0.746977616919 |
| kpi | dryer | Re | 0.181305061943 |
| kpi | dryer | Sc | 0.6026290984 |
| kpi | dryer | Sh | 2.21579410729 |
| kpi | dryer | T_out | 366.288022771 |
| kpi | dryer | T_wetbulb | 318.053221267 |
| kpi | dryer | X_critical | 0.15 |
| kpi | dryer | X_equilibrium | 0.0156916759905 |
| kpi | dryer | X_final | 0.0156916760249 |

*. . . and 36 more reference values in the case's `expected`.*

## sprayDryer03_pressure_nozzle

`tutorials/steady/drying/sprayDryer03_pressure_nozzle`




**What it does:** Spray dryer: 40 wt% sugar solution → powder (PRESSURE-SWIRL nozzle at 40 bar, d32=9575∗dP^-1/3; Ranz-Marshall)

**The story:** spray dryer: pressure-swirl-nozzle atomisation (d32 from nozzle dP).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N2/O2/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a PRESSURE-SWIRL nozzle at 40 bar (d32 = 9575∗dP^(-1/3) um, the class-average design curve) instead of a wheel.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.0138888888889 |
| stream | exhaustAir | F | 0.0148043580604 |
| stream | feed | F | 0.000957611111111 |
| stream | powder | F | 4.2141939629e-05 |
| kpi | dryer | Nu | 2.24532614778 |
| kpi | dryer | Pr | 0.746977616919 |
| kpi | dryer | Re | 0.203070475508 |
| kpi | dryer | Sc | 0.6026290984 |
| kpi | dryer | Sh | 2.2283799688 |
| kpi | dryer | T_out | 366.288022769 |
| kpi | dryer | T_wetbulb | 318.053221267 |
| kpi | dryer | X_critical | 0.15 |
| kpi | dryer | X_equilibrium | 0.0156916759917 |
| kpi | dryer | X_final | 0.0156916759917 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## sprayDryer04_profiles

tutorials/steady/drying/sprayDryer04_profiles




**What it does:** Spray dryer DISTRIBUTED: axial profiles of moisture, size and temperature along the chamber

**The story:** spray dryer: axial profiles of moisture, droplet size and T along the chamber.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N2/O2/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.0138888888889 |
| stream | exhaustAir | F | 0.0148043580604 |
| stream | feed | F | 0.000957611111111 |
| stream | powder | F | 4.2141939629e-05 |
| kpi | dryer | Nu | 2.24583682227 |
| kpi | dryer | Pr | 0.746977616919 |
| kpi | dryer | Re | 0.203916784363 |
| kpi | dryer | Sc | 0.6026290984 |
| kpi | dryer | Sh | 2.22885536787 |
| kpi | dryer | T_out | 366.288022769 |
| kpi | dryer | T_wetbulb | 318.053221267 |
| kpi | dryer | X_critical | 0.15 |
| kpi | dryer | X_equilibrium | 0.0156916759917 |
| kpi | dryer | X_final | 0.0156916759917 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## sprayDryer05_whey

`tutorials/steady/drying/sprayDryer05_whey`




**What it does:** Spray dryer: whey (prob 5.2) – 500 kg/h, X0=5, rotary co-current, sorption-isotherm residual moisture

**The story:** spray dryer: whey feed (500 kg/h, X0=5), rotary co-current, sorption-isotherm residual.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: ~500 kmol/h of hot AIR (N2/O2/Ar) at 180 C – sized to carry the whey water. Whey solids modelled as sucrose (lactose proxy, a disaccharide MW~342).

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.138888888889 |
| stream | exhaustAir | F | 0.145294374409 |
| stream | feed | F | 0.00649261111111 |
| stream | powder | F | 8.71255906462e-05 |
| kpi | dryer | Nu | 2.33731434528 |
| kpi | dryer | Pr | 0.745230561117 |
| kpi | dryer | Re | 0.384509154435 |
| kpi | dryer | Sc | 0.602916461126 |
| kpi | dryer | Sh | 2.31430911286 |
| kpi | dryer | T_out | 378.559527516 |
| kpi | dryer | T_wetbulb | 315.824008236 |
| kpi | dryer | X_critical | 0.5 |
| kpi | dryer | X_equilibrium | 0.0151904301947 |
| kpi | dryer | X_final | 0.0151904301947 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## sprayDryer06_rea

`tutorials/steady/drying/sprayDryer06_rea`




**What it does:** Spray dryer REA (model chen): Reaction Engineering Approach kinetics – smooth rate, no critical-moisture break

**The story:** spray dryer: Reaction Engineering Approach (REA) drying kinetics (Chen model).

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N2/O2/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.0138888888889 |
| stream | exhaustAir | F | 0.0148043580604 |
| stream | feed | F | 0.000957611111111 |
| stream | powder | F | 4.2141939629e-05 |
| kpi | dryer | Nu | 2.24583682227 |
| kpi | dryer | Pr | 0.746977616919 |
| kpi | dryer | Re | 0.203916784363 |
| kpi | dryer | Sc | 0.6026290984 |
| kpi | dryer | Sh | 2.22885536787 |
| kpi | dryer | T_out | 366.288022769 |
| kpi | dryer | T_wetbulb | 318.053221267 |
| kpi | dryer | X_critical | 0.15 |
| kpi | dryer | X_equilibrium | 0.0156916759917 |
| kpi | dryer | X_final | 0.0156916759917 |
| *... and 40 more reference values in the case's `expected`.* | | | |

### sprayDryer07_design

tutorials/steady/drying/sprayDryer07_design




**What it does:** Spray dryer DESIGN: give targets (powder size, residence), the sizing pass inverts to the geometry

**The story:** spray dryer: DESIGN mode – targets in, the sizing pass inverts to geometry.

feed: 100 kg/h of a 40 wt% sucrose solution at 25 °C. 40 kg/h sucrose / 342.30 = 0.11686 kmol/h 60 kg/h water / 18.015 = 3.33056 kmol/h total 3.4474 kmol/h ; z_sucrose = 0.03390

dryingAir: 50 kmol/h of hot air (real air (N2/O2/Ar)) at 200 °C, 1 bar. Dry air (no inlet humidity) for a clean wet-bulb demo.

The dryer atomises the feed with a 0.1 m disc spinning at 20 000 rpm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | dryingAir | F | 0.0138888888889 |
| stream | exhaustAir | F | 0.0148043580604 |
| stream | feed | F | 0.000957611111111 |
| stream | powder | F | 4.2141939629e-05 |
| kpi | dryer | Nu | 2.24583682227 |
| kpi | dryer | Pr | 0.746977616919 |
| kpi | dryer | Re | 0.203916784363 |
| kpi | dryer | Sc | 0.6026290984 |
| kpi | dryer | Sh | 2.22885536787 |
| kpi | dryer | T_out | 366.288022769 |
| kpi | dryer | T_wetbulb | 318.053221267 |
| kpi | dryer | X_critical | 0.15 |
| kpi | dryer | X_equilibrium | 0.0156916759917 |
| kpi | dryer | X_final | 0.0156916759917 |
| *... and 52 more reference values in the case's `expected`.* | | | |

## economics

**The theory you need.** Sizing (stirred tanks, shell-and-tube areas) feeds costing (Turton bare modules); prices are CASE data (`constant/economics`, dated and cited) – the engine refuses to invent a price. Expect honest refusal when a price is missing, never a silent default.

## economics01_esterification_dcf

tutorials/steady/economics/economics01_esterification_dcf




**What it does:** Esterification plant: CAPEX (Turton) → OPEX (COM_d) → discounted-cash-flow appraisal (NPV/IRR/payback)

**The story:** A FIXED (designed) esterification plant, sized for a full economic appraisal rather than an optimisation:

feed -→ reactor (CSTR) -→ heater -→ separator (flash) |→ liqProd (ethyl acetate rich) |→ vapProd

The feed is scaled to plant throughput so the reactor volume and the heat-exchanger area land INSIDE the Turton correlation validity ranges (V in [0.3, 520] m^3, A in [10, 1000] m^2) – the resulting CAPEX is a physically meaningful base for the DCF, not an extrapolated artefact.

Prices (product/raw-material/labour) live in constant/economics, each dated and primary-cited. The postDict chain is sizing → costing → economics; the economics pass prints the COM_d / NPV / IRR / payback appraisal and publishes the headline scalars into the economics KPIs.

*A fixed (designed) ethyl-acetate plant taken all the way from equipment sizing to a discounted-cash-flow economic appraisal:*

*1. sizing — StirredTank for the CSTR, ShellTubeHX for the heater, sized from each unit's KPIs. 2. costing — Guthrie/Turton purchased + bare-module + total-module cost, CEPCI-updated to the target year, in EUR. 3. economics — the new EconomicsPass: it builds the CAPEX ladder (FCI, working capital, TCI), the Turton cost-of-manufacture COM_d, revenue, a straight-line-depreciation cash-flow timeline, then NPV, IRR (bisection bracket + Newton polish), and discounted + simple payback.*

*Every cost factor and DCF assumption is dict-owned and printed — read the appraisal block and you can re-derive every number by hand.*

*Depreciation enters the DCF exactly once (it is ∗not∗ in COM_d) — the classic textbook double-count is avoided. Working capital is an outflow at startup and is recovered untaxed at end-of-life.*

*Market/time-specific data — product and feed prices, labour rate, FX — lives only in constant/economics, each value dated and primary-cited (an index/bulletin/agency named in a provenance{} block). The EconomicsPass refuses to run if a required price is absent, with a remedy message; set economics.refuseOnMissingPrice 0; in the postDict to switch to defaulted-but-loud. No price ever defaults to a literal in code.*

*The economics pass also publishes its headline scalars into result.kpis["economics"] (IRR, NPV, paybackYears, FCI, TCI, COM_d), which is what lets a sweep or optimisation read economics as an ordinary response — see economics02_irr_sweep.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.138888888889 |
| stream | hotStream | F | 0.138888888889 |
| stream | liqProd | F | 0.129188220152 |
| stream | reactorOut | F | 0.138888888889 |
| stream | vapProd | F | 0.00970066873646 |
| kpi | economics | COM_d | 221216158.676 |
| kpi | economics | FCI | 395275.413492 |
| kpi | economics | NPV | -985322163.869 |
| kpi | economics | TCI | 454566.725516 |
| kpi | economics | WC | 59291.3120238 |
| kpi | economics | revenue | 18311056.5908 |
| kpi | heater | F | 0.138888888889 |
| kpi | heater | P | 101325 |
| kpi | heater | Q | 300000 |

*... and 81 more reference values in the case's `expected`.*

## economics02_irr_sweep

tutorials/steady/economics/economics02_irr_sweep




**What it does:** Sweep reactor volume; read economics.IRR/NPV/payback per point (SweepDriver runs the post chain)

**The story:** A FIXED (designed) esterification plant, sized for a full economic appraisal rather than an optimisation:

feed -→ reactor (CSTR) -→ heater -→ separator (flash) |→ liqProd (ethyl acetate rich) |→ vapProd

The feed is scaled to plant throughput so the reactor volume and the heat-exchanger area land INSIDE the Turton correlation validity ranges (V in [0.3, 520] m^3, A in [10, 1000] m^2) – the resulting CAPEX is a physically meaningful base for the DCF, not an extrapolated artefact.

Prices (product/raw-material/labour) live in constant/economics, each dated and primary-cited. The postDict chain is sizing → costing → economics; the economics pass prints the COM_d / NPV / IRR / payback appraisal and publishes the headline scalars into the economics KPIs.

*The same esterification plant as economics01, but driven by a sensitivity sweep over the reactor volume that reads economic responses at every point:*

*The SweepDriver now runs the full post-processing chain (sizing → costing → economics) on each converged point, so the headline economic scalars the EconomicsPass publishes into result.kpis["economics"] resolve as ordinary unit.kpi responses. A sweep can finally ∗see cost∗:*

*On this case's declared price basis every point LOSES money: the yearly cash flow is negative, so there is no rate at which the NPV is zero (the IRR is published as null, never as a number) and no payback. The sweep still shows the trade-off: a bigger reactor raises conversion and CAPEX together, and over this range the NPV becomes less negative as the reactor grows. (This table was regenerated 2026-11-03 from the run; the one it replaced showed a profitable plant that the golden had stopped describing before then.)*

*> The range floors at 4 $m^3$ on purpose: below ~3 $m^3$ the lower conversion > drops the mixture bubble point and the fixed-duty heater would cross the > saturation dome (a heater is sensible-only) — the deliberate validity edge.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.138888888889 |
| stream | hotStream | F | 0.138888888889 |
| stream | liqProd | F | 0.127980647247 |
| stream | reactorOut | F | 0.138888888889 |
| stream | vapProd | F | 0.010908241642 |
| kpi | economics | COM_d | 221239267.448 |
| kpi | economics | FCI | 448083.733462 |
| kpi | economics | NPV | -972143541.209 |
| kpi | economics | TCI | 515296.293481 |
| kpi | economics | WC | 67212.5600193 |
| kpi | economics | revenue | 21059471.3188 |
| kpi | heater | F | 0.138888888889 |
| kpi | heater | P | 101325 |
| kpi | heater | Q | 300000 |

*. . . and 78 more reference values in the case's `expected`.*

## economics03_startup_years

tutorials/steady/economics/economics03_startup_years

**What it does:** The start-up years of a discounted cash flow (DEV.md 4c C63): ammonia02_full_plant's plant and costing, appraised over a THREE-year construction with a declared capital schedule, a two-year production ramp-up and a start-up cost expensed in year 1 – the cash flow is printed year by year and checked by hand in system/postDict. Half the fixed capital is borrowed (DEV.md 4c C110 G4): the loan's interest is capitalised during construction and repaid as an annuity, and the owners' EQUITY NPV/IRR are printed BESIDE the project's, which the loan does not move.

**The story:** the plant below is ammonia02_full_plant's, copied unchanged; only system/postDict's 'economics {}' block differs.

ammonia02_full_plant – the complete Haber-Bosch plant, ~59 t/h NH3

makeup 25 bar [makeupComp 25→200] recycle [recycleComp] [mixer] mixed (cool) cold side [FEHE] hot effluent (heat recovery) preheatedFeed (~683 K) [converter] gibbsReactor mode ADIABATIC N2+3H2<→2NH3 exothermic → self-heats to ~840 K hotEffluent hot side of FEHE → cooledEffluent [waterCooler] (cooling water) ~310 K [separator] flash 250 K/200 bar (refrig.) rawLiquid unreactedGas [letdown] [purgeSplit] productNH3 gas purge recycle (tear)

THE REACTOR HEAT. N2+3H2→2NH3 is EXOTHERMIC – there is NO external reactor heating. The converter runs ADIABATIC: the exotherm raises the gas T (feed 683 K in, 840 K out), and the feed-effluent exchanger (FEHE) recovers that heat to preheat the incoming feed to catalyst light-off. Self-sustaining once lit. The compressors set the 200-bar synthesis pressure; the water cooler (cooling-water utility) + refrigerated flash condense the NH3. Every machine is explicit and its duty/work is reported.

CAPACITY ~59 t/h NH3 (~1400 t/day) at 98.6% product purity – a world-scale single train.

DISSOLVED GASES BY HENRY'S LAW. H2 and N2 are supercritical at the separator (no physical Psat), so their dissolved amounts follow MEASURED Henry constants (pairs H2-NH3 from Wiebe 1934, N2-NH3 from Larson & Black 1925 as evaluated in IUPAC SDS Vol 5/6), with the SRK vapour fugacity at 200 bar: K = H(T)/(phi_V P). The raw liquid carries ~0.5% H2 + 0.2% N2 – physical, vs the ~13% a Psat extrapolated above Tc used to fake. KNOWN RESIDUAL: Ar has no measured Ar-NH3 Henry pair yet, so its dissolved ~1% is a Raoult-extrapolation artefact (~7x over by analogy with N2) – the product's main impurity is overstated.

THE SIMULATION STRATEGY (how you build a plant like this). 1. PICK THE THERMO BY THE PHYSICS, per phase ("the two worlds"): the 200-bar GAS side needs real-gas fugacity → a cubic EoS (SRK) declared once in the package; the reactor folds phi into its equilibrium automatically. The DISSOLVED gases are supercritical solutes in a condensable solvent → the unsymmetric convention (Henry / Krichevsky-Kasarnovsky), NOT Raoult: pair files H2-NH3 + N2-NH3 fitted to measured solubility, 'solvent NH3;' in the package, 'role solute;' overlays on H2/N2 in this case's constant/. 2. TEAR THE LOOP WHERE INFORMATION FLOWS BACKWARD: one MASS tear (recycle, guessed then Newton-converged) and one ENERGY tear (hotEffluent – the FEHE preheats the feed with the reactor's own product heat, so the converter's outlet is needed upstream of the converter). Both declared in solverDict tearStreams; the solver announces every iteration. 3. LET THE EXOTHERM DO THE HEATING: the converter is ADIABATIC (mode adiabatic; T is a guess, not a spec) – the reaction self-heats 683 → 840 K and the FEHE recycles that heat to light off the feed. No phantom reactor heater exists anywhere. 4. MAKE EVERY MACHINE EXPLICIT so it can be SIZED AND COSTED: compressors by shaft power, exchangers by area, the converter + drums by vessel design bases (space velocity / residence time) in system/postDict → Turton costs → the DCF economics in constant/economics. Every unit operation carries its own sizing – the doctrine. 5. VALIDATE AGAINST MEASUREMENT AT EVERY JOINT: the reactor equilibrium against the ideal-vs-SRK shift (gibbs10), the dissolved gas against Wiebe 1934 / Larson-Black 1925 (x_H2 at 400 atm: 0.0287 computed vs 0.0282 measured), the balances against 100% element closure.

HONEST SIMPLIFICATIONS. (1) The converter is a SINGLE adiabatic bed, so it runs hotter (~567 C) than a real multi-bed converter with interbed quench cooling (which stays near the 400-500 C equilibrium sweet spot for more NH3 per pass). (2) The makeup compressor is one stage (real: multi-stage with

intercooling), so its discharge is hot. (3) The let-down gas and the loop pressure drops are as in ammonia01. (4) 'water' is in the package ONLY as the cooling-water utility fluid (it is not a reaction species).

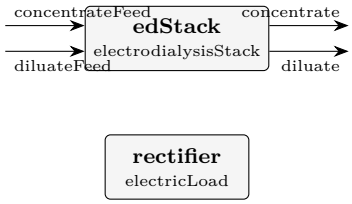
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | chilledEffluent | F | 10.2992003164 |
| stream | chilledEffluent | T | 311.655519488 |
| stream | compMakeup | F | 2.22222222222 |
| stream | compMakeup | T | 656.394257943 |
| stream | compRecycle | F | 9.04024011142 |
| stream | compRecycle | T | 263.877015971 |
| stream | cooledEffluent | F | 10.2992003164 |
| stream | cooledEffluent | T | 485.640252687 |
| stream | cw | F | 138.888888889 |
| stream | cw | T | 298.15 |
| stream | cwOut | F | 138.888888889 |
| stream | cwOut | T | 303.664081245 |
| stream | hotEffluent | F | 10.2991998088 |
| stream | hotEffluent | T | 839.643638219 |
| *... and 305 more reference values in the case's `expected`.* | | | |

## electrodialysis

## ed01_nacl_desalination

tutorials/steady/electrodialysis/ed01_nacl_desalination




**What it does:** Electrodialysis stack: NaCl diluate desalination, 100 cell pairs

**The story:** ED01 NaCl desalination by electrodialysis

An electrodialysis stack of 100 cell pairs desalinates a dilute NaCl diluate stream (0.1 mol/kg) against a more-concentrated concentrate (0.5 mol/kg). A specified stack current (BELOW the limiting current) drives Na+ through the CMX (cation-exchange) membranes and Cl- through the AMX (anion-exchange) membranes, OUT of the diluate and INTO the concentrate.

diluateFeed → +————————+ → diluate (desalinated) | ED stack | concentrateFeed → | (CMX/AMX, N=100)| → concentrate (enriched) +————————+ | (power, energy wire) v rectifier (electricLoad)

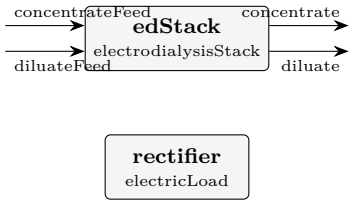
The stack PRINTS (verbosity 3) the Faraday transfer, the Nernst membrane potential from REAL Davies activities, the Cowan-Brown limiting current i_lim (and i/i_lim), the ohmic drop and the stack voltage/power.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.00501492454831 |
| stream | concentrateFeed | F | 0.005 |
| stream | diluate | F | 0.00498507545169 |
| stream | diluateFeed | F | 0.005 |
| kpi | edStack | E_mem_pair | 0.0794573845707 |
| kpi | edStack | I | 8 |
| kpi | edStack | N_cellpairs | 100 |
| kpi | edStack | R_pair | 0.00507753048751 |
| kpi | edStack | U | 13.5077628471 |
| kpi | edStack | W_electric | 108.062102777 |
| kpi | edStack | W_electric_kW | 0.108062102777 |
| kpi | edStack | current_efficiency | 0.9 |
| kpi | edStack | demin_ratio | 0.831450327631 |
| kpi | edStack | i_density | 40 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## ed02_over_limiting_current

tutorials/steady/electrodialysis/ed02_over_limiting_current




**What it does:** Electrodialysis stack driven ABOVE its limiting current (loud warning)

**The story:** ED02 over-limiting current electrodialysis

Same stack as ED01, but run on a MORE DILUTE diluate (0.02 mol/kg) at a LOW cross-flow velocity and a HIGH current. The diluate-side limiting current density i_lim (Cowan-Brown) is small; the applied current density EXCEEDS it, so the stack runs in the OVER-LIMITING regime.

Choupo does NOT silently clamp the current (the "no silent crutch" credo): it computes the unclamped Faraday transfer and prints a LOUD warning that i > i_lim – the diluate boundary layer is ion-depleted, water splitting / pH swings set in, and the real current efficiency would collapse. The student SEES the regime and the i/i_lim ratio, then chooses to reduce the current or raise the flow / feed concentration.

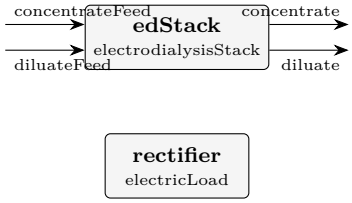
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.00500205212539 |
| stream | concentrateFeed | F | 0.005 |
| stream | diluate | F | 0.0199979478746 |
| stream | diluateFeed | F | 0.02 |
| kpi | edStack | E_mem_pair | 0.156590406589 |
| kpi | edStack | I | 5.5 |
| kpi | edStack | N_cellpairs | 20 |
| kpi | edStack | R_pair | 0.0130038886532 |
| kpi | edStack | U | 6.06223588363 |
| kpi | edStack | W_electric | 33.34229736 |
| kpi | edStack | W_electric_kW | 0.03334229736 |
| kpi | edStack | current_efficiency | 0.9 |
| kpi | edStack | demin_ratio | 0.142508707778 |
| kpi | edStack | i_density | 27.5 |

*... and 17 more reference values in the case's `expected`.*

## ed03_stack_record

tutorials/steady/electrodialysis/ed03_stack_record




**What it does:** Electrodialysis: the stack named as a record, the crossflow velocity derived

**The story:** ED03 the stack is a RECORD, and the velocity is derived

The SAME NaCl chemistry as ed01 (0.1 mol/kg diluate against a 0.5 mol/kg concentrate, Neosepta CMX/AMX), run through a stack that the case NAMES instead of typing:

– the bench unit of Geraldes & Afonso, J. Membr. Sci. 360 (2010) 499-508, as a 'kind edStack' record in data/standards/assets/. What the record supplies, and what the operation therefore no longer declares:

membranes CMX_AMX (the IEM pair) cellPairs 7 activeArea 0.140 m2 THE CELL-PAIR AREA OF THE WHOLE STACK channel 0.77 mm x 0.114 m x 0.175 m, mesh spacer, porosity 0.90 hydraulicPasses 1 massTransfer Sh = 0.29 Re^0.5 Sc^0.33 (fitted to THIS stack) limits i_max, T_max

WHAT A STUDENT SEES HERE.

(1) THE VELOCITY IS NOT A KNOB. It used to be typed as 'linearVelocity 0.05 m/s;' beside a flow the unit already had on its inlet stream – two homes for one fact, and nothing checked them against each other, while the velocity sets the mass-transfer coefficient and therefore the limiting current. Here the diluate feed carries 175 L/h; the record says seven channels run side by side in one hydraulic pass, each 0.114 m wide and 0.77 mm high; and the run prints u = Q / (n W h) = 175 L/h / (7 x 0.114 x 7.7e-4) = 0.0791 m/s which is the paper's own 0.080 m/s for that flow rate. Change the feed and the velocity follows. Declare a velocity beside 'stack' and the run REFUSES by name.

(2) VITOR'S RULE ON THE AREA, printed as arithmetic: 'activeArea' is the CELL-PAIR area of the whole stack, so this 0.140 m2 stack carries 0.140 m2 of cation-exchange membrane AND 0.140 m2 of anion-exchange membrane (0.280 m2 in total), and the current density is taken on the 0.020 m2 one cell pair presents. Nothing stores any of those three derivatives.

(3) THE ESTIMATES ANNOUNCE THEMSELVES. Four values in the record are not in the paper – the spacer porosity, the Schmidt exponent (which the paper POSTULATES rather than fits), and both limits – and each prints an '[estimate]' line with the note saying how to verify it, and reaches the end-of-run caveat block.

(4) THE LIMITING CURRENT IS PREDICTED, not read off a declared transport number: the run prints D_eff, k_c,eff, the limiting transport number of every ion at each membrane, both membranes' limiting current, and – because NaCl is a single salt – the classical Eq. 16 beside the general Eq. 15, which agree.

diluateFeed → +————————+ → diluate | EUR2C-7P18, 7 pairs| concentrateFeed → | CMX / AMX | → concentrate +————————+ | (power, energy wire) v rectifier (electricLoad)

A bench stack of seven cell pairs in ONE pass removes a few per cent of the diluate's sodium, which is why a real bench run RECIRCULATES the diluate through a tank. That batch arrangement is not modelled here.

*The same NaCl chemistry as ed01_nacl_desalination (0.1 mol/kg diluate against a 0.5 mol/kg concentrate, Neosepta CMX/AMX), run through a stack the case names instead of typing:*

*EUR2C-7P18 is a kind edStack record in data/standards/assets/ — the bench unit of Geraldes & Afonso, ∗J. Membr. Sci.∗ 360 (2010) 499–508.*

*Declaring any of the left-hand keys beside stack refuses by name.*

*It was a second home. The unit already receives the diluate flow on its inlet stream; the crossflow velocity is that flow divided by the section of the channels running side by side. Nothing checked the two against each*

*other, and the velocity is what sets the mass-transfer coefficient and therefore the limiting current — so a typo in a velocity moved an answer silently. The run now prints the arithmetic:*

*175 L/h is one of the paper's own three flow rates, and 0.0791 m/s is its own 0.080 m/s for that flow (section 3.3). The correlation in this record was fitted on the SUPERFICIAL velocity, which the record declares (velocityBasis superficial;) rather than leaving to a reader to guess; the interstitial value is printed beside it so the two are never confused.*

*Change the feed flow and the velocity, the Reynolds number, the Sherwood number and the limiting current all follow. That is the exercise.*

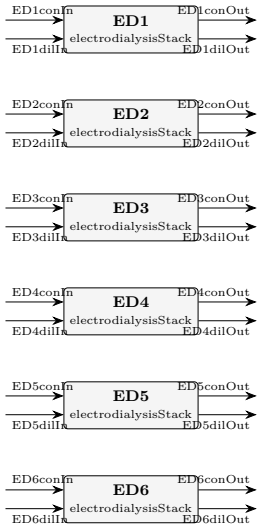
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.00274719599904 |
| stream | concentrate | T | 298.15 |
| stream | concentrateFeed | F | 0.00274693481944 |
| stream | concentrateFeed | T | 298.15 |
| stream | diluate | F | 0.00270778475096 |
| stream | diluate | T | 298.15 |
| stream | diluateFeed | F | 0.00270804593056 |
| stream | diluateFeed | T | 298.15 |
| kpi | edStack | D_eff | 1.60708333333e-09 |
| kpi | edStack | E_mem_pair | 0.0794451090256 |
| kpi | edStack | I | 2 |
| kpi | edStack | N_cellpairs | 7 |
| kpi | edStack | R_pair | 0.0636141624947 |
| kpi | edStack | Re | 68.4552367603 |

*. . . and 37 more reference values in the case's `expected`.*

## ed04_limiting_current_multiionic

tutorials/steady/electrodialysis/ed04_limiting_current_multiionic




**What it does:** Limiting current density, MgCl2 + MgSO4: engine against Geraldes & Afonso (2010) Table 4

**The story:** ED04 the limiting current, PREDICTED, against a measurement

AN EXTERNAL-REFERENCE CASE. Every number below comes from ONE primary source, end to end, and the case says which:

V. Geraldes, M.D. Afonso, "Limiting current density in the electrodialysis of multi-ionic solutions", J. Membr. Sci. 360 (2010) 499-508, doi:10.1016/j.memsci.2010.05.054.

WHAT IS BEING TESTED. The limiting current density of an ED stack used to be a number a student SUPPLIED, through a counter-ion transport number declared on the membrane record. It is now PREDICTED: the linearised Nernst-Planck equations in the diluate film, closed by electroneutrality, give the limiting transport numbers explicitly from the ion diffusivities and the bulk composition (the paper's Eqs. 12 and 13) and the limiting current in closed form (its Eq. 15). Nothing is fitted to this system; the only equipment datum is the stack's own Sherwood correlation, which the record carries.

THE PROVENANCE OF EVERY NUMBER IN THIS CASE.

THE STACK. 'stack EUR2C-7P18;' – the paper's own bench unit, as a 'kind edStack' record in data/standards/assets/ (section 3.1: seven cells, 0.020 m2 per membrane, channels 0.175 x 0.114 m with a 7.7e-4 m net-like spacer; section 3.4: Sh = 0.29 Re^0.5 Sc^0.33). The record marks its four non-source values as estimates and the run announces each one.

THE SOLUTION. Equimolar-in-equivalents MgCl2 + MgSO4 (the paper's "5 + 5" and "10 + 10 equiv./m3" assays, Tables 2 and 4). Each unit below is declared at the AVERAGE cation concentration of its row – the paper computes its prediction on the average of the inlet and outlet bulk concentrations along the channel (section 4.1 and the footnote to Table 4), and that average is what Table 4 lists.

THE FLOW. 145, 175 and 230 L/h (section 3.3). The velocity is NOT declared: the engine derives it from this flow and the record's geometry, u = Q/(7 W h), giving 0.0655, 0.0791 and 0.1040 m/s against the paper's own 0.066, 0.080 and 0.11 m/s.

THE TEMPERATURE. 293.15 K (20 C, section 3.3). The ion diffusivities curated in data/standards/species/ are referenced at 25 C; the engine corrects them to 20 C through Stokes-Einstein on the declared liquid-viscosity model, which is exactly how the paper's Table 3 was built, and the run prints the corrected values.

THE CURRENT. 0.2 A (10 A/m2) on every unit – FAR below every limiting current here. This case measures i_lim; it does not approach it. The applied current changes nothing about i_lim, which is computed from the diluate bulk composition.

THE SIX UNITS ARE THE SIX MgCl2 + MgSO4 ROWS OF TABLE 4:

unit flow Re C_1,in C_1,average i_lim (paper) (equiv./m3) (equiv./m3) (A/m2) ED1 145 L/h 50.0 10.0 9.38 34.3 ED2 175 L/h 61.0 10.0 9.43 37.6 ED3 230 L/h 86.0 10.0 9.54 42.9 ED4 145 L/h 50.0 20.0 18.70 71.2 ED5 175 L/h 61.0 20.0 18.80 81.9 ED6 230 L/h 86.0 20.0 19.00 93.5

The ‘i_lim_cem‘ golden rows are ANCHORS against that last column – a MEASUREMENT, not a self-recorded value – with a band of 13 %, which is the largest average relative deviation the paper itself reports between its model and its data (13 % for MgCl2, 9 % for MgSO4 + MgCl2, section 4.3). ‘i_lim‘ is pinned separately as an ordinary regression row. What the engine actually returns, and the two findings that came out of it, are in the case README and in docs/design/a-stack-is-a-record-and-its-limiting-current-is-predicted.md.

The diluate and the concentrate carry the SAME solution here, which is the paper's own arrangement (section 3.3: both streams were recycled back to a single 6 L tank to hold the concentrations nearly constant), so the Nernst back-EMF is zero by construction.

*An external-reference case: every number in it comes from one primary source, end to end, and the case says which.*

*> V. Geraldes, M.D. Afonso, ∗"Limiting current density in the electrodialysis > of multi-ionic solutions"∗, J. Membr. Sci. 360 (2010) 499–508, > doi:10.1016/j.memsci.2010.05.054.*

*The limiting current density used to be computed from a declared counter-ion transport number — a parameter the student supplied, on a membrane record, measured in a different solution at a different temperature. It is now predicted. The Nernst–Planck equations in the diluate film, linearised on nearly-linear profiles and closed by electroneutrality, give the limiting transport numbers explicitly from the ion diffusivities and the bulk composition (Eqs. 12/13) and the limiting current in closed form (Eq. 15).*

*This removes a parameter. Nothing in the prediction is fitted to this system; the only equipment datum is the stack's own Sherwood correlation, which lives on the stack record where it belongs.*

*Average 6.9 %, against the 9 % the paper reports for this system (section 4.3). The i_lim_cem golden rows are anchors against the measured column with a band of 13 % — the larger of the two average relative deviations the paper itself reports, ∗its∗ number and not one chosen here. i_lim is pinned separately and tightly as an ordinary regression row.*

*1. The paper's own Reynolds number at its highest flow does not follow from its own flow rate. Section 3.3 lists 145, 175 and 230 L/h against superficial velocities 0.066, 0.080 and 0.11 m/s and Reynolds numbers 50, 61 and 86. The first two close exactly on the stated geometry with all seven diluate channels in parallel (0.0655 and 0.0791 m/s); the third does not — the same division gives 0.1040 m/s, and 86/61 = 1.41 is not 230/175 = 1.31 either. The engine derives the velocity from the flow, so ED3 and ED6 run at Re 79.98 rather than 86, which lowers their predicted i_lim by 3.7 % relative to a run at the paper's printed Re. Nothing was adjusted to close the gap: the case declares the flow, which is the measurable.*

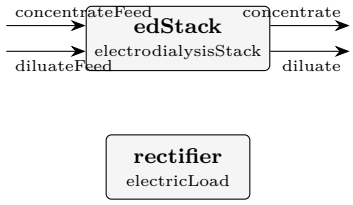
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | ED1conIn | F | 0.00223622618722 |
| stream | ED1conIn | T | 293.15 |
| stream | ED1conOut | F | 0.00223624351959 |
| stream | ED1conOut | T | 293.15 |
| stream | ED1dilIn | F | 0.00223622618722 |
| stream | ED1dilIn | T | 293.15 |
| stream | ED1dilOut | F | 0.00223620885485 |
| stream | ED1dilOut | T | 293.15 |
| stream | ED2conIn | F | 0.00269889671236 |
| stream | ED2conIn | T | 293.15 |
| stream | ED2conOut | F | 0.00269891404762 |
| stream | ED2conOut | T | 293.15 |
| stream | ED2dilIn | F | 0.00269889671236 |
| stream | ED2dilIn | T | 293.15 |

*. . . and 298 more reference values in the case's `expected`.*

## ed05_industrial_stack

tutorials/steady/electrodialysis/ed05_industrial_stack




**What it does:** Electrodialysis: a 100-cell-pair, 50 m2 industrial stack, two hydraulic passes

**The story:** ED05 an INDUSTRIAL stack beside the bench one

ed03 runs a BENCH stack: seven cell pairs, 0.140 m2 of cell-pair area, 175 L/h, and it removes 2.7 % of the diluate's sodium in one pass. This is the same machine grown up:

stack EurodiaED-100P-50; // 100 cell pairs, 50 m2, TWO passes membrane CMX_AMX; // the CASE names the pair – see below targetDemin 0.5; // and Choupo solves for the current

a 100-cell-pair, 50 m2 industrial Eurodia unit that Vitor Geraldes has OPERATED. Its record's facts are his own recollection, marked 'origin measured; reviewStatus unverified;' and announced on every run – they are not a data sheet.

THREE THINGS A STUDENT SHOULD READ OFF THE LOG.

(1) THE CASE NAMES THE MEMBRANE PAIR, and the record does not. He did not say which membranes the unit carried, so the record leaves the field out – a stack FRAME takes any ion-exchange pair, exactly as a laboratory flat-sheet cell takes any coupon. Omit 'membrane' here and the run REFUSES by name. (ed03's EUR2C-7P18 DOES name its pair, and declaring one beside it refuses instead. The record decides which way the refusal points; the engine never guesses a pair.)

(2) THE CHANNEL LENGTH IS DERIVED, AND SAYS SO. Nobody stated the channel width or its length. Exactly ONE of the two is stored, and it is the ESTIMATE: 'channelWidth 0.298 m', back-calculated from a superficial velocity of 0.080 m/s at the design flow of 3000 L/h, with a note saying it must be verified by measuring a membrane sheet. The length is then activeArea/(cellPairs x channelWidth) = 1.680 m, derived where it is used and printed as DERIVED. Storing a back-calculated length beside a back-calculated width would be two homes for one guess, and the second would read like a declaration.

(3) TWO HYDRAULIC PASSES. 100 cell pairs over 2 passes is 50 channels side by side, so the same 3000 L/h gives twice the velocity it would in a single pass – and a parcel of diluate travels 1.680 m twice, 3.360 m in all. The pass belongs to the EQUIPMENT (Vitor's ruling), so it is on the record and a case that declares it is refused.

WHY THE MASS-TRANSFER CORRELATION MAY BE USED HERE. It is the BENCH stack's, fitted over Re 50-86. At the 0.066-0.110 m/s band that bench stack was run at, this stack sits at Re 46-77 on its 0.70 mm channel against the bench stack's 51-84 on its 0.77 mm – the two bands very nearly coincide, because Re is built on the channel height and the two spacer thicknesses very nearly coincide. So the fit is APPLIED INSIDE its own band, not extrapolated. The record declares that band ('validity { Re ( 50 86 ); }') and the engine ANNOUNCES any run that leaves it; at the design flow this case sits at Re 62.8.

The permission is about the REYNOLDS NUMBER only: this spacer is ZIG-ZAG and the bench stack's was a MESH, no correlation in this tree selects on the spacer type, and the record says so.

diluateFeed → +——————+ → diluate | EurodiaED-100P-50 | concentrateFeed → | 100 pairs, CMX/AMX | → concentrate +——————+ | (power, energy wire) v rectifier (electricLoad)

*ed03_stack_record runs a bench stack: seven cell pairs, 0.140 m² of cell-pair area, 175 L/h, and it removes 2.7 % of the diluate's sodium in one pass. This is the same machine grown up — a 100-cell-pair, 50 m² industrial Eurodia unit that V?tor Geraldes has operated, at 3000 L/h, taking 50 % of the sodium out.*

*data/standards/assets/EurodiaED-100P-50.dat is not a data sheet. Its numbers — 100 cell pairs, 50 m², a channel height of ∗"aprox"∗ 0.7 mm, a zig-zag spacer, two hydraulic passes, a design flow of 3000 L/h —*

*are what their owner remembers of a unit he ran, and each carries origin measured; reviewStatus unverified; with a note saying how to verify it. They are announced on every run.*

*He did not say which membranes the unit carried, so the record leaves the field out. A stack ∗frame∗ takes any ion-exchange pair, exactly as the SEPA CF laboratory cell takes any coupon. Omit membrane here and the run refuses by name; declare one beside stack EUR2C-7P18; in ed03, whose record ∗does∗ name its pair, and that refuses instead. The record decides which way the refusal points. Neither side guesses a pair on the owner's behalf.*

*Nobody stated the channel width or its length. Exactly one of the two is stored, and it is the estimate:*

*Storing a back-calculated length ∗beside∗ a back-calculated width would be two homes for one guess, and the second would read like a declaration. (ed03's record does declare its length, because its paper states the width, the length ∗and∗ the effective area as three separate facts — and the engine cross-checks them.)*

*100 cell pairs over 2 passes is 50 channels side by side, so the same 3000 L/h gives twice the velocity a single pass would, and a parcel of diluate travels 1.68 m twice. The pass belongs to the equipment (V?tor's ruling), so it is on the record; a case that declares it is refused.*

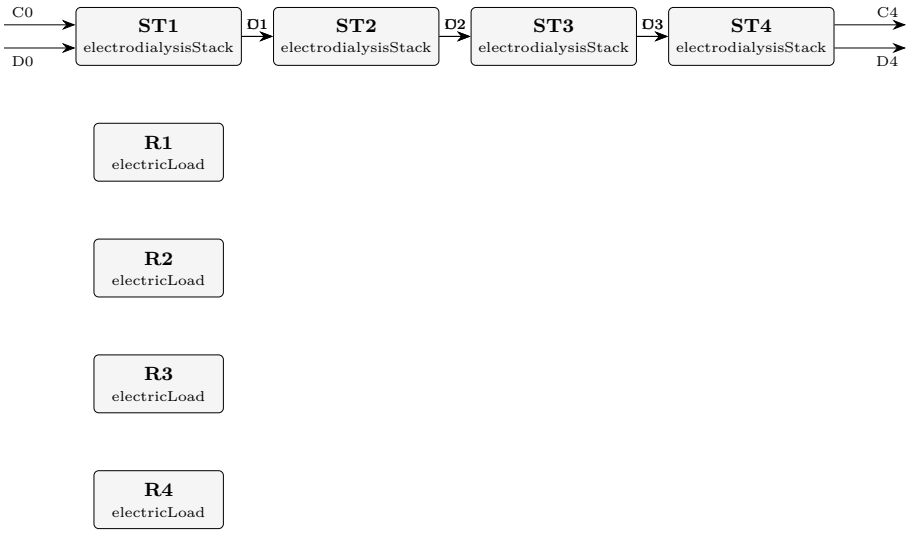
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.0467153111984 |
| stream | concentrate | T | 298.15 |
| stream | concentrateFeed | F | 0.0466736445318 |
| stream | concentrateFeed | T | 298.15 |
| stream | diluate | F | 0.0462986445318 |
| stream | diluate | T | 298.15 |
| stream | diluateFeed | F | 0.0463403111984 |
| stream | diluateFeed | T | 298.15 |
| kpi | edStack | C_ion_Cl | 50 |
| kpi | edStack | C_ion_Na | 50 |
| kpi | edStack | D_eff | 1.60708333333e-09 |
| kpi | edStack | D_ion_Cl | 2.03e-09 |
| kpi | edStack | D_ion_Na | 1.33e-09 |
| kpi | edStack | E_mem_pair | 0.0772348315003 |

*. . . and 37 more reference values in the case's* `expected`*.*

## ed06_stages_in_series

tutorials/steady/electrodialysis/ed06_stages_in_series




**What it does:** Four electrodialysis stages in series: the limiting current falls 15.6x along the train, so each stage is given its own current

**The story:** FOUR electrodialysis stages in series, and why a plant steps the current DOWN along the train.

THE NETWORK NEEDS NO ELECTRICAL TOPOLOGY, and that is the point worth stating first. An ELECTRICAL STAGE is a set of cell pairs served by ONE electrode pair on its own rectifier. So stages in series are simply UNITS in series: four ‘electrodialysisStack‘ units, each with its own ‘electricLoad‘ rectifier on its own energy wire, the diluate passing from one to the next and the concentrate loop passing through all four. Nothing in the engine models a busbar, and nothing needs to.

D0 –[ ST1 ]– D1 –[ ST2 ]– D2 –[ ST3 ]– D3 –[ ST4 ]– D4 C0 –[ ]– C1 –[ ]– C2 –[ ]– C3 –[ ]– C4 | | | | R1 R2 R3 R4 (four rectifiers)

Each stage is the industrial frame of ‘ed05_industrial_stack‘ (EurodiaED-100P-50: 100 cell pairs, 50 m2 of cell-pair area, 0.5 m2 each), and each is asked for ‘targetDemin 0.6‘ – sixty per cent of ITS OWN inlet – so the ENGINE solves the current every stage needs instead of the case declaring four numbers.

WHAT THE RUN MEASURES (printed by this run; nothing here is a target):

stage I [A] i [A/m2] i_lim [A/m2] i/i_lim 1 26.80 53.60 338.88 0.158 2 10.72 21.44 135.55 0.158 3 4.29 8.58 54.22 0.158 4 1.72 3.43 21.69 0.158

overall demineralisation 97.44 % (0.05 → 0.00128 mol/kg)

THE LESSON. The limiting current is a PREDICTION from the diluate composition (Geraldes & Afonso, J. Membr. Sci. 360 (2010) 499-508), so as the diluate is stripped down the train i_lim falls with it – 338.88 A/m2 at the first stage, 21.69 A/m2 at the fourth, a factor of 15.6. A plant therefore CANNOT hold one current density along the train: stage 1 runs at 53.60 A/m2, and that same density in stage 4 would be 53.60 / 21.69 = 2.47 times ITS limiting current – deep in the water-splitting regime, with the current efficiency collapsing and the membranes stressed. Running each stage at its own current keeps the same 0.158 margin everywhere, and the current falls by the same factor the inlet concentration does.

This is the steady twin of ‘edbatch01_constant_current‘: there ONE stack sees a falling concentration in TIME and crosses its own limiting current; here FOUR stacks see it in SPACE and each is given a current that suits where it sits. A batch rig and a series train are the same physics read along two different axes.

WHAT IS PINNED AND WHAT IS NOT. The golden pins the answer this model gives. Nothing here reproduces a measured four-stage plant; the stack record is Vitor Geraldes’ own industrial unit as he remembers it (its own header says so, value by value) and its mass-transfer correlation is BORROWED from the bench stack and applied inside the Reynolds band it was fitted over. The concentrate loop is

CO-CURRENT and single-pass; no back-diffusion, no water transport and no inter-stage pumping are modelled.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | C0 | F | 0.0466736445318 |
| stream | C0 | T | 298.15 |
| stream | C1 | F | 0.0467236445318 |
| stream | C1 | T | 298.15 |
| stream | C2 | F | 0.0467436445318 |
| stream | C2 | T | 298.15 |
| stream | C3 | F | 0.0467516445318 |
| stream | C3 | T | 298.15 |
| stream | C4 | F | 0.0467548445318 |
| stream | C4 | T | 298.15 |
| stream | D0 | F | 0.0463403111984 |
| stream | D0 | T | 298.15 |
| stream | D1 | F | 0.0462903111984 |
| stream | D1 | T | 298.15 |
| *... and 178 more reference values in the case's `expected`.* | | | |

## ed07_feed_and_bleed

tutorials/steady/electrodialysis/ed07_feed_and_bleed







**What it does:** Feed-and-bleed electrodialysis: the stack never sees the feed, and its limiting current is set by the LOOP concentration

**The story:** one stack, a recycle loop, and the concentration the stack actually sees.

THE ARRANGEMENT. A feed-and-bleed loop: fresh feed joins a recycled fraction of the stack's own product, the mixture passes once through the stack, and a BLEED equal to the fresh feed leaves as product while the rest goes round again.

Feed —→[ MIX ]—→ StackIn —→[ ED ]—→ StackOut ^ | | [ SPLIT ] +————————- Recycle <————+

ConcFeed —→[ ED ]—→ ConcOut (single-pass concentrate)

It is how a plant reaches a deep demineralisation with ONE stack instead of the four of 'ed06_stages_in_series': the recycle does in time what the train does in space.

THE TRAP IT EXISTS TO SHOW. A student sizing this loop reads the FEED analysis – 0.05 mol/kg – looks up the limiting current at that concentration, and sizes the rectifier against it. The stack never sees 0.05 mol/kg. It sees the MIXED loop concentration, which sits between the feed and the product and is much nearer the product, because four fifths of what enters it has already been through.

WHAT THE RUN MEASURES (printed by this run; nothing here is a target):

fresh feed 0.050000 mol/kg at 0.6 m3/h what the STACK sees 0.018667 mol/kg at 3.0 m3/h (the mixed loop) product (the bleed) 0.010831 mol/kg at 0.6 m3/h OVERALL demineralisation 78.34 % PER PASS demineralisation 41.98 % i 14.00 A/m2 (7.0 A over 0.5 m2) i_lim 126.52 A/m2 at the LOOP concentration i_lim was 338.88 A/m2 at the FEED concentration (ed05, the same stack and flow) i/i_lim 0.1107 stack power 0.1379 kW

So the limiting current in this loop is 338.88 / 126.52 = 2.68 times LOWER than the feed analysis suggests, and the per-pass duty is 41.98 % rather than the 78.34 % the plant delivers. Both errors point the same way – towards an under-designed stack run too hard – which is why they are worth meeting before a design sheet is written.

A THIRD READING OF THE SAME TRAP, and it is on the printed sheet. The stack's 'specific_energy_kWh_per_m3' is 0.0460: the run's own kW divided by what passes THROUGH the stack, 3.0 m3/h. The plant makes 0.6 m3/h of product, so its specific energy is 0.1379 kW / 0.6 m3/h = 0.230 kWh/m3 – five times the number on the unit's own line, because four fifths of what the stack treats it has treated before. A per-pass figure of merit is not a plant figure of merit whenever there is a recycle.

WHY THE ENGINE NEEDS A TEAR, and what that costs. 'Recycle' is produced by the splitter, which is declared LAST, and consumed by the mixer, which is declared FIRST; the solver executes units in DECLARED ORDER, so it cannot reach the recycle before it needs it. The cut is declared in 'system/solverDict' ('tearStreams ( Recycle );') and a Newton on the tear variables closes it. Omit the declaration and the engine refuses by name with the cycle chain, rather than quietly reading a stale seed.

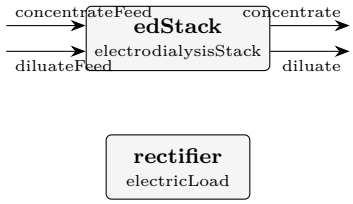
WHAT IS PINNED AND WHAT IS NOT. The golden pins the answer this model gives. As in ed05 and ed06 the stack record is an industrial unit described from operating experience, its mass-transfer correlation is borrowed from the bench stack and applied inside its own Reynolds band, and no measured feed-and-bleed plant is reproduced anywhere in this tree. The concentrate is single-pass; back-diffusion and water transport are not modelled.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | ConcFeed | F | 0.0466736445318 |
| stream | ConcFeed | T | 298.15 |
| stream | ConcOut | F | 0.0466867035115 |
| stream | ConcOut | T | 298.15 |
| stream | Feed | F | 0.00926806223969 |
| stream | Feed | T | 298.15 |
| stream | Product | F | 0.00925500599929 |
| stream | Product | T | 298.15 |
| stream | Recycle | F | 0.0370200239972 |
| stream | Recycle | T | 298.15 |
| stream | StackIn | F | 0.0462880889762 |
| stream | StackIn | T | 298.15 |
| stream | StackOut | F | 0.0462750299965 |
| stream | StackOut | T | 298.15 |
| *. . . and 48 more reference values in the case's `expected`.* | | | |

## ed08_wine_kht_model_solution

tutorials/steady/electrodialysis/ed08_wine_kht_model_solution




**What it does:** Electrodialysis for wine tartrate stabilisation, reduced to a potassium hydrogen tartrate model solution: 20 % of the conductivity removed on the industrial stack (no ethanol, no other wine ion)

**The story:** ED08 tartrate stabilisation of wine, REDUCED to a potassium hydrogen tartrate model solution

THE OPERATION. A young wine can deposit potassium hydrogen tartrate (KHT, potassium bitartrate) crystals in the bottle. Electrodialysis prevents it by taking part of the wine's potassium and bitartrate out through a stack of cation- and anion-exchange membranes, and the cellar controls the operation by the drop of the wine's electrical CONDUCTIVITY, because that is what it can measure on line. This case runs that operation the way a cellar states it:

stack EurodiaED-100P-50; // 100 cell pairs, 50 m2 membrane CMX_AMX; // the record names no pair targetConductivityRemoval 0.20; // remove 20 % of kappa // → Choupo solves I

(1) WHAT THE "WINE" HERE IS, AND WHY IT IS ONLY THAT. The diluate is KHT dissolved in water – ONE 1:1 salt, K+ and HC4H4O6-, nothing else. When this case was built (2026-10-05) the steady stack could not divide the current among a real wine's dozen ions: it gave EVERY counter-ion the whole Faraday share, xi I N / (|z_i| F), whatever each one brought, which stripped the minor ions to zero and left the diluate charged. On ONE monovalent salt that arithmetic is exact – one cation, one anion, charge conserved – so the wine was reduced to it. The rule was supplied the same day (C35: the counter-ions of each membrane SHARE its current by z_i D_i c_i), and the thirteen-ion wine it unblocked is 'ed09_wine_multiionic'. This case stays as the reduction a student can check by hand: on one salt the shared rule IS the single-salt Faraday law, and nothing below moved.

(2) WHAT THIS MODEL IGNORES, said plainly. ETHANOL. It is not in the feed because it cannot be: the stack sends every non-water component through the aqueous-species bridge, and 'ethanol' declares none, so adding it REFUSES the run by name (measured on this case: 'aqueousMapping: component 'ethanol' declares no aqueous mapping'). The solvent is pure water, so ethanol's effect on the ion activities, on the solution viscosity and dielectric constant, and therefore on every ion mobility, the conductivity and the limiting current, is ABSENT. EVERY OTHER WINE SOLUTE. No sodium, calcium, magnesium, chloride, sulfate or phosphate; no organic acid but the bitartrate (no malate, lactate, tartrate dianion); no sugars, phenolics or polysaccharides. In a real wine those anions carry part of the anion current, so this model OVERSTATES the bitartrate removed per ampere. ACID-BASE CHEMISTRY. No pH, no H+, no undissociated tartaric acid, no tartrate dianion: the stack carries ions as they are declared and speciates nothing, so the pH change a real treatment causes is not predicted. TARTRATE STABILITY ITSELF. The engine computes no KHT solubility in a hydroalcoholic medium and no saturation, so it cannot say whether 20 % is ENOUGH. The target is the cellar's input, not the engine's answer. The stack's own simplifications, as in ed01-ed07: no water transport across the membranes, co-ion leakage lumped into xi, membrane resistances measured in 0.5 M NaCl (the CMX_AMX record) applied to a 15 mol/m3 KHT solution, a lumped electrode term, no thermal effect (the energy balance is UNAVAILABLE: the ion components carry no enthalpy datum, and the run says so).

(3) THE PROVENANCE OF EVERY NUMBER. THE STACK. data/standards/assets/EurodiaED-100P-50.dat – Vitor Geraldes' recollection of an industrial unit he operated, marked 'origin measured; reviewStatus unverified;' (100 cell pairs, 50 m2, two hydraulic passes, 0.7 mm channel, design flow 3000 L/h); its three estimates (channel width, spacer porosity, Schmidt exponent) are ANNOUNCED on every run. Its mass-transfer fit is the bench stack's, Geraldes & Afonso, J. Membr. Sci. 360 (2010) 499-508, used inside its own Re band (this run: Re 62.85, band 50-86). THE MEMBRANES. data/standards/assets/CMX_AMX.dat, cited in its own record. The stack record names no pair, so the case must. K+. data/standards/species/K.dat

(D0 1.96e-9 m2/s, cited there). HC4H4O6-. The catalogue record carries NO diffusivity, so this case carries a CASE-LOCAL copy, constant/species/HTart.dat, whose D0 = 7.568e-10 m2/s is an ESTIMATE by Stokes-Einstein on a crystal-volume radius derived from the tree's own records, 'origin estimated; reviewStatus unverified;', ANNOUNCED on every run. No measured limiting conductance was read, so none is cited. AUTHOR-SET ASSUMPTIONS, in those words: the diluate at 15 mol/m3 KHT and the concentrate at 20 mol/m3 KHT (round numbers, not a measured wine or a plant's bleed set-point); both flows at 3000 L/h (the record's design flow); the 20 % conductivity target; xi = 0.9; E_electrodes = 2.0 V; P = 2.5 bar; and T = 298.15 K. WHY 25 C AND NOT A CELLAR TEMPERATURE. The ion diffusivities are referenced at 25 C. The limiting-current model corrects them to the run temperature by Stokes-Einstein, but the conductivity does not: it divides the 25 C D0 by R T at the run temperature, an engine inconsistency named and not fixed (it moves every ED golden). At 298.15 K the correction is exactly 1 and both read the SAME diffusivities, so the case avoids the inconsistency instead of teaching it.

(4) WHAT TO READ OFF THE LOG – every headline number can be checked by hand: the CURRENT. A 1:1 salt, one counter-ion per membrane, so 20 % of the conductivity is 20 % of the salt: 0.2 x 0.045 kmol/h = 2.5e-3 mol/s of each ion, and I = 2.5e-3 x 96485 / (0.9 x 100) = 2.680 A. the CONDUCTIVITY. kappa = c (lambda_K + lambda_HT), lambda = F^2 D0 / (R T): 15 x (7.361e-3 + 2.842e-3) = 0.1530 S/m in, 0.1224 out. The two ions are equal in number and K+ carries 72 % of the current in the solution, because it diffuses 2.6 times faster. the LIMITING CURRENT. 65.7 A/m2, and it is set by the ANION membrane: the slow bitartrate depletes its film first (the cation-side value is 170.2 A/m2). At 5.36 A/m2 the stack runs at i/i_lim = 0.082 – nowhere near the limit on this model. the VOLTAGE. U = 7.98 V, of which 2.0 V is the lumped electrode term and 5.98 V the 100 cell pairs; P = 21.4 W, 0.0071 kWh per m3 of model wine.

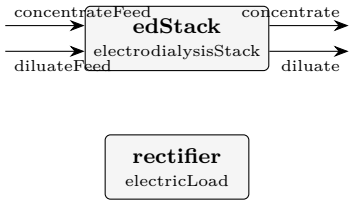
NOTHING HERE IS VALIDATED AGAINST A MEASURED WINE. The golden pins this model's answer. A full wine composition was run on 2026-09-19 and was parked, not shipped, for the reason in (1); DEV.md C4 records what a faithful wine case still needs. The rectifier below the stack only receives the stack's electrical power on an energy wire.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.0462953111984 |
| stream | concentrate | T | 298.15 |
| stream | concentrateFeed | F | 0.0462903111984 |
| stream | concentrateFeed | T | 298.15 |
| stream | diluate | F | 0.0462769778651 |
| stream | diluate | T | 298.15 |
| stream | diluateFeed | F | 0.0462819778651 |
| stream | diluateFeed | T | 298.15 |
| kpi | edStack | C_ion_HTart | 15 |
| kpi | edStack | C_ion_K | 15 |
| kpi | edStack | D_eff | 1.09196702002e-09 |
| kpi | edStack | D_ion_HTart | 7.568e-10 |
| kpi | edStack | D_ion_K | 1.96e-09 |
| kpi | edStack | E_mem_pair | 0.013980745806 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## ed09_wine_multiionic

tutorials/steady/electrodialysis/ed09_wine_multiionic




**What it does:** Electrodialysis for wine tartrate stabilisation on a thirteen-ion wine: 20 % of the conductivity removed on the industrial stack, each membrane's current shared between its counter-ions by z D c (no ethanol)

**The story:** ED09 tartaric stabilisation of a wine carried as THIRTEEN IONS, on a CONDUCTIVITY target

WHAT THIS CASE IS, AND WHAT ed08 IS NOT. ed08_wine_kht_model_solution reduces the wine to its stability-determining salt, KHT, because on ONE 1:1 salt the old Faraday transfer (every counter-ion given the WHOLE current, xi I N/(|z| F)) was exact. On a real wine it was not: the parked 2026-09-19 attempt at this case removed H+, Cl-, HMal- and Mal 2- ENTIRELY while K+ fell 6 %, and left the diluate carrying net charge (DEV.md C4). Vitor ruled the missing rule on 2026-10-05 (C35): the counter-ions of each membrane SHARE its current by

t_i = |z_i| D_i c_i / SUM_j |z_j| D_j c_j (no new parameter)

– the transport numbers of Geraldes & Afonso, J. Membr. Sci. 360 (2010) 499-508, Eqs. (12)/(13), applied at every current – integrated along the pass, with D_i each ion's D0. Each membrane then moves exactly xi I N / F equivalents and the diluate leaves electroneutral. This case is the full wine that rule unblocks.

1. ETHANOL IS NOT IN THIS WINE, AND THAT IS THE FIRST THING TO KNOW. A dry red carries about 13 % v/v of ethanol and the stack cannot be told so: every non-water component passes through the aqueous-species bridge, and declaring 'ethanol' in the components list REFUSES – measured on a twin of this case on 2026-10-05: "aqueousMapping: component 'ethanol' declares no aqueous mapping", exit 2. So the engine prices the solvent as PURE WATER: Re and Sc on water's viscosity, every ion's conductivity on its infinite-dilution D0 in water. Mobilities scale with 1/viscosity, and a water-ethanol mixture is more viscous than water, so kappa_in and kappa_out are both overestimated by a common factor this case cannot compute; the RELATIVE target (20 %) is insensitive to it, the absolute voltage and power are not. NAMED AND NOT BUILT: a neutral co-solvent carried through the mass balance and priced into the solvent viscosity and dielectric constant (DEV.md C4 (b)).

2. THE WINE IS AN AUTHOR-SET COMPOSITION, closed on sulfate – the table, the closure and what it stands for are in 0/diluateFeed. No number in it is transcribed from a measured wine and no source is cited for it; it is a round thirteen-ion composition of the order a dry red at pH about 3.6 carries, chosen so that the stack meets a wine's real spread of mobilities: H+ (D0 9.31e-9 m2/s) beside K+ (1.96e-9) beside Mg2+ (0.71e-9) on the cation membrane, Cl- (2.03e-9) beside six organic and inorganic anions on the anion membrane. The speciation is FIXED as declared: the stack does not speciate, so the H+ it removes is not replaced by the undissociated acids a real wine would re-dissociate (DEV.md C4 (d)).

3. SIX DIFFUSIVITIES ARE ESTIMATES, AND THE RUN SAYS SO SIX TIMES. K, Na, Ca, Mg, H, Cl and SO4 carry curated D0 in the catalogue; HTart, Tart, H2PO4, HMal, Mal and Lactate are supplied CASE-LOCALLY under constant/species/, each 'origin estimated; reviewStatus unverified;' with its derivation in the record (a Stokes-Einstein radius from the KHT crystal volume, scaled by molar mass; H2PO4- set equal to sulfate) and ANNOUNCED on every run. With the C35 rule these numbers now SET the split – the bitartrate's estimated D0 decides how much of the anion current it carries – so they matter more here than in ed08.

4. THE TARGET IS CONDUCTIVITY, BECAUSE THAT IS WHAT IS MEASURED (Vitor: "na eletrodialise aponta para a condutividade, porque isso e que e medido"). 'targetConductivityRemoval 0.20;' asks for 20 % of the diluate's conductivity removed – an AUTHOR-SET operating point, the same one ed08 uses – and Choupo solves the current. The conductivity counts every ion with its own z^2 D0 c, so

the target does not depend on the order of the components list. T = 298.15 K, as in ed08 and for ed08's reason (at 25 C the conductivity and the limiting current read the same diffusivities; DEV.md C4 (f)).

WHAT THE RUN PRINTS (every number is this model's; the golden pins it):

I = 6.710 A on 0.5 m2 per cell pair (i = 13.4 A/m2), U = 11.40 V, P = 76 W, 0.0255 kWh per m3 of wine; kappa 0.4789 → 0.3831 S/m; i_lim = 225.2 A/m2, set by the ANION membrane, so i/i_lim = 0.060.

ion membrane share of its removed membrane's current K+ CEM 0.746 22.4 % Na+ CEM 0.105 15.8 % Ca2+ CEM 0.032 9.7 % Mg2+ CEM 0.093 8.7 % H+ CEM 0.023 70.0 % Cl- AEM 0.051 35.1 % SO4 2- AEM 0.324 20.2 % HTart- AEM 0.159 14.9 % Tart 2- AEM 0.059 14.9 % HMal- AEM 0.021 15.4 % Mal 2- AEM 0.002 15.5 % Lactate- AEM 0.209 17.4 % H2PO4- AEM 0.175 20.2 %

THE LESSON IS IN THE LAST COLUMN. The fraction of an ion removed depends on its D0 ALONE – ln(1 - removed) / D0 is one number on each membrane – so ions of equal D0 leave in equal FRACTION whatever their charge or concentration (HTart- and Tart 2-, which share an estimated D0; SO4 2- and H2PO4-, which share one by construction), the fast H+ and Cl- are stripped hardest, and the slow divalent cations least. The SHARE of the current, on the other hand, is set by z D c: potassium, the dominant cation, carries three quarters of the cation current. The equivalents moved through the two membranes are printed beside the table and are equal.

WHAT IS PINNED AND WHAT IS NOT. The golden pins this model's answer. Nothing here is validated against a measured wine or a measured stack run: the composition is author-set, six D0 are announced estimates, the sulfate is a closure, the solvent is water, the speciation is frozen, and the share z D c is a RULING that leaves out any selectivity the real membranes add between counter-ions. Whether 20 % of the conductivity removes enough KHT to stabilise the wine is NOT computed: that needs a KHT solubility in a hydroalcoholic medium (DEV.md C4 (e)), which the tree does not carry.

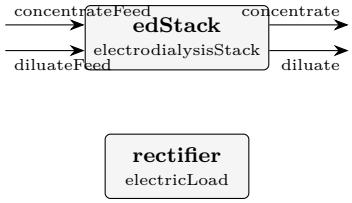
diluateFeed (the wine) → +——————+ → diluate (stabilised wine) | EurodiaED-100P-50 | concentrateFeed (brine) → | 100 pairs, CMX/AMX | → concentrate +——————+ | (power, energy wire) v rectifier (electricLoad)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.0463012284647 |
| stream | concentrate | T | 298.15 |
| stream | concentrateFeed | F | 0.0462903111984 |
| stream | concentrateFeed | T | 298.15 |
| stream | diluate | F | 0.0463032897655 |
| stream | diluate | T | 298.15 |
| stream | diluateFeed | F | 0.0463142070318 |
| stream | diluateFeed | T | 298.15 |
| kpi | edStack | C_ion_Ca | 1.25 |
| kpi | edStack | C_ion_Cl | 1.1 |
| kpi | edStack | C_ion_H | 0.25 |
| kpi | edStack | C_ion_H2PO4 | 6.5 |
| kpi | edStack | C_ion_HMal | 1 |
| kpi | edStack | C_ion_HTart | 8 |
| *... and 111 more reference values in the case's `expected`.* | | | |

## ed10_passthrough_benzene

tutorials/steady/electrodialysis/ed10_passthrough_benzene




**What it does:** Electrodialysis stack: NaCl desalination with benzene DECLARED to pass through

**The story:** ED10 a component the stack does not model, DECLARED to pass through it

ed01's stack and ed01's NaCl diluate, with ONE change: the diluate feed also carries a little benzene (0.01 kmol/h against 17.9 kmol/h of water, about 0.78 kg/h – above benzene's solubility in water, so in the plant it would ride as droplets of a second liquid phase).

diluateFeed (+ benzene) → +————————+ → diluate (+ benzene) | ED stack | concentrateFeed → | (CMX/AMX, N=100)| → concentrate +————————+

WHAT THIS SHOWS. The stack models the aqueous solvent and the IONS: every other component must cross its declared bridge to one catalogued ion (‘choupoSolve –describe-units‘ prints that contract). Benzene is not an ion, so without the ‘passThrough‘ block below the run is REFUSED before any unit runs, naming benzene and both remedies. With the block, benzene is ACCEPTED as a DECLARED APPROXIMATION: it enters none of the stack's equations, every mole of it leaves in the outlet the case names (‘fate diluate‘), and it is in every balance (mass, elements, energy).

THE PASS-THROUGH IS A DECLARED APPROXIMATION, NOT A RESULT. The run records it on the problem-divergence channel – printed ABOVE the KPIs, in the result JSON and in converged/problemDivergence – with the reason below. What it sets aside is real: an organic film on the membranes, a change in the channel's conductivity, any benzene the membranes would sorb. The stack computes none of them, and says so.

Benzene moves no stack number: the molalities are taken on the water and benzene is not an ion of the channel, so the same case run WITHOUT benzene gives every KPI (voltage, demin ratio, limiting current, specific energy) to the last printed digit – check_unit_acceptance runs that twin. (Read against ed01 the voltage differs in the fourth figure: this case was sealed against the Choupo-2608 catalogue, whose Na and Cl records carry different curated diffusivities from the 2607 records ed01 was sealed with. That is the catalogue, not the benzene.)

The thermophysical package is ed01's ideal liquid with benzene added (see constant/thermoPhysPropDict): it prices the streams for the balances and does not represent the second liquid phase either.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.00501492454869 |
| stream | concentrate | T | 298.15 |
| stream | concentrateFeed | F | 0.00500000000039 |
| stream | concentrateFeed | T | 298.15 |
| stream | diluate | F | 0.00498785323076 |
| stream | diluate | T | 298.15 |
| stream | diluateFeed | F | 0.00500277777906 |
| stream | diluateFeed | T | 298.15 |
| kpi | edStack | E_mem_pair | 0.0794573845787 |
| kpi | edStack | I | 8 |
| kpi | edStack | N_cellpairs | 100 |
| kpi | edStack | R_pair | 0.00507540958664 |
| kpi | edStack | U | 13.5060661272 |
| kpi | edStack | W_electric | 108.048529017 |

*... and 17 more reference values in the case's `expected`.*

## evaporation

**The theory you need.** An evaporator boils solvent off a solution. Two pieces of physics beyond a flash: boiling-point elevation, $T_{\mathrm{boil}} = T_{\mathrm{sat}}(P) + \mathrm{BPE}$ with the BPE from the electrolyte activity of water ($\ln a_w = -\varphi\,\nu\,m\,M_w/1000$); and the heat of dilution/concentration (the relative apparent molar enthalpy $L_\varphi(m,T)$) entering the duty. Expect: the BPE to GROW as the brine concentrates, steam economy below 1 for a single effect, and – for NaCl – the full Silvester–Pitzer $T$-dependence of $\gamma_\pm$ shaping the answer between 25 and 200°C. *Full derivation: Theory Guide, the electrolyte chapters.*

## evaporator01_brine

tutorials/steady/evaporation/evaporator01_brine




**What it does:** Single-effect evaporator: 35 g/L brine at 1 bar; the fixed steam duty/UA evaporates ~82 % of the feed water (V/F=0.82), enriching NaCl ~5.5x (x 0.011 → 0.059) – duty-driven, not a target (BPE included)

**The story:** single-effect brine concentrator (NaCl-water), PHYSICAL-EQUIPMENT model (v0.29+).

We say WHAT THE HARDWARE IS (heat-exchange area, overall HTC, operating pressure on the process side, heating-steam temperature) and the simulator tells us WHAT IT DOES (V/F, duty Q, T_boil with BPE, steam demand, economy). There is no "vaporFraction" or "duty" spec to choose — the equipment shape itself IS the spec.

Feed: seawater-equivalent 35 g/L NaCl in water, 100 kmol/h (~ 1.8 kg/s of seawater, the volume of a small medium-capacity desalination unit), 25 °C. At MW_NaCl = 58.44 g/mol and rho_water ~ 1 g/cm$^3$:

35 g/L / 58.44 g/mol = 0.599 mol NaCl per litre water = 0.599 mol/kg water (dilute approx) x_NaCl_feed ≈ 0.599 / (0.599 + 55.5) = 0.01067

Heating side: saturated steam at 130 °C (= 403.15 K, P ≈ 2.7 bar).

Equipment: heat-exchange area = 20 m$^2$ overall U = 2000 W/(m$^2$·K) operating pressure = 1 bar (process side)

Expected order-of-magnitude (computed by the solver, not specified): T_sat(water, 1 bar) ≈ 373 K (~ 100 °C) ΔT_drive (steam - process) ≈ 30 K Q = U · A · ΔT ≈ 2000 · 20 · 30 ≈ 1200 kW V evaporated ≈ Q / ΔHvap ≈ 110 kmol/h ≈ 1980 kg/h water V/F (mol-basis) ≈ 0.82 (the CONVERGED run; the rough hand-estimate above is order-of-magnitude only) Steam consumption ≈ Q / ΔHvap_at_steam ≈ 2000 kg/h Economy ≈ V_evap / F_steam ≈ 0.82 (the CONVERGED run; the hand estimate above would give ~0.99, and the difference is the estimate's, not the run's) x_NaCl_outlet ≈ 0.059 (~5.5x the feed, matching the run header)

Industrial relevance: this is the building block of multi-effect evaporators in seawater desalination (MED, MSF), brine crystallisers, and the sugar industry. Real plants run under vacuum (T_boil ~ 60 °C at 0.2 bar) to minimise scaling and use BPE-loaded multi-effect cascades to recover the latent heat — see evaporator02_triple_effect_sugar.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.00498887089392 |
| stream | condensate | F | 0.0277777777778 |
| stream | feed | F | 0.0277777777778 |
| stream | saturatedSteam130C | F | 0.0277777777778 |
| stream | vapour | F | 0.0227889068839 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 1.79842779271 |
| kpi | evap | F_conc | 0.00498887089392 |
| kpi | evap | F_conc_kmol_h | 17.9599352181 |
| kpi | evap | F_conc_mass | 0.101856029987 |
| kpi | evap | F_cond | 0.500416666667 |
| kpi | evap | F_cond_kg_h | 1801.5 |
| kpi | evap | F_in | 0.0277777777778 |

*... and 24 more reference values in the case's `expected`.*

## evaporator02_triple_effect_sugar

tutorials/steady/evaporation/evaporator02_triple_effect_sugar




**What it does:** Triple-effect cocurrent evaporator: 22500 kg/h sugar solution 5%→25% (textbook Example 10.4)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. ('sucrose.dat' does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

textbook Example 10.4: triple-effect cocurrent evaporator concentrating a 22 500 kg/h sucrose solution from 5 wt% sucrose toward ~ 25 wt%, using 200 kPa saturated steam to drive Effect 1 and a 15 kPa vacuum on Effect 3.

Cascade (cocurrent: liquid 1→2→3, generated vapour drives the next effect's heating side):

F=22 500 kg/h L1 L2 L3 x=0.05 →[1] ——→ [2] ——→ [3] —-→ ⌢ ⌢ ⌢ | | | Steam V1 V2 V3 → condenser 200 kPa (Effect 1's (Effect 2's T_sat~393K vapour drives vapour drives Effect 2) Effect 3) P1 = 80 kPa P2 = 45 kPa P3 = 15 kPa

PHYSICAL-EQUIPMENT model (v0.29+). Each effect is its own physical evaporator with an 'area' and a 'U'; the simulator computes V/F, Q, T_boil and steam demand from the energy balance. Effect 1 consumes utility steam; the vapour V1 that Effect 1 produces is declared as the heating-side input to Effect 2, and similarly V2 → Effect 3. The flowsheet runs SEQUENTIALLY left-to-right: no outer iteration is needed because the inputs of each effect are produced by the preceding one (or by the utility stream).

Equipment we picked (the "spec"): A_1 = A_2 = A_3 = 100 $m^2$ U_1 = 2319 W/($m^2$·K) (Effect 1, clean steam) U_2 = 2194 W/($m^2$·K) (Effect 2) U_3 = 1296 W/($m^2$·K) (Effect 3, lowest pressure → lowest k)

The textbook target L_3 ~ 4500 kg/h at x = 0.25 corresponds to evaporating ~ 18 000 kg/h total water; that target is NOT enforced here — the equipment dimensions ARE the spec, the outlet concentration is the RESULT. A later tutorial wraps this case in a DesignSpec block that manipulates A_i (or, more commonly, U_i at fixed A_i = A) to drive L_3 to the design target.

1. Cocurrent multi-effect literally just chains single-effect evaporators. Choupo's flowsheet engine handles the cascading naturally — the V_i stream of Effect i appears as the heating-side input of Effect i+1, and the V_i mass flow is computed by Effect i's energy balance.

2. Economy: in single-effect (evaporator01_brine) the economy was ~ 0.83 kg evap / kg steam. Running this cascade gives total V_evap / utility steam ~ 2.0 (= 17 972 / 8 917) — more than a doubling, simply by chaining the heat through three effects. Industrial design targets ~ 2.5 with optimised area allocation and pressure cascade.

3. BPE matters more in Effect 3 (most concentrated solute, lowest U); the student SEES T_boil deviate further from T_sat(water, P_3) in the per-effect run header.

Honest simplification for v0.29:

The Evaporator unit op reads only T from its heating-steam input (and ignores the F). This means the F of V_1 that effect 2 actually CONSUMES on its heating chest is computed from effect 2's own energy balance and is NOT constrained to equal the F of V_1 that effect 1 PRODUCED. In a real plant those two flows must close (all of V_1 condenses in effect 2's chest, with no other path). Here the v0.29 model treats the three energy balances independently — there is a residual mismatch in the V_i mass that the cascade quietly absorbs.

Closing the inter-effect mass balance turns the cascade into a coupled system (one constraint per link) that wants an outer DesignSpec wrap: manipulate (T_boil_i) or (A_i) to enforce "V_i produced == V_i consumed at effect i+1's chest". That is the v0.30 multi-effect tutorial.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | L1 | F | 0.240916518138 |
| stream | L2 | F | 0.145314913232 |
| stream | L3 | F | 0.0443422438223 |
| stream | V1 | F | 0.0895828875602 |
| stream | V2 | F | 0.0956016049063 |
| stream | V3 | F | 0.10097266941 |
| stream | cond1 | F | 0.1375 |
| stream | cond2 | F | 0.0895828875602 |
| stream | cond3 | F | 0.0956016049063 |
| stream | feed | F | 0.330499405699 |
| stream | utilitySteam_200kPa | F | 0.1375 |
| kpi | effect1 | A | 100 |
| kpi | effect1 | A_m2 | 100 |
| kpi | effect1 | BPE | 0.10830859768 |
| *... and 95 more reference values in the case's `expected`.* | | | |

## evaporator03_co_current

tutorials/steady/evaporation/evaporator03_co_current




**What it does:** Co-current forward-feed triple-effect evaporator, user-defined components (textbook problem 10.38, case a)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. (‘sucrose.dat‘ does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

triple-effect evaporator, forward-feed, user-defined components, textbook Example 10.38 (a).

Problem (textbook givens): Feed 1.25 kg/s, 10 wt% solids, 294 K Live steam 170 kPa (saturated vapour) P_3 such that effect 3 boils at 325 K => P_3 = 13.5 kPa U_1 3100 W/(m$^2$·K) U_2 2500 W/(m$^2$·K) U_3 1700 W/(m$^2$·K) Spec productConc = 40 wt% solids Equal areas A_1 = A_2 = A_3 = A

Question (in system/outerDict): find A, P_1, P_2 and the live- steam consumption.

Wiring (liquid + vapour BOTH flow 1 → 2 → 3):

feed (294 K, 10 %) productConc (40 %) | ^ v | +———+ L1to2 +———+ L2to3 +———+ | effect1 |———→| effect2 |———→| effect3 | | P_1 | | P_2 | | P_3 | +—-+—-+ +—-+—-+ +—-+—-+ ^ ^ ^ liveSteam V1 (from 1) V2 (from 2) (170 kPa) chest of 2 chest of 3

V_3 → condenser (outside the dict) cond1, cond2, cond3 → condensate terminals

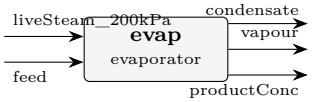
Notes for the student: Thermo properties (MW, Cp, Antoine, ...) of ‘agua‘ and ‘solidos‘ live in constant/-components/<name>.dat — these are USER-DEFINED components, NOT from the standard catalogue. Open them to see the textbook’s Cp = 4.18 kJ/(K·kg) assumption baked in, and to confirm BPE is set to zero (no K_b on ‘agua‘). Forward-feed with COLD feed (294 K) penalises effect 1: most of its duty goes into sensible heating, not evaporation. Case (b) of the textbook pre-heats the feed to 355 K — swap T_feed below to compare economies.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | L1to2 | F | 0.0469536109199 |
| stream | L2to3 | F | 0.0295647943065 |
| stream | V1 | F | 0.0161887935578 |
| stream | V2 | F | 0.0173888166134 |
| stream | V3 | F | 0.018462595509 |
| stream | cond1 | F | 0.0262363636123 |
| stream | cond2 | F | 0.0161887935578 |
| stream | cond3 | F | 0.0173888166134 |
| stream | feed | F | 0.0631424044778 |
| stream | liveSteam | F | 0.0262363636123 |
| stream | productConc | F | 0.0111021987975 |
| kpi | effect1 | A | 16.5383245281 |
| kpi | effect1 | A_m2 | 16.5383245281 |
| kpi | effect1 | BPE | 0 |
| *... and 92 more reference values in the case’s `expected`.* | | | |

## evaporator04_orange_juice_simple

`tutorials/steady/evaporation/evaporator04_orange_juice_simple`



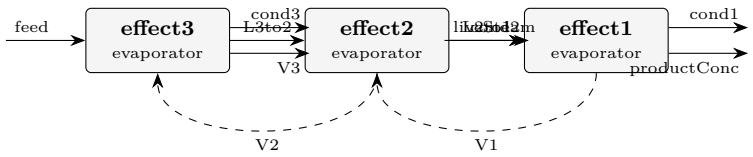

**What it does:** Single-effect evaporator concentrating orange juice 10% to 50% solids (textbook Example 10.2 a-c, without BPE)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | condensate | F | 0.360686102898 |
| stream | feed | F | 0.350791135988 |
| stream | liveSteam_200kPa | F | 0.360686102898 |
| stream | productConc | F | 0.0424059629635 |
| stream | vapour | F | 0.308385173024 |
| kpi | evap | A | 75.5133330137 |
| kpi | evap | A_m2 | 75.5133330137 |
| kpi | evap | BPE | 0 |
| kpi | evap | F_conc | 0.0424059629635 |
| kpi | evap | F_conc_kmol_h | 152.661466669 |
| kpi | evap | F_conc_mass | 1.38888555242 |
| kpi | evap | F_cond | 6.49776014371 |
| kpi | evap | F_cond_kg_h | 23391.9365174 |
| kpi | evap | F_in | 0.350791135988 |
| *… and 24 more reference values in the case's `expected`.* | | | |

## evaporator05_counter_current

tutorials/steady/evaporation/evaporator05_counter_current




**What it does:** Counter-current backward-feed triple-effect evaporator (textbook problem 10.38 variant)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. ('sucrose.dat' does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

COUNTER-CURRENT (BACKWARD-FEED) triple-effect evaporator, Mode-2, credo-pure.

Problem (textbook givens, textbook problem 10.38 numerics, counter-current variant): Feed 1.25 kg/s, 10 wt% solids, 294 K Live steam 170 kPa (saturated vapour) entering EFFECT 1 (hot end) P_3 (cold end) such that T_3 = 325 K → 13.5 kPa U_1 3100 W/(m²·K) U_2 2500 W/(m²·K) U_3 1700 W/(m²·K) Spec productConc = 40 wt% solids (exits from effect 1) Equal areas A_1 = A_2 = A_3 = A (single $A by reference)

Wiring (LIQUID flows 3 → 2 → 1 ; VAPOUR flows 1 → 2 → 3 ; OPPOSITE directions => counter-current):

feed (294 K, 10 %) productConc (40 %, exits effect 1) | ^ v | +———+ L3to2 +———+ L2to1 +———+ | effect3 |<———-| effect2 |<———-| effect1 | | COLD | | MID | | HOT | +—-+—-+ +—-+—-+ +—-+—-+ ^ ^ ^ | | | V2 (from 2) V1 (from 1) liveSteam chest of 3 chest of 2 (170 kPa)

v V3 → condenser cond1, cond2, cond3 → condensate terminals

Both V_1 and V_2 flow OPPOSITE to the liquid: effect 1 produces V_1 which feeds effect 2's chest; effect 2 produces V_2 which feeds effect 3's chest. In the directed graph this creates a CYCLE:

effect3 –L3to2-→ effect2 –L2to1-→ effect1 ^ | | V2 V1 | | v +– effect2 <– effect1 (V1 chest) +

So we DECLARE V_1 and V_2 as TEAR STREAMS. The flowsheet executive runs effect3 → effect2 → effect1 once per pass, with the initial V_1 / V_2 guesses from the streams block; at end of pass the newly-produced V_1, V_2 are compared with the old. Wegstein (or plain direct substitution) iterates until they match.

In Mode 2 each effect computes T_boil and P_op locally from the chest T and the hardware (A, U) — so the only thing that the Wegstein loop actually iterates is V_i.F (the produced flows). T and P_op stabilise IMMEDIATELY (single pass). Convergence in ~3-5 Wegstein passes.

Design unknowns (in outerDict): $A common heat-exchange area $F_steam live-steam flow

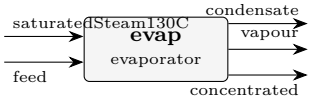
Constraints: productConc.F_mass = 0.3125 kg/s (40 wt%) effect3.P = 13.5 kPa (textbook terminal vacuum)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | L2to1 | F | 0.0319496046345 |
| stream | L3to2 | F | 0.0496904316429 |
| stream | V1 | F | 0.020847035226 |
| stream | V2 | F | 0.0177408270084 |
| stream | V3 | F | 0.0134519728349 |
| stream | cond1 | F | 0.0227907734916 |
| stream | cond2 | F | 0.0208470355984 |
| stream | cond3 | F | 0.0177408244446 |
| stream | feed | F | 0.0631424044778 |
| stream | liveSteam | F | 0.0227907734916 |
| stream | productConc | F | 0.0111025694085 |
| kpi | effect1 | A | 17.1719452001 |
| kpi | effect1 | A_m2 | 17.1719452001 |
| kpi | effect1 | BPE | 0 |
| *... and 92 more reference values in the case's `expected`.* | | | |

## evaporator06_nacl_pitzer

tutorials/steady/evaporation/evaporator06_nacl_pitzer




**What it does:** Single-effect NaCl evaporator with the boiling-point elevation from the PITZER electrolyte model (a_w), not the ideal K_b∗m. Concentrated brine → BPE several degrees, the real value.

**The story:** / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the PITZER water activity (a_w), not the ideal K_b∗m: concentrated brine is a REAL electrolyte, and the BPE follows from the osmotic coefficient. The thermo arrives via the aqueousNaCl_pitzer property package (see constant/thermoPhysPropDict) – swap to evaporator07's eNRTL to compare.

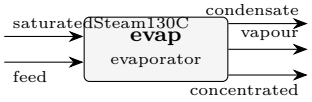
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.0198874983274 |
| stream | condensate | F | 0.0125 |
| stream | feed | F | 0.0277777777778 |
| stream | saturatedSteam130C | F | 0.0125 |
| stream | vapour | F | 0.00789027945038 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 5.19956015536 |
| kpi | evap | F_conc | 0.0198874983274 |
| kpi | evap | F_conc_kmol_h | 71.5949939786 |
| kpi | evap | F_conc_mass | 0.414419115701 |
| kpi | evap | F_cond | 0.2251875 |
| kpi | evap | F_cond_kg_h | 810.675 |
| kpi | evap | F_in | 0.0277777777778 |

*. . . and 29 more reference values in the case's* `expected`.

## evaporator07_nacl_enrtl

tutorials/steady/evaporation/evaporator07_nacl_enrtl




**What it does:** Single-effect NaCl evaporator with the boiling-point elevation from the eNRTL water activity (electrolyte-NRTL) – the same slot as model pitzer, one word changed. eNRTL is the model for salt in a non-ideal molecular solvent.

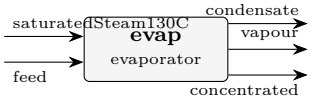
**The story:** / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the eNRTL water activity (a_w), not the ideal K_b∗m – the SAME flowsheet as evaporator06, with only the electrolyte model swapped (pitzer → eNRTL): run both and compare the BPE the two G^E surfaces predict.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.019637506972 |
| stream | condensate | F | 0.0125 |
| stream | feed | F | 0.0277777777778 |
| stream | saturatedSteam130C | F | 0.0125 |
| stream | vapour | F | 0.00814027080581 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 3.94504785681 |
| kpi | evap | F_conc | 0.019637506972 |
| kpi | evap | F_conc_kmol_h | 70.6950250991 |
| kpi | evap | F_conc_mass | 0.409915521433 |
| kpi | evap | F_cond | 0.2251875 |
| kpi | evap | F_cond_kg_h | 810.675 |
| kpi | evap | F_in | 0.0277777777778 |
| *... and 28 more reference values in the case's `expected`.* | | | |

## evaporator08_naoh_dilution_heat

tutorials/steady/evaporation/evaporator08_naoh_dilution_heat




**What it does:** Single-effect NaOH evaporator: the heat of dilution IN the duty (fitted L_phi, Parker-validated)

**The story:** the heat of dilution, IN the duty

Single-effect caustic (NaOH) evaporator – the McCabe enthalpy-concentration classic. NaOH is the FIRST calorimetrically fitted salt (Silvester-Pitzer dbeta/dT vs Parker NSRDS-NBS 2, rel-AAD 4.97%), so this evaporator's duty CLIMBS THE MEASURED L_phi CURVE: concentrating caustic absorbs extra heat beyond sensible+latent, and the run prints it as its own named line:

heat of dilution = X kW (L_phi: m a → b mol/kg, calorimetric fit)

Contrast evaporator06 (whose CASE-LOCAL pair overlay omits the T-slots, so ITS NaCl runs calorimetricFit false – the standards Na-Cl record is fitted since the Parker refit): same machinery, but there the dilution heat is OMITTED and the run says so loudly. The feed (m ~ 2.0) concentrates to m ~ 4-5 mol/kg – inside the Parker fit window (<= 6), so no extrapolation flag; push the steam rate up and watch the [electrolyte] EXTRAPOLATED advisory fire.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.0113820861346 |
| stream | condensate | F | 0.0208333333333 |
| stream | feed | F | 0.0277777777778 |
| stream | saturatedSteam130C | F | 0.0208333333333 |
| stream | vapour | F | 0.0163956916432 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 6.7542776432 |
| kpi | evap | F_conc | 0.0113820861346 |
| kpi | evap | F_conc_kmol_h | 40.9755100845 |
| kpi | evap | F_conc_mass | 0.226419670604 |
| kpi | evap | F_cond | 0.3753125 |
| kpi | evap | F_cond_kg_h | 1351.125 |
| kpi | evap | F_in | 0.0277777777778 |

*... and 29 more reference values in the case's `expected`.*

### evaporator09_nacl_sucrose_brine

tutorials/steady/evaporation/evaporator09_nacl_sucrose_brine




**What it does:** Evaporating a brine that also carries sugar: the electrolyte BPE is computed on the salt molality, and the rest is announced, not absorbed

**The story:** / flowsheetDict Single-effect NaCl evaporator whose boiling-point elevation comes from the PITZER water activity (a_w), not the ideal K_b∗m: concentrated brine is a REAL electrolyte, and the BPE follows from the osmotic coefficient. The thermo arrives via the aqueousNaCl_pitzer property package (see constant/thermoPhysPropDict) – swap to evaporator07's eNRTL to compare.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.0206862565421 |
| stream | condensate | F | 0.0125 |
| stream | feed | F | 0.0281111111111 |
| stream | saturatedSteam130C | F | 0.0125 |
| stream | vapour | F | 0.00742485456904 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 5.04671954326 |
| kpi | evap | F_conc | 0.0206862565421 |
| kpi | evap | F_conc_kmol_h | 74.4705235515 |
| kpi | evap | F_conc_mass | 0.536902744939 |
| kpi | evap | F_cond | 0.2251875 |
| kpi | evap | F_cond_kg_h | 810.675 |
| kpi | evap | F_in | 0.0281111111111 |

*. . . and 29 more reference values in the case's* `expected`.

## flash

**The theory you need.** An isothermal flash splits a feed of composition $z_i$ into vapour $y_i$ and liquid $x_i$ at fixed $(T, P)$. Phase equilibrium is $y_i = K_i x_i$ with $K_i = \gamma_i P_i^{\mathrm{sat}}/P$ ($\gamma$–$\varphi$ formulation), and the vapour fraction $\beta$ solves the Rachford–Rice equation

$$\sum_i \frac{z_i\,(K_i - 1)}{1 + \beta\,(K_i - 1)} = 0,$$

monotone in $\beta$ – Newton converges in a handful of iterations, all of them visible at verbosity 3. Expect: a bubble point when the equation's root hits $\beta = 0$, a dew point at $\beta = 1$, and non-ideal systems ($\gamma_i \neq 1$: NRTL/Wilson/UNIQUAC/UNIFAC) bending $K_i$ with composition. Liquid–liquid and three-phase cases replace successive-substitution with direct Gibbs-energy minimisation (the symmetric $\gamma$-model trap: Newton collapses onto the trivial $K = 1$ root). *Full derivation: Theory Guide, the VLE and flash chapters.*

## adiabaticFlash01_benzene_toluene

`tutorials/steady/flash/adiabaticFlash01_benzene_toluene`




**What it does:** Adiabatic flash, benzene/toluene 40/60, 380K/3bar → 1atm

**The story:** Adiabatic flash: sub-cooled 40/60 benzene/toluene liquid at 380 K, 3 bar expanded to 1 atm. ΔT and V/F come from the energy balance.

The outlet pressure is a plain editable value – tweak it in the GUI's Properties box (transient tinkering) and re-run, or edit it here.

*A benzene/toluene liquid (40/60 mol, 100 kmol/h) at 380 K and 3 bar is let down to 1 atm across a drum with no heat added. Nothing tells the drum what temperature to settle at. The energy balance does: the outlet comes out at 368.66 K, an 11.34 K drop, with 5.19 % of the feed flashed to vapour (5.19 kmol/h vapour, 94.81 kmol/h liquid). Every number here is the case's own golden (expected), which bin/runTests re-checks.*

*1. Q = 0 is a definition, not a setting. The dict declares only the outlet pressure (system/flowsheetDict, operation { P 1.01325 bar; }). There is no T to give; an adiabatic drum has none to receive. 2. Two Newtons, one inside the other. The outer one walks T until the outlet enthalpy equals the inlet's; at every trial T the inner one solves the Rachford-Rice flash. Watch it in the Log tab — each outer step prints its H-residual, and the last one prints H_residual at the round-off floor (-2.2e-3 J/mol; over the whole 100 kmol/h stream that is 6.2e-5 kW against a 1091 kW throughput). 3. Where the 11 K went. Vaporising 5 % of the feed costs latent heat, and the only place it can come from is the sensible heat of the liquid. Open Streams: the vapour is benzene-rich, the liquid barely moved in composition, and both leave at the same 368.66 K. 4. Compare with flash01_benzene_toluene — the same feed held at a ∗fixed∗ 370 K. There the temperature is an input and the drum needs a duty to hold it. The two cases are the same split asked two ways.*

*5. Read the caveat the engine prints — it is about THIS case. The feed is at 380 K, and benzene's Antoine fit is declared for 287.7–354.07 K. The run says so on the console ([psat] ? OUTSIDE its declared Trange) and again in the caveat block at the end: the vapour pressure was ∗extrapolated∗, still returned, and the answer stands with that stated. A number you were warned about is a number you can defend; one you were not is the dangerous kind.*

*Click the drum and edit P in the Properties box on the right — the Case tab is a read-only view of the files, so nothing is typed there. Set 0.5 bar and run again: a deeper let-down flashes more, and the drum ends colder. The edit is transient — it never touches the file, the node shows it in amber until you reset it, and Reset puts the declared value back.*

*Now try the other direction, because it teaches more than it looks. Raise P to 1.5 bar and nothing happens: V/F goes to 0 and the temperature does not drop at all. That is not a broken run — this feed's bubble pressure at 380 K is 1.4088 bar, so 1.5 bar leaves it 0.09 bar subcooled and there is nothing to flash. The engine says so in as many words (Regime: subcooled liquid). The wall is much closer above than below. Then look at the stream table printed ∗before∗ the solve — it shows the 0/ seed values, which look like an answer and are not one; the converged table further down is the answer.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | flashAdiab | T_out | 368.661803634 |
| kpi | flashAdiab | V_over_F | 0.0518811133842 |
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0263366357393 |
| stream | vapor | F | 0.00144114203845 |
| stream | feed | T | 380 |
| stream | liquid | T | 368.661803634 |
| stream | vapor | T | 368.661803634 |
| kpi | flashAdiab | F_in | 0.0277777777778 |
| kpi | flashAdiab | F_liquid | 0.0263366357393 |
| kpi | flashAdiab | F_vapor | 0.00144114203845 |
| kpi | flashAdiab | H_residual | -0.00215364377073 |
| kpi | flashAdiab | P | 101325 |
| kpi | flashAdiab | T_drop | 11.3381963657 |
| *… and 4 more reference values in the case's `expected`.* | | | |

## analysis01_water_analysis_inlet

tutorials/steady/flash/analysis01_water_analysis_inlet




**What it does:** A laboratory water analysis as a first-class inlet: mg/L with alkalinity as CaCO3, a measured density, and a +0.78 % ion imbalance reconciled onto chloride within a declared 10 % limit – measurement, reconciled inventory and equilibrium recorded as three separate layers

**The story:** A LABORATORY ANALYSIS AS AN INLET.

The inlet of this flash is not a composition somebody worked out: it is a water analysis as a laboratory sends it – mg/L, alkalinity reported as CaCO3, a measured density, per-line uncertainties, and an ion balance that is off by 0.78 %.

THREE LAYERS, SEPARATELY RECORDED, and the separation is the whole point:

1 0/feed the MEASUREMENT. Never rewritten by a run (ratified 2026-08-09, O1) – a reconciliation that edited it would destroy the thing it was reconciling, and the next run would reconcile the reconciliation. 2 converged/feed calculated { analysisReconciliation { ? } conservedInventory { ? } } the RECONCILED conserved inventory, with the imbalance before, the imbalance after, and every correction that was applied. This is the layer the engine did not have: an answer never says what it was corrected from. 3 the flash the EQUILIBRIUM solved from layer 2.

WHAT IS MEASURED HERE (the numbers the gate re-derives independently):

ion balance BEFORE +0.7813 % (cations minus anions, over the sum) ion balance AFTER 0.0000 % chloride adjusted +5.1208 % of the measured 45.0 mg/L limit declared 10 % – exceeded, it REFUSES with the numbers

FIVE REFUSALS GUARD IT, each by name, and check_aqueous_analysis.py fires every one through the real reader: no DENSITY ROUTE – mg/L is per volume, a stream is a flow; an ITERATIVE density is a declared A1 gap, refused rather than done silently; a correction OVER THE LIMIT – past the declared bound it stops being a reconciliation and becomes the analysis rewritten to fit; pH NAMED AS THE BALANCING VARIABLE – pH may participate only under an explicit declared rule, and that grammar is A2's; 'method weightedLeastSquares;' – THE NAMED NEXT SLICE. Refused by name rather than silently approximated by the one method A1 has; TWO MATERIAL FORMS in one file – the seventh canonical form is exclusive with the other six, same as they are with each other.

And with no 'reconciliation {}' block at all: an analysis already closing within the tolerance passes through with an announcement stating what was measured; one that does not REFUSES, naming the two legal remedies. What never happens is an adjustment nobody asked for.

Contract: docs/design/aqueous-analysis-inlet-scope.md (slice A1). Compare: flash18_water_analysis_basis – the same water as an INVENTORY, where charge closure is a contract and a violation is refused.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0307824830015 |
| stream | liquid | F | 0.0307824830015 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.0307824830015 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.0307824830015 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | pH | 7.90774504802 |
| kpi | flash01 | p_eq_sum_atm | 0.0747733964818 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 313.15 |
| stream | liquid | T | 313.15 |
| *... and 1 more reference values in the case's* ***expected****.* | | | |

## analysis02_weighted_least_squares

`tutorials/steady/flash/analysis02_weighted_least_squares`




**What it does:** The same laboratory water as analysis01, reconciled by constrained weighted least squares over four measured quantities under electroneutrality plus an elemental-conservation redundancy: the correction spreads across all four at under 0.29 sigma each instead of shoving 5.12 % into chloride

**The story:** THE SAME WATER, RECONCILED PROPERLY.

analysis01 closes a +0.78 % ion balance by moving ONE ion: chloride absorbs the whole residue and shifts 5.12 %. That is a legitimate convention and it is what a water chemist does by hand – but it is a CHOICE dressed as an answer. Nothing about the analysis says chloride was the wrong number; what the sheet actually says is that FOUR quantities were measured, each with its own uncertainty, and that they cannot all be right at once.

A2 asks the question the uncertainties make askable:

what is the composition CLOSEST to what was measured – measuring closeness in each quantity's OWN declared uncertainty – that satisfies the conservation laws exactly?

That is constrained weighted least squares, and it is a one-way PRE-STEP:

laboratory analysis → reconciliation → ONE conserved admissible composition → aqueous equilibrium / speciation

The arrow never reverses. The reconciler lives in its own translation unit (src/streams/AnalysisReconciler.{H,cpp}) which includes NOTHING from thermo/, so it cannot name a speciation solver, an activity model or a thermodynamic package – the equilibrium is structurally unable to reach back into the measurements and make the chemistry easier to satisfy.

THE SHEET. A calcium bicarbonate groundwater, the same water analysis01 carries, plus one line a real laboratory almost always also sends:

Ca 84.0 mg/L ICP-OES 2 % Cl 45.0 mg/L IC 5 % alkalinity 143.0 mg/L as CaCO3 titration 4 % totalHardness 208.0 mg/L as CaCO3 EDTA titration 3 % <– NEW

The hardness line is the interesting one. It measures the SAME CALCIUM the ICP measured, by a completely different method, and the two disagree by 0.84 % – which is unremarkable laboratory life and exactly why nobody can say which is right. It is declared 'element Ca;': it carries NO material of its own (that would count the calcium twice), it is a REDUNDANT determination, and its only effect is the law that ties it to the Ca row.

Note what is NOT declared on it. 'alkalinity ... as CaCO3' needs 'perFormulaUnit 2' because nothing in the formula CaCO3 says a titration counts two bicarbonate per unit – that is a CONVENTION. 'totalHardness { element Ca; as CaCO3; }' needs no such key and is refused if it carries one: how many Ca there are in a CaCO3 is ARITHMETIC, and the formula already says it.

TWO LAWS ARE ENFORCED, and both are declared rather than assumed:

electroneutrality Sum z_i n_i = 0 elementalConservation the measured Ca total = the Ca the species rows carry

...plus non-negativity, which is not in the list because it is not optional.

WHAT THE FIT DOES. Two residues have to disappear – +0.0650 meq/L of charge and +0.0177 mmol/L of calcium disagreement – and the fit spends them where the measurements are weakest:

Ca -0.2735 sigma (-0.547 %) Cl +0.1562 sigma (+0.781 %) alkalinity +0.2813 sigma (+1.125 %) totalHardness +0.1000 sigma (+0.300 %) Sum ((x-m)/sigma)^2 = 0.1883

FOUR quantities move, none by a third of its own uncertainty, against analysis01's single 5.12 % shove. The declared limit is 3 sigma and the worst correction is 0.28 – and the run REFUSES with the numbers if any quantity would go past it.

READ 'converged/feed' FOR THE POINT OF THE SLICE. Every corrected quantity states what was reported, what it became, its sigma and weight, its correction IN SIGMA, which law caused it (exactly – the per-law parts sum to the correction, they are not shares invented afterwards), and how much of that law's own residue it removed. And the block opens with a legend saying that NOTHING in it is a chemistry result: the ions and the solved pH are in 'speciation {}', keyed by species rather than by the author's sheet labels, and carrying no 'reportedValue' because nothing in them was reported.

Provenance: a REPRESENTATIVE analysis composed for this witness, not a transcription of a published one. The disagreements are deliberate and are what the case exists to exercise.

Contract: docs/design/aqueous-analysis-inlet-scope.md (slice A2). Compare: analysis01_water_analysis_inlet – the same water, one ion moved; flash18_water_analysis_basis – the same water as an INVENTORY, where charge closure is a contract and a violation is refused.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0307825083878 |
| stream | liquid | F | 0.0307825083878 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.0307825083878 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.0307825083878 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | pH | 7.90867882055 |
| kpi | flash01 | p_eq_sum_atm | 0.074797651023 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 313.15 |
| stream | liquid | T | 313.15 |
| *... and 1 more reference values in the case's `expected`.* | | | |

## bubbleT01_ethanol_water

tutorials/steady/flash/bubbleT01_ethanol_water




**What it does:** Bubble-T of 30/70 ethanol/water at 1 atm (NRTL)

**The story:** Bubble-T of a 30/70 ethanol/water liquid at 1 atm (NRTL gamma, non-ideal). A saturation calc, not a flash: the unit solves sum(K_i x_i) = 1 for T at the given P (no product streams). Expected: T_bubble ~ 354.4 K with y_ethanol ~ 0.58 – ethanol enriched in the incipient vapour well beyond its 0.30 in the liquid.

*30/70 ethanol/water (100 kmol/h) at 1 atm: at what temperature does the first bubble form, and what is in it? The answer from the golden: T_bubble = 354.42 K (81.27 °C), and the first vapour is 58.2 mol% ethanol from a liquid that is 30 % — the enrichment every distillation column lives on.*

*1. Raoult would get this wrong, so the case declares NRTL. Open constant/thermoPhysPropDict: the liquid activity model is named, and the run announces it at the top of the log (liquid activity.NRTL; vapour idealGas). Nothing is chosen for you. 2. One equation, one unknown. Σ y_i = Σ K_i x_i = 1, solved for T by Newton. The Log tab prints the whole table: five iterations from a 366.6 K guess, driven to 4.4e-12. Count them; read ΔT shrink. 3. K carries the non-ideality. The result table gives K_ethanol = 1.940 and K_water = 0.597 at 354.4 K. With Raoult alone (K = Psat/P) ethanol's K would be far smaller here — the difference is the activity coefficient, and it is why the vapour is ethanol-rich. 4. This is one point on a curve. dewT01_ethanol_water asks the mirror question (Σ x = Σ y/K = 1) for a vapour of the same composition, and the two together bracket the two-phase region at 1 atm.*

*Change the feed to 90/10 ethanol/water in 0/feed and run again: the bubble point moves toward ethanol's boiling point, and the vapour is barely richer than the liquid — you are near the azeotrope, which is what column03_azeotrope_mesh is about.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| kpi | bubble01 | P | 101325 |
| kpi | bubble01 | T_bubble | 354.424819099 |
| kpi | bubble01 | converged | 1 |
| kpi | bubble01 | y_ethanol | 0.581915403358 |
| kpi | bubble01 | y_water | 0.418084596646 |
| stream | feed | T | 350 |
| boundary | global | H_feeds_kW | 0 |
| boundary | global | Q_boundary_kW | 0 |
| boundary | global | H_products_kW | 0 |

## bubbleT02_uniquac_ethanol_water

tutorials/steady/flash/bubbleT02_uniquac_ethanol_water




**What it does:** Bubble-T of 30/70 ethanol/water at 1 atm (UNIQUAC) – validates the UNIQUAC model against the NRTL/Wilson twins

**The story:** Bubble-T of a 30/70 ethanol/water liquid at 1 atm (UNIQUAC).

Same conditions as bubbleT01 (NRTL): a strongly non-ideal binary whose bubble temperature and vapour composition are set by the activity coefficients. The UNIQUAC result should land close to the NRTL/Wilson twins (T_bubble ~ 353.5 K, the vapour enriched in ethanol towards the azeotrope, y_ethanol ~ 0.58).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| kpi | bubble02 | P | 101325 |
| kpi | bubble02 | T_bubble | 353.469206954 |
| kpi | bubble02 | converged | 1 |
| kpi | bubble02 | y_ethanol | 0.582711434226 |
| kpi | bubble02 | y_water | 0.417288565785 |
| stream | feed | T | 350 |
| boundary | global | H_feeds_kW | 0 |
| boundary | global | Q_boundary_kW | 0 |
| boundary | global | H_products_kW | 0 |

## dewT01_ethanol_water

tutorials/steady/flash/dewT01_ethanol_water




**What it does:** Dew-T of 30/70 ethanol/water vapour at 1 atm (NRTL)

**The story:** Dew-T of a 30/70 ethanol/water vapour at 1 atm (NRTL gamma, non-ideal). The mirror of bubbleT01: the unit solves sum(y_i / K_i) = 1 for T at the given P (a saturation calc, no product streams); the incipient liquid is water-enriched. Dew-T sits ABOVE the bubble-T of the same overall composition – the two bound the two-phase window.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 370 |
| boundary | global | H_feeds_kW | 0 |
| boundary | global | Q_boundary_kW | 0 |
| boundary | global | H_products_kW | 0 |

## flash01_benzene_toluene

tutorials/steady/flash/flash01_benzene_toluene




**What it does:** Isothermal flash benzene/toluene at 370 K, 1 bar (Raoult)

**The story:** THE first tutorial: isothermal flash of a 40/60 benzene/toluene feed (100 kmol/h) at 370 K, 1 bar, ideal Raoult K-values (K_i = Psat_i/P). 370 K sits inside the two-phase window (bubble ~368 K, dew ~374 K), so the feed splits into a benzene-enriched vapour and a toluene-enriched liquid – the Rachford-Rice solve is printed iteration by iteration at verbosity 3.

*The same 40/60 benzene/toluene feed (100 kmol/h) as adiabaticFlash01, this time held at 370 K and 1 bar. Raoult's law, ideal vapour, the Rachford-Rice equation solved by Newton in one unknown. The golden: V/F = 0.3040 (30.40 kmol/h vapour, 69.60 kmol/h liquid), K_benzene = 1.651, K_toluene = 0.674.*

*1. type → model → operation. The unit is an isothermalFlash, the machine is named (model rachfordRice;), and the numbers live in operation { T 370.0 K; P 1.0 bar; }. That order is the grammar of every unit in Choupo. 2. K is a ratio of pressures here. With Raoult's law K_i = Psat_i(T) / P. The Log tab prints both vapour pressures with the K they give (benzene Psat 1.6514 bar K = 1.6514), so you can check the two K's by hand — a two-line calculation a student should do once. (This line used to send you to the Props tab. It cannot answer: that view is a ?/? coverage matrix — its "Psat" column means ∗a vapour-pressure model exists∗, not a value — plus one pill per operation the case declares in a propsDict, and this case has none. For Psat as a NUMBER at a temperature you choose, the Property surfaces view is the other home.) 3. Rachford-Rice has one root in (0, 1) when the feed is between its bubble and dew points. The Log tab prints each Newton iterate of V/F; count them. 4. This drum adds no heat — and that is worth understanding. The golden reports Q_kW = 0. The feed is declared at 370 K and 1 bar, the drum holds 370 K and 1 bar, and enthalpy is a state function of (T, P, z): the feed stream ∗already is∗ the two-phase mixture the drum separates. Move T away from the feed's 370 K and a duty appears — that is the isothermal flash earning its name. (This README first said "holding T costs a duty"; the case's own golden said otherwise.)*

*Move T between the feed's bubble point and dew point and watch V/F sweep from 0 to 1; flash06_sweep_T does exactly that automatically with an outerDict, which is the next case to open.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | flash01 | V_over_F | 0.30398357314 |
| kpi | flash01 | T | 370 |
| stream | vapor | F | 0.00844398814279 |
| stream | liquid | F | 0.019333789635 |
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 370 |
| stream | liquid | T | 370 |
| stream | vapor | T | 370 |
| kpi | flash01 | F_alpha | 0.019333789635 |
| kpi | flash01 | F_beta | 0.00844398814279 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | K_benzene | 1.65136809583 |
| kpi | flash01 | K_toluene | 0.673501829388 |
| kpi | flash01 | P | 100000 |

*... and 6 more reference values in the case's `expected`.*

## flash02_ethanol_water

tutorials/steady/flash/flash02_ethanol_water




**What it does:** Isothermal flash 50/50 ethanol/water at 355 K (NRTL)

**The story:** isothermal flash, ethanol/water, NRTL

NOTE (golden re-recorded): the operation pressure formerly read a BARE ‘P 1.01325;‘ (no unit) → parsed as 1.01325 Pa, not bar. At ~1 Pa the K-values blow up to O(1e5), so the flash silently reported a fully superheated vapour (V/F = 1, phaseSet = 0) and "Converged: yes" with no warning – a wrong all-vapour answer for a 50/50 ethanol/water at 355 K, 1 atm, which MUST be two-phase. Adding the ‘bar‘ unit restores the physically-correct partial vaporisation; the ‘expected‘ golden was re-recorded accordingly (was the wrong all-vapour result).

*A 50/50 ethanol/water feed (50 kmol/h of each) held at 355 K and 1.01325 bar in one isothermalFlash. Structurally this is flash01_benzene_toluene again: one unit, one operation { T ... P ... }, Rachford-Rice in one unknown. Everything that is different is in constant/thermoPhysPropDict, which declares NRTL with exactly the parameters compare_activity_etoh_water plotted — a_ij -0.8009, b_ij 246.18, a_ji 3.4578, b_ji -586.0809, alpha 0.30.*

*The golden: V/F = 0.7976 (79.76 kmol/h vapour, 20.24 kmol/h liquid), K_ethanol = 2.2274, K_water = 0.5850, and Q_kW = 0.*

*1. K stopped being a ratio of pressures. In flash01 Raoult gave K_i = Psat_i(T) / P and you could check both K's against the two vapour pressures the log prints. Here the liquid is non-ideal, so K_i = gamma_i ∗ Psat_i(T) / P, and the activity coefficient is the whole of the difference between this case and the last one. The Log tab still prints the vapour pressures, so the gap between Psat/P and the reported K is the gamma, read off the screen. 2. The model arrived from the case, not from the unit. The unit spec in flowsheetDict carries no thermodynamics at all — type, in/outputs, and operation { T 355.0 K; P 1.01325 bar; }. Swapping Raoult for NRTL touched one file, and it was not the one describing the equipment. 3. The gamma that acts here is not the gamma you plotted. Case 2 scanned an ∗equimolar∗ liquid; the liquid leaving this drum is not equimolar. K_ethanol 2.2274 is greater than 1 and K_water 0.5850 is less than 1, so ethanol concentrates in the vapour and the liquid left behind is water-rich — at a composition the previous case never evaluated. A curve at fixed composition is a slice through a surface the flash walks across. 4. Q_kW = 0, for the same reason as flash01. The feed file declares only T = 355 K and P = 1.01325 bar — a mixture's (T, P) fix its split, so the engine resolves the feed at its own state and the stream table shows vf = 0.798 — and the drum holds that same T and P. So the feed stream ∗already is∗ the two-phase mixture the drum separates, and the answer V/F = 0.797563408329 is the feed's own resolved fraction. Enthalpy is a state function; nothing has to be added to hold a state you were already in. 5. Read the flowsheetDict header before you move anything. It records that the operating pressure once read a bare P 1.01325; with no unit — parsed as 1.01325 Pa — which drove the K's to O(1e5) and reported a fully vaporised feed, converged, with no warning. Scalars carry units in this grammar; that header is what a silent wrong answer costs.*

*Set the liquid's activityModel back to ideal in constant/thermoPhysPropDict and run again. The K's collapse to Psat/P, the split moves, and Q_kW stops being zero — because the feed's declared vapour fraction is no longer the answer at 355 K. That difference is case 2's three curves, arriving in a unit. Then open column01_benzene_toluene, where this same equilibrium is solved on fifteen stages at once.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.00562323865754 |
| stream | vapor | F | 0.0221545391202 |
| kpi | flash02 | F_alpha | 0.00562323865754 |
| kpi | flash02 | F_beta | 0.0221545391202 |
| kpi | flash02 | F_in | 0.0277777777778 |
| kpi | flash02 | P | 101325 |
| kpi | flash02 | T | 355 |
| kpi | flash02 | V_over_F | 0.797563408329 |
| kpi | flash02 | phaseSet | 0 |
| stream | feed | T | 355 |
| stream | liquid | T | 355 |
| stream | vapor | T | 355 |
| kpi | flash02 | K_ethanol | 2.22741301451 |
| *… and 6 more reference values in the case's `expected`.* | | | |

## flash03_wilson_ethanol_water

tutorials/steady/flash/flash03_wilson_ethanol_water




**What it does:** Isothermal flash 50/50 ethanol/water at 355 K (Wilson activity model)

**The story:** same conditions as flash02, but Wilson activity model

NOTE (golden re-recorded): the operation pressure formerly read a BARE ‘P 1.01325;‘ (no unit) → parsed as 1.01325 Pa, not bar. At ~1 Pa the K-values blow up to O(1e5), so the flash silently reported a fully superheated vapour (V/F = 1, phaseSet = 0) and "Converged: yes" with no warning – a wrong all-vapour answer for a 50/50 ethanol/water at 355 K, 1 atm, which MUST be two-phase. Adding the ‘bar‘ unit restores the physically-correct partial vaporisation; the ‘expected‘ golden was re-recorded accordingly (was the wrong all-vapour result).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0063767199082 |
| stream | vapor | F | 0.0214010578696 |
| kpi | flash03 | F_alpha | 0.0063767199082 |
| kpi | flash03 | F_beta | 0.0214010578696 |
| kpi | flash03 | F_in | 0.0277777777778 |
| kpi | flash03 | P | 101325 |
| kpi | flash03 | T | 355 |
| kpi | flash03 | V_over_F | 0.770438083305 |
| kpi | flash03 | phaseSet | 0 |
| stream | feed | T | 355 |
| stream | liquid | T | 355 |
| stream | vapor | T | 355 |
| kpi | flash03 | K_ethanol | 2.07217514574 |

*. . . and 6 more reference values in the case’s* ***expected****.*

## flash04_molarFlows_input

tutorials/steady/flash/flash04_molarFlows_input




**What it does:** Isothermal flash, feed declared via molarFlows (benzene 40 + toluene 60 kmol/h)

**The story:** same benzene/toluene flash as flash01, but the feed is declared via per-species molar flows.

Instead of writing F = 100 kmol/h + composition { benzene 0.40; toluene 0.60; }, this case writes the absolute molar flow of each species:

molarFlows { benzene 40 kmol/h; toluene 60 kmol/h; }

The parser sums to F_total = 100 kmol/h and derives x_benzene = 0.4, x_toluene = 0.6. Physically equivalent to flash01 – and the result table is identical to printer precision (V/F = 0.304, L = 69.6 kmol/h, V = 30.4 kmol/h).

Demonstrates the (d) form of stream declaration; see userGuide "Composition basis" for the five accepted forms.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.019333789635 |
| stream | vapor | F | 0.00844398814279 |
| kpi | flash04 | F_alpha | 0.019333789635 |
| kpi | flash04 | F_beta | 0.00844398814279 |
| kpi | flash04 | F_in | 0.0277777777778 |
| kpi | flash04 | P | 100000 |
| kpi | flash04 | T | 370 |
| kpi | flash04 | V_over_F | 0.30398357314 |
| kpi | flash04 | phaseSet | 0 |
| stream | feed | T | 370 |
| stream | liquid | T | 370 |
| stream | vapor | T | 370 |
| kpi | flash04 | K_benzene | 1.65136809583 |
| *... and 6 more reference values in the case's `expected`.* | | | |

## flash05_ternary_massFlows

tutorials/steady/flash/flash05_ternary_massFlows




**What it does:** Ternary flash benzene/toluene/n-hexane, feed in mass flows (5000 kg/h)

**The story:** ternary benzene/toluene/n-hexane flash, feed declared via per-species MASS flows.

This is the (e) form of stream input from the v0.27 grammar (see userGuide "Composition basis"): instead of writing F total plus a composition, the student writes the absolute mass flow of each species and the parser sums + converts to molar:

m_i [kg/s] → F_mol,i [kmol/s] = m_i / MW_i F_total [kmol/s] = Sigma F_mol,i x_i = F_mol,i / F_total

Feed composition by mass (5000 kg/h total): benzene 1500 kg/h (30 wt%) MW = 78.11 kg/kmol toluene 2500 kg/h (50 wt%) MW = 92.14 kg/kmol n-hexane 1000 kg/h (20 wt%) MW = 86.18 kg/kmol

Conversion to molar: F_benzene = 1500 / 78.11 = 19.203 kmol/h F_toluene = 2500 / 92.14 = 27.131 kmol/h F_nHexane = 1000 / 86.18 = 11.604 kmol/h F_total = 57.938 kmol/h x_benzene = 0.3315 ; x_toluene = 0.4682 ; x_nHexane = 0.2003

At T = 365 K, P = 1 bar the n-hexane (Tb = 342 K) is well above its boiling point and goes preferentially to the vapour; toluene (Tb = 384 K) is sub-cooled and stays preferentially in the liquid; benzene (Tb = 353 K) partitions. Result:

molar basis F = 57.94 kmol/h V/F = 0.494 L = 29.34 kmol/h V = 28.60 kmol/h

mass basis F = 5000.0 kg/h V/F_mass = 0.487 L = 2567.3 kg/h V = 2432.7 kg/h

Mass fractions tell the partition story directly:

feed: w_benzene 0.300 w_toluene 0.500 w_nHexane 0.200 liquid: w_benzene 0.244 w_toluene 0.624 w_nHexane 0.133 vapor: w_benzene 0.360 w_toluene 0.369 w_nHexane 0.271

Mass balance closes per species — e.g. n-hexane: feed: 5000 ∗ 0.200 = 1000 kg/h liquid: 2567 ∗ 0.133 = 341 kg/h vapor: 2433 ∗ 0.271 = 659 kg/h → 341 + 659 = 1000.

Toggle the units menu in the GUI to Mass-basis to see the F and composition flip back to kg/h + w_i, matching the input statement symmetrically.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0160941453232 |
| stream | liquid | F | 0.00814913402453 |
| stream | vapor | F | 0.00794501129864 |
| kpi | flash05 | F_alpha | 0.00814913402453 |
| kpi | flash05 | F_beta | 0.00794501129864 |
| kpi | flash05 | F_in | 0.0160941453232 |
| kpi | flash05 | P | 100000 |
| kpi | flash05 | T | 365 |
| kpi | flash05 | V_over_F | 0.493658478851 |
| kpi | flash05 | phaseSet | 0 |
| stream | feed | T | 365 |
| stream | liquid | T | 365 |
| stream | vapor | T | 365 |
| kpi | flash05 | K_benzene | 1.43515088002 |

*... and 7 more reference values in the case's `expected`.*

## flash06_sweep_T

tutorials/steady/flash/flash06_sweep_T




**What it does:** Flash benzene/toluene at 1 bar: T swept 360→385 K across the bubble→dew window (V/F S-curve + duty)

**The story:** the SAME 40/60 benzene/toluene isothermal flash as flash01 (100 kmol/h, 1 bar, Raoult); the T value below is only the BASE point – system/outerDict sweeps it 360 → 385 K across the bubble→dew window (~368..374 K) and reads the V/F S-curve + duty as responses.

*The flash01 benzene/toluene drum, with its temperature swept from 360 K to 385 K in 1 K steps by a system/outerDict — the first time the simulator is run ∗many times∗ rather than once. The result is the V/F S-curve and, beside it, the duty each temperature costs.*

*The sweep table is written to sweep_flashT.csv in the case folder. Read down it:*

*1. The knob is a dict path. parameter { target units[0].operation.T; range ( 360 385 ); nPoints 26; } — ∗any∗ scalar a unit's operation block carries can be swept the same way. The dict is the schema; there is no separate list of "sweepable" things. 2. Bubble and dew fall out of the curve. V/F leaves 0 between 367 and 368 K and reaches 1 between 374 and 375 K. You did not ask for the bubble and dew points; you read them off. 3. Q = 0 at 370 K is not a coincidence. The feed is declared at 370 K: at the feed's own temperature the drum adds nothing (the point flash01 makes on its own). Below it the drum ∗removes∗ heat to condense; above it the drum ∗supplies∗ heat to boil. 4. Between two points the curve is not a line. V/F goes 0.30 → 0.60 → 0.96 over 370–374 K — steepest near the dew point. A hand calculation at two temperatures would miss the shape entirely.*

*Sweep pressure instead: change target to units[0].operation.P with a range across 0.5–2 bar and watch the same feed move from all-vapour to all-liquid. flash07_gridsweep_TP sweeps both at once and draws the envelope.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0110077140444 |
| stream | vapor | F | 0.0167700637334 |
| kpi | flash01 | F_alpha | 0.0110077140444 |
| kpi | flash01 | F_beta | 0.0167700637334 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | K_benzene | 1.74456473693 |
| kpi | flash01 | K_toluene | 0.716223951412 |
| kpi | flash01 | P | 100000 |
| kpi | flash01 | Q | 284075.447071 |
| kpi | flash01 | Q_kW | 284.075447071 |
| kpi | flash01 | T | 372 |
| kpi | flash01 | V_over_F | 0.603722294402 |
| kpi | flash01 | phaseSet | 0 |

*... and 7 more reference values in the case's `expected`.*

## flash07_gridsweep_TP

tutorials/steady/flash/flash07_gridsweep_TP




**What it does:** 2-D grid sweep: flash V/F over T x P (the flash envelope)

**The story:** one isothermal flash, benzene/toluene 40/60.

The base point is 370 K, 1 bar. The gridSweep in system/outerDict overrides BOTH operation.T and operation.P across a grid, so V/F traces the flash envelope (V/F=0 bubble line, V/F=1 dew line) over the T x P plane.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| csv | gridsweep_TP.csv | 8:flashDrum.V_over_F | 0.235178 |
| csv | gridsweep_TP.csv | 23:flashDrum.V_over_F | 0.303984 |
| csv | gridsweep_TP.csv | 38:flashDrum.V_over_F | 0.641142 |
| csv | gridsweep_TP.csv | 46:flashDrum.V_over_F | 0.385974 |
| csv | gridsweep_TP.csv | 0:flashDrum.V_over_F | 1 |
| csv | gridsweep_TP.csv | 1:flashDrum.V_over_F | 0 |

## flash08_co2_water_package

tutorials/steady/flash/flash08_co2_water_package

**What it does:** Carbonated water flash whose thermo is an INLINE property-package MANIFEST: constant/thermoPhysPropDict IS the full record (components, methods per phase, solvent+solutes, the CO2-water Henry pair declared+verified). CO2 dissolves by the unsymmetric Henry convention; water stays Raoult. Read the run log: the assembly announces itself.

**The story:** the property-package convention, minimal.

Soda water at 298 K: 1 bar flash of a CO2-rich feed. The THERMO is not a pile of flat keys – the case's constant/thermoPhysPropDict IS the full INLINE manifest (name flash08_co2Water) and the ThermoPackageBuilder assembles everything from it: liquid = solution.henryDilute (water on the Raoult rung, CO2 on the infinite-dilution Henry rung – ONE Gibbs surface, two conventions), vapour = ideal gas, the CO2-water Henry pair declared and VERIFIED at assembly (delete the pair path from the package and the run REFUSES, naming the file). Expect K_CO2 = H(298)/P ~ 1.6e3 (Sander): a huge K sets the vapour's PURITY, not the split – the vapour is nearly pure CO2 (y_CO2 ~ 0.97) but its AMOUNT is tiny (V/F ~ 4e-4), so only ~40% of the fed CO2 leaves in it; the liquid keeps x_CO2 ~ y_CO2/K ~ 6e-4.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | flatWater | F | 0.000277662593734 |
| stream | gasOff | F | 1.15184043824e-07 |
| stream | sodaWater | F | 0.000277777777778 |
| kpi | node | F_alpha | 0.000277662593734 |
| kpi | node | F_beta | 1.15184043824e-07 |
| kpi | node | F_in | 0.000277777777778 |
| kpi | node | P | 101325 |
| kpi | node | Q | 0 |
| kpi | node | Q_kW | 0 |
| kpi | node | T | 298.15 |
| kpi | node | V_over_F | 0.000414662557767 |
| kpi | node | phaseSet | 0 |
| stream | flatWater | T | 298.15 |
| stream | gasOff | T | 298.15 |
| *... and 6 more reference values in the case's `expected`.* | | | |

## flash09_n2ch4_stryjek

tutorials/steady/flash/flash09_n2ch4_stryjek

**What it does:** phi-phi VLE VALIDATION vs Stryjek, Chappelear & Kobayashi (1974), J. Chem. Eng. Data 19:334 (DOI 10.1021/je60063a023), Table III, -190 F = 149.82 K. The case carries its thermo as an INLINE manifest: liquid+vapour both eos.SRK (one cubic, two roots: K = phi_L/phi_V) with the N2-CH4 interaction constant kij = 0.0289 (a small measured correction to the mixing rule, from the Knapp/DECHEMA data compilation). Measured at 299 psia: x_N2=0.1503, y_N2=0.4579 – compare the Flash Result table in the log with these measured values (agreement ~2%).

**The story:** the phi-phi world pinned to a MEASURED isotherm

Nitrogen-methane at 149.82 K (-190.00 F), 20.616 bar (299 psia) – a point of Stryjek, Chappelear & Kobayashi (1974), J. Chem. Eng. Data 19:334, Table III (DOI 10.1021/je60063a023). MEASURED there: x_N2 = 0.1503 y_N2 = 0.4579 K_N2 = 3.047 K_CH4 = 0.638 The feed z_N2 = 0.30 sits inside the envelope, so the flash must split into exactly those phases if the thermodynamics is right. The K-values come from K = phi_L/phi_V – the SAME SRK cubic evaluated at its liquid and vapour roots, with kij(N2,CH4) = 0.0289 – a small MEASURED correction to the cubic mixing rule, from the Knapp/DECHEMA VLE data compilation (announced in this log). Neighbouring measured points bracket the model: K_N2 = 3.047 at 299 psia and 2.241 at 396.5 psia – the engine interpolates between them consistently (try P 22 bar in this dict: K_N2 ~ 2.9). This is the quantitative anchor of the phi-phi world, the same standard as Wiebe-1934 for the Henry world.

THE AGREEMENT IS EXECUTABLE (2026-08-10). Everything above was prose: the golden pinned F/P/T/Q – all SELF-RECORDED – so a model that drifted off Stryjek's isotherm would still have passed while this header went on claiming a measured anchor. 'expected' now carries two 'anchor' rows holding the PAPER's K_N2 = 3.047 and K_CH4 = 0.638; they are compared on every run and are never refreshed by '–record' (a published number is not the model's to rewrite). Measured agreement: K_N2 +2.25 %, K_CH4 -1.56 %, x_N2 -0.52 %, y_N2 +1.73 % – inside the declared 3 % band, with margin.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | liq | F | 0.000145628387466 |
| stream | vap | F | 0.000132149390312 |
| kpi | node | F_alpha | 0.000145628387466 |
| kpi | node | F_beta | 0.000132149390312 |
| kpi | node | F_in | 0.000277777777778 |
| kpi | node | P | 2061600 |
| kpi | node | Q | 0 |
| kpi | node | Q_kW | 0 |
| kpi | node | T | 149.82 |
| kpi | node | V_over_F | 0.475737805121 |
| kpi | node | phaseSet | 0 |
| kpi | node | K_N2 | 3.047 |
| kpi | node | K_CH4 | 0.638 |

*... and 6 more reference values in the case's `expected`.*

## flash09_nh3_water_reactive

tutorials/steady/flash/flash09_nh3_water_reactive




**What it does:** Reactive NH3/water flash: SIMULTANEOUS aqueous speciation (NH4+/H+/OH-, pH solved) + VLE at 368 K, 1 atm – ions never reach the vapour or the streams

**The story:** the volatile weak electrolyte, done honestly.

Ammonia in water IONISES (NH3 + H2O <→ NH4+ + OH-) and VOLATILISES (Henry) at the same time: the ionised fraction is unavailable to the vapour, so speciation and phase split must be solved TOGETHER – never flash-once/speciate-once. The flash unit implements NO chemistry: it delegates to the package's reactive equilibrium (unit-local speciation, property authority section 6b). Streams – including 0/ and converged/ – carry ONLY the apparent components NH3 and water; the aqueous species (NH4+, H+, OH-) exist in the internal state and the report.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0224533469429 |
| stream | vapor | F | 0.00532443083488 |
| kpi | flash01 | F_alpha | 0.0224533469429 |
| kpi | flash01 | F_beta | 0.00532443083488 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | Q | 0 |
| kpi | flash01 | Q_kW | 0 |
| kpi | flash01 | T | 368.15 |
| kpi | flash01 | V_over_F | 0.191679510056 |
| kpi | flash01 | pH | 9.899823444 |
| kpi | flash01 | p_eq_sum_atm | 1 |
| kpi | flash01 | phaseSet | 0 |

*... and 8 more reference values in the case's `expected`.*

## flash10_acetic_water_reactive

tutorials/steady/flash/flash10_acetic_water_reactive




**What it does:** Reactive acetic acid/water flash: SIMULTANEOUS aqueous speciation (Acetate-/H+/OH-, pH solved) + VLE – the weak ACID mirror of flash09; ions never reach the vapour or the streams

**The story:** the volatile weak ACID, done honestly.

Acetic acid in water IONISES (CH3COOH <→ CH3COO- + H+, pKa 4.756) and VOLATILISES at the same time – and its VAPOUR DIMERISES (2 HAc = (HAc)2, carboxylic hydrogen bonding), so even the saturation check must price the dimer's own partial pressure. Speciation and phase behaviour are solved TOGETHER: the flash unit implements no chemistry, it delegates to the package's reactive equilibrium (unit-local speciation, section 6b). Streams – 0/ and converged/ included – carry ONLY the apparent components aceticAcid and water; the aqueous species (CH3COO-, H+, OH-) exist in the internal state and the report.

THE OPERATING POINT IS DELIBERATELY SUBSATURATED (T fixed at 368.15 K, 1 atm; sum p_eq = 0.818 atm < P): the honest regime for a dilute, barely-volatile acid on the Davies rung. The unit is an ISOTHERMAL (T,P) flash – T is the specification, not a result. Temperature still ACTS, through BOTH van't Hoff legs (dissolution dH = -51.5 kJ/mol, dimerisation dH = -63.2 kJ/mol), and the flash publishes what it moves as KPIs: pH, p_eq_sum_atm, p_mono_atm, p_dimer_atm, dimer_share. flash11_acetic_T_sweep sweeps T across the subsaturated window so the student SEES the total volatility climb while the dimer share falls; flash13_acetic_ethanol_vacuum_flash crosses into the TWO-PHASE regime, where the vapour mole balance is re-weighted by the dimer explicitly (apparent moles = n_mono + 2 n_dim) and its association heat is priced into the duty – never a silent approximation.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0277777777778 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.0277777777778 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | Q | 0 |
| kpi | flash01 | Q_kW | 0 |
| kpi | flash01 | T | 368.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | dimer_share | 0.271969921802 |
| kpi | flash01 | pH | 2.15260602547 |
| kpi | flash01 | p_dimer_atm | 0.0138783233614 |

*... and 9 more reference values in the case's `expected`.*

## flash10_ch4propane_pcsaft

tutorials/steady/flash/flash10_ch4propane_pcsaft

**What it does:** PC-SAFT phi-phi binary flash, CH4/propane at 273.15 K / 20 bar, kij = 0 (announced): the PREDICTIVE baseline of the SAFT world – K = phi_L/phi_V from ONE a_res surface at its two density roots. Sanity anchor carried by the case itself: propane's curated Antoine gives Psat(273.15 K) = 4.7 bar, so the Raoult-magnitude K_propane ~ 0.24 – the PC-SAFT K_propane = 0.32 sits near it (the deviation IS the fugacity correction of a liquid holding ~11% dissolved CH4, visible); K_CH4 » 1 (supercritical light key). The published-kij refinement is a NAMED follow-up: it needs the cited pair record (data/standards/parameters/PCSAFT/), never an invented number.

**The story:** the PC-SAFT phi-phi world on a BINARY, kij = 0

Methane/propane, 273.15 K, 20 bar, z = 0.50/0.50. CH4 is supercritical (Tc = 190.6 K) – no Raoult route exists for it; the phi-phi K comes from the SAME PC-SAFT a_res at the liquid and vapour density roots. With NO binary correction (kij = 0, announced by the EoS), everything is PREDICTED from the two components' (m, sigma, epsilon/k) alone (Gross & Sadowski 2001, Table 1; CH4 and propane parameters independently cross-checked against the NIST teqp reference implementation's test values).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | liq | F | 9.97838549688e-05 |
| stream | vap | F | 0.000177993922809 |
| kpi | node | F_alpha | 9.97838549688e-05 |
| kpi | node | F_beta | 0.000177993922809 |
| kpi | node | F_in | 0.000277777777778 |
| kpi | node | P | 2000000 |
| kpi | node | Q | 0 |
| kpi | node | Q_kW | 0 |
| kpi | node | T | 273.15 |
| kpi | node | V_over_F | 0.640778122112 |
| kpi | node | phaseSet | 0 |
| stream | feed | T | 273.15 |
| stream | liq | T | 273.15 |

*... and 6 more reference values in the case's* `expected`.

## flash11_acetic_T_sweep

tutorials/steady/flash/flash11_acetic_T_sweep




**What it does:** Reactive acetic/water flash: T swept 328→368 K in the subsaturated window (p_eq climbs, dimer share falls, pH drifts)

**The story:** the same reactive flash as flash10, but the temperature specification is OWNED by system/outerDict: the sweep driver rewrites units[0].operation.T pass by pass (setScalarAtPath – the dict IS the schema) and records the flash's reactive KPIs at each point. Topology, chemistry, and feed are identical to flash10_acetic_water_reactive; see that case's header for the physics and for why the two-phase corner is refused.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0277777777778 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.0277777777778 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | Q | -43266.6666667 |
| kpi | flash01 | Q_kW | -43.2666666667 |
| kpi | flash01 | T | 348.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | dimer_share | 0.317465631471 |
| kpi | flash01 | pH | 2.15093435013 |
| kpi | flash01 | p_dimer_atm | 0.0065733081438 |
| *... and 10 more reference values in the case's `expected`.* | | | |

## flash12_nh3_acetic_ethanol_reactive

tutorials/steady/flash/flash12_nh3_acetic_ethanol_reactive




**What it does:** Reactive 4-component flash: NH3 + HAc neutralise into an ammonium-acetate buffer (pH ~ pKa_HAc); ethanol classified nonionising from its record fact, ideal VLE authorised; subsaturated at 358.15 K

**The story:** One isothermal flash, four components, two aqueous families + one authorised molecular co-volatile:

feed (2 HAc + 1 NH3 + 1 EtOH + 96 H2O, kmol/h) → [flash01, 358.15 K / 1 atm] → liquid (everything) + vapor (0)

The chemistry does the interesting work in the LIQUID: NH3 + HAc → NH4+ + Ac- consumes the base completely (K = Ka_HAc/Ka_NH4 ~ 10^4.5), leaving an equimolar Ac-/HAc buffer at pH ~ 4.6. The saturation check prices each volatile over THAT speciated liquid – ionised acid and ammonium are unavailable to the vapour, the acetic dimer is priced, and ethanol contributes x∗psat (authorised ideal). KPIs: pH, p_eq_sum_atm, p_mono_atm / p_dimer_atm / dimer_share, p_ethanol_atm.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0277777777778 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.0277777777778 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | Q | 0 |
| kpi | flash01 | Q_kW | 0 |
| kpi | flash01 | T | 358.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | dimer_share | 0.0762211005883 |
| kpi | flash01 | pH | 4.6182465052 |
| kpi | flash01 | p_dimer_atm | 0.000380469802896 |

*. . . and 10 more reference values in the case's `expected`.*

## flash13_acetic_ethanol_vacuum_flash

tutorials/steady/flash/flash13_acetic_ethanol_vacuum_flash




**What it does:** Reactive VACUUM flash on mixed-solvent v1 (NRTL backbone + Davies ions): V/F = 0.58 at 358.15 K / 0.65 atm, gamma_EtOH = 2.97, the buffer holds the ammonia, dimer heat priced exactly

**The story:** One isothermal VACUUM flash, four components, REAL vapour flow:

feed (2 HAc + 1 NH3 + 12 EtOH + 85 H2O, kmol/h) → [flash01, 358.15 K / 0.65 atm] → liquid (42.3 kmol/h, pH 4.98) + vapor (57.7 kmol/h)

On the NRTL backbone the feed's bubble pressure is ~1.1 atm (gamma_EtOH = 2.97 triples the ideal ethanol volatility): 0.65 atm strips well over half the feed without wringing the liquid dry. The vapour balance carries the acetic dimer explicitly (v_HAc = t_mono + 2 t_dim, K_dim(T) van't Hoff) and the buffer chemistry keeps the ammonia liquid-bound. KPIs: pH, p_mono_atm, p_dimer_atm, dimer_share, p_ethanol_atm, p_eq_sum_atm (= P, two-phase).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0116941006641 |
| stream | vapor | F | 0.0160836771137 |
| kpi | flash01 | F_alpha | 0.0116941006641 |
| kpi | flash01 | F_beta | 0.0160836771137 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | P | 65861.25 |
| kpi | flash01 | Q | 671656.06605 |
| kpi | flash01 | Q_kW | 671.65606605 |
| kpi | flash01 | T | 358.15 |
| kpi | flash01 | V_over_F | 0.579012376094 |
| kpi | flash01 | dimer_share | 0.0870838206127 |
| kpi | flash01 | pH | 5.00900567054 |
| kpi | flash01 | p_dimer_atm | 0.000508532811379 |

*... and 19 more reference values in the case's `expected`.*

## flash14_calcite_carbonate_basis

tutorials/steady/flash/flash14_calcite_carbonate_basis




**What it does:** Reactive calcite/carbonate flash: the FIRST component whose aqueous bridge spans TWO master families (CaCO3 → Ca+2 AND HCO3-), so the component-to-species map is a matrix, not a renaming

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0275033888889 |
| stream | liquid | F | 0.0269197611327 |
| stream | vapor | F | 0.000583627756178 |
| kpi | flash01 | F_alpha | 0.0269197611327 |
| kpi | flash01 | F_beta | 0.000583627756178 |
| kpi | flash01 | F_in | 0.0275033888889 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0.0212202124813 |
| kpi | flash01 | pH | 6.01561221041 |
| kpi | flash01 | p_eq_sum_atm | 1 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 313.15 |
| stream | liquid | T | 313.15 |
| *... and 4 more reference values in the case's `expected`.* | | | |

## flash15_refused_salt_basis_rank

tutorials/steady/flash/flash15_refused_salt_basis_rank




**What it does:** DELIBERATE REFUSAL: K/Na x Cl/SO4 – four salt components onto four ion families with rank 3, so the component basis has exactly (c-1)(a-1) = 1 degree of freedom and the converged state cannot be read back as these components uniquely

## flash16_calcite_precipitation

tutorials/steady/flash/flash16_calcite_precipitation




**What it does:** Reactive flash that PRECIPITATES: the calcite/carbonate basis of flash14 fed past saturation, with calcite admitted – the speciation drives SI to 0 and the freed proton drops the pH from 6.63 to 6.02 – ONE proton per mole of calcite, and exactly once: the crystal is neutral, so it moves no charge, and the acidification arrives through the master balances that electroneutrality closes

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0275083333333 |
| stream | liquid | F | 0.0269247487752 |
| stream | vapor | F | 0.000583584558179 |
| kpi | flash01 | F_alpha | 0.0269247487752 |
| kpi | flash01 | F_beta | 0.000583584558179 |
| kpi | flash01 | F_in | 0.0275083333333 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0.0212148279253 |
| kpi | flash01 | pH | 6.02023176857 |
| kpi | flash01 | p_eq_sum_atm | 1 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 313.15 |
| stream | liquid | T | 313.15 |
| *... and 9 more reference values in the case's `expected`.* | | | |

## flash17_two_liquids_reactive

tutorials/steady/flash/flash17_two_liquids_reactive




**What it does:** Reactive flash with TWO liquids: vapour + speciated aqueous + benzene-rich organic at 313.15 K / 0.35 atm – V/F = 0.324, organic 1.7 % of the liquid, pH 2.89

**The story:** One isothermal flash, four components, THREE phases:

feed (60 H2O + 20 benzene + 8 EtOH + 0.1 HAc, kmol/h) → [flash01, 313.15 K / 0.35 atm] → liquid (59.6 kmol/h: aqueous + organic) + vapor (28.5 kmol/h)

The two liquids leave as ONE apparent liquid stream. That is deliberate and it is the same contract the speciation already lives under: the split is INTERNAL state – reported in full, never persisted – exactly as the ions are. A stream file carries apparent components; it does not carry phases, and it does not carry species.

KPIs: V/F, pH, p_benzene_atm, p_ethanol_atm, p_eq_sum_atm (= P, two-phase), p_mono_atm / p_dimer_atm / dimer_share (the acetic vapour dimer).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0244722222222 |
| stream | liquid | F | 0.0165443840142 |
| stream | vapor | F | 0.007927838208 |
| kpi | flash01 | F_alpha | 0.0165443840142 |
| kpi | flash01 | F_beta | 0.007927838208 |
| kpi | flash01 | F_in | 0.0244722222222 |
| kpi | flash01 | P | 35463.75 |
| kpi | flash01 | Q | 86.8677617756 |
| kpi | flash01 | Q_kW | 0.0868677617756 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0.323952526093 |
| kpi | flash01 | dimer_share | 0.0232924752345 |
| kpi | flash01 | pH | 2.89048667096 |
| kpi | flash01 | p_benzene_atm | 0.232354234286 |

*... and 20 more reference values in the case's `expected`.*

## flash18_water_analysis_basis

tutorials/steady/flash/flash18_water_analysis_basis




**What it does:** A stream written in the AQUEOUS-SPECIES basis: a calcium-bicarbonate water analysis (Ca 0.0122, HCO3 0.0244 kmol/h) inverted onto the case's components by m = A n – the engine formulates the salts, the author does not

**The story:** Everything else in this corpus writes a stream in apparent components. This one writes it in the AQUEOUS SPECIES a laboratory actually measures, and lets the engine do the formulation:

speciesMolarFlows { network carbonate; basis analytical; Ca 0.0122; HCO3 0.0244; water 97.0; }

– m = A n, against the components' DECLARED bridges –

componentMolarFlows { CO2 0.0122; CaCO3 0.0122; water 97.0; }

WHY IT MATTERS. Converting an analysis into components by hand is a choice of LABELS, not a calculation: the same ions can be formulated as different component sets, and picking one silently is the arbitrariness the basis-rank test exists to refuse. Done here, the inversion is performed by the engine and REFUSED when it is not unique – the same refusal flash15_refused_salt_basis_rank raises, reached from the input side.

THREE REFUSALS GUARD IT, and check_stream_basis.py exercises all three: an analysis that does not close in CHARGE (a measurement to fix, never a residue for the solved pH to absorb); a 'network' naming a chemistry set this system does not declare; a missing 'basis' – because analytical and stoichiometric sets differ exactly in whether charge must close (H/OH are excluded mediators), so there is no safe default.

The answer is IDENTICAL to the same water written in components: the species basis is a change of coordinates, not a change of model. pH 7.757 – a natural calcium-bicarbonate water.

Contract: docs/design/aqueous-stream-basis-proposal.md.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0269512222222 |
| stream | liquid | F | 0.0269512222222 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.0269512222222 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.0269512222222 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | pH | 7.75731867771 |
| kpi | flash01 | p_eq_sum_atm | 0.0881612782912 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 313.15 |
| stream | liquid | T | 313.15 |

*... and 1 more reference values in the case's `expected`.*

## flash19_organic_and_precipitate

tutorials/steady/flash/flash19_organic_and_precipitate




**What it does:** FOUR phases at once: vapour + speciated aqueous + benzene-rich organic + solid calcite. flash16's feed plus benzene and ethanol, nothing else changed – the co-solvent moves the precipitation from 729 to 1096 mg/kg

*One isothermal flash at 313.15 K / 1 atm, and four phases coexist:*

*The three condensed phases leave through one bottoms nozzle and are named inside that stream's phases {} block; the vapour is its own stream.*

*Until it was written, the reactive path had been exercised with an organic liquid (flash17) and with a precipitate (flash16) — but never both at once. The file format supported it and the solver was believed to; no run proved it. This is that run, and the joint residual closes to = 6.6e-14.*

*The feed is flash16's, unchanged, plus benzene 2.0 and ethanol 1.0 kmol/h. Same water, same $CO_2$, same $CaCO_3$, same T, same P. So the two cases differ by exactly those two components:*

*Adding a co-solvent pushed half again as much calcite out of solution.*

*That number is not a mixed-solvent antisolvent calculation, and a student who reads it as one will be wrong about why.*

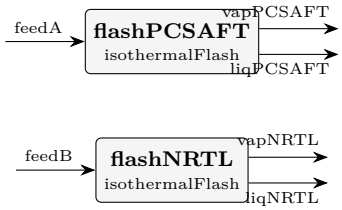
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0283416666667 |
| stream | liquid | F | 0.0275385746121 |
| stream | vapor | F | 0.000803092054611 |
| kpi | flash01 | F_alpha | 0.0275385746121 |
| kpi | flash01 | F_beta | 0.000803092054611 |
| kpi | flash01 | F_in | 0.0283416666667 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0.0283360913124 |
| kpi | flash01 | pH | 6.10723545836 |
| kpi | flash01 | p_benzene_atm | 0.238716176589 |
| kpi | flash01 | p_eq_sum_atm | 1 |
| kpi | flash01 | p_ethanol_atm | 0.011504063078 |
| kpi | flash01 | phaseSet | 0 |

*... and 13 more reference values in the case's `expected`.*

## flash20_ethanol_water_pcsaft

tutorials/steady/flash/flash20_ethanol_water_pcsaft




**What it does:** Ethanol/water flash: predictive PC-SAFT (assoc, kij=0) vs fitted NRTL

**The story:** flash20 – ethanol/water VLE, PREDICTIVE vs FITTED on the same state

The same 30/70 mol% ethanol/water feed flashed at 358.15 K / 1 atm through TWO thermodynamic worlds, side by side:

flashPCSAFT the GLOBAL world: phi-phi PC-SAFT with the ASSOCIATION term active on both components (2B water + 2B ethanol, Gross & Sadowski 2002 Table 1 – the paper lists water with TWO sites; the mixture witness of the association programme, validation item 4). The cross-association follows the Wolbach-Sandler combining rules and kij = 0 (announced) – so the MIXTURE behaviour is fully PREDICTED: nothing in this route was fitted to any ethanol/water datum. Only the PURE parameters are fits.

flashNRTL a per-unit thermo {} override: the classic gamma-phi route on the curated NRTL pair (fitted TO ethanol/water VLE data) + Antoine. This is the case's experimental anchor: the fitted route reproduces the measured azeotropic system, so the DIFFERENCE between the two units' K-values and splits is an honest measurement of what pure-component association physics alone can and cannot predict for a strongly hydrogen-bonded pair.

What the student compares, measured at the feed composition z (illustrative – what is goldened are the CONVERGED K's: PC-SAFT 11.47 / 0.581 vs NRTL 3.896 / 0.602 for ethanol / water):

predictive PC-SAFT fitted NRTL (gamma-phi) K_ethanol (at z) 2.185 2.238 K_water (at z) 0.769 0.695 V/F 0.649 0.512 x_ethanol / y_ethanol 0.038 / 0.441 0.121 / 0.471

The predictive route lands the K's within ~2 % (ethanol) and ~11 % (water) of the data-fitted route – from PURE-component parameters alone, a genuinely strong showing for a hydrogen-bonded pair – and the residual K_water error still moves V/F from 0.51 to 0.65: the pedagogical point, stated up front rather than hidden. Predictive means "no mixture data used", not "as accurate as a fit"; the fitted gamma-phi route stays the design tool, the predictive EoS is the screen for systems with NO data to fit.

This case is ALSO the witness that catches a curation error the pure witnesses cannot: with water mis-curated as 4C (the same eps/kappa, the wrong scheme) the pure density read +0.6 % – fine – while THIS flash collapsed to K_water = 0.0044. The scheme is part of the fit.

A model boundary note: the two units run DIFFERENT thermo worlds on the same global component set (the general heterogeneous-solver contract); each stream leaves in its own unit's world.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedA | F | 0.0277777777778 |
| stream | feedB | F | 0.0277777777778 |
| stream | liqNRTL | F | 0.0135537277127 |
| stream | liqPCSAFT | F | 0.00975103940319 |
| stream | vapNRTL | F | 0.014224050065 |
| stream | vapPCSAFT | F | 0.0180267383746 |
| kpi | flashNRTL | F_alpha | 0.0135537277127 |
| kpi | flashNRTL | F_beta | 0.014224050065 |
| kpi | flashNRTL | F_in | 0.0277777777778 |
| kpi | flashNRTL | K_ethanol | 3.89559152547 |
| kpi | flashNRTL | K_water | 0.602023174315 |
| kpi | flashNRTL | P | 101325 |
| kpi | flashNRTL | Q | 0 |
| kpi | flashNRTL | Q_kW | 0 |
| *… and 27 more reference values in the case's `expected`.* | | | |

## flash21_brine_co2_pitzer

tutorials/steady/flash/flash21_brine_co2_pitzer




**What it does:** NaCl brine + CO2 reactive flash under the pitzerHMW ionic rung – seam S5(b) witness: the declared model IS the served model.

**The story:** the S5(b) witness: a 2 mol/kg NaCl brine with dissolved CO2, flashed at 313.15 K / 1 atm, with the aqueous speciation priced by pitzerHMW (curated Na-Cl / Na-HCO3 / Na-CO3 / Na-OH / H-Cl pairs) instead of the charge-only Davies correlation that has no business at I = 2. The salt stays in the liquid by construction (nonvolatile, complete dissociation through its declared bridge); CO2 partitions against its ionised fraction exactly as in flash09's ammonia lesson – speciation and phase split solved TOGETHER, never in sequence.

WHAT THE MODEL CHOICE MOVES, measured at build time (2026-08-24) against an otherwise identical davies twin: the liquid pH (3.84 here vs 3.96 under davies – the charge-only bracket is out of its band at I = 2 and the Davies advisory says so on that run), while V/F barely moves (0.0211 vs 0.0210): the vapour is set by the NEUTRAL CO2aq, which Davies prices at exactly gamma = 1 and HMW at its real specific-interaction value.

A CAVEAT THIS CASE ANNOUNCES AND DOES NOT HIDE: the energy report tries to re-resolve every stream's own equilibrium, and the VAPOUR stream (93 % CO2, 7 % water, zero salt) read as an aqueous liquid has an implied molality near 770 mol/kg – outside any aqueous model's domain. pitzerHMW fails there loudly (singular Jacobian) and the report falls back to the carried state, announced in the caveat block. Davies returns a number at the same absurd state, which is not a virtue: a charge-only correlation cannot know it is out of its depth. The answer's own streams are unaffected.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.0271904240052 |
| stream | vapor | F | 0.000587353772572 |
| kpi | flash01 | F_alpha | 0.0271904240052 |
| kpi | flash01 | F_beta | 0.000587353772572 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | K_CO2 | 3299.5218183 |
| kpi | flash01 | K_NaCl | 0 |
| kpi | flash01 | K_water | 0.0696646141595 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 313.15 |
| kpi | flash01 | V_over_F | 0.0211447358126 |
| kpi | flash01 | pH | 3.83985178539 |
| kpi | flash01 | p_eq_sum_atm | 1 |

*... and 4 more reference values in the case's `expected`.*

## flash21_freeze_concentration

tutorials/steady/flash/flash21_freeze_concentration




**What it does:** Freeze-concentration SLE flash: ice crystallises from a water/ethanol liquor at -15 C; the liquor concentrates until gamma_w * x_w = exp(-dG_fus/RT)

*What this witnesses (migration S4b, 2026-08-09): the flash's LIQUID+SOLID branch. A C3 uniform phases ( ? ) declaration — one liquid, one crystallizing solid (component water;), no vapour — dispatches to the SLE solve, where the ice phase becomes a SolidCandidate on THE ONE solid-equilibrium mechanism (solidEq::equilibrate, ruling R1's active-set + simultaneous damped Newton). The candidate's lnSI is ln(x_w · fEff_liq / fEff_solid) and its remove pulls water from the liquid inventory — the one ledger.*

*The acceptance is a closed form, not a golden alone. At equilibrium the K = 1 condition collapses to the freezing-point-depression identity*

*gamma_w · x_w = exp(-dG_fus / (R T)), dG_fus = dHfus · (1 - T/Tfus)*

*with every quantity from the record's own data (water.dat Hfus 6008 J/mol, triple point 273.16 K). At 258.15 K the right side is 0.857434120855; the run publishes the liquor's activity as KPI a_water = 0.857434120931 — agreement to 9e-11. check_solid_service arm A10 recomputes the closed form independently in Python on every suite run.*

*The numbers: feed 90/10 water/ethanol (mol) at 25 °C, 1 atm; the freezer operates at -15 °C. Ice forms at solidFraction 0.476 mol per mol of feed — a function of the operating (T, P) alone, not of the feed temperature; the liquor concentrates to x_ethanol ≈ 0.19, exactly where the NRTL water activity meets the saturation value. The liquor and crystal leave through ONE outlet (liquor): its overall material equals the feed and the phases {} block decomposes it; vent is the structurally-required second outlet and carries F = 0 (no vapour phase is declared).*

*The duty (gap closed 2026-08-09 by docs/design/tp-stream-energy-coherence.md slice 1, R-E3/R-E4): h_formation gained its gas-natural → solid view (Hf - Hvap(298) - Hfus + ?cp_solid, from the record's own declared transition data), and the flash prices every crystallised mole on that solid formation rung — the SAME leg the balance report reads. The warm feed makes the duty visible as physics: Q ≈ -167.4 kW = sensible cooling of 100 kmol/h by 40 K (≈ -88 kW) + fusion of 47.6 kmol/h of ice (≈ -79 kW). Cooling is negative by the sign convention; gate arm A10 asserts the published Q is finite and negative.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquor | F | 0.0277777777778 |
| stream | vent | F | 0 |
| kpi | freezer01 | F_alpha | 0.0277777777778 |
| kpi | freezer01 | F_beta | 0 |
| kpi | freezer01 | F_in | 0.0277777777778 |
| kpi | freezer01 | P | 101325 |
| kpi | freezer01 | Q | -167415.991569 |
| kpi | freezer01 | Q_kW | -167.415991569 |
| kpi | freezer01 | T | 258.15 |
| kpi | freezer01 | V_over_F | 0 |
| kpi | freezer01 | aEq_water | 0.857434120855 |
| kpi | freezer01 | a_water | 0.857434120931 |
| kpi | freezer01 | phaseSet | 0 |

*... and 11 more reference values in the case's `expected`.*

## flash_polymer_devolatilisation

tutorials/steady/flash/flash_polymer_devolatilisation




**What it does:** Devolatilise a polystyrene melt: flash off residual toluene; the polymer (fixed-MW pseudo-component, role nonvolatile) stays liquid

**The story:** polymer melt + residual solvent → flash (devolatiliser).

The flash boils off the residual toluene (vapour) and leaves the polymer in the liquid. polystyrene is 'role nonvolatile;', so its K = 0: it stays 100 % in the liquid by construction (no Psat needed for a nonvolatile lump).

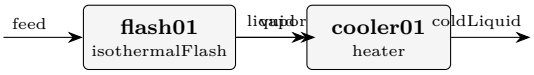
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | liquid | F | 0.00246249927126 |
| stream | meltFeed | F | 0.00277777777778 |
| stream | vapor | F | 0.000315278506519 |
| kpi | devol01 | F_alpha | 0.00246249927126 |
| kpi | devol01 | F_beta | 0.000315278506519 |
| kpi | devol01 | F_in | 0.00277777777778 |
| kpi | devol01 | K_polystyrene | 0 |
| kpi | devol01 | K_toluene | 24.2878386162 |
| kpi | devol01 | P | 20000 |
| kpi | devol01 | Q | 0 |
| kpi | devol01 | Q_kW | 0 |
| kpi | devol01 | T | 450 |
| kpi | devol01 | V_over_F | 0.113500262347 |
| kpi | devol01 | phaseSet | 0 |

*... and 3 more reference values in the case's `expected`.*

## flash_pseudoComponent_petCut

tutorials/steady/flash/flash_pseudoComponent_petCut




**What it does:** Flash of a petroleum pseudo-component (Riazi-Daubert lump) + toluene, with a heater closing the energy balance

**The story:** feed → isothermal flash → heater on the liquid.

The flash splits the feed into vapour + liquid (the pseudo-component participates in VLE through its critical{} + Ambrose-Walton Psat). The heater unit then REMOVES sensible heat from the liquid (Q -30 kW – a heater with negative duty is a cooler); energyBalance reports the implied duty, and it CLOSES because the lump carries gasIdeal + liquidPure Cp. No reaction → no formation datum needed (matching the lump's omitted standardThermochemistry).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | coldLiquid | F | 0.00758956122005 |
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.00758956122005 |
| stream | vapor | F | 0.0201882165577 |
| kpi | cooler01 | F | 0.00758956122005 |
| kpi | cooler01 | P | 100000 |
| kpi | cooler01 | Q | -30000 |
| kpi | cooler01 | Q_kW | -30 |
| kpi | cooler01 | Q_per_mol | -3952.79768226 |
| kpi | cooler01 | T_in | 420 |
| kpi | cooler01 | T_out | 405.894884539 |
| kpi | cooler01 | dT | -14.1051154607 |
| kpi | cooler01 | vf_out | 0 |
| kpi | flash01 | F_alpha | 0.00758956122005 |

*. . . and 16 more reference values in the case's `expected`.*

## leaf01_flash

tutorials/steady/flash/leaf01_flash

**What it does:** Leaf-node form: isothermal flash benzene/toluene (sectored-layout base case)

**The story:** / flowsheetDict (LEAF NODE – sectored-layout base case)

A leaf carries ‘type‘ + ‘operation‘ (the unit) + ‘boundary‘ (its interface) + ‘streams‘ (the default feeds, so it runs isolated). No ‘units (...)‘ list. Compare flash01_benzene_toluene, which writes the SAME flash as a one-element units list – identical physics, two roupas.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.019333789635 |
| stream | vapor | F | 0.00844398814279 |
| kpi | flash | F_alpha | 0.019333789635 |
| kpi | flash | F_beta | 0.00844398814279 |
| kpi | flash | F_in | 0.0277777777778 |
| kpi | flash | P | 100000 |
| kpi | flash | T | 370 |
| kpi | flash | V_over_F | 0.30398357314 |
| kpi | flash | phaseSet | 0 |
| stream | feed | T | 370 |
| stream | liquid | T | 370 |
| stream | vapor | T | 370 |
| kpi | flash | K_benzene | 1.65136809583 |

*... and 6 more reference values in the case's* `expected`.

## marcilla01_lls_tie_triangle

tutorials/steady/flash/marcilla01_lls_tie_triangle




**What it does:** LLS anchor contact vs Marcilla 1995 (Table 10A row 4 charge): brine + solid halite reproduced; the organic vertex is NOT – the corpus UNIFAC (VLE table) keeps water/1-butanol miscible, a named model finding, never tuned

**The story:** THE FLAGSHIP LLS WITNESS (solid-equilibrium migration S5).

One isothermal flash at the paper's own conditions: 25.0 C, 1 atm. The feed is Marcilla, Ruiz & Olaya's Table 10A initial mixture, row 4 – a published experimental charge, not an invented one – and the anchor is the tie-triangle that charge produced (Table 10B row 4): an aqueous brine, a butanol-rich organic liquid, and solid NaCl, all simultaneously at equilibrium. PRIMARY: A. Marcilla, F. Ruiz, M. M. Olaya, Fluid Phase Equilibria 105 (1995) 71-91, DOI 10.1016/0378-3812(94)02595-R.

What the engine must do at once here, through the ONE mechanism each: dissolve NaCl onto the Na/Cl master families (the typed dissociatesTo bridge), split the molecular backbone into two liquids by equality of activity, and hold the halite complementarity through the solid- equilibrium service while that split resolves. No unit below declares any of that chemistry – the case does (thermoPhysPropDict + chemistryDict), and the flash reads it.

WHAT THIS CASE CURRENTLY WITNESSES, honestly (2026-08-10): the brine + solid halite half of the anchor, and a MODEL FINDING on the other half. The corpus' UNIFAC is the VLE table (Hansen 1991), and under it the water/1-butanol pair is fully miscible at 25 C – g_mix stays convex (independently confirmed by a propertyScanBinary sweep), so the Gibbs decision correctly reports ONE liquid where the paper measures two. The organic vertex of the tie-triangle is therefore NOT reproduced, and no number here was tuned to change that. The named remedy is curation: a UNIFAC-LLE parameter set (Magnussen, Rasmussen & Fredenslund, Ind. Eng. Chem. Process Des. Dev. 20 (1981) 331 – the parameterisation created precisely because the VLE table misses binodals), or fitted NRTL pairs for water/1-butanol and ethanol/1-butanol. See README.md.

*PRIMARY: A. Marcilla, F. Ru?z, M. M. Olaya, ∗Fluid Phase Equilibria∗ 105 (1995) 71–91, DOI 10.1016/0378-3812(94)02595-R. Water–ethanol–1-butanol–NaCl at 25.0 ± 0.1 °C. Anchor values staged in [docs/design/solid-migration-witness-data.md](../../../../docs/design/solid-migration-witness-data.md) ?2 (reviewStatus interim; this case cites the primary).*

*The feed is the paper's own Table 10A row 4 initial mixture — a published experimental charge, not an invented one (wt%):*

*That charge produced the paper's Table 10B row 4 tie-triangle — three phases simultaneously at equilibrium (wt%):*

*(The lever-rule check on the paper's own numbers closes to $\Sigma\lambda = 1.010$.)*

*Davies ions on water-referenced molality + UNIFAC molecular backbone (the corpus table — the VLE parameterisation of Hansen et al. 1991) + halite complementarity through the solid-equilibrium service + a declared WET organic (water is a member; its cross-liquid equality carries the ionic a_w factor).*

*Reproduced: the saturated brine + solid halite coexistence. The run converges with solid NaCl present and a single fluid liquid; Davies is announced out of its validity band ($I \approx 3.6$ mol/kg against a ~0.5 band), so the predicted saturation is ∗indicative∗ — the model holds ~3.6 mol/kg against Pinho & Macedo's measured 6.14, and that gap is Davies', reported.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000849885277778 |
| stream | liquid | F | 0.000849885277778 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.000849885277778 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.000849885277778 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 298.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | pH | 7.35672575928 |
| kpi | flash01 | p_eq_sum_atm | 0.0415604200493 |
| kpi | flash01 | p_ethanol_atm | 0.0111811211227 |
| kpi | flash01 | p_nButanol_atm | 0.00389708966162 |
| kpi | flash01 | phaseSet | 0 |
| *… and 3 more reference values in the case's `expected`.* | | | |

## marcilla02_lls_ternary_invariant

tutorials/steady/flash/marcilla02_lls_ternary_invariant




**What it does:** The ternary LLS invariant vs Marcilla 1995 Table 4B: aqueous brine + WET butanol-rich organic + solid halite, all three at once – the first PRESENT wet-organic witness (UNIQUAC backbone, Winkelman pair)

**The story:** THE TERNARY LLS INVARIANT (solid-equilibrium migration S5b).

Water + 1-butanol + NaCl at 25.0 C, 1 atm: with solid NaCl present the two-liquid tie-line is INVARIANT (F = 0 at fixed T, P), so the paper reports ONE tie-triangle for the whole three-phase region – Table 4B: aqueous 73.14 / 0.72 / 26.14 (W / B / NaCl, wt%) organic 7.53 / 92.28 / 0.19 solid NaCl PRIMARY: A. Marcilla, F. Ruiz, M. M. Olaya, Fluid Phase Equilibria 105 (1995) 71-91, DOI 10.1016/0378-3812(94)02595-R.

This is the state marcilla01 could not reach: there the UNIFAC (VLE) backbone kept water/1-butanol miscible. Here the backbone is UNIQUAC on the CITED Winkelman 2009 water/1-butanol set – the same values vlle01 consumes inline, where they demonstrably open the gap – so the wet organic (water a declared member) is PRESENT, and all three phases coexist through their one mechanism each: the halite complementarity, the activity-equality split with the ionic a_w factor on water, and the Davies speciation of the brine.

*PRIMARY: A. Marcilla, F. Ru?z, M. M. Olaya, ∗Fluid Phase Equilibria∗ 105 (1995) 71–91, DOI 10.1016/0378-3812(94)02595-R, Table 4B — water + 1-butanol + NaCl at 25.0 ± 0.1 °C. With solid NaCl present the two-liquid tie-line is invariant, so one tie-triangle describes the whole three-phase region. Anchors staged in docs/design/solid-migration-witness-data.md ?2a (interim).*

*This is the state marcilla01 could not reach (its UNIFAC-VLE backbone keeps water/1-butanol miscible). The backbone here is UNIQUAC on the cited Winkelman 2009 water/1-butanol set (case-local record — the same values vlle01_waterButanol consumes inline, where they demonstrably open the gap). The engine's first PRESENT wet organic: all three phases coexist, each through its one mechanism — halite complementarity, activity-equality split with the ionic a_w factor on water, Davies speciation of the brine.*

*Every deviation is a NAMED model approximation, announced by the run:*

*∗ Aqueous NaCl low (16.2 vs 26.1 wt%): Davies saturates halite at 3.65 mol/kg against the measured 6.14 — the same number marcilla01 measured, seven times outside Davies' band, announced. ∗ Aqueous butanol high (7.7 vs 0.72 wt%): Davies prices neutral solutes at $\gamma = 1$, so the brine's salting-out of butanol — an order of magnitude in the anchor — is invisible to the ionic leg. ∗ Organic too wet (13.6 vs 7.53 wt% water): Davies' $\varphi = 1$ water activity ($a_w \approx 0.88$ vs the real $\approx 0.75$ over saturated brine) under-prices the salt's pull on the water, so too much crosses.*

*The internal identities the gate pins: the cross-liquid equality residual including water's a_w factor is at machine precision ( ~ 1e-15 per member), and the dissolved salt over the AQUEOUS water alone reproduces the Davies saturation molality — the conversion-basis bug this case caught (a total-water basis overstated the crystal by exactly the organic water's share, while every mass balance still closed: a wrong basis conserves mass perfectly).*

*Energy balance: UNAVAILABLE (the apparent salt has no elements-datum route in this world — named diagnostic; mass and element balances close).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000820413055556 |
| stream | liquid | F | 0.000820413055556 |
| stream | vapor | F | 0 |
| kpi | flash01 | F_alpha | 0.000820413055556 |
| kpi | flash01 | F_beta | 0 |
| kpi | flash01 | F_in | 0.000820413055556 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 298.15 |
| kpi | flash01 | V_over_F | 0 |
| kpi | flash01 | pH | 7.35672525831 |
| kpi | flash01 | p_eq_sum_atm | 0.032687682758 |
| kpi | flash01 | p_nButanol_atm | 0.00579877206855 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 298.15 |
| *... and 2 more reference values in the case's `expected`.* | | | |

## vlle01_waterButanol

tutorials/steady/flash/vlle01_waterButanol




**What it does:** Liquid-liquid flash of water + 1-butanol at 320 K, 1.013 bar – the real miscibility gap from the cited Winkelman et al. (2009) UNIQUAC LLE fit: water-rich ~98.3 mol% water, butanol-rich ~46 mol% butanol (their Figs 1-2). See README.

**The story:** liquid-liquid flash of water + 1-butanol with the cited Winkelman et al. (2009) UNIQUAC LLE parameters.

Water + 1-butanol is THE textbook partially-miscible binary. At 320 K and 1.013 bar an 80/20 water/butanol mixture is well below the bubble point (water boils at 373 K, 1-butanol near 390 K), so the equilibrium is GENUINELY two LIQUIDS – no vapour. A simple 'phaseSet LL' flash (Michelsen TPD pre-check, then direct Gibbs-energy minimisation, multi-start Nelder-Mead) finds the split.

THE PARAMETERS (cited, NOT fabricated – see constant/thermoPhysPropDict and README.md): the UNIQUAC interaction parameters are the LLE-specific fit of

J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, Fluid Phase Equilibria 284 (2009) 71-79, Table 2 (p.73); structural r,q from their Table 1 (p.72).

with the full temperature dependence A_ij(T) = a_ij + b_ij∗T + c_ij∗T^2 (their Eq. 10), tau_ij = exp(-A_ij/T) (their Eq. 7), valid 273-363 K. The numbers are inline here; the record is STAGED CASE-LOCALLY (reviewStatus interim) under marcilla02's constant/parameters/UNIQUAC/, NOT curated into data/standards/ – promotion is the maintainer's batch review.

a REAL liquid-liquid miscibility gap, reproduced from a cited primary source – the water-rich phase is ~98.3 mol% water (~1.7 mol% butanol) and the butanol-rich phase is ~54 mol% water (~46 mol% butanol), matching Winkelman's Figs 1-2; the IsothermalFlash LL branch: a Michelsen TPD stability test that flags the feed as unstable, then a direct Gibbs-minimisation that resolves the two coexisting compositions; honest phase labelling – the two output streams are named by their actual composition (waterRich / butanolRich), with alphaRich/betaRich hints steering which liquid lands in which slot.

HISTORICAL NOTE: an earlier version of this case carried only a DECHEMA VLE-fit NRTL set, which (being a VLE fit) has NO miscibility gap, so the flash found a single liquid – it was kept as an honest "VLE-fit-misses-LLE" lesson pending a cited LLE set. Winkelman et al. (2009) IS that cited set, so the case now shows the genuine split. (The VLE-blind NRTL lesson lives on in the NRTL catalogue file's provenance.)

*Water + 1-butanol (n-butanol) is ∗the∗ textbook partially-miscible binary. Below the bubble point it splits into two coexisting liquids: a water-rich phase carrying only a couple of mol % butanol, and a butanol-rich phase holding roughly half water. So a flash of a feed ∗inside the miscibility gap∗ should give two liquids — and here it does.*

*At 320 K, 1.013 bar (well below the bubble point, so no vapour) a feed of 80 mol % water / 20 mol % butanol splits into:*

*with split fraction $\beta \approx 0.41$ into the butanol-rich phase. The solver reports the Michelsen TPD instability (tm_min ≈ -0.45, ∗unstable = YES∗) and then a direct Gibbs-energy minimisation that lands the two compositions.*

*An independent re-implementation of the UNIQUAC $\gamma$ (fresh code, isoactivity Newton) reproduces the engine's split to 4 decimals across 298–350 K, confirming the parameters are entered correctly. The agreement with the published figures is essentially exact — the UNIQUAC fit is doing its job.*

*constant/thermoPhysPropDict carries the LLE-specific UNIQUAC for water/1-butanol, cited per value:*

*- Primary source: J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, ∗"Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"∗, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013. - Binary interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation A_ij(T) = a_ij +*

*b_ij·T + c_ij·T² (their Eq. 10) and τ_ij = exp(-A_ij/T) (their Eq. 7): - A_1,3 (butanol→water): a = 155.31, b = 1.0822, 10?·c = -43.711 - A_3,1 (water→butanol): a = -579.36, b = 2.7517, 10?·c = -6.7700 - Structural r, q — their Table 1 (p.72) (value Magnussen et al.): 1-butanol r = 3.9243, q = 3.668; water r = 0.9200, q = 1.400.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | butanolRich | F | 0.0114525507776 |
| stream | feed | F | 0.0277777777778 |
| stream | waterRich | F | 0.0163252270002 |
| kpi | decanter01 | F_alpha | 0.0163252270002 |
| kpi | decanter01 | F_beta | 0.0114525507776 |
| kpi | decanter01 | F_in | 0.0277777777778 |
| kpi | decanter01 | P | 101300 |
| kpi | decanter01 | T | 320 |
| kpi | decanter01 | betaFraction | 0.412291827994 |
| kpi | decanter01 | phaseSet | 1 |
| stream | butanolRich | T | 320 |
| stream | feed | T | 320 |
| stream | waterRich | T | 320 |
| boundary | global | H_feeds_kW | -8114.08370112 |

*… and 2 more reference values in the case's `expected`.*

## vlle03_audit_artificial

tutorials/steady/flash/vlle03_audit_artificial




**What it does:** Algorithm-audit synthetic VLLE: 3 components with identical Psat and exaggerated NRTL. Should converge to 3-phase by construction.

**The story:** algorithm-correctness audit for the VLLE Gibbs-min branch.

Three synthetic components A, B, C with identical Antoine (Psat ≈ 0.5 bar at 350 K). NRTL parameters chosen to GUARANTEE three-phase coexistence at T = 350 K, P = 1 bar:

A-B: $\tau$_AB = $\tau$_BA = 5 (very strong, near-immiscible) A-C: $\tau$_AC = $\tau$_CA = 5 (very strong, near-immiscible) B-C: $\tau$_BC = $\tau$_CB = 0 (ideal, fully miscible)

Expected behaviour by symmetry and construction: one liquid phase rich in A (the "isolated" component); another liquid phase rich in B+C (they pair together); a vapour phase carrying some of each (Psat is equal so all are equally volatile, ln(P/Psat) ≈ +0.301 at 1 bar).

Equimolar feed (1/3, 1/3, 1/3). This case is the litmus test for the VLLE solver: if it cannot find 3-phase here, the bug is in the algorithm, not in any real-system NRTL parameter set.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | liquidA | F | 0.011307571384 |
| stream | feed | F | 0.0277777777778 |
| stream | liquidB | F | 0.0113091924076 |
| stream | vapor | F | 0.00516101398624 |
| kpi | decanter01 | F_alpha | 0.011307571384 |
| kpi | decanter01 | F_beta | 0.0113091924076 |
| kpi | decanter01 | F_in | 0.0277777777778 |
| kpi | decanter01 | F_vapor | 0.00516101398624 |
| kpi | decanter01 | P | 100000 |
| kpi | decanter01 | T | 350 |
| kpi | decanter01 | V_over_F | 0.185796503505 |
| kpi | decanter01 | beta_vapor | 0.185796503505 |
| kpi | decanter01 | phaseSet | 2 |
| stream | feed | T | 350 |

*. . . and 11 more reference values in the case's* `expected`.

## vlle04_two_liquids_decanter

tutorials/steady/flash/vlle04_two_liquids_decanter




**What it does:** Structural witness: a phaseSet VLLE flash whose search finds TWO LIQUIDS and no vapour (ethanol + cyclohexane at 340 K, 3 bar, with Choupo's own case-local regressed NRTL pair, which predicts a liquid split its measurements do not show). The second liquid must leave on liquidBeta, never on the vapour port. Not a validated LLE.

**The story:** a 'phaseSet VLLE' flash whose three-phase search finds NO vapour and falls back to TWO LIQUIDS.

A STRUCTURAL WITNESS, NOT A VALIDATED LLE. Read this before reading any number it prints.

WHAT IT PINS. A VLLE flash always emits three ports (vapour, liquid alpha, liquid beta); when the Gibbs minimisation finds the vapour fraction below 1 %, the solver demotes the answer to a liquid-liquid split. Until 2026-10-05 the unit decided WHICH two-phase fallback it had by searching the regime's human-readable string for "VL" – and the LL fallback's own string reads "two-phase liquid (LL, VLLE attempt found V beta ~ 0)", whose "VLLE" contains "VL". So the beta LIQUID left on the VAPOUR port with vf = 1, the second liquid port carried zero, and the energy report booked a latent heat that never happened (498.03 kW on this very drum). The decision now reads a typed field of the flash solution ('liquidLiquid'), and this case is the witness that it does:

the 'vapour' port carries ZERO flow; 'liquidAlpha' and 'liquidBeta' both carry flow, both as liquids; the unit speaks the LL vocabulary (betaFraction, no V_over_F, no K-values, no duty – an LL split's liquid enthalpies carry no excess-of-mixing term, the same posture as a 'phaseSet LL' flash); the plant first law closes (no phantom latent heat).

Gate: bin/curate/check_vlle_fallback.py. Record: DEV.md section 5, the entry dated 2026-10-05.

WHY THESE NUMBERS ARE NOT PHYSICS. The NRTL pair is Choupo's OWN regression to measured BUBBLE TEMPERATURES of ethanol + cyclohexane (the case tutorials/props/curation/curate12_nrtl_ethanol_cyclohexane, whose record is copied UNCHANGED into constant/parameters/NRTL/ here). It passed its held-out bubble-temperature test and was deliberately NOT promoted to the catalogue, because it predicts a liquid-liquid split up to about 358.8 K while the measurements it was fitted to show ONE liquid. That spurious split is exactly what this witness needs – a VLLE flash that ends with two liquids and no vapour – and exactly why no composition it prints may be quoted as the behaviour of real ethanol + cyclohexane. The run's own T-x-y diagnostic at 3 bar boils every composition at or above about 370 K, thirty kelvin above the drum, so the ABSENCE of vapour is the model's honest answer at these conditions; the SECOND LIQUID is what its own evidence does not support.

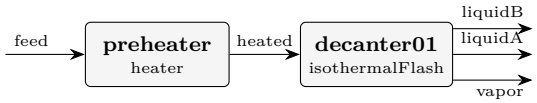
The copied record names 'fittedInCase', so check_regressed_pairs re-runs curate12 and holds this copy to the fit: it cannot drift.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 340 |
| stream | liquidAlpha | F | 0.0143891231219 |
| stream | liquidAlpha | T | 340 |
| stream | liquidBeta | F | 0.0133886546559 |
| stream | liquidBeta | T | 340 |
| stream | vapour | F | 0 |
| stream | vapour | T | 340 |
| kpi | decanter01 | F_alpha | 0.0143891231219 |
| kpi | decanter01 | F_beta | 0.0133886546559 |
| kpi | decanter01 | F_in | 0.0277777777778 |
| kpi | decanter01 | P | 300000 |
| kpi | decanter01 | T | 340 |
| kpi | decanter01 | betaFraction | 0.481991567611 |
| *… and 5 more reference values in the case's `expected`.* | | | |

## vlle05_produced_feed

tutorials/steady/flash/vlle05_produced_feed




**What it does:** Synthetic VLLE decanter fed by a heater: a produced stream keeps its producer's vapour-liquid answer (DEV.md C37 item 4).

**The story:** a 'phaseSet VLLE' decanter whose feed ANOTHER unit produced: whose search does that stream mean?

A STRUCTURAL WITNESS on vlle03's synthetic system (compA, compB, compC: not real substances). Its three component records are ADOPTED copies that carry an INVENTED formation datum (dHf_298 = 0, s_298 = 200 for all three) so that the energy report runs; only differences of it are read. No number here is a property of any real mixture.

THE QUESTION. An unpinned (T, P, z) means its own equilibrium (R-E2), and for a feed to a unit that admits a vapour and TWO liquids that equilibrium is the three-phase one – which is why, since DEV.md C35 item 4, a stream fed to a VLLE flash is resolved with the flash's own search by the energy report and the stream table. But a PRODUCED stream keeps its PRODUCER's answer (CLAUDE.md ?3). 'preheater' is a heater: it solves for the outlet whose enthalpy is H_in + Q on the VAPOUR-LIQUID surface, and publishes 'heated' at 352.33 K with vf = 0.0089 – one liquid and a whisker of vapour. Under the decanter's three-phase search the same (T, P, z) is two liquids and a fifth of vapour.

WHAT WENT WRONG (measured 2026-10-05, before C37 item 4). The report re-read 'heated' with the decanter's search while the heater had priced it vapour-liquid: the preheater's row left 193.92 kW unattributed and the plant's first law missed by -193.92 kW, at exit 0.

THE RULE (DEV.md C37 item 4). A stream means its CONSUMER's search only when it is an AUTHORED inlet or a three-phase search itself published it ('flashState::consumerSearchApplies'). 'heated' is a heater's outlet, so the report, the stream table and the decanter's own duty all read the state the heater published; the demixing it undergoes in the drum is the decanter's duty (169.8 kW at 350 K), and every row closes.

What to read: the energy balance by unit (both rows close) and the plant first law; the decanter's 'Q_kW', which is the cost of bringing the heater's vapour-liquid outlet to the drum's three-phase state. Gate: check_vlle_fallback arm (f).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 340 |
| stream | heated | F | 0.0277777777778 |
| stream | heated | T | 352.326629907 |
| stream | liquidA | F | 0.011307571384 |
| stream | liquidA | T | 350 |
| stream | liquidB | F | 0.0113091924076 |
| stream | liquidB | T | 350 |
| stream | vapor | F | 0.00516101398624 |
| stream | vapor | T | 350 |
| kpi | decanter01 | F_alpha | 0.011307571384 |
| kpi | decanter01 | F_beta | 0.0113091924076 |
| kpi | decanter01 | F_in | 0.0277777777778 |
| kpi | decanter01 | F_vapor | 0.00516101398624 |

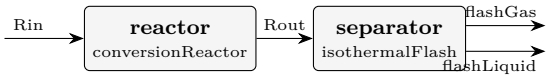
*... and 33 more reference values in the case's `expected`.*

## flowsheets

**The theory you need.** Recycle turns a flowsheet into a fixed-point problem: guess the tear stream, march the units, compare. Successive substitution converges linearly (and diverges past unit gain); Wegstein accelerates it; Newton-on-tears solves the tear residuals with a Jacobian. Choupo announces the tear choice, every iteration's residual, and any bound that binds – no silent crutch. Expect: purge streams controlling inert build-up (close the loop with no purge and watch the accumulation), and convergence behaviour that CHANGES with the tear location. *Full derivation: Theory Guide, the flowsheet-convergence chapter.*

## acetone03_luyben_reaction_section

tutorials/steady/flowsheets/acetone03_luyben_reaction_section




**What it does:** Luyben's acetone REACTION SECTION: reactor → cooler → flash, against his Rout and the flash split

**The story:** Three units of Luyben's Figure 1 wired together – reactor, cooler, flash – and the first piece of the plant whose answer DOES depend on the mixture thermodynamics. acetone02 isolated the reaction; this adds the phase split that sends hydrogen and acetone one way and the aqueous liquid the other.

W. L. Luyben, Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. Feed Rin as derived in acetone02 (57.82 kmol/h, IPA 0.668 / water 0.332, 389 K, 2.6 atm).

WHAT IS INDEPENDENT HERE, and it is the point. acetone02's outlet composition agreed with the paper because the feed had been DERIVED from that same outlet – a closed loop, and its golden says so. The flash split is not: nothing in 0/Rin was built from it. Luyben's Figure 1 gives

flash vapour 39.76 kmol/h H2 0.8742 A 0.0634 IPA 0.0002 W 0.0622 flash liquid 52.84 kmol/h (92.6 - 39.76; his F1 of 72.85 kmol/h is this stream AFTER the absorber bottoms of 20.05 kmol/h rejoin it)

and those are a real test of the thermodynamics this case declares.

THE THERMODYNAMICS IS DECLARED AND WEAK, said before the numbers rather than after. Luyben runs UNIQUAC; Choupo ships no UNIQUAC pair for any of these binaries, so this case runs PREDICTIVE UNIFAC on a lake-estimated isopropanol. The sibling props case measures what that costs on the same binary: the IPA boiling point is 4.5 K high and the IPA/water azeotrope lands at x 0.590 against a published 0.673. A separation built on that will not reproduce a separation built on fitted pairs, and the interesting question is HOW MUCH, not WHETHER.

Plus the one declared approximation, in constant/thermoPhysPropDict: H2 runs outside the activity model at gamma = 1, authorised by name, recorded in the answer.

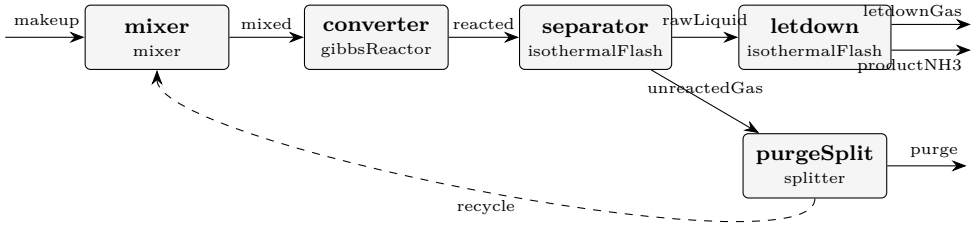
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Rin | F | 0.0160611111111 |
| stream | Rout | F | 0.0257186111111 |
| stream | flashGas | F | 0.0123053585397 |
| stream | flashLiquid | F | 0.0134132525714 |
| kpi | reactor | F_in_kmol_h | 57.82 |
| kpi | reactor | F_out_kmol_h | 92.587 |
| kpi | reactor | Q_kW | 840.263234652 |
| kpi | reactor | T | 623 |
| kpi | reactor | conversion | 0.9 |
| kpi | reactor | dHrxn_kJ_per_mol | 48.7173210118 |
| kpi | reactor | extent_kmol_h | 34.767 |
| kpi | separator | F_alpha | 0.0134132525714 |
| kpi | separator | F_beta | 0.0123053585397 |
| kpi | separator | F_in | 0.0257186111111 |

*... and 28 more reference values in the case's `expected`.*

## ammonia01_synthesis_loop

tutorials/steady/flowsheets/ammonia01_synthesis_loop




**What it does:** Complete ammonia synthesis loop (Haber-Bosch) with recycle and purge: makeup 1:3 N2:H2 (+ Ar inert) → mix with recycle → SRK Gibbs converter at 700 K / 200 bar (real-gas fugacity) → chill-flash at 250 K to condense the NH3 product → split the unreacted gas into a small Ar-bleeding PURGE and the recycle. The recycle tear + purge are the whole point: high overall N2/H2 utilisation despite a modest per-pass conversion, with the inert Ar held at steady state by the purge.

**The story:** the classic Haber-Bosch loop, recycle + purge

makeup (N2:H2 = 1:3 + trace Ar) [mixer] recycle [converter] gibbsReactor (SRK) N2 + 3H2 <→ 2NH3, 700 K, 200 bar (fugacity-corrected equilibrium ~30% NH3) [separator] isothermalFlash 250 K, 200 bar (chill: NH3 condenses out) rawLiquid unreactedGas [purgeSplit] splitter [letdown] flash 20 bar purge recycle (bleeds Ar) (~95% back to the mixer) productNH3 letdownGas (97.8% NH3) (H2-rich)

THE LOOP. Per-pass conversion is equilibrium-limited (~30% NH3 at the reactor outlet), but the RECYCLE lifts overall N2/H2 utilisation high – the reason the loop exists. The reactor equilibrium AND the separator both use SRK FUGACITY (real gas at 200 bar): switch the package to idealGas and the conversion falls. Mass closes to 100% on N, H and Ar.

THE PURGE. Ar enters with the makeup and would accumulate without end; the purge is what makes the loop finite, and Ar reaches steady state (Ar_in_makeup = Ar_out, closing at 100.00 %). But the purge is NOT the exit that carries most of it out. Measured from this case's own converged/ streams: of 0.01 kmol/h of Ar in, the purge takes 16.2 %, the let-down gas 9.6 %, and 74.1 % leaves DISSOLVED IN THE LIQUID AMMONIA PRODUCT – argon is 1.56 % of that product. This paragraph used to say the purge bleeds it, which the diagram makes easy to believe and the streams do not support. Delete the purge and the recycle still fills with Ar until the loop chokes – try it.

THE LET-DOWN. At 200 bar the separator leaves N2/H2 DISSOLVED in the liquid (a real high-pressure effect, over-stated here because there is no N2/H2-NH3 binary interaction parameter – SRK over-predicts light-gas solubility). Measured from converged/: the raw separator liquid is 97.676 % NH3 (1.752 % Ar, 0.385 % H2, 0.187 % N2), and the let-down drum (flash to 20 bar) boils the dissolved gas off to give 98.390 % NH3 – a gain of 0.7 percentage points, almost all of it H2 and N2. These two numbers read '~80%' and '97.8%' until 2026-09-24; the first was wrong by 17 points and made the let-down look like the main separation. What it does NOT remove is the argon, which stays at 1.56 % of the product.

HONEST SIMPLIFICATIONS. (1) The loop is isobaric at 200 bar; the makeup compression and the feed/effluent heat integration are folded into the unit duties (reported), not drawn as separate machines. (2) The let-down gas (letdownGas, H2-rich) is RECOMPRESSED AND RECYCLED in a real plant; here it exits as a secondary stream to keep ONE clean recycle tear – so the H2 loss reads higher than industrial. Recovering it (a recompressor + a second tear) is the efficiency refinement. The topology keeps the loop's ESSENCE: reaction, separation, recycle, purge, product finishing.

*The Haber-Bosch loop, five units: a mixer, a gibbsReactor converter at 700 K and 200 bar, an isothermalFlash separator chilled to 250 K at 200 bar, a let-down drum flashing the separator's liquid to 20 bar at the same 250 K, and a splitter that sends 5 % of the unreacted gas to purge and 95 % back to the mixer. The makeup is 1 kmol/h — N2 0.2475, H2 0.7425, Ar 0.01 kmol/h — a 1:3 stoichiometric ratio with a trace of inert argon.*

*Every case before this one could be solved by running its units in declared order. This one cannot: recycle leaves the splitter and re-enters the mixer, so the mixer's inlet depends on an answer that is four units downstream of it.*

*The golden: the converter reaches y_NH3 = 0.3108 at equilibrium (y_H2 0.4998, y_N2 0.1632, y_Ar 0.0262); the separator condenses V/F = 0.6931 with K_NH3 = 0.01626; the let-down boils off V/F = 0.008021 with K_NH3 = 0.09469; the mixer leaves at 274.38 K.*

*1. The cut is declared, and the seed is yours. system/solverDict says tearStreams ( recycle ); with recycleSolver Newton;, recycleTol 1e-5;, recycleMaxIter 200; — the topology stays in flowsheetDict and the numerical strategy sits here. 0/recycle carries the starting guess, and its own header says in so many words that it is an estimate and not a specification: the solver converges it and announces what it did. Nothing about closing this loop happens behind your back. 2. The recycle is visible in the arithmetic of the pinned numbers. One kmol/h of makeup enters, and the converter sees 2.02566484668 kmol/h. The difference is the recycle, and it is what turns a per-pass equilibrium of about 31 % ammonia into a loop that recovers far more of the makeup than any single pass could. The reason the loop exists is a subtraction you can do in your head. 3. The converter is told elements, not a conversion. gibbsReactor takes elements ( N H Ar ) and a species list with their atom counts, and minimises Gibbs energy: no rate constant, no declared extent, no conversion specification anywhere. What comes back is y_NH3 0.310807360284 plus the element potentials lambda_N, lambda_H, lambda_Ar — the Lagrange multipliers of the atom balances, printed because they are the answer's own arithmetic. 4. Two flashes, one temperature, two pressures. The separator at 250 K and 200 bar condenses ammonia out of the loop gas (K_NH3 0.01626); the let-down drum holds the same 250 K at 20 bar and boils dissolved light gas back out of that liquid (K_NH3 0.09469, V/F 0.008021). The liquid world is declared as a dilute solution — ammonia as the solvent on the pure-liquid rung, N2 and H2 as Henry solutes on the infinite-dilution rung with their own pair files in constant/parameters/Henry/ — while the vapour is fugacityModel SRK;, a real gas at 200 bar. Read constant/thermoPhysPropDict: one stream, two reference rungs in its liquid, and both of them declared. 5. The purge is a splitter and nothing else, and it is what makes the loop finite. Argon enters with the makeup at 0.01 kmol/h, takes part in no reaction, and would otherwise circulate for ever. fractions ( 0.05 0.95 ) is the entire mechanism. Then do the balance yourself: add the argon on every stream that leaves the plant boundary — purge, letdownGas and productNH3 — and check it against the 0.01 kmol/h that came in. It closes, and the exit carrying most of it may not be the one you would have guessed from the diagram.*

*Change the splitter to fractions ( 0.02 0.98 ); — a smaller purge. The recycle grows, the argon in the loop climbs, and the converter's y_Ar rises at the expense of everything it is diluting. Push it far enough and the loop chokes on its own inert, which is the whole argument for spending reactants on a purge. Then open tutorials/plant/acetonePlant, the last case on this path: a published industrial process, closed, with a recycle, and a written account of where its errors come from.*

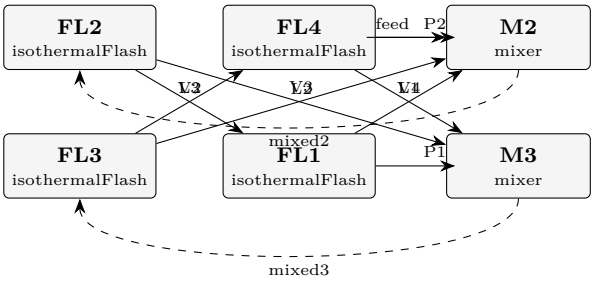
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | letdownGas | F | 1.06533052494e-06 |
| stream | makeup | F | 0.000277777777778 |
| stream | mixed | F | 0.000562684679632 |
| stream | productNH3 | F | 0.000131750445273 |
| stream | purge | F | 1.49951001055e-05 |
| stream | rawLiquid | F | 0.000132815775798 |
| stream | reacted | F | 0.000432717777908 |
| stream | recycle | F | 0.000284906902004 |
| stream | unreactedGas | F | 0.000299902002109 |
| kpi | converter | F_in_kmol_h | 2.02566484668 |
| kpi | converter | F_out_kmol_h | 1.55778400047 |
| kpi | converter | N_in_mol_s | 0.562684679632 |
| kpi | converter | N_out_mol_s | 0.432717777908 |
| kpi | converter | P | 20000000 |

*... and 65 more reference values in the case's `expected`.*

## cavett01_recycle_train

tutorials/steady/flowsheets/cavett01_recycle_train




**What it does:** Cavett (1963) four-flash recycle benchmark on the primary Rosen-Pauls spec: 2 tears, Newton 7 it (Wegstein defeated). NO published product comparison is carried – see the flowsheetDict header

**The story:** / flowsheetDict – the Cavett problem on its PRIMARY specification (Cavett 1963; Rosen & Pauls 1977; Rosen, CACHE News Fall 2005; Seader & Henley 1998, Exercise 10.35).

The flowsheet is a four-theoretical-stage NEAR-ISOTHERMAL distillation (pressure, not temperature, changes stage to stage – the Gunther arrangement, US 3,575,077): FL1 is the partial condenser, FL4 the partial reboiler, and the FEED enters at the second stage. Each stage's liquid cascades DOWN; each stage's vapour climbs back UP – those three up-going vapours are the recycles R1, R2, R3 of the literature.

P1 (vapour product) FL1 (100 F, 800 psia) V2 L1 (R1) feed M2 FL2 (120 F, 270 psia) V3 L2 (R2) M3 FL3 (96 F, 49 psia) V4 L3 (R3) FL4 (85 F, 13 psia) P2 (liquid product)

HISTORY OF THIS DICT: the first commit of this case carried a reconstructed wiring (feed into FL1's mixer; V4 into FL2's) built from paraphrases, with the header saying so. The primary source then became available and showed BOTH edges wrong – the feed enters FL2 and V4 recycles to FL3. The reconstruction is gone; what you read is the documented flowsheet. A topology built from paraphrase and one built from the source differed in 2 edges out of 11 – worth remembering when citing anything from memory.

THE MISSING ANCHOR, named rather than implied (2026-08-10). This case's controlDict used to advertise "SRK vs published APR/FLOWTRAN products". It carries NO such comparison: no published product table is staged anywhere in the case, and nothing checks a Choupo number against one. What IS anchored here is the SPECIFICATION – the feed, the four stage conditions and the topology, all from the primary – plus the solver behaviour (Wegstein defeated on the real conditions; Newton converges in 7 iterations). That is a reproduction of the PROBLEM, not agreement with a published ANSWER, and the description now says so.

To become a real external anchor this case needs Rosen & Pauls' (or FLOWTRAN's) product compositions transcribed from the primary and pinned as ‘anchor‘ rows in ‘expected‘ – published values, never refreshed by ‘–record‘. Until someone reads that paper, the gap stays visible: a claim of agreement nobody can check is worse than an admitted absence.

TWO tears still close all three loops: cut the two mixer outlets and every stream is computable in the declared order (validated by the sequential-plan contract, not asserted here).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | L1 | F | 23.6955550394 |
| stream | L2 | F | 11.4542917866 |
| stream | L3 | F | 5.71821815965 |
| stream | P1 | F | 3.68430363786 |
| stream | P2 | F | 3.91030744687 |
| stream | V2 | F | 27.3798586772 |
| stream | V3 | F | 7.54398434784 |
| stream | V4 | F | 1.80791071278 |
| stream | feed | F | 7.59461111111 |
| stream | mixed2 | F | 38.8341504983 |
| stream | mixed3 | F | 13.2622024993 |
| kpi | FL1 | F_alpha | 23.6955550394 |
| kpi | FL1 | F_beta | 3.68430363786 |
| kpi | FL1 | F_in | 27.3798586772 |
| *. . . and 138 more reference values in the case's `expected`.* | | | |

## composite01_two_flashes

`tutorials/steady/flowsheets/composite01_two_flashes`

**What it does:** Composite node: two flashes in series (sectored-layout step 2)

**The story:** / flowsheetDict (COMPOSITE node)

children : the child unit ops (folders flash1/, flash2/ – leaf nodes). connections : the wiring. 'from'/'to' endpoints are either a bare name (this node's boundary inlet/outlet) or '<child>/<port>'. The engine flattens to namespaced streams (flash1.liquid → flash2.feed).

Two flashes in series, each a GENUINE vapour/liquid split. flash1 (370 K) splits the feed; its liquid (benzene-lean) feeds flash2 (373 K), which is still below that liquid's dew point, so flash2 produces a real vapour AND a real liquid 'bottoms' – not a total vaporiser.

*A 100 kmol/h equimolar benzene/toluene feed at 365 K and 1 bar into flash1 (370 K, 1 bar); flash1's liquid goes on into flash2 (373 K, 1 bar). Raoult's law throughout — the same ideal thermodynamics as flash01_benzene_toluene. What is new is the shape of the case: the root flowsheetDict holds no units at all, only sectors ( flash1 flash2 ); and a connections {} block. Each child is a folder with its own system/flowsheetDict.*

*The golden: flash1 splits V/F = 0.7638 (76.38 kmol/h vapour, 23.62 kmol/h liquid) taking 686.96 kW; flash2 splits that liquid V/F = 0.4359 into 10.30 kmol/h of vapour and 13.33 kmol/h of bottoms, taking 97.36 kW.*

*1. A composite node is a flowsheet of flowsheets. sectors names the children; connections wires them, each endpoint being either a bare name (this node's own boundary) or a child and one of its ports. The engine flattens the whole tree into ONE flat solver problem with namespaced names — the hierarchy is authoring and namespace, never a solver running inside a solver. A child is a leaf here, but the same grammar nests as deep as a plant needs. 2. The stream between them is a file, like every other stream. liquid1 is produced by flash1 and consumed by flash2, which is the only thing that makes it internal; its state lives in 0/liquid1 beside 0/feed and 0/bottoms. Nothing about being an intermediate gives a stream a different kind of home. 3. flash1's K-values are flash01's K-values, to the last digit. K_benzene 1.65136809583 and K_toluene 0.673501829388 — the same two numbers that case pins, because T, P, the components and the model are the same. Yet flash01 splits at V/F = 0.3040 and this one at 0.7638. For an isothermal flash the K's come from the thermodynamics and V/F comes from Rachford-Rice on those K's and the feed composition: 40/60 there, 50/50 here. Two different answers from one identical equilibrium — worth sitting with, because it is the difference between a property and a balance. 4. The feed's state does not move the split, but it pays for the duty. flash01's feed is already at 370 K and reports Q_kW = 0; this feed arrives at 365 K and flash1 charges 686.96 kW to get it there. Same K's, same V/F logic, different heat bill. 5. The boundary closes over both boxes at once. The plant-level Q_boundary_kW is 784.320784601, which is exactly flash1's 686.964038241 plus flash2's 97.3567463602; add it to H_feeds_kW 1154.14146199 and you get H_products_kW 1938.46221389. The first law is drawn across the composite, not per box — check that sum by hand, it is two additions.*

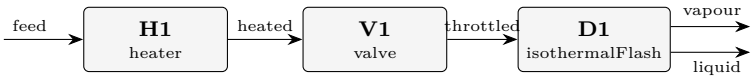
*Raise flash2's temperature in flash2/system/flowsheetDict and watch bottoms shrink while vapor2 grows; push it far enough and the second drum stops being a separator at all. Then open ammonia01_synthesis_loop, where one stream runs backwards and the units can no longer be solved in order.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.00370154472079 |
| stream | feed | F | 0.0277777777778 |
| stream | vapor1 | F | 0.0212163063101 |
| stream | vapor2 | F | 0.00285992674691 |
| kpi | flash1 | F_alpha | 0.0065614714677 |
| kpi | flash1 | F_beta | 0.0212163063101 |
| kpi | flash1 | F_in | 0.0277777777778 |
| kpi | flash1 | P | 100000 |
| kpi | flash1 | T | 370 |
| kpi | flash1 | V_over_F | 0.763787027163 |
| kpi | flash1 | phaseSet | 0 |
| kpi | flash2 | F_alpha | 0.00370154472079 |
| kpi | flash2 | F_beta | 0.00285992674691 |
| kpi | flash2 | F_in | 0.0065614714677 |
| *… and 29 more reference values in the case's `expected`.* | | | |

## credo01_valve_heater_drum

tutorials/steady/flowsheets/credo01_valve_heater_drum




**What it does:** Info-follows-streams credo: heater (Q) + valve (P) set the state; the flash drum (no knobs) just separates

**The story:** "Information follows the streams" — every change of state is a VISIBLE box on the stream, and the separator has no operating knobs. Read the chain top-to-bottom, the way the plant is built:

feed (10 bar liquid) → [H1 heater : Q] the heat box. Its only knob is the duty; T_out | is a result. Stays liquid at 10 bar (well below | the bubble point). → [V1 valve : P] the pressure box. Its only knob is the | downstream pressure; the let-down to 1 bar is | isenthalpic, so the hot liquid flashes and T and | vf come out as RESULTS. → [D1 flash : {} ] the separator. NO knobs — 'operation {}' is | empty. It inherits T and P from the stream V1 | hands it and just splits the phases. → liquid + vapour

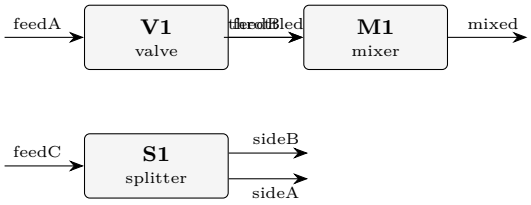
Compare with flash01_benzene_toluene, where the same separation is asked as 'isothermalFlash; operation { T 370; P 1 bar; }'. Those two numbers on the flash were, physically, a hidden heater (T) and a hidden valve (P). Here they are visible boxes, and the drum is honestly knob-less.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | H1 | T_out | 413.08978161 |
| kpi | V1 | T_out | 369.263388422 |
| kpi | V1 | vf | 0.199748128548 |
| kpi | V1 | dP | 900000 |
| stream | liquid | F | 0.0222292186514 |
| stream | vapour | F | 0.00554855912634 |
| utility | H1 | heating.steamMP.duty_kW | 260 |
| utility | H1 | heating.steamMP.T | 413.089781612 |
| utility | H1 | heating.steamMP.kg_s | 0.1305876444 |
| utility | H1 | heating.steamMP.eur_h | 14.04 |
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 350 |
| stream | heated | F | 0.0277777777778 |
| stream | heated | T | 413.089781612 |
| *... and 41 more reference values in the case's `expected`.* | | | |

## inletState01_undeclared_vapour

tutorials/steady/flowsheets/inletState01_undeclared_vapour




**What it does:** Witness: undeclared superheated vapours are read as vapours by the valve, the mixer and the splitter

**The story:** THE STATE A UNIT COMPUTES WITH MUST BE THE STATE ITS INLET MEANS.

feedA (450 K, 3 bar) -→ [V1 valve: 1.5 bar] -→ throttled –+ +-→ [M1 mixer] -→ mixed feedB (420 K, 1.5 bar) ————————————+

feedC (420 K, 1.5 bar) -→ [S1 splitter: 0.3 / 0.7] -→ sideA + sideB

The three feeds are benzene/toluene, each well above its dew point at its own pressure, and NONE declares 'phase'. A stream file that declares no phase does not thereby declare a liquid: its (T, P, z) means its own equilibrium, and here that equilibrium is all vapour. The stream record still carries a default 'vf = 0' until something resolves it, and for a domain inlet nothing upstream does – so each unit that reads its inlet's phase resolves it itself, at the inlet's own (T, P, z), in its own thermo world ('flashState::resolvedInletVaporFraction', the one home):

V1 (valve). The isenthalpic let-down prices H_in on the inlet's phase. Read as a liquid, H_in is a liquid's and the outlet search lands tens of kelvin colder, part-condensed – a valve that "condenses" a vapour. Read as what it is, an ideal-gas vapour has no Joule-Thomson effect and T_out = T_in. M1 (mixer). The inlets' phases price the inlet enthalpies. feedB is the LARGER inlet (150 against 100 kmol/h) on purpose: read as a liquid, its enthalpy would be a liquid's, and the adiabatic outlet – the state of equal enthalpy, resolved at its own equilibrium since 2026-10-24 – would leave part-condensed and colder (until that date the inlets also VOTED for a dominant-phase label, and feedB's vote would have labelled the vapour a liquid). S1 (splitter). A splitter preserves the phase – the phase the inlet MEANS. Its branches are PRODUCED streams that nobody re-resolves, so copying the carried default would publish two vapours as liquids.

The plant's first law closes only if the units and the energy report read the same states. check_inlet_resolution arm (i) runs this case and holds all three readings and the closure.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedA | F | 0.0277777777778 |
| stream | feedA | T | 450 |
| stream | feedB | F | 0.0416666666667 |
| stream | feedB | T | 420 |
| stream | feedC | F | 0.0166666666667 |
| stream | feedC | T | 420 |
| stream | mixed | F | 0.0694444444444 |
| stream | mixed | T | 432.508746466 |
| stream | sideA | F | 0.005 |
| stream | sideA | T | 420 |
| stream | sideB | F | 0.0116666666667 |
| stream | sideB | T | 420 |
| stream | throttled | F | 0.0277777777778 |
| stream | throttled | T | 450 |

*. . . and 16 more reference values in the case's* `expected`*.*

## plant01_two_sectors

tutorials/steady/flowsheets/plant01_two_sectors

**What it does:** Recursive plant: factory of two composite sectors (sectored-layout step 3)

**The story:** the RECURSIVE node (sectored-layout step 3).

A FACTORY whose children are SECTORS that are themselves composite (each a flowsheet of one flash). Recursion: plant → sector → unit. Tests nested namespacing (secA.fa, secB.fb) and multi-level cabling (feed → secA → secA/liquid → secB). The whole plant runs from one command.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0 |
| stream | feed | F | 0.0277777777778 |
| stream | vaporA | F | 0.0212163063101 |
| stream | vaporB | F | 0.0065614714677 |
| utility | secA.fa | heating.steamLP.duty_kW | 686.964038241 |
| utility | secA.fa | heating.steamLP.T | 370 |
| utility | secA.fa | heating.steamLP.kg_s | 0.314256193157 |
| utility | secA.fa | heating.steamLP.eur_h | 29.676846452 |
| utility | secB.fb | heating.steamLP.duty_kW | 226.733908578 |
| utility | secB.fb | heating.steamLP.T | 378 |
| utility | secB.fb | heating.steamLP.kg_s | 0.103720909688 |
| utility | secB.fb | heating.steamLP.eur_h | 9.79490485056 |
| stream | bottoms | T | 378 |
| stream | feed | T | 365 |
| *... and 5 more reference values in the case's `expected`.* | | | |

## plant02_basename_twins

tutorials/steady/flowsheets/plant02_basename_twins

**What it does:** basename twins: a top-level ‘feed‘ and a DISTINCT sector-internal ‘A.feed‘ coexist

**The story:** the ADVERSARIAL alias case of design forum #87-P0.

A top-level stream ‘feed‘ (the domain inlet) and a sector-INTERNAL edge also named ‘feed‘ (A.feed – a completely different stream between A’s two heaters) coexist. Identity is the PATH, never the basename: the manifest must carry BOTH (0/feed and 0/A/feed). The old basename filter silently dropped the top-level one from the manifest, so a legitimate case reported ORPHAN on its own inlet.

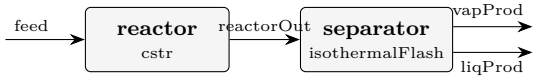
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00100000000002 |
| stream | hot | F | 0.00100000000002 |
| stream | out | F | 0.00100000000002 |
| kpi | h | F | 0.00100000000002 |
| kpi | h | P | 100000 |
| kpi | h | Q | 2000 |
| kpi | h | Q_kW | 2 |
| kpi | h | Q_per_mol | 1999.99999997 |
| kpi | h | T_in | 298.15 |
| kpi | h | T_out | 366.647115308 |
| kpi | h | dT | 68.497115308 |
| utility | A.h1 | heating.steamMP.duty_kW | 1 |
| utility | A.h1 | heating.steamMP.T | 400.742542906 |
| utility | A.h1 | heating.steamMP.kg_s | 0.000502260170768 |

*. . . and 16 more reference values in the case’s `expected`.*

## process01_reactor_flash

tutorials/steady/flowsheets/process01_reactor_flash




**What it does:** Two-unit flowsheet: CSTR → isothermal flash

**The story:** Two-unit flowsheet: feed → CSTR → flash → (liquid, vapour)

feed reactorOut reactor separator (CSTR) (T 365, P 1 bar) liqProd vapProd

Reaction kinetics are in constant/reactions Solver defaults are in system/solverDict

*A 1 kmol/h equimolar ethanol / acetic-acid feed enters a 5-litre CSTR at 350 K; the reactor outlet goes straight into a flash held at 365 K and 1 bar. Two units, three streams, and one idea: a flowsheet is a graph of streams, and information follows them.*

*The golden: conversion of ethanol X = 0.5259 at a residence time of 310.3 s (Damk?hler number 1.109, k = 3.575e-3 s?*[1] *from the Arrhenius law), the reactor removing a small 0.154 kW (the liquid esterification is only mildly exothermic); the separator flashes 23.8 % of what arrives (0.238 kmol/h vapour, 0.762 kmol/h liquid) with K_ethanol 1.660, K_ethylAcetate 1.573, K_aceticAcid 0.302, K_water 0.751, taking 2.93 kW to hold 365 K. Acetic acid's K is the low one: its liquid is referenced to the monomer's saturation pressure, because its vapour associates into dimers (the theory guide's paragraph ∗Which vapour the latent heat speaks∗).*

*1. The topology is the whole of flowsheetDict. Two units, each naming its in and outputs; the stream reactorOut is produced by one and consumed by the other, and that is what makes it internal. No stream values live there — they live in 0/, one file per stream. 2. The reaction is a named record, constant/reactions (esterification_etac): stoichiometry, which reactant is limiting, and an Arrhenius A/Ea. Its own header says the kinetics are ILLUSTRATIVE pseudo-first-order — read that before quoting a rate. 3. Units run in declared order, reactor first. Reverse the two entries and the engine refuses, naming the stream that would be read before it was produced. 4. Every balance is drawn from the engine, by default. The Reports tab carries the mass closure and the plant-boundary first law (Q_boundary 2.7793 kW — the NET heat crossing the boundary, not either unit's duty); the flash's 2.93 kW appears there under Utilities, as the steamLP that serves it. The per-element closure is drawn in the Plots tab, not in Reports. The reactor's -0.154 kW is in neither: it is a per-unit duty, and the engine writes it to reports/balances/energyBalance_byUnit.csv, which the CLI leaves in the case directory. Three numbers, three surfaces — worth knowing before you go looking for one in the wrong place.*

*Double the reactor volume (operation { V_R 0.010; }): conversion rises, the flash feed changes, and the separator's duty follows — one edit, three consequences, all visible in Streams. Then open process03_recycle, where a stream loops back and the solver has to iterate the whole graph.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | liqProd | F | 0.000211795205482 |
| stream | reactorOut | F | 0.000277777777778 |
| stream | vapProd | F | 6.59825722955e-05 |
| kpi | reactor | Da_kTau | 1.10929132154 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -0.153690679658 |
| kpi | reactor | T | 350 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | X_limiting | 0.52590711876 |
| kpi | reactor | k | 0.00357499942014 |
| kpi | reactor | minus_r_ethanol | 14.6085310767 |
| kpi | reactor | tau_s | 310.291329081 |
| *... and 29 more reference values in the case's `expected`.* | | | |

## process02_reactor_hx_flash

tutorials/steady/flowsheets/process02_reactor_hx_flash




**What it does:** Three-unit flowsheet: CSTR → heater → flash

**The story:** Three-unit flowsheet — physically realistic operation:

feed (350 K) hotStream (~359 K, 0.9 % vap) reactorOut CSTR 350 K, X=52% flash [reactor] [heater] [separator] vapProd liqProd

The flash inherits T from its input stream (no explicit operation.T) — propagation through the flowsheet.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | hotStream | F | 0.000277777777778 |
| stream | liqProd | F | 0.000251250707192 |
| stream | reactorOut | F | 0.000277777777778 |
| stream | vapProd | F | 2.65270705859e-05 |
| kpi | heater | F | 0.000277777777778 |
| kpi | heater | P | 101325 |
| kpi | heater | Q | 1000 |
| kpi | heater | Q_kW | 1 |
| kpi | heater | Q_per_mol | 3600 |
| kpi | heater | T_in | 350 |
| kpi | heater | T_out | 363.644080563 |
| kpi | heater | dT | 13.6440805633 |
| kpi | heater | vf_out | 0.0529680604206 |
| *... and 44 more reference values in the case's `expected`.* | | | |

## process02_with_design

tutorials/steady/flowsheets/process02_with_design




**What it does:** process02 with full reports: balances + sizing + Turton costing

**The story:** Three-unit flowsheet — physically realistic operation:

feed (350 K) hotStream (~362 K, 0.02 % vap) reactorOut CSTR 350 K, X=52% flash [reactor] [heater] [separator] vapProd liqProd

The flash inherits T from its input stream (no explicit operation.T) — propagation through the flowsheet.

*process01 with a heater between the reactor and the flash, and the two reports a project needs most: equipment sizing and Turton costing. The chain runs from system/postDict; reports { design {} economics {} } in the controlDict ∗serialise∗ what it produced into reports/design/sizing.csv and reports/economics/costs.csv.*

*The golden (process side): heater 0.4 kW → 361.95 K; reactor X = 0.526; flash at 361.95 K with V/F 0 – below this liquid's bubble point, so nothing vaporises (acetic acid's vapour is associated, which lowers the volatility an ideal monomer vapour would give it). The design side, from the run: reactor V_R = 0.005 $m^3$, wall 3.15 mm, 3.7 kg of SS316; heater A = 0.0667 $m^2$ of SS304; costing totals ?155,475 purchased / ?917,756 bare module / ?1,082,952 total module, CEPCI 820/397 = 2.0655, 0.92 EUR/USD.*

*1. The rules are declared, once, in postDict. sizing { units ( { unitName reactor; type stirredTank; material SS316; designRules { L_over_D 2.5; pressureDesign 3.0; ? } } ? ) } and costing { method Turton; year 2026; cepci 820; cepci2001 397; usdToEur 0.92; }. Every number in the tables descends from those lines and the run's own results — nothing else. 2. The costing table prints its own arithmetic. Below the totals: the log-quadratic correlation, K1–K3 per item, B1/B2, F_M, the CEPCI ratio, the currency, the 1.18 contingency — and the source (Turton App. A). Recompute one line; check_cost_provenance does exactly that on every suite run and requires 0.1 %. 3. READ THE TWO WARNINGS — they are the real lesson of this case. [validity] WARNING: stirredTank 'reactor': size V_R = 0.005 is OUTSIDE the correlation range [0.3, 520] and the heater's 0.0667 $m^2$ against [10, 1000]. A 1 kmol/h process is laboratory scale; Turton's correlations were fitted to plant scale, and the engine ∗extrapolates and says so∗ rather than clamping. The ?1.06 M for a 0.4 kW exchanger is what an extrapolated correlation returns — a number to distrust because you were told to. 4. A report draws, it does not recompute. Until 2026-09-02 this case carried the rules inside the report blocks (an older grammar), the reports failed by name on every run, and the golden — which pins only the process KPIs — never noticed. The README you are reading is why it was found.*

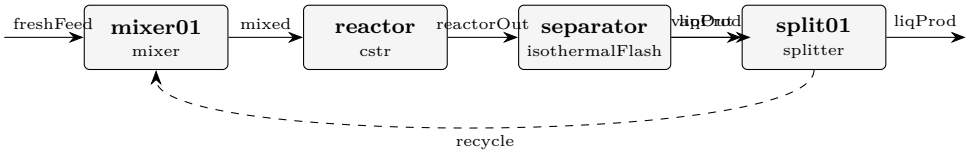
*Scale the feed to 100 kmol/h in 0/feed: the reactor volume the design rule wants lands inside the correlation window and the warning goes away. Then open economics01_esterification_dcf, where the same chain continues into OPEX, NPV and payback.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | hotStream | F | 0.000277777777778 |
| stream | liqProd | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| stream | vapProd | F | 0 |
| kpi | heater | F | 0.000277777777778 |
| kpi | heater | P | 101325 |
| kpi | heater | Q | 400 |
| kpi | heater | Q_kW | 0.4 |
| kpi | heater | Q_per_mol | 1440 |
| kpi | heater | T_in | 350 |
| kpi | heater | T_out | 361.946251976 |
| kpi | heater | dT | 11.9462519763 |
| kpi | heater | vf_out | 0 |
| *... and 59 more reference values in the case's `expected`.* | | | |

## process03_recycle

tutorials/steady/flowsheets/process03_recycle




**What it does:** Recycle loop: the tear stream is solved by Newton (2 iterations, |r|2 3.9e-6)

**The story:** the classic Reactor-Separator-Recycle architecture

freshFeed vapProd (product) [mixer] [reactor] [separator] (flash) liqProd [splitter] recycle purge

(the splitter is fed by liqProd)

Tear stream: ‘recycle‘ — declared explicitly; the Flowsheet runs NEWTON on the tear variables (F, z..., T) until the post-pass value matches the input guess to a tolerance. (‘recycleSolver Wegstein;‘ selects the fixed-point accelerator instead — try both and compare the iteration counts.)

The CSTR converts ~81% per pass at the converged tear (k.tau ~ 4.2 in 0.020 m3); the splitter returns 60% of liqProd, so the ethanol that escapes one pass gets further passes instead of leaving in liqProd, and the overall conversion is higher than the per-pass figure.

*process01 with a loop: reactor → flash → splitter, and 60 % of the flash liquid is sent ∗back∗ to a mixer ahead of the reactor. The reactor is four times larger (20 litres). The golden: the reactor now sees 1.758 kmol/h (1.0 fresh + 0.758 recycle), converts 79.8 % per pass at τ = 740 s, and the loop closes with the recycle stream at 0.7582 kmol/h, 364 K.*

*1. A cycle needs a cut, and the author declares it. system/solverDict says tearStreams ( recycle ); — the engine detects cycles, but ∗which∗ stream to tear is a choice, so it is written down. Reverse the units or forget the tear and the run refuses by name (INVALID ORDER, MISSING TEAR), never silently. 2. The tear starts from 0/recycle, a guess, and the log says so. Tear streams (initial guesses): recycle F = 0.7876 kmol/h ? — then a Newton over five variables (F, three compositions, T). The Log tab prints 2 per iteration: 1.34e-1 → 4.25e-3 → converged in 2 iterations at 3.9e-6. The seed was 60 % too high and it did not matter. 3. [plan] announces the cut before anything runs. material recycle: tear ’recycle’ cuts mixer01 → reactor → separator → split01 –recycle-→ mixer01 — read that line first; it is the whole topology in one sentence. 4. The recycle changes what the reactor is asked to do. Conversion per pass is 79.8 % because the reactor is larger and the feed is richer in what came back; overall, less ethanol leaves in liqProd. Compare the Streams tab against process01 line by line.*

*Change fractions ( 0.6 0.4 ) to ( 0.8 0.2 ): more recycle, a bigger loop flow, and watch whether the Newton still closes in two steps. recycle_autoinit_tear shows what happens when you give it ∗no∗ guess at all.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | freshFeed | F | 0.000277777777778 |
| stream | liqOut | F | 0.000351011838293 |
| stream | liqProd | F | 0.000140404735317 |
| stream | mixed | F | 0.000488384602904 |
| stream | reactorOut | F | 0.000488384602904 |
| stream | recycle | F | 0.000210607102976 |
| stream | vapProd | F | 0.000137372764611 |
| kpi | mixer01 | F_out | 0.000488384602904 |
| kpi | mixer01 | P_out | 100000 |
| kpi | mixer01 | T_out | 355.958546044 |
| kpi | mixer01 | n_in | 2 |
| kpi | mixer01 | vf | 0 |
| kpi | reactor | Da_kTau | 3.95873112738 |
| kpi | reactor | F_in_kmol_h | 1.75818457045 |
| *... and 40 more reference values in the case’s `expected`.* | | | |

## process04_research_workflow

tutorials/steady/flowsheets/process04_research_workflow




**What it does:** Case-local binary pair file (research workflow pattern)

**The story:** isothermal flash of ethanol/water at 355 K, 1 atm

Identical to flash02_ethanol_water, but this case demonstrates the RESEARCH WORKFLOW: parameters loaded from case-local parameters file rather than declared inline. See README.md for the full discussion.

*The research workflow for managing thermodynamic parameters: instead of declaring NRTL pairs (...) inline in the thermoPhysPropDict, the case declares only the model and lets the simulator load the parameters from a file inside the case directory.*

*This is the pattern used when you have your own fitted parameters that you do not want to mix into the standards database. The case remains self-contained and reproducible: the parameters travel with the case.*

*When the simulator builds the NRTL model and discovers there are no inline pairs in thermoPhysPropDict, it falls back to file lookup:*

*1. Case-local – <case>/constant/parameters/NRTL/ethanol-water.dat ? used here 2. Standards – $CHOUPO_HOME/data/standards/parameters/NRTL/ethanol-water.dat*

*In this case the local file exists, so it wins. If you delete the local file, the run still succeeds because the standards file is the same DECHEMA set.*

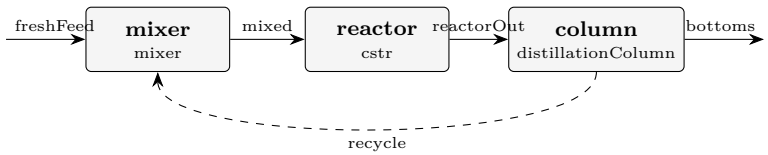
*The result is numerically identical to flash02_ethanol_water (same NRTL parameters; just a different way of supplying them).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | liquid | F | 0.00562323865754 |
| stream | vapor | F | 0.0221545391202 |
| kpi | flash01 | F_alpha | 0.00562323865754 |
| kpi | flash01 | F_beta | 0.0221545391202 |
| kpi | flash01 | F_in | 0.0277777777778 |
| kpi | flash01 | P | 101325 |
| kpi | flash01 | T | 355 |
| kpi | flash01 | V_over_F | 0.797563408329 |
| kpi | flash01 | phaseSet | 0 |
| stream | feed | T | 355 |
| stream | liquid | T | 355 |
| stream | vapor | T | 355 |
| kpi | flash01 | K_ethanol | 2.22741301451 |
| *... and 6 more reference values in the case's `expected`.* | | | |

## process05_isomerization_recycle

tutorials/steady/flowsheets/process05_isomerization_recycle




**What it does:** Isomerization neoPentane→nPentane: mixer + CSTR + distillation with recycle

**The story:** Reactor + distillation with recycle of the unconverted reactant — the canonical flowsheet for exercising the reports{} layer (stream table, mass balance with closure, energy balance) on a non-trivial topology with a tear stream.

freshFeed (neoPentane) [mixer] recycle (neoPentane) [reactor] neoPentane → nPentane (CSTR) [column] (distillation, alpha ~ 2.5) distillate bottoms (neoPentane, (nPentane, recycled) (recycle) the PRODUCT

Chemistry: neoPentane → nPentane is an isomerization — 1:1 in moles AND mass (identical MW 72.15). So at steady state the fresh feed (1 kmol/h neoPentane) equals the product (1 kmol/h nPentane in the bottoms), and the overall MASS BALANCE CLOSES TO 100 %. The recycle carries per-pass unconverted neoPentane back to the reactor, so the OVERALL conversion is ~100 % (product is 99 % nPentane) even though the per-pass conversion is only ~80 %.

Tuning notes (honest): The loop runs at 5 bar so every neoPentane/nPentane stream is a genuine LIQUID at its temperature. neoPentane's Antoine here boils at ~275 K at 1 bar (Psat(320 K) ~4.3 bar), so a 1 bar loop would flash the feed + recycle to VAPOUR — the phase is a CONSEQUENCE of (T,P,z) (Duhem), not a silent liquid default. At 5 bar the bubble point clears ~326 K, so the warm reactor (~327 K) and the recycled distillate stay liquid, as the flowsheet intends. The reaction uses a LOW activation energy on purpose — a high-Ea rate would make the per-pass conversion collapse against the recycle temperature and destabilise the loop. Low Ea decouples conversion from the recycle temperature. 'distillateRate' is FIXED here (the column's spec). It cannot perfectly track the per-pass unconverted neoPentane, so the recycle carries some nPentane (product) along — exactly why real recycle loops need a purge. Closing the recycle PURITY would be a DesignSpec manipulating distillateRate; left out here to keep the case a single pass for reviewing the reports.

This is the case to open in the GUI and to read the reports/ CSVs from: streamTable.csv, massBalance.csv (+ _byUnit), energyBalance_byUnit.csv.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.000277777777778 |
| stream | freshFeed | F | 0.000277777777778 |
| stream | mixed | F | 0.000363888888889 |
| stream | reactorOut | F | 0.000363888888889 |
| stream | recycle | F | 8.61111111111e-05 |
| kpi | column | B | 0.000277777777778 |
| kpi | column | D | 8.61111111111e-05 |
| kpi | column | L_rect | 0.000172222222222 |
| kpi | column | L_strip | 0.000618116666857 |
| kpi | column | R | 2 |
| kpi | column | T_bottom | 365.015971991 |
| kpi | column | T_top | 348.431788528 |
| kpi | column | V_rect | 0.000258333333333 |
| kpi | column | V_strip | 0.000340338889079 |

*... and 43 more reference values in the case's `expected`.*

## proxy01_gas_loop

tutorials/steady/flowsheets/proxy01_gas_loop

**What it does:** Proxy step 1: gas WGS loop with one recycle (mixer + Gibbs reactor + splitter)

**The story:** The GUI/credo bench in one case: TWO recycles + a shaft-work energy wire (W) + a utility STREAM (cooling water). Toggle the stream classes in the canvas: energy (the W wire), recycle (the two tears), utility (the cooling-water edge).

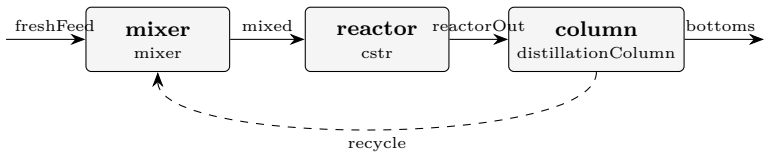
feed M1 mixer recycle1 (hot) recycle2 (cold, dry) R1 gibbsReactor (WGS) S1 splitter recycle1 (30 % hot gas back) toFlash CW (cooling water, UTILITY) HX1 warmCW (HX1: sensible cooling of the gas) F1 flash { } (no knobs: inherits T,P; condenses the water out) productWater (liquid) product dryGas (vapour) S2 splitter recycle2 (30 % dry gas back) productGas T1 turbine (power recovery) G1 electricLoad (W wire) expandedProduct

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | CW | F | 0.0166666666667 |
| stream | cooledGas | F | 0.000389139226554 |
| stream | dryGas | F | 0.000371204829226 |
| stream | expandedProduct | F | 0.000259843380458 |
| stream | feed | F | 0.000277777777778 |
| stream | mixed | F | 0.000555913180791 |
| stream | productGas | F | 0.000259843380458 |
| stream | productWater | F | 1.79343973278e-05 |
| stream | reacted | F | 0.000555913180791 |
| stream | recycle1 | F | 0.000166773954237 |
| stream | recycle2 | F | 0.000111361448768 |
| stream | toFlash | F | 0.000389139226554 |
| stream | warmCW | F | 0.0166666666667 |
| kpi | F1 | F_alpha | 1.79343973278e-05 |
| *... and 96 more reference values in the case's `expected`.* | | | |

## recycle02_dominant_eigenvalue

tutorials/steady/flowsheets/recycle02_dominant_eigenvalue




**What it does:** Isomerization recycle closed by the dominant-eigenvalue extrapolation (Orbach-Crowe) – iterations vs Newton, Wegstein, direct substitution

**The story:** The recycle of process05_isomerization_recycle closed by 'recycleSolver DominantEigenvalue;' (see system/solverDict for the iteration comparison). Reactor + distillation with recycle of the unconverted reactant — the canonical flowsheet for exercising the reports{} layer (stream table, mass balance with closure, energy balance) on a non-trivial topology with a tear stream.

freshFeed (neoPentane) [mixer] recycle (neoPentane) [reactor] neoPentane → nPentane (CSTR) [column] (distillation, alpha ~ 2.5) distillate bottoms (neoPentane, (nPentane, recycled) (recycle) the PRODUCT

Chemistry: neoPentane → nPentane is an isomerization — 1:1 in moles AND mass (identical MW 72.15). So at steady state the fresh feed (1 kmol/h neoPentane) equals the product (1 kmol/h nPentane in the bottoms), and the overall MASS BALANCE CLOSES TO 100 %. The recycle carries per-pass unconverted neoPentane back to the reactor, so the OVERALL conversion is ~100 % (product is 99 % nPentane) even though the per-pass conversion is only ~80 %.

Tuning notes (honest): The loop runs at 5 bar so every neoPentane/nPentane stream is a genuine LIQUID at its temperature. neoPentane's Antoine here boils at ~275 K at 1 bar (Psat(320 K) ~4.3 bar), so a 1 bar loop would flash the feed + recycle to VAPOUR — the phase is a CONSEQUENCE of (T,P,z) (Duhem), not a silent liquid default. At 5 bar the bubble point clears ~326 K, so the warm reactor (~327 K) and the recycled distillate stay liquid, as the flowsheet intends. The reaction uses a LOW activation energy on purpose — a high-Ea rate would make the per-pass conversion collapse against the recycle temperature and destabilise the loop. Low Ea decouples conversion from the recycle temperature. 'distillateRate' is FIXED here (the column's spec). It cannot perfectly track the per-pass unconverted neoPentane, so the recycle carries some nPentane (product) along — exactly why real recycle loops need a purge. Closing the recycle PURITY would be a DesignSpec manipulating distillateRate; left out here to keep the case a single pass for reviewing the reports.

This is the case to open in the GUI and to read the reports/ CSVs from: streamTable.csv, massBalance.csv (+ _byUnit), energyBalance_byUnit.csv.

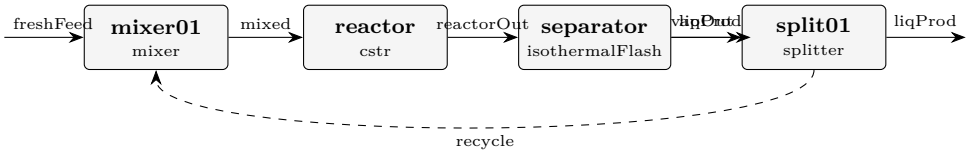
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.000277777777778 |
| stream | freshFeed | F | 0.000277777777778 |
| stream | mixed | F | 0.000363888888889 |
| stream | reactorOut | F | 0.000363888888889 |
| stream | recycle | F | 8.61111111111e-05 |
| kpi | column | B | 0.000277777777778 |
| kpi | column | D | 8.61111111111e-05 |
| kpi | column | L_rect | 0.000172222222222 |
| kpi | column | L_strip | 0.000618116666279 |
| kpi | column | R | 2 |
| kpi | column | T_bottom | 365.015972007 |
| kpi | column | T_top | 348.431788793 |
| kpi | column | V_rect | 0.000258333333333 |
| kpi | column | V_strip | 0.000340338888501 |

*... and 44 more reference values in the case's `expected`.*

### recycle_autoinit_tear

tutorials/steady/flowsheets/recycle_autoinit_tear




**What it does:** Recycle with NO authored tear guess – the solver auto-seeds the tear from the feed aggregate (announced), then converges

**The story:** same Reactor-Separator-Recycle as process03_recycle; this case's LESSON is the TEAR-SEED CONTRACT of the stream-state architecture:

the tear 'recycle' has an AUTHORED state file, 0/recycle – the seed is a first-class case file, owned by the author; completeness is FATAL before the solve: delete 0/recycle and choupoSolve REFUSES (MISSING, zero iterations run); 'bin/choupo-init0' refuses to INVENT a recycle seed (propagation is only honest on an acyclic graph) – it names the missing file and leaves the decision to you.

Compare the seed in 0/recycle with process03_recycle's: a recycle seed is a NUMERICAL aid the student owns, honest and visible ("no silent crutch", CLAUDE.md). (Historical note, marked legacy: on the retired 'streams {}' reader path a missing tear guess was auto-seeded from the feed aggregate, announced on the log; the 0/ contract replaced that with the authored, refuse-if-missing seed above.)

freshFeed vapProd (product) [mixer] [reactor] [separator] (flash) liqProd [splitter] recycle purge

(the splitter is fed by liqProd)

Tear stream: 'recycle' — declared explicitly; the Flowsheet runs Wegstein on (F, z..., T) until the post-pass value matches the input guess to a tolerance.

The CSTR converts ~25% per pass (small volume); without recycle most ethanol would simply leave in liqProd. With ~80% recycle, the overall ethanol conversion approaches 100% at steady state.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | freshFeed | F | 0.000277777777778 |
| stream | liqOut | F | 0.000351011838141 |
| stream | liqProd | F | 0.000140404735256 |
| stream | mixed | F | 0.000488384602712 |
| stream | reactorOut | F | 0.000488384602712 |
| stream | recycle | F | 0.000210607102884 |
| stream | vapProd | F | 0.000137372764571 |
| kpi | mixer01 | F_out | 0.000488384602712 |
| kpi | mixer01 | P_out | 100000 |
| kpi | mixer01 | T_out | 355.958546039 |
| kpi | mixer01 | n_in | 2 |
| kpi | mixer01 | vf | 0 |
| kpi | reactor | Da_kTau | 3.95873112774 |
| kpi | reactor | F_in_kmol_h | 1.75818456976 |
| *... and 40 more reference values in the case's `expected`.* | | | |

## gas-solid

## bagFilter01_dust

tutorials/steady/gas-solid/bagFilter01_dust




**What it does:** Bag filter (baghouse) on a dusty gas: >99% capture + cake pressure drop

**The story:** The high-efficiency complement to cyclone01 (same dusty gas: N2 carrying 50 kg/h of silica, PSD 1..50 um). Where the cyclone captured ~82 % by SIZE (fines escape), the fabric filter's dust cake captures ~99.9 % across all sizes — the price is a much larger pressure drop, dominated by the cake (dP = mu V (K1 + K2 W)).

Face velocity V = Q_gas / filterArea is the air-to-cloth ratio (~1.5 m/min here); with a cake load W = 0.5 kg/m^2 the pressure drop is ~2.7 kPa. The classic train is cyclone (cheap pre-cleaner) THEN bag filter (polish) — this case is the polish stage on the same dust.

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, PSD 1..50 um Filter: 40 m^2 cloth

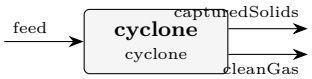
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000231019341285 |
| stream | cleanGas | F | 0.0416668205759 |
| stream | feed | F | 0.0418978399171 |
| kpi | baghouse | airToCloth_m_min | 1.53857549963 |
| kpi | baghouse | arealDustLoad | 0.5 |
| kpi | baghouse | dP_filter | 2745.33935569 |
| kpi | baghouse | efficiency | 0.999334225794 |
| kpi | baghouse | faceVelocity | 0.0256429249938 |
| kpi | baghouse | filterArea | 40 |
| kpi | baghouse | mu_gas | 1.78433840676e-05 |
| kpi | baghouse | penetration | 0.000665774205525 |
| kpi | baghouse | rho_gas | 1.13794383208 |
| kpi | baghouse | solidsCaptured_mass | 0.0138796420244 |
| kpi | baghouse | solidsEscaped_mass | 9.24686396528e-06 |

*... and 7 more reference values in the case's `expected`.*

## cyclone01_dust_removal

tutorials/steady/gas-solid/cyclone01_dust_removal




**What it does:** Cyclone dust removal: silica from N2 (Lapple, D_c = 0.5 m)

**The story:** A dusty gas (N2 carrying 50 kg/h of silica) passes through a cyclone. The cyclone separates BY SIZE: coarse particles spin to the wall and fall out (capturedSolids), fine particles follow the gas (cleanGas). This is the whole point of the PSD — a cyclone is a size classifier, not a simple splitter.

Lapple cut diameter d50 ~ 2.7 um for this geometry/flow. The grade efficiency $\eta$(d) = 1/(1+(d50/d)^2): ~12 % at 1 um, ~93 % at 10 um. Overall collection ~ 82 % of the dust mass; the escaping fines are the sub-d50 tail (why a cyclone is a pre-cleaner, followed by a bag filter).

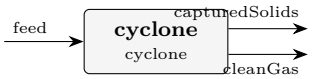
Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, ?_p = 2650 kg/m^3, PSD 1..50 um Cyclone: body diameter 0.5 m, 5 effective turns

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000189667114932 |
| stream | cleanGas | F | 0.0417081728022 |
| stream | feed | F | 0.0418978399171 |
| kpi | cyclone | bodyDiameter | 0.5 |
| kpi | cyclone | d50 | 2.71094994999e-06 |
| kpi | cyclone | d50_micron | 2.71094994999 |
| kpi | cyclone | efficiency | 0.820454419117 |
| kpi | cyclone | inletVelocity | 32.8229439921 |
| kpi | cyclone | mu_gas | 1.78433840676e-05 |
| kpi | cyclone | numberOfTurns | 5 |
| kpi | cyclone | rho_gas | 1.13794383208 |
| kpi | cyclone | rho_particle | 2650 |
| kpi | cyclone | solidsCaptured_mass | 0.0113952002651 |
| kpi | cyclone | solidsEscaped_mass | 0.00249368862328 |
| *... and 11 more reference values in the case's `expected`.* | | | |

## cyclone02_leith_licht

tutorials/steady/gas-solid/cyclone02_leith_licht




**What it does:** Cyclone dust removal – Leith-Licht back-mixing model

**The story:** The same dusty gas as cyclone01, but the grade efficiency uses the Leith-Licht back-mixing model instead of Lapple. Accounting for turbulent re-entrainment in the vortex, Leith-Licht predicts a SMALLER cut diameter (finer collection) for the same geometry/flow.

Leith-Licht cut diameter d50 ~ 1.11 um here. The grade efficiency is ~48 % at 1 um, ~94 % at 10 um. Overall collection ~ 88 % of the dust mass — higher than Lapple's 82 % because more of the sub-micron tail is captured.

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, ?_p = 2650 kg/m^3, PSD 1..50 um Cyclone: body diameter 0.5 m, 5 effective turns

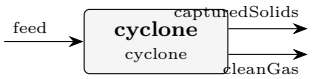
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000203443637145 |
| stream | cleanGas | F | 0.04169439628 |
| stream | feed | F | 0.0418978399171 |
| kpi | cyclone | bodyDiameter | 0.5 |
| kpi | cyclone | d50 | 1.10984825815e-06 |
| kpi | cyclone | d50_micron | 1.10984825815 |
| kpi | cyclone | efficiency | 0.880048347849 |
| kpi | cyclone | inletVelocity | 32.8229439921 |
| kpi | cyclone | mu_gas | 1.78433840676e-05 |
| kpi | cyclone | numberOfTurns | 5 |
| kpi | cyclone | rho_gas | 1.13794383208 |
| kpi | cyclone | rho_particle | 2650 |
| kpi | cyclone | solidsCaptured_mass | 0.0122228937197 |
| kpi | cyclone | solidsEscaped_mass | 0.0016659951687 |

*... and 11 more reference values in the case's `expected`.*

## cyclone03_iozia_leith

tutorials/steady/gas-solid/cyclone03_iozia_leith




**What it does:** Cyclone dust removal – Iozia-Leith data-fitted logistic (beta)

**The story:** The same dusty gas as cyclone01, now through the REAL Iozia-Leith (1989) FLOW-PATTERN model: the cut diameter d50 is NOT the Lapple residence-time surrogate but falls out of the gas flow pattern – the critical particle sits at the core radius where the tangential velocity peaks, and d50 = sqrt(9 mu Q /(pi z_c rho_p Vt_max^2)) (their eq 4), with Vt_max, the core diameter and its length regressed from the cyclone's OWN geometry (eqs 5-7). The grade curve stays logistic, eta(d) = 1/(1+(d50/d)^beta), beta = 3.5. (See the forum note: the 1989 eq-5 prefactor 6.1 is a decimal typo for 0.61 – 6.1 gives a supersonic Vt_max, against the paper's own Figure 4; we use the Figure-4-consistent 0.61.)

Flow-pattern d50 ~ 1.0 um – FINER than Lapple's 2.71 um: a Stairmand-shaped cyclone spins harder than the residence-time picture assumes, so it collects smaller particles. Overall collection RISES accordingly. Geometry defaults to the Stairmand high-efficiency ratios (add an operation.geometry {} block to override a/b/De/S/h/H/B).

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 50 kg/h silica, ?_p = 2650 kg/m^3, PSD 1..50 um Cyclone: body diameter 0.5 m, 5 effective turns, beta = 3.5

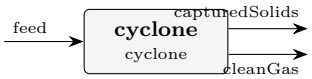
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000223552490081 |
| stream | cleanGas | F | 0.0416742874271 |
| stream | feed | F | 0.0418978399171 |
| kpi | cyclone | Q_gas | 1.02571699975 |
| kpi | cyclone | bodyDiameter | 0.5 |
| kpi | cyclone | d50 | 9.86636018962e-07 |
| kpi | cyclone | d50_micron | 0.986636018962 |
| kpi | cyclone | dP_cyclone | 4903.83536027 |
| kpi | cyclone | efficiency | 0.96703441953 |
| kpi | cyclone | eulerNumber | 8 |
| kpi | cyclone | inletVelocity | 32.8229439921 |
| kpi | cyclone | loading | 0.0118992372584 |
| kpi | cyclone | mu_gas | 1.78433840676e-05 |
| kpi | cyclone | numberOfTurns | 5 |

*... and 11 more reference values in the case's `expected`.*

## cyclone04_high_loading

tutorials/steady/gas-solid/cyclone04_high_loading




**What it does:** Cyclone high dust loading – Muschelknautz mass-loading effect

**The story:** The same cyclone geometry as cyclone01, but with a HEAVY dust load — 2000 kg/h of silica (inlet loading ~ 0.476 kg dust / kg gas, ~40x the ~0.012 kg/kg of cyclone01). At high loading the Muschelknautz model captures the mass-loading effect: above a critical loading the dust forms a wall strand that sweeps out particles independently of the single-particle cut, so the collection efficiency RISES sharply.

Muschelknautz overall collection ~ 98.4 % of the dust mass — far above the ~82 % a dilute Lapple cyclone gives, because the high solids loading itself enhances separation. Grade efficiency ~ 90 % even at 1 um.

Gas: 150 kmol/h N2, 300 K, 1 atm Dust: 2000 kg/h silica, ?_p = 2650 kg/m^3, PSD 1..50 um Cyclone: body diameter 0.5 m, 5 effective turns, cLimit = 0.05

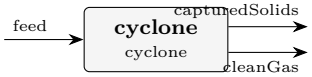
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.00909835478473 |
| stream | cleanGas | F | 0.0418152419014 |
| stream | feed | F | 0.0509135966861 |
| kpi | cyclone | bodyDiameter | 0.5 |
| kpi | cyclone | d50 | 1e-09 |
| kpi | cyclone | d50_micron | 0.001 |
| kpi | cyclone | dP_cyclone | 4903.83536027 |
| kpi | cyclone | efficiency | 0.983932479818 |
| kpi | cyclone | eulerNumber | 8 |
| kpi | cyclone | inletVelocity | 32.8229439921 |
| kpi | cyclone | loading | 0.475969490367 |
| kpi | cyclone | mu_gas | 1.78433840676e-05 |
| kpi | cyclone | numberOfTurns | 5 |
| kpi | cyclone | rho_gas | 1.13794383208 |

*... and 11 more reference values in the case's `expected`.*

## cyclone05_dirgo_leith_validation

tutorials/steady/gas-solid/cyclone05_dirgo_leith_validation




**What it does:** VALIDATION vs Dirgo & Leith (1985): a Stairmand high-efficiency cyclone (D=0.305 m, 0.139 m3/s, ~15 m/s inlet, mineral-oil aerosol) run through the flow-pattern Iozia-Leith model. Swap the model slot to see the ~3x d50 spread across correlations on ONE real cyclone – why you validate against data.

**The story:** VALIDATION against Dirgo & Leith (1985), Aerosol Sci. Technol. 4:401 – the measured fractional-efficiency baseline for a Stairmand high-efficiency cyclone. Their pilot-plant cyclone: body diameter D = 0.305 m, design air flow 0.139 m^3/s (300 cfm) → inlet velocity ~15 m/s; test aerosol a nebulised mineral oil (rho_p = 860 kg/m^3), particles >= 1 um.

This case reproduces THOSE conditions and runs the flow-pattern Iozia-Leith (1989) model. Re-run it with 'model Lapple;' / 'LeithLicht;' / 'Barth;' / 'Muschelknautz;' and compare each model's d50 and grade curve against the paper's measured fractional-efficiency curves (their Figures 5-6). THE LESSON: on this ONE real cyclone the models disagree by ~3x – Iozia-Leith 2.2 um, Leith-Licht 2.6 um (the fine, high-efficiency end) Lapple / Muschelknautz 6.2 um, Barth 6.5 um (the coarse end) – which is EXACTLY why you validate against data rather than trust one correlation. Dirgo & Leith's own finding: the theories BRACKET the measured curve and no single one is universally best (they consistently underpredicted efficiency at the highest inlet velocity, 25 m/s). The geometry defaults to the Stairmand high-efficiency ratios (a/D 0.5, b/D 0.2, De/D 0.5, S/D 0.5, h/D 1.5, H/D 4, B/D 0.375) – exactly Dirgo & Leith's cyclone – so the Iozia-Leith flow pattern (Vt_max, core) is evaluated on the real test geometry; Vi = Q/(ab) ~ 15 m/s reproduces their design point.

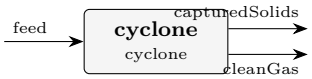
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 6.95101380068e-05 |
| stream | cleanGas | F | 0.00566837296963 |
| stream | feed | F | 0.00573788310764 |
| kpi | cyclone | Q_gas | 0.138813700633 |
| kpi | cyclone | bodyDiameter | 0.305 |
| kpi | cyclone | d50 | 2.24296780859e-06 |
| kpi | cyclone | d50_micron | 2.24296780859 |
| kpi | cyclone | dP_cyclone | 648.673413872 |
| kpi | cyclone | efficiency | 0.702163610002 |
| kpi | cyclone | eulerNumber | 8 |
| kpi | cyclone | inletVelocity | 11.9377544216 |
| kpi | cyclone | loading | 0.00879253984757 |
| kpi | cyclone | mu_gas | 1.78433840676e-05 |
| kpi | cyclone | numberOfTurns | 5 |

*... and 8 more reference values in the case's `expected`.*

## cyclone06_barth

tutorials/steady/gas-solid/cyclone06_barth




**What it does:** WITNESS: the Barth (1956) equilibrium-orbit cyclone model on the Dirgo & Leith Stairmand geometry (same design point as cyclone05); self-recorded golden – exercises a registered model no case selected.

**The story:** WITNESS for the Barth (1956) cyclone model – the static-pressure / equilibrium-orbit correlation – on the SAME Stairmand high-efficiency geometry and design point as cyclone05_dirgo_leith_validation (D = 0.305 m, 0.139 m^3/s, Vi ~ 15 m/s, mineral-oil aerosol rho_p = 860 kg/m^3).

THIS IS A WITNESS, NOT A VALIDATION: its golden is self-recorded, so a PASS means Barth's answer has not moved – never that it is right. The validation discussion (the ~3x d50 spread across the five correlations, and Dirgo & Leith's measured curves) lives in cyclone05's header; this case exists because Barth was a registered model that NO case selected, which invariant I8 names as not-a-capability (2026-08-22 fresh-eyes audit). Until today its only exercise was an invitation in a comment.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 2.14047701787e-05 |
| stream | cleanGas | F | 0.00571647833746 |
| stream | feed | F | 0.00573788310764 |
| kpi | cyclone | Q_gas | 0.138813700633 |
| kpi | cyclone | bodyDiameter | 0.305 |
| kpi | cyclone | d50 | 6.49490874523e-06 |
| kpi | cyclone | d50_micron | 6.49490874523 |
| kpi | cyclone | dP_cyclone | 648.673413872 |
| kpi | cyclone | efficiency | 0.216222426411 |
| kpi | cyclone | eulerNumber | 8 |
| kpi | cyclone | inletVelocity | 11.9377544216 |
| kpi | cyclone | loading | 0.00879253984757 |
| kpi | cyclone | mu_gas | 1.78433840676e-05 |
| kpi | cyclone | numberOfTurns | 5 |
| *... and 8 more reference values in the case's `expected`.* | | | |

## gasSolidSplitter01_ideal

tutorials/steady/gas-solid/gasSolidSplitter01_ideal




**What it does:** Ideal gas-solid splitter: specified 90% solids recovery (no efficiency physics)

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000208055925425 |
| stream | cleanGas | F | 0.0416897839917 |
| stream | feed | F | 0.0418978399171 |
| kpi | separator | gasCarryover | 0 |
| kpi | separator | gasToCleanGas | 0.0416666666667 |
| kpi | separator | gasToSolids | 0 |
| kpi | separator | solidsCaptured_mass | 0.0124999999995 |
| kpi | separator | solidsEscaped_mass | 0.00138888888884 |
| kpi | separator | solidsIn_mass | 0.0138888888884 |
| kpi | separator | solidsRecovery | 0.9 |
| stream | capturedSolids | T | 300 |
| stream | cleanGas | T | 300 |
| stream | feed | T | 300 |
| boundary | global | H_feeds_kW | -208.261349603 |

*. . . and 2 more reference values in the case's* `expected`.

## gasSolidSplitter02_rosinRammler

tutorials/steady/gas-solid/gasSolidSplitter02_rosinRammler




**What it does:** gas-solid split with a FITTED Rosin-Rammler feed PSD (the model is declared, the engine discretises)

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000208055925425 |
| stream | cleanGas | F | 0.0416897839917 |
| stream | feed | F | 0.0418978399171 |
| kpi | separator | gasCarryover | 0 |
| kpi | separator | gasToCleanGas | 0.0416666666667 |
| kpi | separator | gasToSolids | 0 |
| kpi | separator | solidsCaptured_mass | 0.0124999999995 |
| kpi | separator | solidsEscaped_mass | 0.00138888888884 |
| kpi | separator | solidsIn_mass | 0.0138888888884 |
| kpi | separator | solidsRecovery | 0.9 |
| stream | capturedSolids | T | 300 |
| stream | cleanGas | T | 300 |
| stream | feed | T | 300 |
| boundary | global | H_feeds_kW | -208.261349603 |

*. . . and 2 more reference values in the case's* ***expected****.*

## gasSolidSplitter03_tabular_psd

tutorials/steady/gas-solid/gasSolidSplitter03_tabular_psd




**What it does:** WITNESS: tabular (sieve-analysis) particle-size distribution on the gas-solid splitter; sibling of gasSolidSplitter02 (Rosin-Rammler).

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000208055925425 |
| stream | cleanGas | F | 0.0416897839917 |
| stream | feed | F | 0.0418978399171 |
| kpi | separator | gasCarryover | 0 |
| kpi | separator | gasToCleanGas | 0.0416666666667 |
| kpi | separator | gasToSolids | 0 |
| kpi | separator | solidsCaptured_mass | 0.0124999999995 |
| kpi | separator | solidsEscaped_mass | 0.00138888888884 |
| kpi | separator | solidsIn_mass | 0.0138888888884 |
| kpi | separator | solidsRecovery | 0.9 |
| stream | capturedSolids | T | 300 |
| stream | cleanGas | T | 300 |
| stream | feed | T | 300 |
| boundary | global | H_feeds_kW | -208.261349603 |

*. . . and 2 more reference values in the case's* ***expected****.*

### gasSolidSplitter04_lognormal_psd

tutorials/steady/gas-solid/gasSolidSplitter04_lognormal_psd




**What it does:** WITNESS: log-normal particle-size distribution (dgm, sigmaG) on the gas-solid splitter; sibling of gasSolidSplitter02.

**The story:** The same dusty gas as cyclone01 / bagFilter01 (N2 + 50 kg/h silica), but through an IDEAL, spec-driven separator: you SET the recovery (90 % of the solids to capturedSolids), no efficiency physics, no size classification.

Use it when a physics model is overkill ("assume 90 % capture"), as a placeholder, or to bound a design. Contrast: the cyclone COMPUTES ~82 % by size, the bag filter COMPUTES ~99.9 %; here the number is whatever you declare.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | capturedSolids | F | 0.000208055925425 |
| stream | cleanGas | F | 0.0416897839917 |
| stream | feed | F | 0.0418978399171 |
| kpi | separator | gasCarryover | 0 |
| kpi | separator | gasToCleanGas | 0.0416666666667 |
| kpi | separator | gasToSolids | 0 |
| kpi | separator | solidsCaptured_mass | 0.0124999999995 |
| kpi | separator | solidsEscaped_mass | 0.00138888888884 |
| kpi | separator | solidsIn_mass | 0.0138888888884 |
| kpi | separator | solidsRecovery | 0.9 |
| stream | capturedSolids | T | 300 |
| stream | cleanGas | T | 300 |
| stream | feed | T | 300 |
| boundary | global | H_feeds_kW | -208.261349603 |

*. . . and 2 more reference values in the case's* `expected`*.*

## gibbs

**The theory you need.** Equilibrium without a mechanism: minimise total Gibbs energy $G = \sum_i n_i \mu_i$ over ALL species subject to elemental balances $\mathbf{A}\,\mathbf{n} = \mathbf{b}$ (Lagrange multipliers per element – printed as $\lambda_E$, the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH. . . ) are IN the pool, and to overshoot by ∼100–250 K when they are not (dissociation is endothermic). Above ∼1500 K the catalogue's polynomial $c_p$ fits extrapolate WRONG – combustion cases carry case-local NASA-9 overrides (NASA/TP-2002-211556), and the case headers say so.

*Equilibrium maps.* Sweeping the Gibbs solve over a $T \times P$ grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes $T$ and $P$ in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The `temperatureApproach` $\Delta T$ – declared as a magnitude; the engine assigns the sign from the reaction's thermicity and announces it – evaluates the *reaction* equilibrium at $T \pm \Delta T$ while enthalpy and the energy balance stay at the physical $T$ – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on `nh3_equilibrium_map`, at $P = 200$ bar and $\Delta T = 0$, find the temperature at which the equilibrium $NH_3$ mole fraction crosses 30 %; report $T$ to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in $T$ at fixed $P$. *Full derivation: Theory Guide, the Gibbs-equilibrium chapter, §equilibrium maps.*

## claus01_thermal_stage

tutorials/steady/gibbs/claus01_thermal_stage




**What it does:** Claus thermal stage: adiabatic Gibbs reactor over C-H-N-O-S

**The story:** Claus thermal stage: an adiabatic furnace and the boiler behind it, the same twelve species over the same five elements, 900 K apart.

Element order in every atoms( ) list is the order of 'elements' below: ( C H N O S ). Nothing in this file declares a conversion, a temperature for the furnace, or which sulfur allotrope forms – all three are results. See system/controlDict for what comes out and what the model deliberately does not do.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | acidGas | F | 0.0277777777778 |
| stream | acidGas | T | 320 |
| stream | air | F | 0.0595238094444 |
| stream | air | T | 298.15 |
| stream | boilerGas | F | 0.0786432844729 |
| stream | boilerGas | T | 600 |
| stream | feedGas | F | 0.0873015872222 |
| stream | feedGas | T | 305.967920996 |
| stream | furnaceGas | F | 0.0872586259716 |
| stream | furnaceGas | T | 1515.34061493 |
| kpi | boiler | F_in_kmol_h | 314.131053498 |
| kpi | boiler | F_out_kmol_h | 283.115824103 |
| kpi | boiler | N_in_mol_s | 87.2586259716 |
| kpi | boiler | N_out_mol_s | 78.6432844729 |
| *... and 59 more reference values in the case's `expected`.* | | | |

## gibbs01_water_gas_shift

tutorials/steady/gibbs/gibbs01_water_gas_shift




**What it does:** Gibbs reactor: water-gas shift CO + H2O <→ CO2 + H2 at 800 K

**The story:** Water-gas shift reactor at 800 K, 1 bar. CO + water <→ CO2 + H2 Equimolar feed. The Gibbs-minimisation reactor gives CO conversion ~ 67 % (CO 0.50 → 0.164 mole fraction at equilibrium; CO2 = H2 = 0.336), consistent with the independent K(800 K) calculation in Smith, Van Ness, Abbott (Table 13.4).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | products | F | 0.000277777777778 |
| kpi | wgs | F_in_kmol_h | 1 |
| kpi | wgs | F_out_kmol_h | 1 |
| kpi | wgs | N_in_mol_s | 0.277777777778 |
| kpi | wgs | N_out_mol_s | 0.277777777778 |
| kpi | wgs | P | 100000 |
| kpi | wgs | T | 800 |
| kpi | wgs | converged | 1 |
| kpi | wgs | lambda_C | 5549.05961227 |
| kpi | wgs | lambda_H | -60095.3190791 |
| kpi | wgs | lambda_O | -294758.913033 |
| kpi | wgs | y_CO | 0.16409231685 |
| kpi | wgs | y_CO2 | 0.33590768315 |

*. . . and 12 more reference values in the case's `expected`.*

## gibbs02_steam_reforming

tutorials/steady/gibbs/gibbs02_steam_reforming




**What it does:** Gibbs reactor: steam reforming of methane at 1000 K

**The story:** Industrial methane steam reforming feed: methane plus excess steam (steam-to-carbon ratio = 3, i.e. 1 mol CH4 + 3 mol H2O per pass). Expected at 1000 K, 1 bar: near-complete CH4 conversion, hydrogen yield ~ 2.7 mol H2 per mol CH4 (reforming + WGS combined).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00111111111111 |
| stream | syngas | F | 0.0016586091548 |
| kpi | reformer | F_in_kmol_h | 4 |
| kpi | reformer | F_out_kmol_h | 5.97099295728 |
| kpi | reformer | N_in_mol_s | 1.11111111111 |
| kpi | reformer | N_out_mol_s | 1.6586091548 |
| kpi | reformer | P | 100000 |
| kpi | reformer | T | 1000 |
| kpi | reformer | converged | 1 |
| kpi | reformer | lambda_C | -33527.5812363 |
| kpi | reformer | lambda_H | -75161.3018601 |
| kpi | reformer | lambda_O | -309185.611918 |
| kpi | reformer | y_CH4 | 0.0024289965616 |
| kpi | reformer | y_CO | 0.0979819433622 |

*... and 10 more reference values in the case's `expected`.*

## gibbs03_methane_combustion

tutorials/steady/gibbs/gibbs03_methane_combustion




**What it does:** Gibbs reactor: methane combustion in stoichiometric air at 1800 K

**The story:** Stoichiometric combustion of methane in air: CH4 + 2 (O2 + 3.76 N2) -→ CO2 + 2 H2O + 7.52 N2 (ideally) At 1800 K some CO and H2 appear from partial dissociation.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00292222222222 |
| stream | flueGas | F | 0.00292404303636 |
| kpi | burner | F_in_kmol_h | 10.52 |
| kpi | burner | F_out_kmol_h | 10.5265549309 |
| kpi | burner | N_in_mol_s | 2.92222222222 |
| kpi | burner | N_out_mol_s | 2.92404303636 |
| kpi | burner | P | 100000 |
| kpi | burner | T | 1800 |
| kpi | burner | converged | 1 |
| kpi | burner | lambda_C | -357484.903199 |
| kpi | burner | lambda_H | -201140.892694 |
| kpi | burner | lambda_N | -201371.845813 |
| kpi | burner | lambda_O | -268835.220885 |
| kpi | burner | y_CH4 | 7.51542148893e-20 |
| *... and 16 more reference values in the case's `expected`.* | | | |

## gibbs04_wgs_temperature_sweep

`tutorials/steady/gibbs/gibbs04_wgs_temperature_sweep`




**What it does:** Water-gas shift equilibrium swept over T = 600..1200 K

**The story:** / flowsheetDict Water-gas shift equilibrium swept over T = 600..1200 K (outerDict sweep on the Gibbs reactor): CO conversion falls as T rises – exothermic equilibrium – reproducing the textbook K(T) curve point by point.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | products | F | 0.000277777777859 |
| kpi | wgs | F_in_kmol_h | 1 |
| kpi | wgs | F_out_kmol_h | 1.00000000029 |
| kpi | wgs | N_in_mol_s | 0.277777777778 |
| kpi | wgs | N_out_mol_s | 0.277777777859 |
| kpi | wgs | P | 100000 |
| kpi | wgs | Q_kW | -2.0000513466 |
| kpi | wgs | T | 900 |
| kpi | wgs | converged | 1 |
| kpi | wgs | lambda_C | -12610.3166837 |
| kpi | wgs | lambda_H | -69027.9700281 |
| kpi | wgs | lambda_O | -299577.53726 |
| kpi | wgs | y_CO | 0.199139146827 |
| *... and 9 more reference values in the case's `expected`.* | | | |

## gibbs05_adiabatic_flame

tutorials/steady/gibbs/gibbs05_adiabatic_flame




**What it does:** Adiabatic flame of lean CH4 in air ($\varphi = 0.5$)

**The story:** Lean methane combustion at $\varphi = 0.5$ (100 % excess air): 1 mol CH4 + 4 mol O2 + 15.04 mol N2 Feed at ambient 298.15 K, 1 bar. Reactor finds its own adiabatic flame temperature.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00556666666667 |
| stream | flueGas | F | 0.00556667001791 |
| kpi | burner | F_in_kmol_h | 20.04 |
| kpi | burner | F_out_kmol_h | 20.0400120645 |
| kpi | burner | N_in_mol_s | 5.56666666667 |
| kpi | burner | N_out_mol_s | 5.56667001791 |
| kpi | burner | P | 100000 |
| kpi | burner | T | 1491.58454904 |
| kpi | burner | converged | 1 |
| kpi | burner | lambda_C | -432004.832024 |
| kpi | burner | lambda_H | -204860.137156 |
| kpi | burner | lambda_N | -162961.775795 |
| kpi | burner | lambda_O | -186456.945423 |
| kpi | burner | outerIterations | 3 |
| *... and 12 more reference values in the case's `expected`.* | | | |

## gibbs06_h2_flame_radicals

tutorials/steady/gibbs/gibbs06_h2_flame_radicals




**What it does:** Stoichiometric H2/air adiabatic flame – the radical pool at equilibrium

**The story:** THE RADICAL DATABASE AT WORK: stoichiometric H2/air adiabatic flame, equilibrium by Gibbs minimisation over the FULL radical pool (H, O, OH, HO2, H2O2 – the catalogue's NASA-7-derived radical records) plus equilibrium thermal NO.

1 H2 + 0.5 O2 + 1.88 N2 at 298.15 K, 1 atm, adiabatic.

THE ANSWER (golden-locked): T_ad = 2385.8 K – the literature gives 2380-2390 K (Glassman, Combustion, 4th ed.; Law, Combustion Physics). Equilibrium pool: OH 0.55 %, H 0.18 %, O 0.06 %, NO 0.27 % – all inside the published equilibrium ranges. NO here is the Zeldovich CEILING (what kinetics approaches with residence time, never exceeds).

TWO LESSONS THIS CASE TEACHES: 1. THE RADICALS' WORK: endothermic dissociation (H2O → OH + H, O2 → 2 O) soaks up enthalpy near 2400 K. Without the radical species the same reactor answers ~2640 K – the first-year textbook overshoot. 2. THE EXTRAPOLATION TRAP (why constant/components/ overrides exist here): the catalogue's poly3 Cp fits are 250-1500 K PROCESS fits; extrapolated to 2641 K, N2's cubic turns NEGATIVE (-33 J/mol/K) and water's reads 45 instead of ~56. The case-local overrides switch the four stable species to the NASA-9 polynomials of NASA/TP-2002-211556 (public domain, valid 200-6000 K; until 2026-10-25 they were GRI-Mech 3.0's NASA-7, which grants no redistribution). Run it once with the overrides removed to SEE the 250 K error a silent extrapolation buys.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | burnt | F | 0.000808562123724 |
| stream | feed | F | 0.000938888888889 |
| kpi | flame | F_in_kmol_h | 3.38 |
| kpi | flame | F_out_kmol_h | 2.91082364541 |
| kpi | flame | N_in_mol_s | 0.938888888889 |
| kpi | flame | N_out_mol_s | 0.808562123724 |
| kpi | flame | P | 101325 |
| kpi | flame | T | 2385.83679448 |
| kpi | flame | converged | 1 |
| kpi | flame | lambda_H | -240335.177161 |
| kpi | flame | lambda_N | -277509.500244 |
| kpi | flame | lambda_O | -344071.838893 |
| kpi | flame | outerIterations | 3 |
| kpi | flame | y_H | 0.00182173501092 |

*... and 14 more reference values in the case's `expected`.*

## gibbs06_thermal_nox

tutorials/steady/gibbs/gibbs06_thermal_nox




**What it does:** Methane combustion at 1800 K with thermal NOx formation

**The story:** / flowsheetDict Methane combustion at 1800 K with thermal NOx: the Gibbs reactor finds the full equilibrium including NO/NO2 – the EQUILIBRIUM ceiling on NOx, to contrast with the KINETIC path (nox01_zeldovich_postflame, where the rate, not equilibrium, controls).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00292222222222 |
| stream | flueGas | F | 0.00292416183331 |
| kpi | burner | F_in_kmol_h | 10.52 |
| kpi | burner | F_out_kmol_h | 10.5269825999 |
| kpi | burner | N_in_mol_s | 2.92222222222 |
| kpi | burner | N_out_mol_s | 2.92416183331 |
| kpi | burner | P | 100000 |
| kpi | burner | T | 1800 |
| kpi | burner | converged | 1 |
| kpi | burner | lambda_C | -355589.713611 |
| kpi | burner | lambda_H | -200666.346513 |
| kpi | burner | lambda_N | -201373.229482 |
| kpi | burner | lambda_O | -269787.239957 |
| kpi | burner | y_CH4 | 9.68349314225e-20 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## gibbs07_wgs_cooled

tutorials/steady/gibbs/gibbs07_wgs_cooled




**What it does:** Cooled WGS Gibbs reactor (350 K): water condenses, shifting the equilibrium (multi-phase)

**The story:** multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <→ CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The ‘elementPotential‘ method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated (y_H2O.P > Psat(350 K) ~ 0.42 bar), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: ‘gas‘ (the dry-ish product gas) + ‘condensate‘ (liquid water).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | condensate | F | 3.76244802832e-05 |
| stream | feed | F | 0.000277777777778 |
| stream | gas | F | 0.000240153299998 |
| kpi | wgs | F_condensate_kmol_h | 0.135448129019 |
| kpi | wgs | F_in_kmol_h | 1 |
| kpi | wgs | F_out_kmol_h | 0.864551879994 |
| kpi | wgs | N_in_mol_s | 0.277777777778 |
| kpi | wgs | N_liq_mol_s | 0.0376244802832 |
| kpi | wgs | N_out_mol_s | 0.277777780282 |
| kpi | wgs | P | 100000 |
| kpi | wgs | T | 350 |
| kpi | wgs | converged | 1 |
| kpi | wgs | lambda_C | 50085.9480936 |
| kpi | wgs | lambda_H | -24736.1727481 |
| *... and 20 more reference values in the case's `expected`.* | | | |

## gibbs08_wgs_cooled_reactiveflash

tutorials/steady/gibbs/gibbs08_wgs_cooled_reactiveflash




**What it does:** Cooled WGS via reactiveFlash (B1) – cross-checks the elementPotential V+L result

**The story:** the gibbs07_wgs_cooled problem solved by 'model reactiveFlash;' (route B1: an equilibrium-reaction + isothermal-flash inner loop) instead of the default elementPotential (A). Cross-checks the two-phase V+L answer by an independent method.

The physics is gibbs07's: multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <→ CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The 'elementPotential' method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated (y_H2O.P > Psat(350 K) ~ 0.42 bar), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: 'gas' (the dry-ish product gas) + 'condensate' (liquid water).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | condensate | F | 3.76244802827e-05 |
| stream | feed | F | 0.000277777777778 |
| stream | gas | F | 0.000240153299999 |
| kpi | wgs | F_condensate_kmol_h | 0.135448129018 |
| kpi | wgs | F_in_kmol_h | 1 |
| kpi | wgs | F_out_kmol_h | 0.864551879995 |
| kpi | wgs | N_in_mol_s | 0.277777777778 |
| kpi | wgs | N_liq_mol_s | 0.0376244802827 |
| kpi | wgs | N_out_mol_s | 0.277777780281 |
| kpi | wgs | P | 100000 |
| kpi | wgs | T | 350 |
| kpi | wgs | converged | 1 |
| kpi | wgs | lambda_C | 50085.9480935 |
| kpi | wgs | lambda_H | -24736.1727482 |

*... and 20 more reference values in the case's `expected`.*

## gibbs09_wgs_cooled_directmin

tutorials/steady/gibbs/gibbs09_wgs_cooled_directmin




**What it does:** Cooled WGS via directMin (B2, Nelder-Mead direct G-min) – cross-checks A/B1 to ~0.3%

**The story:** the gibbs07_wgs_cooled problem solved by 'model directMin;' (route B2: direct Nelder-Mead minimisation of the total Gibbs energy G) – the third independent route, cross-checking A (elementPotential, gibbs07) and B1 (reactiveFlash, gibbs08) to ~0.3%.

The physics is gibbs07's: multi-phase (gas + liquid) Gibbs reactor.

CO + H2O <→ CO2 + H2, excess steam (1 CO + 3 H2O), 350 K, 1 bar.

The 'elementPotential' method (default) does a gas-only Gibbs solve, sees that the leftover water is supersaturated (y_H2O.P > Psat(350 K) ~ 0.42 bar), and switches to the two-phase V+L solve. The condensing water is pulled out of the gas, which drives the WGS further forward (Le Chatelier) than a gas-only reactor would — the coupling a single-phase Gibbs misses.

Outlets: 'gas' (the dry-ish product gas) + 'condensate' (liquid water).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | condensate | F | 3.77165683291e-05 |
| stream | feed | F | 0.000277777777778 |
| stream | gas | F | 0.000240061209449 |
| kpi | wgs | F_condensate_kmol_h | 0.135779645985 |
| kpi | wgs | F_in_kmol_h | 1 |
| kpi | wgs | F_out_kmol_h | 0.864220354015 |
| kpi | wgs | N_in_mol_s | 0.277777777778 |
| kpi | wgs | N_liq_mol_s | 0.0377165683291 |
| kpi | wgs | N_out_mol_s | 0.277777777778 |
| kpi | wgs | P | 100000 |
| kpi | wgs | T | 350 |
| kpi | wgs | converged | 1 |
| kpi | wgs | liquidFraction | 0.135779645985 |
| kpi | wgs | twoPhase | 1 |

*... and 20 more reference values in the case's `expected`.*

## gibbs10_ammonia_fugacity

tutorials/steady/gibbs/gibbs10_ammonia_fugacity




**What it does:** Ammonia synthesis equilibrium at 200 bar with the REAL-gas fugacity correction: N2 + 3 H2 <→ 2 NH3 at 700 K, 200 bar over an SRK equation of state. At this pressure the ideal-gas equilibrium (30.0 % NH3) is wrong – the Gibbs reactor now folds the SRK fugacity coefficients into the equilibrium (Kphi ~ 0.75, favouring the fewer-moles NH3 side), giving 32.4 % NH3, toward the measured ~35-40 %. Switch the equationOfState to idealGas and watch NH3 drop back to 30.0 % – the fugacity correction is the whole point of a high-pressure reactor.

**The story:** the high-pressure equilibrium done RIGHT

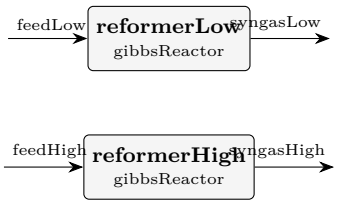
N2 + 3 H2 <→ 2 NH3 at 700 K and 200 bar, a stoichiometric feed. The Gibbs reactor minimises the REAL-gas Gibbs energy: it pulls the fugacity coefficients from the package EoS (SRK here) and adds ln(phi_i) to each species' chemical potential. At 200 bar this MATTERS – the fugacity ratio Kphi = phi_NH3^2/(phi_N2 phi_H2^3) ~ 0.75 shifts the equilibrium toward NH3 (the fewer-moles side): 30.0 % (ideal) → 32.4 % (SRK), toward the measured ~35-40 %. Set the package EoS to idealGas and NH3 falls back to 30.0 %. (A cornerstone of the classic ammonia synthesis loop: the reactor is only quantitative at pressure once fugacity is in.)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | products | F | 0.000209823048593 |
| kpi | ammonia | F_in_kmol_h | 1 |
| kpi | ammonia | F_out_kmol_h | 0.755362974933 |
| kpi | ammonia | N_in_mol_s | 0.277777777778 |
| kpi | ammonia | N_out_mol_s | 0.209823048593 |
| kpi | ammonia | P | 20000000 |
| kpi | ammonia | Q_kW | -3.68175302075 |
| kpi | ammonia | T | 700 |
| kpi | ammonia | converged | 1 |
| kpi | ammonia | lambda_H | -34945.3954982 |
| kpi | ammonia | lambda_N | -59416.3950887 |
| kpi | ammonia | y_H2 | 0.507099864715 |
| kpi | ammonia | y_N2 | 0.169033288238 |
| *... and 10 more reference values in the case's `expected`.* | | | |

## gibbs11_carbon_deposition

tutorials/steady/gibbs/gibbs11_carbon_deposition




**What it does:** Gibbs reactor with a pure solid phase: graphite forms at S/C 0.8, not at S/C 3

**The story:** CARBON DEPOSITION IN A STEAM REFORMER – a Gibbs reactor with a PURE SOLID.

Two reformers see the same methane at 1000 K and 1 bar; only the steam-to-carbon ratio differs:

reformerLow S/C = 0.8 (steam-lean: the reformer COKES) reformerHigh S/C = 3.0 (a steam margin: no carbon – the same feed and conditions as gibbs02)

Six species over three elements (C H O), so R = N - M = 3 independent reactions. A convenient set is

CH4 + H2O <→ CO + 3 H2 (reforming) CO + H2O <→ CO2 + H2 (water-gas shift) CH4 <→ C(gr) + 2 H2 (methane cracking)

and the Boudouard reaction, 2 CO <→ C(gr) + CO2, is a combination of the three (cracking - reforming - shift, rearranged), so it adds nothing. A Gibbs reactor needs no reaction list at all: the element balance is the whole constraint.

GRAPHITE IS NOT A GAS. Its record says 'referenceState pureSolid;', so the reactor gives it its OWN phase: activity 1, chemical potential equal to its pure-solid standard potential, amount >= 0. The reactor first solves the gas WITHOUT the solid and reads the carbon activity there,

a_C = exp( pi_C - g_C(gr)/RT ) (pi_C: the carbon element potential) = K_cracking * p_CH4 / p_H2^2 (partial pressures in bar),

a_C > 1 means depositing carbon LOWERS the Gibbs energy, so the solid appears and the gas is re-equilibrated to a_C = 1; a_C <= 1 means it does not. Both verdicts are printed with the activity that decided them and published as KPIs (n_solid_graphite_mol_s, activity_graphite).

Expected, recomputed INDEPENDENTLY by bin/curate/check_gibbs_solid_phase.py from a reaction-based formulation on this case's own records: S/C 0.8: a_C (carbon-free) = 2.33 → about 0.25 mol graphite / mol CH4 S/C 3.0: a_C (carbon-free) = 0.08 → no graphite; gibbs02's answer.

The deposit leaves in the reactor's outlet as its SOLID phase, priced on the solid rung by the energy report and counted by the mass and element balances – a coke deposit does not leave by a pipe of its own.

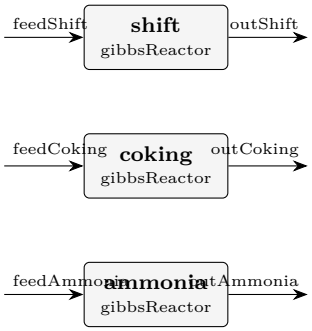
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedHigh | F | 0.00111111111111 |
| stream | feedHigh | T | 1000 |
| stream | feedLow | F | 0.0005 |
| stream | feedLow | T | 1000 |
| stream | syngasHigh | F | 0.0016586091548 |
| stream | syngasHigh | T | 1000 |
| stream | syngasLow | F | 0.000967550025636 |
| stream | syngasLow | T | 1000 |
| kpi | reformerHigh | F_in_kmol_h | 4 |
| kpi | reformerHigh | F_out_kmol_h | 5.97099295728 |
| kpi | reformerHigh | F_solid_kmol_h | 0 |
| kpi | reformerHigh | N_in_mol_s | 1.11111111111 |
| kpi | reformerHigh | N_out_mol_s | 1.6586091548 |
| kpi | reformerHigh | P | 100000 |

*... and 39 more reference values in the case's `expected`.*

## gibbs12_element_potential_newton

tutorials/steady/gibbs/gibbs12_element_potential_newton




**What it does:** The element-potential Newton, logged: four Gibbs reactors – a gas, a condensing liquid, a depositing solid, and a feed far from its equilibrium

**The story:** HOW a Gibbs reactor SOLVES, logged.

The witness of the EduTool 'element-potential'. Three isothermal Gibbs reactors side by side, each on its own feed, each a problem the corpus already solves, and each writing its method's structured log ('newtonLog <file>;', GibbsReactor.cpp / GibbsMethod.H): every Newton iterate's element potentials pi_k, ln N, element-balance residuals, step and ln n_i, the seed route, and the phase searches.

shift gibbs01's problem: CO + H2O <→ CO2 + H2, equimolar, 800 K, 1 bar. Four species over three elements → FOUR unknowns (pi_C, pi_H, pi_O, ln N). The least-squares seed converges in six Newton iterations; the first step is cut to 1/64 by the backtracking line search. coking gibbs11's steam-lean reformer: CH4 : H2O = 1 : 0.8, 1000 K, 1 bar, graphite declared. The fluid Newton runs WITHOUT the solid, reads a_C = exp(pi_C - g_C/RT) = 2.33 > 1, and bisects the deposited amount (37 fluid solves) until a_C = 1. ammonia equil03's feed (10 % NH3 in air) at 700 K and 5 bar, where the least-squares seed at the feed composition makes the Newton's Jacobian singular at iteration 0; the engine re-seeds from a primal Gibbs iteration on ln n and SAYS so (GibbsMethod.cpp).

Each reactor reproduces its source case's golden to every printed digit (the package here is the union of the three source packages – ideal gas, ideal liquid – and a Gibbs reactor reads only its own species list). The LIQUID step of the lesson is read from gibbs07_wgs_cooled, which declares its own 'newtonLog': a condensing reactor was left out of this witness because its energy balance is a pinned open debt (check_energy_closure.KNOWN_OPEN) and a new case must not copy one.

THE LOG IS AN OBSERVER. Remove every 'newtonLog' line and every number this case publishes is unchanged; the CSVs it writes are run outputs, gitignored like every CSV under tutorials/.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedAmmonia | F | 0.0277777777778 |
| stream | feedAmmonia | T | 700 |
| stream | feedCoking | F | 0.0005 |
| stream | feedCoking | T | 1000 |
| stream | feedShift | F | 0.000277777777778 |
| stream | feedShift | T | 800 |
| stream | outAmmonia | F | 0.0284722222222 |
| stream | outAmmonia | T | 700 |
| stream | outCoking | F | 0.000967550025636 |
| stream | outCoking | T | 1000 |
| stream | outShift | F | 0.000277777777778 |
| stream | outShift | T | 800 |
| kpi | ammonia | F_in_kmol_h | 100 |
| kpi | ammonia | F_out_kmol_h | 102.5 |

*... and 61 more reference values in the case's `expected`.*

## gibbs13_adiabatic_approach_direction

tutorials/steady/gibbs/gibbs13_adiabatic_approach_direction




**What it does:** An adiabatic ammonia synthesis bed fed a gas that already carries 16 % NH3 (the third bed of a staged converter), 150 bar, 10 K approach to equilibrium, seeded 80 K hotter than its answer: the engine reads the approach's SIGN from the dT = 0 adiabatic outlet it solves first, not from the seed, so the bed ends BELOW its own equilibrium whichever seed the case declares

**The story:** the SIGN of an adiabatic approach is read from the bed's own answer, never from the seed.

THE BED. The third bed of a staged ammonia converter: the feed already carries 16 % NH3 (N2 19, H2 57, NH3 16, Ar 8 mol%), enters at 415 C and 150 bar, and the bed is ADIABATIC with a 10 K approach to equilibrium (US 5,352,428's design band is 1-10 C; see ammoniaReactor03_equilibriumApproach for the source). 'operation.T' is the SEED of the outer Newton on the outlet temperature, and it is set 80 K HOTTER than the answer on purpose.

WHAT THIS CASE WITNESSES (DEV.md 5, 2026-10-01; fixed 2026-10-03). The engine assigns the approach's direction from the thermicity of the transformation feed → equilibrium. Until 2026-10-03, in adiabatic mode, it read that thermicity AT THE SEED. A feed that already carries product is itself AT equilibrium at some temperature (here ~487 C); seeded hotter than that, the equilibrium at the seed holds LESS ammonia than the feed, the transformation reads as DECOMPOSITION, the approach takes T - 10 K, and the bed ends MORE converted than its own equilibrium outlet – the aid helped instead of hurting, and a 60 K colder seed on the SAME case gave the other sign. Measured on this feed with the old rule: seed 768 K → 734.2 K and 19.32 % NH3; seed 708 K → 721.1 K and 18.36 %; the dT = 0 outlet is 727.6 K and 18.84 %, from either seed.

THE RULE NOW. The bed is solved ONCE with no approach – its own dT = 0 adiabatic outlet T0, which no seed can move – the thermicity of feed → THAT state decides the sign (exothermic here: the bed heats from 688 to T0), the announcement names T0 and the enthalpy change it read, and the bed is re-solved from T0 with the signed approach. So this case ends at the SAME outlet as its cold-seeded twin, BELOW its dT = 0 equilibrium in both NH3 and temperature (the reaction is exothermic), which is what a real bed does. Gate: check_adiabatic_approach_direction (it runs this case under a second seed and under dT = 0).

THE COST, published: 'approachProbeOuterIterations' is the dT = 0 solve's outer Newton count; 'outerIterations' the re-solve's, now seeded from T0. The physics is nothing new – the SAME Gibbs minimisation and the SAME energy balance as gibbs05 – which is the point: an aid's sign must not depend on where the solver started.

TOPOLOGY ONLY. Stream state lives on disk in 0/, one file per stream.

*THE CASE. One adiabatic gibbsReactor: the third bed of a staged ammonia converter, fed a gas that already carries 16 % NH3 (N2 19, H2 57, NH3 16, Ar 8 mol%; author-set to be representative of a last bed, not a plant measurement), entering at 415 °C and 150 bar, with a 10 K approach to equilibrium (the band US 5,352,428 calls "preferably 1 C. to 10 C."; see ammoniaReactor03_equilibriumApproach for the source). operation.T is the SEED of the outer Newton on the outlet temperature, and it is set to 495 °C — 80 K hotter than the answer — on purpose. The 0/out file is an initial estimate of the stream and is NOT that seed; the unit reads its seed from its own dict.*

*temperatureApproach is declared as a MAGNITUDE and the engine assigns its sign from the thermicity of the transformation feed → equilibrium (ruled 2026-09-26). In mode adiabatic the physical temperature is the answer of an outer Newton, and until 2026-10-03 the thermicity was read at the seed. A feed that already carries product is itself at equilibrium at some temperature (here about 487 °C). Seeded hotter than that, the equilibrium at the seed holds LESS ammonia than the feed, the transformation reads as decomposition (endothermic), the approach takes T - 10 K, and the bed ends MORE converted than its own equilibrium outlet — the aid helps instead of hurting. A 60 K colder seed on the SAME case took the other sign.*

*The hot-seeded bed published more ammonia than equilibrium allows at its own outlet, at exit 0, with the sign announced as if it were a finding about the chemistry.*

*The bed is solved once with no approach — its own dT = 0 adiabatic outlet T0, which no seed can move — the thermicity of feed → THAT state decides the sign (exothermic here: the bed heats from 688 K to T0), the announcement names T0, the seed the probe started from and the enthalpy change it read, and the bed is re-solved from T0 with the signed approach. So this case ends at the same outlet as its cold-seeded twin, 721.05 K and 18.36 % NH3, BELOW its dT = 0 equilibrium in both ammonia and temperature, which is what a real bed does. approachProbeOuterIterations publishes the cost of the dT = 0 solve (one extra outer Newton; outerIterations is the re-solve's count, now seeded from T0).*

*Isothermal mode is untouched: there the physical T is operation.T and the reading was at the answer all along.*

*check_adiabatic_approach_direction runs this case under its hot seed and under a cold seed and requires ONE outlet; runs it at dT = 0 and requires the approach outlet below it in NH3 and in T; reads T0 off the announcement and holds it to the dT = 0 run's own temperature. Record: docs/design/the-approach-direction-is-read-from-the-answer.md.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 2.77777777778 |
| stream | feed | T | 688.15 |
| stream | out | F | 2.72249892313 |
| stream | out | T | 721.051945619 |
| kpi | bed3 | F_in_kmol_h | 10000 |
| kpi | bed3 | F_out_kmol_h | 9800.99612328 |
| kpi | bed3 | N_in_mol_s | 2777.77777778 |
| kpi | bed3 | N_out_mol_s | 2722.49892313 |
| kpi | bed3 | P | 15000000 |
| kpi | bed3 | T | 721.051945619 |
| kpi | bed3 | approachProbeOuterIterations | 5 |
| kpi | bed3 | converged | 1 |
| kpi | bed3 | lambda_Ar | -100923.402463 |
| kpi | bed3 | lambda_H | -36944.0037435 |
| *... and 10 more reference values in the case's `expected`.* | | | |

## heat

**The theory you need.** Heat exchange couples streams without mixing them: $Q = U\,A\,\Delta T_{\mathrm{lm}}$, effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

*Exchanger DESIGN vs rating.* A shell-and-tube poses two inverse questions: RATE it (geometry given → find $U$, area, $\Delta P$, duty – `model geometry;`) or DESIGN it (duty given → size the geometry – `model design;`). $U$ is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` → `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, §heat-exchanger design.*

## condenser01_film_nusselt

tutorials/steady/heat/condenser01_film_nusselt




**What it does:** Horizontal-tube steam condenser: Nusselt film HTC → duty (geometry mode)

**The story:** A horizontal-tube steam condenser whose duty is NOT specified – it EMERGES from the physics of the condensing film, exactly as a real condenser's does.

Saturated steam at 1 bar enters a bundle of horizontal tubes cooled by cooling water. The 'phaseChanger' runs in 'model geometry;' (the seventh resolution): a Nusselt (1916) LAMINAR FILM coefficient h_cond is computed on the tube, the wall conducts, and the cooling-water film carries the heat away. Because the condensing-film coefficient itself depends on the driving dT_film = Tsat - T_wall, the unit solves a 1-D energy balance at the wall (q_cond == q_cool) – the wall temperature is the unknown. Then the duty is Q = h_cond ∗ A ∗ dT_film and the outlet is the dome-flash at that duty.

steam (saturated vapour, 1 bar, declared 'phase gas;') → condenser (model geometry: NusseltFilm + cooling-water film) → condensate (two-phase, vf ~ 0.55) + Q (RESULT)

PARTIAL CONDENSATION: with this tube area the duty Q ~ -507 kW removes only part of the latent heat, so the outlet is NOT fully condensed — it leaves at the dome as a wet two-phase stream, vf ~ 0.55 (about 45 % of the vapour condensed) at Tsat. Fully condensing it would need more area; here the area is fixed and the outlet quality is the result.

Method (glass-box): h_cond : Nusselt film, horizontal tube, coefficient 0.729 h = 0.729 [k_l^3 rho_l (rho_l-rho_v) g h_fg' /(mu_l D dT_film)]^1/4 h_fg' = h_fg + 0.68 cp_l dT_film (Rohsenow subcooling corr.) (Incropera & DeWitt, Fund. of Heat & Mass Transfer, Ex. 10.4) wall : carbon-steel cylindrical resistance r_o ln(r_o/r_i)/k coolant: a cooling-water film coefficient h (Dittus-Boelter order) the three resistances + which CONTROLS are printed; the duty is cross-checked against U∗A∗dT.

All film/saturation properties are IAPWS-IF97 water. The case carries its own constant/components/water.dat (case-local copy). WHY THE STEAM DECLARES ITS PHASE: 372.76 K is IF97's Tsat(1 bar), 372.756 K – the curve the film model AND, since DEV.md C39, the package's K-values read (the declared IF97 method owns water's dome). On its own saturation curve a pure stream's side is the last digits of T, which the engine refuses (DEV.md C36 item 1); 'phase gas;' states the saturated vapour this case means. ASSUMPTION stated plainly: the coolant is treated as ISOTHERMAL at its inlet T (its ~10 K heat-up is neglected, so dT = Tsat - T_in slightly OVERSTATES the true LMTD – the computed area errs a few % small). The utility allocation still reports the real cooling-water flow and duty.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | condensate | F | 0.0277777777778 |
| stream | steam | F | 0.0277777777778 |
| kpi | condenser | F | 0.0277777777778 |
| kpi | condenser | P | 100000 |
| kpi | condenser | Q | -506617.311982 |
| kpi | condenser | Q_UAdT_kW | 506.617311982 |
| kpi | condenser | Q_kW | -506.617311982 |
| kpi | condenser | R_coolant | 0.00014880952381 |
| kpi | condenser | R_film | 0.000349003587291 |
| kpi | condenser | R_wall | 4.35883467862e-05 |
| kpi | condenser | Re_film | 15.1166609728 |
| kpi | condenser | T_in | 372.76 |
| kpi | condenser | T_out | 372.755918611 |
| kpi | condenser | T_wall | 325.855275434 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## coolingTower01_merkel

tutorials/steady/heat/coolingTower01_merkel




**What it does:** Water cooling tower, Merkel enthalpy driving force – rating: the packing is given, the cold-water T is the result

**The story:** A counter-current evaporative cooling tower in RATING mode: the packing characteristic is declared (Me = KaV/L = 1.5) and the cold-water temperature is the RESULT. Hot water at 45 C meets ambient air at 25 C (Y = 0.01286 kg/kg dry, about 62 % RH on the N2 carrier), L/G ~ 0.99.

The whole construction lives on the Merkel diagram the unit publishes as its profile: air enthalpy h against water temperature T, the saturation curve h∗(T) above, the operating line h1 + (L cpL / G)(T - T_out) below, and Me the integral of cpL dT over the gap between them.

Method hypotheses (each announced at run time): Lewis = 1; the water loss is neglected INSIDE the integral but reported and honoured at the boundary (the coolWater outlet carries L_in minus the evaporation); the exit air is assumed saturated; cp at mean film temperatures.

What to look at in the output: T_water_out against the wet-bulb floor (T_wb ~ 293 K): the approach is the tower's real currency – a tower can never cool BELOW T_wb. merkelNumber_chebyshev4 beside the converged integral: the CTI four-point hand method against Simpson on 200 intervals. The gap is the price of the hand method, visible, not hidden. evaporation_kg_s = makeup_kg_s: the water the cooling COSTS. A tower cools ~1 % of L per ~7 K of range by evaporating it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | air | F | 0.0184166666667 |
| stream | coolWater | F | 0.027110090426 |
| stream | hotWater | F | 0.0277777777778 |
| stream | humidAir | F | 0.0190843540184 |
| kpi | tower01 | L_over_G | 0.98937581178 |
| kpi | tower01 | Q_kW | 36.4315830585 |
| kpi | tower01 | T_air_out | 307.613817692 |
| kpi | tower01 | T_water_out | 300.778649138 |
| kpi | tower01 | T_wb_in | 293.021697982 |
| kpi | tower01 | Y_in | 0.0128618855531 |
| kpi | tower01 | Y_out | 0.0366432593178 |
| kpi | tower01 | approach_K | 7.75695115581 |
| kpi | tower01 | evaporation_kg_s | 0.0120283876424 |
| kpi | tower01 | evaporation_pct_of_L | 2.40367446642 |
| *... and 14 more reference values in the case's `expected`.* | | | |

## heatExchanger01_water_water

`tutorials/steady/heat/heatExchanger01_water_water`




**What it does:** Two-stream heat exchanger (eps-NTU): hot/cold water recovery

**The story:** Two-stream heat exchanger by the effectiveness-NTU method (the built-in 'heatExchanger'). A hot water stream is cooled by a cold water stream in a counter-current unit — the canonical heat-recovery problem.

Hardware (the credo): area + U. The outlet temperatures and the duty are RESULTS. "I need T_hot_out = X" would be a DesignSpec on $area.

Hot : 100 kmol/h water at 360 K Cold : 120 kmol/h water at 300 K Hardware: A = 10 m^2, U = 500 W/(m^2.K), counter-current

Expected (eps-NTU, Cp_water ~ 75 J/mol/K): C_hot = 100/3.6 * 75 ~ 2083 W/K (= C_min) C_cold = 120/3.6 * 75 ~ 2500 W/K NTU = U.A/C_min ~ 2.4, C_r ~ 0.83, effectiveness ~ 0.75 Q ~ 93 kW T_hot_out ~ 315 K (cooled ~45 K) T_cold_out ~ 337 K (heated ~37 K) LMTD ~ 18.8 K, and U.A.LMTD ~ Q (cross-check: 5000 * 18.76 = 93.8 kW) energy balance: hot stream loses what the cold stream gains.

*Hot water (100 kmol/h, 360 K) against cold water (120 kmol/h, 300 K) in a counter-current exchanger of 10 $m^2$ at U = 500 $W/m^2K$. Nothing about the outlet temperatures is declared; the ε-NTU method finds them. The golden: Q = 93.79 kW, hot out 315.28 K, cold out 337.27 K, NTU 2.384, C_r 0.833, ε = 0.745, LMTD 18.76 K.*

*1. Rating, not design. The hardware (area, U, flow arrangement) is given in operation {}; the question is what it ∗does∗ with these two streams. hxWorkflow1_design_from_duty asks the reverse. 2. ε-NTU in four printed numbers. The Log tab prints NTU, C_r, ε and Q on one line, and then checks itself: U·A·LMTD = 93.79 kW beside the ε-NTU result. Two routes, one answer — that agreement is the method being right, and you can redo either on paper. 3. The smaller heat-capacity rate sets the limit. C_hot = 2097 W/K is the minimum (fewer moles), so ε is measured on the hot stream: (360 - 315.28)/(360 - 300) = 0.745. Read C_hot and C_cold in the KPIs and confirm C_r. 4. No utility is consumed. Both sides are process streams; the 93.79 kW is ∗recovered∗, which is the whole reason to install an exchanger rather than a heater and a cooler.*

*Switch flow counter; to flow co; and run again: same area, same U, smaller ε, less heat — the classic comparison, in one word of the dict.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | cold | F | 0.0333333333333 |
| stream | coldOut | F | 0.0333333333333 |
| stream | hot | F | 0.0277777777778 |
| stream | hotOut | F | 0.0277777777778 |
| kpi | hx | C_cold | 2516.66666667 |
| kpi | hx | C_hot | 2097.22222222 |
| kpi | hx | C_r | 0.833333333333 |
| kpi | hx | LMTD | 18.7584806629 |
| kpi | hx | NTU | 2.38410596026 |
| kpi | hx | Q | 93792.4033146 |
| kpi | hx | Q_kW | 93.7924033146 |
| kpi | hx | T_cold_in | 300 |
| kpi | hx | T_cold_out | 337.268504628 |
| kpi | hx | T_hot_in | 360 |

*... and 13 more reference values in the case's `expected`.*

## heatExchanger02_geometry_U

tutorials/steady/heat/heatExchanger02_geometry_U




**What it does:** Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band (~2000-3000 W/(m2.K), Kern / Sinnott). Tube-side h_i via Gnielinski (Incropera & DeWitt 6th ed.); shell-side h_o via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

**The story:** A shell-and-tube heat exchanger whose overall coefficient U and area are COMPUTED from the bundle geometry plus per-side convective correlations (the ‘model geometry;‘ path), instead of being specified. This is the glass-box answer to "where does U come from?".

Hot water (shell side) is cooled by cold water (tube side), clean service.

Method: tube-side h_i : Gnielinski Nu = (f/8)(Re-1000)Pr/[1+12.7 sqrt(f/8)(Pr^2/3-1)] (Incropera & DeWitt, Fund. of Heat and Mass Transfer, 6th ed.) shell-side h_o: Kern method Nu = 0.36 Re_s^0.55 Pr^1/3 (Kern, Process Heat Transfer, 1950) on the bundle equivalent diameter D_e and crossflow mass flux G_s. U on the OUTSIDE area: 1/U = 1/h_o + r_o ln(r_o/r_i)/k_wall + r_o/(r_i h_i) area = pi d_o L n_tubes PRESSURE DROP (Kern, both sides): tube f = 0.079 Re^-0.25 (Blasius), dP = (4 f L n_p/d_i + 4 n_p)(rho u^2/2) – friction + turnaround losses; shell f = exp(0.576 - 0.19 ln Re_s), dP = f G_s^2 D_s (N_b+1) /(2 rho D_e) over the N_b baffles. (Kern, Process Heat Transfer, 1950.)

Properties are IF97 water (rho, mu, lambda) at each stream temperature.

Expected order of magnitude (clean water-water): U ~ 2000-3000 W/(m2.K), the SHELL side controlling, both sides turbulent (Re ~ 2-4 e4); tube-side dP ~ 9 kPa, shell-side dP ~ 35 kPa (shell > tube: baffled crossflow is more dissipative – the duty-vs-pumping trade-off you design around, NOT just the area). The exact numbers are pinned by –record – they are RESULTS.

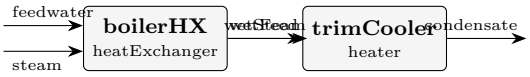
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | coldOut | F | 2.22027777778 |
| stream | coldTube | F | 2.22027777778 |
| stream | hotOut | F | 1.66527777778 |
| stream | hotShell | F | 1.66527777778 |
| kpi | hx | C_cold | 167630.972222 |
| kpi | hx | C_hot | 125728.472222 |
| kpi | hx | C_r | 0.750031277368 |
| kpi | hx | LMTD | 15.4921080564 |
| kpi | hx | NTU | 1.26058061182 |
| kpi | hx | Nu_shell | 171.621364606 |
| kpi | hx | Nu_tube | 287.732215939 |
| kpi | hx | Pr_shell | 3.16405821902 |
| kpi | hx | Pr_tube | 5.87094963164 |
| kpi | hx | Q | 2065604.42087 |

*... and 31 more reference values in the case's `expected`.*

## heatExchanger03_pure_water_plateau

tutorials/steady/heat/heatExchanger03_pure_water_plateau




**What it does:** Pure water on its saturation plateau: a 40 bar boiler, a 100 bar partial condenser and a trim cooler

**The story:** ONE component, two phases, and the one state a temperature and a pressure cannot name.

A pure fluid at fixed P boils at ONE temperature, Tsat(P). Its enthalpy H(T) is therefore not a smooth curve: it JUMPS by the whole latent heat at Tsat. Ask an exchanger for an outlet whose enthalpy lies inside that jump and there is no temperature that answers – the answer is T = Tsat and a VAPOUR FRACTION q = (H - h_L)/(h_V - h_L). And because (T, P) sit on the saturation curve, they do not fix q: the stream must CARRY it, as a pin – the one exception to "a stream is fixed by two variables", and the only stream that may declare T, P AND a vapour fraction.

Until 2026-10-05 the exchanger's bisection on H(T) closed onto Tsat and published the nearer single phase – short of its own duty by the unboiled fraction's latent heat, at exit 0 (measured on 'utility02_hitec_csp_heater' at 40 bar: 3.06 kW, 1.5 % of its duty).

Here (counter-current, A = 2 m^2, U = 1000 W/(m^2.K)): 'boilerHX' – 100 bar superheated steam (700 K) against 40 bar feedwater (480 K). The duty comes from eps-NTU; the steam leaves PARTLY CONDENSED and the feedwater PARTLY BOILED, both at their own Tsat with a pinned quality. A boiler and a partial condenser in one unit. 'trimCooler' – a 'heater' with Q < 0 on the condensing steam. Its inlet is the exchanger's pinned two-phase outlet (the pin is read, not re-resolved) and its outlet lands on the plateau again: the temperature stays at Tsat and only the quality falls.

What to read: the Log's 'PURE fluid on its saturation plateau' lines; the stream table's vf column (two numbers strictly between 0 and 1 at a T that does not move); and the energy balance, where each unit closes to the duty it declares.

STRUCTURAL WITNESS, NOT A VALIDATION. Every number here comes from this case's own water record: an Antoine fit stated for 273-373 K (so Tsat at 40 and 100 bar is EXTRAPOLATED, and announced), an ideal-gas vapour, and the Watson latent heat. Tsat(40 bar) is the record's 515.6 K, not the steam tables' 523.5 K. What the case pins is the MECHANICS – the outlet sits at Tsat with the quality the first law demands, and the balances close – never the steam properties.

*Superheated steam at 100 bar, 700 K against feedwater at 40 bar, 480 K, 100 kmol/h each, in a counter-current exchanger of 2 $m^2$ at U = 1000 $W/m^2K$; then a heater takes 500 kW more out of the steam. Every stream carries ONE component, water.*

*1. A pure fluid boils at one temperature. At fixed pressure its enthalpy H(T) is a step at Tsat(P): every joule between the saturated liquid and the saturated vapour goes in AT Tsat. So when the duty asks for an enthalpy inside that step, the outlet temperature is Tsat and what changes is the vapour fraction, q = (H - h_L)/(h_V - h_L). 2. Both directions in one unit. The exchanger's duty (from ε-NTU) partly condenses the steam (it leaves at Tsat(100 bar) with q ≈ 0.95) and partly boils the feedwater (Tsat(40 bar), q ≈ 0.10). The Log prints a PURE fluid on its saturation plateau line for each side. 3. Why the stream carries q as a pin. For a mixture, T and P between the bubble and dew points fix the split. For ONE component on its curve they do not — any q is consistent with them — so the stream must say which. Open converged/wetFeed: T, P and vaporFraction, the one case where a stream legitimately declares all three. 4. The temperature does not move, the quality does. trimCooler removes 500 kW from the wet steam: T_out equals T_in (to the bisection's precision) and q falls from 0.95 to 0.42. 5. Nothing is lost. The energy balance closes every unit to the duty it declares, and the plant boundary to zero.*

*This is a STRUCTURAL witness. The water record's Antoine fit is stated for 273–373 K, so Tsat at 40 and 100 bar is extrapolated (and announced): Tsat(40 bar) here is 515.6 K, not the steam tables' 523.5 K. The vapour is an ideal gas and the latent heat is Watson's. What the case pins is the mechanics — the outlet sits at Tsat with the quality the first law demands — never the steam properties.*

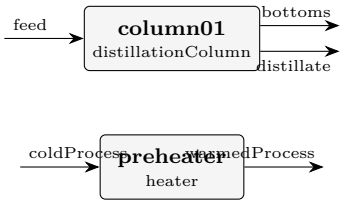
*Raise the area to 4 $m^2$: more feedwater boils, more steam condenses, and both outlet temperatures stay exactly where they were.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | condensate | F | 0.0277777777778 |
| stream | condensate | T | 572.172481697 |
| stream | feedwater | F | 0.0277777777778 |
| stream | feedwater | T | 480 |
| stream | steam | F | 0.0277777777778 |
| stream | steam | T | 700 |
| stream | wetFeed | F | 0.0277777777778 |
| stream | wetFeed | T | 515.579742002 |
| stream | wetSteam | F | 0.0277777777778 |
| stream | wetSteam | T | 572.172481734 |
| kpi | boilerHX | C_cold | 2097.22222222 |
| kpi | boilerHX | C_hot | 1016.46772691 |
| kpi | boilerHX | C_r | 0.484673353227 |
| kpi | boilerHX | H_closure_gap_kW | -8.24229573482e-13 |
| *. . . and 29 more reference values in the case's `expected`.* | | | |

## heatlink01_condenser_to_heater

tutorials/steady/heat/heatlink01_condenser_to_heater




**What it does:** Forward heat-link: column condenser preheats a downstream stream

**The story:** feed column01 distillate bottoms (condenser heat) preheater warmedProcess coldProcess

The preheater is heated by the column's rejected condenser duty (~1.28 MW), warming a 1500 kmol/h benzene stream from 320 K by ~23 K. column01 is listed FIRST so it solves before the preheater reads its condenser KPI (forward heat-link).

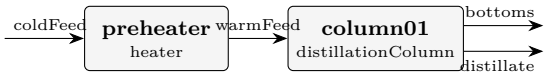
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | coldProcess | F | 0.416666666667 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| stream | warmedProcess | F | 0.416666666667 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0277777777778 |
| kpi | column01 | L_strip | 0.0361877253979 |
| kpi | column01 | Q_condenser_kW | -1315.28783372 |
| kpi | column01 | Q_reboiler_kW | 679.510516909 |
| kpi | column01 | R | 2 |
| kpi | column01 | T_bottom | 381.514100345 |
| kpi | column01 | T_top | 355.689111404 |

*... and 30 more reference values in the case's `expected`.*

## heatlink02_condenser_feed_preheat

tutorials/steady/heat/heatlink02_condenser_feed_preheat




**What it does:** Feedback heat-link (energy tear): column condenser preheats its OWN feed

**The story:** FEEDBACK heat-link == an ENERGY TEAR.

coldFeed preheater column01 distillate bottoms (condenser heat)

The preheater (UPSTREAM of the column) is heated by the column's CONDENSER duty (DOWNSTREAM) → circular. The Flowsheet detects the feedback, turns the recycle outer loop on, and converges the duty by successive substitution (Wegstein). Sized so the ~1 MW condenser preheats the large 1000 kmol/h feed by a modest ~25 K (stays liquid).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.263888888889 |
| stream | coldFeed | F | 0.277777777778 |
| stream | distillate | F | 0.0138888888889 |
| stream | warmFeed | F | 0.277777777778 |
| kpi | column01 | B | 0.263888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0208333333333 |
| kpi | column01 | L_strip | 0.320764193737 |
| kpi | column01 | Q_condenser_kW | -1090.04792065 |
| kpi | column01 | Q_reboiler_kW | 1844.93950415 |
| kpi | column01 | R | 1.5 |
| kpi | column01 | T_bottom | 365.997911437 |
| kpi | column01 | T_top | 354.386818521 |
| kpi | column01 | V_rect | 0.0347222222222 |

*... and 28 more reference values in the case's `expected`.*

## hx01_heater

tutorials/steady/heat/hx01_heater




**What it does:** Heater: 50/50 ethanol/water from 300 K to ~338 K (sensible)

**The story:** Stand-alone heater — credo-pure (v0.34+).

Hardware spec: Q = 100 kW (the absolute thermal power delivered by the heater – think of it as the electrical rating).

Streams IN → OP (Q only) → Streams OUT

Feed is 100 kmol/h of 50/50 ethanol/water at 300 K. The simulator solves the enthalpy balance H_out = H_in + Q/F to compute T_out as a RESULT. With Q = 100 kW the outlet temperature lands near 338 K (Cp_mix ~ 94 J/(mol∗K) at this composition); the exact value falls out of the Newton-1D in T.

NOTE (duty reduced 160 → 100 kW). At 160 kW the outlet reached ~360 K, ABOVE the 50/50 ethanol/water bubble point at 1 atm – the old single-phase heater silently reported a physically-impossible "superheated liquid". The new saturation guard refuses that (a heater is sensible-only); a duty that boils belongs to a ‘phaseChanger‘ / ‘boiler‘ (see steady/heat/phasechange01_partial_condenser and steady/power/rankine02_water). 100 kW keeps this the clean sensible-heat lesson it was meant to be.

If you instead need a specific T_out (say "I want exactly 338 K"), wrap the heater in a DesignSpec outerDict that manipulates $Q with a target on ‘heater01.T_out‘. The unit op itself never specifies its outlet state – that is the credo we adopted in v0.33.

*50/50 ethanol/water (100 kmol/h) at 300 K enters a heater that delivers 100 kW. The dict says how much heat; the energy balance says how hot: T_out = 338.34 K, a 38.3 K rise, and the stream stays liquid (vf_out = 0).*

*1. Hardware is declared, results are solved. operation { Q 100.0 kW; } is the only number the unit takes. T_out is ∗found∗ — the Log tab shows a Newton on the outlet temperature closing the enthalpy balance H_out = H_in + Q. 2. 3600 J/mol is the whole story in one number. Q per mole of feed (Q_per_mol) divided by the mixture's liquid Cp gives the rise; read the Cp's the run used in the Props tab and check 38 K by hand. 3. Somebody has to supply those 100 kW. The run's result records which catalogue utility was picked to serve the duty at the outlet temperature: low-pressure steam, 0.0457 kg/s, at the catalogue's price 4.32 ?/h. That choice is the same rule every steady report uses (pickForDuty); the price is a catalogue number, not a market one, and data/standards/utilities/ is where it lives. 4. Where the stream came from matters. The feed is a subcooled liquid; heat it far enough and it starts to boil — the heater will say so with a vapour fraction, not by refusing.*

*Raise Q until vf_out leaves zero: the outlet temperature stops climbing as fast, because the heat is now going into vaporisation. phasechange01_partial_condenser is the same balance run in reverse.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | hotStream | F | 0.0277777777778 |
| kpi | heater01 | F | 0.0277777777778 |
| kpi | heater01 | P | 101325 |
| kpi | heater01 | Q | 100000 |
| kpi | heater01 | Q_kW | 100 |
| kpi | heater01 | Q_per_mol | 3600 |
| kpi | heater01 | T_in | 300 |
| kpi | heater01 | T_out | 338.338658147 |
| kpi | heater01 | dT | 38.338658147 |
| utility | heater01 | heating.steamLP.duty_kW | 100 |
| utility | heater01 | heating.steamLP.T | 338.338658146 |
| utility | heater01 | heating.steamLP.kg_s | 0.0457456541629 |
| utility | heater01 | heating.steamLP.eur_h | 4.32 |
| *… and 6 more reference values in the case's `expected`.* | | | |

## hxWorkflow1_design_from_duty

tutorials/steady/heat/hxWorkflow1_design_from_duty




**What it does:** PART 1 of 2 – DESIGN: the DUTY TARGET is given (cool the hot shell 330→315 K) and the exchanger GEOMETRY is SOLVED for (model design;). Glass-box Kern sizing: fix the tube choices, then find the tube count (and the shell diameter it implies, Kern/Sinnott bundle) so U∗A = Q_req/LMTD; U/dP then rated with Gnielinski (tube) + Kern (shell). The detailed-design (sizing) step, glass-box. Part 2 (hxWorkflow2_rate_designed) rates the bundle this case produces.

**The story:** PART 1 of 2: DESIGN from the duty

THE HEAT-EXCHANGER DESIGN WORKFLOW (the standard design sequence), part 1. DESIGN mode (‘model design;‘) – detailed sizing, glass-box. Instead of GIVING the tube count and shell diameter (that is ‘model geometry;‘, the RATING problem: geometry → U), here we give the DUTY TARGET (cool the hot shell stream from 330 K to 315 K) and the engine SOLVES for the geometry: fix the tube choices (OD/ID/length/pitch/passes/material), then find the number of tubes – and the shell diameter it implies (Kern/Sinnott bundle correlation, the chosen pattern + passes) so the exchanger DELIVERS THE DUTY – the design loop sizes N by the eps-NTU duty (incl. the 1-2 penalty for a 2-pass U-tube), not merely U∗A = Q/LMTD, so the outlet actually reaches the target. It then rates that designed bundle exactly like a given one (same Gnielinski tube + Kern shell correlations) and reports U, area, and the pressure drop.

DO I HAVE TO SPECIFY EVERYTHING? No – and this is the whole point. A HX poses two OPPOSITE questions, and this is the DESIGN one:

RATING (‘model geometry;‘): you GIVE the full hardware (tube count, shell diameter, ...) and ask "what outlet T do I get?". The temperature is a RESULT. DESIGN (‘model design;‘, THIS case): you GIVE the outlet T you NEED and ask "what hardware delivers it?". The GEOMETRY is the result.

So HOW does the outlet reach the temperature you need? You do NOT tune the geometry by hand until it works. You DECLARE the target and the engine sizes the hardware to hit it. Look at the two blocks below:

‘design { spec { stream hotShell; T 315 K; } }‘ <– WHAT YOU WANT: the required outlet temperature. THIS is where the final temperature is set. ‘geometry { tubeID ...; passes 2; tubePattern ...; wallMaterial ...; }‘ <– the FIXED CHOICES a designer commits to (a standard tube, the number of passes, the layout, the metal). Notice what is MISSING: there is NO ‘nTubes‘ and NO ‘shellID‘. Those are exactly what the engine SOLVES.

So the student specifies the REQUIREMENT (the outlet T) + the design CHOICES (tube size, passes, pattern, material) – NOT the answer. The engine returns the tube count, the shell diameter, U, the pressure drop, and confirms the outlet lands on the target. You said "cool this to 315 K with 1/2-inch steel tubes in a 2-pass U-tube"; it answers "then you need 105 tubes in a 0.31 m shell – here is U, the dP, and yes it leaves at 315 K".

This is the workflow that matches industrial practice: establish the DUTY, THEN design the exchanger from it – you do NOT invent the geometry up front. (Standard practice splits this into a shortcut that gets the duty + a detailed-design step that gets the geometry; Choupo’s design mode folds both into one honest step – it computes the duty from your target and ANNOUNCES it, then sizes.) PART 2 (hxWorkflow2_rate_designed) then RATES the bundle this case produced – a detailed-rating step – to verify it and probe off-design (turn the coolant down and watch the outlet drift off the target; the geometry is now FIXED, only the performance moves).

Water-water, clean service, 2-pass U-tube (the general arrangement). Expect: ~100 tubes, a ~0.3 m shell; rated U ~ 4000 W/(m2.K) (2 tube passes → high tube velocity), and the delivered hot outlet 315 K (the target, reproduced by the 1-2 eps-NTU on the designed geometry). Numbers are RESULTS pinned by –record. Single-phase; the 1-shell/2-tube-pass eps-NTU carries the built-in LMTD F-correction (F<1), so a 2-pass needs more area than pure counter-current.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | coldOut | F | 2.22027777778 |
| stream | coldTube | F | 2.22027777778 |
| stream | hotOut | F | 1.66527777778 |
| stream | hotShell | F | 1.66527777778 |
| kpi | hx | C_cold | 167630.972222 |
| kpi | hx | C_hot | 125728.472222 |
| kpi | hx | C_r | 0.750031277368 |
| kpi | hx | LMTD | 16.7490788864 |
| kpi | hx | NTU | 1.0053658663 |
| kpi | hx | Nu_shell | 195.331265319 |
| kpi | hx | Nu_tube | 416.603846461 |
| kpi | hx | Pr_shell | 3.16405821902 |
| kpi | hx | Pr_tube | 5.87094963164 |
| kpi | hx | Q | 1888506.69951 |

*. . . and 31 more reference values in the case's* `expected`*.*

## hxWorkflow2_rate_designed

tutorials/steady/heat/hxWorkflow2_rate_designed




**What it does:** Shell-and-tube heat exchanger sized from GEOMETRY: the overall U and area are COMPUTED from the tube bundle + per-side convective correlations (Gnielinski tube-side, Kern shell-side), not specified. Water-water clean service. The computed U lands in the textbook clean water-water band (~2000-3000 W/(m2.K), Kern / Sinnott). Tube-side h_i via Gnielinski (Incropera & DeWitt 6th ed.); shell-side h_o via the Kern method (Process Heat Transfer, 1950). U and area are RESULTS that then feed the unchanged eps-NTU duty.

**The story:** PART 2 of 2: RATE the designed exchanger

THE HEAT-EXCHANGER DESIGN WORKFLOW (the standard design sequence), part 2.

Part 1 (hxWorkflow1_design_from_duty) gave the engine a DUTY TARGET (cool the hot shell 330 → 315 K) and it SIZED the bundle: 105 tubes, shell ID 0.306 m (Kern/Sinnott, 2-pass U-tube). That is the detailed-design step.

Part 2 – THIS case – is the detailed-rating / checking step: take the DESIGNED geometry as a FIXED given ('model geometry;') and RATE it – confirm it delivers the target outlet, and read the honest pressure drop. This is the verification a real project always does after sizing: the design loop used a bundle correlation for the shell diameter and rounded the tube count UP, so the rated exchanger is slightly OVER-surfaced – the outlet comes out a touch below 315 K (good: margin), and that is what you check here.

OFF-DESIGN (try it): drop the cold-tube flow F to ~6000 kmol/h and re-run – less coolant → the hot stream leaves WARMER than 315 K, and the shell dP falls. That is the rating question ("will my exchanger cope when the plant turns down?") the design step cannot answer. The geometry is now FIXED; only the performance moves.

Numbers are RESULTS pinned by –record. Same clean water-water service, same Gnielinski (tube) + Kern (shell) correlations, single-phase, 2-pass U-tube (1-2 eps-NTU).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | coldOut | F | 2.22027777778 |
| stream | coldTube | F | 2.22027777778 |
| stream | hotOut | F | 1.66527777778 |
| stream | hotShell | F | 1.66527777778 |
| kpi | hx | C_cold | 167630.972222 |
| kpi | hx | C_hot | 125728.472222 |
| kpi | hx | C_r | 0.750031277368 |
| kpi | hx | LMTD | 16.7498740107 |
| kpi | hx | NTU | 1.00522927337 |
| kpi | hx | Nu_shell | 195.283521062 |
| kpi | hx | Nu_tube | 416.603846461 |
| kpi | hx | Pr_shell | 3.16405821902 |
| kpi | hx | Pr_tube | 5.87094963164 |
| kpi | hx | Q | 1888394.47754 |

*... and 31 more reference values in the case's `expected`.*

## mheatx01_two_hot_one_cold

tutorials/steady/heat/mheatx01_two_hot_one_cold




**What it does:** Multi-stream heat exchanger: 2 hot + 1 cold, balance closes

**The story:** Multi-stream heat exchanger (the commercial multi-stream detailed-rating analogue): TWO hot water streams are cooled by ONE cold water stream inside a single adiabatic shell — the canonical multi-stream heat-recovery problem.

THE SPEC (glass-box, "information follows the streams"): every inlet carries its OWN outlet target temperature. The unit does NOT rate hardware; it VERIFIES the first law:

Q_k = F_k (h_k,out - h_k,in) per stream Q_hot = Σ Q_k over the streams that cool (< 0, released) Q_cold = Σ Q_k over the streams that heat (> 0, absorbed) imbalance = Σ Q_k over ALL streams (external duty)

For a truly adiabatic shell the imbalance is ~0; a non-zero value is the external duty the topology would have to supply, REPORTED honestly (never a silently-invented duty). The unit also builds the hot/cold composite curves and reports the INTERNAL minimum-approach temperature (the in-unit pinch).

Streams (water, liquid): hot1 : 100 kmol/h, 380 → 330 K (gives up ~105 kW) hot2 : 60 kmol/h, 360 → 330 K (gives up ~38 kW) cold : 340 kmol/h, 300 → 320 K (takes up ~143 kW)

The cold flow is sized so the bundle is adiabatic: cold absorbs what the two hot streams release. Both hot outlets (330 K) stay above the cold outlet (320 K), so the composite curves never cross — dT_min > 0.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | cold | F | 0.0944444444444 |
| stream | coldOut | F | 0.0944444444444 |
| stream | hot1 | F | 0.0277777777778 |
| stream | hot1Out | F | 0.0277777777778 |
| stream | hot2 | F | 0.0166666666667 |
| stream | hot2Out | F | 0.0166666666667 |
| kpi | mhx | Q_cold | 142611.111111 |
| kpi | mhx | Q_cold_kW | 142.611111111 |
| kpi | mhx | Q_hot | -142611.111111 |
| kpi | mhx | Q_hot1_kW | -104.861111111 |
| kpi | mhx | Q_hot2_kW | -37.75 |
| kpi | mhx | Q_hot_kW | -142.611111111 |
| kpi | mhx | balanceCloses | 1 |
| kpi | mhx | dT_min | 30 |

*... and 13 more reference values in the case's `expected`.*

## phasechange01_partial_condenser

tutorials/steady/heat/phasechange01_partial_condenser




**What it does:** Partial condenser: benzene/toluene vapour cooled to quality 0.4

**The story:** A superheated benzene/toluene vapour enters a condenser that is asked to land its outlet at a vapour QUALITY of 0.4 (40 % of the moles stay vapour, 60 % condense). The duty Q is the RESULT – the unit solves it as a built-in self-target (no outerDict, no variables block) and ANNOUNCES that it did so. Because the mixture is multicomponent, the condensed liquid is toluene-RICH and the remaining vapour benzene-RICH – the partial- condensation lesson: a partial condenser is also a SEPARATION.

feedVapour (superheated, 1 bar, 390 K, vf = 1) → condenser (outletQuality 0.4) → wetOut (two-phase, 0 < vf < 1)

Same outlet vocabulary as rankine02_water's IF97 boiler/condenser, on the GENERIC (Antoine + Raoult) backend.

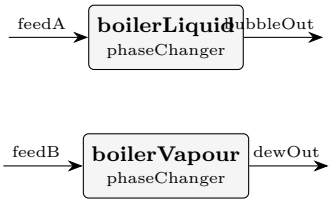
A NOTE THIS CASE EARNED THE HARD WAY (2026-09-12). The line above has said "feedVapour (superheated, 1 bar, 390 K, vf = 1)" since the case was written, and '0/feedVapour' did not say it – the file declared T and P and no phase. Prose is not a channel the engine reads. So 'phaseChanger' took vf = 0, and a unit whose entire subject is condensation published Q_latent_kW = +370.98: a POSITIVE latent term, with the sensible term carrying -1001.49 to make the total duty come out right. The total was right, both halves were wrong, and the plant's first law was out by -898.64 kW at exit 0 – pinned at 142.526 % in 'check_energy_closure' and looked at by nobody. '0/feedVapour' now carries 'phase gas;', the run closes at 0.0000 kW, and that pin is GONE rather than re-measured. Record: docs/design/the-word-that-was-not-there.md

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | condenser | vf_out | 0.399999998746 |
| kpi | condenser | T_out | 367.494289759 |
| kpi | condenser | Q_kW | -630.508868406 |
| kpi | condenser | Tsat | 352.826063711 |
| stream | wetOut | F | 0.0277777777778 |
| stream | wetOut | vf | 0.399999998746 |
| utility | condenser | cooling.coolingWater.duty_kW | -630.508868406 |
| utility | condenser | cooling.coolingWater.T | 367.494289759 |
| utility | condenser | cooling.coolingWater.kg_s | 15.0839442202 |
| utility | condenser | cooling.coolingWater.eur_h | 0.907932770504 |
| stream | feedVapour | F | 0.0277777777778 |
| stream | feedVapour | T | 390 |
| stream | wetOut | T | 367.494289759 |
| kpi | condenser | F | 0.0277777777778 |
| *... and 10 more reference values in the case's `expected`.* | | | |

## phasechange02_saturated_mixture

tutorials/steady/heat/phasechange02_saturated_mixture




**What it does:** Saturated liquid and vapour of a benzene-toluene mixture: bubble and dew point

**The story:** a saturated state of a MIXTURE is its bubble point or its dew point

The same 40/60 benzene/toluene liquid (298 K, 2.2 bar) goes to two phaseChangers:

feedA → boilerLiquid (outletState saturatedLiquid) → bubbleOut feedB → boilerVapour (outletState saturatedVapour) → dewOut

A pure fluid is saturated at ONE temperature; a mixture is saturated over a RANGE, between its bubble point (the first vapour, vf = 0) and its dew point (the last liquid, vf = 1). "Saturated liquid" is the bubble point and "saturated vapour" is the dew point – two different temperatures, here 397.3 K and 403.4 K at 2.2 bar (ideal Raoult). The ‘Tsat(P_out)‘ line the unit prints is the DOMINANT component's own saturation T (413.5 K, pure toluene), a dome centre for the search – it is neither of the two. The outlet must land ON them: the saturated liquid at the bubble point is HEATED from the 298 K feed (Q > 0), never cooled.

WHY THIS CASE EXISTS (DEV.md 4c C202). The unit once located the vapour fraction target by scanning T for a sign change of vf(T) - target. For a target of 0 that function is EXACTLY zero below the bubble point, so the first scan interval was accepted as a root and the outlet landed about 119 K BELOW the dome centre (103 K below the true bubble point) – a sub-cooled liquid labelled saturated, with the duty the wrong sign (-15 kW), at exit 0. The saturation points now come from the package's bubble-point and dew-point solvers, and the result is checked against the unit's own flash. The gate check_phase_changer_saturation reads this case and holds T_out to an independent bubble/dew point.

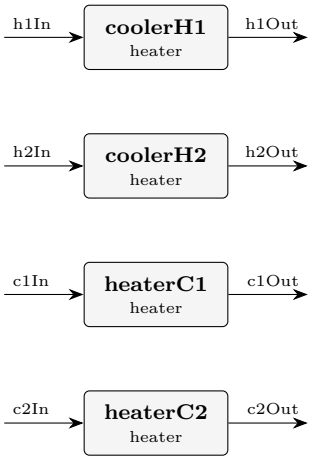
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bubbleOut | F | 0.0277777777778 |
| stream | bubbleOut | T | 397.261090159 |
| stream | dewOut | F | 0.0277777777778 |
| stream | dewOut | T | 403.434181212 |
| stream | feedA | F | 0.0277777777778 |
| stream | feedA | T | 298 |
| stream | feedB | F | 0.0277777777778 |
| stream | feedB | T | 298 |
| kpi | boilerLiquid | F | 0.0277777777778 |
| kpi | boilerLiquid | P | 220000 |
| kpi | boilerLiquid | Q | 409065.981556 |
| kpi | boilerLiquid | Q_kW | 409.065981556 |
| kpi | boilerLiquid | Q_sensible_kW | 409.065981556 |
| kpi | boilerLiquid | T_in | 298 |

*... and 25 more reference values in the case's `expected`.*

## pinch01_four_stream_classic

tutorials/steady/heat/pinch01_four_stream_classic




**What it does:** Pinch P1 targets: classic 4-stream problem table (dTmin 20 K)

**The story:** Four independent duty units realise the classic four-stream pinch population. The working fluid (hxFluid, case-local SYNTHETIC teaching fluid) has a CONSTANT liquid cp = 100 J/(mol K), so each stream's CP = F∗cp is exact and the targets are hand-checkable:

stream F [mol/s] CP [kW/K] T_in [C] T_out [C] Q [kW] H1 hot 20 2.0 150 60 -180.0 H2 hot 80 8.0 90 60 -240.0 C1 cold 25 2.5 20 125 +262.5 C2 cold 30 3.0 25 100 +225.0

Problem table at dTmin = 20 K (shift hot -10 K, cold +10 K; shifted boundaries in C, descending: 140 135 110 80 50 35 30; a positive dH is a heat SURPLUS cascading downward):

interval T∗ CP_hot CP_cold dH [kW] cascade [kW] 140 → 135 2 0 +10.0 +10.0 135 → 110 2 2.5 -12.5 -2.5 110 → 80 2 5.5 -105.0 -107.5 <- minimum 80 → 50 10 5.5 +135.0 +27.5 50 → 35 0 5.5 -82.5 -55.0 35 → 30 0 2.5 -12.5 -67.5

Targets: Q_H,min = -min(cascade) = 107.5 kW, Q_C,min = final + Q_H,min = -67.5 + 107.5 = 40 kW,

pinch at T∗ = 80 C → hot 90 C / cold 70 C.

Each unit is a 'heater' with a SIGNED hardware duty Q; the outlet T is the enthalpy-balance RESULT (with cp constant it lands exactly on the table's target temperatures). The four units are thermally independent here – the pinchPass only TARGETS and ANALYSES; the network is never rewritten.

P2 candidate-match table (hand-worked; pinch hot 90 C / cold 70 C, segments clipped at the pinch; Q_max = min(Q_hot, Q_cold, min(CP_h,CP_c)∗(Th_hi - Tc_lo - dTmin))):

region hot cold CP rule at the pinch Q_max [kW] limited by above H1 C1 2.0 <= 2.5 holds 120 approach (and H1's 120 kW) above H1 C2 2.0 <= 3.0 holds 90 C2's above-duty below H1 C1 2.0 >= 2.5 VIOLATED 60 H1's below-duty (away from pinch only) below H1 C2 2.0 >= 3.0 VIOLATED 60 H1's below-duty (away from pinch only) below H2 C1 8.0 >= 2.5 holds 125 approach == C1's below-duty below H2 C2 8.0 >= 3.0 holds 135 approach == C2's below-duty

Violation diagnostics (each unit is utility-served here, so every duty on the wrong side of the pinch is a coursework violation): cooler above the pinch: H1's 120 kW above 90 C; heaters below the pinch: C1's 125 kW + C2's 135 kW below 70 C. Identity check: 120 + 260 = 380 kW = Q_heat_current - Q_H_min = 487.5 - 107.5 – the excess over target IS the sum of the violations.

*Four streams, the textbook problem: two hot (H1 150 → 60 °C at CP 2 kW/K, H2 90 → 60 °C at CP 8 kW/K) and two cold (C1 20 → 125 °C at CP 2.5 kW/K, C2 25 → 100 °C at CP 3 kW/K), each modelled as a heater or cooler with a declared duty, on a synthetic constant-Cp fluid (hxFluid, 100 J/mol K, case-local) so that CP = F·cp exactly. With ΔT_min = 20 K the pinch pass finds the targets the whole subject is named after: Q_H,min = 107.5 kW, Q_C,min = 40 kW, pinch at shifted T∗ = 353.15 K (hot 363.15 K / cold 343.15 K). Today's unintegrated flowsheet buys 487.5 kW of heating and rejects 420 kW — the gap is what integration could recover.*

*1. The problem table is printed, interval by interval. Six shifted intervals, $\Sigma CP$ hot and cold, $\Delta H$ per interval and the cascade — [pinch] lines in the Log tab. Redo the cascade by hand (start at 0, carry each $\Delta H$ down, then add the most negative value back); the golden is what you get. 2. The targets are a consequence, not a design. Nothing was matched yet. reports/pinch/candidateMatches.csv lists every admissible hot–cold pair per region (6 candidates, 6 admissible) with its counter-current end-approach bound. The word ∗optimal∗ never appears — the pass recommends, it never rewrites the flowsheet. 3. The three violations name themselves. coolerH1 rejects 120 kW ∗above∗ the pinch; heaterC1 and heaterC2 draw 125 + 135 kW ∗below∗ it. Their sum is exactly the current-minus-target excess. Those are the coursework rules, read off your own flowsheet. 4. Composite curves are a file, not a claim. reports/pinch/compositeCurves.csv — plot it; the two curves touch at $\Delta T_min$ and the overhangs are $Q_H,min$ and $Q_C,min$.*

*Change dTmin in system/postDict to 10 K and rerun: both targets fall and the pinch moves. Then look at energyBalance for this case — it says UNAVAILABLE, because hxFluid carries no formation enthalpy on purpose: a pinch analysis needs only CP, and the case declares nothing it does not need.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| `stream` | `c1In` | `F` | `0.025` |
| `stream` | `c1Out` | `F` | `0.025` |
| `stream` | `c2In` | `F` | `0.03` |
| `stream` | `c2Out` | `F` | `0.03` |
| `stream` | `h1In` | `F` | `0.02` |
| `stream` | `h1Out` | `F` | `0.02` |
| `stream` | `h2In` | `F` | `0.08` |
| `stream` | `h2Out` | `F` | `0.08` |
| `kpi` | `coolerH1` | `F` | `0.02` |
| `kpi` | `coolerH1` | `P` | `101325` |
| `kpi` | `coolerH1` | `Q` | `-180000` |
| `kpi` | `coolerH1` | `Q_kW` | `-180` |
| `kpi` | `coolerH1` | `Q_per_mol` | `-9000` |
| `kpi` | `coolerH1` | `T_in` | `423.15` |
| *... and 69 more reference values in the case's `expected`.* | | | |

## reboiler_water_copper

tutorials/steady/heat/reboiler_water_copper




**What it does:** Steam-heated copper-surface pool reboiler: Rohsenow nucleate flux → duty, Zuber CHF ceiling (geometry mode)

**The story:** A steam-heated copper-surface POOL REBOILER whose duty is NOT specified – it EMERGES from the physics of NUCLEATE BOILING, exactly as a real reboiler's does.

Saturated-liquid water at 1 bar sits in a pool boiling on a bundle of horizontal copper tubes heated INSIDE by condensing steam. The 'phaseChanger' runs in 'model geometry;' with a 'boiling { ... }' block (the boiling resolution): a Rohsenow (1952) NUCLEATE-boiling flux q" is computed on the tube surface, the copper wall conducts, and the heating-medium film delivers the heat. Because the nucleate flux itself depends on the wall superheat dT_excess = T_surface - Tsat (and on its CUBE), the unit solves a 1-D energy balance at the surface (q_boil == q_heat) – the surface temperature is the unknown. Then the duty is Q = q" ∗ A and the outlet is the dome-flash at that duty.

water (saturated liquid, 1 bar, vf = 0) → reboiler (model geometry: Rohsenow nucleate flux + steam film) → wet vapour (partially boiled) + Q (RESULT)

HONESTY-AS-STRUCTURE (the binding constraint): the nucleate flux is +/-100% uncertain – C_sf is surface-FINISH lab data (polish, oxidation), NOT a fluid property. There is NO default C_sf and NO catalogue in v1: it is STUDENT-SUPPLIED here (water-copper 0.013, polished) WITH a mandatory citation. the result carries the [q/2, 2q] scatter band (C_sf +/-25%, cubed). the RELIABLE figure is the Zuber (1959) CHF ceiling, printed FIRST. A design above CHF is HARD-REFUSED (burnout) – a pool boiler cannot operate above the critical heat flux.

Method (glass-box): q_boil : Rohsenow nucleate flux, water-copper, C_sf 0.013, s 1.0 q" = mu_l h_fg sqrt(g(rho_l-rho_v)/sigma) [cp_l dT_excess / (C_sf h_fg Pr_l^s)]^3 (Rohsenow 1952; Incropera & DeWitt, Ch.10 Table 10.1) q_CHF : Zuber 1959, q"_max = 0.149 h_fg rho_v^0.5 [sigma g (rho_l-rho_v)]^0.25 (~1.26 MW/m^2 for water at 1 atm) wall : copper cylindrical resistance r_o ln(r_o/r_i)/k medium : a condensing-steam film coefficient h (high) the CHF margin SAFE/BURNOUT + the cube sensitivity are printed.

All film/saturation properties are IAPWS-IF97 water (incl. sigma via R1-76). The case carries its own constant/components/water.dat (case-local copy).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | wetVapour | F | 0.0277777777778 |
| kpi | reboiler | F | 0.0277777777778 |
| kpi | reboiler | P | 100000 |
| kpi | reboiler | Q | 111836.570636 |
| kpi | reboiler | Q_kW | 111.836570636 |
| kpi | reboiler | T_in | 372 |
| kpi | reboiler | T_out | 372.755918611 |
| kpi | reboiler | T_surface | 381.390508143 |
| kpi | reboiler | Tsat | 372.755918611 |
| kpi | reboiler | area | 1.25663706144 |
| kpi | reboiler | chf_margin | 0.0709724846287 |
| kpi | reboiler | dT_excess | 8.6345895313 |
| kpi | reboiler | q_CHF | 1253960.82217 |

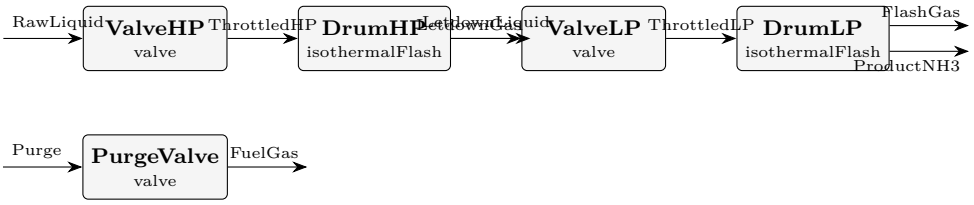
*. . . and 13 more reference values in the case's* `expected`*.*

## hydraulics

**The theory you need.** Pressure drop and flow distribution: Darcy–Weisbach with friction factors, fittings as equivalent lengths, pump curves against system curves. Expect the operating point at the curves' intersection.

## letdown01_ammonia_flashing

tutorials/steady/hydraulics/letdown01_ammonia_flashing




**What it does:** Pressure letdown: the green-ammonia separator liquid throttled 140 → 20 → 1 bar (flashing), and the loop purge 140 → 2 bar (choked gas)

**The story:** A LARGE PRESSURE LETDOWN, WITH THE ENGINE RUNNING (DEV.md 4c C64, the "Large pressure letdown" EduTool).

THE QUESTION. The liquid ammonia that leaves a synthesis loop's separator at 140 bar has to reach an atmospheric store. That is a pressure ratio of about 140, and it is taken in two throttles, each followed by a drum that lets the released gas go. What does the stream look like on the far side of each valve – its temperature, how much of it is vapour, and how much SPACE it now occupies? And what does the same throttle do to a gas – the loop purge – let down to a fuel header?

WHERE THE STATES COME FROM. RawLiquid (0/RawLiquid) – the converged separator liquid of tutorials/plant/greenAmmonia03_basicDesign (converged/RawLiquid), copied to the digit: 140 bar, 253.15 K, 2927 kmol/h of ammonia carrying the H2, N2 and Ar the separator dissolved in it. gui/tests/letdown.test.ts holds the copy to the plant's own file. the two pressures, 20 bar and 1.01325 bar – the plant's own LetdownHP and StorageFlash drums (greenAmmonia03_basicDesign/LetdownHP/system/flowsheetDict, .../StorageFlash/system/flowsheetDict). Purge (0/Purge) – the same plant's converged purge gas (140 bar, 253.15 K, mostly H2 and N2), copied to the digit. 2 bar for the purge's fuel header – AUTHOR-SET for the lesson. The plant sends its purge to a boundary and models no letdown of it; no real fuel system is described by this number.

WHAT IS DIFFERENT FROM THE PLANT, AND WHY. The plant takes each drop as an ISOTHERMAL flash whose temperature it declares (253.15 K at 20 bar, 238.15 K at 1 atm) – a hidden valve and a hidden heat exchanger in one box (see credo01_valve_heater_drum). Here each drop is a 'valve', an isenthalpic throttle (src/unitOperations/valve/Valve.H): the outlet temperature and vapour fraction are RESULTS, and the drum after it has no knobs at all, inheriting T and P from the valve. The plant's own StorageFlash header names the adiabatic letdown as the real one.

WHAT THE VALVE PUBLISHES. Besides T_out and vf, the valve's volumetric flow on both sides (Q_in, Q_out, volumeRatio, rho_in, rho_out), built from the package's own phase densities – homogeneous, no slip. The liquid density is the NH3 record's Vliq carried in T by Rackett, and the dissolved H2, N2 and Ar, which have no liquid density at 253 K, are priced at the density of the rest; the run says so.

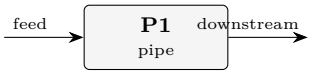
Package: the plant's own (Henry for the dissolved gases in ammonia, SRK for the vapour), copied with its case-local Ar-NH3 pair.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | FlashGas | F | 0.0415445974779 |
| stream | FlashGas | T | 238.884926778 |
| stream | FuelGas | F | 0.00339264003654 |
| stream | FuelGas | T | 248.677720887 |
| stream | LetdownGas | F | 0.00354665692591 |
| stream | LetdownGas | T | 253.01994131 |
| stream | LetdownLiquid | F | 0.813252311193 |
| stream | LetdownLiquid | T | 253.01994131 |
| stream | ProductNH3 | F | 0.771707755979 |
| stream | ProductNH3 | T | 238.884926778 |
| stream | Purge | F | 0.00339264003654 |
| stream | Purge | T | 253.15 |
| stream | RawLiquid | F | 0.816798968119 |
| stream | RawLiquid | T | 253.15 |
| *… and 84 more reference values in the case's `expected`.* | | | |

## pipe01_water_line

tutorials/steady/hydraulics/pipe01_water_line




**What it does:** Pipe pressure drop: 2000 kmol/h water in a 0.1 m x 50 m commercial-steel line (Churchill friction)

**The story:** A single PIPE element. Water enters at 5 bar and flows through 50 m of 0.1 m commercial-steel pipe with two elbows, a globe valve, and a 2 m rise. The pressure DROP is a COMPUTED RESULT of the geometry and the flow — the pipe has NO pressure knob (information follows the streams: a pipe only dissipates, a pump would raise P, a valve would set a let-down P).

feed (5 bar liquid, 2000 kmol/h) → [P1 pipe] D, L, roughness, dz, fittings → ΔP (result) → downstream (P reduced by the computed ΔP, T held)

Friction factor: Churchill (default) — one explicit expression for every Re, continuous from laminar to turbulent. Through the transition band (2300 <= Re < 4000) its value is an interpolation, not a prediction: the regime there is not determined by Re, and a pipe in the band is announced in the caveat block. Re ~ 1.5e5 here, so the flow is turbulent and all three ΔP terms (friction, fittings, elevation) matter.

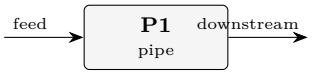
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | downstream | F | 0.555555555556 |
| stream | feed | F | 0.555555555556 |
| kpi | P1 | F | 0.555555555556 |
| kpi | P1 | P_in | 500000 |
| kpi | P1 | P_out | 462986.302076 |
| kpi | P1 | dP_elevation | 19520.1545706 |
| kpi | P1 | dP_fittings | 9626.38419279 |
| kpi | P1 | dP_friction | 7867.15916089 |
| kpi | P1 | deltaP | 37013.6979243 |
| kpi | P1 | density | 995.250904772 |
| kpi | P1 | frictionFactor | 0.0192870918591 |
| kpi | P1 | head_loss_m | 1.79235705234 |
| kpi | P1 | regime | 2 |
| kpi | P1 | reynolds | 149291.31749 |

*... and 7 more reference values in the case's `expected`.*

## pipe02_airwater_twophase

tutorials/steady/hydraulics/pipe02_airwater_twophase




**What it does:** Gas-liquid two-phase pressure drop: air (N2) + water in a horizontal 0.05 m x 10 m line, Lockhart-Martinelli. The two-phase path activates automatically because the feed flashes two-phase ($0 < V/F < 1$).

**The story:** GAS-LIQUID two-phase pressure drop

feed (air N2 + water, two-phase at 2 bar / 25 C) → [P1 pipe] D, L, roughness, horizontal → ΔP (result) → downstream (P reduced by the two-phase ΔP)

The two-phase path activates AUTOMATICALLY: the feed flashes two-phase at (T, P) (N2 is supercritical → all vapour; water → mostly liquid), so $0 < V/F < 1$ and the pipe routes to the gas-liquid models instead of the single-phase liquid path. Model: Lockhart-Martinelli (Chisholm multiplier + Butterworth holdup). Try 'model homogeneous;' in twoPhase to compare.

The Lockhart-Martinelli correlation was developed and VALIDATED on isothermal air-water (and similar gas-liquid) pipe data – Lockhart & Martinelli, Chem. Eng. Prog. 45 (1949) 39 – so air-water is exactly the case it is trusted on.

Basis: N2 3 kmol/h (the fast light gas) + water 200 kmol/h (the slow heavy liquid) → a liquid-dominated two-phase flow (j_G ~ a few m/s, j_L ~ 0.5 m/s).

*Air ($N_2$) + water flowing through a horizontal 0.05 m × 10 m commercial-steel line. The pipe's two-phase path activates automatically: the feed flashes two-phase at 2 bar / 25 °C ($N_2$ is supercritical → all vapour; water → mostly liquid), so $0 < V/F < 1$ and the pipe routes to the gas-liquid models instead of the single-phase liquid path.*

*The two teaching points: 1. The multiplier $\varphi_L^2 = 8.6$: two-phase friction is far above either phase alone — the classic Lockhart-Martinelli amplification, visible in the printout via X and the Chisholm C. 2. Slip → holdup: the homogeneous (no-slip) model would put the liquid fraction at $j_L/(j_L+j_G) \approx 0.12$; L-M gives 0.36 because the liquid moves slower than the gas and accumulates. Switch twoPhase { model homogeneous; } to see the difference for yourself.*

*The Lockhart-Martinelli correlation (Chem. Eng. Prog. 45 (1949) 39) was developed and validated on isothermal air-water (and similar gas-liquid) pipe data — so air-water is exactly the system it is trusted on. The model structure here (X, $\varphi_L^2$, the tt/vt/tv/vv C-selection, the Butterworth 1975 void fraction) reproduces that framework.*

*Switch twoPhase { model ...; } and compare — the spread IS a lesson (two-phase ΔP correlations routinely disagree by ~50 %):*

*Layer 2 (Friedel, BeggsBrill) needs surface tension — supplied here by transport { surfaceTension { model BrockBird; } } (from water's Tc/Pc/Tb). Beggs-Brill also classifies the flow pattern (segregated / intermittent / distributed) and corrects holdup for pipe inclination (here horizontal, so $\psi = 1$).*

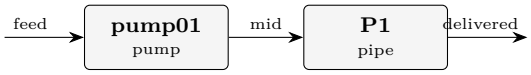
*- Adiabatic: the inlet vapour fraction is held along the pipe (no flashing / condensation in the line). Flashing flow is a future increment. - Liquid density: ?_L = 877 kg/m³ comes from the Rackett branch, ~12 % low for water (a known anomaly, also in pipe01_water_line). It biases the ∗absolute∗ ΔP; the correlation ∗structure∗ is unaffected. - Liquid viscosity via operation { viscosity ...; } (water's value): the flash leaves a trace of dissolved $N_2$ in the liquid and the Vogel model has no $N_2$ entry.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | downstream | F | 0.0563888888889 |
| stream | feed | F | 0.0563888888889 |
| kpi | P1 | F | 0.0563888888889 |
| kpi | P1 | P_in | 200000 |
| kpi | P1 | P_out | 193989.686272 |
| kpi | P1 | Re_Gs | 26035.4493901 |
| kpi | P1 | Re_Ls | 28789.8441141 |
| kpi | P1 | X_martinelli | 2.63535511116 |
| kpi | P1 | dP_elevation | 0 |
| kpi | P1 | dP_friction | 6010.31372773 |
| kpi | P1 | deltaP | 6010.31372773 |
| kpi | P1 | holdup_liquid | 0.357791422881 |
| kpi | P1 | jG | 4.07152675186 |
| kpi | P1 | jL | 0.514023769077 |
| *... and 12 more reference values in the case's `expected`.* | | | |

### pumpSystem01_operating_point

tutorials/steady/hydraulics/pumpSystem01_operating_point




**What it does:** Pump vs system curve: sweep the feed flow, the operating point is the intersection

**The story:** A pump and its piping system IN SERIES, swept over the feed flow (see system/outerDict). The classical operating-point construction:

PUMP characteristic – the shaft delivers a FIXED 2 kW at eta 0.65, so the pressure rise FALLS with flow: dP_pump = eta W / Q_vol. This is the constant-shaft-power characteristic of the pump MODEL, not a manufacturer's tested H(Q) curve – the engine says what it models, never pretends to a datasheet it was not given.

SYSTEM curve – the same flow through 50 m of 0.1 m commercial-steel pipe with two elbows, a globe valve and a 2 m rise: dP_system = friction(Q^2) + fittings(Q^2) + rho g dz.

Where the two curves CROSS, the pump's rise exactly pays the system's demand – the operating point. Hand estimate (water, v ~ 1.807e-5 m3/mol, A = 7.854e-3 m2, f ~ 0.018, sum K = 11.8):

pump: dP ~ 1300 / (F 1.807e-5) ~ 7.19e4 / F Pa system: dP ~ 5.45e4 F^2 + 1.956e4 Pa (F in kmol/s) cross: F* ~ 0.99 kmol/s (~3.6e3 kmol/h), dP* ~ 0.73 bar

The sweep straddles that crossing (0.4..1.6 kmol/s); each point is a complete converged pass, and the curves are read off the pump's and the pipe's own published KPIs – nothing is fitted or interpolated here.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | delivered | F | 1 |
| stream | feed | F | 1 |
| stream | mid | F | 1 |
| kpi | P1 | F | 1 |
| kpi | P1 | P_in | 371942.446043 |
| kpi | P1 | P_out | 297097.627131 |
| kpi | P1 | dP_elevation | 19553.4351728 |
| kpi | P1 | dP_fittings | 31136.3992358 |
| kpi | P1 | dP_friction | 24154.9845033 |
| kpi | P1 | deltaP | 74844.818912 |
| kpi | P1 | density | 996.947743257 |
| kpi | P1 | frictionFactor | 0.0183083994383 |
| kpi | P1 | head_loss_m | 5.65541381864 |
| kpi | P1 | regime | 2 |
| *... and 23 more reference values in the case's `expected`.* | | | |

## membranes

**The theory you need.** Solution-diffusion (Wijmans & Baker 1995): inside a dense membrane the pressure is uniform, so permeation is concentration-driven. Linearised: $J_w = A_w(\Delta P - \Delta\pi)$ and $J_s = B_s(c_m - c_p)$ – water driven by NET pressure, solute blind to pressure; $(A_w, B_s)$ are MATERIAL constants from `data/standards/assets/`. Concentration polarisation (film model) makes the wall concentration $c_m = c_b\, e^{J_w/k}$ exceed the bulk – the flux–polarisation–osmotic feedback is THE nonlinearity of RO/NF, solved per module segment. Expect: rejection improving with pressure ($c_p = B_s c_m/(J_w + B_s)$), polarisation eating driving force at high flux, and charged-membrane (Donnan) effects on multivalent ions that plain solution-diffusion cannot see – that is what the DSPM-DE sibling model is for. *Full derivation: Theory Guide, the membrane chapter.*

## membrane01_RO_NaCl_seawater

tutorials/steady/membranes/membrane01_RO_NaCl_seawater




**What it does:** Spiral-wound RO of 35 g/L NaCl feed at 55 bar (canonical seawater desalination)

**The story:** the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With c_NaCl = 35/58.44 = 0.599 kmol/m$^3$ and c_water in seawater ≈ 1025 kg/m$^3$ ? 18 kg/kmol = 56.94 kmol/m$^3$, the mole fractions are z_NaCl = 0.599 / (0.599 + 56.94) = 0.01040 and z_water = 0.9896. Feed molar flow F = 575 kmol/h (≈ 10 m$^3$/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m$^2$ active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome ~30 bar of osmotic pressure at 35 g/L plus channel ΔP plus a net driving force of ~22 bar). Permeate pressure 1 bar.

Mass transfer: k_film = 5e-5 m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives ~5 × 10?? m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

Expected (with the SW30HR parameters): water recovery ≈ 18 % (single pass) observed NaCl rej. ≈ 99.5 % (typical for SW30HR) average water flux ≈ 20–25 LMH feed ΔP = 2 bar (set linearly via dP_feed_total)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | ROunit | R_obs_NaCl | 0.997858419447 |
| kpi | ROunit | water_recovery | 0.157724991775 |
| kpi | ROunit | J_w_avg | 1.3269386659e-05 |
| stream | feed | F | 0.159722222222 |
| stream | feed | T | 298.15 |
| stream | permeate | F | 0.0257788426835 |
| stream | permeate | T | 298.15 |
| stream | retentate | F | 0.133943379539 |
| stream | retentate | T | 298.15 |
| kpi | ROunit | dP_feed | 200000 |
| kpi | ROunit | mass_closure_rel | 0 |
| kpi | ROunit | Gamma_NaCl_avg | 0.303248552028 |
| kpi | ROunit | solute_closure_rel | 2.6107878401e-16 |

## membrane02_NF_sugar

tutorials/steady/membranes/membrane02_NF_sugar




**What it does:** NF270 partial rejection of glucose at 10 bar (canonical NF: 10 g/L sugar feed)

**The story:** loose-NF separation of glucose from water, the canonical NF test that pedagogically distinguishes NF from RO:

RO tight membrane, high pressure, near-100 % rejection NF (here) loose membrane, low pressure, PARTIAL rejection (~50 %)

Feed: 10 g/L glucose (a typical pharma / food-processing feed). c_glucose = 10 / 180.16 = 0.0555 kmol/m$^3$ With pure-water c_water = 1000/18 = 55.56 kmol/m$^3$, z_glucose = 0.0555 / (0.0555 + 55.56) = 9.97e-4, z_water = 0.999003. Feed F = 200 kmol/h (≈ 3.6 m$^3$/h of dilute solution).

Module: NF270-style spiral-wound element, 35 m$^2$ area, 1 m length. Feed pressure 10 bar (an order of magnitude less than the SW30HR seawater case — the price of loose rejection is that membrane permeability A_w is 10× higher, so the same operating flux is achieved with 10× less driving pressure). Permeate at atmospheric. Same k_film = 5e-5 m/s.

Expected (with the NF270 parameters): water recovery ≈ 20-25 % observed glucose rej.≈ 50-65 % (? the SW30HR's 99 %+ !) average water flux ≈ 30-40 LMH

The partial rejection is the defining NF feature: the membrane is not tight enough to fully reject glucose (MW 180, NF270 nominal MWCO 200 Da), so a fraction of the sugar leaks through with the water — and that leakage is what makes NF useful for fractionation rather than just total rejection.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0555555555556 |
| stream | permeate | F | 0.0149590663407 |
| stream | retentate | F | 0.0405964892149 |
| kpi | NFunit | J_w_avg | 5.40596785995e-05 |
| kpi | NFunit | R_obs_glucose | 0.662718515801 |
| kpi | NFunit | dP_feed | 100000 |
| kpi | NFunit | water_recovery | 0.267671368561 |
| stream | feed | T | 298.15 |
| stream | permeate | T | 298.15 |
| stream | retentate | T | 298.15 |
| kpi | NFunit | mass_closure_rel | 2.19886558047e-16 |
| kpi | NFunit | Gamma_glucose_avg | 1.3572543181 |
| kpi | NFunit | solute_closure_rel | 6.11698818479e-16 |
| boundary | global | H_feeds_kW | -15965.9347526 |

*... and 2 more reference values in the case's `expected`.*

## membrane03_concentration_polarization

tutorials/steady/membranes/membrane03_concentration_polarization




**What it does:** Concentration polarization study: sweep k_film over a SW30HR seawater RO case

**The story:** Same SW30HR seawater RO case as membrane01_RO_NaCl_seawater, but here we use the SweepDriver (see outerDict) to vary the mass-transfer coefficient k_film over a decade and plot how observed rejection and average flux change.

The physics: c_m / c_b = exp ( J_w / k_film ) (film model)

When k_film is large (good cross-flow mixing, e.g. high-velocity channel + good spacer), the boundary layer is thin and the concentration at the membrane wall is close to the bulk: c_m ≈ c_b. $\Delta\pi$ then reflects the bulk concentration, fluxes are high, and observed rejection is close to the intrinsic value.

When k_film is small (poor mixing, low velocity, dead zones), the solute concentrates at the wall: c_m ? c_b. $\Delta\pi$ rises sharply, J_w collapses, and a larger fraction of the wall solute leaks through, dropping observed rejection.

This sweep illustrates why membrane-module designers obsess over spacer geometry and feed velocity — they are not adjusting the membrane itself, they are adjusting k_film.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.159722222222 |
| stream | permeate | F | 0.0312174168627 |
| stream | retentate | F | 0.12850480536 |
| kpi | ROunit | J_w_avg | 1.60686528345e-05 |
| kpi | ROunit | R_obs_NaCl | 0.998356067578 |
| kpi | ROunit | dP_feed | 200000 |
| kpi | ROunit | mass_closure_rel | 0 |
| kpi | ROunit | water_recovery | 0.19099813739 |
| stream | feed | T | 298.15 |
| stream | permeate | T | 298.15 |
| stream | retentate | T | 298.15 |
| kpi | ROunit | Gamma_NaCl_avg | 0.185043964393 |
| kpi | ROunit | solute_closure_rel | 2.6107878401e-16 |

## membrane04_spacer_schock_miquel

tutorials/steady/membranes/membrane04_spacer_schock_miquel




**What it does:** Seawater RO with k_film computed from spacer-channel hydraulics (Schock-Miquel Sherwood); k_film falls ~6.6 to ~5.6e-5 m/s along the channel

**The story:** the same seawater RO as membrane01, but k_film is COMPUTED from the spacer-channel hydraulics (Schock-Miquel Sherwood correlation) instead of being a fixed number. The 'massTransfer' block makes k_film a selectable model: Sh = 0.065 Re^0.875 Sc^0.25 with Re from the local crossflow velocity u = Q_b(z)/A_channel — so k_film FALLS along the channel as the retentate flow drops (the profile column 'k_film' shows this). At the inlet u ~ 0.26 m/s gives k_film ~ 7e-5 m/s, a touch above membrane01's fixed 5e-5 (better mixing → less polarisation → slightly higher recovery/rejection). Declare a bare 'k_film <value>;' (no massTransfer block) to recover membrane01's fixed-coefficient behaviour – the ONE way to pin it (the redundant 'model constant;' route was removed 2026-08-23).

membrane01_RO_NaCl_seawater – the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With c_NaCl = 35/58.44 = 0.599 kmol/m$^3$ and c_water in seawater ≈ 1025 kg/m$^3$ ? 18 kg/kmol = 56.94 kmol/m$^3$, the mole fractions are z_NaCl = 0.599 / (0.599 + 56.94) = 0.01040 and z_water = 0.9896. Feed molar flow F = 575 kmol/h (≈ 10 m$^3$/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m$^2$ active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome ~30 bar of osmotic pressure at 35 g/L plus channel ΔP plus a net driving force of ~22 bar). Permeate pressure 1 bar.

Mass transfer: k_film = 5e-5 m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives ~5 × 10?? m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

Expected (with the SW30HR parameters): water recovery ≈ 18 % (single pass) observed NaCl rej. ≈ 99.5 % (typical for SW30HR) average water flux ≈ 20–25 LMH feed ΔP = 2 bar (set linearly via dP_feed_total)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.159722222222 |
| stream | permeate | F | 0.0286293996445 |
| stream | retentate | F | 0.131092822578 |
| kpi | ROunit | J_w_avg | 1.47365990468e-05 |
| kpi | ROunit | R_obs_NaCl | 0.998101402811 |
| kpi | ROunit | dP_feed | 26066.7335536 |
| kpi | ROunit | water_recovery | 0.175164837924 |
| stream | feed | T | 298.15 |
| stream | permeate | T | 298.15 |
| stream | retentate | T | 298.15 |
| kpi | ROunit | mass_closure_rel | 3.01635071855e-16 |
| kpi | ROunit | Gamma_NaCl_avg | 0.268477357047 |
| kpi | ROunit | solute_closure_rel | 2.6107878401e-16 |

## membrane05_train

tutorials/steady/membranes/membrane05_train




**What it does:** Multi-element membrane train: 3 SW30HR elements in series (seawater RO)

**The story:** a MULTI-ELEMENT TRAIN: 3 SW30HR spiral-wound elements in series, seawater RO (the same feed + per-element geometry as membrane04, which is the single element). 'elements 3;' chains them: the retentate of element 1 is the feed of element 2, and so on; the permeate is collected over the whole train. Both the spacer k_film and the spacer pressure drop (Schock-Miquel) are computed from the local hydraulics, so they fall element by element as the flow drops.

The pedagogical point is the per-element DEGRADATION reported in the KPIs (recovery_elemK, c_ret_elemK): element 1 sees fresh feed (high P, low c) and recovers the most; the last element sees concentrated, lower-pressure brine (P dropped by friction + the inter-element connector loss) and recovers far less. The train recovery is the cumulative sum — higher than a single element, but with diminishing returns down the train.

Feed: 35 g/L NaCl seawater, z_NaCl = 0.01040, F = 575 kmol/h (~10 $m^3$/h), 55 bar, 298.15 K — the same feed as membrane01/04. Each of the 3 elements is 35 $m^2$ / 1 m, SW30HR, permeate at 1 bar, with a 0.2 bar inter-element connector loss.

What this run produces (golden master): train water recovery 36.1 % (3 elements, cumulative) observed NaCl rejection 99.69 % train-average water flux 36.47 LMH (1.013e-5 m/s) total feed-side ΔP ~1.01 bar (Schock-Miquel friction + 2 connector losses)

The per-element DEGRADATION (the teaching point) reads off the KPIs: recovery_elem1 17.5 % c_ret_elem1 0.684 (fresh feed) recovery_elem2 29.0 % c_ret_elem2 0.794 (cumulative) recovery_elem3 36.1 % c_ret_elem3 0.882 (cumulative) The retentate concentration climbs element to element while each successive element adds less recovery: a single SW30HR element at this feed recovers ~17.5 %, the second adds ~11.5 %, the third only ~7 % — the diminishing return of stacking elements in series at fixed feed.

The material balance CLOSES, per solute, to round-off (the module refuses otherwise). Until 2026-09-29 it did not: the element balance lost 0.07 % of the chlorine, because the axial march advanced the concentration rather than the solute flow – see membrane02's element balance for the same defect, and DEV.md C27. The figures above are this run's, rounded (the header had drifted from its own golden before the repair: 35.3 % against 36.2 %).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.159722222222 |
| stream | permeate | F | 0.0590481156677 |
| stream | retentate | F | 0.100674106554 |
| kpi | ROunit | J_w_avg | 1.01316760818e-05 |
| kpi | ROunit | R_obs_NaCl | 0.996942163831 |
| kpi | ROunit | c_ret_elem1 | 0.683704881229 |
| kpi | ROunit | c_ret_elem2 | 0.793651571912 |
| kpi | ROunit | c_ret_elem3 | 0.882255603046 |
| kpi | ROunit | dP_feed | 101156.272291 |
| kpi | ROunit | elements | 3 |
| kpi | ROunit | recovery_elem1 | 0.175164837924 |
| kpi | ROunit | recovery_elem2 | 0.28969612966 |
| kpi | ROunit | recovery_elem3 | 0.361286900686 |
| kpi | ROunit | water_recovery | 0.361286900686 |
| *... and 6 more reference values in the case's `expected`.* | | | |

## membrane06_pitzer

tutorials/steady/membranes/membrane06_pitzer




**What it does:** Seawater RO with the Pitzer osmotic model (vs membrane01 van't Hoff)

**The story:** the canonical seawater RO (same as membrane01) but with the PITZER osmotic-pressure model instead of van't Hoff. van't Hoff (phi = 1) over-estimates the osmotic pressure of a real brine: for NaCl the osmotic coefficient is phi ~ 0.92 at seawater concentration (and dips further before rising at high molality). The Pitzer ion-interaction model captures phi(I), so the osmotic back-pressure is ~7-8 % LOWER → a slightly HIGHER water flux and recovery than membrane01 (15.55 %). The 'osmotic' block makes it a selectable model with the NaCl coefficients (calibratable); remove the block to recover van't Hoff.

membrane01_RO_NaCl_seawater – the canonical first-membrane case.

Feed: 35 g/L NaCl (seawater concentration), water makes up the rest. With c_NaCl = 35/58.44 = 0.599 kmol/m$^3$ and c_water in seawater ≈ 1025 kg/m$^3$ ? 18 kg/kmol = 56.94 kmol/m$^3$, the mole fractions are z_NaCl = 0.599 / (0.599 + 56.94) = 0.01040 and z_water = 0.9896. Feed molar flow F = 575 kmol/h (≈ 10 m$^3$/h of seawater).

Module: a single SW30HR-style spiral-wound element, 35 m$^2$ active area, 1 m channel length. Feed pressure 55 bar (typical for seawater RO — enough head to overcome ~30 bar of osmotic pressure at 35 g/L plus channel ΔP plus a net driving force of ~22 bar). Permeate pressure 1 bar.

Mass transfer: k_film = 5e-5 m/s (typical spacer-enhanced value; a Schock-Miquel correlation at this flow gives ~5 × 10?? m/s; we hardcode the value here and explore the effect of changing it in tutorial membrane03).

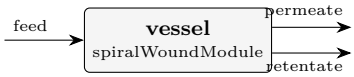
Expected (with the SW30HR parameters): water recovery ≈ 18 % (single pass) observed NaCl rej. ≈ 99.5 % (typical for SW30HR) average water flux ≈ 20–25 LMH feed ΔP = 2 bar (set linearly via dP_feed_total)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.159722222222 |
| stream | permeate | F | 0.0284513881228 |
| stream | retentate | F | 0.131270834099 |
| kpi | ROunit | J_w_avg | 1.46450086344e-05 |
| kpi | ROunit | R_obs_NaCl | 0.997985747803 |
| kpi | ROunit | dP_feed | 200000 |
| kpi | ROunit | water_recovery | 0.174076159342 |
| stream | feed | T | 298.15 |
| stream | permeate | T | 298.15 |
| stream | retentate | T | 298.15 |
| kpi | ROunit | mass_closure_rel | 4.52452607783e-16 |
| kpi | ROunit | Gamma_NaCl_avg | 0.339633563471 |
| kpi | ROunit | solute_closure_rel | 0 |

## membrane07_scaling_si

tutorials/steady/membranes/membrane07_scaling_si




**What it does:** Brackish RO, 6 x 8-inch modules: per-module scaling audit (SI at wall vs bulk)

**The story:** WHERE along the vessel does scaling start?

Brackish RO on the SAME representative groundwater analysis as tutorials/props/electrolyte/scaling_ro_brackish (~1500 mg/L TDS):

ion mg/L mol/kg w ion mg/L mol/kg w Ca+2 84 0.0021 Cl- 510 0.0144 Mg+2 27 0.0011 SO4-2 250 0.0026 Na+ 363 0.0158 HCO3- 183 0.0030 K+ 12 0.0003

Here the analysis is a PROCESS STREAM: the ions are flowsheet components (case-local constant/components/, MW from ions.dat) and the mg/L totals become mole fractions against water (x_i = m_i / 55.55; m_i = (mg/L)/MW at rho ~ 1 kg/L):

Ca 3.7731e-5 Mg 1.9998e-5 Na 2.8425e-4 K 5.5253e-6 Cl 2.5897e-4 SO4 4.6852e-5 HCO3 5.3989e-5 water = balance

HARDWARE BY STANDARD SIZE (no ‘area‘): one pressure vessel with 6 x 8-inch standard 40-inch elements – ‘module Generic-8040; elements 6;‘. The generic element is a RECORD (data/standards/assets/Generic-8040.dat, 37.2 m2, the 400 ft2 class): every value in it is an estimate, announced on every run, and the membrane is the case’s. The k_film and the dP are FIXED here, so the record’s spacer and leaf length shape no number (until 2026-10-24 this case read a hand table in the engine, ‘moduleDiameter 8 in; nModules 6;‘, with the same area).

THE SCALING AUDIT (‘scaling {}‘): at each module exit the local BULK concentrate AND the membrane-WALL composition (c_m from the module’s own film-model polarisation) are speciated (Davies) and the calcite / gypsum saturation indices recorded. The WALL crosses SI=0 modules before the bulk does – concentration polarisation, normally an invisible flux penalty, is seen here as THE scaling driver.

pH 7.0 GIVEN, not solved: the raw well water is already calcite- supersaturated (scaling_ro_brackish solves the raw feed to pH 7.72, SI_calcite = +0.24 – it would scale in the PIPE, let alone the vessel). Real brackish plants dose acid for exactly that reason; here the feed is dosed and CONTROLLED at pH 7.0, and the audit asks the honest follow-up: with the dosing in place, where along the vessel does the concentrate STILL reach calcite saturation? (‘pH solve;‘ is also accepted – it re-solves electroneutrality per module, the undosed scenario.)

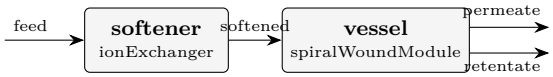
Expected story (KPIs locked by the golden master; recovery 55 %): calcite WALL crosses SI=0 at module 3 of 6 (r_cum = 0.31); calcite BULK is still undersaturated there (-0.27) and only crosses in module 6 (+0.03) – the wall leads the bulk by three full modules: polarisation IS the scaling driver; gypsum stays undersaturated everywhere (max wall SI = -0.73).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.194444444431 |
| stream | permeate | F | 0.107182811802 |
| stream | retentate | F | 0.0872616326288 |
| kpi | vessel | J_w_avg | 8.65099863579e-06 |
| kpi | vessel | R_obs_Ca | 0.999473918004 |
| kpi | vessel | R_obs_Cl | 0.994763171341 |
| kpi | vessel | R_obs_HCO3 | 0.996067396405 |
| kpi | vessel | R_obs_K | 0.993462247168 |
| kpi | vessel | R_obs_Mg | 0.999605388292 |
| kpi | vessel | R_obs_Na | 0.994763171341 |
| kpi | vessel | R_obs_SO4 | 0.999736892045 |
| kpi | vessel | SI0_module_calcite | 3 |
| kpi | vessel | SI0_module_gypsum | -1 |
| kpi | vessel | SI0_recovery_calcite | 0.305149502926 |
| *... and 35 more reference values in the case's `expected`.* | | | |

## membrane08_softened_scaling

tutorials/steady/membranes/membrane08_softened_scaling




**What it does:** Softener + brackish RO: ion-exchange removes the scaling driver upstream of the vessel

**The story:** SOFTEN the feed, and the scaling is GONE.

THE CONTRAST (read membrane07_scaling_si first – the BEFORE):

membrane07 the SAME brackish groundwater (~1500 mg/L TDS, Ca 84 / Mg 27 mg/L) goes straight into a 6 x 8-inch brackish-RO vessel. As the concentrate builds along the train the membrane-WALL composition crosses calcite SI = 0 at MODULE 3 of 6 (r_cum = 0.31) – concentration polarisation is THE scaling driver; the vessel would foul.

membrane08 here a cation-exchange SOFTENER ('ionExchanger', SAC resin in the Na form, Gaines-Thomas) sits UPSTREAM of the SAME vessel. Ca and Mg drop ~99% onto the resin (Na rises eq-for-eq – the salt penalty), so the water entering the vessel is essentially hardness-free. With Ca gone the calcite IAP collapses and the per-module WALL SI_calcite stays NEGATIVE through ALL 6 modules – the scaling driver is removed at source.

ZERO change to the membrane. The spiralWoundModule is byte-for-byte the membrane07 vessel (module Generic-8040; elements 6; the same scaling{} audit); it simply receives a BETTER feed. Removing the hardness, not chasing the supersaturation in the vessel, is the cure – the engineering lesson.

HONESTY (printed, non-suppressible): the softener returns the LIMITING (fully-equilibrated) effluent – the best a fresh / fully-loaded contactor does per pass, NOT a bed in service (breakthrough / regeneration are TRANSIENT and not modelled). See the banner on every run.

Data: curated USGS PHREEQC records (speciation, minerals, ions, exchange; public domain) resolved from data/standards/ at runtime + case-local ion .dats (constant/components/), the SAC_Na resin nameplate and the SW30HR membrane (constant/membranes/) – the same records as membrane07 / softener_brackish; self-DESCRIBING (inline manifest), seal with bin/choupo-import for a runtime-self-contained copy.

Expected story (golden master): softener removes ~99.99% hardness (Ca/Mg leakage ~ 1e-2 mg/L, Na added ~ 6.7 meq/L at the given pH 7.0 closure – the eq-for-eq salt penalty for the full hardness load); the vessel's maxSI_calcite_wall is NEGATIVE (-3.68: no SI = 0 crossing in any module, SI0_module_calcite = -1), vs +0.27 / crossing at module 3 in membrane07.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.194444444431 |
| stream | permeate | F | 0.105421189691 |
| stream | retentate | F | 0.0890344782693 |
| stream | softened | F | 0.19445566796 |
| kpi | softener | CEC_utilised_pct | 100 |
| kpi | softener | Ca_leakage_mgL | 0.00999051830577 |
| kpi | softener | I | 0.024397591267 |
| kpi | softener | Mg_leakage_mgL | 0.00549118647587 |
| kpi | softener | Na_added_meqL | 6.71875413026 |
| kpi | softener | Na_added_mgL | 154.464157455 |
| kpi | softener | bedVolume_m3 | 1.5 |
| kpi | softener | dH_exchange_Jkgw | 21.9905023247 |
| kpi | softener | hardness_in_mgL_CaCO3 | 320.962820378 |
| kpi | softener | hardness_out_mgL_CaCO3 | 0.0475632817167 |

*... and 52 more reference values in the case's `expected`.*

## membrane09_index_vs_rigorous

tutorials/steady/membranes/membrane09_index_vs_rigorous




**What it does:** Seawater RO at high recovery: industry calcite indices (LSI / Stiff-Davis / Ryznar) vs the rigorous activity SI at the membrane WALL

**The story:** the bag-of-indices industry shortcut is NOT safe at the membrane wall in a high-recovery concentrate.

THE LESSON (activity vs concentration – the whole point of the case):

The industry calcite-scaling indices read every plant data-sheet ships – LSI Langelier saturation index (Langelier 1936) Stiff-Davis brine-corrected LSI (Stiff & Davis 1952) RSI Ryznar stability index (Ryznar 1944) are built from ION CONCENTRATIONS plus an EMPIRICAL ionic-strength correction (a published pK-shift / chart-fit K). Choupo's rigorous SI_calcite instead forms real ION ACTIVITIES a_i = gamma_i m_i from the Davies model:

LSI = pH - logK_cc(T) - p[Ca2+] - p[HCO3-] (CONCENTRATIONS) SI = pH - logK_cc(T) - p{a_Ca2+}- p{a_HCO3-} (ACTIVITIES)

They share the SAME thermodynamic anchor logK_cc(T) (the HCO3-basis calcite log K, = pK2 - pKsp) – so the ONLY difference is concentration vs activity. At LOW ionic strength gamma → 1 and LSI == SI (validation); as the activity coefficients fall away from 1 the two DIVERGE.

WHERE IT BITES – THE WALL. Concentration polarisation piles solute against the membrane wall: the LOCAL ionic strength at the wall runs far above the bulk. There the empirical index over-predicts calcite saturation (it cannot see the gamma reduction), while the rigorous activity SI does not. Reading the index off the bulk analysis – as the industry shortcut does – therefore MISLEADS at exactly the place scaling actually starts.

THE CASE: standard surface seawater (S = 35, the major-ion subset), one pressure vessel of 7 x 8-inch SW elements, pushed to high recovery. Acid is dosed and CONTROLLED at pH 7.0 (real SWRO practice). At each module exit the BULK concentrate AND the membrane-WALL composition are speciated (Davies) and BOTH the rigorous SI_calcite AND the empirical LSI / Stiff-Davis / Ryznar are recorded, bulk + wall.

THE STORY (locked by the golden master): - This is already a SEAWATER concentrate (bulk I ~ 0.68 mol/kg at module 1, far above the I → 0 regime where LSI == SI), so the empirical index OVER-PREDICTS calcite saturation at EVERY module, including the first: at module 1 LSI(wall) = +0.98 vs rigorous SI_calcite(wall) = +0.20 (gap ~ 0.78); LSI(bulk) = +0.56 vs SI(bulk) = -0.28 (gap ~ 0.84). At this ionic strength there is no point where index and activity-SI agree to hundredths – to SEE that limit you must drop to a brackish / low-I feed (the index family is only validated there). - the gap widens further into the vessel: at the deepest module LSI(wall) over-predicts the rigorous SI_calcite(wall) by ~ 0.78 – the gap (LSI - SI) printed in the audit footer is the headline number. Stiff-Davis pulls PART of the way back toward the rigorous SI (its empirical brine K), but not all the way – it is still a chart fit, not the real gammas. - Ryznar (RSI = 2 pHs - pH) reads the same saturation pHs as LSI on the practical 0..14 stability scale.

Engineering take-away (Choupo's transparency differentiator): the concentration-based index family is convenient but UNSAFE for high-recovery membrane scaling – the rigorous activity SI, visible term-by-term, is the honest call.

Sibling cases: membrane07_scaling_si (bulk vs wall SI, brackish); tutorials/props/electrolyte/pitzer_vs_davies_ro_concentr (Davies vs Pitzer activity fork). Index kernel + primary sources: src/thermo/electrolyte/ScalingIndices.{H,cpp}.

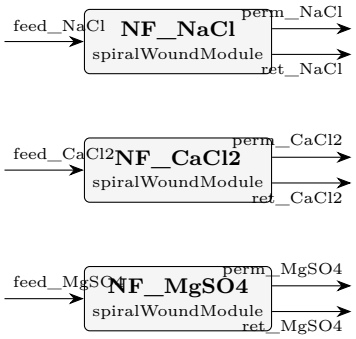
PRIMARY SOURCES: Langelier, W.F. (1936) J. AWWA 28(10) 1500-1521. Stiff, H.A. & Davis, L.E. (1952) Trans. AIME 195, 213-216. Ryznar, J.W. (1944) J. AWWA 36(4) 472-486.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.250000000004 |
| stream | permeate | F | 0.0675333162948 |
| stream | retentate | F | 0.182466683709 |
| kpi | vessel | J_w_avg | 4.67251910678e-06 |
| kpi | vessel | LSI_calcite_bulk_end | 0.765554629083 |
| kpi | vessel | LSI_calcite_wall_end | 0.985373688063 |
| kpi | vessel | RSI_calcite_wall_end | 5.02925262387 |
| kpi | vessel | R_obs_Ca | 0.999254953387 |
| kpi | vessel | R_obs_Cl | 0.992599618056 |
| kpi | vessel | R_obs_HCO3 | 0.994439330533 |
| kpi | vessel | R_obs_K | 0.990766762524 |
| kpi | vessel | R_obs_Mg | 0.999441109966 |
| kpi | vessel | R_obs_Na | 0.992599618056 |
| kpi | vessel | R_obs_SO4 | 0.999627336568 |
| *… and 37 more reference values in the case's `expected`.* | | | |

## membrane10_dspmde_divalent

tutorials/steady/membranes/membrane10_dspmde_divalent




**What it does:** DSPM-DE NF: divalent (MgSO4) > mixed (CaCl2) > monovalent (NaCl) rejection at 10 bar

**The story:** the charge-aware NF signature, glass-box.

THREE parallel NF270 elements, one per salt, all on the SAME membrane (NF270_dspmde, with the DSPM-DE charged-pore tier) at the SAME 10 bar / 5 mM feed. The point is the REJECTION ORDERING that the Donnan-Steric Pore Model with Dielectric Exclusion produces from first principles:

R_obs(MgSO4) > R_obs(CaCl2) > R_obs(NaCl) (2:2 divalent) (2:1 mixed) (1:1 monovalent)

WHY (Bowen-Welfoot 2002): the membrane carries a NEGATIVE fixed charge (X_d = -5 mol/m3, the record's value). The CO-ION (the anion, here Cl- / SO4–) is excluded by Donnan; a DIVALENT co-ion (SO4–) is excluded far more strongly than a monovalent one (z^2 in the Donnan + Born partition). So MgSO4 – whose co-ion is divalent – is rejected most; NaCl – whose co-ion is the monovalent Cl- – least. This is the classic NF fingerprint that plain solution-diffusion (one B_s per salt) cannot predict mechanistically.

Data: the salts (case-local constant/components/) carry an electrolyte{} ion decomposition and the membrane (constant/membranes/) carries the poreModel{} block; the ions' radius/D0 transport tier resolves from data/standards/ at runtime – self-DESCRIBING (inline manifest), seal with bin/choupo-import for a runtime-self-contained copy.

'transport DSPM-DE;' selects the charged-pore law; drop it (or set 'transport solutionDiffusion;') to compare against the baseline.

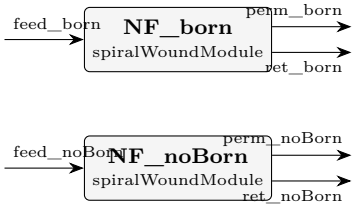
THE MODEL, with nothing declared (DSPM-DE's defaults since 2026-10-24, DEV.md 4c C141, each announced in the run): the hindrance pair of Dechadilok & Deen 2006 (K_d = H/Phi – the earlier default counted the steric partition twice in diffusion); the pore SOLVED by finite differences on 101 nodes, electroneutral at every node with zero current (NanoFiltran's model, solved since C142 by one Newton per station that carries the pore, the wall and the water flux together); and the wall closed with that pore's own permeate. membrane20_dspmde_pore_finite_difference declares the same model and sets the solution-diffusion wall closure beside it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed_CaCl2 | F | 0.0555555555556 |
| stream | feed_MgSO4 | F | 0.0555555555556 |
| stream | feed_NaCl | F | 0.0555555555556 |
| stream | perm_CaCl2 | F | 0.0198924018864 |
| stream | perm_MgSO4 | F | 0.0200291397136 |
| stream | perm_NaCl | F | 0.0215682162363 |
| stream | ret_CaCl2 | F | 0.0356631536691 |
| stream | ret_MgSO4 | F | 0.035526415842 |
| stream | ret_NaCl | F | 0.0339873393193 |
| kpi | NF_CaCl2 | J_w_avg | 7.16824231251e-05 |
| kpi | NF_CaCl2 | R_obs_CaCl2 | 0.696509463412 |
| kpi | NF_CaCl2 | R_obs_MgSO4 | 1 |
| kpi | NF_CaCl2 | R_obs_NaCl | 1 |
| kpi | NF_CaCl2 | dP_feed | 100000 |
| *… and 40 more reference values in the case's `expected`.* | | | |

## membrane11_dspmde_born_toggle

tutorials/steady/membranes/membrane11_dspmde_born_toggle




**What it does:** DSPM-DE NF: Born dielectric-exclusion toggle (same NaCl feed, poreDielectric on vs off)

**The story:** SEE the dielectric-exclusion (Born) term.

The SAME NaCl feed (50 mM, 10 bar) through TWO NF270 elements that differ in ONE thing only:

NF_born : membrane NF270_dspmde – poreDielectric 35 (Born ON) NF_noBorn : membrane NF270_dspmde_noBorn – poreDielectric absent (Born OFF)

Everything else – pore radius, porosity/thickness, fixed charge X_d, feed, pressure, area, k_film – is IDENTICAL. The ONLY physics that changes is the Born dielectric-exclusion term

dW_i = (z_i^2 e^2 / 8 pi eps0 r_i)(1/eps_p - 1/eps_b)

(Bowen & Welfoot 2002, Eq. 16). With eps_p < eps_b an ion pays an extra solvation-energy penalty to enter the low-dielectric pore, so it is excluded MORE strongly. The student reads the Born contribution straight off the two R_obs(NaCl) KPIs: R_obs(NF_born) > R_obs(NF_noBorn).

This is the pedagogical core of DSPM-"DE": the DE (dielectric exclusion) is what distinguishes the 2002 Bowen-Welfoot model from the 1997 steric-Donnan DSPM. Toggling poreDielectric is a single, visible switch.

Both units run DSPM-DE's defaults (since 2026-10-24, DEV.md 4c C141, each announced): the Dechadilok & Deen 2006 hindrance (K_d = H/Phi), the pore SOLVED on 101 nodes, and the wall closed with that pore's own permeate. The Born term is what separates the two observed rejections; read the wall polarisation (Gamma_avg) beside them – the film, not only the pore, sets what the permeate sees.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed_born | F | 0.0555555555556 |
| stream | feed_noBorn | F | 0.0555555555556 |
| stream | perm_born | F | 0.0159093373676 |
| stream | perm_noBorn | F | 0.0201361870707 |
| stream | ret_born | F | 0.0396462181879 |
| stream | ret_noBorn | F | 0.0354193684848 |
| kpi | NF_born | J_w_avg | 5.74022297607e-05 |
| kpi | NF_born | R_obs_NaCl | 0.300852023184 |
| kpi | NF_born | dP_feed | 100000 |
| kpi | NF_born | water_recovery | 0.286194183657 |
| kpi | NF_noBorn | J_w_avg | 7.26834214911e-05 |
| kpi | NF_noBorn | R_obs_NaCl | 0.0938863676869 |
| kpi | NF_noBorn | dP_feed | 100000 |
| kpi | NF_noBorn | water_recovery | 0.362382655966 |
| *... and 14 more reference values in the case's `expected`.* | | | |

## membrane12_sdem_nf270_nacl_traces

tutorials/steady/membranes/membrane12_sdem_nf270_nacl_traces




**What it does:** SDEM on NF270: dominant NaCl 0.01 M + NaNO3/NH4Cl traces – NO3- slips through, the field says why

**The story:** one NF270 element under the SOLUTION-DIFFUSION-ELECTROMIGRATION law ('transport SDEM;'): per-ion permeances, ONE electric field, ZERO current, and therefore an electroneutral permeate with no parameter the uncoupled law does not already have. See the case README for the lesson and the paper the permeances come from.

*The question. Why does nitrate slip through a nanofiltration membrane that rejects chloride, when the two ions have almost the same hydrated size?*

*The lesson. Under transport SDEM; every ion keeps its own permeance, but the ions share ONE electric field, fixed by zero current. NF270 is less permeable to Cl- than to Na+ (the record: 3.9 against 113 um/s), so a field arises that slows the cation and speeds the anions – and a trace anion more permeable than Cl- (NO3-, 13.8 um/s) is pushed through faster than the dominant salt: its rejection sits BELOW the salt's and, at small fluxes, goes negative. The run publishes the field (membranePotential_mV_avg, and psi_mV along the channel) so the mechanism is a number, not a sentence.*

*Source of the numbers. Fernandez de Labastida & Yaroshchuk, Membranes 11 (2021) 272, CC BY 4.0, Table 1 (row NaCl) – ion permeances the authors fitted with the SDEM model to CP-corrected rejections on NF270 at 0.01 M NaCl + 2e-4 M traces, 20 C. What the paper reports for this case: NaCl and NH4+ rejected 40-80 %, NO3- below 50 % and lightly negative at small fluxes. This run is intrinsic (large k_film) at one operating point; the gate check_sdem holds the permeate electroneutral to machine precision and the dominant-salt rejection to the single-salt law R = J_v/(J_v + P_s) with the AMBIPOLAR P_s, recomputed independently.*

*The figures, digitised. constant/experimental/fdl2021_figures.csv holds the 106 measured points and the authors' fitted lines of the paper's Figures 1-4, read off the published panels by bin/curate/digitise_fdl2021.py (the reading error is declared per point; the images are not in the tree). check_sdem drives the engine to each figure's flux and holds it, in the trace limit the authors' lines assume, to those lines within a stated band per series – and pins that at the paper's own 2e-4 M the traces are NOT traces (under a sulfate they carry the anion current and relax the field).*

*What is NOT validated here. The measured SYMBOLS are compared by nobody: the model is held to the authors' FIT, and the symbols scatter about it by a residual that is theirs to explain. NH4+ here is a bound in the paper ('>60 um/s, the fit became insensitive'); the engine meets that line only near 1000 um/s and the record keeps the bound.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0154246666667 |
| stream | feed | T | 293.15 |
| stream | permeate | F | 0.000143506830367 |
| stream | permeate | T | 293.15 |
| stream | retentate | F | 0.0152811598484 |
| stream | retentate | T | 293.15 |
| kpi | NF | J_w_avg | 2.58542546558e-05 |
| kpi | NF | R_obs_Cl | 0.774930206622 |
| kpi | NF | R_obs_NH4 | 0.783943147317 |
| kpi | NF | R_obs_NO3 | 0.307702586566 |
| kpi | NF | R_obs_Na | 0.765592156411 |
| kpi | NF | dP_feed | 5000 |
| kpi | NF | mass_closure_rel | 7.98890233342e-16 |
| kpi | NF | membranePotential_mV_avg | 34.2157491014 |

*... and 1 more reference values in the case's `expected`.*

## membrane13_sdem_nf270_mgso4_traces

tutorials/steady/membranes/membrane13_sdem_nf270_mgso4_traces




**What it does:** SDEM on NF270: dominant MgSO4 0.01 M + NaNO3/NH4Cl traces – both trace anions negatively rejected

**The story:** one NF270 element under the SOLUTION-DIFFUSION-ELECTROMIGRATION law ('transport SDEM;'): per-ion permeances, ONE electric field, ZERO current, and therefore an electroneutral permeate with no parameter the uncoupled law does not already have. See the case README for the lesson and the paper the permeances come from.

*The question. A membrane that rejects MgSO4 almost completely – how can it let MORE nitrate into the permeate than there is in the feed?*

*The lesson. SO4(2-) is the least permeable ion here (0.13 um/s against 0.9 for Mg2+ and 16-80 for the traces). Zero current forces the sulfate and the magnesium fluxes to match, so a strong field arises already at small fluxes; every trace ANION is dragged through it and every trace CATION is held back. Cl- and NO3- come out with NEGATIVE rejections – concentrated in the permeate – while Na+ and NH4+ are rejected. Same permeances as the record, no parameter for the effect: it is what the field does.*

*Source of the numbers. Fernandez de Labastida & Yaroshchuk, Membranes 11 (2021) 272, CC BY 4.0, Table 1 (row MgSO4), NF270 at 0.01 M MgSO4 + 2e-4 M traces, 20 C. What the paper reports: the dominant salt highly rejected, trace cations rejected, Cl- from -70 % at small flux up to +60 %, NO3- negative over the whole flux range (-151 % to -5 %). check_sdem holds the signs the text states at this run's flux, the permeate electroneutral, and the MgSO4 rejection to the single-salt ambipolar law.*

*Against the paper's Figure 4 (digitised into the NaCl witness's constant/experimental/fdl2021_figures.csv): in the TRACE LIMIT the engine reproduces the authors' SDEM lines for Na+, Cl- and NO3- within 2-9 %; at this case's 2e-4 M it does not, and that is the lesson – here 2e-4 M of NO3- at 41.9 um/s carries more anion current than 0.01 M of SO4 at 0.13 um/s, the field relaxes, NO3- is rejected LESS negatively (f 1.37x the trace-limit value at 34.7 um/s) and the trace cations less (0.45x). The authors' analytical fit assumes the traces do not touch the field; their text says they do.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0154246666667 |
| stream | feed | T | 293.15 |
| stream | permeate | F | 0.000137923803682 |
| stream | permeate | T | 293.15 |
| stream | retentate | F | 0.0152867429149 |
| stream | retentate | T | 293.15 |
| kpi | NF | J_w_avg | 2.48474629942e-05 |
| kpi | NF | R_obs_Cl | 0.400430661135 |
| kpi | NF | R_obs_Mg | 0.981477848116 |
| kpi | NF | R_obs_NH4 | 0.665213383817 |
| kpi | NF | R_obs_NO3 | -0.152388078224 |
| kpi | NF | R_obs_Na | 0.665547359434 |
| kpi | NF | R_obs_SO4 | 0.99230502972 |
| kpi | NF | dP_feed | 5000 |

*... and 3 more reference values in the case's `expected`.*

## membrane14_polarisation_multiionic

tutorials/steady/membranes/membrane14_polarisation_multiionic




**What it does:** NF270 + SDEM on a real spacer channel: per-ion film, one interface field (Geraldes & Afonso 2007) – Gamma per ion, the trace anion below the bulk

**The story:** membrane12's feed and law (NF270, NaCl 0.01 M with NO3-/NH4+ traces, 'transport SDEM;') on a REAL spacer channel instead of an intrinsic film: the film coefficient is priced PER ION from the Schock-Miquel correlation on each ion's own diffusivity, corrected for suction (Geraldes & Afonso 2006), and the four ions are COUPLED at the interface by electroneutrality – the multi-ionic polarisation model of Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20-27, with no film thickness and no parameter the intrinsic twin does not have. See the README.

*The question. membrane12 keeps its film out of the way (k_film 1e-2, wall = bulk) so the SDEM rejections are intrinsic. In a module the wall is NOT the bulk: what does concentration polarisation look like when the feed is four ions of different mobility and charge, and how do they pile up?*

*The lesson. The film model needs ONE diffusivity and a mixture of ions has none; applied ion by ion it lets each ion polarise alone, so a fast anion runs ahead and charge separates. Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20-27, keep the module's own mass-transfer correlation, evaluate it PER ION on the ion's own D (k_film_<ion> columns, Cl- above Na+), correct it for the suction of the permeation (the 2006 factor), and add ONE unknown – the electrical potential gradient at the interface, fixed by electroneutrality at the wall. No film thickness, no new parameter. This case declares it:*

*massTransfer { model SchockMiquel; channelHeight 0.7 mm; spacerPorosity 0.9; viscosity 1e-3; } pressureDrop { model SchockMiquel; } polarisation { suctionCorrection GeraldesAfonso2006; ionCoupling electroneutral; }*

*What a student sees: the polarisation index Gamma = (c_m - c_b)/c_b differs BY ION (Gamma_<ion>_avg: Na+ 0.163, Cl- 0.163, NO3- 0.064, NH4+ 0.118 – the dominant pair driven together by the field, the traces each on their own rejection); the interface field xi_V_per_m_avg (~34 V/m) beside the membrane's own potential membranePotential_mV_avg (~31 mV) – two fields, one in the solution and one across the active layer; R_obs BELOW the intrinsic witness (Cl- 68.9 % here against membrane12's 77.5 %), because the wall is richer than the bulk. Set ionCoupling none; and each ion polarises alone: Na+ 0.187, Cl- 0.135, NO3- 0.043, NH4+ 0.143 – the coupling moves the trace anion's Gamma by half. (The README that briefed this case said the trace anion's Gamma might go negative; on this feed it does not, and the number above is what the run says.)*

*Geometry. One nominal 4-inch element (7.9 m2, 1.016 m) fed at 4 m3/h, so the spacer-channel crossflow is ~0.45 m/s (a realistic velocity) and the recovery ~16 %. Membrane12's composition (NaCl 0.01 M, traces 2e-4 M) and its NF270 permeance record, unchanged.*

*Source of the numbers. Permeances: Fernandez de Labastida & Yaroshchuk, Membranes 11 (2021) 272, Table 1 (CC BY 4.0), as in membrane12. Ion diffusivities: the species records' D0 – Na+/Cl- from the standards; NO3-/NH4+ from case-local copies under constant/species/ that add a Nernst-Einstein D0 from lambda0 (Robinson & Stokes 1959, App. 6.1), cited in the record and not promoted. Correlation: Schock & Miquel 1987.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0616986666667 |
| stream | feed | T | 293.15 |
| stream | permeate | F | 0.00968605216978 |
| stream | permeate | T | 293.15 |
| stream | retentate | F | 0.0520126144969 |
| stream | retentate | T | 293.15 |
| kpi | NF | Gamma_Cl_avg | 0.163444876237 |
| kpi | NF | Gamma_NH4_avg | 0.117861790906 |
| kpi | NF | Gamma_NO3_avg | 0.0644251901895 |
| kpi | NF | Gamma_Na_avg | 0.162479502895 |
| kpi | NF | J_w_avg | 2.20896192514e-05 |
| kpi | NF | R_obs_Cl | 0.688552189835 |
| kpi | NF | R_obs_NH4 | 0.712239505944 |
| kpi | NF | R_obs_NO3 | 0.183725671006 |
| *… and 7 more reference values in the case's `expected`.* | | | |

## membrane15_module_nf270_4040

tutorials/steady/membranes/membrane15_module_nf270_4040




**What it does:** NF270-4040 element as a RECORD: NaCl 2 g/L at the rated 4.8 bar – the geometry, limits and estimates come from data/standards/assets/NF270-4040.dat

**The story:** a commercial element named as a RECORD. 'module NF270-4040;' replaces 'membrane', 'area', 'length', 'channelHeight' and 'spacerPorosity': the record in data/standards/assets/NF270-4040.dat carries the data sheet's active area and limits, and three EDUCATED values the sheet does not state (spacer thickness, spacer porosity, leaf length), each announced as an estimate on every run. NaCl 2 g/L at the sheet's rated 4.8 bar; the rated MgSO4 test itself is run by the gate check_membrane_modules and printed beside the sheet. See the README.

*The question. What does a student have to type to run a commercial membrane element, and where do the numbers the data sheet does not print come from?*

*Before this case a spiral-wound module was typed by hand: membrane NF270; area 7.6 m2; length 1 m; and, inside massTransfer {}, channelHeight 0.7 mm; spacerPorosity 0.9; – five numbers from four places, and nothing said which were the manufacturer's and which were guessed. Here one line, module NF270-4040;, names the record data/standards/assets/NF270-4040.dat (kind membraneModule), which carries the DuPont FilmTec NF270 Element data sheet's active area (82 ft2, 7.6 m2) and its limits (41 bar, 45 C, 1.0 bar drop, 3.6 m3/h feed, pH 3-10), the membrane the element is wound with (NF270), and three values the sheet does NOT state.*

*What the student sees at the top of the log. Every run that reads the record announces its estimates, one line each, with how to verify them:*

*[estimate] module 'NF270-4040': channelHeight ... is an ESTIMATE, reviewStatus unverified – 28 mil is the feed spacer DuPont states for its 8-inch SW30HR-380 ...; the NF270-4040 sheet states none ... [estimate] module 'NF270-4040': spacerPorosity ... – 0.90 is typical for commercial diamond feed spacers; to be verified by weighing a spacer coupon ... [estimate] module 'NF270-4040': leafLength ... – element length 1.016 m per the data sheet includes the end caps; the leaf length is not published ...*

*They also reach the end-of-run caveat block and the result JSON. A record with no estimate prints nothing: silence keeps meaning "nothing was assumed". The [spec] lines say what channel the record gave, and that this is a spiral leaf with TWO membrane faces (W = A/(2L)); compare membrane17, a flat cell with one.*

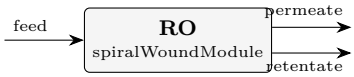
*The limits speak, they do not stop the run. Raise the feed pressure in 0/feed above 41 bar, or the feed flow above 3.6 m3/h, and the log prints a [limit] module 'NF270-4040': feed P ... EXCEEDS the maximum operating pressure ... line and continues – a student may run an element outside its rating on purpose, and the run then says so.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0400358277778 |
| stream | feed | T | 298.15 |
| stream | permeate | F | 0.0079811998844 |
| stream | permeate | T | 298.15 |
| stream | retentate | F | 0.0320546278934 |
| stream | retentate | T | 298.15 |
| kpi | NF | Gamma_NaCl_avg | 0.219567749358 |
| kpi | NF | J_w_avg | 1.89258351319e-05 |
| kpi | NF | R_obs_NaCl | 0.723443537339 |
| kpi | NF | dP_feed | 25181.9130567 |
| kpi | NF | mass_closure_rel | 4.61155269823e-16 |
| kpi | NF | u_crossflow_inlet | 0.282091538818 |
| kpi | NF | water_recovery | 0.199151845893 |
| kpi | NF | solute_closure_rel | 0 |

## membrane16_module_sw30hr_8040

tutorials/steady/membranes/membrane16_module_sw30hr_8040




**What it does:** SW30HR-380 element as a RECORD: seawater 32 g/L NaCl at the rated 55 bar – membrane01's separation with the hardware read from data/standards/assets/SW30HR-380.dat

**The story:** the 8-inch seawater element named as a RECORD. membrane01 typed 'area 35 m2; length 1 m; k_film 5e-5'; here 'module SW30HR-380;' reads the 35 m2, the 28 mil spacer (a data-sheet fact on this element) and the two estimates (porosity, leaf length) from data/standards/assets/SW30HR-380.dat, and the film is COMPUTED on the record's channel. Seawater 32 g/L NaCl at the sheet's 55 bar (800 psi), 8 % recovery being the sheet's own test point. See the README.

*The question. membrane01 runs "an SW30HR-style element, 35 m2, 1 m channel, k_film 5e-5 m/s". What changes when the element is named instead of typed?*

*Compare with membrane01. Same membrane record (SW30HR, the solution-diffusion fit), the same seawater-class feed (32 g/L NaCl here – the data sheet's own test salinity – against membrane01's 35 g/L), the same 55 bar. membrane01 types area 35 m2; length 1 m; and a CONSTANT k_film 5e-5. Here module SW30HR-380; reads the record data/standards/assets/SW30HR-380.dat: 380 ft2 (35 m2), the 28 mil feed spacer the sheet states (a data-sheet fact on this element, where the NF270-4040 sheet states none), and two ESTIMATES the sheet does not give (spacer porosity 0.90, leaf length 0.95 m) – each announced on every run with how to verify it. The film is then COMPUTED (Schock & Miquel on the record's channel, ~0.32 m/s crossflow) instead of declared, and the polarisation index Gamma_NaCl_avg says what the film did.*

*What the student sees. Two [estimate] lines (not three: the spacer is measured here), the [spec] channel lines (W = A/(2L) = 18.4 m of leaf width in one 8-inch element), the record's rated test printed beside the result (32,000 ppm NaCl, 55 bar, 8 %: 26 m3/d, 99.7 %), and the KPIs. Headline: NaCl rejection ~99.84 %, recovery ~16 %, ~62 LMH. The sheet's own point is 26 m3/d over 35 m2 = 31 LMH: the engine's A_w (a fit to membrane01's teaching envelope) is about twice the sheet's – the gate check_membrane_modules runs the rated test and prints that ratio as a measurement (see docs/design/a-module-is-a-record.md).*

*The limits. 83 bar, 45 C, 1.0 bar drop per element, pH 2-11; the sheet states no maximum feed flow for this element, so none is declared and none is checked – an absent limit is absent, never a zero. Set P 90 bar; in 0/feed to see the [limit] line; the run continues.*

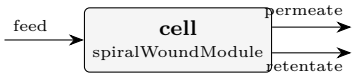
*Try. elements 6; beside module – a six-element pressure vessel of this record (the count is not geometry, so it is allowed); the per-element pressure drop is checked against the sheet's 1.0 bar.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.203553611111 |
| stream | feed | T | 298.15 |
| stream | permeate | F | 0.0332041779473 |
| stream | permeate | T | 298.15 |
| stream | retentate | F | 0.170349433164 |
| stream | retentate | T | 298.15 |
| kpi | RO | Gamma_NaCl_avg | 0.240392810527 |
| kpi | RO | J_w_avg | 1.70912638753e-05 |
| kpi | RO | R_obs_NaCl | 0.998416486976 |
| kpi | RO | dP_feed | 32650.3893093 |
| kpi | RO | mass_closure_rel | 3.55268001518e-16 |
| kpi | RO | u_crossflow_inlet | 0.318043615679 |
| kpi | RO | water_recovery | 0.159516951325 |
| kpi | RO | solute_closure_rel | 2.11179646744e-16 |

## membrane17_sepa_cf_flat_cell

tutorials/steady/membranes/membrane17_sepa_cf_flat_cell




**What it does:** A laboratory cell in five lines: SEPA CF flat-sheet cell + an NF270 coupon, membrane12's NaCl + traces feed under SDEM at 0.2 m/s crossflow, the coupled Geraldes & Afonso film

**The story:** a LABORATORY CELL a student can build a case for in five lines: 'module SEPA_CF;' (Sterlitech's flat-sheet crossflow cell, data/standards/assets/SEPA_CF.dat) with the NF270 coupon of membrane12 ('membrane NF270_sdem_NaCl;', the case-local record with the paper's per-ion permeances), membrane12's feed (NaCl 0.01 M with NO3-/NH4+ traces) under 'transport SDEM;', the film per ion on Schock & Miquel and coupled at the interface (Geraldes & Afonso 2007). The cell has ONE membrane face, so the channel width is A/L, not A/(2L) – the engine reads the format and says so. The feed flow is the one that gives 0.2 m/s in the 31 mil channel (see 0/feed). See the README.

*The question. A student has a Sterlitech SEPA CF cell on the bench and a coupon of NF270. How many lines of case does that take?*

*module SEPA_CF; // the cell membrane NF270_sdem_NaCl; // the coupon transport SDEM; P_permeate 1.0 bar; massTransfer { model SchockMiquel; viscosity 1.0e-3; }*

*plus the polarisation policy (Geraldes & Afonso 2006 suction correction, ions coupled at the interface) and a feed. The cell record data/standards/assets/SEPA_CF.dat (kind membraneModule, format flatSheetCell) carries the manual's Table 1: 140 cm2 active area, the 95.3 mm x 1.09 mm slot, 70 ml hold-up, 69 bar and – the binding limit – the polypropylene spacer's 82 C. A cell takes ANY coupon, so the membrane is the CASE's to declare beside the module (a spiral element is wound with its own; there membrane beside module is refused).*

*One membrane face. In a spiral leaf the channel has membrane on both sides, so the module reads the leaf width as A/(2L). A flat cell has the coupon on its permeate carrier and a steel lid above the channel: ONE face, W = A/L. The engine reads the record's format and says which it used in the [spec] line – and the channel LENGTH is not stored anywhere: it is activeArea / slotWidth = 146.9 mm, derived where it is used (the tree never stores a derivative).*

*The channel height is yours. The slot is 1.09 mm deep and shimmed down to the spacer you install, so the operating channel is a fact about your stack, not about the cell. The record's 31 mil (0.787 mm), one of the four spacers Sterlitech ships, is an EDUCATED DEFAULT marked origin estimated; reviewStatus unverified; and announced on every run: copy the record to constant/assets/ and set your own.*

*The crossflow is declared, and published. The manual (section 5.1) says spacer channels are usually run at 0.1-0.5 m/s. 0/feed states the arithmetic: channel section = 95.3 mm x 0.787 mm x 0.90 = 6.75e-5 m2, so 0.2 m/s needs 48.6 L/h, and membrane12's feed (0.01 M NaCl + 2e-4 M NaNO3 and NH4Cl) is scaled to it. The run publishes u_crossflow_inlet = 0.200 m/s so the case can be held to the number it claims.*

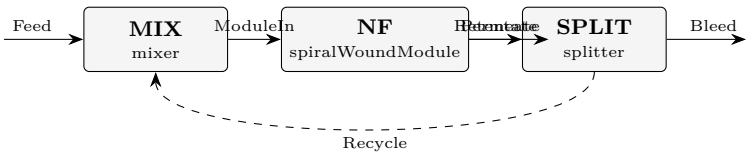
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000749639133333 |
| stream | feed | T | 293.15 |
| stream | permeate | F | 1.90808514124e-05 |
| stream | permeate | T | 293.15 |
| stream | retentate | F | 0.000730558285788 |
| stream | retentate | T | 293.15 |
| kpi | cell | Gamma_Cl_avg | 0.379935512008 |
| kpi | cell | Gamma_NH4_avg | 0.2721421923 |
| kpi | cell | Gamma_NO3_avg | 0.146470492872 |
| kpi | cell | Gamma_Na_avg | 0.377501162862 |
| kpi | cell | J_w_avg | 2.455497741e-05 |
| kpi | cell | R_obs_Cl | 0.67551068628 |
| kpi | cell | R_obs_NH4 | 0.714109096783 |
| kpi | cell | R_obs_NO3 | 0.180801606137 |

*... and 7 more reference values in the case's `expected`.*

## membrane18_nf_feed_and_bleed

tutorials/steady/membranes/membrane18_nf_feed_and_bleed




**What it does:** Feed-and-bleed NF: the module never sees the feed, it sees the LOOP concentration

**The story:** one NF module, a recycle loop, and the concentration the module actually works on.

THE ARRANGEMENT. A feed-and-bleed loop, the way a membrane skid is run when the recovery is set by recirculating the retentate: fresh feed joins the recycled RETENTATE, the mixture crosses the module once, the permeate leaves as product, and the retentate is split – 95 % goes round again, the other 5 % is the BLEED (the concentrate).

Feed –→[ MIX ]–→[ NF ]–→ Permeate (product) ^ | | Retentate | | +– Recycle <–[ SPLIT ]-→ Bleed (concentrate)

It is the pressure-driven analogue of ed07_feed_and_bleed: the same loop, the same trap. A student sizes the module from the FEED analysis. The module never sees the feed: it sees the MIXED loop concentration, and everything that depends on concentration is evaluated there.

WHAT THE RUN MEASURES (printed by this run; nothing here is a target):

loop (0.95/0.05) single pass (0.0/1.0) glucose, fresh feed 9.970e-4 (mole fr.) 9.970e-4 glucose, what the MODULE 1.806e-3 (1.81x) 9.970e-4 is fed glucose, bleed 1.940e-3 2.669e-3 glucose, permeate 5.617e-4 4.548e-4 overall recovery, by mass 68.16 % 75.15 % (permeate / fresh feed) water flux (average) 165.2 LMH 182.1 LMH R_obs, module 68.54 % 54.16 % rejection of the PLANT, 43.7 % 54.4 % 1 - x_perm / x_feed

The single-pass column is this same case with ‘fractions ( 0.0 1.0 )‘ in SPLIT: nothing recycles, and the module is fed the fresh feed.

Read off the module’s own profile (the Plot tab, or the axialProfile in converged/internalStates/NF – columns z, J_w, Q_b):

loop single pass flux at the module INLET 168.1 LMH 213.1 LMH (= at the fresh-feed conc.) flux at the module OUTLET 162.2 LMH 135.3 LMH volumetric flow through 7.1e-3 m3/s 1.0e-3 m3/s the module, inlet outlet flow / inlet flow 0.90 0.25

THREE READINGS OF THE SAME LOOP. The module works at 1.81 times the feed concentration. Its flux at the fresh-feed concentration is 213 LMH (the single pass’s first node); the loop gets 165. A module sized from the feed analysis is short of area by 213/165 = 1.29, nearly a third. R_obs = 68.54 % is the module’s rejection AGAINST WHAT IT IS FED. The permeate is 43.7 % cleaner than the fresh feed, not 68.5 %: a rejection is a property of the membrane at a concentration, and the plant is judged against the feed. In the single pass the two numbers nearly coincide (54.2 % and 54.4 %), because there the module IS fed the fresh feed. The loop recovers less and makes a dirtier permeate than one pass of the same area, and that is PHYSICS, not a defect of the model: a feed-and-bleed loop runs its whole membrane near the concentration of the bleed, while a single pass spends its first metres on the dilute feed. It is the known price of feed-and-bleed, and the reason a large plant stages its modules instead. What this model does NOT show is why a loop is used at all. The single pass here takes 75 % recovery out of one element, so the flow leaving the element is a quarter of the flow entering it; a real element at that recovery would lose its cross-flow velocity, its film coefficient and its polarisation control, and element makers state a maximum recovery per element and a minimum concentrate flow for exactly that reason. The loop pushes seven times the flow through the module and loses only a tenth of it – that is what it buys. Here ‘k_film‘ is DECLARED, 5.0e-5 m/s, whatever the flow, so the single pass pays nothing for its collapsing velocity and the comparison flatters it. (membrane04 computes k_film and the pressure drop from the spacer hydraulics; see WHY NO PUMP for why that is not used here.)

WHY NO PUMP, AND WHY dP_feed_total IS ZERO. A real loop has a recirculation pump: the module loses feed-side pressure on every pass and the pump makes it up. Here the mixer takes the LOWEST inlet

pressure, so a loop whose recycle arrives a bar low and has no pump loses that bar again each pass and has no steady state. The ‘pump‘ unit would restore it, and it now gets further than it did: glucose's record carries a liquid molar volume (Seitz et al. 2019, a partial molar volume at infinite dilution), which the incompressible pump model needs. It still refuses, one datum later: the outlet temperature needs the liquid ENTHALPY of the mixture and glucose has no ‘liquidHeatCapacity‘ (a nonvolatile's enthalpy must not route through the ideal-gas reference). The heat capacity of glucose in solution needs a primary source of its own, and a typed number would be a number with no source. So the feed-side pressure drop is DECLARED ZERO, the loop sits at 10 bar throughout, and the pump's work is not counted anywhere. Both are idealisations of this case, not of the module.

THE ORDER OF THE UNITS IS THE ORDER THEY RUN. ‘Recycle‘ is produced by the splitter (declared LAST) and consumed by the mixer (declared FIRST), so the plan cannot reach it without a cut: ‘system/solverDict‘ declares ‘tearStreams ( Recycle );‘ and a Newton on the tear variables closes it. Omit the declaration and the engine refuses by name with the cycle chain.

WHAT IS PINNED AND WHAT IS NOT. The golden pins the answer this model gives. The feed is membrane02's 10 g/L glucose; the NF270 permeability set is the record's own pedagogical calibration to that case (it declares itself ‘fittedToCase‘), so this is a structural witness of the loop and not a validation of NF270 on glucose. No measured feed-and-bleed plant is reproduced anywhere in this tree; fouling is not modelled; and the energy balance is UNAVAILABLE, as it is for every glucose case, because the glucose record carries only its solid formation datum.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Bleed | F | 0.0175408330548 |
| stream | Bleed | T | 298.15 |
| stream | Feed | F | 0.0555555555556 |
| stream | Feed | T | 298.15 |
| stream | ModuleIn | F | 0.388831247128 |
| stream | ModuleIn | T | 298.15 |
| stream | Permeate | F | 0.0380145860324 |
| stream | Permeate | T | 298.15 |
| stream | Recycle | F | 0.333275828041 |
| stream | Recycle | T | 298.15 |
| stream | Retentate | F | 0.350816661096 |
| stream | Retentate | T | 298.15 |
| kpi | MIX | F_out | 0.388831247128 |
| kpi | MIX | P_out | 1000000 |
| *... and 14 more reference values in the case's* `expected`*.* | | | |

## membrane19_uf_gel_polarisation

tutorials/steady/membranes/membrane19_uf_gel_polarisation




**What it does:** UF of a protein by gel polarisation: past a modest pressure the flux is set by mass transfer, not by the pump

**The story:** ULTRAFILTRATION of a protein, and the flux that stops answering the pump.

THE PLANT. 1 m3/h of a 10 kg/m3 bovine serum albumin solution (BSA) is fed at 3 bar to 5 m2 of a 30 kDa UF membrane; the permeate leaves at 1 bar, so the membrane sees 2 bar.

THE LAW ('transport gelPolarisation', Michaels 1968; Blatt et al. 1970). The water passes by viscous flow, J = A_w dP, and the protein is pushed to the wall by that flow, where it piles up by the film law (c_m - c_p)/(c_b - c_p) = exp(J/k), c_p = S c_m. A protein solution does not concentrate without limit: at the GEL concentration c_g the wall layer stops being a solution. From there the wall is pinned at c_g and the flux is whatever the film can carry away:

J = k ln( c_g (1 - S) / (c_b - S c_g) ) ~ k ln(c_g / c_b)

– independent of pressure. Here A_w dP would be 360 L/(m2 h), while the film carries k ln(c_g/c_b) = 1e-5 ln(25) m/s = 116 L/(m2 h) at the inlet: the module is GEL-LIMITED from its first node, and doubling the pump pressure would buy nothing (check_uf_gel sweeps it). Along the channel the protein concentrates, c_b rises, and the flux FALLS as ln(c_g/c_b): the profile shows it.

WHAT IS DECLARED AND WHERE IT COMES FROM – read before quoting any number: BSA (constant/components/BSA.dat): MW and formula DERIVED from the UniProt sequence P02769, the derivation in the record's header. Its charge is not modelled. the membrane (constant/assets/UF30_hypothetical.dat): a HYPOTHETICAL teaching membrane, A_w 5e-5 m/(s bar) = 180 L/(m2 h bar), not a product; c_gel 250 kg/m3, S = 0.01 and k_film 1e-5 m/s: HYPOTHETICAL teaching values of the order the UF literature reports for albumin, not a measurement (the ruling of 2026-09-30: a declared sieving coefficient, no MWCO model, the case saying it is a hypothesis). So the SHAPE is the lesson – the plateau, the flux falling along the channel – and no number here may be quoted as a property of BSA or of any real membrane.

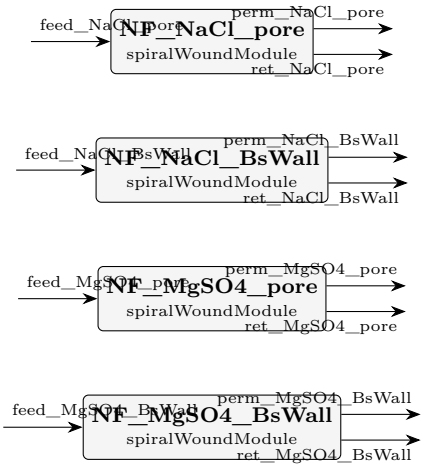
NOT MODELLED, said plainly: the osmotic pressure of the protein (the gel model's own assumption), fouling and its decline in time (see the batch diafilter's Hermia laws), the protein's charge and the pH, and the temperature dependence of k and c_g. No first law is published: the BSA record carries no heat capacity or formation enthalpy.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0153611529458 |
| stream | feed | T | 298.15 |
| stream | permeate | F | 0.00868072508216 |
| stream | permeate | T | 298.15 |
| stream | retentate | F | 0.00668042786368 |
| stream | retentate | T | 298.15 |
| kpi | UF | Gamma_BSA_avg | 18.0522439361 |
| kpi | UF | J_w_avg | 3.13550187502e-05 |
| kpi | UF | R_obs_BSA | 0.748442407672 |
| kpi | UF | dP_feed | 0 |
| kpi | UF | mass_closure_rel | 3.97205898188e-16 |
| kpi | UF | solute_closure_rel | 1.58180682192e-16 |
| kpi | UF | water_recovery | 0.560896239275 |

## membrane20_dspmde_pore_finite_difference

tutorials/steady/membranes/membrane20_dspmde_pore_finite_difference




**What it does:** DSPM-DE NF: the charged pore solved node by node (the default) on NaCl and MgSO4 at 10 bar, its wall closed with the pore's own permeate against the solution-diffusion B_s closure

**The story:** the charged pore SOLVED, and the wall closed with the permeate it gives, or with one it does not.

FOUR NF270 elements, two salts – every unit has the SAME membrane (NF270_dspmde: r_p 0.6 nm, X_d = -5 mol/m3, Born on), the same 10 bar, the same 5 mM feed of its salt, and the same SOLVED pore (101 nodes):

NF_NaCl_pore NF_MgSO4_pore wallClosure porePermeate; NF_NaCl_BsWall NF_MgSO4_BsWall wallClosure solutionDiffusion;

The '_pore' units DECLARE what has been DSPM-DE's default since 2026-10-24 (DEV.md 4c C141, C142) – the pore solved, the wall closed with that pore's own permeate – so the declaration is visible; membrane10 and membrane11 run the same model with nothing declared.

WHAT THE SOLVED PORE IS. The extended Nernst-Planck equation of every ion on 101 nodes across the pore, electroneutrality with the fixed charge at EVERY node, zero net current, and the Donnan + steric + Born partitions at both faces; and, in the SAME Newton (C142), the film's interface balance at the wall and the water flux J_w = A_w (dP - dpi) – one Newton per station on the declared block-tridiagonal structure, with an analytic Jacobian. The problem is STIFF – a co-ion held back by the Donnan field carries a migration and a diffusion flux far larger than the flux they net, and the run prints that ratio ("stiffness") beside its residuals. NanoFiltran (Geraldes & Alves, "Computer program for simulation of mass transport in nanofiltration membranes", J. Membr. Sci. 321 (2008) 172-182) is the reference it is compared with, in tutorials/props/membrane/ nanofiltran_geraldesAlves2008 (the paper's one case).

WHAT THE PAIR TEACHES: THE WALL MUST BE CLOSED WITH THE PERMEATE THE MEMBRANE MAKES. The film's interface balance (Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20, Eq. 1) says the solute leaving the wall into the membrane is J_v c_p. 'porePermeate' puts there the permeate THIS pore makes, so the film delivers to the wall exactly what the pore carries away – solute is conserved across the interface. 'solutionDiffusion' puts there c_p = B_s c_m/(J_v + B_s) with the record's solution-diffusion B_s – a permeate from ANOTHER model, which the pore does not make – so the film is solved for a solute flux that does not cross the membrane. (It is the wall the retired closed form had, and C138 traced a NEGATIVE NaCl rejection on membrane11 to it.) The gap measures how far that B_s is from what the pore transmits: NaCl B_s 5e-6 m/s says c_p ~ 0.06 c_m (94 % at the wall); the pore transmits about 0.16 c_m (its R_obs 49.7 % over a wall 3.07 times the bulk), nearly three times more. The B_s wall believes almost all the salt is held back, so its film must carry more salt back to the bulk and over-polarises (Gamma 2.79 against 2.07); the pore is fed that richer wall, and R_obs falls from 49.73 % to 30.80 %. MgSO4 both permeates are a few parts in ten thousand of the wall, the film carries back essentially everything either way, and the two R_obs agree to 1e-6 (99.9751 % / 99.9752 %). So the

closure is a modelling decision exactly where the permeate is not negligible – which, in nanofiltration, is the monovalent salt.

THE RETIRED CLOSED FORM. Until 2026-10-24 this case carried a ‘method closedForm;‘ unit (the default until C141): the pore made electroneutral at the feed face only, each ion carried across ALONE by a closed-form transmission with no field coupling the ions inside. With the corrected hindrance (K_d = H/Phi, C141) that transmission tends to K_c > 1 at low Peclet number – it put more NaCl in the permeate than the wall held and exhausted the bulk, and gave MgSO4 58 % where the solved pore gives 99.98 %. It is RETIRED (DEV.md 4c C142): the word refuses by name with that reason.

WHAT TO LOOK AT. The four R_obs and Gamma, and the [DSPM-DE] lines: the node count, the first station’s Newton iterations and residual, the block-tridiagonal structure and the analytic Jacobian VERIFIED on the first solve, the pore summary per unit (flux constancy, zero current and electroneutrality, each measured on the answer), and the continuation the MgSO4 units need at their first station (announced and counted).

WHAT THIS CASE DOES NOT CLAIM. Nothing in it is validated against any measurement. The grid, the residuals, the analytic Jacobian and the analytic limit are held by bin/curate/check_pore_solver.py. Shared by all four units: the Davies activity at the feed face, an ideal permeate, van’t Hoff osmotic pressure.

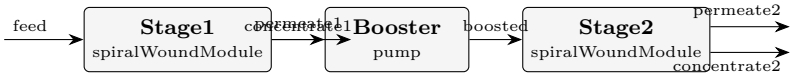
Data: as membrane10_dspmde_divalent – the salts carry their ion decomposition, the membrane its poreModel{} tier, the ions their radius and D0 from the species records.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed_MgSO4_pore | F | 0.0555555555556 |
| stream | feed_MgSO4_pore | T | 298.15 |
| stream | feed_NaCl_pore | F | 0.0555555555556 |
| stream | feed_NaCl_pore | T | 298.15 |
| stream | perm_MgSO4_pore | F | 0.0200796563488 |
| stream | perm_MgSO4_pore | T | 298.15 |
| stream | perm_NaCl_pore | F | 0.0216109651417 |
| stream | perm_NaCl_pore | T | 298.15 |
| stream | ret_MgSO4_pore | F | 0.0354758992068 |
| stream | ret_MgSO4_pore | T | 298.15 |
| stream | ret_NaCl_pore | F | 0.0339445904139 |
| stream | ret_NaCl_pore | T | 298.15 |
| kpi | NF_MgSO4_pore | Gamma_CaCl2_avg | 0 |
| kpi | NF_MgSO4_pore | Gamma_MgSO4_avg | 3.23444105173 |
| *... and 58 more reference values in the case’s `expected`.* | | | |

## membrane21_two_stage_nf_array

tutorials/steady/membranes/membrane21_two_stage_nf_array




**What it does:** A two-stage NF array, 4:2 vessels x 6 generic 8-inch elements, interstage booster: one unit per stage, the spacer and the elements as records, SDEM + the multi-ionic film per ion and per element

**The story:** a TWO-STAGE nanofiltration ARRAY, 4:2 pressure vessels of 6 generic 8-inch elements each, with the interstage booster pump a unit of its own. Multi-ion throughout: membrane14's NaCl feed with NO3-/NH4+ traces, written ion by ion, through the SDEM transport law and the Geraldes & Afonso multi-ionic film. See the README.

ONE UNIT IS ONE STAGE. 'vessels 4; elements 6;' – four identical pressure vessels in parallel, each holding six elements in series. The vessels see the same feed state, so the engine solves ONE vessel on F/4 and scales its retentate and permeate by 4 (said on every run). Stage 2 takes stage 1's concentrate, BOOSTED by the pump, on half the vessels: the "tapered" array that keeps the crossflow up as the feed shrinks.

THE HARDWARE IS RECORDS. 'module Generic-8040;' (data/standards/assets/) is a 37.2 m2 element of 16 leaves, 0.854 m long, wound with the FilmTec FT30 feed spacer Schock & Miquel measured (assets/SchockMiquel- FilmTecFT30.dat): its thickness is the channel height, its Eq. (17) hydraulic diameter (0.952 mm) is what the correlations read, and its correlations (Sh = 0.065 Re^0.875 Sc^0.25, lambda = 6.23 Re^-0.3) come WITH their measured window, 100 < Re < 1000 – outside which the run announces the extrapolation. Every value of the generic element is an estimate and is announced; the membrane is the case's.

PER ION, PER ELEMENT. The film coefficient is priced for each ion on its own Sc_i = nu/D_i, the four ions are coupled at the wall by electroneutrality, and every element publishes its own flux, friction drop, feed, inlet Re and the polarisation modulus of each ion ('J_w_avg_elemK', 'dP_elemK', 'feedFlow_elemK', 'Re_in_elemK', 'Gamma_<ion>_elemK').

*The question. A real membrane plant is not one element: it is an ARRAY of pressure vessels, several elements in series in each, the vessels of a stage in parallel, the concentrate of one stage boosted and fed to the next on fewer vessels. What does each stage, and each element of a vessel, actually do – and where does the film correlation stop being a measurement?*

*The case. 48 m3/h of membrane14's solution (NaCl 0.01 mol/L with NO3-/NH4+ traces at 2e-4 mol/L, written ion by ion) through a 4:2 array of Generic-8040 elements, six per vessel:*

*feed (2.4 bar) → Stage1 (4 vessels x 6) → concentrate1 → Booster (+1.5 bar) → Stage2 (2 vessels x 6) → concentrate2 permeate1, permeate2 leave as two product streams*

*spiralWoundModule with vessels <n>; elements <m>; IS a stage: the vessels are identical, so the engine solves ONE vessel on F/n and scales its outlets by n, and says so ([array]). The booster is a pump of its own, so its shaft work is on the stream table.*

*The hardware is records. module Generic-8040; is a generic 8-inch x 40-inch element (data/standards/assets/Generic-8040.dat): 37.2 m2, 16 leaves, 0.854 m leaf length – every value an ANNOUNCED estimate – and the leaf width DERIVED, W = A/(2 n L) = 1.361 m. It is wound with the FilmTec FT30 feed spacer of Schock & Miquel, Desalination 64 (1987) 339-352, Tab. 1 (assets/SchockMiquel-FilmTecFT30.dat): h = 0.77 mm, d_F = 0.385 mm, eps = 0.89, so S_v = 4/d_F (Eq. 23) and the hydraulic diameter is their Eq. 17, d_h = 4 eps/(2/h + (1 - eps) S_v) = 0.952 mm – not the 2 h eps = 1.37 mm an inline channel would use. The spacer record carries the two correlations (Sh = 0.065 Re^0.875 Sc^0.25, Eq. 30; lambda = 6.23 Re^-0.3, Eq. 24) WITH their window, 100 < Re < 1000: the case names the model, the record supplies the numbers, and a run outside the window is announced.*

*What a student sees. - Per stage: stage 1 recovers 37.7 % of its feed, stage 2 35.1 % of its own (about 60 % overall); the observed rejections rise in stage 2 (Na+ 35.1 % → 40.2 %) because its flux is lower against a richer feed. - Per element (J_w_avg_elemK, dP_elemK, feedFlow_elemK, Re_in_elemK, Gamma_<ion>_elemK): the first element of each vessel does the most work and polarises the most (Gamma_Na 0.090 in element 1 of stage 1, 0.014 in element 6); Re falls from 213 to 132 along a stage-1 vessel, still inside the spacer's window – the 4:2 taper keeps stage 2's inlet Re at 265. - Per ion: the film*

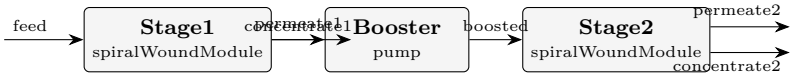
*coefficient is the spacer correlation evaluated on each ion's own D (k_film_<ion>, k proportional to D^0.75 through Sc^0.25), and the four ions are coupled at the wall by electroneutrality (Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20-27): the trace nitrate passes NEGATIVELY rejected under the dominant NaCl (R_obs -4.7 % in stage 1), as in membrane12/14.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | boosted | F | 0.461180543856 |
| stream | boosted | T | 293.165385998 |
| stream | concentrate1 | F | 0.461180543856 |
| stream | concentrate1 | T | 293.15 |
| stream | concentrate2 | F | 0.299341194167 |
| stream | concentrate2 | T | 293.165385998 |
| stream | feed | F | 0.740384 |
| stream | feed | T | 293.15 |
| stream | permeate1 | F | 0.279203456144 |
| stream | permeate1 | T | 293.15 |
| stream | permeate2 | F | 0.161839349689 |
| stream | permeate2 | T | 293.165385998 |
| kpi | Booster | F | 0.461180543856 |
| kpi | Booster | P_in | 133388.103411 |
| *... and 187 more reference values in the case's `expected`.* | | | |

## membrane22_two_stage_nf_array_dspmde

tutorials/steady/membranes/membrane22_two_stage_nf_array_dspmde




**What it does:** membrane21 with the charged-pore membrane model: the same 4:2 array of generic 8-inch elements and booster, DSPM-DE with its solved pore, the Geraldes & Afonso film, D(T) by Stokes-Einstein

**The story:** membrane21's TWO-STAGE array (4:2 pressure vessels of 6 generic 8-inch elements, the interstage booster a unit of its own) with a DIFFERENT MEMBRANE MODEL: DSPM-DE, the charged pore with steric, Donnan and dielectric exclusion (Bowen & Welfoot 2002), SOLVED across the pore at every station (the extended Nernst-Planck equations on 101 nodes, one Newton with the film's interface balance and the water flux – the default since C141). See the README.

THE HARDWARE IS THE SAME RECORD. 'module Generic-8040;' with the FilmTec FT30 spacer and Schock & Miquel's correlations, exactly membrane21's; only 'membrane' and 'transport' change. NF270_dspmde (data/standards/assets/) carries the pore: radius 0.60 nm, A_k/dx, a fixed charge -5 mol/m3, a pore dielectric 35.

THE FEED IS SALTS, THE MODEL WORKS IN IONS. DSPM-DE decomposes NaCl, CaCl2 and MgSO4 into Na+, Ca2+, Mg2+, Cl- and SO4 2- (their charges, Stokes radii and D0 from the species records) and solves them together in the pore; the film couples the SAME ions at the wall by electroneutrality ('ionCoupling electroneutral;', Geraldes & Afonso 2007), each ion's k on its own D on Sc_i. The D's are corrected to 20 C by Stokes-Einstein on the package's declared liquid viscosity (constant/thermoPhysPropDict).

*The question. membrane21 runs a two-stage array of generic 8-inch elements with the SDEM membrane model. What changes when the SAME hardware carries a different membrane model – the charged pore of DSPM-DE?*

*What is the same. The array (4:2 vessels of six Generic-8040 elements, the interstage booster a unit of its own), the FT30 feed spacer and Schock & Miquel's correlations, the Geraldes & Afonso multi-ionic film (ionCoupling electroneutral; with the 2006 suction correction), 20 C, 12 m3/h per stage-1 vessel. Only two lines of each stage differ:*

*membrane NF270_dspmde; // the pore tier: r_p 0.60 nm, X_d -5 mol/m3, eps_p 35 transport DSPM-DE; // the pore SOLVED at every station*

*What the model does. DSPM-DE (Bowen & Welfoot 2002) carries each ion through the pore by the extended Nernst-Planck equations – hindered diffusion, convection and migration – with steric, Donnan and dielectric exclusion at both faces, on 101 nodes, in ONE Newton per station with the film's interface balance and the water flux. The feed is declared by SALTS (NaCl, CaCl2, MgSO4); the law decomposes them into Na+, Ca2+, Mg2+, Cl- and SO4 2-, and the film couples those same ions at the wall. Each stage prints a pore summary: 61 and 43 pore solves, at most 8 Newton iterations each. The whole array runs in under a second.*

*D(T). The ion D0 are 25 C values; this case DECLARES the liquid viscosity (transport { liquid { viscosity { model Vogel; } } } in its thermoPhysPropDict), so every D is corrected by Stokes-Einstein to the stage's temperature – 0.874 at 20 C, printed on the run as [D(T)] (the booster's dissipation warms stage 2 by 0.015 K, and its factor is 0.8744). membrane21, which declares none, runs on the 25 C values and says so.*

*What the student sees – and the limit it shows. In ions, the pore rejects SO4 2- almost completely (Donnan and dielectric exclusion of a divalent co-ion) while Mg2+ and Ca2+ pass in part, and Na+ and Cl- more. But the module marches SALTS, and a pore conserves CHARGE, not salts: Mg2+ crosses with Cl-, not with SO4 2-. So a salt's two legs differ in a mixture – at the first station MgSO4's cation leg passes 0.18 mol/m3 and its anion leg 3e-4 – and the salt basis can carry only one. The engine carries the cation leg and ANNOUNCES every salt whose legs differ ([DSPM-DE] salt 'MgSO4' in a MIXED feed ...): the permeate stream's sulfate and chloride are NOT the pore's. The published R_obs_MgSO4 (0.69 in stage 1) is therefore the rejection of Mg2+, not of the salt. The ion-by-ion permeate is what the props bench poreTransport prints (its ion rows). Mapping the ion permeate back onto the salt components – or marching the module in ions – is a model decision that has not been taken.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | boosted | F | 0.508750920568 |
| stream | boosted | T | 293.165385998 |
| stream | concentrate1 | F | 0.508750920568 |
| stream | concentrate1 | T | 293.15 |
| stream | concentrate2 | F | 0.410594876442 |
| stream | concentrate2 | T | 293.165385998 |
| stream | feed | F | 0.739665555556 |
| stream | feed | T | 293.15 |
| stream | permeate1 | F | 0.230914634987 |
| stream | permeate1 | T | 293.15 |
| stream | permeate2 | F | 0.0981560441261 |
| stream | permeate2 | T | 293.165385998 |
| kpi | Booster | F | 0.508750920568 |
| kpi | Booster | P_in | 126335.527185 |
| *… and 173 more reference values in the case's `expected`.* | | | |

## membrane23_constant_rejection_nf8040

tutorials/steady/membranes/membrane23_constant_rejection_nf8040




**What it does:** Constant intrinsic rejection per salt (Geraldes & Afonso 2007, Eq. 1): NaCl + MgSO4 through one vessel of six generic 8-inch NF elements

**The story:** the SIMPLEST membrane model a spiral-wound element can carry: a CONSTANT INTRINSIC REJECTION per solute, c_p,i = (1 - R_i) c_m,i at the wall (Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20, Eq. 1, where R_i is the paper's INPUT). One pressure vessel of six generic 8-inch elements; NaCl and MgSO4, 1 g/L each.

THE SAME ELEMENT TAKES ANY MEMBRANE MODEL. The module (Generic-8040, its FT30 feed spacer and Schock & Miquel's correlations) is the hardware; the 'transport' word is the membrane: 'solutionDiffusion' (B_i per solute), 'SDEM' (per-ion B_i under zero current, membrane21), 'DSPM-DE' (the charged pore, membrane22) or, here, 'constantRejection'. The membrane record gives the water permeance A_w only.

INTRINSIC IS NOT OBSERVED. R_int is measured against the WALL. The run publishes R_obs_<salt> = 1 - c_p/c_b against the FEED, which the polarisation lowers: the element sees a wall richer than its feed, so a constant R_int still passes more salt where the film is thicker. Compare R_obs_NaCl with the 0.60 declared below.

WHY BY SALTS. A salt is neutral, so one R per salt cannot charge the permeate. A feed written ion by ion with one R per ion generally does – the permeate charge is sum z_i (1 - R_i) c_m,i – and the law REFUSES it by name (see the README for the measurement on membrane21's feed).

*The question. A spiral-wound element is hardware; the membrane model is a choice. What is the simplest model it can carry, and what does it leave out?*

*The model. transport constantRejection; – each solute leaves through the membrane with a DECLARED intrinsic rejection, measured against the wall:*

*c_p,i = (1 - R_i) c_m,i, N_i = J_v c_p,i, J_v = A_w (dP - dpi)*

*This is Eq. (1) of Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20-27, which takes R_i as an input. The wall concentration c_m comes from the module's polarisation model (here Schock & Miquel's correlation on the FT30 spacer with the Geraldes & Afonso 2006 suction correction) and the water flux from the same Newton the solution-diffusion law uses. The membrane record (NF270) supplies A_w only.*

*transport constantRejection; transportParameters { rejection { NaCl 0.60; MgSO4 0.98; } source "..."; }*

*The same module Generic-8040; runs solutionDiffusion (B per solute), SDEM (per-ion B under zero current, membrane21) and DSPM-DE (the charged pore, membrane22): only the transport line changes.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.184745844444 |
| stream | feed | T | 298.15 |
| stream | permeate | F | 0.080576329433 |
| stream | permeate | T | 298.15 |
| stream | retentate | F | 0.104169515011 |
| stream | retentate | T | 298.15 |
| kpi | NF | Gamma_MgSO4_avg | 0.14208197636 |
| kpi | NF | Gamma_MgSO4_elem1 | 0.235594911845 |
| kpi | NF | Gamma_MgSO4_elem2 | 0.196754783898 |
| kpi | NF | Gamma_MgSO4_elem3 | 0.157820648895 |
| kpi | NF | Gamma_MgSO4_elem4 | 0.119172795219 |
| kpi | NF | Gamma_MgSO4_elem5 | 0.0829956168512 |
| kpi | NF | Gamma_MgSO4_elem6 | 0.0508018079009 |
| kpi | NF | Gamma_NaCl_avg | 0.0815896543444 |
| *… and 57 more reference values in the case's `expected`.* | | | |

## membrane24_fit_then_simulate

tutorials/steady/membranes/membrane24_fit_then_simulate




**What it does:** One case, two modes: NF270's DSPM-DE pore fitted to published intrinsic rejections (Properties mode), then the fitted membrane in a spiral-wound stage (Process mode)

**The story:** PROPERTIES & DATA MODE of membrane24 (runCase <case> props): the DSPM-DE pore of NF270 FITTED to published measurements, and written as the membrane record the Process mode's spiral-wound stage reads.

THE DATA. Fernandez de Labastida & Yaroshchuk, "Nanofiltration of Multi-Ion Solutions: Quantitative Control of Concentration Polarization and Interpretation by Solution-Diffusion-Electro-Migration Model", Membranes 11 (2021) 272, doi:10.3390/membranes11040272 (CC BY 4.0): NF270 in a rotating-disk cell at 20 C, four dominant salts at 0.01 mol/L, INTRINSIC rejections (the authors corrected them for concentration polarisation) at seven permeate fluxes each, 6-56 um/s. The symbols of their Figures 1a-4a, digitised by bin/curate/digitise_fdl2021.py and written as four datasets by bin/curate/export_fdl2021_rejections.py (constant/experimental/fdl2021_<salt>.dat). The reading error is in each file's header; the authors publish no measurement uncertainty, so the fit is UNWEIGHTED (a rejection as a fraction). Not in the feeds: the 2e-4 mol/L NaNO3 + NH4Cl traces they added (2 % of the dominant salt).

THE PARTITION. Fitted: the two CHLORIDES, NaCl and MgCl2 – one co-ion, a monovalent and a divalent counter-ion. HELD OUT, never seen by the fitter: the two SULFATES, Na2SO4 and MgSO4, whose divalent co-ion the fitted pore must exclude by its charge and its size alone. CHOSEN AFTER MEASURING THE ALTERNATIVES, and said here because it is what a reader must weigh: on these four series the identifiability of (r_p, X_d) depends on WHICH salts are fitted, and three partitions were run before this one – fit NaCl + MgSO4 r_p ran to its upper bound (3 nm, X_d -404 mol/m3): a Donnan-only pore, the two not separable; held-out AAD 16 points (notValidated) fit NaCl + Na2SO4 correlation(r_p, X_d) = -1.0000: not identifiable fit NaCl + MgCl2 identifiable (this case) so the verdict below is that of a partition chosen for identifiability, not one fixed blind; and the held-out sulfates sit at R 0.977-0.997, where a tight pore of any charge predicts them closely – it is a weak test of X_d. A blind partition would be a separate study.

THE MODEL. DSPM-DE (Bowen & Welfoot 2002), its pore SOLVED across 101 nodes (the default since C141), the Dechadilok-Deen hindrance, ideal activities. The film is NanoFiltran's "no polarisation" (k_film 10 m/s) because the data are intrinsic: R = 1 - c_p/c_wall is what the authors report and what 'rejection intrinsic;' compares. FREE: the pore radius r_p and the fixed charge X_d (either sign); HELD at the base record NF270_dspmde's values: the porosity over thickness A_k/dx 5e6 1/m, the pore dielectric constant 35 (Born ON) and A_w (not read at an imposed J_v). Freeing A_k/dx or eps_p as well was measured and is not identifiable from these data (|correlation| >= 0.99). The ion radii and D0 are the species records' (hydrated radii, the convention NF270_dspmde's own header states), corrected to 20 C by Stokes-Einstein on the declared liquid viscosity.

THE RECORD. 'output { record ...; }' writes constant/assets/ NF270_FdL2021.dat with the fitted values, their 95 % half-widths, the verdict and the fingerprints of the four datasets, of this operation's block and of the base record. The Process mode (system/flowsheetDict) reads that record and REFUSES it when any fingerprint no longer matches.

WHAT IS PINNED (expected.props). The fitted values and their uncertainties, the held-out AAD and the verdict: self-recorded rows – they pin that the fit has not MOVED, not that NF270 is right. The held-out verdict is the validation claim, and it is the paper's data that decide it.

*The question. A researcher has measured the rejections of his membrane in the laboratory. Can he fit a pore model to them, judge the fit against data it never saw, and then use the fitted membrane in a process – without the data, the fit and the process drifting apart?*

*This case is the template for that. Copy it, replace the measured files with yours, and run it. The section "Use it with your own data" below says which file is which. The page that explains the models, what goes in and what comes out is docs/ai/membrane-research.md.*

*Two modes, one case. This case carries BOTH a system/propsDict (Properties & data mode) and a system/flowsheetDict (Process mode) over one constant/:*

*runCase tutorials/steady/membranes/membrane24_fit_then_simulate props # the fit runCase tutorials/steady/membranes/membrane24_fit_then_simulate # the stage*

*The fit. NF270's DSPM-DE pore – its radius r_p and fixed charge X_d – fitted to Fernandez de Labastida & Yaroshchuk's (Membranes 11 (2021) 272, CC BY 4.0) measured intrinsic rejections of NaCl and MgCl2; Na2SO4 and MgSO4 held out. Result: r_p = 0.675 +- 0.034 nm, X_d = -10.5 +- 4.1 mol/m3, held-out AAD 0.93 points of rejection, verdict validated against a band of 10 points declared before the fit. Read the propsDict header for how the partition was chosen: other partitions were measured first and do NOT identify the pore, and the sulfates are an easy test.*

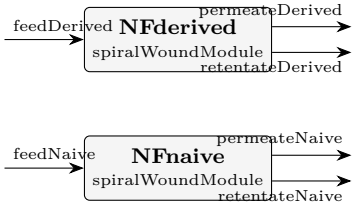
*The record. The fit writes constant/assets/NF270_FdL2021.dat: the fitted values, their uncertainty, the verdict, and the fingerprints of the datasets, of the fit block and of the base record.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrate | F | 0.107699137018 |
| stream | concentrate | T | 293.15 |
| stream | feed | F | 0.184936666667 |
| stream | feed | T | 293.15 |
| stream | permeate | F | 0.0772375296488 |
| stream | permeate | T | 293.15 |
| kpi | Stage | Gamma_MgCl2_avg | 0 |
| kpi | Stage | Gamma_MgCl2_elem1 | 0 |
| kpi | Stage | Gamma_MgCl2_elem2 | 0 |
| kpi | Stage | Gamma_MgCl2_elem3 | 0 |
| kpi | Stage | Gamma_MgSO4_avg | 0.448895665879 |
| kpi | Stage | Gamma_MgSO4_elem1 | 0.464530286351 |
| kpi | Stage | Gamma_MgSO4_elem2 | 0.443415485733 |
| kpi | Stage | Gamma_MgSO4_elem3 | 0.437177763507 |
| *... and 46 more reference values in the case's `expected`.* | | | |

## membrane25_rated_test_inversion_nf270_4040

tutorials/steady/membranes/membrane25_rated_test_inversion_nf270_4040




**What it does:** A data sheet's OBSERVED rejection inverted to the INTRINSIC one: NF270-4040 with MgSO4 fromRatedTest, beside the same element with the sheet's 0.97 pasted as intrinsic

**The story:** a data sheet's OBSERVED rejection is not the INTRINSIC rejection 'constantRejection' takes.

The NF270-4040 sheet quotes "stabilized salt rejection > 97.0 %" for 2,000 ppm MgSO4 at an applied 70 psi (4.8 bar), 25 C, 15 % recovery – OBSERVED, feed to permeate. An applied pressure on a data sheet is GAUGE unless the sheet says otherwise (DEV.md C189): the record declares 'pressureBasis feedGauge;', so the test is run at 4.8 + 1.01325 = 5.81 bar absolute, and the run says so. The law's R_int is measured against the WALL, which is richer than the feed: the film polarises it and the bulk concentrates along the leaf. Two copies of the same element, fed the same 2,000 ppm MgSO4 at 8 bar absolute (away from the rated 5.81 bar):

NFderived 'rejection { MgSO4 fromRatedTest; }' – the engine runs THIS element at the sheet's rated test and finds the R_int whose predicted observed rejection is 0.97: R_int(MgSO4) = 0.98641 (KPI R_int_fromRatedTest_MgSO4) c_m/c_b at the test = 1.874 (ratedTest_cm_over_cb_MgSO4, the channel average) NFnaive 'rejection { MgSO4 0.97; }' – the sheet's number pasted as if it were intrinsic.

At 8 bar the two permeates carry (KPI R_obs_MgSO4; the permeate streams): NFderived R_obs 0.95453, permeate 1124 kg/h at 91 ppm MgSO4 NFnaive R_obs 0.90065, permeate 1145 kg/h at 199 ppm MgSO4 – the naive element passes 2.18x the salt, and at the sheet's own test it would report 93.4 %, not the 97.0 % the sheet measured. (Both R_obs fall below 0.97 at 8 bar: 43 % of the water leaves, so the bulk concentrates, and the film is thicker at the higher flux.)

NOT a validation: the NF270 record's A_w is a fit to a teaching case – at the rated test the engine's permeate is 18.9 m3/d against the sheet's 9.5 (recovery 29.8 % against 15 %), printed by the [ratedTest] line, so the polarisation the inversion corrects for is the engine's, not the bench's – and a constant R_int stays constant away from the test. See the README.

*The question. The NF270-4040 data sheet says "stabilized salt rejection > 97.0 %" (2,000 ppm MgSO4, an applied 70 psi (4.8 bar), 25 C, 15 % recovery). An applied pressure on a data sheet is GAUGE unless the sheet says otherwise, so the record declares pressureBasis feedGauge; and the test runs at 4.8 + 1.01325 = 5.81 bar absolute. transport constantRejection; asks for an intrinsic rejection R_int. Can the student type 0.97?*

*No, and this case shows by how much. The sheet's number is OBSERVED, 1 - c_p/c_feed; the law's R_int is measured against the WALL, 1 - c_p/c_m. The wall is richer than the feed twice over: the film polarises it (c_m/c_b > 1), and the bulk concentrates along the leaf as the water leaves. So 0.97 typed as R_int passes MORE salt than the sheet measured.*

*What the case does. Two copies of the same element, fed the same 2,000 ppm MgSO4 at 8 bar absolute:*

*NFderived rejection { MgSO4 fromRatedTest; } NFnaive rejection { MgSO4 0.97; }*

*fromRatedTest makes the engine run THIS element (its record, its spacer channel, its Schock & Miquel film, its osmotic model) at the sheet's rated test – one element, the sheet's solute at its mass fraction, its P and T, the case's permeate pressure, and the feed the sheet's figures imply (permeate / recovery = 9.5 / 0.15 = 63.3 m3/d) – and solve, by a bracketed 1-D search, for the R_int whose predicted observed rejection is 0.97. The run says so on stdout, on stderr and in the caveat block:*

*[ratedTest] R_int(MgSO4) = 0.986415 DERIVED from the rated test of module 'NF270-4040': observed 97 % at 2000 ppm, 5.81 bar (permeate 1 bar), ...; the module predicts c_m/c_b = 1.8742 (channel average), recovery 29.83 % and permeate 18.89 m3/d there; bracketed solve residual ... after 5 probe(s)*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedDerived | F | 0.0406204850803 |
| stream | feedDerived | T | 298.15 |
| stream | feedNaive | F | 0.0406204850803 |
| stream | feedNaive | T | 298.15 |
| stream | permeateDerived | F | 0.0173300456402 |
| stream | permeateDerived | T | 298.15 |
| stream | permeateNaive | F | 0.0176496756356 |
| stream | permeateNaive | T | 298.15 |
| stream | retentateDerived | F | 0.0232904394401 |
| stream | retentateDerived | T | 298.15 |
| stream | retentateNaive | F | 0.0229708094447 |
| stream | retentateNaive | T | 298.15 |
| kpi | NFderived | Gamma_MgSO4_avg | 1.60657778823 |
| kpi | NFderived | J_w_avg | 4.10822257679e-05 |
| *… and 27 more reference values in the case's `expected`.* | | | |

## membrane26_class_sulphate_removal

tutorials/steady/membranes/membrane26_class_sulphate_removal




**What it does:** A membrane chosen by WHAT IT DOES: sulphate removal on the illustrative functional class NF_DIVALENT_SELECTIVE, its intrinsic rejections derived from the class's observed ones

**The story:** a membrane chosen by WHAT IT DOES.

THE TASK. A brackish water (3 g/L NaCl, 2 g/L MgSO4) must lose its sulphate and may keep its sodium chloride. Before choosing a product, a student chooses a FUNCTION: 'module NF_DIVALENT_SELECTIVE;' names a functional CLASS (data/standards/assets/NF_DIVALENT_SELECTIVE.dat, kind membraneClass) – "high sulphate retention with NaCl passage".

WHAT THE CLASS IS. An ILLUSTRATIVE teaching profile, not a membrane anyone sells and not a measurement: an OBSERVED rejection per salt at its own reference test (2,000 ppm, 4.8 bar gauge, 25 C, 15 % recovery) – NaCl 30 %, MgSO4 95 %, each the middle of the class's screening window (10-50 %, > 90 %) – a water permeance (10 L m-2 h-1 bar-1) and the generic element it is deployed in (Generic-4040, by reference). The run says so on every line that matters: '[illustrative]', and the element's own '[estimate]' lines.

WHAT THE ENGINE DERIVES. 'constantRejection' needs the INTRINSIC rejection, against the wall; the class states the OBSERVED one, against the feed. 'fromRatedTest' runs THIS element at the class's reference test and finds the R_int whose predicted observed rejection is the class's (src/unitOperations/membrane/RatedTestInversion): R_int(NaCl) = 0.37566 (KPI R_int_fromRatedTest_NaCl) R_int(MgSO4) = 0.96342 (KPI R_int_fromRatedTest_MgSO4)

THE ANSWER, three elements in series at 7 bar absolute: water recovery 32.45 %, permeate 811 kg/h; R_obs(NaCl) 0.2865 → permeate 2140 ppm NaCl (feed 3000 mg/L) R_obs(MgSO4) 0.9466 → permeate 107 ppm MgSO4 (feed 2000 mg/L), i.e. 85 ppm as sulphate. The sodium chloride passes, the sulphate stays: what the class was chosen to do. Both observed rejections sit a little below the class's stated ones (30 %, 95 %): at 7 bar a third of the water leaves, so the bulk concentrates along the vessel, which the reference test at 15 % recovery barely saw.

WHAT A STUDENT SHOULD CHANGE. The CLASS: 'module NF_OPEN;' passes more salt of every kind (it rates NaCl only, so MgSO4 then needs a typed R_int with its source – a class says nothing about a salt it does not rate); 'module NF_TIGHT;' holds more NaCl too. The PRESSURE and the number of ELEMENTS: recovery against permeate sulphate. The CLASS's assumptions themselves: copy the record into this case's constant/assets/ and move its MgSO4 rejection inside its band – the intrinsic value is re-derived; typing a number here instead is refused, because a rated salt's intrinsic rejection is derived. Then the PRODUCT: a commercial element ('module NF270-4040;', its data sheet's rated test) – the shortlist from a class to products is the props op 'membraneShortlist' (tutorials/props/membrane/shortlist01_class_to_modules, DEV.md 4c C189 slice 5).

NOT a validation: no number in this case is measured.

*The question. A brackish water (3 g/L NaCl, 2 g/L MgSO4) must lose its sulphate and may keep its sodium chloride. Which membrane? Before a product, a student chooses a FUNCTION.*

*The choice. module NF_DIVALENT_SELECTIVE; names a functional CLASS – data/standards/assets/NF_DIVALENT_SELECTIVE.dat, kind membraneClass – one of six (NF_OPEN, NF_DIVALENT_SELECTIVE, NF_TIGHT, BWRO_HIGH_FLUX, BWRO_HIGH_REJECTION, SWRO). A class is an ILLUSTRATIVE teaching profile, not a product and not a measurement: an OBSERVED rejection per salt at its own reference test (here NaCl 30 % and MgSO4 95 %, each the middle of the class's screening window), a water permeance, and the generic element it is deployed in (Generic-4040, by reference). The run says so: [illustrative] module 'NF_DIVALENT_SELECTIVE': a FUNCTIONAL CLASS, not a product ..., on stdout, on stderr and in the end-of-run caveat block.*

*What the engine derives. transport constantRejection; takes the INTRINSIC rejection (against the wall); the class states the OBSERVED one (against the feed). rejection { NaCl fromRatedTest; MgSO4 fromRatedTest;*

*} runs this element at the class's reference test and inverts it (DEV.md 4c C189 slice 1): R_int(NaCl) = 0.37566, R_int(MgSO4) = 0.96342.*

*The answer. Three elements at 7 bar: recovery 32.45 %, the permeate at 2140 ppm NaCl and 107 ppm MgSO4 (85 ppm as sulphate).*

*What a student should change. The class (NF_OPEN, NF_TIGHT); the pressure and the number of elements; the class's own assumptions, by copying the record into constant/assets/ and moving a rejection inside its band (typing a number for a rated salt in the case is refused – the intrinsic value is derived); then a product (module NF270-4040;).*

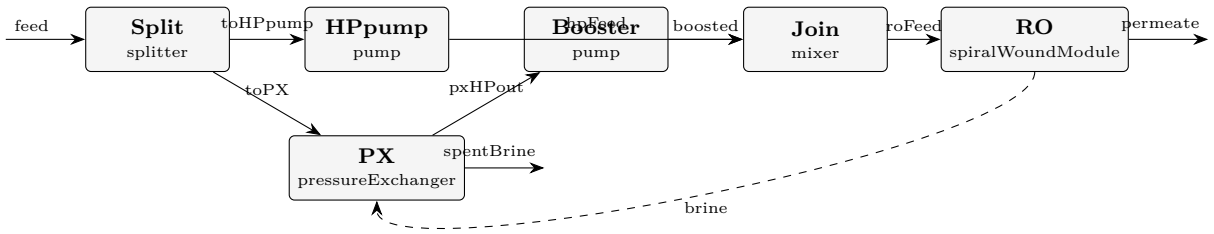
*Not a validation. No number here is measured. Gate: bin/curate/check_membrane_classes.py. Record: docs/design/an-observed-rejection-is-not-an-intrinsic-one.md section 8.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0384027444444 |
| stream | feed | T | 298.15 |
| stream | permeate | F | 0.0124896798589 |
| stream | permeate | T | 298.15 |
| stream | retentate | F | 0.0259130645856 |
| stream | retentate | T | 298.15 |
| kpi | NF | Gamma_MgSO4_avg | 0.222310130994 |
| kpi | NF | Gamma_MgSO4_elem1 | 0.221704274787 |
| kpi | NF | Gamma_MgSO4_elem2 | 0.222726991067 |
| kpi | NF | Gamma_MgSO4_elem3 | 0.222529419939 |
| kpi | NF | Gamma_NaCl_avg | 0.0763298682296 |
| kpi | NF | Gamma_NaCl_elem1 | 0.0761466561845 |
| kpi | NF | Gamma_NaCl_elem2 | 0.0764559181868 |
| kpi | NF | Gamma_NaCl_elem3 | 0.0763961909198 |
| *... and 43 more reference values in the case's `expected`.* | | | |

## membrane27_swro_energy_recovery

tutorials/steady/membranes/membrane27_swro_energy_recovery




**What it does:** SWRO with energy recovery: high-pressure pump + rotary pressure exchanger + booster + one SW30HR-380 stage on membrane01's seawater – the specific energy decomposed

**The story:** WHERE THE ENERGY OF SEAWATER RO GOES.

THE PLANT. membrane01's seawater (35 g/L NaCl, 25 C), 16 m3/h at 2 bar after pretreatment, split between a HIGH-PRESSURE PUMP (to 55 bar) and the low-pressure side of a ROTARY PRESSURE EXCHANGER (the ERI PX class: a ceramic rotor whose ducts are filled in turn by the high-pressure brine, which pushes the seawater they held out at nearly its own pressure, and by low-pressure seawater, which pushes the spent brine to drain). The pressurised seawater the exchanger returns is lifted the last 1.6 bar by a BOOSTER pump, joins the high-pressure pump's discharge, and the two feed ONE SW30HR-380 stage (2 vessels x 7 elements, the record data/standards/assets/SW30HR-380.dat). The stage's brine is the exchanger's high-pressure inlet – a recycle, cut at 'brine'.

THE EXCHANGER'S THREE NUMBERS are a maker's and are declared, never defaulted (Stover, Desalination 203 (2007) 168-175, the ranges he states for isobaric rotary devices): pressure-transfer efficiency 0.96 (95-98 %), leakage 0.015 (1-2 %), both the middle of their ranges; mixing 0.03, the LOW end of his 3-6 %, because under this unit's definition (the brine share of the pressurised outlet's VOLUME) 3 % on this brine (6.3 % w/w against a 3.3 % feed) already raises the feed's salinity 2.8 %, the top of the 1-3 % he quotes for that penalty; the middle of his mixing range would give 4.3 %. His two ranges are not consistent under this definition at 47 % recovery – a finding about the definition, recorded and not tuned away.

THE ANSWER (read off the run, 2026-11-01): plant recovery 47.2 % of the RO feed (16.2 m3/h), 48.3 % of the seawater drawn (16.4 m3/h); permeate 7.94 m3/h, 99.23 % observed NaCl rejection brine 8.31 m3/h at 54.12 bar, 6.3 % w/w exchanger P_HPout 53.37 bar (solved from eta): 51.4 bar recovered on 8.18 m3/h of seawater, 11.7 kW hydraulic; HP-side loss 0.76 bar; feed salinity 3.30 → 3.39 % w/w (+2.83 %); flows balanced to 1.0 % (LP in 8.39 vs HP in 8.31 m3/h); dissipated 0.52 kW high-pressure pump 14.47 kW shaft (8.06 m3/h, 2 → 55 bar, eta 0.82) booster 0.46 kW shaft (8.18 m3/h, 53.37 → 55 bar, eta 0.80)

THE SPECIFIC ENERGY, DECOMPOSED (pump SHAFT powers over the permeate; no motor or drive efficiency is modelled, no pretreatment, no permeate delivery): with the exchanger (14.47 + 0.46) kW / 7.94 m3/h = 1.88 kWh/m3 WITHOUT it, the same plant pumping its whole RO feed (16.24 m3/h, 8.06 + 8.18) through the high-pressure pump and throttling the brine: (16.24/3600) x 53e5 Pa / 0.82 = 29.2 kW → 3.67 kWh/m3 so the device halves the plant's pumping energy, at the price of 2.8 % more salinity on the membrane's feed. THE THERMODYNAMIC MINIMUM for this feed and this recovery, from the osmotic pressure the package gives (the module's own van't Hoff model, pi = 2 c R T; c = 9.568 kmol/h NaCl in 16.45 m3/h = 0.582 kmol/m3, so pi_0 = 28.8 bar) at the plant recovery r = 0.483: w_min = pi_0 (-ln(1 - r)) / r = 28.8 x 1.366 = 39.3 bar = 3.93 MJ/m3 = 1.09 kWh/m3. The 0.79 kWh/m3 between the minimum and the plant is the membrane's overpressure (55 bar fed against 28.8-54 bar of osmotic pressure along the vessel), the pumps' efficiencies, the exchanger's 4 % and its booster. A student should move the pressure, the recovery (the split fraction) and the three device numbers and watch the terms.

THE KNIFE EDGE, which is real. The exchanger returns 98.5 % of the brine's VOLUME to the membrane feed, so the loop's gain is 0.985 and the recovery is set by the 1.5 % that leaks and by how the permeate moves with the feed: the high-pressure pump's flow IS the permeate (plus the leak), and the split fraction here (0.49 to the pump) was tuned by hand until the device ran balanced – what a real train's control does with that pump's drive. Newton on the tear overshoots on such a loop (a trial brine with no water, a

booster asked to pump down); system/solverDict selects Wegstein with its full secant step, which converges in five.

WHAT THE RUN SAYS AND CANNOT SAY. The mixer ‘Join‘ is DECLARED isothermal at the feed temperature: an adiabatic mix would need NaCl’s liquid heat capacity, which its record does not carry (membrane01’s component set), so the pumps’ 0.28 K and 0.01 K dissipation rises are discarded there and no mixing duty is priced – announced. FOR THE SAME REASON THE PLANT’S FIRST LAW IS UNAVAILABLE on this component set (‘energyBalance UNAVAILABLE‘, as on membrane01): the exchanger’s own row closes by construction (T unchanged, Q = 0), but nothing here can show it, and the pumps’ useful work v dP is not carried by the liquid enthalpy surface in any case (Pump.cpp says why). An ion-written twin (water, Na, Cl) would have a first law and would show the pumps’ residual; it is not built here. The exchanger’s three balances ARE checked, independently of the engine, by bin/curate/check_pressure_exchanger.py on this case’s result JSON.

THE SHEET. system/postDict sizes the exchanger (design/PX/ pressureExchanger): one rotor at a DECLARED rated flow of 60 m3/h per unit (an author-set order of magnitude for the PX-Q300 class, to be replaced by a maker’s sheet), the pressure drops and the hydraulic booster duty read from the unit, NOT COSTED (no Turton set). The GUI draws it: the rotor nominal, the four ports with their pressures, the booster beside the high-pressure outlet.

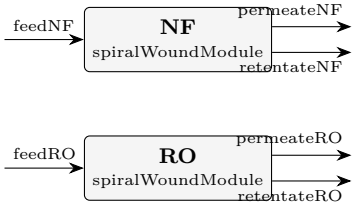
NOT a validation: no number here is measured. NOT modelled: the rotor’s hydraulics, the duct-by-duct unsteadiness, any dependence of the mixing on flow imbalance or salinity, the motor and drive, pretreatment.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | boosted | F | 0.127146448001 |
| stream | boosted | T | 298.159780601 |
| stream | brine | F | 0.13035446605 |
| stream | brine | T | 298.15 |
| stream | feed | F | 0.255555 |
| stream | feed | T | 298.15 |
| stream | hpFeed | F | 0.12522195 |
| stream | hpFeed | T | 298.428449039 |
| stream | permeate | F | 0.122013931952 |
| stream | permeate | T | 298.15 |
| stream | pxHPout | F | 0.127146448001 |
| stream | pxHPout | T | 298.15 |
| stream | roFeed | F | 0.252368398001 |
| stream | roFeed | T | 298.15 |
| *... and 160 more reference values in the case’s `expected`.* | | | |

## membrane28_water_permeance_from_sheets

tutorials/steady/membranes/membrane28_water_permeance_from_sheets




**What it does:** The water permeance a data sheet implies: NF270-4040 and SW30HR-380 with waterPermeance fromRatedTest, reproducing each sheet's permeate flow and observed rejection at its rated test

**The story:** the WATER PERMEANCE a data sheet implies, found the way its observed rejection is (DEV.md 4c C189 slice 5b).

THE FINDING THIS CASE ANSWERS. At the rated tests of the two shipped elements the engine's permeate was 2.16x the NF270-4040 sheet and 2.08x the SW30HR-380 sheet. Not the inversion's fault, and not the active area: the membrane records NF270 and SW30HR carry an A_w that is a fit to a TEACHING case (their 'provenance fittedToCase;', 36 and 3.6 L/m2/h/bar), while a data sheet measures THE ELEMENT's permeate at a stated pressure. Two things that share a symbol and are not the same fact.

THE DECISION. 'waterPermeance fromRatedTest;' in the unit's operation (beside 'rejection { <salt> fromRatedTest; }', which it needs) runs the module at the record's rated test and finds, together, the A_w that reproduces the sheet's PERMEATE flow and the R_int that reproduces its observed REJECTION. They are coupled (the wall concentration depends on the flux, the osmotic pressure on the rejection), so the two solves alternate until A_w is stationary. The membrane record is not edited: its fit still serves the case it was fitted to (membrane02, membrane01), and nothing else moves. The derived A_w is a MODEL parameter of the element as the film model reads it – not a manufacturer's number – and the unit publishes it beside the record's (A_w_fromRatedTest_LMH_per_bar, A_w_record_LMH_per_bar).

TWO UNITS, one per sheet: NF270-4040 (MgSO4, rated 4.8 bar gauge) fed at 8 bar absolute, SW30HR-380 (NaCl, rated 55 bar gauge) fed at 56 bar.

*The question. At their own rated tests the engine gave 2.16x the NF270-4040 sheet's permeate and 2.08x the SW30HR-380 sheet's (the C189 slice-2 finding). Whose number is wrong?*

*The answer. The membrane records'. NF270 and SW30HR carry an A_w that is a FIT to a teaching case (provenance fittedToCase;: 36 and 3.6 L/m2/h/bar), while a data sheet measures THE ELEMENT's permeate at a stated pressure. Two facts sharing a symbol. The inversion that turns the sheet's observed rejection into the intrinsic one (membrane25) is not at fault, and neither is the active area.*

*The decision. waterPermeance fromRatedTest; in the unit's operation, beside rejection { <salt> fromRatedTest; } (which it needs), runs the module at the record's rated test and finds, TOGETHER, the A_w that reproduces the sheet's permeate and the R_int that reproduces its observed rejection (the two are coupled through the wall concentration and the osmotic pressure, so the solves alternate until A_w is stationary). The membrane records are NOT edited: their fit still serves the cases it was fitted to.*

*(A_w in L/m2/h/bar; KPIs A_w_fromRatedTest_LMH_per_bar, A_w_record_LMH_per_bar.) The derived value is a MODEL parameter of the element as this film model reads it, not a manufacturer's number.*

*Not a validation. One test per element fixes one A_w and one R_int; away from the test a constant R_int stays constant (the real one rises with flux). Gate: bin/curate/check_rated_tests.py arm (h) – an independent run with both values typed reproduces each sheet to 1e-7. The shortlist (tutorials/props/membrane/shortlist01_class_to_modules) re-rates every candidate with this derivation.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedNF | F | 0.0406204850803 |
| stream | feedNF | T | 298.15 |
| stream | feedRO | F | 0.203553611111 |
| stream | feedRO | T | 298.15 |
| stream | permeateNF | F | 0.00942283109071 |
| stream | permeateNF | T | 298.15 |
| stream | permeateRO | F | 0.0166865465761 |
| stream | permeateRO | T | 298.15 |
| stream | retentateNF | F | 0.0311976539896 |
| stream | retentateNF | T | 298.15 |
| stream | retentateRO | F | 0.186867064535 |
| stream | retentateRO | T | 298.15 |
| kpi | NF | A_w_alternations | 6 |
| kpi | NF | A_w_fromRatedTest_LMH_per_bar | 14.8806573771 |
| *... and 44 more reference values in the case's `expected`.* | | | |

## optimisation

**The theory you need.** The outer driver wraps the whole simulation in $f(\mathbf{x})$: sweeps map it, Nelder–Mead descends it, parameter estimation fits it to data (Levenberg–Marquardt, with identifiability diagnostics – $\chi^2_{\text{red}}$, correlation, conditioning). Expect every function evaluation to be a full, honest flowsheet pass.

## designSpec01_triple_equal_areas

tutorials/steady/optimisation/designSpec01_triple_equal_areas




**What it does:** DesignSpec: equal-area triple-effect sugar evaporator (manipulate $A so L3 = 4500 kg/h)

**The story:** SOURCE. The exercise number below is the numbering of the course notes this case was built from; the TEXTBOOK those notes drew on is not recorded anywhere in this case. It used to be labelled with a course acronym, and that acronym named the WRONG course (a CFD one), so it was removed on 2026-08-12 rather than replaced with a guess. The gap is left visible: a missing citation a reader can see beats a plausible one nobody can check. (‘sucrose.dat‘ does carry a primary for the Cp correlation – Coulson & Richardson 6th ed. Example 14.4.)

Same physics as evaporator02_triple_effect_sugar (textbook Example 10.4 — three cocurrent effects on a 22 500 kg/h, 5 wt% sucrose feed). The twist: instead of HARD-CODING each effect’s area at 100 $m^2$, we declare A as a CASE-LEVEL VARIABLE and use the outer DesignSpec driver to find the equal-areas value that delivers L_3 ~ 4500 kg/h (25 wt% sucrose), the textbook design target.

This tutorial showcases two new pieces of v0.30 machinery:

1. $references in the dict (‘variables { ... }‘ block + ‘$A‘ syntax inside any scalar). The same A is shared by the three effects without copy-paste — which makes it trivially a single degree of freedom for the outer solver.

2. The ‘designSpec‘ outer driver: a Newton-ND wrap on Flowsheet::solve, manipulating the case’s $variables to satisfy named targets. Here it is a 1-D Newton in $A on the residual L3.F - 0.001249 kmol/s. Convergence in 4-5 evaluations.

F=22 500 kg/h L1 L2 L3 (target ~ 4500 kg/h) x=0.05 →[1] ——→ [2] ——→ [3] —-→ ^ ^ ^ | V1 V2 V3 → condenser Steam (drives 2) (drives 3) 200 kPa P1 = 80 P2 = 45 P3 = 15 kPa

Constraints we leave fixed: - P_i pressures (textbook cascade) - U_i overall HTCs per effect (textbook values) - Feed and steam conditions - Equal areas: enforced trivially by the single $A variable

Read the run log to see the Newton converge in <10 evaluations. At the optimum, the case ends with the converged final state on stdout, exactly like a hand-tuned evaporator02 run — but the area was DERIVED, not specified.

Initial guess A_0 = 100 $m^2$ (the educated guess from evaporator02); the solver should home in on ~ 110-120 $m^2$ to hit the target.

Target tolerance: ±5 kg/h on L_3, encoded via tol = 1.4e-6 kmol/s (= 5 kg/h ? 3600 s/h ? 18 kg/kmol).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | L1 | F | 0.243341003299 |
| stream | L2 | F | 0.150708221741 |
| stream | L3 | F | 0.0529528345186 |
| stream | V1 | F | 0.0871584023993 |
| stream | V2 | F | 0.0926327815587 |
| stream | V3 | F | 0.097755387222 |
| stream | cond1 | F | 0.136219356305 |
| stream | cond2 | F | 0.0871584023993 |
| stream | cond3 | F | 0.0926327815587 |
| stream | feed | F | 0.330499405699 |
| stream | utilitySteam_200kPa | F | 0.136219356305 |
| kpi | effect1 | A | 109.204790334 |
| kpi | effect1 | A_m2 | 109.204790334 |
| kpi | effect1 | BPE | 0.107225420283 |
| *... and 95 more reference values in the case's `expected`.* | | | |

## fitNRTL01_ethanol_water

tutorials/steady/optimisation/fitNRTL01_ethanol_water




**What it does:** Fit NRTL ethanol/water pair against 1 atm bubble-T data (Levenberg-Marquardt)

**The story:** fitNRTL01 – Levenberg-Marquardt regression of the ethanol-water NRTL pair against 1 atm bubble-T data, on the CANONICAL fit engine (fitParameters, choupoProps). This case previously ran the condemned FitBinaryPair outer driver in choupoSolve, which abused a dummy flowsheet and steered the feed composition through the legacy streams{} block per data row – the design forum (#53) ordered it moved here, WITH a golden: its chi^2 once degraded 100x with nothing going red, because a fit without numerical verification does not belong in a green corpus.

Data: transcribed from constant/experiments/ethanol-water-101kPa (the raw- data home; DECHEMA / Gmehling-Onken Vol. I Part 1, pedagogical subset). Starting point: the catalogue NRTL values in constant/thermoPhysPropDict.

*The parameter-estimation workflow: regress the four NRTL binary interaction parameters of the ethanol-water pair (a_ij, b_ij, a_ji, b_ji; $\alpha$ held fixed) against experimental bubble-temperature data at 1.01325 bar, using hand-rolled Levenberg-Marquardt inside the fitParameters engine (successor of the retired fitBinaryPair driver).*

*$*$ See an outer Marquardt loop (3-up/5-down trust region on $\lambda$) wrapped around an inner Newton-1D solving $\Sigma\ K\ x = 1$ for each ($x_i$, T) pair. $*$ Watch $\lambda$ oscillate as the algorithm switches between Gauss-Newton ($\lambda$ small) and gradient-descent ($\lambda$ large) regimes. $*$ Watch some LM steps get rejected — chi$^2$ gets worse, the step is reversed, $\lambda$ is multiplied by 5, retry. The whole "trust region" idea visible in one column of the log.*

*Note how minimal the forward case is: one stream, one bubbleT unit. The outer driver is what does the work — iterating over the dataset internally and looping LM over the four parameters.*

*Four keywords (plus optional maxIter, tolerance, fdStep, lambda0). The driver:*

*1. locates the pair in thermoPhysPropDict.activityModel.pairs by name, 2. reads the dataset (a two-column flat list of $x_i$ $T_i$), 3. finds the bubbleT unit's input stream from the flowsheetDict, 4. for each LM step: - 8 forward evals to build a central-difference Jacobian over 4 params $\times$ N data points, - solve the 4×4 normal equations with Marquardt damping, - try the step; accept if chi$^2$ decreased, otherwise raise $\lambda$ and retry.*

*After the run header you get (the iteration trace below is illustrative – it shows the SHAPE of an LM run, accept/reject and $\lambda$ adaptation; the actual run converges to chi$^2$ = 1.0426, RMS = 0.3079 K with fitted a_ij = 9.328, b_ij = -3609.18):*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_NRTL_ethanol_water | chi2 | 1.04279 |
| diag | fit_NRTL_ethanol_water | chi2_reduced | 0.14897 |
| diag | fit_NRTL_ethanol_water | cond_JtJ | 3.55369e+11 |
| diag | fit_NRTL_ethanol_water | converged | 1 |
| diag | fit_NRTL_ethanol_water | corr_computed | 1 |
| diag | fit_NRTL_ethanol_water | dof | 7 |
| diag | fit_NRTL_ethanol_water | identifiable | 0 |
| diag | fit_NRTL_ethanol_water | invertible | 1 |
| diag | fit_NRTL_ethanol_water | iter | 24 |
| diag | fit_NRTL_ethanol_water | max_abs_corr | 0.999967 |
| diag | fit_NRTL_ethanol_water | max_abs_resid | 0.625729 |
| diag | fit_NRTL_ethanol_water | n_data | 11 |
| diag | fit_NRTL_ethanol_water | n_params | 4 |
| diag | fit_NRTL_ethanol_water | params_at_bound | 0 |
| *... and 7 more reference values in the case's `expected`.* | | | |

## fitNRTL02_thermoml_isobars

tutorials/steady/optimisation/fitNRTL02_thermoml_isobars

| fit__NRTL__across_ | score__catalogue__pair_ | score__catalogue__pair__at__vacuum |
|---|---|---|
| fitParameters | fitParameters | fitParameters |

**What it does:** Fit the ethanol/water NRTL pair across three measured isobars, then predict a fourth held out from a different laboratory

**The story:** fitNRTL02 – A GOOD FIT THAT PREDICTS WORSE. The ethanol/water NRTL pair regressed across three measured vacuum isobars, then asked about a fourth isobar, at atmospheric pressure, from a different laboratory.

THE RESULT FIRST, BECAUSE IT IS NOT THE ONE THE CASE WAS BUILT EXPECTING. The regression converges in 19 iterations to a reduced chi-square of 0.023 over 45 points – an in-sample rms of 0.144 K, which is at the scatter of the measurements. Every number on the fit's own report says it went well. Then the fitted pair meets 21 points it never saw, at 101.3 kPa, and its average absolute deviation is 0.386 K. The pair the case STARTED from – the catalogue's DECHEMA set, adjusted by nothing – manages 0.082 K on the same 21 points. The fit made the model 4.7 times worse where it was not looking.

The two control operations below make that symmetric rather than a one-sided accusation. All four numbers, each measured by this run:

vacuum data atmospheric data (Voutsas, 13-33 kPa) (Kamihama, 101.3 kPa) catalogue pair rms 0.1619 K aad 0.0821 K pair fitted here rms 0.1438 K (in-sample) aad 0.3857 K

Each pair is at its best where its evidence was taken. The regression bought an 11 % improvement where it looked, and lost a factor of 4.7 where it did not. A student who reads only 'converged 1', 'chi2_reduced 0.023' and 'rms 0.144 K' would ship the new pair; the held-out column is the only thing in the run that says not to.

WHY IT HAPPENS, AND THE SECOND LESSON. NRTL's interaction is tau = a + b/T, so identifying 'a' and 'b' separately needs data that span 1/T. Three isobars from 13 to 33 kPa span about 28 K around 320 K, which moves 1/T by roughly 9 % – better than the 12 K of a single isobar (fitNRTL01), and it shows: the condition number of J'J falls from 3.6e11 to 5.6e9, two orders of magnitude. But 'max|correlation|' is still 1.000 and the run still reports 'identifiable 0'. Two orders of magnitude of conditioning did NOT buy identifiability, and the confidence intervals say why: b_ij = -539 +/- 456, b_ji = 682 +/- 768. The parameters are a correlated pair that happens to describe the vacuum data; nothing pins either one, so nothing constrains what they do at 101 kPa. The catalogue record's own declared validity window is 298-373 K – it was fitted where it is being asked, and this fit was not.

THE EVIDENCE, AND WHERE IT CAME FROM. Both datasets were read out of the NIST/TRC ThermoML Archive by 'bin/choupo-thermoml extract-vle' and carry the citation of the article they were measured in, as a field the engine reads rather than a comment it discards:

FIT Voutsas, Pamouktsis, Argyris & Pappa, Fluid Phase Equilib. (2011), doi:10.1016/j.fluid.2011.06.009 – 45 points on three isobars, 13.15 / 19.71 / 32.86 kPa.

HELD OUT Kamihama, Matsuda, Kurihara, Tochigi & Oba, J. Chem. Eng. Data (2012), doi:10.1021/je2008704 – 21 points at 101.3 kPa.

The held-out set is not a slice withheld from the same measurement: it is a DIFFERENT LABORATORY, a different journal, a different year, at a pressure a full decade above anything the fitter was shown. That is what makes its number mean something the in-sample one cannot. The identity collision the engine CAN settle (the same DOI in both roles) is settled by construction here, because the DOIs differ; whether two studies are scientifically independent stays a human judgement, and the run announces what it knows rather than adjudicating it.

ITS NEIGHBOUR, AND WHAT IS DIFFERENT. 'tutorials/props/curation/ curate02_vle_heldout_ethanol_water' already holds out three of eleven measured points on this same system, and already reports the same identifiability finding – it was the first place in the corpus where a fitted binary pair met real evidence it had not been handed, and this case does not claim that. What is new here is the SHAPE of the test: curate02 withholds points from ONE series at ONE pressure, so it asks whether the model interpolates

within a study. This case withholds an ENTIRE INDEPENDENT STUDY at a pressure a decade away, which asks whether the model transfers at all – and the two questions get different answers. Read curate02 first; it is the cleaner statement of the contract.

NO ACCEPTANCE BAND IS DECLARED, deliberately. This run REPORTS the held-out residual and claims no pass or fail on it, because a limit chosen after seeing the residual is not a limit. Every number quoted above is pinned in 'expected', so none of them can move unnoticed – including the ones that make the fit look bad. Nothing here was tuned to produce this outcome: the fit is the catalogue pair as its starting point, the bounds are fitNRTL01's, and the two datasets were the only ethanol+water bubble-temperature studies in the mirror that suited the roles.

WHAT THE RUN'S TWO CAVEATS ARE ABOUT, said plainly. The end-of-run block reports ethanol's Antoine correlation evaluated at 371.8 K and 372.8 K, outside its declared 273-369 K window. Both are the bubble-point Newton's STARTING temperature – the mole-fraction-weighted normal boiling point of the first datum in each set – and neither is a state any published answer contains. The mechanism that would say so ('AdvisoryFrame') is opened around this operation's parameter search but NOT yet inside the bubble-point Newton itself, so those two lines are reported as though they qualified the answer. That gap is named in docs/design/advisory-attribution.md rather than papered over here.

*Regress the ethanol/water NRTL pair against 45 bubble points measured on three vacuum isobars, then ask the fitted pair about 21 points at atmospheric pressure, measured by a different laboratory in a different year. Both datasets come out of the NIST/TRC ThermoML Archive and carry the citation of the article they were published in.*

*The regression converges in 19 iterations to a reduced chi-square of 0.023 over 45 points — an in-sample rms of 0.144 K, which is at the scatter of the measurements. Everything on the fit's own report says it went well.*

*The regression bought an 11 % improvement where it looked, and lost a factor of 4.7 where it did not. A reader who saw only converged 1, chi2_reduced 0.023 and rms 0.144 K would ship the new pair.*

*Each pair is at its best where its evidence was taken. The two evaluate operations in system/propsDict are what make that symmetric rather than an accusation: they score the ∗catalogue∗ pair on both datasets, so all four numbers come from this one run.*

*NRTL's interaction is tau = a + b/T, so separating a from b needs data that span 1/T. Three isobars from 13 to 33 kPa span about 28 K around 320 K — better than the 12 K of the single-isobar twin (fitNRTL01), and it shows: the condition number of J'J falls from 3.6e11 to 5.6e9. But max is still 1.000 and the run still reports identifiable 0. Two orders of magnitude of conditioning did not buy identifiability, and the confidence intervals say why — b_ij = -539 ± 456, b_ji = 682 ± 768. The parameters are a correlated pair that happens to describe the vacuum data; nothing pins either one, so nothing constrains what they do at 101 kPa.*

*The catalogue record's own declared validity window is 298–373 K. It was fitted where it is being asked; this fit was not.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_NRTL_across_isobars | aad_heldout_K | 0.385655690401 |
| diag | fit_NRTL_across_isobars | aad_heldout_pct | 0.108817419527 |
| diag | fit_NRTL_across_isobars | chi2 | 0.930325161527 |
| diag | fit_NRTL_across_isobars | chi2_in_sample | 0.930325161527 |
| diag | fit_NRTL_across_isobars | chi2_reduced | 0.0226908575982 |
| diag | fit_NRTL_across_isobars | cond_JtJ | 5604973889.67 |
| diag | fit_NRTL_across_isobars | converged | 1 |
| diag | fit_NRTL_across_isobars | corr_computed | 1 |
| diag | fit_NRTL_across_isobars | dof | 41 |
| diag | fit_NRTL_across_isobars | identifiable | 0 |
| diag | fit_NRTL_across_isobars | invertible | 1 |
| diag | fit_NRTL_across_isobars | iter | 19 |
| diag | fit_NRTL_across_isobars | max_abs_corr | 0.999542472648 |
| diag | fit_NRTL_across_isobars | max_abs_resid | 0.367751438189 |
| *... and 33 more reference values in the case's `expected`.* | | | |

## optim01_column_reflux

`tutorials/steady/optimisation/optim01_column_reflux`




**What it does:** Optimise reflux ratio + feedStage of benzene/toluene column to minimise reboiler vapour V_strip

**The story:** Benzene/toluene distillation, used by the optimisation driver. Same column as sensitivity01_column_reflux; we add looser tolerances on the outer loop so the simplex can probe odd feedStages without blowing up.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0416666666667 |
| kpi | column01 | L_strip | 0.0500766142867 |
| kpi | column01 | R | 3 |
| kpi | column01 | T_bottom | 383.044555298 |
| kpi | column01 | T_top | 354.047300008 |
| kpi | column01 | V_rect | 0.0555555555556 |
| kpi | column01 | V_strip | 0.0361877253979 |
| kpi | column01 | feedStage | 9 |
| kpi | column01 | nStages | 15 |
| *... and 15 more reference values in the case's `expected`.* | | | |

## optim02_process_cost

tutorials/steady/optimisation/optim02_process_cost




**What it does:** Minimise total module Turton cost of reactor + heater by varying reactor V_R and heater Q

**The story:** Same reactor → heater → separator flowsheet as process02_with_design. The optimisation driver varies two design variables:

reactor V_R in [0.002, 0.010] m^3 heater Q in [ 0, 500] W (credo-pure: hardware, not outlet T)

In v0.34 the heater is credo-pure: it specifies Q (the absolute thermal power, hardware), not Tout (which would be a DesignSpec disguised as a unit-op parameter). The Nelder-Mead simplex therefore varies the heater's Q directly; the heater's T_out becomes a KPI the student can plot, not a control variable.

Initial guesses are placeholders – overwritten by Nelder-Mead.

*A small reactor → heater → separator process. The OptimizationDriver (Nelder-Mead) tunes two design variables to minimise the total Turton module cost of the two costed units (reactor + heater):*

*The heater is credo-pure: it specifies Q (the thermal power, hardware), not an outlet T. T_out is a reported KPI, not a control variable.*

*The optimum is not at the lower corner. Nelder-Mead converges (8 moves, simplexInit 0.15) to:*

*Increasing Q monotonically lowers the objective: the simplex walks the cost from ~2.0e6 EUR at Q≈170 W down to ~9.35e5 EUR at Q=500 W.*

*The dominant cost is the heat exchanger (~916 k of the ~935 k total). Its area follows A = Q / (U·LMTD) = Q / 6000 $m^2$, so Q=500 W gives only A ≈ 0.083 $m^2$. The Turton shell-and-tube correlation used by the costing model is valid for A ? [10, 1000] $m^2$; our exchanger is two orders of magnitude below Smin. In that extrapolated tail the correlation*

*turns ∗upward∗ as A shrinks (the quadratic term dominates for log10 A < 0), so a larger area is cheaper. The optimiser correctly exploits this: pushing Q to its upper bound grows A toward the correlation's valid range and slides down the cost curve. The reactor volume settles at an interior 0.00807 $m^3$ — large enough for adequate conversion (X ≈ 0.64), small enough that its vessel cost (~19 k) stays a rounding error next to the exchanger.*

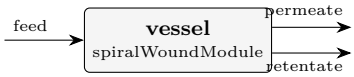
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | hotStream | F | 0.000277777777778 |
| stream | liqProd | F | 0.000261961581415 |
| stream | reactorOut | F | 0.000277777777778 |
| stream | vapProd | F | 1.58161963624e-05 |
| kpi | heater | F | 0.000277777777778 |
| kpi | heater | P | 101325 |
| kpi | heater | Q | 500 |
| kpi | heater | Q_kW | 0.5 |
| kpi | heater | Q_per_mol | 1800 |
| kpi | heater | T_in | 350 |
| kpi | heater | T_out | 362.582514603 |
| kpi | heater | dT | 12.582514603 |
| kpi | heater | vf_out | 0.00752893531985 |

*... and 64 more reference values in the case's `expected`.*

## optim04_membrane_recovery_scaling

tutorials/steady/optimisation/optim04_membrane_recovery_scaling




**What it does:** Brackish RO: maximise water recovery up to the scaling limit (SQP, scaling constraints active at the optimum)

**The story:** HOW HARD can we push recovery before the membrane scales?

The flagship CONSTRAINED-OPTIMISATION case. Same brackish-RO vessel and scaling physics as tutorials/steady/membranes/membrane07_scaling_si (6 x 8-inch elements, the ~1500 mg/L groundwater analysis, the bulk-vs-wall saturation-index audit), but now we ASK the engine the design question the audit only diagnosed:

maximise water_recovery varying feed pressure and the (continuous) module count subject to NO mineral scaling at the membrane wall: maxSI_gypsum_wall <= 0 maxSI_calcite_wall <= 0 (rigorous Davies saturation index – smoother than the Stiff-Davis/Langelier chart fit)

Recovery and wall saturation pull AGAINST each other: every extra % recovery concentrates the retentate and drives the WALL composition (bulk x polarisation) further into supersaturation. The optimum therefore sits ON the scaling limit – at least one of the two constraints is ACTIVE, with a strictly positive Lagrange multiplier (the shadow price: how much more recovery one unit of scaling head-room would buy). That is exactly what the SQP trace prints.

GOLDEN-CHECKED regression: the SQP now converges to a CERTIFIED KKT optimum (||grad L||inf ~ 3e-8, all feasibility 0, complementarity ~1e-8, calcite-wall constraint ACTIVE with lambda ~ 1.14 > 0). Because this is a certified stationary point – not a guessed answer – its KPIs are pinned in 'expected', which ALSO guards against a regression to the singular-QP failure a readiness review caught earlier (a "QP subproblem failed: KKT system singular – LICQ failure" at iteration 1 that left the design infeasible). The KPIs still inherit the polarisation + Davies model assumptions; the golden locks the answer THIS model gives at the certified optimum, not an independent ground truth. (If the engine ever fails to converge again, the optimum block prints an explicit "NO KKT certificate ... NOT a certified optimum" block – never the misleading all-zeros certificate it used to forge on the failure path.)

Two design variables, written through $references so the SQP can drive them and the trace shows their canonical-SI values.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.194444444431 |
| stream | permeate | F | 0.0840585502161 |
| stream | retentate | F | 0.110385894215 |
| kpi | vessel | J_w_avg | 5.94615857264e-06 |
| kpi | vessel | LSI_calcite_bulk_end | 0.408440184659 |
| kpi | vessel | LSI_calcite_wall_end | 0.5923502588 |
| kpi | vessel | RSI_calcite_wall_end | 5.8152994824 |
| kpi | vessel | R_obs_Ca | 0.99939574394 |
| kpi | vessel | R_obs_Cl | 0.99398983415 |
| kpi | vessel | R_obs_HCO3 | 0.995485652703 |
| kpi | vessel | R_obs_K | 0.992498464432 |
| kpi | vessel | R_obs_Mg | 0.999546740094 |
| kpi | vessel | R_obs_Na | 0.99398983415 |
| kpi | vessel | R_obs_SO4 | 0.999697781476 |
| *... and 37 more reference values in the case's `expected`.* | | | |

## optim05_reactor_npv

tutorials/steady/optimisation/optim05_reactor_npv




**What it does:** Esterification CSTR sized & operated for MAX NPV by hand-rolled SQP, subject to a reactor-size cap and a capital-budget cap (active-constraint shadow prices)

**The story:** CONSTRAINED ECONOMIC optimisation: size & operate an esterification CSTR plant for MAXIMUM NPV.

Same esterification plant as the economics01 flagship –

feed -→ reactor (CSTR) -→ heater -→ separator (flash) |→ vapProd (ethyl-acetate-rich | overhead, the prime | product @ 1.55 EUR/kg) |→ liqProd (water/acid-rich bottoms, low-value byproduct @ 0.30 EUR/kg)

– but instead of a FIXED design we ask the engine the design question:

maximise economics.NPV varying V_R (reactor volume, m^3) T_feed (feed/reactor T, K) subject to reactor.V_R <= V_max (a physical SIZE cap) economics.FCI <= FCI_max (a CAPITAL-BUDGET cap)

THE ECONOMIC TRADE-OFF (why this is a genuine optimum, not a corner)

The heater runs at a FIXED partial duty, so the downstream flash SPLITS the reactor effluent into a light, ethyl-acetate-rich VAPOUR overhead (sold high) and a heavy, water/acid-rich LIQUID bottoms (sold cheap). Esterification conserves total mass; what conversion buys you is the RESHUFFLING of that mass toward the valuable overhead. Both knobs raise conversion – and the CSTR is ISOTHERMAL at the feed temperature (CSTR.cpp: T = feed.T), so:

V_R (reactor volume) – bigger V_R raises the residence time tau = V_R/Q, so conversion X = k.tau/(1 + k.tau) climbs, more acetic acid + ethanol turn into ethyl acetate, and the priced overhead grows and enriches (REVENUE up). But a bigger vessel costs more steel: the StirredTank sizing → Turton costing drives FCI (CAPEX) UP. Revenue rises with DIMINISHING returns as X saturates toward 1, while CAPEX keeps climbing – so an NPV maximum exists.

T_feed (feed/reactor temperature) – higher T raises the Arrhenius rate constant k = A.exp(-Ea/RT), so X climbs too. T_feed is the cheap conversion ACCELERATOR; its bounds are kept inside the window where the flash splits cleanly (below the temperature that boils the bottoms dry).

We deliberately set the size cap TIGHTER than where the NPV-greedy design would otherwise stop, so at the optimum the reactor-size constraint BINDS.

HOW TO READ THE SHADOW PRICES (the whole point of the tutorial)

The SQP prints, at the certified optimum, a Lagrange multiplier (lambda) for each inequality. A constraint with lambda > 0 is ACTIVE (binding) and its multiplier is the SHADOW PRICE – the marginal NPV you would gain by relaxing that cap by one unit:

reactor.V_R cap ACTIVE → lambda = d(NPV)/d(V_max) [EUR of NPV per extra m^3 of reactor head-room]. economics.FCI cap → lambda = d(NPV)/d(FCI_max) [EUR of NPV per extra EUR of capital you may spend].

A constraint with lambda ~ 0 is SLACK (not binding) – relaxing it buys nothing. Here the FCI cap is deliberately generous: it stays SLACK so the student sees BOTH an active and an inactive constraint side by side. The trace marks the active set and the KKT certificate proves the point is a true constrained optimum (||grad L|| ~ 0, complementarity ~ 0).

The two design variables are case-level $references so the SQP can drive them through ‘variables.<name>‘ and the trace shows their canonical-SI values.

THE ECONOMICS INVERTED ON 2026-08-09, AND THE OPTIMISER DID NOT

Everything above about the SQP is unchanged and still true: the run converges, the reactor-size cap is ACTIVE with a positive multiplier, the FCI cap stays SLACK, and the KKT certificate closes. What changed is the SIGN of the money.

The ‘heater‘ used to invert its fixed 200 kW on a SENSIBLE enthalpy relation, which walked the stream 11.8 K to 364.27 K without paying any latent heat. The flash then found 98.8 % vapour, and the ethyl-acetate-rich overhead – the whole revenue of this case – existed because the heater had crossed the bubble point for free. The heater now solves for the outlet STATE whose enthalpy is H_in + Q: 200 kW reaches 356.55 K and 7.7 % vapour, the flash splits V/F 0.077, revenue falls 321.7 → 88.2 M EUR against an essentially unchanged 221 M EUR cost of manufacture, and the optimum NPV is -646 M EUR (IRR and payback are UNDEFINED and published as null – the project never breaks even).

So this case still demonstrates constrained SQP, shadow prices and a KKT certificate, and it now demonstrates them on a project that loses money. Restoring the intended economics is a DESIGN decision, not a code fix: the duty (or the pressure, or the product spec) has to be chosen so the overhead is actually made. It is deliberately NOT chosen here – picking a number to make the old answer come back would be fitting the case to a result nobody re-derived. Record: docs/design/tp-stream-energy-coherence.md.

*Take the economics01 ethyl-acetate plant and stop ∗fixing∗ the design — let the engine choose it. The hand-rolled SQP sizes and operates the reactor to maximise the net present value, subject to inequality caps, and prints the shadow prices (Lagrange multipliers) of the binding caps.*

*The objective economics.NPV is a KPI published by the EconomicsPass in the system/postDict chain (sizing → costing → economics). The optimiser runs that whole post chain on every evaluation, so the economics scalars (economics.NPV, economics.FCI, ?) are reachable as ordinary objective/constraint paths — exactly like a sweep reads them.*

*The heater runs at a fixed partial duty, so the flash ∗splits∗ the reactor effluent into a light ethyl-acetate-rich vapour overhead (the prime product at 1.55 ?/kg) and a heavy water/acid-rich liquid bottoms (a low-value byproduct at 0.30 ?/kg). Esterification conserves total mass; what conversion buys you is the reshuffling of that mass toward the valuable overhead.*

*∗ V_R — a bigger reactor raises the residence time τ = V_R/Q, so the isothermal-CSTR conversion X = k·τ/(1+k·τ) climbs and the priced overhead grows. But a bigger vessel costs more steel — the StirredTank sizing → Turton costing drives FCI (CAPEX) up. Revenue rises with diminishing returns (X saturates toward 1) while CAPEX keeps climbing → an NPV maximum exists. ∗ T_feed — a higher feed temperature raises the Arrhenius rate k = A·exp(-E?/RT), the cheap conversion ∗accelerator∗. Its bounds stay inside the window where the flash splits cleanly (below the ~353 K edge that boils the bottoms dry).*

*We deliberately set the size cap tighter than where the NPV-greedy design would otherwise stop, so the reactor-size constraint binds at the optimum.*

*The optimiser drives the money-losing initial design (V_R = 4 $m^3$, T_feed = 350 K → NPV ≈ -164 M?) to a profitable, certified optimum:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.138888888889 |
| stream | hotStream | F | 0.138888888889 |
| stream | liqProd | F | 0.128000396484 |
| stream | reactorOut | F | 0.138888888889 |
| stream | vapProd | F | 0.0108884924051 |
| kpi | economics | COM_d | 221226973.083 |
| kpi | economics | FCI | 509710.813033 |
| kpi | economics | NPV | -651375292.92 |
| kpi | economics | TCI | 586167.434987 |
| kpi | economics | WC | 76456.6219549 |
| kpi | economics | revenue | 87139786.3507 |
| kpi | heater | F | 0.138888888889 |
| kpi | heater | P | 101325 |
| kpi | heater | Q | 200000 |

*. . . and 78 more reference values in the case's `expected`.*

## optim06_integer_stages

tutorials/steady/optimisation/optim06_integer_stages




**What it does:** The classic N-vs-R trade solved in ONE run: nStages declared integer → exhaustive enumeration, one full SQP per stage count, ALL values on the table

**The story:** sensitivity01_column_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0402689234701 |
| kpi | column01 | L_strip | 0.0486788710901 |
| kpi | column01 | Q_condenser_kW | -1699.3833409 |
| kpi | column01 | Q_reboiler_kW | 1065.59910237 |
| kpi | column01 | R | 2.89936248984 |
| kpi | column01 | T_bottom | 382.835751971 |
| kpi | column01 | T_top | 354.273376618 |
| kpi | column01 | V_rect | 0.0541578123589 |
| kpi | column01 | V_strip | 0.0347899822012 |
| *... and 18 more reference values in the case's `expected`.* | | | |

## pareto01_purity_energy

tutorials/steady/optimisation/pareto01_purity_energy




**What it does:** The purity-energy PARETO FRONT by epsilon-constraint: minimise V_strip subject to x_D_LK >= eps, eps swept 0.96..0.994 – each front point one auditable SQP run

**The story:** sensitivity01_column_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0618441580991 |
| kpi | column01 | L_strip | 0.0702541057191 |
| kpi | column01 | Q_condenser_kW | -2369.28314516 |
| kpi | column01 | Q_reboiler_kW | 1736.51498286 |
| kpi | column01 | R | 4.45277938313 |
| kpi | column01 | T_bottom | 383.4896836 |
| kpi | column01 | T_top | 353.562993958 |
| kpi | column01 | V_rect | 0.0757330469879 |
| kpi | column01 | V_strip | 0.0563652168302 |

*... and 18 more reference values in the case's `expected`.*

## sensitivity01_column_reflux

tutorials/steady/optimisation/sensitivity01_column_reflux




**What it does:** Sweep reflux ratio R = 1.5..5.0 (10 pts) across the column01 case; responses: x_D purity + V_strip (reboiler-duty proxy)

**The story:** the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0424382716049 |
| kpi | column01 | L_strip | 0.050848219225 |
| kpi | column01 | Q_condenser_kW | -1764.83717155 |
| kpi | column01 | Q_reboiler_kW | 1131.55472916 |
| kpi | column01 | R | 3.05555555556 |
| kpi | column01 | T_bottom | 383.160243614 |
| kpi | column01 | T_top | 353.921741298 |
| kpi | column01 | V_rect | 0.0563271604938 |
| kpi | column01 | V_strip | 0.0369593303361 |

*. . . and 15 more reference values in the case's `expected`.*

### sensitivity02_feasibility_map

tutorials/steady/optimisation/sensitivity02_feasibility_map




**What it does:** Feasibility MAP: the reflux sweep with declared constraints – every row marked (value, bound, residual, satisfied), nothing filtered; the feasible window between the purity spec and the boilup budget read off the CSV

**The story:** sensitivity01_column_reflux – the column01 benzene/toluene column (classical Raoult textbook problem, no azeotrope), reused UNCHANGED as the inner problem of a sensitivity sweep: system/outerDict varies units[0].operation.refluxRatio over R = 1.5 .. 5.0 (10 points) and reads T_top, T_bottom, x_D_LK, x_B_HK and V_strip (the reboiler-duty proxy) – the classic purity-vs-energy trade-off, one CSV row per R.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0138888888889 |
| stream | distillate | F | 0.0138888888889 |
| stream | feed | F | 0.0277777777778 |
| kpi | column01 | B | 0.0138888888889 |
| kpi | column01 | D | 0.0138888888889 |
| kpi | column01 | L_rect | 0.0424382716049 |
| kpi | column01 | L_strip | 0.050848219225 |
| kpi | column01 | Q_condenser_kW | -1764.83717155 |
| kpi | column01 | Q_reboiler_kW | 1131.55472916 |
| kpi | column01 | R | 3.05555555556 |
| kpi | column01 | T_bottom | 383.160243614 |
| kpi | column01 | T_top | 353.921741298 |
| kpi | column01 | V_rect | 0.0563271604938 |
| kpi | column01 | V_strip | 0.0369593303361 |
| *... and 18 more reference values in the case's `expected`.* | | | |

## power

**The theory you need.** Gas (Brayton) and steam (Rankine) cycles, alone and combined. Compressor and turbine hardware is specified by SHAFT WORK and isentropic efficiency – outlet pressure is a RESULT, not an input (the anti-DesignSpec credo). Expect: pressure ratio setting Brayton efficiency, the combined cycle's gain coming from bottoming the gas turbine's exhaust exergy, and every energy flow visible in the ledger. *Full derivation: Theory Guide, the power-cycle chapter.*

## brayton01_gas_turbine

tutorials/steady/power/brayton01_gas_turbine




**What it does:** Closed Brayton (gas-turbine) cycle: compress air → combust (gas-aware heater) → expand to feed P

**The story:** a closed-loop Brayton (gas-turbine) cycle, the showcase for the gas-aware heater (v0.65).

feed (air, 1 bar, 298 K) → compressor (+10 kW, eta 0.80) : 1 → ~9.9 bar, 298 → 633 K → combustor (heater, +20 kW) : add heat at constant P; the air is gas (vf=1) so the heater integrates the ideal-gas H_ig (N2/O2 carry no liquid Cp) → 633 → ~1246 K → turbine (-W_turb, eta 0.85) : expand back to 1.013 bar

The turbine shaft power is NOT guessed: a DesignSpec (system/outerDict) manipulates $W_turb until the turbine discharges at the feed pressure, closing the cycle. Net shaft work and thermal efficiency are computed $variables (v0.64), evaluated after the solve from the unit KPIs.

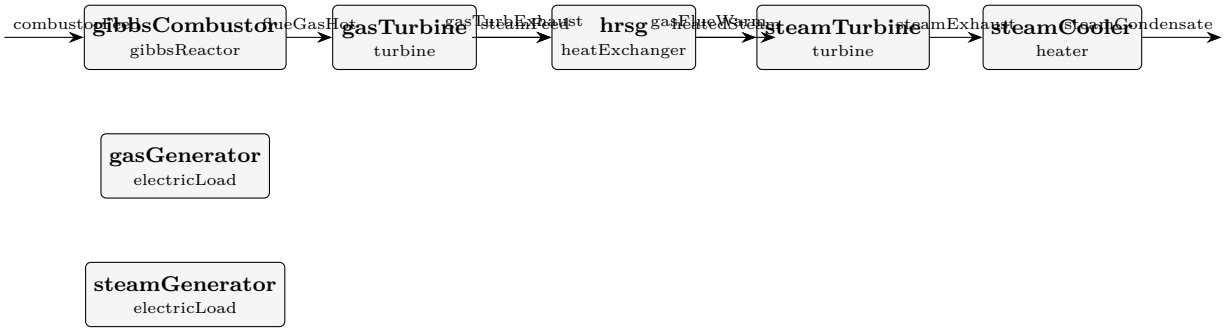
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | compressed | F | 0.001 |
| stream | exhaust | F | 0.001 |
| stream | feed | F | 0.001 |
| stream | hot | F | 0.001 |
| kpi | combustor | F | 0.001 |
| kpi | combustor | P | 991200.306811 |
| kpi | combustor | Q | 20000 |
| kpi | combustor | Q_kW | 20 |
| kpi | combustor | Q_per_mol | 20000 |
| kpi | combustor | T_in | 633.374172563 |
| kpi | combustor | T_out | 1246.40394079 |
| kpi | combustor | dT | 613.029768231 |
| kpi | compressor | F | 0.001 |
| kpi | compressor | P_in | 100000 |

*... and 42 more reference values in the case's `expected`.*

## combined01_brayton_rankine

tutorials/steady/power/combined01_brayton_rankine




**What it does:** Combined-cycle: Brayton (CH4 + air, Gibbs combustor) + HRSG + Rankine + 2 generators

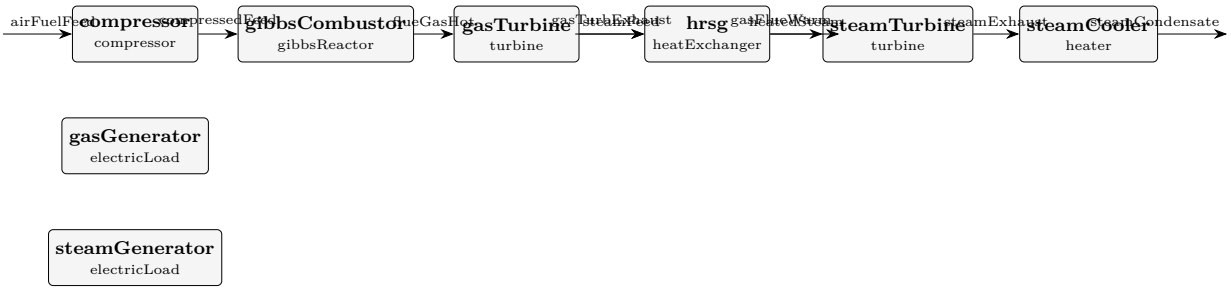
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | combustorFeed | F | 0.1025 |
| stream | flueGasHot | F | 0.1025 |
| stream | gasFlueWarm | F | 0.1025 |
| stream | gasTurbExhaust | F | 0.1025 |
| stream | heatedSteam | F | 0.01 |
| stream | steamCondensate | F | 0.01 |
| stream | steamExhaust | F | 0.01 |
| stream | steamFeed | F | 0.01 |
| kpi | gasGenerator | W_electric | 582000 |
| kpi | gasGenerator | W_electric_kW | 582 |
| kpi | gasGenerator | W_losses | 18000 |
| kpi | gasGenerator | W_shaft | 600000 |
| kpi | gasGenerator | W_shaft_kW | 600 |
| kpi | gasGenerator | eta_generator | 0.97 |

*. . . and 104 more reference values in the case's `expected`.*

## combined02_brayton_rankine_shaft

tutorials/steady/power/combined02_brayton_rankine_shaft




**What it does:** Combined cycle with SHAFT-SPLIT: gas turbine drives the compressor + net to the generator

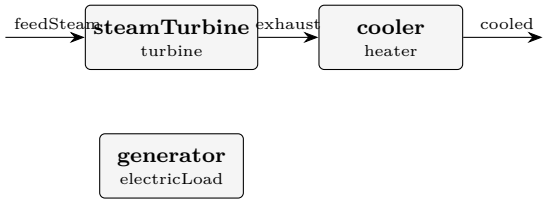
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | airFuelFeed | F | 0.1025 |
| stream | compressedFeed | F | 0.1025 |
| stream | flueGasHot | F | 0.1025 |
| stream | gasFlueWarm | F | 0.1025 |
| stream | gasTurbExhaust | F | 0.1025 |
| stream | heatedSteam | F | 0.01 |
| stream | steamCondensate | F | 0.01 |
| stream | steamExhaust | F | 0.01 |
| stream | steamFeed | F | 0.01 |
| kpi | compressor | F | 0.1025 |
| kpi | compressor | P_in | 100000 |
| kpi | compressor | P_out | 828817.033759 |
| kpi | compressor | T_in | 300 |
| kpi | compressor | T_out | 592.090873473 |

*. . . and 121 more reference values in the case's `expected`.*

## rankine01_water

tutorials/steady/power/rankine01_water




**What it does:** Simplified gas-phase Rankine: superheated steam → turbine (-10 kW) → cooler (gas-only, no phase change)

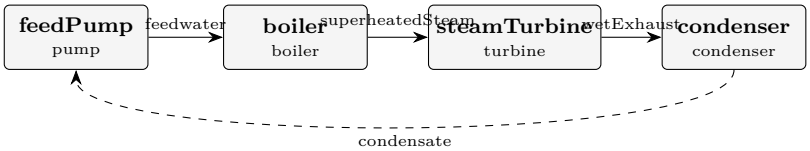
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | cooled | F | 0.001 |
| stream | exhaust | F | 0.001 |
| stream | feedSteam | F | 0.001 |
| kpi | cooler | F | 0.001 |
| kpi | cooler | P | 52827.3976295 |
| kpi | cooler | Q | -5000 |
| kpi | cooler | Q_kW | -5 |
| kpi | cooler | Q_per_mol | -5000 |
| kpi | cooler | T_in | 530.878445241 |
| kpi | cooler | T_out | 388.087132345 |
| kpi | cooler | dT | -142.791312896 |
| kpi | generator | W_electric | 9700 |
| kpi | generator | W_electric_kW | 9.7 |
| kpi | generator | W_losses | 300 |

*. . . and 34 more reference values in the case's `expected`.*

### rankine02_water

tutorials/steady/power/rankine02_water




**What it does:** Closed IF97 Rankine cycle: pump → boiler → turbine → condenser → recycle

**The story:** a CLOSED Rankine steam cycle on IAPWS-IF97

recycle (condensate) [feedPump] [boiler] [steamTurbine] [condenser] raise P cross the extract work cross the dome to 80 bar dome up, (W_shaft, eta), DOWN, land on (liquid) superheat exhaust ~0.1 bar saturated liquid

A real Rankine: the working fluid crosses the saturation dome TWICE – up in the boiler (subcooled liquid → superheated vapour) and down in the condenser (wet vapour → saturated liquid). Pure water on IF97 supplies the steam-table h/s/v surface, so:

the boiler is a dome-crossing phaseChanger driven to superheatedVapour (its duty Q_boiler is the result), the turbine expands on the IF97 entropy surface (H_real/S_real routed), the condenser is a dome-crossing phaseChanger driven to saturatedLiquid (its duty Q_condenser is the result), the pump returns the condensate to boiler pressure (closed-form v∗dP).

The loop is CLOSED: the condensate is the tear stream, recycled to the pump. Wegstein/Newton drives the post-pass condensate to match the guess.

First-law check (KPIs): Q_boiler + Q_condenser + W_pump + W_turbine ~ 0.

Contrast: rankine01_water runs the SAME architecture on the ideal gas, in the gas phase only (no liquid leg, no phase change) – the pedagogical baseline. rankine02 is the steam-table truth.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | boiler | T_out | 773.15 |
| kpi | boiler | vf_out | 1.0 |
| kpi | boiler | Tsat | 568.159121229 |
| kpi | boiler | Q_kW | 47.8559413747 |
| kpi | condenser | vf_out | 0.0 |
| kpi | condenser | Tsat | 446.515894355 |
| kpi | condenser | Q_kW | -39.017165456 |
| kpi | steamTurbine | P_out | 858590.397228 |
| kpi | steamTurbine | W_gen_kW | 9.0 |
| stream | superheatedSteam | vf | 1.0 |
| stream | condensate | vf | 0.0 |
| utility | boiler | heating.-.unserved | 1 |
| utility | condenser | cooling.coolingWater.duty_kW | -39.0172479639 |
| utility | condenser | cooling.coolingWater.T | 446.515894355 |
| *... and 59 more reference values in the case's `expected`.* | | | |

## reactors

**The theory you need.** Batch reactors integrate $\mathrm{d}n_i/\mathrm{d}t = \nu_i\, r\, V$ from a charged initial state – no in/out flows; time replaces residence time. Adiabatic runs couple the energy balance $\sum_i n_i \bar{c}_{p,i}\, \mathrm{d}T/\mathrm{d}t = -\Delta H_{\mathrm{rxn}} r V$, so composition and temperature evolve together. Stiffness appears the moment fast and slow reactions coexist (radical chains: $\mu$s radical lifetimes vs ms induction) – the explicit RK4 integrator then needs $\Delta t$ at the FASTEST scale or blows up, while Rosenbrock23 (L-stable, linearly implicit) steps at the scale of the SOLUTION. The stiffness ratio is printed; watch it. *Full derivation: Theory Guide, the batch/ODE chapter.*

## acetone02_luyben_reactor

tutorials/steady/reactors/acetone02_luyben_reactor




**What it does:** Luyben's acetone reactor: IPA dehydrogenation at 90 % conversion, 623 K – outlet composition and duty against the published Rout

**The story:** the FIRST PROCESS slice of the acetone plant

W. L. Luyben, "Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol", Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a – the Turton base case of his Figure 1.

ONE UNIT, chosen because it is the piece of the flowsheet whose answer does NOT depend on the mixture thermodynamics. At 623 K and 2.6 atm everything is vapour; no activity model is consulted. What is exercised is stoichiometry, the elements-datum heat of reaction and the energy balance – and those can be held against published numbers without first solving the harder problem the separations pose (acetone01_ipa_water_azeotrope measures how hard).

THE FEED IS DERIVED FROM THE PAPER'S OWN STREAM TABLE, and the derivation is a check in itself. Luyben gives Rin = 57.83 kmol/h at 389 K, 2.6 atm and Rout = 92.6 kmol/h with H2 0.3755, acetone 0.3755, IPA 0.0417, water 0.2073. Water is inert and hydrogen is produced one-for-one with acetone, so:

H2 produced = 92.6 x 0.3755 = 34.77 kmol/h = IPA converted IPA out = 92.6 x 0.0417 = 3.86 kmol/h IPA in = 34.77 + 3.86 = 38.63 kmol/h water in = 92.6 x 0.2073 = 19.19 kmol/h (inert) total in = 38.63 + 19.19 = 57.82 kmol/h vs the stated 57.83 conversion = 34.77 / 38.63 = 90.0 % vs the stated 90 %

Two independent closures on a table transcribed by hand. The feed in 0/ is those numbers; the conversion spec is the paper's 90 %.

WHAT IS BEING COMPARED (see 'expected'): the outlet composition, four mole fractions from Luyben's Rout; the reactor DUTY against his 0.960 MW.

THE DUTY IS THE INTERESTING ONE. Choupo derives the heat of reaction from each species' own formation enthalpy – it is never read from the reactions dict – and isopropanol's dHf here is a JOBACK ESTIMATE, because Choupo has no curated isopropanol. So the published duty tests a chain that runs from a group-contribution estimate of one molecule all the way to a megawatt of furnace heat. That is the whole point of building this rather than asserting it: a case says where in the chain the error entered.

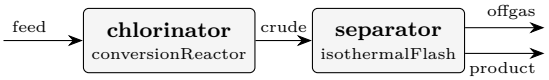
NOT MODELLED, and named rather than implied: the reactor's SIZE. Luyben specifies 450 tubes with a 624 K molten-salt jacket, and the paper does not give the tube dimensions this record was built from, so a 'pfr' on his kinetics cannot be dimensioned from what is in hand. A 'conversionReactor' at his stated 90 % asks the thermodynamic question without pretending to the kinetic one. The kinetics (Luyben Table 1) are transcribed in the design record and remain unused here, on purpose.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Rin | F | 0.0160611111111 |
| stream | Rout | F | 0.0257186111111 |
| kpi | reactor | F_in_kmol_h | 57.82 |
| kpi | reactor | F_out_kmol_h | 92.587 |
| kpi | reactor | Q_kW | 840.263234652 |
| kpi | reactor | T | 623 |
| kpi | reactor | conversion | 0.9 |
| kpi | reactor | dHrxn_kJ_per_mol | 48.7173210118 |
| kpi | reactor | extent_kmol_h | 34.767 |
| stream | Rout | isopropanol | 0.0417229200644 |
| stream | Rout | acetone | 0.375506280579 |
| stream | Rout | H2 | 0.375506280579 |
| stream | Rout | water | 0.207264518777 |
| kpi | reactor | Q_reaction_kW | 470.487527672 |
| *… and 11 more reference values in the case's `expected`.* | | | |

## chlorophenol01_chlorination

tutorials/steady/reactors/chlorophenol01_chlorination




**What it does:** Chlorophenol: classic phenol chlorination (PhOH + Cl2 → 4-CP + HCl) + HCl flash

**The story:** the CLASSIC chlorophenol process.

Direct chlorination of phenol with chlorine gas – the industrial route to the chlorophenols (here the mono, 4-chlorophenol):

chlorinator : C6H5OH + Cl2 → C6H4ClOH + HCl (exothermic, ~-130 kJ/mol) separator : a flash at 80 C, 1 atm splits the volatile HCl (+ excess Cl2) off the liquid phenol / chlorophenol.

Feed: 1.1 kmol/h phenol + 1.0 kmol/h Cl2 (slight phenol excess → Cl2 limiting). phenol + 4-chlorophenol are JOBACK ESTIMATES (glass-box, unvalidated).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | crude | F | 0.000583333333333 |
| stream | feed | F | 0.000583333333333 |
| stream | offgas | F | 0.000275331266235 |
| stream | product | F | 0.000308002067098 |
| kpi | chlorinator | F_in_kmol_h | 2.1 |
| kpi | chlorinator | F_out_kmol_h | 2.1 |
| kpi | chlorinator | T | 353 |
| kpi | chlorinator | conversion | 0.9 |
| kpi | chlorinator | extent_kmol_h | 0.9 |
| kpi | separator | F_alpha | 0.000308002067098 |
| kpi | separator | F_beta | 0.000275331266235 |
| kpi | separator | F_in | 0.000583333333333 |
| kpi | separator | K_Cl2 | 26.0852015979 |
| kpi | separator | K_HCl | 122.317563764 |

*. . . and 12 more reference values in the case's `expected`.*

## conversion01_hda

tutorials/steady/reactors/conversion01_hda




**What it does:** Conversion reactor: HDA toluene+H2 → benzene+CH4 at 75%/pass (gas-basis, no kinetics, no Vliq)

**The story:** A single gas-phase conversion reactor: toluene + H2 → benzene + CH4, 75 % of the limiting reactant (toluene) per pass. H2 is fed in excess (3:1), as in the real HDA.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.111111111111 |
| stream | product | F | 0.111111111111 |
| kpi | reactor | F_in_kmol_h | 400 |
| kpi | reactor | F_out_kmol_h | 400 |
| kpi | reactor | Q_kW | -1007.67392146 |
| kpi | reactor | T | 900 |
| kpi | reactor | conversion | 0.75 |
| kpi | reactor | dHrxn_kJ_per_mol | -48.3683482303 |
| kpi | reactor | extent_kmol_h | 75 |
| kpi | reactor | Q_reaction_kW | -1007.67392146 |
| kpi | reactor | Q_sensible_kW | 0 |
| stream | feed | T | 900 |
| stream | product | T | 900 |
| boundary | global | H_feeds_kW | 6015.44639582 |

*. . . and 6 more reference values in the case's `expected`.*

## conversion02_parallel_selectivity

tutorials/steady/reactors/conversion02_parallel_selectivity




**What it does:** Conversion reactor, PARALLEL network A→B / A→C: selectivity as a SPEC (75 %)

**The story:** a stoichiometric (conversion) reactor with a PARALLEL network, A → B / A → C.

Multi-reaction grammar ‘reactions ( a_to_b a_to_c );‘. Each reaction’s extent is a SPEC, not a result: ‘conversions ( { reaction ...; conversion ...; } ... )‘ gives the fraction of the FEED A that each leg consumes. 60 % of A goes the wanted way and 20 % the side way, so:

total A conversion = 80 % selectivity B/(B+C) = 0.36 / 0.48 = 75 %

This is the honest reactor when you have neither rate data (cstr/pfr) nor an equilibrium (equilibriumReactor) – just a catalyst whose split you measured. A single-reaction conversionReactor cannot express a selectivity at all.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | product | F | 0.000188888888889 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 0.68 |
| kpi | reactor | T | 350 |
| kpi | reactor | extent_a_to_b_kmol_h | 0.12 |
| kpi | reactor | extent_a_to_c_kmol_h | 0.04 |
| kpi | reactor | nReactions | 2 |
| stream | feed | T | 350 |
| stream | product | T | 350 |

## cstr01_first_order

tutorials/steady/reactors/cstr01_first_order




**What it does:** CSTR with pseudo-first-order Arrhenius esterification (AcOH + EtOH <→ EtOAc + H2O), ~53% conversion of ethanol

**The story:** CSTR single-reaction tutorial. The reaction kinetics live in constant/reactions; here the unit references it by name.

*A 1 kmol/h equimolar ethanol / acetic-acid feed at 350 K enters a 5-litre CSTR. One reaction (Fischer esterification), pseudo-first-order in ethanol, Arrhenius kinetics from constant/reactions. The golden: X = 52.59 % conversion of ethanol, extent ? = 0.0730 mol/s, residence time τ = 310.3 s, rate constant k = 3.575e-3 s?[1] at 350 K, and the reactor removing only 0.154 kW to stay isothermal (the reaction is only mildly exothermic; see lesson 4).*

*1. One equation, one unknown, and the unknown is the extent. The Log tab prints Newton in extent ? (?_max = 1.389e-01 mol/s) and a two-row table: from a guess of ?_max/2 the residual goes -7.6e-3 → -2.4e-13 in one step — a first-order rate law in a CSTR is linear in ?, so Newton lands exactly. 2. Damk?hler is the whole design in one number. Da = kτ = 1.109, and for first order X = Da/(1 + Da) = 0.526. Do that division by hand; it is the golden to three figures. 3. k comes from A and Ea, at this T. The reaction record says the pair is ILLUSTRATIVE. Read the Arrhenius line in the log (k_fwd: 3.5750e-03) and reproduce it from A exp(-Ea/RT). 4. Isothermal means somebody removes the heat. The liquid-phase esterification is only mildly exothermic: calorimetry puts it at about -3 to -4 kJ/mol, and Choupo prices it from the four liquid formation enthalpies at about -2.1 kJ/mol at 350 K (dHrxn_kJ_per_mol in the log; Q_kW = -0.154). Acetic acid's liquid is bridged to the ideal-gas monomer datum through the heat of vaporising to the MONOMER (the measured latent heat at Tb plus the dissociation of the dimers its vapour carries, carried to 298 K on the record's own heat capacities; announced in the log); with the calorimetric heat alone the reaction would read -25.7 kJ/mol, which is the heat of making dimers, not of esterifying. cstr04_adiabatic removes nothing and lets T answer.*

*Double V_R to 0.010: τ doubles, Da doubles, and X moves to Da/(1+Da) — check it before you run. Then open pfr01_first_order: the same volume, the same kinetics, and a different answer.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | X_limiting | 0.52590711876 |
| kpi | reactor | T | 350 |
| kpi | reactor | k | 0.00357499942014 |
| stream | feed | F | 0.000277777777778 |
| stream | feed | T | 350 |
| stream | reactorOut | F | 0.000277777777778 |
| stream | reactorOut | T | 350 |
| kpi | reactor | Da_kTau | 1.10929132154 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -0.153690679658 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | minus_r_ethanol | 14.6085310767 |
| kpi | reactor | tau_s | 310.291329081 |
| *... and 8 more reference values in the case's `expected`.* | | | |

## cstr02_reversible_wgs

tutorials/steady/reactors/cstr02_reversible_wgs




**What it does:** Reversible water-gas shift (CO + H2O <→ CO2 + H2) in a CSTR

**The story:** water-gas-shift reaction in a CSTR with reverse rate from thermodynamic K_eq.

Feed: equimolar CO + H2O at 600 K, 10 bar. The forward Arrhenius kinetics drives a high conversion in the early residence times; the reverse term (auto-derived from g_pure_ig of the four species at T = 600 K) caps the conversion at the Le Chatelier equilibrium.

At T = 600 K: dG_rxn = g_CO2 + g_H2 - g_CO - g_H2O ≈ -16.3 kJ/mol K_eq = exp(16300 / (R · 600)) ≈ 26.5 For equimolar CO + H2O, K_eq = X^2 / (1-X)^2 -→ X_eq ≈ 0.84

V_R is picked large enough (1e-3 m^3 for a 1 kmol/h feed) that the kinetics are fast relative to residence; we see the conversion approach (but not exceed) X_eq ≈ 0.84 — the thermodynamic ceiling.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | feed | vf | 1 |
| stream | products | F | 0.000277777777778 |
| stream | products | vf | 1 |
| kpi | wgs | Da_kTau | 145.857299066 |
| kpi | wgs | F_in_kmol_h | 1 |
| kpi | wgs | F_out_kmol_h | 1 |
| kpi | wgs | T | 600 |
| kpi | wgs | V_R | 0.001 |
| kpi | wgs | X_limiting | 0.841431955275 |
| kpi | wgs | k | 1.08521882096 |
| kpi | wgs | tau_s | 134.403584096 |
| kpi | wgs | xi_mol_per_s | 0.116865549344 |
| stream | feed | T | 600 |
| *... and 10 more reference values in the case's `expected`.* | | | |

## cstr03_series_selectivity

tutorials/steady/reactors/cstr03_series_selectivity




**What it does:** CSTR with a SERIES network A→B→C: multi-reaction, the selectivity problem

**The story:** a CSTR with a SERIES reaction network, A → B → C, solved as ONE coupled system.

The multi-reaction grammar ‘reactions ( a_to_b b_to_c );‘ names both steps from constant/reactions; the reactor solves the R coupled design equations xi_j = r_j(xi) V_R together by the multivariate Newton. A single-reaction CSTR cannot express this at all: the intermediate B is BOTH a product and a reactant, so its yield peaks in residence time and SELECTIVITY becomes the design question.

Feed: pure A at 350 K. Try sweeping V_R (an outerDict sweep) and watch the B yield rise, peak, and fall as the over-reaction B → C takes over.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | product | F | 0.000199218293117 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 0.717185855222 |
| kpi | reactor | T | 350 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | k_a_to_b | 0.00214522777035 |
| kpi | reactor | k_b_to_c | 0.000536306942589 |
| kpi | reactor | nReactions | 2 |
| kpi | reactor | newtonResidual | 6.93889390391e-18 |
| kpi | reactor | tau_s | 225 |
| kpi | reactor | xi_a_to_b_mol_per_s | 0.0392797423303 |
| kpi | reactor | xi_b_to_c_mol_per_s | 0.0126884461066 |
| stream | feed | T | 350 |
| *... and 2 more reference values in the case's* ***expected****.* | | | |

## cstr04_adiabatic

tutorials/steady/reactors/cstr04_adiabatic




**What it does:** ADIABATIC CSTR: T_out solved WITH the extents; Q=0 by construction (elements datum); the liquid esterification is nearly thermoneutral, T_out only ~5 K above the feed

**The story:** an ADIABATIC continuous stirred-tank reactor.

The outlet temperature is an UNKNOWN, solved WITH the reaction extents:

xi_j = r_j(xi, T) V_R (R design equations) H_out(T, xi) = H_in (one enthalpy balance)

There is NO dH_rxn source term. With the enthalpy on the ELEMENTS datum, the heat of reaction already lives inside H: as reactants become products the formation enthalpies differ and T rises on its own. Q_kW = 0 by construction.

Since the acetic-acid correction (docs/design/acetic-acid-latent-heat-closure.md) this LIQUID esterification is nearly thermoneutral (dH_rxn about -2.7 kJ/mol at 298 K, NBS-consistent; the engine used to print about -25.7 kJ/mol), so the adiabatic rise is a few K (T_out 355.3 K from the 350 K feed) and the energy balance has ONE steady state: the scan finds one. A non-isothermal CSTR CAN have up to three steady states, but that needs a strongly exothermic reaction; this one does not. 'T_guess' is kept (it is the declared aid that selects the branch where there are several) and here it cannot matter.

The original teaching point of this case (an adiabatic reactor that heats itself by ~100 K) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed to recover the old rise.

*The cstr01 reactor with no heat removal. The esterification is only mildly exothermic in the liquid (dH_rxn about -2.7 kJ/mol at 298 K, after the acetic-acid correction recorded in docs/design/acetic-acid-latent-heat-closure.md), so the contents heat only a little: the golden lands at T_out = 355.34 K, a 5.34 K rise from the 350 K feed, with the extent ? = 0.0853 mol/s (?_max is 0.1389 mol/s, so conversion is about 61 %) at a rate constant of 5.1e-3 s?*[1].

*Earlier versions of this case printed a 104 K rise and complete conversion; they were written around a liquid heat of reaction about ten times too large. The original teaching point (an adiabatic reactor heating itself by ~100 K) needs a genuinely exothermic reaction; a replacement case is planned. Nothing in this case was tuned to recover it.*

*The feed is still declared at 15 bar, a leftover from the old design, where a 454 K product had to be held liquid. At the corrected 355 K the outlet is liquid even at 1 atm.*

*1. Two unknowns now, solved together. The extent from the design equation and T from H_out(T, ?) = H_in: the dict says so in its own header, and Q_kW in the golden is ~1e-11 — zero by construction, not by declaration. 2. An adiabatic CSTR can have more than one answer — here it has one. Before solving, the engine ∗scans∗ the energy balance across T (the Log line [cstr] scanning the energy balance on T in [325.0, 383.5] K for steady states) and reports how many it found — 1. The scan is the van Heerden diagram; with a nearly thermoneutral reaction the generation curve is too small to cut the removal line more than once. 3. T_guess is a declared aid, and the log says where it landed from. operation { T_guess 350; } picks the branch the Newton starts on; the result line prints landed from T_guess = 350.00 K. With one steady state it cannot matter; with several it would decide the answer, which is why it is in the dict and not hidden. 4. Read the caveats the run prints. Acetic acid's Cp polynomial is declared to 390 K and the ethanol vapour-pressure fit to 369 K; the scan's upper end asks for slightly more, so the path is ∗extrapolated∗, announced, and still returned — a warned number is a defensible one. 5. "Newton 742 it" is the scan's total, not the answer's. The log says so on the line. Every trial temperature of the scan re-enters the extent Newton; the final solve took a handful.*

*Set T_guess 360; — same single answer. Then run cstr06_jacketed, where a coolant stream carries heat away and the outlet T sits below the feed.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -9.63495949691e-12 |
| kpi | reactor | T | 355.339972307 |
| kpi | reactor | T_guess | 350 |
| kpi | reactor | T_out | 355.339972307 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | dT_rise | 5.33997230743 |
| kpi | reactor | k_esterification_etac | 0.00513176885398 |
| kpi | reactor | nReactions | 1 |
| kpi | reactor | newtonResidual | 1.38777878078e-17 |
| kpi | reactor | tau_s | 310.291329081 |
| *… and 7 more reference values in the case's `expected`.* | | | |

## cstr05_multiplicity

tutorials/steady/reactors/cstr05_multiplicity




**What it does:** CSTR steady-state scan: with the corrected (nearly thermoneutral) liquid esterification the reactor has ONE steady state (~300 K); the three-state multiplicity needs an exothermic reaction (replacement case planned)

**The story:** an ADIABATIC CSTR fed COLD, scanned for steady states.

Eliminate the extents (at a fixed T the design equations are monotone) and what remains is ONE equation in the temperature:

phi(T) = H_out(T) - H_in - Q_ext(T) = 0

the textbook "heat generated equals heat removed". The generation curve G(T) is a sigmoid (the rate saturates once the reactant is gone); the removal line R(T) = F cp (T - T_in) is straight (adiabatic: the only sink is the sensible heat that leaves with the product). A straight line can cut a sigmoid THREE times: the extinguished state, an unstable middle state, and the ignited state – IF the sigmoid is tall enough, i.e. IF the reaction releases enough heat. Choupo scans, finds them all, prints them, and reports the one nearest ‘T_guess‘. Change T_guess and the reactor moves to another branch – when there is more than one.

This case was written for exactly that: a 300 K feed, a negligible rate there, and an ignited branch reached by ‘T_guess‘. It no longer shows it. After the acetic-acid correction (docs/design/acetic-acid-latent-heat-closure.md) the liquid esterification is nearly thermoneutral (dH_rxn about -2.7 kJ/mol, where the engine used to print about -25.7), the complete-conversion adiabatic temperature is only 310.7 K, and G(T) is far too small to cut the removal line more than once: the scan finds ONE steady state, 300.04 K, whatever ‘T_guess‘ is (380 here, and the log says it landed from it).

The original teaching point (multiplicity, ignition and extinction, a start-up procedure) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed to recover the three states. The van Heerden construction itself is unchanged: the engine still scans, counts and prints every root it finds.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | 8.78230821399e-12 |
| kpi | reactor | T | 300.040632256 |
| kpi | reactor | T_guess | 380 |
| kpi | reactor | T_out | 300.040632256 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | dT_rise | 0.0406322555593 |
| kpi | reactor | k_esterification_etac | 1.17023320484e-05 |
| kpi | reactor | nReactions | 1 |
| kpi | reactor | newtonResidual | 5.0990028172e-16 |
| kpi | reactor | steadyStates | 1 |
| *... and 7 more reference values in the case's `expected`.* | | | |

## cstr06_jacketed

tutorials/steady/reactors/cstr06_jacketed




**What it does:** JACKETED CSTR: the coolant carries heat away – Q = UA (T_c - T_out); the nearly thermoneutral esterification leaves the outlet below the feed (replacement exothermic case planned)

**The story:** the same esterification as cstr04, now with a COOLING JACKET instead of an adiabatic wall.

The energy balance gains one term, and only one:

H_out(T) = H_in + UA (T_coolant - T_out)

There is still NO dH_rxn source: on the elements datum the heat of reaction is already inside H. What the jacket does is remove heat, so the outlet settles at 338.8 K – BELOW the 350 K feed, because the reaction (dH_rxn about -2.7 kJ/mol after the acetic-acid correction, docs/design/acetic-acid-latent-heat-closure.md) releases far less heat than the 25 W/K jacket takes from a stream 30 K above the coolant. (Before the correction the reaction heated the contents to 394 K against a 454 K adiabatic limit; that was the wrong exotherm.)

Check it by hand from the printed KPIs: Q_kW should equal UA (T_c - T_out), and it is NEGATIVE – heat leaves the reactor (-0.470 kW). The scan finds ONE steady state. Nothing here is switched on by a flag; it is the same one equation in T, with one more term.

The original teaching point of this case (a jacket pulling a hot, ignited reactor down the generation curve, and extinction when it cools too hard) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed. The feed pressure (15 bar) is a leftover of the old design; the outlet is liquid at 1 atm as well.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -0.469673699205 |
| kpi | reactor | T | 338.786947968 |
| kpi | reactor | T_guess | 400 |
| kpi | reactor | T_out | 338.786947968 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | dT_rise | -11.2130520322 |
| kpi | reactor | k_esterification_etac | 0.00161255283829 |
| kpi | reactor | nReactions | 1 |
| kpi | reactor | newtonResidual | 6.93889390391e-18 |
| kpi | reactor | steadyStates | 1 |

*... and 11 more reference values in the case's `expected`.*

## cstr07_lhhw_methylAcetate

tutorials/steady/reactors/cstr07_lhhw_methylAcetate




**What it does:** LHHW on activities: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

**The story:** an isothermal CSTR packed with Amberlyst 15.

The rate constants are reported per GRAM of dry catalyst, so the reactor needs to know how much catalyst a cubic metre of it holds: 'catalystLoading'. It is a declared, visible number, not a fudge – change the packing and the reactor changes with it.

Run it and read the KPI 'k_esterification_meoac' against the conversion: the forward rate constant says one thing, the adsorption denominator says another, and the outlet is where they settle.

This reactor is KINETICALLY limited, which is the point – if it sat at equilibrium the denominator would be invisible. Sweep V_R and watch it climb to the activity-based equilibrium and stop:

V_R = 0.005 m3 X = 49.02 % <- this case V_R = 0.020 m3 X = 63.74 % V_R = 0.100 m3 X = 72.92 % V_R = 1.000 m3 X = 76.18 % <- equilibrium

The plateau is not asserted anywhere in the dict. It is where k1 a'_HOAc a'_MeOH = k-1 a'_MeOAc a'_H2O, i.e. exactly the K_a = 25.50 that the authors' own Eq 17 recovers from the same four rate constants and four adsorption constants at 333.15 K. Two independent equations of the paper agreeing is a cheap and honest check that the parameters were entered right.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -0.0517242743531 |
| kpi | reactor | T | 333.15 |
| kpi | reactor | T_out | 333.15 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | k_esterification_meoac | 0.00280769603355 |
| kpi | reactor | nReactions | 1 |
| kpi | reactor | newtonResidual | 1.2490009027e-16 |
| kpi | reactor | tau_s | 366.524129505 |
| kpi | reactor | xi_esterification_meoac_mol_per_s | 0.0680930737737 |
| stream | feed | T | 333.15 |
| *... and 9 more reference values in the case's `expected`.* | | | |

## equil01_reforming

tutorials/steady/reactors/equil01_reforming




**What it does:** Equilibrium reactor (REquil): SMR + WGS to SIMULTANEOUS equilibrium at 1100 K

**The story:** the EQUILIBRIUM reactor (REquil) with MULTIPLE simultaneous reactions.

Steam-methane reforming: two independent reactions are driven to SIMULTANEOUS chemical equilibrium at 1100 K, 1 bar – no kinetics, no residence time. The two extents are a RESULT of each reaction's Kp(T) (from the standardThermochemistry data), solved together by the multivariate Newton:

smr : CH4 + H2O <→ CO + 3 H2 (strongly endothermic) wgs : CO + H2O <→ CO2 + H2 (mildly exothermic)

The stoichiometry lives in constant/reactions and is referenced BY NAME – the same 'reactions ( r1 r2 );' grammar the batch / dynamic reactors take.

Feed: 1 kmol/h CH4 + 3 kmol/h steam (steam-to-carbon = 3). At 1100 K, 1 bar the reforming is near-complete (Le Chatelier favours the mole-increasing SMR at low pressure); the shift then sets the CO/CO2 split. This is the classic syngas problem – the reactor a 'conversionReactor' cannot do (it needs the extents GIVEN) and a 'cstr'/'pfr' cannot do without rate data.

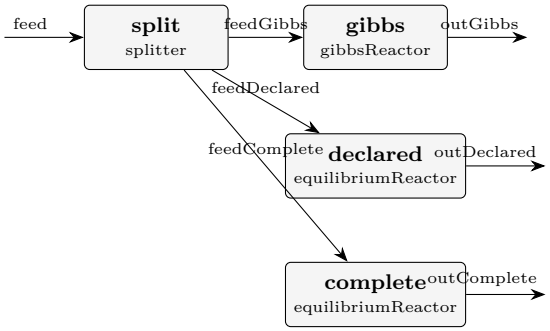
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00111111111111 |
| stream | syngas | F | 0.00166593958576 |
| kpi | reformer | F_in_kmol_h | 4 |
| kpi | reformer | F_out_kmol_h | 5.99738250873 |
| kpi | reformer | Kp_smr | 312.740523794 |
| kpi | reformer | Kp_wgs | 0.978425617548 |
| kpi | reformer | Q_kW | 59.6424648629 |
| kpi | reformer | T | 1100 |
| kpi | reformer | conversion_smr | 0.998691254363 |
| kpi | reformer | conversion_wgs | 0.109768673641 |
| kpi | reformer | extent_smr_kmol_h | 0.998691254363 |
| kpi | reformer | extent_wgs_kmol_h | 0.329306020923 |
| kpi | reformer | nReactions | 2 |
| kpi | reformer | newtonResidual | 8.881784197e-16 |

*. . . and 7 more reference values in the case's `expected`.*

## equil02_methanol_declared_pathways

tutorials/steady/reactors/equil02_methanol_declared_pathways




**What it does:** Declared pathways: one syngas, three equilibria – Gibbs over every species, a declared reaction subset (methanation excluded), and the complete reaction set

**The story:** ONE feed, three equilibrium answers: what a DECLARED reaction list is.

The same synthesis gas (70 H2 : 25 CO : 5 CO2, 525 K, 50 bar, ideal gas) goes to three equilibrium reactors side by side. A splitter divides it; every answer below is read as MOLE FRACTIONS, which do not depend on how much of the feed each reactor received, so the split is bookkeeping.

gibbs gibbsReactor over EVERY species (CO H2 methanol water CO2 CH4): no reaction is declared, the mixture finds the minimum of G subject to the element balances. Every pathway the species list permits is open. declared equilibriumReactor over a DECLARED subset: methanolSynthesis CO + 2 H2 <→ CH3OH waterGasShift CO + H2O <→ CO2 + H2 and NOTHING else. Methanation (CO + 3 H2 <→ CH4 + H2O) is left OUT of the list. complete equilibriumReactor over all three reactions, methanation included – a COMPLETE set of independent reactions for these six species over three elements (6 - 3 = 3).

THE PREMISE IS YOURS. Leaving methanation out states that, over the catalyst this reactor carries and at these conditions, methane formation is too slow to matter. That is kinetic knowledge; the engine cannot check it and does not try. It computes what FOLLOWS from it: equilibrium, but only along the pathways you declared possible.

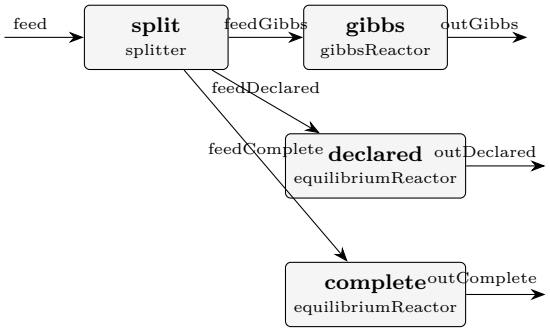
WHAT THE THREE ANSWERS SHOW (measured with this case, 2026-09-27): gibbs y_CH4 = 0.4513, y_methanol = 1.76e-10: with every pathway open, equilibrium at 525 K and 50 bar is methane and water. complete the SAME outlet as gibbs (y_methanol 1.76e-10 again, the flow 13.14 kmol/h per 25 in, as gibbs has 26.28 per 50): a complete set of independent reactions IS the Gibbs reactor. So the difference below is the exclusion, not the solver. declared methanol conversion 0.610 of the CO: remove ONE reaction and the answer is a different product. And the water-gas shift runs BACKWARD in ‘declared‘ (the feed carries CO2 and no water), which the approach knob below reads.

THE SECOND KNOB. ‘declared‘ carries ‘temperatureApproach 0;‘ – a MAGNITUDE the engine gives a sign PER REACTION from that reaction’s own thermicity as it runs (src/unitOperations/reactor/TemperatureApproach.H): methanol synthesis is exothermic and runs forward, so its Kp is taken at (T + dT); the shift runs backward, so as it runs it is endothermic and its Kp is taken at (T - dT). Both push against the direction the reaction runs. With 20 K: methanol conversion 0.610 → 0.442. The key is declared at 0 so the EduTool ‘declared-pathways‘ can move it, and every key a knob moves sits on its own line (the EduTool’s override channel matches a key at the start of a line).

NOT here, and said: carbon deposition (no carbon species in this list; both reactors carry a pure-solid phase since 2026-09-27 – the declared pathways with graphite are equil04_reforming_carbon_declared_pathways, the Gibbs reactor alone is gibbs11_carbon_deposition), non-ideal fugacities at 50 bar (every Kp is on the ideal-gas rung; the case declares ‘fugacityModel idealGas‘), and any reactor that is not isothermal.

## equil03_ammonia_oxidation_declared_pathways

tutorials/steady/reactors/equil03_ammonia_oxidation_declared_pathways




**What it does:** Declared pathways, a reaction that runs to completion: ammonia in air, three equilibria – Gibbs over every species, the NO route alone (the N2 route excluded), and the complete reaction set

**The story:** a declared reaction that runs to COMPLETION.

The companion of equil02 (same shape, same lesson, a harder number). Ten per cent ammonia in air, 1100 K, 5 bar, ideal gas, goes to three equilibrium reactors side by side; every answer is read as MOLE FRACTIONS, so the split is bookkeeping.

gibbs gibbsReactor over EVERY species (NH3 O2 N2 NO water): no reaction declared, the minimum of G under the N, H and O balances. declared equilibriumReactor over ONE reaction: ammoniaToNO 4 NH3 + 5 O2 <→ 4 NO + 6 H2O and nothing else. The N2 route (4 NH3 + 3 O2 <→ 2 N2 + 6 H2O) is left OUT – and with it every path to N2, including the decomposition of NO itself (2 NO <→ N2 + O2 is a combination of the two library reactions). complete equilibriumReactor over both reactions: a COMPLETE set of independent reactions for five species over three elements (5 - 3 = 2).

THE PREMISE IS YOURS. Leaving the N2 route out states that, over the catalyst this reactor carries, ammonia burns to NO and neither ammonia nor NO finds its way to nitrogen. That is kinetic knowledge; the engine cannot check it and does not try. It computes what FOLLOWS from it.

WHAT THE THREE ANSWERS SHOW (measured with this case, 2026-09-27): gibbs y_N2 = 0.7424, y_NO = 6.78e-5, y_NH3 = 5.66e-18: with every pathway open, ammonia burns to NITROGEN and water. complete the SAME outlet, every species to the last printed digit, NH3 at 5.66e-18 included: a complete set of independent reactions IS the Gibbs reactor. declared y_NO = 0.0976, every mole of ammonia to NO (conversion 1 to the printed digits), and y_NH3 = 1.68e-14 left unconverted. Recompute it by hand from the declared equilibrium: 4 ln y_NH3 = 4 ln y_NO + 6 ln y_H2O - 5 ln y_O2 + ln(P/P0) - ln Kp, ln Kp = 121.515

WHY THIS CASE EXISTS. ln Kp = 121.5 puts the unconverted ammonia near 1e-14 of the outlet. Until 2026-09-27 the equilibrium reactor solved for the reaction EXTENTS, so every amount was the feed minus a multiple of an extent, and a species that small is the difference of two nearly equal numbers; it also held every species above a floor of 1e-12 of the feed. Both the declared and the complete reactor here stalled ("Newton stalled at residual 16.26" and "69.77") and refused. The unknowns are now the LOGARITHMS of the amounts (src/unitOperations/reactor/EquilibriumReactor.cpp, record docs/design/an-equilibrium-that-runs-to-completion.md), and 1e-14, or 1e-300, is an ordinary number for a logarithm.

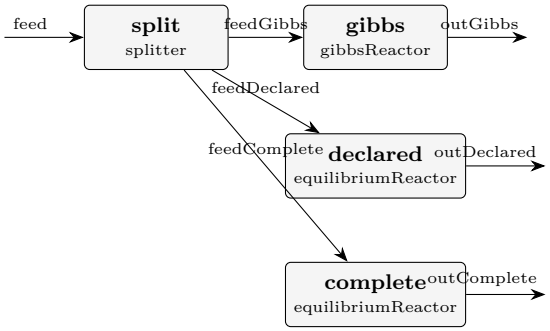
NOT here, and said: the NO → NO2 oxidation downstream of the burner (NO2 is not in this species list), non-ideal fugacities, any rate, and any reactor that is not isothermal.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0277777777778 |
| stream | feed | T | 1100 |
| stream | feedComplete | F | 0.00694444444444 |
| stream | feedComplete | T | 1100 |
| stream | feedDeclared | F | 0.00694444444444 |
| stream | feedDeclared | T | 1100 |
| stream | feedGibbs | F | 0.0138888888889 |
| stream | feedGibbs | T | 1100 |
| stream | outComplete | F | 0.00711805555556 |
| stream | outComplete | T | 1100 |
| stream | outDeclared | F | 0.00711805555556 |
| stream | outDeclared | T | 1100 |
| stream | outGibbs | F | 0.0142361111111 |
| stream | outGibbs | T | 1100 |
| *… and 54 more reference values in the case's `expected`.* | | | |

## equil04_reforming_carbon_declared_pathways

tutorials/steady/reactors/equil04_reforming_carbon_declared_pathways




**What it does:** Declared pathways, a reformer that cokes: one feed, three equilibria – Gibbs over every species (graphite included), the complete reaction set with methane cracking, and the declared set with Boudouard and cracking left out

**The story:** a reformer that cokes at equilibrium, and a declared list that says it does not.

The third of the declared-pathways witnesses (equil02 methanol, equil03 ammonia oxidation), and the one with a SOLID. Methane and steam at a steam-to-carbon ratio of 1.0, 900 K, 1 bar, ideal gas, go to three equilibrium reactors side by side; every gas answer is read as MOLE FRACTIONS of the fluid and every solid answer per mole of methane fed, so the split is bookkeeping.

gibbs gibbsReactor over EVERY species (CH4 water CO CO2 H2 graphite): no reaction declared, the minimum of G under the C, H and O balances, graphite given its own pure-solid phase. complete equilibriumReactor over a COMPLETE set of independent reactions for six species over three elements (6 - 3 = 3): steamReforming CH4 + H2O <→ CO + 3 H2 waterGasShift CO + H2O <→ CO2 + H2 methaneCracking CH4 <→ C(gr) + 2 H2 Boudouard (2 CO <→ C(gr) + CO2) is NOT added: it is cracking - reforming + shift, and a fourth reaction would be refused as linearly dependent. Declaring it IN PLACE of cracking gives the same outlet – an equilibrium does not care which route reaches it. declared equilibriumReactor over reforming and the shift ONLY: Boudouard and methane cracking are left OUT, and with them every pathway to carbon.

THE PREMISE IS YOURS. Leaving the carbon routes out states that, over the catalyst this reactor carries, carbon does not form at these conditions. That is kinetic knowledge; the engine cannot check it and does not try. It does NOT say carbon cannot form thermodynamically – the outlet says the opposite, and says how much the premise is asserting.

WHAT THE THREE ANSWERS SHOW (measured with this case, 2026-09-27): gibbs graphite DEPOSITS: 0.2168 mol per mol of methane fed (21.7 % of the carbon). The solid-free gas had carbon activity a = 1.580 > 1, so it was not the minimum of G. Gas left: y_CH4 0.1146, y_H2O 0.1374, y_CO 0.0923, y_CO2 0.0476, y_H2 0.6081. complete the SAME outlet, gas and graphite, to the printed digits (a solid participant: absent in the first solve, activity 1.580 read off it, then present at activity 1). declared NO graphite, and the gas is the solid-free equilibrium: y_CH4 0.1568, y_H2O 0.1056, y_CO 0.1204, y_CO2 0.0512, y_H2 0.5660. Its carbon activity is published, activity_graphite = 1.580: the outlet is SUPERSATURATED in carbon, a metastable state the declared list asserts. Recompute it by hand, by either route the list leaves out: a_C = Kp_crack ∗ p_CH4 / p_H2^2 (p in bar) = Kp_boud ∗ p_CO^2 / p_CO2 with Kp_crack = 3.2283 and Kp_boud = 5.5887 at 900 K – both give 1.580, because reforming and the shift are the complete set for the five gases, so the gas is fully equilibrated. Raise the steam to S/C 3.0 and a_C falls to 0.294: carbon is absent everywhere and the three reactors agree – the declared list then costs nothing.

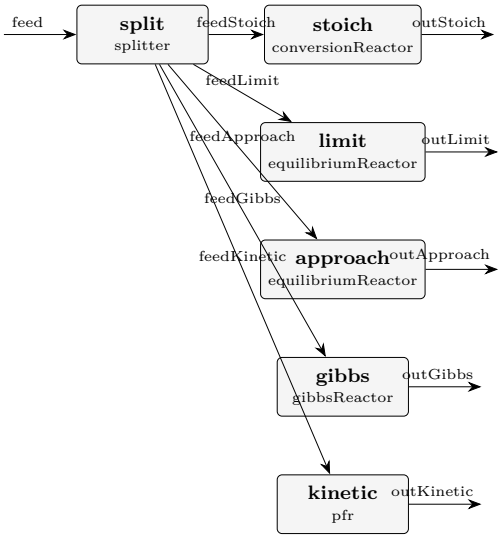
NOT here, and said: any rate (whisker and encapsulating carbon and their kinetics are not modelled), a second solid, a solid in the feed (refused by name), non-ideal fugacities, and any reactor that is not isothermal. Record: docs/design/a-solid-in-a-declared-reaction-set.md.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.00555555555556 |
| stream | feed | T | 900 |
| stream | feedComplete | F | 0.00138888888889 |
| stream | feedComplete | T | 900 |
| stream | feedDeclared | F | 0.00138888888889 |
| stream | feedDeclared | T | 900 |
| stream | feedGibbs | F | 0.00277777777778 |
| stream | feedGibbs | T | 900 |
| stream | outComplete | F | 0.00228803991324 |
| stream | outComplete | T | 900 |
| stream | outDeclared | F | 0.00211460671012 |
| stream | outDeclared | T | 900 |
| stream | outGibbs | F | 0.00457607982775 |
| stream | outGibbs | T | 900 |
| *. . . and 61 more reference values in the case's `expected`.* | | | |

## ladder01_wgs_three_rungs

tutorials/steady/reactors/ladder01_wgs_three_rungs




**What it does:** The reactor ladder: one water-gas-shift feed through the stoichiometric, the equilibrium and the kinetic rung

**The story:** THE REACTOR LADDER on one feed.

CO + H2O <→ CO2 + H2 equimolar, 600 K, 10 bar, ideal gas

The feed is split five ways and each fifth meets a different MODEL of the same reactor. They are ordered as a design proceeds, not as a textbook does: what the stoichiometry allows, then what thermodynamics allows, then what the rate delivers.

stoich conversionReactor rung 1 – a DECLARED conversion. Balances only. X = 0.90 is declared on purpose: it is ABOVE this feed's equilibrium, and this rung has no way to know. limit equilibriumReactor rung 2 – the thermodynamic LIMIT at 600 K. K(T) from the species' Gibbs energies; no rate, no volume. approach equilibriumReactor rung 2 with a 25 K temperature APPROACH: the exothermic shift is asked at 625 K, so the design conversion sits below the limit by a margin a designer chose, not one a rate law computed. The outlet stays at 600 K. gibbs gibbsReactor rung 2 by the other road – no reaction written, only elements and species. Over exactly these four species it must agree with "limit"; add a species (CH4) and it would not (equil02 is that lesson). kinetic pfr rung 3 – the rate law. Its axial profile rises toward the "limit" conversion and never crosses it. The volume is NOT a physical gas volume: this reactor prices concentrations on a liquid molar volume (src/unitOperations/reactor/PFR.cpp, "Inlet (mol/s) + liquid-basis volumetric flow"), so read its profile against V_R as a SHAPE, never as a size.

What each rung needs is the other half of the lesson: rung 1 a number someone must supply, rung 2 formation data the catalogue already has, rung 3 a rate law somebody had to MEASURE. The ladder is climbed only as far as the decision in hand requires.

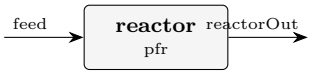
EduTool: gui/src/ui/methods/ReactorLadderTool.tsx.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000555555555556 |
| stream | feed | T | 600 |
| stream | feedApproach | F | 0.000111111111111 |
| stream | feedApproach | T | 600 |
| stream | feedGibbs | F | 0.000111111111111 |
| stream | feedGibbs | T | 600 |
| stream | feedKinetic | F | 0.000111111111111 |
| stream | feedKinetic | T | 600 |
| stream | feedLimit | F | 0.000111111111111 |
| stream | feedLimit | T | 600 |
| stream | feedStoich | F | 0.000111111111111 |
| stream | feedStoich | T | 600 |
| stream | outApproach | F | 0.000111111111111 |
| stream | outApproach | T | 600 |
| *… and 85 more reference values in the case's `expected`.* | | | |

## pfr01_first_order

tutorials/steady/reactors/pfr01_first_order




**What it does:** PFR with the same kinetics as cstr01 (RK4 integration)

**The story:** PFR with the same kinetics and conditions as cstr01. For a first-order reaction the PFR gives X = 1 - exp(-k$\tau$). With k$\tau$ = 1.10: X_PFR = 1 - exp(-1.10) = 0.667 (vs CSTR X = 0.524)

*The reactor of cstr01 rebuilt as a plug-flow tube: the same 5 litres, the same 1 kmol/h feed at 350 K, the same Arrhenius pair. The golden: X = 67.02 % against the CSTR's 52.59 %, at the same $\tau$ = 310.3 s and the same Da = 1.109.*

*1. No back-mixing is the whole difference. In the CSTR every molecule reacts at the ∗outlet∗ concentration, the lowest in the vessel. In the tube the concentration falls along V, so most of the volume works at a higher rate. For first order: CSTR X = Da/(1+Da) = 0.526; PFR X = 1 - e^(-Da) = 0.670. Both from the same Da; both in the goldens. 2. The profile is printed, not just the endpoint. nSteps 100; writeInterval 10; in operation {} — the Log tab shows the axial table, eleven rows from V = 0 to 5e-3 $m^3$, each component's molar flow falling or rising along the tube. The integrator is named (RK4, 100 uniform steps, dV = 5.0e-05 $m^3$); halve nSteps and see what moves. 3. The heat of reaction is a KPI here. dHrxn_kJ_per_mol = -2.10 (at 350 K) is computed from the species' formation enthalpies (the elements datum — the one base every reactor in Choupo prices on), and the tube's isothermal duty Q_kW = -0.196 follows from it and the extent. The real liquid esterification is only mildly exothermic (about -3 to -4 kJ/mol at 298 K, where Choupo reads -2.7). 4. Same Da, different X, is the pedagogical point of the pair. Put the two READMEs side by side; the numbers are the lecture.*

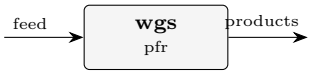
*Set nSteps 10 and rerun: X barely moves, because RK4 is fourth order and the profile is smooth. Then try pfr04_adiabatic, where the tube heats itself and T(V) becomes part of the profile.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| stream | feed | T | 350 |
| stream | reactorOut | T | 350 |
| kpi | reactor | Da_kTau | 1.10929132154 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -0.195860880897 |
| kpi | reactor | T | 350 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | X_limiting | 0.670207404765 |
| kpi | reactor | dHrxn_kJ_per_mol | -2.10412229473 |
| kpi | reactor | tau_s | 310.291329081 |
| boundary | global | H_feeds_kW | -104.131976653 |
| boundary | global | Q_boundary_kW | -0.195860880897 |
| *... and 5 more reference values in the case's `expected`.* | | | |

## pfr02_reversible_wgs

tutorials/steady/reactors/pfr02_reversible_wgs




**What it does:** Reversible water-gas shift (CO + H2O <→ CO2 + H2) in a PFR; approaches the same X_eq as the CSTR

**The story:** the reversible water-gas shift in a PLUG-FLOW reactor, the PFR sibling of cstr02_reversible_wgs.

Same reaction, same feed (equimolar CO + H2O at 600 K, 10 bar), same reverse term auto-derived from g_pure_ig. The pedagogical point: a reversible reaction relaxes toward the THERMODYNAMIC equilibrium conversion regardless of reactor type — with enough volume the PFR approaches the very same ceiling the CSTR hit,

K_eq(600 K) ~ 26.5 → X_eq ~ 0.84 (equimolar feed),

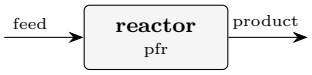
not the X = 1 a forward-only PFR would chase. V_R is taken large enough that the axial profile flattens onto X_eq well before the exit.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | feed | vf | 1 |
| stream | products | F | 0.000277777777778 |
| stream | products | vf | 1 |
| stream | feed | T | 600 |
| stream | products | T | 600 |
| kpi | wgs | Da_kTau | 0.00175028758879 |
| kpi | wgs | F_out_kmol_h | 1 |
| kpi | wgs | Q_kW | -4.55086758232 |
| kpi | wgs | T | 600 |
| kpi | wgs | V_R | 0.002 |
| kpi | wgs | X_limiting | 0.841431577908 |
| kpi | wgs | dHrxn_kJ_per_mol | -38.9410707335 |
| kpi | wgs | tau_s | 268.807168191 |
| *... and 7 more reference values in the case's expected.* | | | |

## pfr03_series_selectivity

tutorials/steady/reactors/pfr03_series_selectivity




**What it does:** PFR with the SAME series network A→B→C: no back-mixing → MORE of the intermediate B

**The story:** the SAME series network A → B → C as cstr03, now in a PFR. Compare them.

Multi-reaction grammar ‘reactions ( a_to_b b_to_c );‘; the PFR marches dF_i/dV = SUM_j nu_ij r_j(F) by RK4 – an IVP, no Newton. At the SAME residence time (tau = 225 s):

species CSTR PFR compA (feed) 40.8 % 14.1 % compB (want) 52.8 % 79.2 % <– the point compC (over) 6.4 % 6.8 %

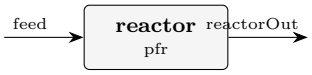
The PFR converts far more A AND keeps far more of the intermediate B: with no back-mixing, the freshly made B is not held in a vessel at low A concentration where the over-reaction B → C dominates. For positive-order kinetics the PFR beats the CSTR – the classic result a single-reaction reactor cannot show.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | product | F | 0.000176612457804 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 0.635804848095 |
| kpi | reactor | T | 350 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | k_a_to_b | 0.00214522777035 |
| kpi | reactor | k_b_to_c | 0.000536306942589 |
| kpi | reactor | nReactions | 2 |
| kpi | reactor | nSteps | 200 |
| kpi | reactor | tau_s | 225 |
| stream | feed | T | 350 |
| stream | product | T | 350 |
| kpi | reactor | T_out | 350 |

## pfr04_adiabatic

tutorials/steady/reactors/pfr04_adiabatic




**What it does:** ADIABATIC PFR: T(V) is a RESULT; the nearly thermoneutral liquid esterification warms the stream only ~6.6 K (replacement exothermic case planned)

**The story:** an ADIABATIC plug-flow reactor. The temperature is a RESULT, marched with the species by the same RK4:

dF_i/dV = SUM_j nu_ij r_j(F,T) dT/dV = SUM_j (-dH_j(T)) r_j / SUM_i F_i cp_i(T)

The esterification is only mildly exothermic in the liquid (dH_rxn about -2.7 kJ/mol after the acetic-acid correction, docs/design/acetic-acid-latent-heat-closure.md), so the stream warms by only 6.6 K (356.6 K out, the hot spot is the outlet). The Arrhenius feedback is present but weak. T(V) is in the axial profile: plot it.

The original teaching point of this case (positive feedback, a hot spot ~100 K above the feed) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed.

'thermalMode' takes 'isothermal' (default – T imposed, duty a result), 'adiabatic' (here), or 'heatExchange' (a jacket: U, areaPerVolume, T_coolant). A non-isothermal reactor needs the heat of reaction, so every reacting species must carry 'standardThermochemistry' – a fictitious toy species cannot be used.

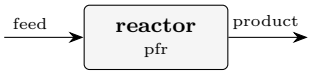
The feed is declared at 15 bar, a leftover of the old design (a 454 K product had to be held liquid); at the corrected ~357 K the outlet is liquid at 1 atm too.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | reactorOut | F | 0.000277777777778 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | Q_kW | -1.42108547152e-14 |
| kpi | reactor | T | 350 |
| kpi | reactor | T_max | 356.618955495 |
| kpi | reactor | T_min | 350 |
| kpi | reactor | T_out | 356.618955495 |
| kpi | reactor | V_R | 0.005 |
| kpi | reactor | dT_rise | 6.61895549509 |
| kpi | reactor | hotSpot_dT | 6.61895549509 |
| kpi | reactor | k_esterification_etac | 0.00357499942014 |
| kpi | reactor | nReactions | 1 |
| *... and 7 more reference values in the case's `expected`.* | | | |

## pfr05_cooled_multitubular

tutorials/steady/reactors/pfr05_cooled_multitubular




**What it does:** A COOLED MULTITUBULAR PFR: o-xylene oxidation in salt-cooled tubes – the hot spot is a RESULT, and the tube count is SOLVED against it

**The story:** A COOLED MULTITUBULAR PLUG-FLOW REACTOR.

Catalyst in N parallel tubes, molten salt in the shell. A strongly exothermic gas-phase oxidation heats each tube faster than the wall can take the heat away near the inlet, so the temperature RISES, peaks – the HOT SPOT – and falls as the reactant is used up. That peak is the number a designer of such a reactor watches: a few kelvin more at the inlet, or a few tenths of a kilopascal more o-xylene, and the peak runs away (the source's Table 4 gives the critical inlet pressures).

WHAT IS SOLVED. The bed is a one-dimensional pseudo-homogeneous plug flow (the source's model), marched by RK4 along the volume: the species, the enthalpy (T is read back from it), the pressure (Ergun, per tube) and the COOLANT's temperature, which here changes along the tubes by its own energy balance (coolant { arrangement ...; }). The wall coefficient is NOT typed: Li & Finlayson's correlation for a packed tube computes the bed side at the LOCAL gas, in series with the tube wall and the salt's film. system/outerDict then SOLVES the number of tubes for a hot spot at its limit – see there for why fewer tubes run COOLER here.

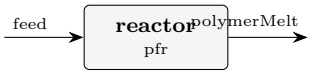
WHERE EACH NUMBER COMES FROM. Zuluaga-Botero, Dobrosz-Gomez & Gomez-Garcia, Catalysts 10 (2020) 626 (CC BY 4.0), Table 2: tube diameter 0.025 m, tube length 2 m, 3000 tubes (the starting guess here), particle 2 mm, bed porosity 0.5, bulk density 1300 kg/m3, 101.325 kPa, coolant 72 000 kg/h of cp 0.3105 kcal/(kg K) entering at 625 K; Table 3 the rate. AUTHOR-SET, none cited: the o-xylene inlet pressure (0/feed), the tube wall (1.65 mm of a 16 W/(m K) steel), the salt's film (1000 W/(m2 K)), the spherical pellet, and the hot-spot limit in system/outerDict. The source's own overall U, 82.7 kcal/(m2 h K) = 96.2 W/(m2 K), is NOT used: compare it with the U_in / U_out KPIs this run computes.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0639658783333 |
| stream | feed | T | 625 |
| stream | product | F | 0.0639658783333 |
| stream | product | T | 655.871241388 |
| kpi | reactor | A_bed_m2 | 1.90026269868 |
| kpi | reactor | A_wall_m2 | 608.084063579 |
| kpi | reactor | F_in_kmol_h | 230.277162 |
| kpi | reactor | F_out_kmol_h | 230.277162 |
| kpi | reactor | GHSV_normal_h | 1358.08204294 |
| kpi | reactor | G_tube_kg_m2_s | 1.00201918836 |
| kpi | reactor | L_tube_m | 2 |
| kpi | reactor | Nu_wall_in | 36.5561173269 |
| kpi | reactor | P_out_bar | 0.772127318262 |
| kpi | reactor | Pr_wall_in | 0.776054107796 |
| *... and 80 more reference values in the case's `expected`.* | | | |

## pfr_polyesterification

tutorials/steady/reactors/pfr_polyesterification




**What it does:** PFR step-growth polyesterification — Carothers/Flory chain statistics

**The story:** Step-growth polyesterification in a PFR. A molten AB-type hydroxy-acid monomer self-condenses to polycaprolactone, releasing water. As the monomer conversion p rises along the reactor, the Carothers relation Xn = 1/(1-p) makes the average chain length climb steeply, and the polydispersity PDI = 1+p approaches 2 (the most-probable distribution).

Try a 'sweep' over operation.V_R and watch Xn, Mn and PDI climb and the Flory-Schulz distribution broaden – the whole story in one glass-box run.

*A molten AB-type hydroxy-acid monomer (6-hydroxyhexanoic acid) self- condenses to polycaprolactone in a plug-flow reactor, releasing one water per ester bond:*

*As the monomer conversion p climbs along the reactor, the Carothers relation makes the average chain length rise steeply and the polydispersity approach 2 — the most-probable (Flory–Schulz) distribution.*

*and emits the Flory–Schulz distribution as a profile (chainLength vs n_x, w_x) so the GUI plots it: n(x)=(1-p)p^(x-1) decays monotonically, w(x)=x(1-p)²p^(x-1) peaks near x≈Xn.*

*Run a sweep over operation.V_R (reactor volume → residence time → p) and watch PDI climb toward 2, Xn and Mn shoot up, and the weight distribution broaden. The whole Carothers/Flory story in one glass-box run.*

*Lives inside the reaction. Absent ? the reactor behaves as an ordinary PFR. p is taken from the reactor's own conversion of limitingReactant (single source of truth). The distribution + PDI = 1+p are valid because a PFR has a uniform residence time; on a CSTR the residence-time distribution broadens both, so the distribution is gated to PFR/batch only.*

*- The chain-statistics math is exact and standard: P.J. Flory, ∗Principles of Polymer Chemistry∗, Cornell Univ. Press (1953), Ch. VIII–IX (the most-probable distribution and PDI = 1+p); G. Odian, ∗Principles of Polymerization∗, 4th ed., Ch. 2; H.S. Fogler, ∗Elements of Chemical Reaction Engineering∗, 5th ed., ?9. - The monomer/repeat component properties are ILLUSTRATIVE (clearly labelled in each .dat) — plausible round numbers for a teaching case, not curated literature values. The Arrhenius rate is illustrative, tuned so Da = kτ ≈ 4 gives p ≈ 0.98. Curate these before any quantitative use.*

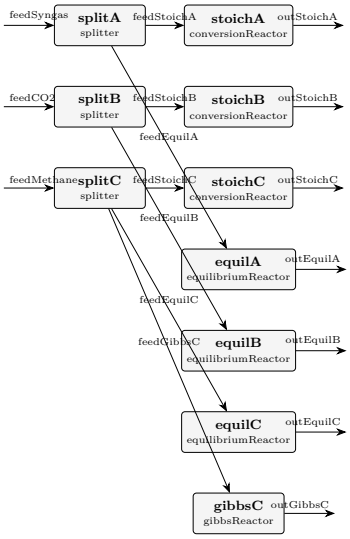
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | polymerMelt | F | 0.000550395491925 |
| kpi | reactor | Da_kTau | 3.98587252785 |
| kpi | reactor | F_out_kmol_h | 1.98142377093 |
| kpi | reactor | Mn_kg_kmol | 6144.41174045 |
| kpi | reactor | Mw_kg_kmol | 12174.6834809 |
| kpi | reactor | PDI | 1.98142377093 |
| kpi | reactor | T | 450 |
| kpi | reactor | V_R | 0.2 |
| kpi | reactor | X_limiting | 0.981423770929 |
| kpi | reactor | Xn | 53.8322388335 |
| kpi | reactor | Xw | 106.664477667 |
| kpi | reactor | p_conversion | 0.981423770929 |
| kpi | reactor | tau_s | 5829.95951417 |

*... and 11 more reference values in the case's `expected`.*

## screen01_methanol_routes

tutorials/steady/reactors/screen01_methanol_routes




**What it does:** Screening three routes to methanol on paper: stoichiometry, equilibrium, heat and selectivity

**The story:** SCREENING ROUTES TO A PRODUCT, ON PAPER.

Three known routes to methanol, each fed at its own stoichiometric ratio at 525 K and 50 bar, ideal gas:

A synthesis gas CO + 2 H2 → CH3OH B CO2 hydrogenation CO2 + 3 H2 → CH3OH + H2O C direct oxidation 2 CH4 + O2 → 2 CH3OH

Each route is asked three questions, by three reactors on one split of its feed, none of which needs a rate law:

stoich<X> conversionReactor, X = 1: the heat of reaction the route would have to remove (KPI dHrxn_kJ_per_mol). equil<X> equilibriumReactor over the route's OWN reaction: how far thermodynamics lets it go at this temperature and pressure. gibbsC gibbsReactor over EVERY species the case carries, on route C's feed: where the atoms go when nothing stops them at methanol. Route C passes the equilibrium test with its own reaction (K is enormous) and fails this one – the product is not where its Gibbs energy is lowest, so the route stands or falls on a catalyst's SELECTIVITY, which no paper screen can supply. (Routes A and B have the same weakness, at a milder scale: equil02 shows syngas going to methane when methanation is allowed.)

The economic and atom-economy filters need only molar masses and prices; the EduTool computes them from the catalogue's MW values (its test holds the table to the records) and from prices the student sets. NO PRICE IS DECLARED HERE: this repository curates none, and a number invented for a price would be a falsehood nobody could detect.

EduTool: gui/src/ui/methods/RouteScreeningTool.tsx.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedCO2 | F | 0.00111111111111 |
| stream | feedCO2 | T | 525 |
| stream | feedEquilA | F | 0.000416666666667 |
| stream | feedEquilA | T | 525 |
| stream | feedEquilB | F | 0.000555555555556 |
| stream | feedEquilB | T | 525 |
| stream | feedEquilC | F | 0.000277777777778 |
| stream | feedEquilC | T | 525 |
| stream | feedGibbsC | F | 0.000277777777778 |
| stream | feedGibbsC | T | 525 |
| stream | feedMethane | F | 0.000833333333333 |
| stream | feedMethane | T | 525 |
| stream | feedStoichA | F | 0.000416666666667 |
| stream | feedStoichA | T | 525 |
| *... and 126 more reference values in the case's `expected`.* | | | |

## rotating

**The theory you need.** Compressors, turbines, pumps: specified by shaft work $W_s$ and isentropic efficiency $\eta_s$; the outlet state follows from $h_{\text{out}} = h_{\text{in}} + W_s/\dot{n}$ with the isentropic reference path. Outlet pressure is a RESULT (hardware, not a spec).

## compressor01_air

tutorials/steady/rotating/compressor01_air




**What it does:** Adiabatic air compressor: 10 kW shaft, eta 0.75 — P_out and T_out are RESULTS

**The story:** The simplest credo-pure rotating-equipment case. An adiabatic compressor takes ambient air (1 bar, 25 degC) and a fixed shaft power of 10 kW with isentropic efficiency 0.75.

Hardware (the only things you specify): W_shaft = 10 kW motor rating delivered to the gas eta = 0.75 isentropic efficiency (machine characteristic)

RESULTS the simulator computes: P_out the discharge pressure the machine reaches T_out the discharge temperature (real, with losses) ratio P_out / P_in T_out_isen isentropic discharge T (diagnostic) dS_gen entropy generated by the irreversibility

Hand check (ideal gas, Cp ~ 29.2 J/(mol*K), gamma ~ 1.40): F = 3.6 kmol/h = 1 mol/s w_real = 10000 W / 1 mol/s = 10000 J/mol w_isen = 0.75 * 10000 = 7500 J/mol T_out_isen = 298.15 + 7500/29.2 ~ 555 K ratio = (T_isen/T_in)^(gamma/(gamma-1)) = (555/298)^3.5 ~ 8.8 T_out = 298.15 + 10000/29.2 ~ 640 K

"I need exactly 8 bar at discharge" is a DESIGN question — it goes in a DesignSpec that manipulates $W_shaft with target compressor.P_out.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.001 |
| stream | out | F | 0.001 |
| kpi | compressor01 | F | 0.001 |
| kpi | compressor01 | P_in | 100000 |
| kpi | compressor01 | P_out | 890211.642995 |
| kpi | compressor01 | T_in | 298.15 |
| kpi | compressor01 | T_out | 633.374206884 |
| kpi | compressor01 | T_out_isen | 551.37943898 |
| kpi | compressor01 | W_isen | 7500 |
| kpi | compressor01 | W_isen_kW | 7.5 |
| kpi | compressor01 | W_shaft | 10000 |
| kpi | compressor01 | W_shaft_kW | 10 |
| kpi | compressor01 | dS_gen | 4.22615345389 |
| kpi | compressor01 | dT | 335.224206884 |
| *... and 13 more reference values in the case's `expected`.* | | | |

## compressor02_designspec_pout

tutorials/steady/rotating/compressor02_designspec_pout




**What it does:** Air compressor sized to hit P_out = 8 bar exactly, via DesignSpec on $W_shaft

**The story:** Same air compressor as compressor01, but now answering the DESIGN question: "what shaft power do I need to discharge at exactly 8 bar?"

The credo forbids specifying P_out on the unit op (it's a result). Instead the outer DesignSpec manipulates the hardware variable $W_shaft until the computed compressor.P_out hits the 8-bar target. This is the rotating-equipment analogue of sizing an evaporator's area to hit a product concentration.

Converges to W_shaft ~ 9.36 kW (slightly below compressor01's 10 kW, which over-shot to 8.90 bar). At the solution T_out ~ 612 K (339 degC) and W_isen ~ 7.02 kW (eta * W_shaft, eta = 0.75).

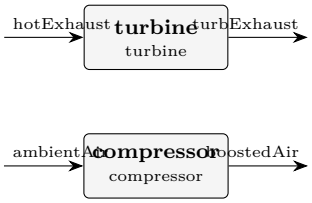
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.001 |
| stream | out | F | 0.001 |
| kpi | compressor | F | 0.001 |
| kpi | compressor | P_in | 100000 |
| kpi | compressor | P_out | 800004.793643 |
| kpi | compressor | T_in | 298.15 |
| kpi | compressor | T_out | 612.41049696 |
| kpi | compressor | T_out_isen | 535.384868081 |
| kpi | compressor | W_isen | 7017.30522226 |
| kpi | compressor | W_isen_kW | 7.01730522226 |
| kpi | compressor | W_shaft | 9356.40696301 |
| kpi | compressor | W_shaft_kW | 9.35640696301 |
| kpi | compressor | dS_gen | 4.08116534481 |
| kpi | compressor | dT | 314.26049696 |

*... and 13 more reference values in the case's `expected`.*

## compressorTurbine01_turbo_booster

tutorials/steady/rotating/compressorTurbine01_turbo_booster




**What it does:** Turbo-booster: turbine shaft drives a compressor (energy wire)

**The story:** turbo-booster: turbine drives compressor through an ENERGY WIRE (v0.80)

Two physically independent gas streams, one shared shaft:

hotExhaust → turbine —-shaft—-+

ambientAir → compressor <———+

Hardware specs (the only knobs): turbine W_shaft = -5 kW (NEGATIVE: work extracted) eta = 0.80 compressor W_shaft is OMITTED (the wire provides it) eta = 0.75

The wire injects +W_turb into the compressor before its solve(), so the compressor's W_shaft is whatever the turbine produces — here, 5 kW (sign flipped: -W_shaft on the producer's expression → the consumer reads +W_shaft, the usual sign convention).

Hand check (ideal gas, Cp ~ 29 J/(mol∗K), gamma ~ 1.40): Turbine: 1 mol/s, 800 K, 5 bar → extract 5 kW (= 5000 J/mol) w_isen = w_real / eta = 5000 / 0.80 = 6250 J/mol T_isen = 800 - 6250/29 ~ 585 K → ratio = (585/800)^3.5 ~ 0.35 P_out ~ 5 ∗ 0.35 ~ 1.74 bar T_out (real) = 800 - 5000/29 ~ 627 K (warmer than T_isen: the losses stay as heat in the gas)

Compressor: 0.5 mol/s, 298 K, 1 bar, receiving 5 kW w_real = 5000 / 0.5 = 10000 J/mol w_isen = 0.75 ∗ 10000 = 7500 J/mol T_isen = 298 + 7500/29 ~ 557 K → ratio = (557/298)^3.5 ~ 8.7 P_out ~ 8.7 bar T_out (real) = 298 + 10000/29 ~ 643 K

The wire makes the energy flow visible at the case level — a single edge in the flowsheet that the GUI can render dashed.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | ambientAir | F | 0.0005 |
| stream | boostedAir | F | 0.0005 |
| stream | hotExhaust | F | 0.001 |
| stream | turbExhaust | F | 0.001 |
| kpi | compressor | F | 0.0005 |
| kpi | compressor | P_in | 100000 |
| kpi | compressor | P_out | 890211.642995 |
| kpi | compressor | T_in | 298.15 |
| kpi | compressor | T_out | 633.374206884 |
| kpi | compressor | T_out_isen | 551.37943898 |
| kpi | compressor | W_isen | 3750 |
| kpi | compressor | W_isen_kW | 3.75 |
| kpi | compressor | W_shaft | 5000 |
| kpi | compressor | W_shaft_kW | 5 |

*. . . and 27 more reference values in the case's* `expected`*.*

## pump01_water

tutorials/steady/rotating/pump01_water




**What it does:** Water pump: 2 kW shaft, eta 0.65 — incompressible, P_out / T_out are RESULTS

**The story:** An incompressible-liquid pump. Liquid water enters at 1 bar, 298 K; the pump shaft delivers 2 kW with efficiency 0.65.

Hardware (the only OP parameters): W_shaft = 2 kW motor power delivered to the liquid eta = 0.65 pump efficiency

Because the liquid is incompressible (v ~ const), the discharge is closed-form — no EoS, no Newton: w_real = W_shaft / F_mol_s w_isen = eta * w_real (the useful v*dP part) dP = w_isen / v_molar P_out = P_in + dP (RESULT) dT = (1 - eta) * w_real / Cp_liq (dissipation)

Hand check (water: v ~ 1.8e-5 m3/mol, Cp_liq ~ 75 J/(mol*K)): F = 36 kmol/h = 10 mol/s w_real = 2000 W / 10 mol/s = 200 J/mol w_isen = 0.65 * 200 = 130 J/mol dP = 130 / 1.8e-5 ~ 7.2e6 Pa = 72 bar... (so P_out ~ 73 bar) dT = 0.35 * 200 / 75 ~ 0.9 K (tiny — pumps barely heat)

"Pump to exactly 50 bar" is a DesignSpec on $W_shaft targeting pump.P_out (converges in one step — the relation is linear).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.01 |
| stream | out | F | 0.01 |
| kpi | pump01 | F | 0.01 |
| kpi | pump01 | P_in | 100000 |
| kpi | pump01 | P_out | 7294244.60432 |
| kpi | pump01 | T_in | 298.15 |
| kpi | pump01 | T_out | 299.077152318 |
| kpi | pump01 | W_hydraulic | 1300 |
| kpi | pump01 | W_shaft | 2000 |
| kpi | pump01 | W_shaft_kW | 2 |
| kpi | pump01 | dP | 7194244.60432 |
| kpi | pump01 | dT | 0.927152317881 |
| kpi | pump01 | eta | 0.65 |
| kpi | pump01 | head_m | 735.848502288 |

*... and 12 more reference values in the case's `expected`.*

## pump02_pressure_spec

tutorials/steady/rotating/pump02_pressure_spec




**What it does:** Water pump specified by DISCHARGE PRESSURE (P_out 30 bar) — the shaft power W_shaft is the RESULT (closed-form inverse of pump01)

**The story:** The SAME incompressible-liquid pump as pump01, specified the OTHER way: by the discharge pressure it must reach. In a real plant you rarely know a pump's shaft power up front — you know the HEADER it feeds (here 30 bar). So you state P_out and the shaft power is the RESULT.

This is NOT a DesignSpec. For an incompressible liquid the pump relation is invertible in CLOSED FORM,

$\Delta P = \text{eta} \cdot \text{W_shaft} / \text{Q_vol} <\!\!\rightarrow \text{W_shaft} = \text{Q_vol} \cdot \Delta P / \text{eta}$ ,

with Q_vol = F · v set by the feed. Give P_out, get W_shaft — one shot, no Newton, no outer loop. (pump01 gives W_shaft and gets P_out; the two cases are exact inverses of one another.)

Specify EXACTLY ONE of { W_shaft | P_out | dP }. A pressure let-down (P_out < P_in) is not a pump — use a 'valve'.

Hand check (water: v ~ 1.8e-5 m3/mol): F = 36 kmol/h = 10 mol/s Q_vol = 10 ∗ 1.8e-5 = 1.8e-4 m3/s dP = 30 - 1 bar = 2.9e6 Pa W_shaft = Q_vol ∗ dP / eta = 1.8e-4 ∗ 2.9e6 / 0.65 ~ 803 W ~ 0.80 kW

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.01 |
| stream | out | F | 0.01 |
| kpi | pump02 | F | 0.01 |
| kpi | pump02 | P_in | 100000 |
| kpi | pump02 | P_out | 3000000 |
| kpi | pump02 | T_in | 298.15 |
| kpi | pump02 | T_out | 298.523735099 |
| kpi | pump02 | W_hydraulic | 524.03 |
| kpi | pump02 | W_shaft | 806.2 |
| kpi | pump02 | W_shaft_kW | 0.8062 |
| kpi | pump02 | dP | 2900000 |
| kpi | pump02 | dT | 0.373735099338 |
| kpi | pump02 | eta | 0.65 |
| kpi | pump02 | head_m | 296.620531272 |

*. . . and 12 more reference values in the case's `expected`.*

### turbine01_air

tutorials/steady/rotating/turbine01_air




**What it does:** Adiabatic gas turbine: extract 10 kW from hot air, eta 0.85 — P_out / T_out are RESULTS

**The story:** The mirror of compressor01. Hot compressed air (10 bar, 900 K) drives an adiabatic turbine; the shaft extracts 10 kW with isentropic efficiency 0.85.

Hardware (the only things you specify): W_shaft = -10 kW shaft power extracted (NEGATIVE = out) eta = 0.85 isentropic efficiency

Sizing note: at 1 mol/s and 900 K inlet, the enthalpy available down to ambient bounds the extractable work to ~15 kW. The 10 kW asked for here expands the gas to ~1.32 bar (ratio ~7.6); asking for much more (e.g. 30 kW) would drive the isentropic discharge below 0 K — physically impossible, and the solver says so.

RESULTS the simulator computes (this run): P_out ~1.32 bar the discharge pressure the gas expands to T_out ~583 K the (real) exhaust temperature T_out_isen ~524 K the ideal (isentropic) discharge T ratio ~7.59 P_in / P_out (expansion ratio) W_generated +10 kW what the shaft delivers to the grid

Per the credo, P_out is a RESULT. If the turbine must exhaust at a fixed back-pressure (a condenser, the atmosphere), wrap the case in a DesignSpec that manipulates $W_shaft until turbine.P_out hits it.

**What to expect (golden-locked):**

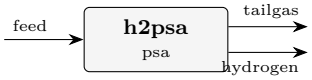
| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | exhaust | F | 0.001 |
| stream | hotAir | F | 0.001 |
| kpi | turbine01 | F | 0.001 |
| kpi | turbine01 | P_in | 1000000 |
| kpi | turbine01 | P_out | 131729.126973 |
| kpi | turbine01 | T_in | 900 |
| kpi | turbine01 | T_out | 582.935206477 |
| kpi | turbine01 | T_out_isen | 524.463853734 |
| kpi | turbine01 | W_gen_kW | 10 |
| kpi | turbine01 | W_generated | 10000 |
| kpi | turbine01 | W_shaft | -10000 |
| kpi | turbine01 | dS_gen | 3.19034018436 |
| kpi | turbine01 | dT | -317.064793523 |
| kpi | turbine01 | dT_isen | -375.536146266 |

*... and 9 more reference values in the case's `expected`.*

## separation

## psa01_h2_psa

tutorials/steady/separation/psa01_h2_psa




**What it does:** H2-PSA: 20-bar syngas (H2/CO2/CH4/CO/N2) over zeolite 5A → high-purity H2 raffinate + CO2-rich tail gas (equilibrium cyclic-steady-state shortcut)

**The story:** the canonical first PSA case: hydrogen recovery from steam-methane-reformer / shifted syngas.

Feed: a typical PSA feed gas (mol fractions) H2 0.70 (the LIGHT product → raffinate) CO2 0.18 (the strongest adsorbate → tail gas) CH4 0.05 CO 0.04 N2 0.03 F = 1000 kmol/h, 20 bar, 313 K.

Adsorbent: zeolite 5A (Ca-LTA), the classic H2-PSA sieve. CO2 » CH4 > CO ~ N2 » H2 on the competitive (extended) Langmuir isotherm, so the heavy impurities load onto the bed at P_high (20 bar) and are blown down with the tail at P_low (1.2 bar), while H2 barely adsorbs and sweeps through as the high-pressure raffinate product.

Operation: P_high 20 bar / P_low 1.2 bar swing; isothermal 313 K bed; purgeRatio 0.15 (a Skarstrom-style countercurrent light-product sweep that cleans the regeneration at the cost of product); eta 0.80 (the equilibrium-utilisation / LUB derating — the equilibrium ceiling is eta = 1); bedCapacity 2.0 kg adsorbent per mol of feed throughput (the lumped throughput knob; cycle time / bed volume are deliberately collapsed into it by the equilibrium shortcut).

Expected (this model, golden-checked): H2 recovery ~ 74 % (single train, with the 0.15 purge sweep) H2 purity ~ 98-99 % in the raffinate tail gas CO2-enriched ~ 18 % → ~ 38 %

The run header prints, out loud, what the equilibrium shortcut does NOT model (breakthrough / cycle time, thermal swing, blowdown gas loss, hydraulics, isotherm-validity window) — "all models are wrong, some useful".

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.277777777778 |
| stream | hydrogen | F | 0.145276064457 |
| stream | tailgas | F | 0.13250171332 |
| kpi | h2psa | Dq_CH4 | 0.0260273101613 |
| kpi | h2psa | Dq_CO | 0.0367785976251 |
| kpi | h2psa | Dq_CO2 | 1.74280135323 |
| kpi | h2psa | Dq_H2 | 0.0186248522203 |
| kpi | h2psa | Dq_N2 | 0.00987823175641 |
| kpi | h2psa | eta_used | 0.8 |
| kpi | h2psa | pressureRatio | 16.6666666667 |
| kpi | h2psa | productivity | 0.0715324198173 |
| kpi | h2psa | purgeRatio | 0.15 |
| kpi | h2psa | purity_H2 | 0.984779152498 |
| kpi | h2psa | recovery_H2 | 0.735762032407 |
| *... and 11 more reference values in the case's `expected`.* | | | |

## tsa01_co2_twin_bed

tutorials/steady/separation/tsa01_co2_twin_bed




**What it does:** Twin-bed TSA shortcut: continuous CO2 removal from helium by alternating adsorption at 298 K and regeneration at 398 K

**The story:** cycle-averaged twin-bed thermal swing.

One column adsorbs while the other regenerates. tCycle is the switch interval and therefore the adsorption duration of one column. The unit refuses a bed that cannot hold every adsorbate mole arriving in that time.

This teaching case uses a CASE-LOCAL synthetic property anchor: cpSolid = 920 J/(kg.K), explicitly assumed for this numerical gate. It is not promoted to the global zeolite13X catalogue.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | cleanHe | F | 0.0002475 |
| stream | regenerationGas | F | 3.02777777778e-05 |
| kpi | co2tsa | recovery_He | 0.9 |
| kpi | co2tsa | purity_He | 1 |
| kpi | co2tsa | Dq_CO2 | 0.408156018947 |
| kpi | co2tsa | capacity_utilisation_CO2 | 0.680567638068 |
| kpi | co2tsa | Q_sensible_per_cycle_kJ | 276 |
| kpi | co2tsa | Q_desorption_per_cycle_kJ | 37.5 |
| kpi | co2tsa | Q_regeneration_kW | 1.045 |
| kpi | co2tsa | Q_adsorption_cooling_kW | -1.045 |
| stream | cleanHe | T | 298 |
| stream | feed | T | 298 |
| stream | regenerationGas | T | 298 |
| *... and 22 more reference values in the case's `expected`.* | | | |

### `tsa02_refused_capacity`

`tutorials/steady/separation/tsa02_refused_capacity`




**What it does:** TSA refusal lesson: one undersized column cannot hold the CO2 loaded during the declared switch interval

## solids

**The theory you need.** Solids carry mass and enthalpy without VLE: crystallisers, dryers, cyclones, gas–solid splitters. A dryer's air is a REAL stream (two inlets, adiabatic mixing, outlet humidity from the psychrometrics); a cyclone's cut size comes from the selected model (Lapple…Muschelknautz). Expect solid streams flagged separately ($F_{\text{solid}}$) and energy balances that close across phase change.

## pneumaticConveyor01_sand

`tutorials/steady/solids/pneumaticConveyor01_sand`

**What it does:** Dilute-phase pneumatic conveying of sand in nitrogen: the glass-box dP breakdown + the saltation-margin safety check. Particle velocity from a mechanistic FORCE BALANCE (drag = wall friction + gravity); solids friction from Yang (1974, AIChE J 20:605); Rizk (1973) saltation. Loading ~2 kg/kg (in the Yang voidage range); solids friction DOMINATES the dP – conveying solids IS a pressure-drop problem. u_g/u_salt = 1.72 (safe margin above plugging).

**The story:** / flowsheetDict (single unit)

Operation block is HARDWARE only (the credo): pipe diameter, length, rise. Every velocity, the suspended density and the pressure drop are RESULTS.

*A single pneumaticConveyor unit: silica sand (~200 μm) carried by nitrogen up a 100 mm line, 50 m long, with a 10 m vertical rise.*

*The total conveying-line pressure drop as a sum of physical contributions (glass-box) — Rhodes, ∗Introduction to Particle Technology∗, Ch. 8:*

*- superficial gas velocity u_g = 21.9 m/s, - particle velocity u_p = 21.1 m/s (Hinkle slip), - single-particle terminal velocity u_t = 1.03 m/s, - saltation velocity u_salt = 3.6 m/s (Rizk) — u_g is well above it, so transport is stable. The conveyor warns aloud (verbosity ≥ 1) if u_g ever falls below u_salt (risk of solids settling and blocking the line) — a first-class, announced design check, never a silent clip.*

*Solids loading = 0.20 kg solid / kg gas (dilute phase), suspended-solids density ?∗ = 0.47 kg/m$^3$.*

*- particle velocity: Hinkle u_p = u_g(1 - 0.68 d_p^0.92 ?_p^0.5 ?_g^-0.2 D^-0.54), with an honest u_g - u_t fallback if the correlation leaves the dilute regime; - gas friction: Fanning (Blasius turbulent / 16/Re laminar); - solids friction: Hinkle f_p = (3/8)(?_g/?_p) C_D (D/d_p)((u_g-u_p)/u_p)$^2$, C_D from Schiller–Naumann at the slip Reynolds number; - saltation: Rizk (1973).*

*Operation block is hardware only (the credo): pipe diameter D, length L, vertical rise dz. Every velocity, the suspended density and the pressure drop are results. Particle density comes from the solid component's solid { rho_p; } (silica = 2650 kg/m$^3$); the size from the inlet PSD (Sauter mean for the drag). The carrier-gas viscosity needs a transport { model Chung; } block in the thermoPhysPropDict (for the friction Reynolds numbers).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Feed | F | 0.0268345909167 |
| stream | conveyed | F | 0.0268345909167 |
| kpi | node | P_out | 234394.335965 |
| kpi | node | accelerationLength | 16.8207521299 |
| kpi | node | deltaP | 65605.6640351 |
| kpi | node | deltaP_bends | 0 |
| kpi | node | deltaP_gasAccel | 361.754574941 |
| kpi | node | deltaP_gasFriction | 1492.10736304 |
| kpi | node | deltaP_solidsAccel | 1383.86777284 |
| kpi | node | deltaP_solidsFriction | 61751.3873821 |
| kpi | node | deltaP_static | 616.546942187 |
| kpi | node | nBends | 0 |
| kpi | node | particleWallFriction | 0.4 |
| kpi | node | rho_gas | 3.39180174393 |

*... and 15 more reference values in the case's `expected`.*

### pneumaticConveyor02_bends

tutorials/steady/solids/pneumaticConveyor02_bends

**What it does:** A SHORT (8 m) conveying line routed through 6 bends – the lesson: A FITTING ADDS PRESSURE DROP WITHOUT ADDING LENGTH. With long-radius sweeps the 6 bends cost ~10% of the total dP; swap them for blind tees (type blindTee;) and their share jumps to ~25% – the erosion-vs-pressure trade made visible. (On this short line the straight-pipe solids friction still carries most of the dP; the bends' share GROWS as lines get shorter and bendier.) Same gas/solids as pneumaticConveyor01 (which runs 30 m / 6 m rise, bend-free) – compare the PER-METRE dP, not the totals.

**The story:** a fitting adds pressure drop without adding length

The same sand-in-N2 service as pneumaticConveyor01 (loading ~2), on a SHORTER line: 100 mm, 8 m, 2 m rise, routed through 6 long-radius bends. (Case 01 runs 30 m / 6 m rise, bend-free – compare dP PER METRE, not totals.) Each bend decelerates the solids to ~0.75 u_p; the gas re-accelerates them downstream – the same momentum work as the line entry, once per bend: dP_bend = mDotS (u_p - u_p,bend) / A. Run both cases and compare the dP breakdown: the bends add ~10% (long-radius) to ~25% (blind tees) of the dP while adding NO length. Swap a bend 'type longRadius;' for 'type blindTee;' and watch its re-acceleration cost jump (ratio 0.75 → 0.25) – the blind tee trades pressure for near-zero erosion. Solids friction: Yang (1974); particle velocity: force balance; Rizk saltation.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Feed | F | 0.0268345909167 |
| stream | conveyed | F | 0.0268345909167 |
| kpi | node | P_out | 279210.165227 |
| kpi | node | accelerationLength | 15.5453372639 |
| kpi | node | deltaP | 20789.8347732 |
| kpi | node | deltaP_bends | 2070.63576095 |
| kpi | node | deltaP_gasAccel | 361.754574941 |
| kpi | node | deltaP_gasFriction | 397.89529681 |
| kpi | node | deltaP_solidsAccel | 1380.42384063 |
| kpi | node | deltaP_solidsFriction | 16373.2628921 |
| kpi | node | deltaP_static | 205.862407813 |
| kpi | node | nBends | 6 |
| kpi | node | particleWallFriction | 0.4 |
| kpi | node | rho_gas | 3.39180174393 |
| *... and 15 more reference values in the case's `expected`.* | | | |

## thermo

**The theory you need.** Consistency tests interrogate DATA, not models: the van Ness point test checks measured $(x, y, T, P)$ sets against the Gibbs–Duhem constraint (residuals in $\ln(\gamma_1/\gamma_2)$); Herington's area test is its integral cousin. A dataset that fails is not "bad" – the test says WHERE and HOW MUCH, and the fit that follows should weight accordingly. Expect plain-language verdicts with severity, not a pass/fail stamp. *Full derivation: Theory Guide, the consistency chapter of the Props Guide.*

## air01_heater_expand

tutorials/steady/thermo/air01_heater_expand




**What it does:** Predefined mixture: heat a stream fed with 'components ( air )', which expands to N2/O2/Ar at load time

**The story:** showcase for TRACK A predefined-mixture expansion.

The thermoPhysPropDict lists 'components ( air )'. At load time the loader splices 'air' into N2 + O2 + Ar (see constant/thermoPhysPropDict). A single heater then adds 5 kW to a 1 mol/s air stream at 1 bar, 298.15 K. The air is gas (vf = 1), so the heater integrates the ideal-gas mixture H_ig over the three real components → a correct MULTICOMPONENT result.

Z-SEEDING NOTE (glass-box honesty): The expansion captures the member mole fractions (N2 0.7808 / O2 0.2095 / Ar 0.0093) as a per-component seed on the ThermoPackage ('mixtureSeedVector()'). The feed below STATES the same fractions explicitly so the case is self-evident and the result is unambiguous – a stream that declares its own composition always wins over the seed. The seed exists so a future feed that OMITS its composition can be filled from the mixture defaults; that wiring lives in the stream-construction path (Flowsheet / Composition), not in this case.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.001 |
| stream | hot | F | 0.001 |
| kpi | airHeater | F | 0.001 |
| kpi | airHeater | P | 100000 |
| kpi | airHeater | Q | 5000 |
| kpi | airHeater | Q_kW | 5 |
| kpi | airHeater | Q_per_mol | 5000 |
| kpi | airHeater | T_in | 298.15 |
| kpi | airHeater | T_out | 468.478847995 |
| kpi | airHeater | dT | 170.328847995 |
| utility | airHeater | heating.steamHP.duty_kW | 5 |
| utility | airHeater | heating.steamHP.T | 468.478847995 |
| utility | airHeater | heating.steamHP.kg_s | 0.00291715285881 |
| utility | airHeater | heating.steamHP.eur_h | 0.324 |

*... and 6 more reference values in the case's `expected`.*

## basis01_two_unit_chain

tutorials/steady/thermo/basis01_two_unit_chain




**What it does:** Basis reconciliation, the two-unit chain: a speciating flash hands its brine to a unit running a DIFFERENT thermo world, which transports the species basis as matter instead of re-asking the chemistry

**The story:** THE TWO-UNIT CHAIN (basis-reconciliation spike S2, ratified 2026-08-03).

Topology only; state lives in 0/.

feed -→ [speciator: isothermalFlash] -→ offgas

brine (carries BOTH bases: the apparent components | that ARE the state, and the species the | network resolved in them) v [transporter: heater, LOCAL molecular thermo world]

WHAT THE CHAIN IS FOR. The ‘speciator‘ runs the case’s electrolyte world and SOLVES the chemistry: its brine outlet carries a speciation block stamped ‘origin "solved:speciator"‘. The ‘transporter‘ runs a DIFFERENT world – a plain molecular gammaPhi declared in its own ‘thermo {}‘ block, which resolves no ions at all. It changes T (a duty) and nothing else: one material in, one material out, F and z untouched.

The question the chain asks: what happens to the species basis at that boundary?

Three answers were available and only one is honest. (a) Re-derive the block with the GLOBAL package. That is a chemistry the unit’s own world does not do, attributed to the unit – silent respeciation across a model boundary. This is what the engine did before the spike, and it is what the transport contract now stops. (b) Drop the block. The outlet then reports one basis where its inlet reported two, and a student comparing them sees chemistry vanish across a heater. (c) CARRY it. The heater moved no atoms and no charge; the ions in the brine are the same ions, in the same amounts, at a higher T. The block crosses as MATTER, stamped ‘origin "transported:transporter"‘ so the reader can see it was carried and not re-solved.

(c) is the contract. ‘converged/warmBrine‘ is where it is visible.

AND THE PART THAT TOOK A SECOND OPINION TO GET RIGHT. A heater is a heater: the outlet is at 330.81 K while the speciation it carries was solved at 313.15 K. The first version of this case stamped ‘solvedAtT‘ and considered the matter closed – the file said where the block came from, so the reader could not be misled. That was wrong, and the counsel of 2026-08-03 named why:

AN EQUILIBRIUM IS NOT A COMPOSITION. It is a RELATION between a composition and a thermodynamic state. A composition solved at 313 K does not become the equilibrium composition at 331 K by being carried there with a note saying where it came from. Stamping the provenance makes the datum TRACEABLE; it does not make it VALID.

So the two claims are now separate, and only one of them travels:

the AMOUNTS are a material inventory – the transport conserved every species exactly, and they stand; the EQUILIBRIUM is a claim about this state, and it dies here.

‘converged/warmBrine‘ therefore carries ‘equilibriumValidHere false‘, both temperatures, and NO pH – the pH is WITHHELD rather than inherited, because inheriting it would publish a number nothing here computed. The same withholding applies to any activity, saturation index or conditional constant that would rest on the old equilibrium.

Staleness is DECIDED, not assumed: the block goes stale because T actually moved. A pump carrying the same brine at constant T and P across the same boundary would leave the equilibrium intact and the pH reported – the boundary alone does not invalidate anything; the change of state does. A reader who wants the equilibrium AT the outlet must put a unit there that can solve one; the molecular world cannot, which is the whole point of the boundary.

THE IDENTITY SECOND WITNESS. Delete the ‘thermo {}‘ block below: the transporter then runs the GLOBAL electrolyte world, which CAN re-solve the chemistry, and does – at 332.23 K, giving a different (and locally correct) block. What must be identical either way is the INTERMEDIATE stream ‘brine‘:

what unit 2 is cannot reach back and change what unit 1 solved. That is the invariant the gate probes; the differing outlet is not a failure, it is the boundary being visible.

AND SINCE 2026-08-09 IT IS VISIBLE IN THE TEMPERATURE TOO, which is worth reading slowly. A ‘heater‘ now solves for the outlet STATE whose enthalpy is H_in + Q – in ITS OWN declared world (R-E1: an unpinned stream means the equilibrium of (T, P, z) in the CONSUMING unit’s thermo world). The molecular world below finds this brine 1.3 % vapour at ~331 K, so 40 kW buys 17.66 K there; the global electrolyte world holds the same volatiles as ions, finds essentially no vapour, and the same 40 kW buys 19.08 K to 332.23 K. Same enthalpy, two temperatures – exactly the settled rule that H is the conserved truth and T is the model-dependent readout. The consequence a reader will meet: the energy-balance report prices EVERY stream in the GLOBAL package, so this unit’s RAW closure reads 92.54 % rather than 100 %. That residual is the model boundary, not a first-law error – and SINCE THE SAME DAY THE REPORT SAYS SO. It was a named gap for exactly one afternoon (the report had no notion of a per-unit thermo world, recorded in docs/design/tp-stream-energy-coherence.md), and it was closed by the model-boundary LEDGER:

raw imbalance -2.9851 kW (dH 37.0149 vs 40.0000 kW of duty) boundary step -2.9851 kW (independently computed, see below) remaining 9.4e-9 kW → adjusted closure 100.00 %, no alarm

reports/balances/energyBalance_byUnit.csv carries all three, side by side and never collapsed, plus a ledger row naming BOTH worlds by their declared models. The step is not taken on trust: the report assembles this unit’s declared world ITSELF, prices both streams in both packages at the same (T, P, z), and credits the step only if its own number reproduces the imbalance – an auditor that reuses the auditee’s arithmetic checks nothing. See docs/design/model-boundary-energy-ledger.md.

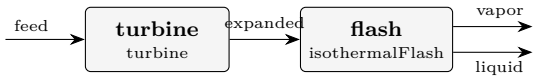
Before the 2026-08-09 heater fix the sensible inversion happened to track the global datum and the gap read 100 %, which was luck, not agreement.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | branchA | F | 0.00689112315881 |
| stream | branchB | F | 0.0206733694764 |
| stream | brine | F | 0.0275644926352 |
| stream | feed | F | 0.0277222222222 |
| stream | offgas | F | 0.000157729586982 |
| stream | warmBrine | F | 0.0275644926352 |
| kpi | speciator | F_alpha | 0.0275644926352 |
| kpi | speciator | F_beta | 0.000157729586982 |
| kpi | speciator | F_in | 0.0277222222222 |
| kpi | speciator | K_CO2 | 4.78835784059 |
| kpi | speciator | K_N2 | 105036.780664 |
| kpi | speciator | K_NH3 | 0.777804159627 |
| kpi | speciator | K_ethanol | 0.180718643037 |
| kpi | speciator | K_water | 0.0721940189402 |
| *... and 38 more reference values in the case’s `expected`.* | | | |

### perUnitThermo01_srk_nrtl

tutorials/steady/thermo/perUnitThermo01_srk_nrtl




**What it does:** Per-unit thermo: turbine SRK (real gas) → flash NRTL (non-ideal liquid)

**The story:** Demonstrates PER-UNIT thermo models: the global package (Raoult) is the default, but each unit can override it with a 'thermo { ... }' block that replaces the model it cares about. Components stay global.

turbine → equationOfState SRK (real-gas expansion, high P) flash → activityModel NRTL (non-ideal ethanol/water liquid)

Pedagogical caveat: the stream that crosses from the turbine (SRK) into the flash (NRTL + idealGas) is re-interpreted by a different model – a small thermodynamic inconsistency at the boundary. Real practice keeps one consistent method per section; mixing is an advanced, deliberate act.

**What to expect (golden-locked):**

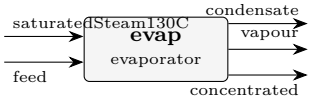
| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | expanded | F | 0.0138888888889 |
| stream | feed | F | 0.0138888888889 |
| stream | liquid | F | 0.00854830793289 |
| stream | vapor | F | 0.005340580956 |
| kpi | flash | F_alpha | 0.00854830793289 |
| kpi | flash | F_beta | 0.005340580956 |
| kpi | flash | F_in | 0.0138888888889 |
| kpi | flash | K_ethanol | 1.4838682811 |
| kpi | flash | K_water | 0.647356121192 |
| kpi | flash | P | 100000 |
| kpi | flash | Q | -413177.313739 |
| kpi | flash | Q_kW | -413.177313739 |
| kpi | flash | T | 353 |
| kpi | flash | V_over_F | 0.384521828832 |

*... and 35 more reference values in the case's `expected`.*

## thermoTest

## model1_lumped_evaporator

tutorials/steady/thermoTest/model1_lumped_evaporator




**What it does:** thermoTest model 1 – NaCl as a LUMPED molecular component: the boiling-point elevation is the IDEAL K_b∗m (van't Hoff), BPE ~ 2.11 K. Compare model2 (same evaporator, Pitzer a_w → BPE ~ 5.07 K): the REPRESENTATION axis, lumped vs dissociated, on one operation.

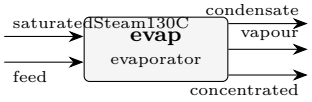
**The story:** thermoTest model1_lumped_evaporator / flowsheetDict NaCl treated as a LUMPED molecular component: the boiling-point elevation is the IDEAL van't Hoff K_b∗m (~2.11 K). Model 1 of the thermoTest ladder – the deliberately-too-simple baseline the pitzer (model 2) and eNRTL rungs correct. Same evaporator, climbing thermo.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.020084561515 |
| stream | condensate | F | 0.0125 |
| stream | feed | F | 0.0277777777778 |
| stream | saturatedSteam130C | F | 0.0125 |
| stream | vapour | F | 0.00769321626283 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 2.11524224539 |
| kpi | evap | F_conc | 0.020084561515 |
| kpi | evap | F_conc_kmol_h | 72.3044214538 |
| kpi | evap | F_conc_mass | 0.417969209025 |
| kpi | evap | F_cond | 0.2251875 |
| kpi | evap | F_cond_kg_h | 810.675 |
| kpi | evap | F_in | 0.0277777777778 |
| *... and 24 more reference values in the case's `expected`.* | | | |

## model2_pitzer_evaporator

tutorials/steady/thermoTest/model2_pitzer_evaporator




**What it does:** thermoTest model 2 – NaCl as a PITZER electrolyte (Na+ + Cl-, aqueous-inf-dilution reference): the boiling-point elevation comes from the real water activity a_w, BPE ~ 5.07 K vs the ideal 2.11 K of model1.

**The story:** BPE NOTE (2026-07-02, a story in three acts): (1) frozen 25 C surface → BPE 5.07 K; (2) LINEAR calorimetric T-slots (Parker fit) → 5.58 K – the 25 C derivative extrapolated straight to the boil; (3) the FULL Silvester & Pitzer 1977 T-functions (eqs 30-32, Table IV verbatim in Na-Cl.dat) + A_phi(T) curated against their Table II → BPE 5.06 K. beta0(T) has a MAXIMUM near 150 C (their Fig 4): the linear slope overshoots where the real curve has already bent over. The lesson: a first derivative is not a function – validity windows are data.

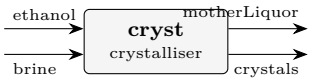
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | concentrated | F | 0.0198874983274 |
| stream | condensate | F | 0.0125 |
| stream | feed | F | 0.0277777777778 |
| stream | saturatedSteam130C | F | 0.0125 |
| stream | vapour | F | 0.00789027945038 |
| kpi | evap | A | 20 |
| kpi | evap | A_m2 | 20 |
| kpi | evap | BPE | 5.19956015536 |
| kpi | evap | F_conc | 0.0198874983274 |
| kpi | evap | F_conc_kmol_h | 71.5949939786 |
| kpi | evap | F_conc_mass | 0.414419115701 |
| kpi | evap | F_cond | 0.2251875 |
| kpi | evap | F_cond_kg_h | 810.675 |
| kpi | evap | F_in | 0.0277777777778 |

*... and 29 more reference values in the case's `expected`.*

## model3_enrtl_antisolvent_crystalliser

tutorials/steady/thermoTest/model3_enrtl_antisolvent_crystalliser




**What it does:** Antisolvent (drowning-out) crystallisation of NaCl: a SATURATED aqueous brine is mixed with ethanol; the eNRTL mixed-solvent term lowers the dielectric, raises gamma_pm, and DROPS the saturation molality m_sat – so NaCl precipitates with NO cooling and NO evaporation. Saturation from the SAME ion-activity product Ksp anchored in pure water, solved with the mixed-solvent gamma_pm.

**The story:** thermoTest/model3_enrtl_antisolvent_crystalliser – step 3 of the thermoTest arc (same {NaCl, CaSO4, ethanol, water} system, five property models): here the package selects eNRTL MIXED-SOLVENT, the only model in the arc that can reach the ethanol-lowered dielectric. The flowsheet is crystalliser06_nacl_antisolvent's, unchanged:

ANTISOLVENT (drowning-out) crystallisation

[cryst] crystals (solid NaCl product) ethanol motherLiquor (saturated supernatant: water + ethanol + residual NaCl)

The crystalliser receives BOTH feeds directly – a saturated aqueous NaCl brine AND the ethanol antisolvent – combines them internally, and discharges its TWO natural products: the crystals and the saturated mother liquor. No separate mixer, no separate filter: an equilibrium crystalliser already KNOWS how much solid forms and what the saturated liquor is.

The physics: ethanol's low dielectric (24.3 vs water's 78.5) lowers the mixture permittivity, which RAISES the mean ionic activity gamma_pm (eNRTL mixed-solvent + Born), so the saturation molality m_sat COLLAPSES (6.14 → ~1.3 mol/kg) and NaCl precipitates with NO cooling and NO boil-off. The saturation is the SAME ion-activity product Ksp = (gamma_pm ∗ m_sat)^2 anchored in pure water, solved with the mixed-solvent gamma_pm at the feed's salt-free ethanol fraction – the physics the aqueous Pitzer model cannot reach (the reason eNRTL exists).

Basis: 100 kmol/h saturated brine (x_water 0.900, x_NaCl 0.100; ~6.14 mol/kg) + 60.0 kmol/h ethanol → salt-free x_ethanol ~ 0.40. Components live case-local in constant/components/; the eNRTL ion/tau records resolve from data/standards/ at runtime.

(Want imperfect separation – a wet cake with clinging liquor, fines loss, washing? Feed the single-output 'magma' form to a downstream 'filter'.)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brine | F | 0.0277777777778 |
| stream | crystals | F | 0.00202035108762 |
| stream | ethanol | F | 0.0166666666667 |
| stream | motherLiquor | F | 0.0424240933568 |
| kpi | cryst | Ksp_activity | 34.8439846653 |
| kpi | cryst | Q_kW | -6.55987377671 |
| kpi | cryst | Q_removed | 6559.87377671 |
| kpi | cryst | T_op | 298.15 |
| kpi | cryst | antisolvent_xfrac | 0.399920015997 |
| kpi | cryst | c_sat | 0.0517444790334 |
| kpi | cryst | cakeLiquidFlow | 0.000329661969914 |
| kpi | cryst | cakeMoisture | 0.1 |
| kpi | cryst | cakeWetness_pct | 9.09090909091 |
| kpi | cryst | crystal_flow | 0.0988038720389 |
| *... and 17 more reference values in the case's `expected`.* | | | |

### `model5_nrtl_flash`

`tutorials/steady/thermoTest/model5_nrtl_flash`




**What it does:** thermoTest model 5 – water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model. This is the antisolvent (ethanol) and the solvent (water) in their own role – the same substances that dissolve/precipitate NaCl in models 2-4, here separated by distillation.

**The story:** thermoTest model5_nrtl_flash / flowsheetDict Water + ethanol as MOLECULAR components (no ions): an isothermal VLE flash on the NRTL activity model, selected through the ethanolWater_NRTL property package. Model 5 of the thermoTest ladder – the molecular gamma-phi anchor beside the electrolyte rungs.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| `stream` | `feed` | `F` | `0.0277777777778` |
| `stream` | `liquid` | `F` | `0.0170966158658` |
| `stream` | `vapor` | `F` | `0.010681161912` |
| `kpi` | `flash` | `F_alpha` | `0.0170966158658` |
| `kpi` | `flash` | `F_beta` | `0.010681161912` |
| `kpi` | `flash` | `F_in` | `0.0277777777778` |
| `kpi` | `flash` | `P` | `100000` |
| `kpi` | `flash` | `Q` | `0` |
| `kpi` | `flash` | `Q_kW` | `0` |
| `kpi` | `flash` | `T` | `353` |
| `kpi` | `flash` | `V_over_F` | `0.384521828832` |
| `kpi` | `flash` | `phaseSet` | `0` |
| `stream` | `feed` | `T` | `353` |
| `stream` | `liquid` | `T` | `353` |
| *... and 6 more reference values in the case's `expected`.* | | | |

## userops

**The theory you need.** User-defined operations: the extension seam. A custom property op or unit compiled into the binary via the explicit factory – the tutorial shows the registration pattern (`registerBuiltins`) end to end.

## userOp01_yield_reactor

tutorials/steady/userops/userOp01_yield_reactor




**What it does:** User-defined yieldReactor (case-local code/): neoPentane → nPentane, X=0.8

**The story:** A single unit op — but a USER-DEFINED one (‘yieldReactor‘), written in this case’s code/ folder, not in the Choupo source tree. Build + run it with:

bin/buildCode tutorials/steady/userops/userOp01_yield_reactor

(NOT runCase: the type ‘yieldReactor‘ does not exist in the stock binary; buildCode compiles the case’s code/ against libchoupo.so.)

The op converts a SPECIFIED fraction of the reactant to the product (no kinetics) — a "yield reactor", which the built-in CSTR/PFR are not. Here: 80 % of neoPentane isomerised to nPentane. Because the two are isomers (same MW), the mass balance closes to 100 %.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.000277777777778 |
| stream | out | F | 0.000277777777778 |
| kpi | reactor | F_in_kmol_h | 1 |
| kpi | reactor | F_out_kmol_h | 1 |
| kpi | reactor | T | 320 |
| kpi | reactor | X | 0.8 |
| kpi | reactor | converted_kmol_h | 0.8 |
| stream | feed | T | 320 |
| stream | out | T | 320 |
| boundary | global | H_feeds_kW | -45.85856841 |
| boundary | global | Q_boundary_kW | 0 |
| boundary | global | H_products_kW | -48.028683133 |
| boundary | global | residual_kW | 2.170114723 |

## userOp02_component_splitter

tutorials/steady/userops/userOp02_component_splitter




**What it does:** User-defined componentSplitter (case-local code/): shortcut benzene/toluene split

**The story:** A USER-DEFINED unit op ('componentSplitter'), written in this case's code/ folder. Build + run with:

runCase tutorials/steady/userops/userOp02_component_splitter

(runCase sees the code/ folder and compiles it first.)

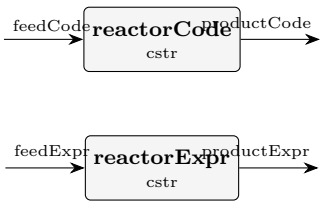
The op is a SHORTCUT separator: a per-component recovery to the overhead. Here a benzene/toluene feed is split so 95 % of the benzene and only 8 % of the toluene leave overhead — an enriched light cut, the kind of stand-in you place before sizing a real column. Two outlet streams (overhead, bottoms); mass closes to 100 % (every component is conserved across the split).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bottoms | F | 0.0134722222222 |
| stream | feed | F | 0.0277777777778 |
| stream | overhead | F | 0.0143055555556 |
| kpi | splitter | F_bottoms_kmol_h | 48.5 |
| kpi | splitter | F_overhead_kmol_h | 51.5 |
| kpi | splitter | overhead_fraction | 0.515 |
| stream | bottoms | T | 360 |
| stream | feed | T | 360 |
| stream | overhead | T | 360 |
| boundary | global | H_feeds_kW | 1108.06863474 |
| boundary | global | Q_boundary_kW | 0 |
| boundary | global | H_products_kW | 1108.06863474 |

## userOp03_case_rate_law

tutorials/steady/userops/userOp03_case_rate_law




**What it does:** ONE rate law written TWICE, by the two routes a case carries its own kinetics: a Michaelis-Menten law in C++ in the case's code/ folder (registered into the rate-law factory, local only) and the same law as 'type expression;' in constant/reactions (runs everywhere) – two CSTRs, one answer, checked against the closed form

**The story:** ONE rate law, written TWICE, by the two routes a case has to carry its own kinetics (DEV.md 4c C67).

reactorCode reads 'kinetics { type michaelisMenten; }' – a law written in C++ in THIS case's code/ folder and registered into the engine's rate-law factory from code/registerUserTypes.cpp. 'bin/buildCode' compiles it into a per-case binary; it runs on a machine with a compiler and NOT on the site (the browser has none).

reactorExpr reads 'kinetics { type expression; }' – the SAME law written as a formula in constant/reactions, evaluated by the engine's own expression evaluator. It runs everywhere, the site included.

The reaction is an ISOMERISATION, isomerA → isomerB, between two synthetic species this case carries in its own constant/components/: one molar mass between them, so each reactor's mass balance closes at 100 % (the witness used to turn the catalogue's compA, MW 30, into compB, MW 50, one-for-one, and the engine reported 133.333 % – correctly), and one molar liquid volume, so the constant-volumetric-flow liquid basis is exact.

The two reactors are fed the same stream and must give the same answer; both must equal the closed-form CSTR solution of Michaelis-Menten kinetics (a quadratic in the outlet concentration). 'bin/curate/check_rate_expression.py' holds all three together.

Why ever write the C++ one, if the formula is shorter? Because some laws are not a formula: a rate that needs an inner solve (a surface coverage found by Newton, a population moment), a loop, a table the case reads from a file. The C++ route is the door for those; this witness walks through it with a law simple enough to check by hand.

*A rate law is a fact about a catalyst, a solvent, an enzyme — about the EQUIPMENT and its contents — so it belongs in the case (the three-axiom layout's axiom 3), not compiled into the engine. Choupo gives a case two ways to carry one (DEV.md 4c C67, 2026-10-12):*

*This case writes ONE law — Michaelis–Menten, r = Vmax c_A / (Km + c_A) — both ways, and feeds two identical CSTRs (reactorCode, reactorExpr) the same stream. The reaction is an isomerisation, isomerA → isomerB, between two synthetic species the case carries itself (see ∗The species∗ below).*

*This case does not run in the browser (it ships C++ in code/); the GUI says so when it is opened. Its reactorExpr half would, on its own.*

*∗ constant/reactions — mm_code names type michaelisMenten; and declares Vmax and Km WITH UNITS, which the C++ reads through the dictionary (so the units are checked there, not trusted); mm_expr writes the same law as a formula over c[isomerA], Vmax and Km. ∗ code/MichaelisMenten.{H,cpp} — the whole law, thirty lines: read the parameters, return the extent rate in the reactor's convention (RateState::amountPerKmol says whether the reactor works in mol or kmol). ∗ code/registerUserTypes.cpp — one ReactionRate::registerType call; the same explicit factory the built-in Arrhenius, LHHW and expression use. ∗ The run log — the [expression] law is printed back AS PARSED, with the dimensions of every definition; the user law prints its own type name.*

*V_R = 0.025 m3, feed 1 kmol/h of A at 12.5 kmol/m3 (liquid basis), Vmax = 0.01 kmol/(m3 s), Km = 5 kmol/m3 (both AUTHOR-SET for this witness — the form is Michaelis & Menten's, the numbers are nobody's measurement). The CSTR balance F_A0 X = V Vmax c_A / (Km + c_A) with c_A = c_A0 (1 - X) is a quadratic in X, and these numbers put its root at X = 0.5 exactly. Both reactors publish xi =*

*0.138889 mol/s, i.e. X = 0.5; bin/curate/ check_rate_expression.py holds the two reactors to each other and to the closed form.*

*Write the formula whenever the law IS a formula: it is shorter, it is checked (an unknown symbol, a dimension mismatch, a non-finite value all refuse by name), and it runs everywhere. Write C++ when it is not — an inner solve, a loop, a property the expression grammar cannot reach. The C++ route has no dimension checker but the dictionary's own (lookupScalar(key, dims)), and it is the author's responsibility to refuse what it does not read, as this one does.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedCode | F | 0.000277777777778 |
| stream | feedCode | T | 350 |
| stream | feedExpr | F | 0.000277777777778 |
| stream | feedExpr | T | 350 |
| stream | productCode | F | 0.000277777777778 |
| stream | productCode | T | 350 |
| stream | productExpr | F | 0.000277777777778 |
| stream | productExpr | T | 350 |
| kpi | reactorCode | F_in_kmol_h | 1 |
| kpi | reactorCode | F_out_kmol_h | 1 |
| kpi | reactorCode | T | 350 |
| kpi | reactorCode | T_out | 350 |
| kpi | reactorCode | V_R | 0.025 |
| kpi | reactorCode | k_mm_code | 0.01 |

*. . . and 12 more reference values in the case's* `expected`.

## utilities

**The theory you need.** Utility streams (steam levels, cooling water, refrigerants, electricity) carry duty into and out of the plant at declared temperature levels and prices; allocation is by $T$-level feasibility, announced.

## utility01_dowtherm_preheat

tutorials/steady/utilities/utility01_dowtherm_preheat




**What it does:** Dowtherm A loop preheats a hydrocarbon feed (eps-NTU)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | hotOil | F | 0.0416666666667 |
| stream | oilReturn | F | 0.0416666666667 |
| stream | processFeed | F | 0.0111111111111 |
| stream | processOut | F | 0.0111111111111 |
| kpi | preheater | C_cold | 2992.03981719 |
| kpi | preheater | C_hot | 17325.8215856 |
| kpi | preheater | C_r | 0.17269252153 |
| kpi | preheater | LMTD | 162.011885097 |
| kpi | preheater | NTU | 5.01330226751 |
| kpi | preheater | Q | 895145.718325 |
| kpi | preheater | Q_kW | 895.145718325 |
| kpi | preheater | T_cold_in | 320 |
| kpi | preheater | T_cold_out | 526.359227681 |
| kpi | preheater | T_hot_in | 623.15 |

*. . . and 10 more reference values in the case's `expected`.*

## utility02_hitec_csp_heater

tutorials/steady/utilities/utility02_hitec_csp_heater




**What it does:** HITEC molten salt loop sensibly heats pressurised (48 bar) boiler feedwater to ~244 C, just below T_sat, for a downstream steam generator (eps-NTU, no phase change), ~208 kW duty

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | boilerFeed | F | 0.0222222222222 |
| stream | boilerOut | F | 0.0222222222222 |
| stream | boilerOut | T | 517.400926398 |
| stream | saltReturn | F | 0.0833333333333 |
| stream | saltSupply | F | 0.0833333333333 |
| kpi | boilerHX | C_cold | 1677.77777778 |
| kpi | boilerHX | C_hot | 12795.4797468 |
| kpi | boilerHX | C_r | 0.131122694184 |
| kpi | boilerHX | LMTD | 306.566828203 |
| kpi | boilerHX | NTU | 0.405298013245 |
| kpi | boilerHX | Q | 208465.443178 |
| kpi | boilerHX | Q_kW | 208.465443178 |
| kpi | boilerHX | T_cold_in | 393.15 |
| kpi | boilerHX | T_cold_out | 517.400926398 |

*... and 10 more reference values in the case's `expected`.*

## utility03_glycol_brine_chiller

tutorials/steady/utilities/utility03_glycol_brine_chiller




**What it does:** PG-30 brine sensibly cools an ethanol vapour stream from 330 K to ~273 K (eps-NTU, no phase change), ~20.8 kW duty

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | brineReturn | F | 0.416666666667 |
| stream | brineSupply | F | 0.416666666667 |
| stream | ethanolOut | F | 0.00555555555556 |
| stream | ethanolVapour | F | 0.00555555555556 |
| kpi | condChiller | C_cold | 57599.6888818 |
| kpi | condChiller | C_hot | 623.888888889 |
| kpi | condChiller | C_r | 0.0108314628256 |
| kpi | condChiller | LMTD | 12.519257418 |
| kpi | condChiller | NTU | 4.48797862867 |
| kpi | condChiller | Q | 35053.9207703 |
| kpi | condChiller | Q_kW | 35.0539207703 |
| kpi | condChiller | T_cold_in | 273.15 |
| kpi | condChiller | T_cold_out | 273.758578301 |
| kpi | condChiller | T_hot_in | 330 |
| *... and 10 more reference values in the case's `expected`.* | | | |

# Category II: Transient process simulation (choupoSemiContinuous, no control loop)

Unit operations connected by streams whose internal states are NON-STATIONARY: the same $\mathrm{d}\mathbf{Y}/\mathrm{d}t = f(\mathbf{Y})$ integration in real time as the control category, with no controller declared – the process is watched, not driven. A discipline of its own (ruled 2026-09-20): the open-loop transient a student studies before any loop is closed on it. The binary is `choupoSemiContinuous`, the fifth problem class (ruled 2026-09-20): the same driver `choupoCtrl` runs, without the control layer, and a `controllers` block is refused there by name; `choupoBatch` cases join this folder later. Two cases today: the cold start-up of a CSTR relaxing to its natural steady state while OpenFOAM-style instant directories fill, and three equal mixing tanks in series answering an impulse with the Erlang-3 residence-time density.

## unsteady01_startup_transient

tutorials/unsteady/unsteady01_startup_transient




**What it does:** Open-loop CSTR START-UP transient with OpenFOAM-style real-time instant dirs: charged cold (320 K), no control, watch T(t) and the holdup approach the natural steady state by scrubbing 0/ 50/ 100/ ... s

**The story:** Process: A continuous CSTR (first-order compA → compB, mild exotherm) is CHARGED cold (320 K, pure compA) and then left to run with NO controller. The jacket sits at a fixed 320 K. From the cold start the reactor warms a little from the reaction exotherm, the feed/ outlet convection competes, and (T, composition) relax to the natural open-loop steady state over ~ several residence times.

What this tutorial shows: The OpenFOAM-style transient solution directories. The case root fills with REAL-TIME directories '0/ 50/ 100/ ... 1000/', each with the layout of 0/: '<t>/internalStates/reactor' is the reactor HOLDUP at that instant, '<t>/feed' and '<t>/product' the two streams. Scrub the times and you watch T(t) and the inventory march to steady state — the same way you scrub a CFD transient in ParaView.

To compare CLOSED-LOOP control, see ctrl01_cstr_temp_control (a PID drives T to a 350 K setpoint) and ctrl05_disturbance_transient (a PID rejecting a feed-temperature disturbance), both also writing time dirs.

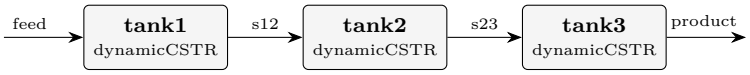
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | T_final | 320.759374011 |
| kpi | reactor | n_compA_final | 0.0117988900072 |
| kpi | reactor | n_compB_final | 0.000201109992805 |
| kpi | reactor | t_end | 1000 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 1.77265609598e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 1.5 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 1.5 |
| kpi | balance | mass_residual_kg | -2.65898414398e-14 |
| kpi | balance | material_available | 1 |
| kpi | reactor | F_in_final | 5e-05 |
| kpi | reactor | z_in_compA_final | 1 |

*... and 1 more reference values in the case's `expected`.*

## unsteady02_tanks_in_series

tutorials/unsteady/unsteady02_tanks_in_series




**What it does:** Tanks-in-series RTD: three equal mixing tanks chained by the ctrl router; impulse in tank 1 → Erlang-3 response at tank 3 (tbar = 3 tau, sigma2/tbar2 = 1/3), Levenspiel closed forms as anchors.

**The story:** the SECOND HALF of the RTD lesson, unlocked by the ctrl router (ruled 2026-08-24): three EQUAL mixing tanks in series, each tau = n/F = 0.004/5e-5 = 80 s, chained by the same 'in'/'outputs' topology grammar every steady flowsheet already uses. The router applies each connection ONE-STEP-EXPLICITLY: at every accepted driver step the downstream tank integrates on the upstream outlet at the step's START – a transport delay of one driver step (1 s here, against tau = 80 s), announced by the [routing] lines at startup.

THE EXPERIMENT. The tracer impulse is an INITIAL-CONDITION substitution in tank 1 only (compB 0.0002 of the 0.004 kmol holdup – substitution, not addition, so every tank holds exactly tau = 80 s and the closed forms are exact). An initial holdup m in tank 1 is mathematically a perfect impulse into tank 1 at t = 0, so tank 3's outlet is the Erlang-3 density:

E(t) = t^2 e^(-t/tau) / (2 tau^3) tbar = 3 tau = 240 s sigma2 = 3 tau^2 = 19200 s^2 (sigma2/tbar2 = 1/3 exactly) peak at 2 tau = 160 s (visible in trajectory.csv)

Levenspiel, Chemical Reaction Engineering, ch. 14 (the tanks-in-series model): the variance ratio 1/N is THE diagnostic the lesson teaches – one tank gives 1, three give 1/3, a PFR gives 0. ctrl18 is the N = 1 control; this case is N = 3 on identical tanks.

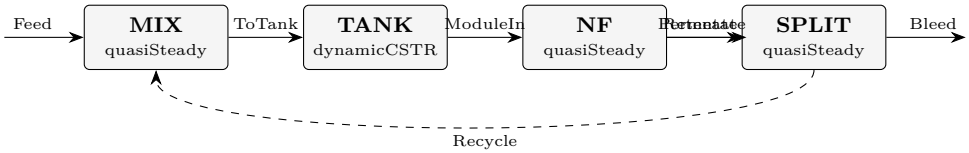
WHAT THE ANCHORS TOLERATE, stated: the engine measures moments by trapezoid on the accepted 1 s grid and truncates at 1600 s (= 20 tau of one tank); both effects are far inside the anchor bands, and the router adds a one-step transport delay per connection (2 s total on tbar's 240 s – inside the 1 % band, and the honest price of an explicit scheme, stated rather than hidden).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.15876699233e-16 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 7.2 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 7.2 |
| kpi | balance | mass_residual_kg | -1.55431223448e-15 |
| kpi | balance | material_available | 1 |
| kpi | rtd | area_kmol | 0.000198752512114 |
| kpi | rtd | sigma2_s2 | 19198.7859557 |
| kpi | rtd | tbar_s | 241.502454857 |
| kpi | rtd | truncation_final_over_peak | 1.54894525802e-06 |
| kpi | tank1 | F_in_final | 5e-05 |
| kpi | tank1 | T_final | 330 |
| *... and 23 more reference values in the case's `expected`.* | | | |

## unsteady03_nf_feed_and_bleed_startup

tutorials/unsteady/unsteady03_nf_feed_and_bleed_startup




**What it does:** Feed-and-bleed NF start-up: the loop concentrates from the feed's to its steady value, the module solved quasi-steady at every instant

**The story:** the START-UP of the feed-and-bleed loop whose steady state is membrane18_nf_feed_and_bleed.

THE PLANT. Fresh feed (1 wt % glucose, 200 kmol/h) joins the recycle in MIX and fills a 1 m3 recirculation tank; the tank feeds an NF270 module (15 m2); the retentate is split, 95 % back to MIX and 5 % bled off. The permeate and the bleed leave. At t = 0 the tank holds feed.

WHAT IS SOLVED HOW. The TANK is a vessel (dynamicCSTR, no reaction, no jacket): it holds an inventory and is integrated in time. MIX, NF and SPLIT hold essentially no liquid, so on the loop's time scale they are at steady state at every instant – each is its steady unit wrapped as 'quasiSteady' and re-solved on every accepted state. The module the run solves is exactly the one membrane18 solves; only the time around it is new.

WHAT TO WATCH. The tank starts at the feed concentration, so the module first sees the feed and delivers its single-pass flux. As the retentate returns, the loop concentrates, the osmotic pressure rises and the flux falls towards the value membrane18 reports at steady state. The time scale is the tank's contents over the flows that carry solute OUT of the loop – the bleed, and the permeate because NF270 passes glucose partly: V / (Q_bleed + Q_permeate (1 - R)) is about 1 m3 / 5.3e-4 m3/s = 1900 s, and the loop is faster than that (measured e-folding about 1340 s, from successive gaps in balanceTrajectory.csv) because the bleed leaves at the RETENTATE concentration, richer than the tank's. 6 h is some 16 of them, so the end state must be membrane18's: the golden holds the module's flux, rejection and the permeate and bleed flows to membrane18's own within 2e-5. That agreement is the case's anchor – a transient that does not end on the steady answer is wrong.

WHAT IS ASSUMED, and printed on the run: the quasi-steady units hold no inventory (nothing accumulates in the module or the pipes), the tank is well mixed, and the loop's pressure is the module's feed pressure throughout (no pump is modelled, as in membrane18 – see its header). The first law is NOT claimed: the dynamic ledger needs a stored energy functional per unit, which the quasi-steady adapter does not provide, and the run says so rather than closing it with a zero.

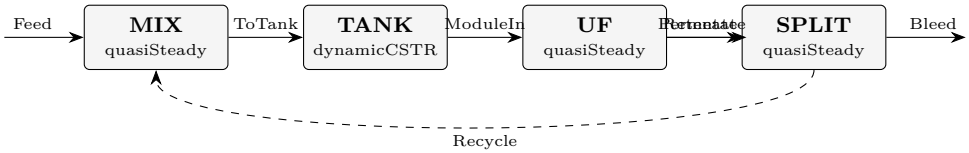
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | MIX | F_ToTank_final | 0.388832449894 |
| kpi | MIX | kpi_F_out_final | 0.388832449894 |
| kpi | MIX | kpi_P_out_final | 1000000 |
| kpi | MIX | kpi_T_out_final | 298.15 |
| kpi | MIX | kpi_n_in_final | 2 |
| kpi | MIX | kpi_vf_final | 0 |
| kpi | MIX | t_end | 21600 |
| kpi | NF | F_Permeate_final | 0.0380146663544 |
| kpi | NF | F_Retentate_final | 0.350817783479 |
| kpi | NF | kpi_Gamma_glucose_avg_final | 1.04649230438 |
| kpi | NF | kpi_J_w_avg_final | 4.58864465983e-05 |
| kpi | NF | kpi_R_obs_glucose_final | 0.685413515081 |
| kpi | NF | kpi_dP_feed_final | 0 |
| kpi | NF | kpi_mass_closure_rel_final | 6.23839085054e-16 |

*... and 25 more reference values in the case's `expected`.*

## unsteady04_uf_feed_and_bleed_startup

tutorials/unsteady/unsteady04_uf_feed_and_bleed_startup




**What it does:** Feed-and-bleed UF start-up: the loop concentrates the protein, and the gel-limited flux falls as ln(c_g/c_b) while it does

**The story:** ultrafiltration of a protein in a feed-and-bleed loop, from start-up.

THE PLANT. 1 m3/h of a 10 kg/m3 BSA solution joins the recycle in MIX and fills a 0.5 m3 recirculation tank; the tank feeds the UF module of membrane19 (5 m2, 3 bar, the same hypothetical membrane and gel parameters); the retentate is split, 80 % back to MIX and 20 % bled off. At t = 0 the tank holds feed.

WHAT IS SOLVED HOW. The TANK is a vessel (dynamicCSTR, no reaction, no jacket), integrated in time, and DECLARED isothermal ('energy isothermal;'): BSA carries no heat capacity, so the tank cannot solve an energy equation, and here nothing would move T anyway – every stream is at 298.15 K with no reaction and no jacket. MIX, UF and SPLIT hold essentially no liquid and are each a steady unit wrapped as 'quasiSteady', re-solved at every accepted state – the module is membrane19's, law and all.

WHAT TO WATCH – the lesson of gel polarisation IN A LOOP. The module is gel-limited, so its flux is k ln(c_g/c_b) whatever the pump does (membrane19 shows the plateau). The loop returns the retained protein, the tank concentrates, c_b rises – and the flux FALLS with it, as a logarithm, until the protein leaving in the bleed and the permeate balances the protein fed. MEASURED on this case: the module's mean flux falls from 3.30e-5 m/s (119 L/(m2 h)) at start-up to 2.88e-5 m/s (104 L/(m2 h)), 13 % less, while the tank goes from 10 to 15.3 kg/m3 of protein, with an e-folding time of about 2500 s – so a plant sized on the FEED's flux is short of area by that 13 %. 6 h is some 8 time constants. Every <t>/ directory the run writes holds each stream's state and the tank's holdup at t (open 600/ against 21600/), and the module's own J_w_avg rides trajectory.csv as a kpi column.

EVERY UF NUMBER HERE IS HYPOTHETICAL, as in membrane19 (the membrane, c_gel, S, k_film – read its header); BSA's MW is derived from its UniProt sequence. The shape is the lesson, never a number. NOT MODELLED: the first law (BSA carries no enthalpy datum, and the quasi-steady adapter claims none), a pump, fouling in time.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | MIX | F_ToTank_final | 0.044910327156 |
| kpi | MIX | kpi_F_out_final | 0.044910327156 |
| kpi | MIX | kpi_P_out_final | 300000 |
| kpi | MIX | kpi_T_out_final | 298.15 |
| kpi | MIX | kpi_n_in_final | 2 |
| kpi | MIX | kpi_vf_final | 0 |
| kpi | MIX | t_end | 21600 |
| kpi | SPLIT | F_Bleed_final | 0.00738729355532 |
| kpi | SPLIT | F_Recycle_final | 0.0295491742213 |
| kpi | SPLIT | t_end | 21600 |
| kpi | TANK | F_in_final | 0.044910327156 |
| kpi | TANK | T_final | 298.15 |
| kpi | TANK | n_BSA_final | 0.000115420024839 |
| kpi | TANK | n_water_final | 27.6499598825 |

*... and 27 more reference values in the case's `expected`.*

## unsteady05_tank_valve_drain

tutorials/unsteady/unsteady05_tank_valve_drain




**What it does:** Water tank drained through a valve, no controller: after a 50 % step in the inflow the level SELF-REGULATES to the closed-form h∗ = (Q/(Cv x))^2/g (0.998 m → 2.245 m)

**The story:** A water tank drained through a valve, with NO controller. The inflow steps from 0.20 to 0.30 kmol/s (a 50 % step, applied at t = 0 by starting the tank at the steady level of the OLD inflow and feeding the NEW one – the initial-condition step of unsteady02), and the level climbs to a new steady level by itself, because the outflow grows with the level:

Q = Cv x opening x sqrt(dP / rho) [m3/s] dP = P + rho g h - P_downstream (static head + vessel pressure - downstream) F_out = Q rho / MW [kmol/s] A rho/MW dh/dt = F_in - F_out (mass balance, the level is volume / area)

Here the vessel is open to the same pressure the valve discharges to (P = P_downstream = 1 atm), so the head alone drives the flow and the steady state F_out = F_in has a closed form:

h∗ = (Q / (Cv x))^2 / g with Q = F_in MW / rho

Cv is the valve's effective flow AREA in m2 (the orifice equation's Cd A), 'opening' the fraction of the stroke (linear), the outlet is at the bottom. Because Q ~ sqrt(h), h∗ grows with the SQUARE of the inflow: a 1.5-fold step in the inflow gives a 2.25-fold step in the level (0.998 m → 2.245 m at rho = 1001.55 kg/m3, the package's own density at 293.15 K: the water record's Vliq at 25 C with Rackett supplying the temperature shape). The engine prints this closed form at start-up; the run's own final level, read from trajectory.csv (column 'level'), agrees with it to < 1e-3.

WHAT TO SEE. A first-order-like response with tau ~ 2600 s at the new level (linearising Q about h∗: tau = A / (dQ/dh) = 2 A h∗ / Q). Compare with ctrl21: with a manipulated OUTFLOW instead of a valve the same tank has NO such tendency – it integrates the difference of the flows without bound, and only a level controller closes the loop.

ENERGY. The tank stores H on the elements datum (the package's mixture liquid surface) and T is a readout; the feed arrives at the tank's own temperature, so T stays at 293.15 K and the first-law ledger closes (see the '[balance]' lines and check_dynamic_tank).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | balance | element_H_closure_rel | 6.61926901636e-08 |
| kpi | balance | element_O_closure_rel | 6.61926901636e-08 |
| kpi | balance | element_worst_closure_rel | 6.61926901636e-08 |
| kpi | balance | energy_balance_available | 1 |
| kpi | balance | energy_closure_rel | 6.61927451875e-08 |
| kpi | balance | energy_functional_H | 1 |
| kpi | balance | energy_kJ_final | -112451371.957 |
| kpi | balance | energy_kJ_heat_cum | 0 |
| kpi | balance | energy_kJ_in_cum | -2064809460.48 |
| kpi | balance | energy_kJ_initial | -50088123.8522 |
| kpi | balance | energy_kJ_out_cum | -2002446075.7 |
| kpi | balance | energy_residual_kJ | 136.675406478 |
| kpi | balance | mass_closure_rel | 6.61926901501e-08 |
| kpi | balance | mass_kg_final | 7064.01381482 |

*. . . and 13 more reference values in the case's `expected`.*

## unsteady06_reactor_fill_and_react

tutorials/unsteady/unsteady06_reactor_fill_and_react




**What it does:** A reactor that starts as a heel and FILLS while a first-order isomerisation runs: volume and level follow the holdup, n_A(t) = F/k + (n0 - F/k) exp(-kt) in closed form

**The story:** a reactor with a LEVEL fills from a heel while it reacts (C205, slice 2: dynamicCSTR with 'volumeMode variable').

The constant-volume dynamicCSTR cannot start empty: its volume is a number you type and F_out = F_in. Here the liquid inventory is the state, the volume is derived from it on the case's package and the level is that volume over a declared vertical cylinder (D = 0.5 m, H = 1.0 m):

dn_i/dt = ndot_in,i - F_out x_i + sum_r nu_i r_r V V = sum n_i MW_i / rho(T, P, x) h = V / (pi D^2 / 4)

The outflow is DECLARED: 'mode flow', F_out = 0, so the vessel only fills. Feed: pure isobutanol, F = 1.0e-3 kmol/s, k = 4.0e-4 1/s (constant: Ea = 0), starting from a 0.02 m heel. An isomerisation changes no moles, and a first-order law makes the extent rate k n_A independent of the volume, so the mole numbers have an EXACT closed form while the volume keeps growing:

n_A(t) = F/k + (n_A0 - F/k) exp(-k t) n_B(t) = n_A0 + n_B0 + F t - n_A(t) M(t) = M0 + MW F t (both isomers have the same MW) V(t) = M(t) / rho h(t) = V(t) / area c_A = n_A / V

check_dynamic_cstr_level recomputes these from the declared numbers and holds the run to them (the density is the run's own column: it moves a little with T, because the isomerisation is mildly exothermic).

ENERGY. Both species carry a formation datum, so the vessel takes the CANONICAL route (announced at start-up): it stores H on the elements datum, the heat of reaction is a consequence of it, the outlet term cancels, and the ledger CLAIMS mass and energy and closes. Compare ctrl22, the same reactor with a level loop and a temperature loop together.

*dynamicCSTR with volumeMode variable (D = 0.5 m, H = 1.0 m) starts from a heel (0.02 m of isobutanol at 300 K, an "empty" vessel — exactly zero is refused by name), is fed 1.0e-3 kmol/s of pure isobutanol, has no outflow, and isomerises it to 2-butanol with a first-order law, k = 4.0e-4 1/s (Ea = 0, so k cannot move with the small temperature rise). 1500 s, 5 s steps; runs under choupoSemiContinuous (no controller).*

*1. The closed form. The reaction changes no moles and its extent rate k·n_A does not depend on the volume, so the mole numbers are exact: n_A(t) = F/k + (n_A0 - F/k)·exp(-kt), n_B = n_A0 + n_B0 + F·t - n_A. The volume is V = M(t)/? with M(t) = M0 + MW·F·t, and the level is V over the 0.196 $m^2$ cross-section. The run reproduces all of them to ~4e-9 (check_dynamic_cstr_level recomputes them). 2. The rate uses the current volume. A first-order law cannot show it; the gate runs a second-order variant (r·V = k·$n_A^2$/V) against an independent integration — a vessel that kept the heel's volume in the rate would be off by 73 %. 3. The ledger claims mass and energy and closes (~1e-14): the canonical route is announced at start-up (the vessel stores H on the elements datum; the heat of reaction is a consequence of it). 4. Change outflow { F 0.0 kmol/s; } to 0.5 and the heel runs dry in the first step (RAN DRY); set endTime to 4000 and it overflows. Both are refused by name, never clamped.*

*The same reactor under a level loop and a temperature loop is ctrl22_reactor_level_and_temperature.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | balance | element_C_closure_rel | 6.33533060921e-15 |
| kpi | balance | element_H_closure_rel | 6.22014277995e-15 |
| kpi | balance | element_O_closure_rel | 6.33533060921e-15 |
| kpi | balance | element_worst_closure_rel | 6.33533060921e-15 |
| kpi | balance | energy_balance_available | 1 |
| kpi | balance | energy_closure_rel | 4.02406944735e-14 |
| kpi | balance | energy_functional_H | 1 |
| kpi | balance | energy_kJ_final | -512056.569424 |
| kpi | balance | energy_kJ_heat_cum | 0 |
| kpi | balance | energy_kJ_in_cum | -498064.493979 |
| kpi | balance | energy_kJ_initial | -13992.0754455 |
| kpi | balance | energy_kJ_out_cum | 0 |
| kpi | balance | energy_residual_kJ | -2.06055119634e-08 |
| kpi | balance | mass_closure_rel | 6.34047758513e-15 |
| *… and 17 more reference values in the case's `expected`.* | | | |

# Validation study: ion rejection in nanofiltration against Fernández de Labastida & Yaroshchuk (2021)

This section compares one transport law of Choupo's membrane module, the solution-diffusion-electromigration (SDEM) law, against one published data set: M. Fernández de Labastida and A. Yaroshchuk, *Nanofiltration of Multi-Ion Solutions: Quantitative Control of Concentration Polarization and Interpretation by Solution-Diffusion-Electro-Migration Model*, Membranes 11 (2021) 272, doi:10.3390/membranes11040272, published under CC BY 4.0 (reference [3] in the list that closes this section; the model itself is Yaroshchuk, Bruening and Licón Bernal 2013 [1], and the analytical trace-ion solution the paper fits with is Yaroshchuk and Bruening 2017 [2]). It is written so that the authors of that paper can check it. Every number in the tables is produced by the engine or read from the published figures by a script; none is typed. The section ends with the commands that regenerate it and with the points on which we would value the authors' reading.

## The model as Choupo implements it

The active layer is a slab of dimensionless thickness $x \in [0,1]$. Each ion $i$ has a membrane permeance $P_i$ (m/s), which already contains its partition coefficient (the solution-diffusion definition $P = DK/\delta$), a charge $z_i$, and a *virtual* concentration $c_i(x)$: the concentration of the solution that would be in local equilibrium with the membrane at $x$. The flux of each ion is Nernst–Planck on the virtual solution,

$$j_i \;=\; -P_i \left( \frac{\mathrm{d}c_i}{\mathrm{d}x} + z_i\, c_i\, \frac{\mathrm{d}\psi}{\mathrm{d}x} \right), \qquad \psi = \frac{F\,\varphi}{RT},$$

the virtual solution is electroneutral at every $x$, $\sum_i z_i c_i(x) = 0$, and no current flows, $\sum_i z_i j_i = 0$. The last two conditions fix the one electric field the ions share,

$$\frac{\mathrm{d}\psi}{\mathrm{d}x} \;=\; -\,\frac{\sum_i z_i\, j_i / P_i}{\sum_i z_i^2\, c_i},$$

so there is exactly one state variable beyond the uncoupled solution-diffusion law, the potential drop $\Delta\psi$ across the layer, and no new parameter. The boundary conditions are the wall concentrations $c_i(0) = c_{m,i}$ (from the film model, per ion) and the permeate condition $j_i = J_v\, c_{p,i}$ with $c_i(1) = c_{p,i}$, where $J_v$ is the transmembrane volume flux. This is the formulation of Yaroshchuk, Bruening and Licón Bernal [1], and it has a property the engine checks rather than assumes: for a single salt it reduces *exactly* to

$$R_s \;=\; \frac{J_v}{J_v + P_s}, \qquad P_s \;=\; \frac{(z_+ + |z_-|)\, P_+\, P_-}{z_+ P_+ + |z_-|\, P_-},$$

the ambipolar permeance, which is what makes per-ion permeances calibratable from single-salt tests.

What the implementation does *not* contain, so that the comparison is read for what it is: no fixed membrane charge (no Donnan partition, no dielectric exclusion — those effects are inside the permeances the authors fitted); ideal dilute virtual solutions; constant permeances; and, in these runs, a concentration-polarisation film that is computed per ion *without* a field (the default, which the engine announces every run). A coupled film — the interface model of Geraldes and Afonso (2007), one potential gradient at the wall fixed by electroneutrality, declared by `polarisation { ionCoupling electroneutral; }` — exists in the engine and is deliberately *not* used here: all runs below use a film coefficient of 1 m/s, so the wall equals the bulk to better than 0.01 % and the rejections are intrinsic — which is what the paper's CP-corrected rejections are, and a film that does not matter has no field to carry. Numerically, each station solves a Newton problem in the permeate composition whose residual is a fourth-order Runge–Kutta shot of the profile *from the permeate face back to the wall* (320 steps); the osmotic back-pressure closes an outer Newton in $J_v$. The source is `src/unitOperations/membrane/transport/SDEM.cpp`.

## What was taken from the paper

**The permeances.** Table 1 of the paper gives, for each dominant salt, the membrane permeance to the salt $P_s$ fitted with Eq. (1) and the ion permeances fitted with the SDEM trace solution (its Eq. (3), the analytical solution of [2]), all in $\mu$m/s. Choupo uses the ion values as they stand. Where the paper gives a *bound* or a *range* the value used is stated in the case record: $Na^+$ under NaCl "20–113" is taken at 113, the end that reproduces $P_s = 7.5$ through the ambipolar identity; $NH_4^+$ "> 60" (NaCl), "> 700" ($MgCl_2$), "> 100" ($Na_2SO_4$), $Na^+$ "> 250" ($MgCl_2$) and "> 0.8" ($Na_2SO_4$) are taken at the bound. The consequences of the bounds are measured below rather than assumed away.

**The feeds.** Dominant salt 0.01 mol/L, the two trace salts $NaNO_3$ and $NH_4Cl$ at $2 \times 10^{-4}$ mol/L each, 20 °C, on a flat NF270 disc in the rotating-disk cell, transmembrane pressure 2–14 bar.

**The figures.** Figures 1–4 plot, for each dominant salt, the reciprocal intrinsic transmission $f = 1/(1-R)$ of the salt (panel a) and of the trace ions (panel b) against $J_v$. The *symbols* are the measurements after the

authors' CP correction; the *lines* are the authors' fits — Eq. (1) for the salt, the analytical trace-limit solution Eq. (3) for a trace — with the Table 1 permeances. The publisher's full-resolution panels were digitised by `bin/curate/digitise_fdl2021.py`: the axes are calibrated from the axis lines and verified against the tick labels (they map back to $0, 10, 20, \ldots$ within a few hundredths of a $\mu$m/s), markers are separated from lines by a morphological opening and located at the centre of their bounding box, the drawn line is read under each marker, and the three arrows of Fig. 3b are blanked as annotations. The reading error is about $\pm 0.07\,\mu$m/s in $J_v$ and $\pm 0.3\,\%$ of the axis span in $f$; a marker partly hidden under another series' symbol carries a larger error and is marked † in the tables. The 106 points live, with their reading errors, in the NaCl witness case:

`tutorials/steady/membranes/membrane12_sdem_nf270_nacl_traces/constant/experimental/fdl2021_figures.csv`

The images are not redistributed. One reading we ask the authors to confirm: the right-hand axis of Fig. 3b ($Cl^-$ and $NO_3^-$ under $Na_2SO_4$) is labelled 0 to 8 but carries nine equal tick intervals; it was read as 0 to 18.

## How Choupo was driven

For each digitised point the engine is run on a one-element NF270 case built from the shipped NaCl witness (`membrane12_sdem_nf270_nacl_traces`) with the salt's Table 1 row as the membrane record, and the feed pressure is adjusted until the engine's $J_v$ equals the figure's to 0.5 %. Two feeds are used for every salt. **Trace limit**: the trace salts at $2 \times 10^{-7}$ mol/L, one thousandth of the paper's, which is the limit the analytical Eq. (3) assumes (a trace that does not feed back on the field). **Paper's feed**: the trace salts at $2 \times 10^{-4}$ mol/L, the concentration actually used in the experiments. The comparison that tests the *implementation* is therefore Choupo-in-the-trace-limit against the authors' *lines* (same model, same parameters, one answer); the comparison that tests the *model* is Choupo-at-the-paper's-feed against the *symbols*, and it inherits the authors' own fit residual.

Figure 1 draws the whole comparison at once; the tables that follow carry the same points with their numbers. Read it series by series: where the solid line sits on the dashed one, Choupo in the trace limit reproduces the authors' line; where the dotted line leaves both, the paper's own trace concentration is no longer a trace for that ion (the section on that point below); and where the markers leave every line, the residual is the authors' fit against their own measurement, which no implementation of the same model can close.

## Table 1 against the ambipolar identity

The first check needs no run: the ion permeances of Table 1 must reproduce the salt permeances of the same table through the ambipolar identity. They do, to the table's rounding.

| salt | cation | $P_+$ | anion | $P_-$ | $P_s$ (Table 1) | $P_s$ ambipolar | dev. (%) |
|---|---|---|---|---|---|---|---|
| NaCl | $Na^+$ | 113 | $Cl^-$ | 3.9 | 7.5 | 7.54 | +0.5 |
| $MgCl_2$ | $Mg^{2+}$ | 3.7 | $Cl^-$ | 9.3 | 6.1 | 6.18 | +1.3 |
| $Na_2SO_4$ | $Na^+$ | 0.8 | $SO_4^{2-}$ | 0.06 | 0.16 | 0.16 | -2.2 |
| $MgSO_4$ | $Mg^{2+}$ | 0.9 | $SO_4^{2-}$ | 0.13 | 0.23 | 0.23 | -1.2 |

Permeances in $\mu$m/s. This is arm (b) of the gate `bin/curate/check_sdem.py`, recomputed in Python independently of the engine.

## Dominant NaCl (Figure 1)

**Table 1:** Dominant NaCl, Figure 1 of the paper. $f = 1/(1-R)$; $J_v$ in $\mu$m/s. "measured" and "paper's fit" are read from the figure (†: marker partly hidden, larger reading error; –: the drawn line stops before the point); "trace limit" and "paper's feed" are Choupo at the same flux with the trace salts at $2 \times 10^{-7}$ and $2 \times 10^{-4}$ mol/L; the last column is Choupo (trace limit) against the paper's fit.

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| $Na^+$ (dominant) | 8.2 | 1.758 | 2.121 | 2.092 | 2.060 | -1.4 |
| | 12.3 | 2.145 | 2.666 | 2.636 | 2.581 | -1.1 |
| | 20.2 | 3.726 | 3.706 | 3.677 | 3.573 | -0.8 |
| | 27.7 | 4.870 | 4.715 | 4.668 | 4.516 | -1.0 |
| | 36.2 | 5.980 | 5.841 | 5.772 | 5.567 | -1.2 |
| | 45.1 | 7.049 | 7.032 | 6.938 | 6.676 | -1.3 |
| | 55.8 | 6.275 | – | 8.336 | 8.007 | – |
| $Cl^-$ (dominant) | 8.2 | 1.758 | 2.121 | 2.092 | 2.109 | -1.4 |
| | 12.3 | 2.145 | 2.666 | 2.636 | 2.660 | -1.1 |
| | 20.2 | 3.726 | 3.706 | 3.677 | 3.710 | -0.8 |
| | 27.7 | 4.870 | 4.715 | 4.668 | 4.708 | -1.0 |

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| | 36.2 | 5.980 | 5.841 | 5.772 | 5.818 | -1.2 |
| | 45.1 | 7.049 | 7.032 | 6.938 | 6.988 | -1.3 |
| | 55.8 | 6.275 | – | 8.336 | 8.391 | – |
| $NH_4^+$ (trace) | 8.3 | 2.041 | 2.020 | 2.185 | 2.152 | +8.2 |
| | 12.3 | 2.396 | 2.481 | 2.775 | 2.719 | +11.9 |
| | 20.1 | 3.461 | 3.388 | 3.945 | 3.838 | +16.5 |
| | 27.8 | 4.361 | 4.260 | 5.076 | 4.919 | +19.2 |
| | 36.2 | 4.997 | 5.220 | 6.343 | 6.129 | +21.5 |
| | 45.1 | 5.533 | 6.219 | 7.687 | 7.414 | +23.6 |
| | 55.9 | 5.153 | – | 9.315 | 8.970 | – |
| $NO_3^-$ (trace) | 8.2 | 0.916 | 1.036 | 0.952 | 0.962 | -8.2 |
| | 12.4 | 0.984 | 1.118 | 1.036 | 1.049 | -7.3 |
| | 20.2 | 1.315 | 1.321 | 1.253 | 1.269 | -5.1 |
| | 27.7 | 1.526 | 1.547 | 1.491 | 1.508 | -3.6 |
| | 36.2 | 1.913 | 1.817 | 1.775 | 1.793 | -2.3 |
| | 45.1 | 2.143 | 2.113 | 2.083 | 2.102 | -1.4 |
| | 55.9 | 1.963 | 2.477 | 2.461 | 2.481 | -0.6 |

For the salt, Choupo's dominant anion follows the paper's line to within 1.5 % over the whole flux range ($Cl^-$ at $2 \times 10^{-4}$ M: the salt rejection is set by the ambipolar permeance, and the traces perturb it little here). For $NO_3^-$ the trace-limit engine sits within 8 % of the line, closest where the line is flattest; at the two lowest fluxes the engine gives a slightly *negative* rejection where the drawn line is slightly positive — and the measured symbols there are negative, as the paper's text says ("lightly negative values at small transmembrane fluxes"). $NH_4^+$ is the case where the permeance is a bound: at 60 $\mu$m/s the engine's $f$ is 8–24 % above the line, and the sensitivity table further down shows the line is met only near 1000 $\mu$m/s; the authors report that their fit "became insensitive to the value of ionic permeance as this increased", which is consistent with a fit that had already saturated.

## Dominant $MgCl_2$ (Figure 2)

**Table 2:** Dominant $MgCl_2$, Figure 2 of the paper. $f = 1/(1 - R)$; $J_v$ in $\mu$m/s. "measured" and "paper's fit" are read from the figure (†: marker partly hidden, larger reading error; –: the drawn line stops before the point); "trace limit" and "paper's feed" are Choupo at the same flux with the trace salts at $2 \times 10^{-7}$ and $2 \times 10^{-4}$ mol/L; the last column is Choupo (trace limit) against the paper's fit.

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| $Mg^{2+}$ (dominant) | 8.7 | 2.237 | 2.068 | 2.416 | 2.471 | +16.8 |
| | 13.3 | 2.757 | 2.603 | 3.156 | 3.267 | +21.2 |
| | 19.8 | 3.230 | 3.292 | 4.189 | 4.407 | +27.2 |
| | 28.4 | 4.040 | 4.143 | 5.562 | 5.975 | +34.3 |
| | 36.5 | 4.757 | 4.874 | 6.865 | 7.512 | +40.8 |
| | 44.4 | 5.665 | 5.520 | 8.129 | 9.043 | +47.3 |
| | 52.5 | 6.125 | 6.120 | 9.422 | 10.6 | +53.9 |
| $Cl^-$ (dominant) | 8.7 | 2.237 | 2.068 | 2.416 | 2.376 | +16.8 |
| | 13.3 | 2.757 | 2.603 | 3.156 | 3.072 | +21.2 |
| | 19.8 | 3.230 | 3.292 | 4.188 | 4.013 | +27.2 |
| | 28.4 | 4.040 | 4.143 | 5.562 | 5.220 | +34.2 |
| | 36.5 | 4.757 | 4.874 | 6.864 | 6.325 | +40.8 |
| | 44.4 | 5.665 | 5.520 | 8.127 | 7.365 | +47.2 |
| | 52.5 | 6.125 | 6.120 | 9.419 | 8.403 | +53.9 |
| $NH_4^+$ (trace) | 8.4 | 0.738† | 0.798 | 0.760 | 0.776 | -4.7 |
| | 13.3 | 0.684 | 0.694 | 0.698 | 0.725 | +0.5 |
| | 19.8 | 0.604 | 0.623 | 0.643 | 0.686 | +3.2 |
| | 28.4 | 0.548 | 0.567 | 0.597 | 0.662 | +5.3 |
| | 36.5 | 0.532 | 0.529 | 0.568 | 0.654 | +7.4 |
| | 44.5 | 0.485 | 0.500 | 0.549 | 0.654 | +9.6 |
| | 52.5 | 0.476 | 0.478 | 0.534 | 0.658 | +11.7 |
| $Na^+$ (trace) | 19.8 | 0.724 | 0.695 | 0.685 | 0.730 | -1.4 |
| | 28.4 | 0.679 | 0.668 | 0.656 | 0.724 | -1.8 |
| | 36.5 | 0.671 | 0.657 | 0.644 | 0.733 | -2.0 |

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| | 44.4 | 0.640 | 0.656 | 0.639 | 0.750 | -2.5 |
| | 52.5 | 0.651 | 0.661 | 0.641 | 0.772 | -3.0 |
| $NO_3^-$ (trace) | 8.7 | 1.603 | 1.891 | 1.933 | 1.899 | +2.2 |
| | 13.3 | 1.856 | 2.309 | 2.396 | 2.325 | +3.8 |
| | 19.7 | 2.830 | 2.887 | 3.029 | 2.886 | +4.9 |
| | 28.4 | 3.421 | 3.650 | 3.859 | 3.589 | +5.7 |
| | 36.5 | 4.243 | 4.354 | 4.636 | 4.220 | +6.5 |
| | 44.4 | 5.063 | 5.025 | 5.381 | 4.805 | +7.1 |
| | 52.5 | 6.093 | 5.674 | 6.136 | 5.382 | +8.1 |

The three trace lines are reproduced in the trace limit within 1–8 % ($NO_3^-$ +2 to +8 %, $Na^+$ −1 to −3 %, $NH_4^+$ −5 to +12 %), with the negative rejections of both trace cations coming out of the same permeances and no parameter for the effect. The salt panel is different, and it is a point for the authors: the drawn line of Fig. 2a reads $f_s = 6.1$ at 52.5 $\mu$m/s and 5.5 at 44.4 $\mu$m/s, whereas Eq. (1) with the Table 1 value $P_s = 6.1\,\mu$m/s gives 9.6 and 8.3 (off the top of the plotted axis); the measured points follow the drawn line, and read on their own they imply an apparent salt permeance rising from about 7 to 10 $\mu$m/s across the range. Choupo, which solves the same equations as Eq. (1) for the dominant ions, agrees with Eq. (1) and the ambipolar identity (both dominant ions, up to the trace coupling discussed below) and therefore *not* with the drawn line. We have not tried to reconcile the two and would welcome the authors' reading of which curve Fig. 2a shows.

## Dominant $Na_2SO_4$ (Figure 3)

**Table 3:** Dominant $Na_2SO_4$, Figure 3 of the paper. $f = 1/(1-R)$; $J_v$ in $\mu$m/s. "measured" and "paper's fit" are read from the figure (†: marker partly hidden, larger reading error; –: the drawn line stops before the point); "trace limit" and "paper's feed" are Choupo at the same flux with the trace salts at $2 \times 10^{-7}$ and $2 \times 10^{-4}$ mol/L; the last column is Choupo (trace limit) against the paper's fit.

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| $Na^+$ (dominant) | 6.4 | 52.2 | 39.8 | 42.0 | 22.8 | +5.5 |
| | 10.5 | 77.2 | 65.1 | 67.6 | 34.9 | +3.9 |
| | 18.9 | 122.8 | 117.0 | 121.2 | 60.8 | +3.6 |
| | 27.2 | 160.2 | 168.4 | 174.0 | 86.6 | +3.4 |
| | 33.6 | 214.7 | 207.3 | 214.3 | 106.4 | +3.3 |
| | 41.8 | 261.7 | 257.3 | 265.5 | 131.8 | +3.2 |
| | 49.5 | 285.0 | 304.7 | 314.0 | 156.0 | +3.0 |
| $SO_4^{2-}$ (dominant) | 6.4 | 52.2 | 39.8 | 42.0 | 49.6 | +5.7 |
| | 10.5 | 77.2 | 65.1 | 67.7 | 80.3 | +4.0 |
| | 18.9 | 122.8 | 117.0 | 121.4 | 144.3 | +3.7 |
| | 27.2 | 160.2 | 168.4 | 174.3 | 207.0 | +3.5 |
| | 33.6 | 214.7 | 207.3 | 214.6 | 254.5 | +3.5 |
| | 41.8 | 261.7 | 257.3 | 265.9 | 315.0 | +3.3 |
| | 49.5 | 285.0 | 304.7 | 314.4 | 372.0 | +3.2 |
| $NH_4^+$ (trace) | 6.5 | 44.8 | 42.5 | 20.4 | 7.475 | -51.9 |
| | 10.5 | 50.9 | 48.9 | 29.9 | 9.290 | -38.7 |
| | 18.9 | 63.8 | 61.1 | 48.0 | 12.7 | -21.5 |
| | 27.2 | 71.2 | 72.1 | 64.0 | 15.6 | -11.2 |
| | 33.6 | 77.8 | 80.3 | 76.0 | 17.7 | -5.4 |
| | 41.7 | 89.0 | 90.0 | 90.2 | 20.1 | +0.2 |
| | 49.5 | 89.5 | 99.1 | 103.4 | 22.4 | +4.3 |
| $Cl^-$ (trace) | 6.5 | 1.337 | 1.113 | 1.127 | 1.349 | +1.2 |
| | 10.5 | 1.972 | 1.663 | 1.761 | 2.061 | +5.9 |
| | 18.9 | 3.384 | 2.896 | 3.119 | 3.587 | +7.7 |
| | 27.3 | 4.429 | 4.130 | 4.467 | 5.098 | +8.1 |
| | 33.6 | 6.735 | 5.094 | 5.490 | 6.244 | +7.8 |
| | 41.7 | 7.354 | 6.274 | 6.796 | 7.705 | +8.3 |
| | 49.5 | 7.036 | 7.422 | 8.040 | 9.096 | +8.3 |
| $NO_3^-$ (trace) | 6.4 | 0.501 | 0.514 | 0.423 | 0.557 | -17.7 |
| | 10.5 | 0.719 | 0.692 | 0.634 | 0.788 | -8.3 |
| | 18.9 | 1.137 | 1.149 | 1.098 | 1.299 | -4.5 |

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| | 27.2 | 1.437 | 1.613 | 1.561 | 1.812 | -3.2 |
| | 33.6 | 2.089 | 1.965 | 1.918 | 2.209 | -2.4 |
| | 41.7 | 2.524 | 2.418 | 2.371 | 2.712 | -2.0 |
| | 49.5 | 2.808 | 2.842 | 2.800 | 3.190 | -1.5 |

Here the traces are, in the authors' words, not authentic traces: sulfate passes at 0.06 $\mu$m/s, the trace anions at 3.4 and 9.8, so the anion current is carried by the traces. In the trace limit Choupo reproduces the $Cl^-$ line within +1 to +8 % and the $NO_3^-$ line within −2 to −8 % from 10 $\mu$m/s upward (−18 % at the lowest point, where $f \approx 0.5$ and the drawn line is thick) (the reversal from negative to positive rejection of $NO_3^-$ falls at about 16 $\mu$m/s in both); at the paper's own feed the engine's $f$ for the trace anions is 8–24 % above the lines and its $f$ for the dominant $Na^+$ well below that of $SO_4^{2-}$ — the effect the paper's text predicts ("the reciprocal transmission of the more permeable ion (such as $Na^+$ in $Na_2SO_4$) could be basically lower than that for $SO_4^{2-}$"). $NH_4^+$ is not a test of the implementation: its own permeance is a bound and, as the sensitivity table shows, its rejection is set mostly by the $Na^+$ permeance, which the paper calls "just orientative".

## Dominant $MgSO_4$ (Figure 4)

**Table 4:** Dominant $MgSO_4$, Figure 4 of the paper. $f = 1/(1-R)$; $J_v$ in $\mu$m/s. "measured" and "paper's fit" are read from the figure (†: marker partly hidden, larger reading error; –: the drawn line stops before the point); "trace limit" and "paper's feed" are Choupo at the same flux with the trace salts at $2 \times 10^{-7}$ and $2 \times 10^{-4}$ mol/L; the last column is Choupo (trace limit) against the paper's fit.

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| $Mg^{2+}$ (dominant) | 7.2 | 42.9 | 32.6 | 32.8 | 20.1 | +0.5 |
| | 11.6 | 53.3 | 52.9 | 51.8 | 28.6 | -2.1 |
| | 19.2 | 92.6 | 87.0 | 85.0 | 43.1 | -2.3 |
| | 26.9 | 120.5 | 121.6 | 118.5 | 58.0 | -2.6 |
| | 34.7 | 152.3 | 157.0 | 152.6 | 73.6 | -2.8 |
| | 43.1 | 184.3 | 194.8 | 189.0 | 90.7 | -3.0 |
| | 53.8 | 200.5 | 242.2 | 235.0 | 113.0 | -3.0 |
| $SO_4^{2-}$ (dominant) | 7.2 | 42.9 | 32.6 | 32.8 | 36.9 | +0.6 |
| | 11.6 | 53.3 | 52.9 | 51.9 | 59.9 | -1.9 |
| | 19.2 | 92.6 | 87.0 | 85.2 | 100.2 | -2.1 |
| | 26.9 | 120.5 | 121.6 | 118.7 | 141.0 | -2.4 |
| | 34.7 | 152.3 | 157.0 | 153.0 | 182.2 | -2.6 |
| | 43.1 | 184.3 | 194.8 | 189.5 | 225.6 | -2.7 |
| | 53.8 | 200.5 | 242.2 | 235.6 | 279.8 | -2.7 |
| $NH_4^+$ (trace) | 7.2 | 3.838† | – | 3.804 | 2.503 | – |
| | 19.2 | 5.212 | 5.429 | 5.618 | 2.838 | +3.5 |
| | 53.8 | 7.651 | 8.416 | 8.686 | 3.825 | +3.2 |
| $Na^+$ (trace) | 7.2 | 3.427 | 3.643 | 3.815 | 2.506 | +4.7 |
| | 11.4 | 4.554 | 4.345 | 4.563 | 2.634 | +5.0 |
| | 19.2 | 5.698 | 5.358 | 5.613 | 2.838 | +4.8 |
| | 26.9 | 6.080 | 6.161 | 6.459 | 3.049 | +4.8 |
| | 34.8 | 6.838† | 6.898 | 7.194 | 3.271 | +4.3 |
| | 43.1 | 7.408 | 7.543 | 7.891 | 3.515 | +4.6 |
| | 53.8 | 7.073 | 8.316 | 8.693 | 3.832 | +4.5 |
| $Cl^-$ (trace) | 7.2 | 0.712 | 0.660 | 0.615 | 0.788 | -6.8 |
| | 11.6 | 0.825 | 0.817 | 0.774 | 1.003 | -5.3 |
| | 19.2 | 1.215 | 1.127 | 1.088 | 1.386 | -3.5 |
| | 26.9 | 1.492 | 1.459 | 1.420 | 1.771 | -2.7 |
| | 34.7 | 1.827 | 1.807 | 1.767 | 2.163 | -2.2 |
| | 43.1 | 2.137 | 2.185 | 2.141 | 2.577 | -2.0 |
| | 53.8 | 2.313 | 2.665 | 2.618 | 3.099 | -1.8 |
| $NO_3^-$ (trace) | 7.2 | 0.419 | 0.440 | 0.401 | 0.558 | -8.9 |
| | 11.6 | 0.478 | 0.471 | 0.434 | 0.631 | -7.9 |
| | 19.2 | 0.587 | 0.559 | 0.528 | 0.767 | -5.7 |
| | 26.9 | 0.687 | 0.668 | 0.640 | 0.904 | -4.2 |
| | 34.7 | 0.788 | 0.785 | 0.761 | 1.042 | -3.1 |

| ion | $J_v$ | $f$ measured | $f$ paper's fit | $f$ Choupo, trace limit | $f$ Choupo, paper's feed | dev. (%) |
|---|---|---|---|---|---|---|
| | 43.1 | 0.913 | 0.916 | 0.894 | 1.189 | -2.4 |
| | 53.8 | 0.955 | 1.096 | 1.067 | 1.374 | -2.7 |

This is the case where the paper's fit determines every permeance, and in the trace limit the engine reproduces all four trace lines within 2–9 % ($Na^+$ +4 to +5 %; $Cl^-$ −2 to −7 %; $NO_3^-$ −2 to −9 %; $NH_4^+$, whose line lies on $Na^+$'s, +3 % on the points visible under the $Na^+$ diamonds). At the paper's own feed the engine departs from the lines by 20–55 %, all in one direction: the trace cations are rejected less and the trace anions less negatively. Both are the signature of a *weaker* field, and the next subsection measures why.

## A trace is not a trace when the dominant anion is a hundred times slower

The authors' Eq. (3), from [2], is the analytical solution for a trace ion in the field set by the dominant salt alone. At $2 \times 10^{-4}$ mol/L against 0.01 mol/L that assumption holds within 3 % for NaCl and within 15 % for $MgCl_2$; under a sulfate it fails, because $2 \times 10^{-4}$ mol/L of $NO_3^-$ at 42 $\mu$m/s carries more anion current than 0.01 mol/L of $SO_4^{2-}$ at 0.13 $\mu$m/s. The traces then relieve the field, and every rejection moves toward its uncoupled value. The table gives, for each salt at its fifth measured flux, the ratio of the engine's $f$ at the paper's feed to its $f$ in the trace limit, per trace ion.

| salt | $J_v$ ($\mu$m/s) | $Na^+$ | $NH_4^+$ | $Cl^-$ | $NO_3^-$ |
|---|---|---|---|---|---|
| NaCl | 36.2 | – | 0.97 | – | 1.01 |
| $MgCl_2$ | 36.5 | 1.14 | 1.15 | – | 0.91 |
| $Na_2SO_4$ | 33.6 | – | 0.23 | 1.14 | 1.15 |
| $MgSO_4$ | 34.7 | 0.45 | 0.45 | 1.22 | 1.37 |

A ratio of 1 means the trace does not feed back on the field; a dash means the ion is a dominant ion of that salt. The paper's own text names the effect ("trace ions cannot be accounted as authentic traces due to the very high rejections of $SO_4^{2-}$"); the table is its size. For a student it is the lesson of these two cases, and it is why the shipped $MgSO_4$ witness, run at the paper's concentrations, does not reproduce the paper's lines while the trace-limit run does.

## The two permeances the paper leaves as bounds

| salt | permeance swept | value ($\mu$m/s) | $J_v$ ($\mu$m/s) | $f$ (Choupo, trace limit) | $f$ (paper's line) | dev. (%) |
|---|---|---|---|---|---|---|
| NaCl | $NH_4^+$ | 60 | 36.2 | 6.343 | 5.220 | +21.5 |
| NaCl | $NH_4^+$ | 120 | 36.2 | 5.741 | 5.220 | +10.0 |
| NaCl | $NH_4^+$ | 300 | 36.2 | 5.379 | 5.220 | +3.1 |
| NaCl | $NH_4^+$ | 1000 | 36.2 | 5.211 | 5.220 | -0.2 |
| $Na_2SO_4$ | $Na^+$ (sets $NH_4^+$) | 0.8 | 27.2 | 64.0 | 72.1 | -11.2 |
| $Na_2SO_4$ | $Na^+$ (sets $NH_4^+$) | 3 | 27.2 | 118.4 | 72.1 | +64.3 |
| $Na_2SO_4$ | $Na^+$ (sets $NH_4^+$) | 10 | 27.2 | 141.1 | 72.1 | +95.6 |
| $Na_2SO_4$ | $Na^+$ (sets $NH_4^+$) | 30 | 27.2 | 148.5 | 72.1 | +106.0 |

Under NaCl the $NH_4^+$ line is met when its permeance is of the order of 1000 $\mu$m/s, an order of magnitude above the stated bound of 60; under $Na_2SO_4$ the $NH_4^+$ reciprocal transmission is a strong function of the $Na^+$ permeance (0.8 → 30 $\mu$m/s moves it from 64 to 149 at 27 $\mu$m/s), so a value for $NH_4^+$ cannot be read off that figure without one for $Na^+$. Neither series is held by the regression gate, which says so in its own claim.

## Points on which we would value the authors' reading

1. Which curve Fig. 2a shows: Eq. (1) with $P_s = 6.1\,\mu$m/s does not pass through the drawn line or the measured points (previous subsection on $MgCl_2$).
2. The right-hand axis of Fig. 3b, read here as 0–18 from its tick marks while its labels stop at 8.
3. Whether the lines of the (b) panels were computed from Eq. (3) (the trace limit) or from a full multi-ion solution at $2 \times 10^{-4}$ M; the tables are consistent with the former, and the difference is the size of the "not a trace" table.
4. The $NH_4^+$ permeance under NaCl: an effective value near 1000 $\mu$m/s reproduces the line, the stated bound of 60 does not.

## What this study does not establish

It does not validate the model against the measured *symbols*: the symbols scatter about the authors' lines by the authors' own fit residual (up to about 15 % in $f$ on the last point of Fig. 1a), and a re-implementation cannot be held to a residual its source accepted. It does not exercise the coupled polarisation film (the runs are intrinsic; that

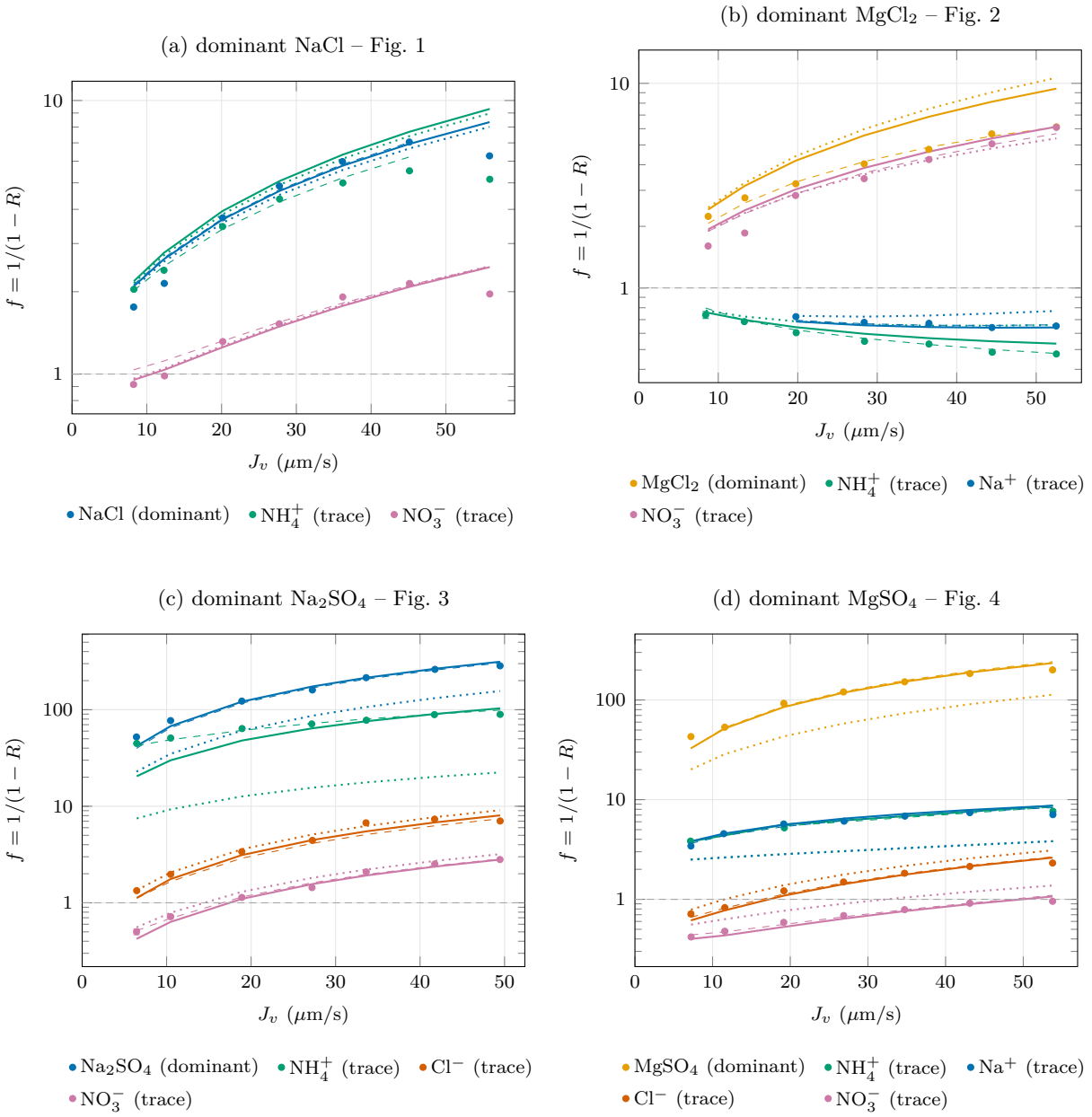



**Figure 1:** Choupo against the paper, every digitised point of Figures 1–4. $f = 1/(1-R)$ on a log axis; below the grey line ($f = 1$) a rejection is negative. Markers: the measured points, the bar being the reading error of the digitisation; dashed: the paper's fitted line (Eq. (1) for the salt, Eq. (3) for a trace); solid: Choupo at the same flux in the trace limit (trace salts at $2 \times 10^{-7}$ mol/L); dotted: Choupo at the paper's feed ($2 \times 10^{-4}$ mol/L). The dominant salt is drawn through its cation; in the trace limit its two ions coincide exactly. One colour per ion across the four panels.

film is the subject of tutorial `membrane14` and of the gate `check_polarisation_coupling`), concentration-dependent permeances, fixed charge, or any membrane but NF270 at 0.01 M. It inherits the digitisation's reading error, stated above and per point in the CSV.

## Reproducing this section

The design record of the law and of this comparison is `docs/design/electroneutrality-without-a-new-parameter.md`; the commands below regenerate every number above.

```
runCase tutorials/steady/membranes/membrane12_sdem_nf270_nacl_traces   # NaCl witness
runCase tutorials/steady/membranes/membrane13_sdem_nf270_mgso4_traces  # MgSO4 witness
bin/curate/digitise_fdl2021.py --download   # the figures -> the CSV (images stay local)
bin/curate/gen_sdem_validation.py           # every table of this section, from the engine
bin/curate/check_sdem.py                    # the regression gate (arms (h)-(j) hold the lines)
```

## References

[1] A. Yaroshchuk, M. L. Bruening and E. E. Licón Bernal, "Solution-diffusion–electro-migration model and its uses for analysis of nanofiltration, pressure-retarded osmosis and forward osmosis in multi-ionic solutions", *Journal of Membrane Science* 447 (2013) 463–476, doi:10.1016/j.memsci.2013.07.047. The model Choupo implements as `transport SDEM;`.

[2] A. Yaroshchuk and M. L. Bruening, "An analytical solution of the solution-diffusion-electromigration equations reproduces trends in ion rejections during nanofiltration of mixed electrolytes", *Journal of Membrane Science* 523 (2017) 361–372, doi:10.1016/j.memsci.2016.09.046. The trace-ion solution behind the fitted lines of [3] (its Eq. (3)).

[3] M. Fernández de Labastida and A. Yaroshchuk, "Nanofiltration of Multi-Ion Solutions: Quantitative Control of Concentration Polarization and Interpretation by Solution-Diffusion-Electro-Migration Model", *Membranes* 11 (2021) 272, doi:10.3390/membranes11040272, CC BY 4.0. The permeances (Table 1) and the rejection figures this section digitises and compares against.

# Category III: The property bench (choupoProps)

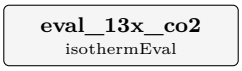
Questions asked of the thermodynamics BEFORE a simulation uses it – CREATE, SEE, TEST, FIT, PROMOTE.

## adsorption

**The theory you need.** The isotherm bench: evaluating a curated adsorption-equilibrium record (Henry-limit, saturation, pin and unit-invariance gates run aloud) and REGRESSING one from data – the multi-temperature van't Hoff Langmuir fit with identifiability verdicts: single-$T$ data refuses $\Delta H_{\mathrm{ads}}$, Henry-regime data yields only the product $q_{\max}b$, and a non-converged fit refuses its verdict by name. *Full derivation: Theory Guide, the adsorption chapter.*

## isotherm01_eval_13x_co2

tutorials/props/adsorption/isotherm01_eval_13x_co2




**What it does:** Adsorption isotherm bench: CO2/zeolite13X q(T,p) grid + Henry/saturation/pin/unit-invariance gates

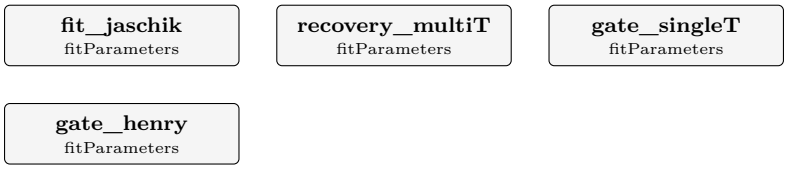
**The story:** One isothermEval operation: the curated CO2/zeolite13X langmuir record on a 3-T x 4-p grid (K, Pa – grid values are RAW canonical SI), with the unit-invariance twin (gate d) declared case-locally in constant/. Temperatures match the Jaschik 2020 dataset the FIT tutorial (isotherm02_fit_jaschik) regresses – eval here, fit there.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | eval_13x_co2 | T0 | 293.15 |
| diag | eval_13x_co2 | T1 | 313.15 |
| diag | eval_13x_co2 | T2 | 333.15 |
| diag | eval_13x_co2 | b_T0 | 12.2661169044 |
| diag | eval_13x_co2 | b_T1 | 3.77235323402 |
| diag | eval_13x_co2 | b_T2 | 1.33660410787 |
| diag | eval_13x_co2 | henry_T0 | 0.000613305845222 |
| diag | eval_13x_co2 | henry_T1 | 0.000188617661701 |
| diag | eval_13x_co2 | henry_T2 | 6.68302053933e-05 |
| diag | eval_13x_co2 | qsat | 5 |
| diag | eval_13x_co2 | tRef | 298.15 |
| diag | eval_13x_co2 | gate_henry | 1 |
| diag | eval_13x_co2 | gate_henry_maxdev | 1.22661154271e-07 |
| diag | eval_13x_co2 | gate_saturation | 1 |
| *... and 9 more reference values in the case's `expected`.* | | | |

## isotherm02_fit_jaschik

tutorials/props/adsorption/isotherm02_fit_jaschik




**What it does:** Isotherm regression: CO2/13X Jaschik 2020 (CC BY) multi-T Langmuir fit + recovery/single-T/Henry identifiability gates

**The story:** Four fitParameters(kind isotherm) operations: the REAL Jaschik fit plus the three identifiability gates on zero-noise synthetics generated from a KNOWN surface (q_max 4.6 mol/kg, b_ref 3.0e-4 1/Pa @ 298.15 K, dH_ads -38000 J/mol) – so every gate has an exact truth to recover or to honestly refuse. Inline synthetic data is canonical SI (K, Pa, mol/kg); the vendored Jaschik dataset declares its own units (bar, mol/kg) and is converted at the data boundary.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_jaschik | converged | 1.00000e+00 |
| diag | fit_jaschik | identifiable | 1.00000e+00 |
| diag | fit_jaschik | n_data | 1.32000e+02 |
| diag | fit_jaschik | n_params | 3.00000e+00 |
| diag | fit_jaschik | dof | 1.29000e+02 |
| diag | fit_jaschik | single_T | 0.00000e+00 |
| diag | fit_jaschik | henry_regime | 0.00000e+00 |
| diag | fit_jaschik | endothermic_suspicious | 0.00000e+00 |
| diag | fit_jaschik | fit.q_max.value | 4.31448e+00 |
| diag | fit_jaschik | fit.q_max.stderr | 5.17246e-02 |
| diag | fit_jaschik | fit.b_ref.value | 4.06758e-04 |
| diag | fit_jaschik | fit.b_ref.stderr | 2.90989e-05 |
| diag | fit_jaschik | fit.dH_ads.value | -3.73835e+04 |
| diag | fit_jaschik | fit.dH_ads.stderr | 1.90453e+03 |
| *... and 69 more reference values in the case's `expected`.* | | | |

### isotherm03_fit_refusals

tutorials/props/adsorption/isotherm03_fit_refusals




**What it does:** Fit refusal gates: non-converged fits refuse their verdicts by name

**The story:** Two ADVERSARIAL fitParameters(kind isotherm) operations – see the controlDict header. Synthetic data generated EXACTLY from the known surface (q_max 4.6 mol/kg, b_298 3.0e-4 1/Pa, dH_ads -38000 J/mol, tRef 298.15 K) – zero noise, so the ONLY failure mode under test is the strangled optimiser, never the data.

## compare

**The theory you need.** Method comparison runs the SAME property through different models side-by-side – UNIFAC vs NRTL, Joback vs measured, ideal vs EOS. The gap between curves IS the modelling uncertainty; picking a method is a decision the case author owns, made with eyes open. *Full derivation: Theory Guide, the Props Guide.*

## aceticWater01_txy_association

tutorials/props/compare/aceticWater01_txy_association




**What it does:** Acetic acid/water T-x-y at 1 atm on UNIFAC with the ASSOCIATED ideal-gas vapour (monomer + dimer), against Calvar, Dominguez & Tojo (2005)

**The story:** WHAT A DIMERISED VAPOUR DOES TO A T-x-y DIAGRAM. Acetic acid's vapour is largely dimers near its boiling point, so the acid is LESS volatile than its vapour pressure alone suggests: the dimers take up partial pressure that the monomer – the species in equilibrium with the liquid – does not. This case evaluates the bubble temperature and the vapour composition at the 24 liquid compositions Calvar, Dominguez & Tojo measured at 101.325 kPa (constant/experimental/calvar2005_txy.csv), plus the two pure boiling points, on UNIFAC (predictive, no fitted pair) and the associated ideal-gas vapour. The gate check_reference_rung arm (k) holds the pure limit and the latent heat at Tb and prints the comparison with these measurements; the numbers are recorded in docs/design/one-record-two-vapours-acetic-acid.md section 7. Nothing here is tuned.

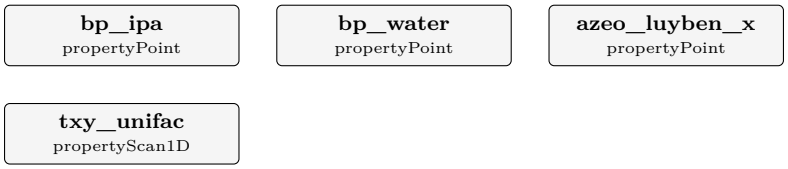
WHAT IS NOT IN IT: no regressed acetic-acid/water pair (the catalogue holds none); the measured points were read from the NIST ThermoML archive copy of the article and NOT checked against the article itself.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bp_aceticAcid | Cp_ig | 80.6641982498 |
| diag | bp_aceticAcid | H_R | -22841.7769643 |
| diag | bp_aceticAcid | H_ig | -425398.226035 |
| diag | bp_aceticAcid | H_real | -448240.002999 |
| diag | bp_aceticAcid | P | 101325 |
| diag | bp_aceticAcid | S_R | -51.4737821304 |
| diag | bp_aceticAcid | S_ig | 302.310579052 |
| diag | bp_aceticAcid | S_real | 250.836796922 |
| diag | bp_aceticAcid | T | 391 |
| diag | bp_aceticAcid | T_bubble | 391.008522453 |
| diag | bp_aceticAcid | Z | 1 |
| diag | bp_aceticAcid | gamma | 1.11492042846 |
| diag | bp_aceticAcid | v_molar | 0.0320844301371 |
| diag | bp_water | Cp_ig | 34.2425437543 |
| *... and 15 more reference values in the case's `expected`.* | | | |

## acetone01_ipa_water_azeotrope

tutorials/props/compare/acetone01_ipa_water_azeotrope




**What it does:** IPA/water T-x-y at 1 atm from PREDICTIVE UNIFAC on a lake-estimated isopropanol, held against Luyben's published azeotrope

**The story:** THE FIRST RUNNABLE SLICE of the acetone-from-2-propanol reference case (docs/design/acetone-ipa-reference-case.md). It answers ONE question, and the question is chosen because it can be answered honestly today:

how close does a component Choupo has never curated, priced by a model that saw no data for it, land on a published azeotrope?

WHAT IS BEING TESTED, and it is a stack of three things at once – said up front, because a single number that agrees can hide two errors that cancel:

1. ISOPROPANOL IS A LAKE ESTIMATE. It is NOT in data/standards/. The record in this case's constant/components/ came from data/groupEstimative/, where every NUMBER is a Joback / Lee-Kesler estimate from the molecular structure – including Tb, Tc, Pc, omega and hence the whole Ambrose-Walton vapour pressure. Its estimated Tb is 359.98 K; the measured normal boiling point Luyben quotes is 355.4 K. The pure component is already 4.6 K out before any mixture model is asked anything. 2. UNIFAC IS PREDICTIVE. No pair was fitted, because Choupo ships no UNIQUAC or NRTL pair for IPA-water and inventing one would convert 'unsourced' into 'falsely sourced'. gamma comes from the groups. 3. WATER IS CURATED. It is the one leg of the three that rests on reviewed data, which is what makes the comparison legible at all.

THE COMPARISON POINTS: W. L. Luyben, "Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol", Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a. He states the IPA/water azeotrope as 67.32 mol % IPA at 353.4 K and 1 atm, and normal boiling points IPA 355.4 K, water 373 K. Those numbers come out of his UNIQUAC model, fitted to VLE data neither this case nor its components have ever seen – so for the predictive route here they are an INDEPENDENT target.

A NOTE ON WHAT KIND OF ROW PINS THIS, because it is not an 'anchor'. Luyben's azeotrope is his UNIQUAC model's value, not a measurement, and the AAD computed here is derived from that number and this model TOGETHER – so nobody published it and no primary can be quoted for it. It is pinned as a self-recorded 'aad' row: what the suite holds is that the agreement has not MOVED. Anchoring properly needs a bubble temperature published as an operation diagnostic, which is a separate gap – see the note at the end.

WHAT THIS CASE IS NOT. It is not a reproduction of Luyben's flowsheet, and it makes no claim about the acetone process. The full reproduction needs three curated UNIQUAC pairs that do not exist; the design record's section 4 lists them and section 5 says which case category that would be. This is the bottom rung: the thermodynamics, alone, against three published points.

A GAP FOUND BUILDING THIS, and left visible rather than worked around. The natural shape for this case is three 'propertyPoint' ops asking for 'T_bubble' at the three compositions, because a propertyPoint publishes into the operation diagnostics where a golden row can read it. It does not work: 'PropertyPoint' never reads its own 'properties ( ... )' list – it emits a fixed set and discards what the case asked for. Both the class header ("properties (all); // or pick a subset") and the published JSON schema (which names 'T_bubble' among the accepted keywords) say otherwise. That is a parsed-and-dropped field, the shape this project has paid for three times, and it is why the comparison below rides the overlay instead.

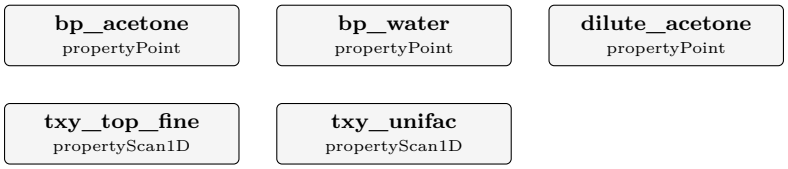
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bp_ipa | Cp_ig | 99.085675 |
| diag | bp_ipa | H_R | 0 |
| diag | bp_ipa | H_ig | -257924.667357 |
| diag | bp_ipa | H_real | -257924.667357 |
| diag | bp_ipa | P | 101325 |
| diag | bp_ipa | S_R | 0 |
| diag | bp_ipa | S_ig | 329.037676662 |
| diag | bp_ipa | S_real | 329.037676662 |
| diag | bp_ipa | T | 350 |
| diag | bp_ipa | T_bubble | 359.907520874 |
| diag | bp_ipa | Z | 1 |
| diag | bp_ipa | gamma | 1.09159801219 |
| diag | bp_ipa | v_molar | 0.0287200781278 |
| diag | bp_water | Cp_ig | 34.055209 |

*. . . and 30 more reference values in the case's `expected`.*

## acetone04_acetone_water_vle

tutorials/props/compare/acetone04_acetone_water_vle




**What it does:** Acetone/water T-x-y at 1 atm on PREDICTIVE UNIFAC – both components curated, so this isolates the mixture model

**The story:** MEASURE THE THERMODYNAMICS BEFORE BUILDING THE UNIT THAT RESTS ON IT.

Luyben's absorber recovers acetone from the reactor's hydrogen-rich offgas by washing it with water, and it is the next unit of the flowsheet to build (docs/design/acetone-ipa-reference-case.md). What that unit costs, and whether it can work at all, is set by ONE thing: how volatile acetone is out of a dilute aqueous solution. So that is measured here first – the same order that made acetone01 useful before acetone03 was written.

WHAT MAKES THIS A CLEANER TEST THAN acetone01. Both components are CURATED: acetone and water are in data/standards/, neither comes from the estimate lake. acetone01 had to measure a mixture model and a lake-estimated component at once and separate them afterwards; here there is nothing to separate. Whatever deviation appears is UNIFAC, plus whatever the pure vapour pressures carry – and the two pure boiling points below say how much that second part is.

WHAT LUYBEN GIVES, and it is less than for the IPA binary: acetone's normal boiling point 329.4 K, and the qualitative statement that acetone/water has NO azeotrope but a pinch at the high-acetone end. A qualitative claim is still falsifiable – a model that invents an azeotrope here would be wrong in a way no AAD would show.

NOTE THE WATER LEG IS KNOWN TO BE 0.70 K LOW before it is measured again here: Choupo's curated water record puts the normal boiling point at 372.4536 K against the definition's 373.15 K, because its Antoine set is being used at the very top of its own declared window (see acetone01_ipa_water_azeotrope/CLAUDE.md). It is repeated in this case rather than assumed, because a number quoted from another case is a number with two homes.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bp_acetone | Cp_ig | 79.38622606 |
| diag | bp_acetone | H_R | 0 |
| diag | bp_acetone | H_ig | -214666.297086 |
| diag | bp_acetone | H_real | -214666.297086 |
| diag | bp_acetone | P | 101325 |
| diag | bp_acetone | S_R | 0 |
| diag | bp_acetone | S_ig | 302.940865079 |
| diag | bp_acetone | S_real | 302.940865079 |
| diag | bp_acetone | T | 330 |
| diag | bp_acetone | T_bubble | 329.417905068 |
| diag | bp_acetone | Z | 1 |
| diag | bp_acetone | gamma | 1.11698686251 |
| diag | bp_acetone | v_molar | 0.0270789308062 |
| diag | bp_water | Cp_ig | 34.217711312 |

*... and 33 more reference values in the case's `expected`.*

## compare_activity_etoh_water

tutorials/props/compare/compare_activity_etoh_water




**What it does:** Compare 3 activity models (ideal / NRTL / Wilson) for $\gamma$_ethanol in equimolar mixture with water vs T

**The story:** COMPARE 3 activity-coefficient models for $\gamma$_ethanol in an equimolar ethanol/water mixture, T = 280 K to 360 K, P = 1 bar (as all three ops below declare; this line said 1 atm, which is 1.01325 bar).

ideal : $\gamma = 1$ always (Raoult's law assumption). NRTL : strong non-ideality, $\gamma$_ethanol $> 1$ (azeotrope behaviour). Wilson : similar to NRTL for this binary, slightly different curve.

Pedagogical point: a student studying ethanol/water VLE will see that "ideal" is a bad approximation here ($\gamma$_ethanol reaches 1.267 under NRTL and 1.377 under Wilson at 280 K, the coldest and most non-ideal end of this scan, against exactly 1 for ideal), and that NRTL/Wilson give similar non-ideal corrections. The choice between the two is then decided by which fits the project's experimental data better (workflow Mode B, see tutorials/props/README.md).

Each op overrides the global activityModel block via 'thermo {}'. The three CSVs (gamma_ideal.csv, gamma_nrtl.csv, gamma_wilson.csv) are auto-overlaid by the GUI.

*An equimolar ethanol/water liquid, held at 1 bar, scanned from 280 K to 360 K in 41 points, and asked for exactly one property: gamma_ethanol. The same scan is run three times — g_ideal, g_nrtl, g_wilson — and the only thing that changes between them is the activity model. There is no unit operation here on purpose: this is the thermodynamics that every unit in the rest of the path will be standing on.*

*Each op writes its own curve (gamma_ideal.csv, gamma_nrtl.csv, gamma_wilson.csv), and the case's system/propsDict says the three are overlaid on one set of axes in the app. The golden pins the SHAPE of the result rather than the numbers: 41 points, 0 failures, 0 empty curves for each of the three ops.*

*1. Ideal is a model, and this case declares it. The case's constant/thermoPhysPropDict says activityModel ideal; with a header that spells out why — ∗Raoult DECLARED, not defaulted∗. The g_ideal curve is flat at 1 across the whole 80 K span not because something was missing but because that is the answer the declared model gives. An absence and a choice would draw the same line; only the dict tells you which one you are looking at. 2. A thermo {} block overrides the models for one op only. All three ops declare the identical vary and state blocks; each carries its own thermo { equilibrium { liquid { activityModel ... } } }. The components stay global — it is the MODEL that is swapped, never the mixture — which is the same per-unit override a flowsheet uses to run one unit in a different thermodynamic world. 3. The dict owns the parameters, in the open. NRTL is declared with a_ij -0.8009, b_ij 246.18, a_ji 3.4578, b_ji -586.0809, alpha 0.30; Wilson with A_ij 1157.95 and A_ji 4081.65 in J/mol. Two models, two different parameter shapes, both written where you can read them. Those four NRTL numbers are worth remembering: they reappear verbatim in flash02_ethanol_water and again in column03_azeotrope_mesh, which is how one thermodynamic decision travels through a curriculum. 4. A property scan is a case like any other. application choupoProps; in controlDict, the work declared as a list of operations in system/propsDict, the same dict grammar, the same .cho marker, the same golden file. Nothing about "just properties" is a lesser kind of run.*

*Change composition in all three ops to ethanol 0.05; water 0.95; and run again: the ideal curve does not move at all, and the other two do. Activity is a function of composition, and that is precisely the information the ideal model does not carry. Then open flash02_ethanol_water, where this same mixture and these same NRTL parameters arrive in a unit operation and change its answer.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | g_ideal | n_emptyCurves | 0 |
| diag | g_ideal | n_failures | 0 |
| diag | g_ideal | n_points | 41 |
| diag | g_nrtl | n_emptyCurves | 0 |
| diag | g_nrtl | n_failures | 0 |
| diag | g_nrtl | n_points | 41 |
| diag | g_wilson | n_emptyCurves | 0 |
| diag | g_wilson | n_failures | 0 |
| diag | g_wilson | n_points | 41 |

## compare_eos_co2

tutorials/props/compare/compare_eos_co2




**What it does:** Compare 3 EoS (idealGas / SRK / PR) for Z of CO2 near critical (T=310 K, P from 1 to 100 bar)

**The story:** COMPARE 3 equations of state for the same property at the same conditions. This is what choupoProps is ∗for∗: helping a student or researcher CHOOSE the model that best describes the system at hand — not just look at pretty Z curves of one EoS at a time.

System: pure CO2 at T = 310 K (just above its critical temperature Tc = 304.13 K), pressure swept from 1 bar to 100 bar. Near the critical pressure (Pc = 7.38 MPa) the compressibility plunges as the fluid turns liquid-like; the three curves read (at P = 73.7 bar, the swept point nearest Pc):

idealGas: Z = 1 always (textbook reference; ignores attraction). SRK : Z = 0.538 at Pc. PR : Z = 0.509 at Pc — systematically lower than SRK (more attraction), and essentially on the experimental value. NIST : the ∗experimental∗ Z near the critical point is ~0.49.

The lesson: at and just above Tc, BOTH cubics track CO2 well — PR sits almost exactly on the ~0.49 datum and SRK is only ~0.05 high. This is a regime where cubic EoS are at their best, not their worst. Both Z curves keep falling past Pc to a shallow minimum (SRK ~0.30, PR ~0.275 near 98 bar) before rising again; the SRK-vs-PR gap is the diagnostic a student should read off the overlay.

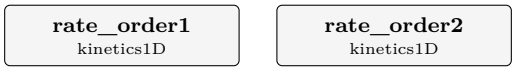
Each operation overrides the global thermoPhysPropDict with a different EoS via the ‘thermo {}‘ per-op block. The three CSVs are independent, so the GUI overlays them on the same Plotly canvas (same X axis "P", same Y "Z"). The result is one figure with three curves, ready for the project report.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | z_ideal | n_emptyCurves | 0 |
| diag | z_ideal | n_failures | 0 |
| diag | z_ideal | n_points | 50 |
| diag | z_srk | n_emptyCurves | 0 |
| diag | z_srk | n_failures | 0 |
| diag | z_srk | n_points | 50 |
| diag | z_pr | n_emptyCurves | 0 |
| diag | z_pr | n_failures | 0 |
| diag | z_pr | n_points | 50 |

## compare_kinetics_order

tutorials/props/compare/compare_kinetics_order




**What it does:** Rate-law order from DATA: 1st- vs 2nd-order fits overlaid on measured c(t) — the data picks the order

**The story:** The "see, then decide" of CHEMICAL KINETICS: fit a 1st- and a 2nd-order rate law to the SAME measured c(t) and overlay both on the data. No ‘k‘ is given, so each op REGRESSES its own k from the points (provenance = fitted). The data here are second-order – the 1st-order curve misses the tail, the 2nd lands on the points. The student picks the order from the evidence.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | rate_order1 | R2 | 0.973397 |
| diag | rate_order1 | chi2 | 0.0105136 |
| diag | rate_order1 | fitted | 1 |
| diag | rate_order1 | k | 0.00065871 |
| diag | rate_order1 | max_abs_resid | 0.0544611 |
| diag | rate_order1 | n_data | 8 |
| diag | rate_order1 | n_points | 61 |
| diag | rate_order1 | order | 1 |
| diag | rate_order1 | rms | 0.0362518 |
| diag | rate_order2 | R2 | 0.999421 |
| diag | rate_order2 | chi2 | 0.000228631 |
| diag | rate_order2 | fitted | 1 |
| diag | rate_order2 | k | 0.000992909 |
| diag | rate_order2 | max_abs_resid | 0.00829088 |

*... and 8 more reference values in the case's `expected`.*

## compare_psat_ambrose_walton

tutorials/props/compare/compare_psat_ambrose_walton




**What it does:** Psat: Ambrose-Walton corresponding-states PREDICTION vs fitted Antoine DATA, for benzene (non-polar, good) and ethanol (associating, degrades) — see the error, then decide

**The story:** Ambrose-Walton corresponding states (Ambrose & Walton 1989; Poling et al. 5th ed., Eq. 7-2.7) predicts vapour pressure from Tc, Pc and omega ALONE – exactly the constants a Joback + Lee-Kesler estimate produces. So an estimated component becomes flashable with NO measured Psat data. But it is an ESTIMATE, and the glass-box rule is: SEE the error before you trust it.

This sweeps T = 270 → 360 K and tabulates four curves: Psat_benzene – fitted Antoine (catalogue DATA) } non-polar pair Psat_benzeneAW – Ambrose-Walton PREDICTION } Psat_ethanol – fitted Antoine (catalogue DATA) } associating pair Psat_ethanolAW – Ambrose-Walton PREDICTION }

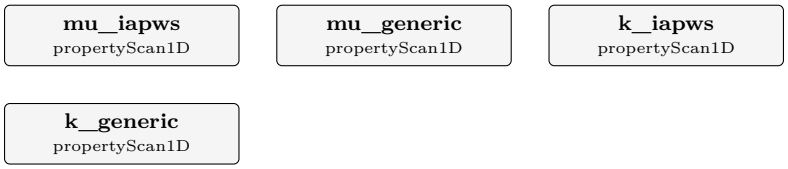
Open 'psat_aw_vs_antoine.csv' (GUI plot, or gnuplot). What you SEE: Near the normal boiling point (~350-360 K) BOTH agree to ~1 % – corresponding states is anchored there and is excellent. Towards the cold end (270 K) the curves separate, and MORE for the associating ethanol than for non-polar benzene: benzene AW vs Antoine: +3.3 % average, +8.9 % at 270 K ethanol AW vs Antoine: +5.2 % average, -16 % at 270 K So corresponding states is a solid FREE first predictor (it makes an estimated component flashable with no data), best near Tb and degrading at low reduced temperature – the degradation worse for hydrogen-bonding species. The lesson: prediction gets you started; for an associating species, or far below Tb, fit Antoine to data (choupoProps vaporPressureFit) before any design number.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| `diag` | `psat_aw_vs_antoine` | `n_emptyCurves` | `0` |
| `diag` | `psat_aw_vs_antoine` | `n_failures` | `0` |
| `diag` | `psat_aw_vs_antoine` | `n_points` | `46` |

## compare_transport_water

tutorials/props/compare/compare_transport_water




**What it does:** Compare water transport: IAPWS (IF97 R12-08/R15-11) vs generic correlations (Andrade / SatoRiedel), 280-370 K

**The story:** COMPARE water transport properties computed TWO WAYS on the SAME water, over the saturated-liquid window T = 280-370 K:

(a) IAPWS reference – water flagged ‘pureFluids { water { method IF97; } }‘ so the transport accessors route to the matching IAPWS releases: viscosity → R12-08 (industrial) thermal conductivity → R15-11 (background term) evaluated at the saturation state (the legacy accessors carry no P, so p = p_sat(T) – saturated liquid). These are THE international standard reference values (verified to 1e-8 by IAPWSTransport::verify(), which fires on every run).

(b) generic correlations – the same property KEY, but with a generic transport block and NO pureFluids flag, so the accessors fall to: viscosity → Andrade ln(mu) = A + B/T thermal conductivity → SatoRiedel fitted from the per-component .dat (Tc/Tb + the Andrade pair).

Because BOTH ops emit the same property key (‘viscosity_liquid‘, ‘thermal_conductivity_liquid‘) the two curves overlay directly – the student SEES where the cheap generic correlation deviates from the IAPWS reference.

THE LESSON (the actual computed gaps): VISCOSITY – Andrade tracks IAPWS within a few percent over the window (~7 % at 280 K, ~2-4 % mid-range, ~3 % at 370 K). The 2-parameter ln(mu)=A+B/T cannot follow the full curvature, but a few % is fine for a Reynolds number; it is NOT good enough for a precision steam duty. CONDUCTIVITY – SatoRiedel is a corresponding-states estimate and it FAILS BADLY for water: it underpredicts IAPWS by ~50-60 % across the window. Water is the textbook SatoRiedel failure (strong hydrogen bonding, anomalous k); this is exactly why "SEE before you commit" matters – a blind generic pick would have halved the heat-transfer coefficient. For water conductivity use IAPWS R15-11, full stop. Generic correlations are CHEAP and GENERAL (any compound with the handful of .dat parameters); IAPWS is EXACT but water-only. For a water-dominated process use the IF97 + IAPWS route; for a screening sweep over many fluids the generic correlations are the right tool. SEE the gap, then decide – do not pick blindly.

The GUI renders the CSVs via the existing compare machinery (overlay of the ∗_iapws / ∗_generic curves vs T) – no new Explorer kind.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | mu_iapws | n_failures | 0 |
| diag | mu_iapws | n_points | 46 |
| diag | mu_generic | n_failures | 0 |
| diag | mu_generic | n_points | 46 |
| diag | k_iapws | n_failures | 0 |
| diag | k_iapws | n_points | 46 |
| diag | k_generic | n_failures | 0 |
| diag | k_generic | n_points | 46 |
| diag | mu_iapws | n_emptyCurves | 0 |
| diag | mu_generic | n_emptyCurves | 0 |
| diag | k_iapws | n_emptyCurves | 0 |
| diag | k_generic | n_emptyCurves | 0 |

## compare_visc_liquid_ammonia

tutorials/props/compare/compare_visc_liquid_ammonia




**What it does:** Liquid-viscosity Andrade for ammonia, 235-300 K (cross-check mu@239.85K ~ 0.255 mPa.s)

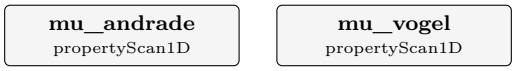
**The story:** Liquid-viscosity Andrade correlation for pure ammonia (NH3), 235-300 K. This case is SEALED: it runs its own copy, constant/components/NH3.dat, whose Andrade pair was fitted to the Engineering Toolbox table – a no-grant source (CLAUDE.md 10) – and, measured against the reference correlation (Monogenidou, Assael & Huber, J. Phys. Chem. Ref. Data 47, 023102 (2018), Table 6), is +10 % at 240 K and +65 % at 300 K. Since 2026-10-11 (DEV.md 4c C65 item c) the CATALOGUE record data/standards/components/NH3.dat carries a fit to that Table 6 (AAD 0.51 % over 200-320 K; 0.253 mPa.s at the normal boiling point, against Table 6's 0.2510 at 240 K). This copy keeps the old pair until the case is re-imported, and check_source_licence pins it until then; the golden pins only the scan's point counts, so the re-import moves no golden row. Note this case scans a SINGLE method (Andrade) – it is a single-correlation curve, not a multi-method comparison. Ammonia is liquid only under pressure in this T window; viscosity_liquid is a pure-T correlation, so the state P is immaterial to the value.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | mu_andrade | n_emptyCurves | 0 |
| diag | mu_andrade | n_failures | 0 |
| diag | mu_andrade | n_points | 27 |

## compare_visc_liquid_water

tutorials/props/compare/compare_visc_liquid_water




**What it does:** Compare 2 liquid viscosity correlations (Andrade vs Vogel) for water, 280-360 K

**The story:** COMPARE 2 liquid-viscosity correlations for pure water, T = 280-360 K.

Andrade: $\ln(\mu) = A + B/T$ — 2-parameter, simplest fit, OK over a narrow T window. Vogel : $\ln(\mu) = A + B/(T + C)$ — 3-parameter, captures the curvature better, recommended for wider T ranges.

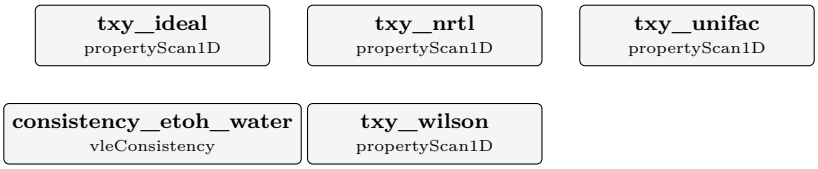
Pedagogical point: for a project using water transport in a non-trivial T window (e.g. 25-90 °C for a heated reactor), the student should know WHICH correlation he is committing to. Both correlations give viscosity FALLING with temperature (the physical trend): ~1.3-1.4 mPa·s near 280 K down to ~0.32 mPa·s at 360 K. This case shows the two models disagree MOST at the cold end (~6 % at 280 K), are within ~2 % at room T, and agree BEST at the hot end (~1 % at 360 K) — the 2-parameter Andrade fit drifts where it is extrapolated coolest. Validate against the textbook value: $\mu$_water @ 298 K $\approx 8.9 \times 10$?? Pa·s — Vogel reads $8.93 \times 10$?? (essentially on it), Andrade $9.09 \times 10$?? (~2 % high).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | mu_andrade | n_emptyCurves | 0 |
| diag | mu_andrade | n_failures | 0 |
| diag | mu_andrade | n_points | 41 |
| diag | mu_vogel | n_emptyCurves | 0 |
| diag | mu_vogel | n_failures | 0 |
| diag | mu_vogel | n_points | 41 |

## compare_vle_etoh_water

tutorials/props/compare/compare_vle_etoh_water




**What it does:** Ethanol/water T-x-y at 1 atm: ideal vs NRTL model curves OVERLAID with measured VLE data — see the data, pick the model

**The story:** The canonical "SEE, then DECIDE" props case: ethanol/water T-x-y at 1 atm, with two MODEL curves and the MEASURED data on ONE plot.

Piece 1 – two scan operations, identical but for the activity model: txy_ideal Raoult (gamma = 1) → a smooth lens, NO azeotrope txy_nrtl NRTL → the real minimum-boiling azeotrope Each emits the (x, T_bubble, y_eq) trio → the T-x-y / x-y / T-x / T-y toggle (x-y is the McCabe-Thiele diagram).

Piece 2 – the 'experimental {}' overlay: lists the two ops as 'models', and points 'dataset' at REAL measured points (constant/experiments/...). The GUI draws both model curves + the lab points together; the student SEES that the ideal model misses the azeotrope and NRTL captures it, and picks NRTL from evidence – not from a recommended-by badge.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | txy_ideal | n_emptyCurves | 0 |
| diag | txy_ideal | n_failures | 0 |
| diag | txy_ideal | n_points | 51 |
| diag | txy_nrtl | n_emptyCurves | 0 |
| diag | txy_nrtl | n_failures | 0 |
| diag | txy_nrtl | n_points | 51 |
| diag | txy_unifac | n_emptyCurves | 0 |
| diag | txy_unifac | n_failures | 0 |
| diag | txy_unifac | n_points | 51 |
| diag | consistency_etoh_water | gamma1_inf | 4.68506 |
| diag | consistency_etoh_water | gamma2_inf | 2.27627 |
| diag | consistency_etoh_water | gd_max_resid | 0.0964023 |
| diag | consistency_etoh_water | gd_mean_resid | 0.05208 |
| diag | consistency_etoh_water | herington_D | 12.0482 |

*... and 14 more reference values in the case's `expected`.*

## evidence01_methanol_water_points

tutorials/props/compare/evidence01_methanol_water_points




**What it does:** Methanol/water bubble T evaluated AT the 13 measured x of a held-out study (doi 10.1021/je200341c): NRTL, UNIFAC and ideal, residual = engine minus measurement at the measured point

**The story:** A MODEL EVALUATED AT THE MEASURED CONDITIONS, NOT ON A GRID (DEV.md 4c C117).

THE QUESTION. A measured dataset is a list of CONDITIONS somebody set and the value they read there. To compare a model with it, the honest residual is model(x_measured) - measurement – the engine asked at the measured x, never a value read off a smooth curve between two grid neighbours. The 'experimental {}' overlay's AAD interpolates the model CSV at the measured x; here the CSV HOLDS the measured x, so that interpolation is exact.

THE GRAMMAR. 'vary { variable x[methanol]; points ( ... ); }' – the 13 interior liquid mole fractions of the held-out study, in its own order, instead of 'from / to / n'. Both in one 'vary {}' is refused by name. The values are canonical SI (a mole fraction here; K or Pa for T or P).

THE EVIDENCE. constant/experiments/methanol-water-bubble-je200341c.dat: Yang, Ma & Yin, J. Chem. Eng. Data (2011), doi 10.1021/je200341c, 13 interior points at 101.32 kPa, the HELD-OUT set of the curate04 regression (tutorials/props/curation/curate04_nrtl_methanol_water), kept here as this case's own copy – a self-contained tutorial keeps its own copy. Its 'reviewStatus transcribedNotCheckedAgainstArticle' is announced by the run.

THE THREE CANDIDATES. The catalogue NRTL pair (regressed by Choupo in curate04 on ANOTHER study, so this set is a validation of it), UNIFAC (predictive, from the components' groups) and ideal mixing. What the curate04 case found holds here to the digit: UNIFAC is closer than the fitted pair on this set, and the fitted pair is 'validated' – inside its declared band, never better than every alternative.

WHAT IS NOT CLAIMED. Nothing is fitted here. Part of every residual is the catalogue water Antoine record's own 0.70 K at 101.32 kPa, which no pair can remove. This is the engine witness of the Property Explorer's Evidence view, which synthesises exactly these operations (one choupoProps run per candidate) and only subtracts.

**What to expect (golden-locked):**

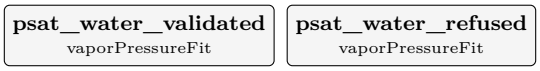
| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | txy_nrtl_points | n_emptyCurves | 0 |
| diag | txy_nrtl_points | n_failures | 0 |
| diag | txy_nrtl_points | n_points | 13 |
| diag | txy_unifac_points | n_emptyCurves | 0 |
| diag | txy_unifac_points | n_failures | 0 |
| diag | txy_unifac_points | n_points | 13 |
| diag | txy_ideal_points | n_emptyCurves | 0 |
| diag | txy_ideal_points | n_failures | 0 |
| diag | txy_ideal_points | n_points | 13 |
| aad | heldout_je200341c | txy_nrtl_points.T_bubble.aadAbs | 0.534750769231 |
| aad | heldout_je200341c | txy_nrtl_points.T_bubble.aadRelPct | 0.150061454094 |
| aad | heldout_je200341c | txy_nrtl_points.y_eq_methanol.aadAbs | 0.0125426852538 |
| aad | heldout_je200341c | txy_unifac_points.T_bubble.aadAbs | 0.366152846154 |
| aad | heldout_je200341c | txy_unifac_points.T_bubble.aadRelPct | 0.10093737486 |

*... and 4 more reference values in the case's `expected`.*

## curation

## curate01_water_evidence_partition

tutorials/props/curation/curate01_water_evidence_partition




**What it does:** The FIT/HELD-OUT partition contract: the same property fitted twice, differing only in whether evidence was held out – one VALIDATED, one VALIDATION REFUSED

**The story:** THE FIT / HELD-OUT VALIDATION CONTRACT, and its refusal, side by side.

TWO OPERATIONS. SAME COMPONENT. SAME PROPERTY. SAME TWO DATASETS. The ONLY thing that differs is whether any evidence was held out – which is exactly the variable the contract governs, and isolating it is why this witness does not use two different properties.

psat_water_validated lowT 'role fit' + highT 'role validation' psat_water_refused lowT 'role fit' + highT 'role fit'

The first can report a held-out residual because the fitter never received the 325-370 K points. The second cannot, and says so by name instead of presenting its in-sample R2 as though it were an external check:

VALIDATION REFUSED: No independent experimental evidence remains after fitting.

THE ANCHOR IS ENFORCED BY CONSTRUCTION. The partition is parsed and frozen before any fitting code runs, and the fitter is handed only the 'role fit' points. It is not forbidden to move a held-out point into the fit – it is never given one. A rule can be broken by the next edit; a data-flow fact cannot. 'role' has NO default, because a defaulted role would make every dataset a fitting dataset and bring the in-sample problem back invisibly.

WHAT THIS CASE DOES NOT SHOW. Both datasets are generated from the SAME curated Antoine correlation (see either dataset header), so the held-out residual is arithmetic rather than evidence about the physics. This is a STRUCTURAL witness – no number in it is validated against measurement.

Contract: src/propertyOps/EvidencePartition.H

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | psat_water_validated | A | 5.40221 |
| diag | psat_water_validated | B | 1838.68 |
| diag | psat_water_validated | C | -31.737 |
| diag | psat_water_validated | R2 | 1 |
| diag | psat_water_validated | aad_heldout_pct | 5.95532e-08 |
| diag | psat_water_validated | dHvap_kJ_per_mol | 43.3787 |
| diag | psat_water_validated | n_heldout | 10 |
| diag | psat_water_validated | n_points | 10 |
| diag | psat_water_validated | rms_heldout_log10P | 2.76024e-10 |
| diag | psat_water_refused | A | 5.40221 |
| diag | psat_water_refused | B | 1838.68 |
| diag | psat_water_refused | C | -31.737 |
| diag | psat_water_refused | R2 | 1 |
| diag | psat_water_refused | dHvap_kJ_per_mol | 42.1162 |
| *... and 4 more reference values in the case's `expected`.* | | | |

## curate02_vle_heldout_ethanol_water

tutorials/props/curation/curate02_vle_heldout_ethanol_water

**nrtl__etoh__water__heldout**
fitParameters

**What it does:** A binary NRTL pair regressed on eight MEASURED bubble temperatures and scored on three it never saw – the first held-out validation in the corpus against real evidence

**The story:** curate02 – THE FIT / HELD-OUT CONTRACT ON A BINARY PHASE EQUILIBRIUM.

The NRTL ethanol-water pair is regressed against eight measured bubble temperatures at 101.325 kPa, and then asked to predict THREE it never saw.

WHY THIS CASE EXISTS BESIDE curate01. curate01 proves the machinery: its two halves come from the same Antoine equation, so its held-out residual is round-off and its acceptance band is a numerical criterion, which its own 'acceptance.origin' says out loud. Here the evidence is MEASURED, so the held-out residual is a statement about the MODEL – the first place in the corpus where a fitted binary pair is scored on real evidence it was never handed.

THE SPLIT IS DECLARED AND ARITHMETIC. Every third point of the eleven-point series (x1 = 0.20, 0.50, 0.80) is held out, spread across the composition range rather than cut contiguously, because a model can interpolate well and extrapolate badly and a contiguous cut tests only one of those. The rule was fixed before anything was fitted.

THE ACCEPTANCE BAND IS DECLARED BEFORE THE FIT, WITH ITS ORIGIN, AND IS NOT ADJUSTED AFTERWARDS. Whatever verdict falls out of it is the result. A band chosen after seeing the residual would make the whole contract decorative.

Read the run's four sections in order: FIT QUALITY is what the model does on what it saw; HELD-OUT EVIDENCE is what it did on what it did not.

TWO THINGS THIS CASE DOES NOT SHOW, BOTH VISIBLE IN ITS OWN GOLDEN.

(1) THE PARAMETERS ARE NOT IDENTIFIABLE. 'identifiable 0', 'cond_JtJ ~4e11' and pairwise correlations of 0.9999 between each a/b pair: eight points at ONE pressure cannot separate the entropic and enthalpic halves of an NRTL parameter, because they only ever appear together over a narrow T span. The model PREDICTS well and the four numbers are individually meaningless – a validated prediction is not a determined parameter set, and a student must be able to read both facts off the same run. This is why the verdict is scoped 'binaryVLE.T_bubble' in the dossier and not to the pair itself.

(2) THE HELD-OUT MARGIN IS NOT LARGE. 0.0737 % against a 0.1 % band is a pass by a factor of 1.4, not by an order of magnitude. The band was declared before the fit and is not adjusted; the margin is reported rather than described as comfortable.

The run also announces an extrapolation: ethanol's vapour pressure is evaluated at 372 K, outside its declared Trange (273-369). Announced, not judged – it rides the caveat surface like every other extrapolation.

WHAT THE STUDENT TAKES AWAY: 'ethanol-water.proposal.dat'.

The console says all of the above and then scrolls. The file is what survives – it is moved into 'constant/parameters/NRTL/etha
water.dat' and simulated with, months later, by someone who never saw this run. So it carries the whole qualification, not the numbers alone: the two datasets and the role each was frozen in, the fingerprint of that declaration, the held-out AAD over the three withheld points, the 0.1 % band with the reason it was set at 0.1 %, the verdict, and – in the header, above everything – the fact that these four coefficients are NOT individually identifiable.

Both facts of (1) are readable off that one file: 'verdict validated' beside 'identifiable false'. A validated prediction is not a determined parameter set, and the record says so in the artefact rather than only in a run log.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | nrtl_etoh_water_heldout | aad_heldout_pct | 0.0736714746429 |
| diag | nrtl_etoh_water_heldout | chi2 | 0.809699405251 |
| diag | nrtl_etoh_water_heldout | chi2_in_sample | 0.809699405251 |
| diag | nrtl_etoh_water_heldout | chi2_reduced | 0.202424851313 |
| diag | nrtl_etoh_water_heldout | cond_JtJ | 394331596992 |
| diag | nrtl_etoh_water_heldout | converged | 1 |
| diag | nrtl_etoh_water_heldout | corr_computed | 1 |
| diag | nrtl_etoh_water_heldout | dof | 4 |
| diag | nrtl_etoh_water_heldout | identifiable | 0 |
| diag | nrtl_etoh_water_heldout | invertible | 1 |
| diag | nrtl_etoh_water_heldout | iter | 29 |
| diag | nrtl_etoh_water_heldout | max_abs_corr | 0.999936039821 |
| diag | nrtl_etoh_water_heldout | max_abs_resid | 0.582462818864 |
| diag | nrtl_etoh_water_heldout | n_data | 8 |
| *… and 9 more reference values in the case's* ***expected****.* | | | |

## curate03_thermoml_fixture_bubble

tutorials/props/curation/curate03_thermoml_fixture_bubble




**What it does:** ThermoML → extract → partition → fit → held-out → dossier, on a SYNTHETIC structural fixture: the plumbing proven end to end, and no physical claim anywhere

**The story:** curate03 – THE THERMOML PATH, END TO END, ON A STRUCTURAL FIXTURE.

ThermoML .xml → extract-vle → EvidencePartition → FIT only → T_bubble regression → held-out → dossier

EVERY NUMBER HERE IS INVENTED. Both datasets were extracted by 'bin/choupo-thermoml extract-vle' from synthetic fixtures under tutorials/props/curation/fixtures/thermoml/, and each carries 'provenance { source synthetic; }' – so the run ANNOUNCES that its evidence was generated, and the dossier records it per dataset. A reader cannot arrive at the verdict without having been told what the numbers are.

WHAT THIS CASE PROVES: that a ThermoML file crosses into the evidence contract without losing its identity – archive file, DOI, system, pressure and checksum all survive into the dossier – and that the partition, the regression and the verdict work on evidence that arrived through the parser rather than being typed into a case.

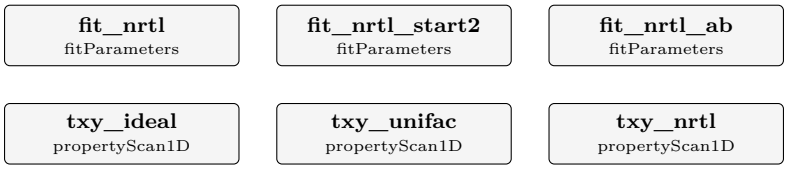
WHAT IT DOES NOT PROVE: anything about ethanol, water, or NRTL. The acceptance band below is a PLUMBING tolerance and says so. The scientific witness on measured evidence is curate02, next door.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | thermoml_fixture_bubble | aad_heldout_pct | 0.0528782856338 |
| diag | thermoml_fixture_bubble | chi2 | 0.669244065892 |
| diag | thermoml_fixture_bubble | chi2_in_sample | 0.669244065892 |
| diag | thermoml_fixture_bubble | chi2_reduced | 0.223081355297 |
| diag | thermoml_fixture_bubble | cond_JtJ | 272780104374 |
| diag | thermoml_fixture_bubble | converged | 1 |
| diag | thermoml_fixture_bubble | corr_computed | 1 |
| diag | thermoml_fixture_bubble | dof | 3 |
| diag | thermoml_fixture_bubble | identifiable | 0 |
| diag | thermoml_fixture_bubble | invertible | 1 |
| diag | thermoml_fixture_bubble | iter | 42 |
| diag | thermoml_fixture_bubble | max_abs_corr | 0.999960515815 |
| diag | thermoml_fixture_bubble | max_abs_resid | 0.656169896008 |
| diag | thermoml_fixture_bubble | n_data | 7 |
| *... and 9 more reference values in the case's `expected`.* | | | |

## curate04_nrtl_methanol_water

tutorials/props/curation/curate04_nrtl_methanol_water




**What it does:** NRTL methanol-water regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. the commonest aqueous alcohol separation in a first distillation course; the fit set spans six isobars (15-95 kPa), so it can ask whether a SECOND temperature coefficient is identified.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search methanol water –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL/methanol-water.dat 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1016/j.jct.2009.11.020 (54 interior points, several isobars). HELD OUT: doi 10.1021/je200341c (13 interior points, 101.32 kPa) – a different article, so the held-out AAD is a validation, not a slice of the fit. Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 165.0 K, b_ji = 41.8 K, in-sample rms 0.50 K over 54 points on six isobars. fit_nrtl_start2, from (800, -400), lands on the same pair to 0.01 K – one minimum, not a start point's choice. HELD OUT (101.32 kPa, another laboratory): AAD 0.53 K (0.150 %) against a band of 0.33 % declared before the fit → 'validated', and the pair is the catalogue record NRTL/methanol-water.dat. fit_nrtl_ab frees a_ij and a_ji too: the run says 'NOT individually identifiable' (a and b trade off, max|corr| 0.996) and its held-out AAD is WORSE (0.64 K). Six isobars over 15-95 kPa do not pay for a second temperature coefficient – which is why the record carries two. THE OVERLAY SAYS SOMETHING THE VERDICT DOES NOT: on the held-out set UNIFAC (0.36 K) is closer than the fitted pair (0.53 K). The pair was regressed below atmospheric pressure and is validated, not best; 'validated' means inside the declared band, never better than every alternative. Ideal mixing misses by 2.8 K. Part of every residual here is not the pair's: the catalogue water Antoine record boils 0.70 K low at 101.32 kPa (measured at this set's own pure endpoint), and no pair can remove that.

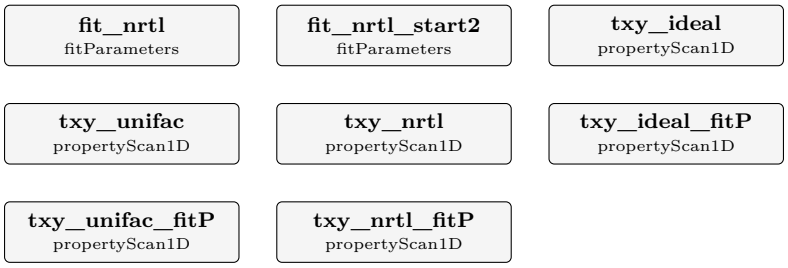
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.534750672125 |
| diag | fit_nrtl | aad_heldout_pct | 0.150061425821 |
| diag | fit_nrtl | chi2 | 315.937277131 |
| diag | fit_nrtl | chi2_in_sample | 315.937277131 |
| diag | fit_nrtl | chi2_reduced | 6.07571686791 |
| diag | fit_nrtl | cond_JtJ | 116.915709849 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 52 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 7 |
| diag | fit_nrtl | max_abs_corr | 0.995750654438 |
| diag | fit_nrtl | max_abs_resid | 2.0802446823 |
| *. . . and 76 more reference values in the case's `expected`.* | | | |

## curate05_nrtl_acetone_methanol

tutorials/props/curation/curate05_nrtl_acetone_methanol




**What it does:** NRTL acetone-methanol regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. a minimum-boiling azeotrope between two components a student meets in the acetone and methanol plants of this corpus.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search acetone methanol –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion NOT TAKEN: the verdict is notValidated, so the record stays in THIS case (constant/parameters/NRTL/acetone-methanol.dat) 8. bin/choupo-import the seal keeps that authored record as the case's own and mirrors the rest

THE EVIDENCE. FIT: doi 10.1021/je5007373 (32 interior points, 101.3 kPa). HELD OUT: doi 10.1021/je0603170 (30 interior points, 100 kPa) – a different article, so the held-out AAD is a validation, not a slice of the fit. Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 369.8 K, b_ji = -92.8 K, in-sample rms 0.13 K; fit_nrtl_start2 lands on the same pair. HELD OUT (100 kPa, another laboratory): AAD 0.32 K (0.096 %) against a band of 0.047 % declared before the fit → 'notValidated'. THE PAIR IS NOT PROMOTED: its record lives in this case ONLY (constant/parameters/NRTL/acetone-methanol.dat, reviewStatus interim, verdict notValidated), so the scans can still draw it. WHY, as far as the run can see: the two studies disagree with each other. UNIFAC, which saw neither, reproduces the held-out study to 0.04 K and misses the fit study by 0.34 K; the fitted pair does the mirror image (0.10 K on its own study, 0.32 K on the other). Two measurements of one mixture, 0.3 K apart where each states about 0.1-0.35 K: a finding about the EVIDENCE, for a curator to settle by reading both articles. Swapping the roles now would be choosing the partition after seeing the residual, which is the tuning the declared partition exists to prevent.

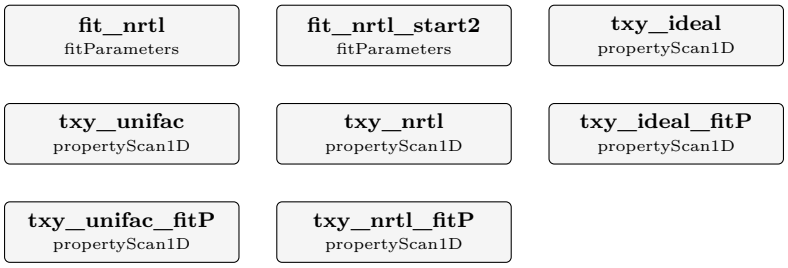
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.315969383935 |
| diag | fit_nrtl | aad_heldout_pct | 0.0960285506587 |
| diag | fit_nrtl | chi2 | 4.45236740196 |
| diag | fit_nrtl | chi2_in_sample | 4.45236740196 |
| diag | fit_nrtl | chi2_reduced | 0.148412246732 |
| diag | fit_nrtl | cond_JtJ | 95.8483654535 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 30 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 11 |
| diag | fit_nrtl | max_abs_corr | 0.994823894221 |
| diag | fit_nrtl | max_abs_resid | 0.242779577201 |
| *... and 70 more reference values in the case's `expected`.* | | | |

## curate06_nrtl_isopropanol_water

tutorials/props/curation/curate06_nrtl_isopropanol_water




**What it does:** NRTL isopropanol-water regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. the azeotrope the acetone-from-isopropanol tutorials have to break; the fit set's stated uncertainty varies tenfold along x, which is what the uncertainty weighting is for.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search isopropanol water –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL water.dat 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1016/j.jct.2017.02.005 (29 interior points, 100 kPa). HELD OUT: doi 10.1021/acs.jced.7b00523 (16 interior points, 101.058 kPa) – a different article, so the held-out AAD is a validation, not a slice of the fit. Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = -69.6 K, b_ji = 968.2 K, in-sample rms 0.52 K; fit_nrtl_start2 lands on the same pair. The fit set's stated uncertainty runs from 0.13 to 1.35 K along x, so the 1/U weighting changes which points the pair is held to. HELD OUT (101.058 kPa, another laboratory): AAD 0.83 K (0.232 %) against a band of 0.63 % → 'validated'; the pair is the catalogue record NRTL/isopropanol-water.dat. Overlay on the held-out set: NRTL 0.83 K, UNIFAC 1.27 K, ideal 8.2 K. The held-out study's own stated uncertainty is 1.6-1.7 K, and the catalogue water record boils 0.52 K low at its pressure.

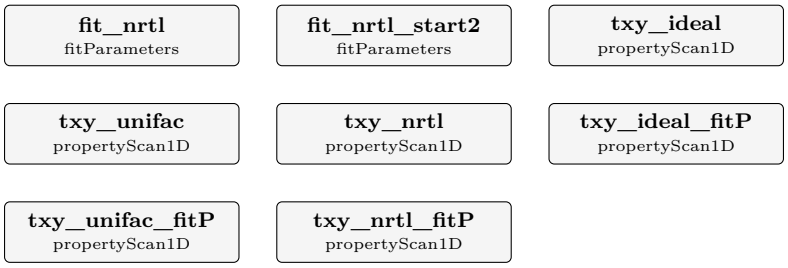
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.84147821571 |
| diag | fit_nrtl | aad_heldout_pct | 0.234200546209 |
| diag | fit_nrtl | chi2 | 90.1257883215 |
| diag | fit_nrtl | chi2_in_sample | 90.1257883215 |
| diag | fit_nrtl | chi2_reduced | 3.33799216005 |
| diag | fit_nrtl | cond_JtJ | 39.3738330039 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 27 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 36 |
| diag | fit_nrtl | max_abs_corr | 0.89819126525 |
| diag | fit_nrtl | max_abs_resid | 1.65678230014 |
| *... and 70 more reference values in the case's `expected`.* | | | |

## curate07_nrtl_ethylacetate_ethanol

tutorials/props/curation/curate07_nrtl_ethylacetate_ethanol




**What it does:** NRTL ethylAcetate-ethanol regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. the esterification product and its reactant, a minimum-boiling azeotrope the esterification tutorials run through.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search ethylAcetate ethanol –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL ethylAcetate.dat 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1016/j.fluid.2005.07.010 (24 interior points, 101.325 kPa). HELD OUT: doi 10.1021/je700322p (28 interior points, 100 kPa) – a different article, so the held-out AAD is a validation, not a slice of the fit. Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 95.4 K, b_ji = 214.8 K, in-sample rms 0.077 K; fit_nrtl_start2 lands on the same pair. HELD OUT (100 kPa, another laboratory): AAD 0.11 K (0.032 %) against a band of 0.052 % → 'validated'; the pair is the catalogue record NRTL/ethanol-ethylAcetate.dat. Overlay on the held-out set: NRTL 0.11 K, UNIFAC 0.69 K, ideal 4.1 K.

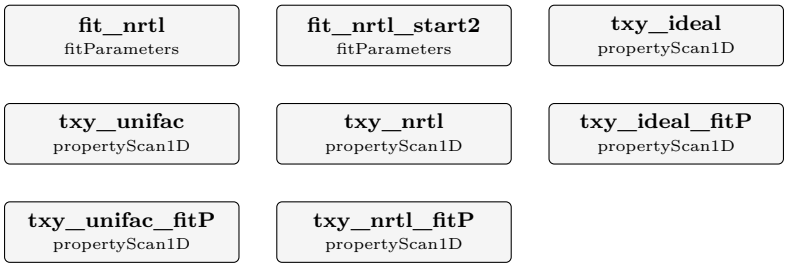
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.11098102751 |
| diag | fit_nrtl | aad_heldout_pct | 0.0321107150642 |
| diag | fit_nrtl | chi2 | 4.34313933595 |
| diag | fit_nrtl | chi2_in_sample | 4.34313933595 |
| diag | fit_nrtl | chi2_reduced | 0.197415424361 |
| diag | fit_nrtl | cond_JtJ | 151.806801216 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 22 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 6 |
| diag | fit_nrtl | max_abs_corr | 0.995865280749 |
| diag | fit_nrtl | max_abs_resid | 0.194183542276 |
| *... and 70 more reference values in the case's `expected`.* | | | |

## curate08_nrtl_methylacetate_methanol

tutorials/props/curation/curate08_nrtl_methylacetate_methanol




**What it does:** NRTL methylAcetate-methanol regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. the methyl acetate reactive-distillation system, where the catalogue already carries a UNIQUAC pair (Popken 2000) – an NRTL pair fitted to independent measurements puts a second model beside it.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search methylAcetate methanol –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL/methanol-methylAcetate.dat 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1021/je600518s (27 interior points, 100 kPa). HELD OUT: doi 10.1016/j.fluid.2015.04.018 (19 interior points, 101.3 kPa) – a different article, so the held-out AAD is a validation, not a slice of the fit. Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 182.9 K, b_ji = 185.6 K, in-sample rms 0.029 K; fit_nrtl_start2 lands on the same pair. HELD OUT (101.3 kPa, another laboratory): AAD 0.15 K (0.045 %) against a band of 0.089 % → 'validated'; the pair is the catalogue record NRTL/methanol-methylAcetate.dat, beside the UNIQUAC record the catalogue already carries for the same system. Overlay on the held-out set: NRTL 0.15 K, UNIFAC 0.22 K, ideal 4.4 K. A third study was set aside BEFORE fitting: the archive's j.fluid.2015.12.029 block repeats every (x, P, T) row twice, and 'extract-vle' refuses a duplicated observation rather than keep one copy on a guess.

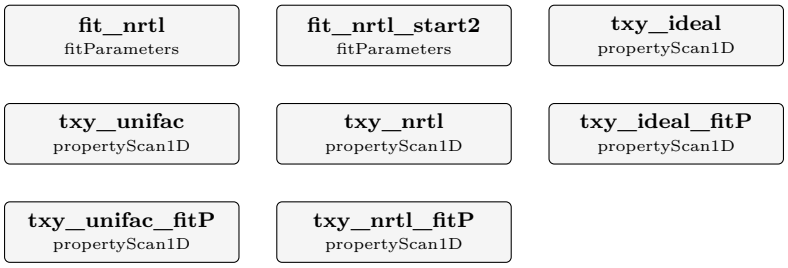
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.150139370506 |
| diag | fit_nrtl | aad_heldout_pct | 0.0454327520711 |
| diag | fit_nrtl | chi2 | 0.7380738708 |
| diag | fit_nrtl | chi2_in_sample | 0.7380738708 |
| diag | fit_nrtl | chi2_reduced | 0.029522954832 |
| diag | fit_nrtl | cond_JtJ | 64.7492559018 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 25 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 5 |
| diag | fit_nrtl | max_abs_corr | 0.991685096395 |
| diag | fit_nrtl | max_abs_resid | 0.0690603448025 |
| *... and 70 more reference values in the case's `expected`.* | | | |

## curate09_nrtl_propanol_water

tutorials/props/curation/curate09_nrtl_propanol_water




**What it does:** NRTL 1Propanol-water regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. the third water + alcohol pair of the catalogue after methanol and isopropanol, and the one with the homologue's minimum-boiling azeotrope (both studies bottom out near 360.5 K): with methanol + water and isopropanol + water it completes the C1-C3 series a first distillation course compares. The archive holds five low-pressure studies of it, the most of any candidate.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search 1Propanol water –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL/1Propanol-water.dat ONLY if the held-out verdict is 'validated' 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1021/je800436u (26 interior points, 100 kPa). HELD OUT: doi 10.1016/j.jct.2017.06.014 (19 interior points, 101.3 kPa). Chosen BEFORE any fit by the slice-2 partition rule: the FIT set is the low-pressure bubble-temperature study with the most interior points; the HELD-OUT set is the next largest that shares NO author with it (the archive carries no affiliations, so a shared author is the dependence the rule can check). Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = -25.1 K, b_ji = 979.8 K, in-sample rms 0.52 K; fit_nrtl_start2 lands on the same pair. The in-sample reduced chi2 is 8.9: two parameters cannot follow the fit set within its own stated U (0.09-0.89 K) – recorded, not hidden by a third parameter. HELD OUT (101.3 kPa, no author in common): AAD 0.21 K (0.057 %) against a band of 0.26 % → 'validated'; the pair is the catalogue record NRTL/1Propanol-water.dat. Overlay on the held-out set: NRTL 0.21 K, UNIFAC 0.36 K, ideal 8.9 K. PHASE STABILITY, checked beside the verdict because the verdict cannot see it: one liquid across the validity span (360.5-368.6 K), but BELOW about 355.5 K the pair splits a liquid that is miscible in all proportions. The record says so, and the validity block is the guard.

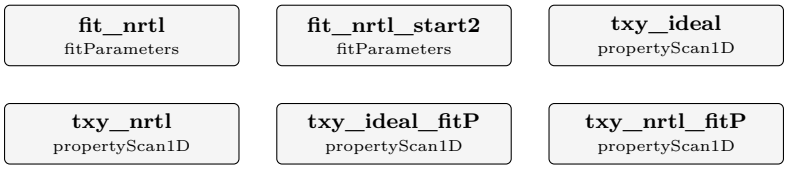
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.205309666797 |
| diag | fit_nrtl | aad_heldout_pct | 0.0565382893231 |
| diag | fit_nrtl | chi2 | 213.201326383 |
| diag | fit_nrtl | chi2_in_sample | 213.201326383 |
| diag | fit_nrtl | chi2_reduced | 8.8833885993 |
| diag | fit_nrtl | cond_JtJ | 45.7035038408 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 24 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 12 |
| diag | fit_nrtl | max_abs_corr | 0.919864189194 |
| diag | fit_nrtl | max_abs_resid | 1.46662361676 |
| *... and 70 more reference values in the case's `expected`.* | | | |

## curate10_nrtl_acetonitrile_water

tutorials/props/curation/curate10_nrtl_acetonitrile_water




**What it does:** NRTL acetonitrile-water regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal mixing

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. a solvent every analytical laboratory recovers, and the textbook system for PRESSURE-SWING distillation: a minimum-boiling azeotrope near 349.6 K at 101.3 kPa (both studies). The catalogue's acetonitrile record carries no UNIFAC group decomposition, so the overlay draws the pair beside ideal mixing only – the predictive comparison is a gap of the COMPONENT record, said here rather than filled by a decomposition this case would have to invent.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search acetonitrile water –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL water.dat ONLY if the held-out verdict is 'validated' 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1021/je400531a (19 interior points, 101.3 kPa). HELD OUT: doi 10.1016/j.jct.2019.07.006 (12 interior points, 101.3 kPa). Chosen BEFORE any fit by the slice-2 partition rule: the FIT set is the low-pressure bubble-temperature study with the most interior points; the HELD-OUT set is the next largest that shares NO author with it (the archive carries no affiliations, so a shared author is the dependence the rule can check). Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 218.3 K, b_ji = 616.3 K, in-sample rms 0.52 K; fit_nrtl_start2 lands on the same pair. HELD OUT (101.3 kPa, no author in common): AAD 0.50 K (0.137 %) against a band of 0.72 % → 'validated'; the pair is the catalogue record NRTL/acetonitrile-water.dat. The band is wide because the held-out study states U95 = 1.69-1.85 K on every point: a held-out set can only validate to the precision it was measured to. Overlay on the held-out set: NRTL 0.50 K, ideal 10.7 K (no UNIFAC: see WHY THIS SYSTEM). PHASE STABILITY, checked beside the verdict because the verdict cannot see it: one liquid across the validity span (350.0-367.6 K), but BELOW about 323.6 K the pair splits a liquid that is miscible at room temperature. The record says so, and the validity block is the guard.

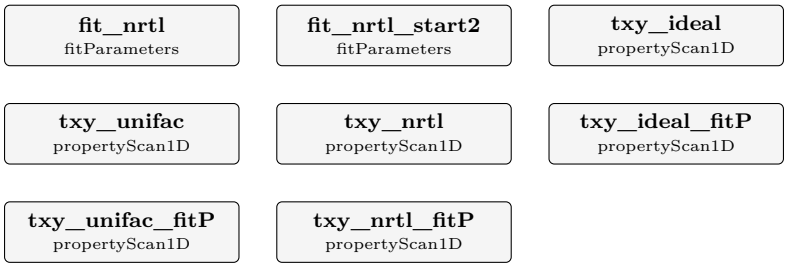
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.495564366362 |
| diag | fit_nrtl | aad_heldout_pct | 0.136733099834 |
| diag | fit_nrtl | chi2 | 27.5888548619 |
| diag | fit_nrtl | chi2_in_sample | 27.5888548619 |
| diag | fit_nrtl | chi2_reduced | 1.6228738154 |
| diag | fit_nrtl | cond_JtJ | 15.0626260021 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 17 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 6 |
| diag | fit_nrtl | max_abs_corr | 0.940389969107 |
| diag | fit_nrtl | max_abs_resid | 1.15866160924 |
| *... and 58 more reference values in the case's `expected`.* | | | |

## curate11_nrtl_acetone_ethanol

tutorials/props/curation/curate11_nrtl_acetone_ethanol




**What it does:** NRTL acetone-ethanol regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. two of the commonest laboratory solvents, and a positive deviation from Raoult's law WITHOUT an azeotrope – the contrast to every other pair of this family, showing that a non-ideal pair need not pinch a column.

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search acetone ethanol –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → data/standards/parameters/NRTL/acetone-ethanol.dat ONLY if the held-out verdict is 'validated' 8. bin/choupo-import the seal mirrors that record here

THE EVIDENCE. FIT: doi 10.1016/j.fluid.2005.01.007 (19 interior points, 101.32 kPa). HELD OUT: doi 10.1021/je400544h (11 interior points, 101.3 kPa) – an article whose title names another system; the acetone + ethanol block is the one the archive files under it, and reading it back is part of the curator's check. Chosen BEFORE any fit by the slice-2 partition rule: the FIT set is the low-pressure bubble-temperature study with the most interior points; the HELD-OUT set is the next largest that shares NO author with it (the archive carries no affiliations, so a shared author is the dependence the rule can check). Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure- component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 149.8 K, b_ji = 68.1 K, in-sample rms 0.072 K; fit_nrtl_start2 lands on the same pair. max|corr| 0.997: the two b's are nearly one direction in the data, as for every near-symmetric pair here. HELD OUT (101.3 kPa, no author in common): AAD 0.093 K (0.028 %) against a band of 0.14 % → 'validated'; the pair is the catalogue record NRTL/acetone-ethanol.dat. Overlay on the held-out set: NRTL 0.093 K, UNIFAC 0.79 K, ideal 3.6 K. PHASE STABILITY: one liquid at every temperature down to 200 K.

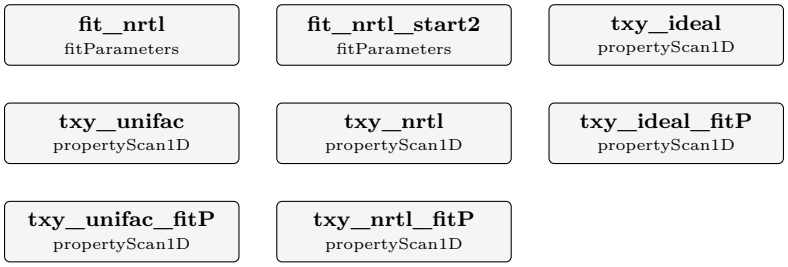
WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.0928861936011 |
| diag | fit_nrtl | aad_heldout_pct | 0.0274764440669 |
| diag | fit_nrtl | chi2 | 1.96410744483 |
| diag | fit_nrtl | chi2_in_sample | 1.96410744483 |
| diag | fit_nrtl | chi2_reduced | 0.115535732049 |
| diag | fit_nrtl | cond_JtJ | 162.970032107 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 17 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 9 |
| diag | fit_nrtl | max_abs_corr | 0.996945998842 |
| diag | fit_nrtl | max_abs_resid | 0.212432153701 |
| *... and 70 more reference values in the case's `expected`.* | | | |

## curate12_nrtl_ethanol_cyclohexane

tutorials/props/curation/curate12_nrtl_ethanol_cyclohexane




**What it does:** NRTL ethanol-cyclohexane regressed to measured bubble temperatures (ThermoML locates, the article sources), held out against an independent study, drawn against both beside ideal and UNIFAC

**The story:** AN NRTL PAIR FROM PUBLISHED MEASUREMENTS, END TO END.

WHY THIS SYSTEM. the binary under the industrial dehydration of ethanol by heterogeneous azeotropic distillation, where cyclohexane is the entrainer: an alcohol + hydrocarbon pair with a strong positive deviation and a minimum-boiling azeotrope near 338 K (both studies).

THE CHAIN, EACH STEP A COMMAND (DEV.md C16; record docs/design/binary-pairs-from-open-measurements.md): 1. bin/choupo-thermoml sync the NIST/TRC ThermoML Archive (ThermoML.v2020-09-30.tgz, doi 10.18434/mds2-2422, sha256-verified) 2. bin/choupo-thermoml identity CAS → InChIKey for the catalogue, two resolvers that must agree 3. bin/choupo-thermoml vle-index every binary VLE block, searchable 4. bin/choupo-thermoml vle-search ethanol cyclohexane –geometry bubbleT 5. bin/choupo-thermoml extract-vle <file> –block N –interior-only → the two files under constant/experiments/ 6. choupoProps (this case) fitParameters → a proposal record 7. promotion (a human act) → NOT promoted – see WHAT THE RUN FOUND below (validated, but it splits the liquid) 8. bin/choupo-import the seal mirrors nothing for the pair (not promoted)

THE EVIDENCE. FIT: doi 10.1016/j.fluid.2018.07.013 (25 interior points, 101.3 kPa). HELD OUT: doi 10.1016/j.fluid.2006.01.023 (14 interior points, 101.325 kPa) – an article whose title names ethanol + water with ionic liquids; the ethanol + cyclohexane block is the one the archive files under it, and reading it back is part of the curator's check. Neither study reports a pure-component bubble point. Chosen BEFORE any fit by the slice-2 partition rule: the FIT set is the low-pressure bubble- temperature study with the most interior points; the HELD-OUT set is the next largest that shares NO author with it (the archive carries no affiliations, so a shared author is the dependence the rule can check). Each file names its article, its archive file and block, the sha256 of the archive file, the two compounds' CAS and InChIKey, and the per-point expanded uncertainty; each says 'reviewStatus transcribedNotCheckedAgainstArticle' because nobody has read the numbers back against the paper yet, and the run announces it. Pure-component endpoints are set aside (they measure the vapour-pressure record, not the pair) and kept in each file under 'selection'.

THE FIT. NRTL with a_ij = a_ji = 0 and alpha = 0.30 fixed: TWO parameters (b_ij, b_ji), because bubble temperatures over the ranges below do not identify more (fit_nrtl_ab in the methanol + water case asks that question directly). Each point is weighted by 1/U, U its OWN stated expanded uncertainty ('weighting uncertainty;'), so a point measured to 0.1 K is not outvoted by one measured to 1 K. The held-out AAD is plain kelvin, unweighted.

WHAT THE RUN FOUND (pinned in 'expected'; measured, not argued). fit_nrtl: b_ij = 354.7 K, b_ji = 565.4 K, in-sample rms 1.30 K (reduced chi2 5.5); fit_nrtl_start2 lands on the same pair. HELD OUT (101.325 kPa, no author in common): AAD 0.32 K (0.094 %) against a band of 0.41 % → 'validated'. Overlay on the held-out set: NRTL 0.33 K, UNIFAC 0.59 K, ideal 9.2 K. AND NOT PROMOTED. The same pair predicts a liquid-liquid split up to about 358.8 K – across the WHOLE of its validity span (337.7-353.5 K), where both studies report one liquid (the archive files each as liquid + gas). The engine's VLLE flash at 340 K and 3 bar, run with this pair, splits a 43/57 feed into two liquids. A bubble-temperature scan solves for ONE liquid by construction, so the held-out verdict cannot see a second one: 'validated' is necessary, not sufficient. The pair stays in this case as the authored record constant/parameters/NRTL/cyclohexane-ethanol.dat (verdict validated, reviewStatus interim); check_regressed_pairs arm (f) refuses it in the catalogue. The remedy – a third parameter constrained by the phase count, or LLE data – is a curation decision, Vitor's.

WHAT IS NOT CLAIMED. The pair is correct at these pressures and in this frame (ideal-gas vapour, the catalogue's Antoine records); nothing here tests it at high pressure, against LLE, or with another vapour model.

**What to expect (golden-locked):**

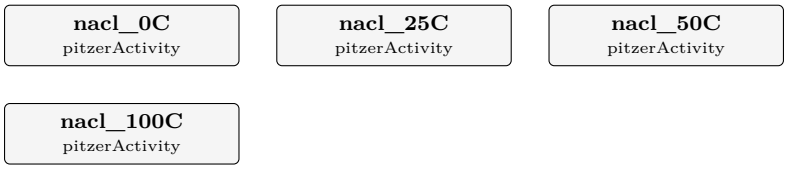
| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fit_nrtl | aad_heldout_K | 0.323443931811 |
| diag | fit_nrtl | aad_heldout_pct | 0.093961584242 |
| diag | fit_nrtl | chi2 | 125.860527228 |
| diag | fit_nrtl | chi2_in_sample | 125.860527228 |
| diag | fit_nrtl | chi2_reduced | 5.47219683598 |
| diag | fit_nrtl | cond_JtJ | 15.8520572952 |
| diag | fit_nrtl | converged | 1 |
| diag | fit_nrtl | corr_computed | 1 |
| diag | fit_nrtl | dof | 23 |
| diag | fit_nrtl | identifiable | 1 |
| diag | fit_nrtl | invertible | 1 |
| diag | fit_nrtl | iter | 6 |
| diag | fit_nrtl | max_abs_corr | 0.940983736236 |
| diag | fit_nrtl | max_abs_resid | 2.85975730621 |
| *... and 70 more reference values in the case's* `expected`*.* | | | |

## electrolyte

**The theory you need.** Ions interact at long range (Debye–Hückel, $\sqrt{I}$ law) and short range (ion–ion, ion–solvent). Choupo's ladder: Davies (charge-only, $I \leq 0.5$), single-salt Pitzer ($\beta_0, \beta_1, C^{\varphi}$; to $I \sim 6$, and with the Silvester–Pitzer $T$-functions to 300°C), multi-ion Pitzer–HMW (per-ion, mixing terms $\theta, \psi$ – seawater-class brines), and eNRTL (local-composition; single-salt, mixed-solvent, and the 2009 symmetric multi-salt with both published and Gibbs–Duhem-consistent formulations). Speciation solves mass action + charge balance by Newton in log-molality space; saturation indices SI = $\log(\mathrm{IAP}/K_{\mathrm{sp}})$ flag scaling. Expect: models to DISAGREE above their validity ($\gamma$ divergence is the lesson, e.g. Davies vs Pitzer at RO recoveries), and every activity model to announce itself. *Full derivation: Theory Guide, the electrolyte + speciation chapters.*

## archer01_nacl_cold_to_hot

tutorials/props/electrolyte/archer01_nacl_cold_to_hot




**What it does:** NaCl Pitzer 0-100 C vs Archer 1992 – the cold edge the freezing curve stands on

**The story:** The catalogue's NaCl Pitzer surface (Appelo 25 C parameters + SP77 T-functions) against Archer, J. Phys. Chem. Ref. Data 21 (1992) 793, DOI 10.1063/1.555915, Tables 9 (phi) and 10 (gamma_pm) – HIS fitted equation's check values ("presented only as a means to test the validity of coding", the paper's own words), at 273, 298, 323 and 373 K.

WHY THE 273 K ROW IS THE POINT: the SP77 temperature extension was fitted ABOVE 25 C, and the freezing-point curve (fpd01_nacl_freezing) leans on that surface BELOW 0 C as an announced extrapolation. This case measures what the extrapolation costs at the last anchor before the ice field – the op announces the T-window violation at 273 K, and the AAD against Archer says whether the announced risk is a real one at the cold edge.

WHAT THIS COMPARISON IS, said precisely: Archer's tables are values of an independent CRITICAL FIT (his global equation, 250-600 K), not raw measurements, and his database overlaps the corpora SP77 drew on. Agreement here is CROSS-EVALUATION CONSISTENCY between two independent fits of overlapping data – stronger than fit-consistency (our parameters never saw Archer's equation), weaker than agreement with an independent measurement. The Hamer & Wu case (pitzer_gamma_hamer_wu) plays the same role at 25 C; this case extends it across T.

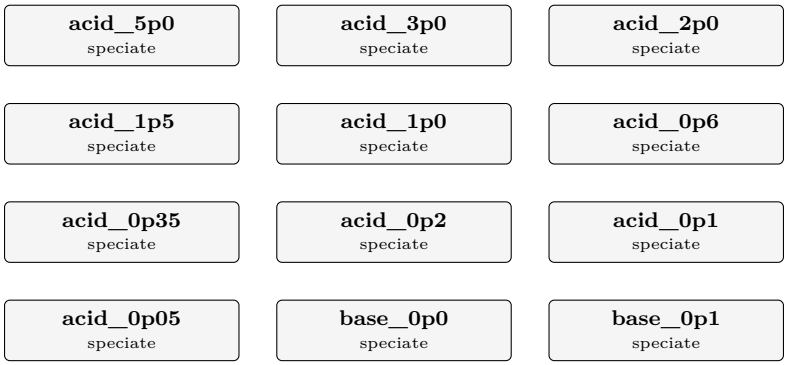
Temperatures are the table's printed values (273 K, not 273.15). If Archer meant T+0.15 K the shift in gamma_pm is below 2e-4 – inside the quoted AADs either way, and stated here so the reading is a recorded choice, not an accident.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | nacl_0C | aw_1m | 0.967382054689 |
| diag | nacl_0C | gamma_AAD_pct | 1.05461170236 |
| diag | nacl_0C | gamma_nMeas | 5 |
| diag | nacl_0C | gamma_pm_1m | 0.641839583122 |
| diag | nacl_0C | gamma_pm_3m | 0.664861748786 |
| diag | nacl_0C | phi_1m | 0.920393250284 |
| diag | nacl_0C | phi_AAD_pct | 0.370645531615 |
| diag | nacl_0C | phi_nMeas | 5 |
| diag | nacl_25C | aw_1m | 0.966828534868 |
| diag | nacl_25C | gamma_AAD_pct | 0.186509289297 |
| diag | nacl_25C | gamma_nMeas | 5 |
| diag | nacl_25C | gamma_pm_1m | 0.657130174321 |
| diag | nacl_25C | gamma_pm_3m | 0.71384939713 |
| diag | nacl_25C | phi_1m | 0.936278540959 |
| *... and 18 more reference values in the case's `expected`.* | | | |

## bjerrum01_carbonate_pH

tutorials/props/electrolyte/bjerrum01_carbonate_pH




**What it does:** Carbonate speciation across the pH range – the Bjerrum plot, titrated

**The story:** THE SPECIATION OF A CARBONATE SOLUTION ACROSS THE WHOLE pH RANGE – the diagram every textbook draws, built the way Choupo has to build it.

THE ONE THING THAT MAKES THIS DIFFERENT FROM THE TEXTBOOK VERSION: here the pH is NOT an axis you set – and that is a CHOICE this case makes, in one word, which you can read three screens down. A textbook draws the fractions against pH by substituting pH into the mass-action expressions, which is arithmetic and always works. Choupo can do that too: 'pH 7.8;' is a legal declaration, it fixes [H+], and it DROPS the electroneutrality row – the run then reports the net charge the composition carries ('pH GIVEN – this is the net charge the analysis carries') instead of forcing it to zero. That mode exists because a laboratory sheet reports a MEASURED pH, which is a datum. It is not what this case wants. Every op below declares 'pH solve;' instead, so [H+] joins the unknowns and the charge balance is the equation that closes the system. The price of that choice is the whole lesson: each point had to be a beaker somebody could weigh out, the 44 of them differ only in what is IN the beaker, and the x-axis is a RESULT, exactly as it is in a laboratory.

WHAT IS ACTUALLY IN EACH BEAKER. Total carbonate is held at 1 mmol/kg throughout. The 44 points differ only in how much strong acid or strong base sits beside it, and each one is a real neutral mixture somebody could weigh out:

Cl / C_T = 5 ... 0.05 hydrochloric acid added to dissolved CO2 Na / C_T = 0 dissolved CO2 alone (H2CO3) Na / C_T = 1 sodium bicarbonate (NaHCO3) Na / C_T = 2 sodium carbonate (Na2CO3) Na / C_T = 2 ... 8 sodium carbonate + caustic soda

That is what 'totalsBasis stoichiometric;' declares below: these totals are neutral-molecule feeds, so the water-analysis charge-imbalance check – which is right for a LAB SHEET, where an imbalance means somebody mismeasured – does not apply. Here the imbalance IS the titrant, and the electroneutrality equation IS the weak-acid pH. Declaring it changes no number: it changes what the engine says about them.

WHAT TO LOOK FOR, in the order a student should look:

1. THE TWO CROSSOVERS LAND ON THE TWO pK's, AND NOBODY PUT THEM THERE. CO2(aq) and HCO3- are equal at pH 6.341; the record's log K1 is 6.352. HCO3- and CO3– are equal at pH 10.262; log K2 is 10.329. Both crossover pH's are INTERPOLATED between the two beakers that bracket them – the axis is 44 sampled points, not a continuum, and no beaker lands exactly on a crossing. Neither pK was used as an input to any point – they are the mass- action constants inside the network, and the crossover is where the titration happens to arrive. The small shift DOWN in both is the activity coefficients, while a textbook plot takes every gamma as 1. The first is moved by gamma_HCO3 alone (about 0.975 there; CO2(aq) is neutral) – gamma_CO3 is 0.90 at that crossover but carbonate is a trace there and moves nothing. The second is moved by the ratio gamma_CO3 / gamma_HCO3, 0.81 against 0.95.

2. THE THREE FRACTIONS DO NOT SUM TO ONE. They dip to 0.99966 through the middle of the range, and the missing carbon is NOT round-off: it is 'NaHCO3aq', the sodium-bicarbonate ION PAIR, which the curated chemistry carries and the three-curve drawing has no place for. The titrant is not a

spectator. Look at the ‘m_NaHCO3aq‘ diagnostic – and note where it is ABSENT: the two beakers with no sodium in them (the acid side, and Na/C_T = 0) have no such species to report, and the engine REFUSES to report a species that is not in the table rather than printing a zero. Absent and zero are different claims.

3. THE ACTIVITY COEFFICIENT OF THE DIVALENT ION MAKES A U. gamma_CO3 runs 0.74 at the acid end, up to 0.98 near Na/C_T = 0 where the ionic strength is lowest, and back down to 0.67 at the alkaline end – because BOTH ends of a titration are loaded with strong electrolyte and only the middle is dilute. A textbook diagram takes every gamma as 1 at every pH, which is the one assumption that is worst exactly where the diagram is usually read.

4. THE SATURATION INDICES ARE THERE ALL ALONG. Every point reports SI_nahcolite / SI_natron / SI_trona – how far the solution is from precipitating a sodium carbonate mineral. At 1 mmol/kg they stay far below zero, which is why this stays a clean acid-base diagram; make C_T a hundred times larger and the diagram acquires a ceiling.

THE CONSTANTS, and their lineage: log K1 = 6.352, log K2 = 10.329 and Kw = 1e-14 at 25 C, from the USGS PHREEQC database (public domain), which is where data/standards/chemistry/{CO2aq,CO3}-formation.dat and water-dissociation.dat come from. Each carries the van’t Hoff dH and the full analytic log K(T), so this whole diagram can be redrawn at another temperature and both pK’s will move – change ‘temperature‘ below and watch which crossover moves further.

NOT VALIDATED AGAINST MEASUREMENT. This case is a STRUCTURAL witness: it shows that the engine’s speciation network, its activity model and its charge balance compose into the diagram the chemistry predicts. No titration data is compared against; the two crossovers landing near the two pK’s is an internal consistency check on the assembly, not agreement with an experiment.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | acid_5p0 | I | 0.00499268 |
| diag | acid_5p0 | aw | 0.999802 |
| diag | acid_5p0 | chargeImposed | 1 |
| diag | acid_5p0 | chargeResidual | -3.02883e-16 |
| diag | acid_5p0 | gamma_CO2aq | 1 |
| diag | acid_5p0 | gamma_CO3 | 0.738626 |
| diag | acid_5p0 | gamma_HCO3 | 0.927056 |
| diag | acid_5p0 | m_CO2aq | 0.000999896 |
| diag | acid_5p0 | m_CO3 | 1.31694e-15 |
| diag | acid_5p0 | m_HCO3 | 1.03591e-07 |
| diag | acid_5p0 | pH | 2.33456 |
| diag | acid_3p0 | I | 0.0029974 |
| diag | acid_3p0 | aw | 0.999874 |
| diag | acid_3p0 | chargeImposed | 1 |
| *... and 635 more reference values in the case’s `expected`.* | | | |

## composition01_nacl

tutorials/props/electrolyte/composition01_nacl




**What it does:** Apparent-salt input: the analysis is given as composition { NaCl 2.0 mol/kg }, expanded to Na + Cl ion totals through NaCl's dissociatesTo, with electroneutrality validated (Sum nu∗charge = 0). Byte-identical to writing totals { Na 2.0; Cl 2.0 } by hand.

*The speciate op expands each salt to ion totals through its dissociatesTo (NaCl → Na 1, Cl 1), and validates electroneutrality (Sum nu∗charge = 0) so a formulated-salts input can never silently unbalance charge. This is byte-identical to writing the ions by hand (totals { Na 2.0; Cl 2.0 }): SI_halite -0.953, feed charge imbalance 0.00%.*

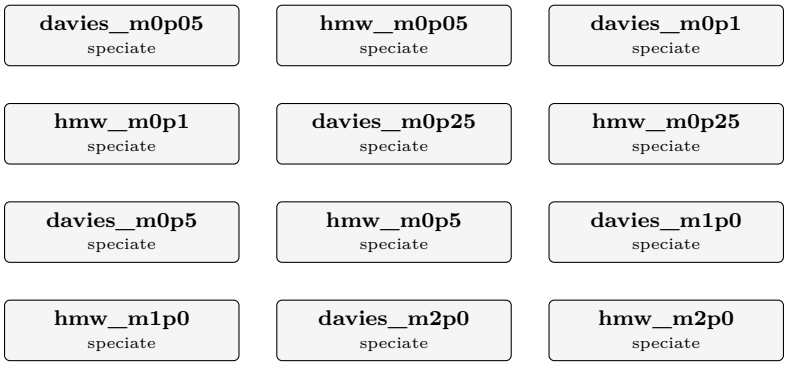
*composition{} is the apparent-salt input mode; analyticalTotals{} / totals{} stay as the mode for waters measured directly in ions/masters (roadmap Phase B).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | naclWater | I | 2 |
| diag | naclWater | SI_halite | -0.953442 |
| diag | naclWater | aw | 0.930475 |
| diag | naclWater | gamma_Cl | 1.01683 |
| diag | naclWater | gamma_Na | 1.01683 |
| diag | naclWater | m_Cl | 2 |
| diag | naclWater | m_Na | 2 |
| diag | naclWater | pH | 7.13303 |
| diag | naclWater | chargeImposed | 1 |
| diag | naclWater | chargeResidual | -6.39009e-17 |

## davies01_band_edge

tutorials/props/electrolyte/davies01_band_edge




**What it does:** The Davies band edge, MEASURED: gamma_pm and a_w for NaCl under Davies beside PitzerHMW on identical brines, m = 0.05..6 mol/kg – the cross-model curve behind every 'indicative beyond I ~ 0.5' advisory

**The story:** THE DAVIES BAND EDGE, measured instead of remembered.

Same NaCl brine, same totals, same solved pH – speciated twice per rung: once under Davies (charge-only, the teaching rung; phi = 1 water activity) and once under PitzerHMW (per-ion virials; rigorous phi). What this measures is CROSS-MODEL divergence between two of Choupo's own engines – weaker than agreement with an independent measurement, and said so; the Pitzer surface itself is anchored elsewhere (pitzer01_nacl, archer01_nacl_cold_to_hot, pitzer_gamma_hamer_wu).

WHY IT EXISTS: every electrolyte run above I ~ 0.5 mol/kg carries the advisory "Davies ... INDICATIVE"; the marcilla01/02 anchor contacts put numbers on what that costs at saturation (halite at 3.65 vs the measured 6.14 mol/kg). This case turns the advisory into a CURVE – per rung, gamma_Na, gamma_Cl and a_w under both closures, side by side. No threshold is invented: the divergence grows smoothly and there is no principled cut, so the table reports and the reader judges (the same posture as the neglected-dCp advisory in SolidPhase).

*Same NaCl brine speciated twice per rung: Davies (teaching rung, charge-only, $\varphi = 1$) beside PitzerHMW (per-ion virials), 298.15 K. Cross-model divergence between two in-tree engines — weaker than an independent measurement and said so; the Pitzer surface itself is anchored elsewhere (pitzer01_nacl, archer01_nacl_cold_to_hot, pitzer_gamma_hamer_wu).*

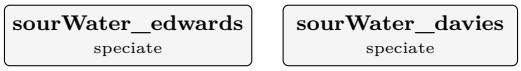
*∗ The "I ~ 0.5 trust band" every advisory quotes is real: the $\gamma\pm$ error crosses ~2–8 % over m = 0.25–0.5 and explodes beyond it (Davies' +0.3·I term has no physics above the band — it is the fit's tail). ∗ At m = 6, Davies reports SI_halite = +1.10 where PitzerHMW reports -0.022 — the rigorous rung sits essentially AT saturation beside Archer's peritectic (6.096 mol/kg), and its a_w = 0.7592 beside the literature ≈ 0.753. This is the whole quantitative story of the marcilla01/02 anchor gaps (Davies saturating at 3.65 vs 6.14 mol/kg; the too-wet organic from a_w ≈ 0.88): the model, not the machinery.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | davies_m0p05 | I | 0.0500001 |
| diag | davies_m0p05 | SI_halite | -4.34316 |
| diag | davies_m0p05 | aw | 0.9982 |
| diag | davies_m0p05 | gamma_Cl | 0.821203 |
| diag | davies_m0p05 | gamma_Na | 0.821203 |
| diag | davies_m0p05 | m_Cl | 0.05 |
| diag | davies_m0p05 | m_Na | 0.05 |
| diag | davies_m0p05 | pH | 7.00292 |
| diag | hmw_m0p05 | I | 0.0500001 |
| diag | hmw_m0p05 | SI_halite | -4.34351 |
| diag | hmw_m0p05 | aw | 0.998302 |
| diag | hmw_m0p05 | gamma_Cl | 0.820867 |
| diag | hmw_m0p05 | gamma_Na | 0.820867 |
| diag | hmw_m0p05 | m_Cl | 0.05 |
| *… and 146 more reference values in the case's `expected`.* | | | |

## edwards01_sour_water_activity

tutorials/props/electrolyte/edwards01_sour_water_activity




**What it does:** Edwards et al. (1978) activity model on a sour-water NH3/CO2 brine – the molecular solutes get a real gamma from the model their parameters were regressed in, where Davies can only return 1.

**The story:** the WITNESS for the ‘edwardsPitzer‘ activity model.

WHAT THIS CASE IS FOR. Choupo already carries two aqueous activity models, and each fails this fluid in its own way:

‘davies‘ is a charge-only correlation. Every singly-charged ion gets the same gamma and every NEUTRAL solute gets gamma = 1 – exactly, by construction, at every ionic strength. A sour-water stripper’s whole difficulty is the volatility of dissolved NH3 and CO2, i.e. of two NEUTRAL molecules, and Davies cannot say anything about either. ‘pitzerHMW‘ is the Harvie-Moller-Weare parameterisation: a DIFFERENT truncation, with a different parameter set (beta2, C-phi, theta, psi), fitted to strong electrolytes. Its numbers are not Edwards’ numbers, and Edwards’ numbers are not valid inside its equation.

‘edwardsPitzer‘ is the truncation Edwards, Maurer, Newman & Prausnitz (AIChE J. 24(6):966-976, 1978) REGRESSED HIS NH3/CO2/H2S/SO2 PARAMETERS IN: Pitzer’s expansion cut after the second virial coefficient, extended to molecular solutes. Using his parameters anywhere else would be quoting his numbers under someone else’s equation.

THE COMPOSITION is the sour-water region his Table 7 anchors: ammonia in large excess over carbon dioxide, the carbonate and carbamate networks both live, at an ionic strength around 1 mol/kg where a dilute correlation has no business being trusted.

WHAT TO READ IN THE OUTPUT. Run the two operations side by side and compare ‘gamma_NH3aq‘ and ‘gamma_CO2aq‘:

under ‘davies‘ they are 1.000000 and 1.000000. Not approximately – exactly, because the model’s gamma is a function of charge alone and the charge is zero. A student who has only ever seen Davies may not realise this is a STRUCTURAL silence rather than a small correction. under ‘edwardsPitzer‘ they are 1.0224 and 1.0101: both SALTED OUT, NH3 rather more than CO2. Both rise because both Table 4 self-terms are positive at 25 C – Eq 14 gives beta0(NH3,NH3) = 0.01522 and beta0(CO2,CO2) = 0.00822 kg/mol – and NH3 rises further because it is the more abundant of the two by three orders of magnitude, so its own self-term dominates its Eq 8 sum.

THE SPECIATION MOVES TOO, and by far more than the gammas: the answer is I = 2.78 against Davies’ 1.78, with m_NH3aq 0.41 against 1.01. That is not a molecular effect at all – it is the IONIC one. Davies at I ~ 2 returns gamma_NH4 = 0.956, because its bracket saturates and then turns back up; Edwards’ Eq 8 returns 0.301 at the same charge. The NH4+/NH3aq equilibrium reads those gammas, so the split follows. The lesson is the one the advisory prints on the Davies run: a dilute correlation used far outside its band does not merely lose accuracy, it changes the answer’s SHAPE.

Neither number here is validated against measurement – see below.

WHAT THIS CASE IS ∗NOT∗. It is not the Table 7 anchor. Reproducing Table 7 needs the vapour side – Henry’s constants on the paper’s own convention, the two-phase equilibrium, the total pressure – and that slice is BUILT: ‘edwards02_table7_vle‘ (2026-08-23; record in docs/design/sour-water-stripper-scope.md section 6c). This case validates the ACTIVITY SURFACE alone, which is the piece the model owns.

WHAT IS AND IS NOT VALIDATED HERE. This case is a STRUCTURAL witness: it proves the model is wired, that its five parameter rules fire and are labelled, and that a neutral solute gets a gamma. It does NOT validate any number against data – there is no measurement in it. The kernel itself is separately checked against two closed forms the paper supplies (the Eq 8a low-dissociation limit and the Debye-Huckel limiting law), and the measured-vs-computed comparison is Table 7 – edwards02_table7_vle. A green run here means "the machine is right", never "the answer is right".

HONESTY ABOUT THE PARAMETERS. Edwards measured exactly one family: beta0 for LIKE molecule pairs (Eq 14, Table 4). Every other entry in the matrix Eq 8 sums over is ESTIMATED from those by his Eqs 21-24, and the paper calls both of its parameter tables "preliminary results subject to change as more and better data become available". The engine prints, at verbosity >= 3, WHICH RULE produced each number – one measured family and four estimation routes – so nothing here can be mistaken for a measured constant.

DATA SCOPE: SEALED (bin/choupo-import). The case carries its own species, reactions and EdwardsPitzer parameter records under constant/, each named and sha256-verified by constant/propertyManifest, and the runtime is FORBIDDEN the installation catalogue.

Sealing this case is where the importer's dependency closure was found not to know the EdwardsPitzer parameter home. The first seal kept 9 of the 28 pair parameters – the Bronsted like-sign zeros, which need no record at all – and the case still RAN, still converged, and answered with 19 terms missing: gamma_HCO3 and gamma_NH4 identical (the charge-only Debye-Huckel fallback) and both neutral gammas exactly 1, which is what a Davies run prints. Nothing refused. The importer now compares the staged sealed run against this case's own 'expected' golden, because running is not agreeing.

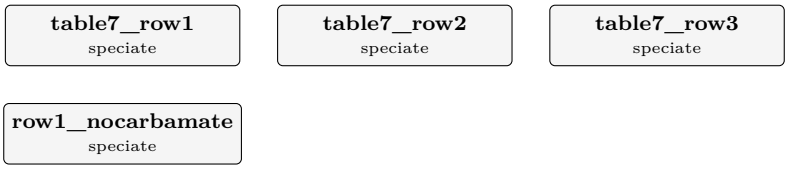
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | sourWater_edwards | I | 2.77652 |
| diag | sourWater_edwards | aw | 0.954591 |
| diag | sourWater_edwards | gamma_CO2aq | 1.01011 |
| diag | sourWater_edwards | gamma_CO3 | 0.0181147 |
| diag | sourWater_edwards | gamma_HCO3 | 0.227994 |
| diag | sourWater_edwards | gamma_NH3aq | 1.02239 |
| diag | sourWater_edwards | gamma_NH4 | 0.30091 |
| diag | sourWater_edwards | m_CO2aq | 0.000484785 |
| diag | sourWater_edwards | m_CO3 | 0.537903 |
| diag | sourWater_edwards | m_HCO3 | 0.911612 |
| diag | sourWater_edwards | m_NH3aq | 0.41019 |
| diag | sourWater_edwards | m_NH4 | 2.48981 |
| diag | sourWater_edwards | pH | 9 |
| diag | sourWater_davies | I | 1.77822 |

*... and 16 more reference values in the case's `expected`.*

## edwards02_table7_vle

tutorials/props/electrolyte/edwards02_table7_vle




**What it does:** Edwards (1978) Table 7 anchor: NH3-CO2-H2O VLE at 100 degC – the paper's own predictions reproduced from its own equations, the experiment quoted as context.

**The story:** the LITERATURE ANCHOR for the edwardsPitzer model: Table 7 of Edwards, Maurer, Newman & Prausnitz, AIChE J. 24(6):966-976 (1978) – NH3-CO2-H2O at 100 degC, three points.

edwards01 is the STRUCTURAL witness: it proves the activity surface is wired, and its own header says none of its numbers is validated against anything. This case is the validation it points at. For each Table 7 point the paper states the total molalities of ammonia and carbon dioxide in the LIQUID and predicts the speciation (molecular NH3 and CO2, ionic strength) and the VAPOUR above it (y_NH3, y_CO2, total pressure). The whole model stack here is the paper's own:

liquid activities Eq 8/10 (edwardsPitzer, the case surface) equilibria K(T) Eq 7 / Table 1 + Eq 20 (case-local chemistry/) Henry's constants Eq 13 / Table 3 (parameters/EdwardsPitzer/henry-∗) pressure correction Eq 11 / Table 2 (vInfinity in the same records) vapour side Eq 5/6 (the speciate op's vapour{} block)

WHAT IS COMPARED, AND AGAINST WHAT. Table 7 carries TWO different columns and conflating them would be dishonest:

the paper's own PREDICTIONS (column I) – a code-to-code comparison. Those are the golden values here, with ONE declared gap: the paper computes vapour-phase fugacity coefficients from a model its transcription does not carry, so this case declares 'fugacity ideal;' (phi = 1) and the residual disagreement is expected to be of the size of that correction (a few percent at 2-3 atm), plus the numerical difference between two implementations of Eqs 7-24.

the EXPERIMENT (Otsuka et al. 1960) – quoted below as CONTEXT, never as a target. The paper's OWN agreement with it is rough (first row: y_NH3 0.066 measured vs 0.123 predicted – a factor of two) and the paper says so plainly. A student reading this case sees a correlation reproduced closely and its real-world agreement stated honestly, which is the opposite of a tool that prints one number.

Table 7, the paper's column I (the golden targets here): row mNH3tot mCO2tot m_NH3 m_CO2 I y_NH3 y_CO2 P[atm] 1 2.90 1.45 1.245 0.014 1.50 0.123 0.494 2.51 2 3.71 1.14 2.279 0.0056 1.26 0.293 0.230 1.98 3 4.30 0.907 3.087 0.0030 1.01 0.407 0.116 1.97 The experiment (Otsuka 1960, context only): 1 0.066 0.506 3.15 2 0.274 0.202 2.08 3 0.355 0.095 2.00

THE CARBAMATE ENTERS THE CATALOGUE HERE (case-local): NH2COO- with Eq 20's equilibrium. Without it, an ammonia-rich carbonated water reports molecular molalities the paper's own predictions contradict – it holds a substantial share of the dissolved CO2 at these conditions. Eq 20 is fitted 20-60 degC and used at 100 degC exactly as the paper uses it, with the extrapolation ANNOUNCED and the paper's own sensitivity argument in the record header (halving K20 moves P by under 2 %).

DECLARED APPROXIMATIONS, each announced by the run: 'fugacity ideal;' – phi = 1 on all vapour species (see above); the solvent Poynting factor of Eq 6 is omitted (< 0.2 % below 4 atm); Eq 20 extrapolated 60 → 100 degC (the paper's own move); water's Psat comes from the case catalogue's own curated Antoine, not from a value the paper does not print.

DATA SCOPE: sealed (bin/choupo-import); every record named and verified by constant/propertyManifest.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | table7_row1 | I | 1.547 |
| diag | table7_row1 | P_atm | 2.51337 |
| diag | table7_row1 | aw | 0.946304 |
| diag | table7_row1 | chargeImposed | 1 |
| diag | table7_row1 | chargeResidual | 2.5541e-13 |
| diag | table7_row1 | gamma_CO2aq | 0.919147 |
| diag | table7_row1 | gamma_CO3 | 0.022685 |
| diag | table7_row1 | gamma_HCO3 | 0.268386 |
| diag | table7_row1 | gamma_NH2COO | 0.528413 |
| diag | table7_row1 | gamma_NH3aq | 0.989831 |
| diag | table7_row1 | gamma_NH4 | 0.318905 |
| diag | table7_row1 | m_CO2aq | 0.0141073 |
| diag | table7_row1 | m_CO3 | 0.0555284 |
| diag | table7_row1 | m_HCO3 | 1.09498 |
| *. . . and 99 more reference values in the case's* `expected`*.* | | | |

## edwards03_table8_vle

tutorials/props/electrolyte/edwards03_table8_vle




**What it does:** Edwards (1978) Table 8 anchor: NH3-H2S-H2O VLE at 80 degC – the paper's own predictions reproduced from its own equations, the experiment quoted as context.

**The story:** the SECOND literature anchor for the Edwards (1978) model stack: Table 8 of the paper, ammonia + hydrogen sulfide in water at 80 degC, against the paper's OWN prediction column I.

WHY A SECOND ANCHOR. edwards02 (Table 7) pins the NH3-CO2 system, where the difficulty is the carbonate/carbamate network. Table 8 exercises what Table 7 cannot: the SULFIDE family (the mi-H2Saq-∗ molecule-ion records curated 2026-08-24 from the paper's Table 6, including its two TABULATED ZEROS), a different temperature (80 degC, so every T-polynomial is read off its 25 degC calibration point), and ionic strengths up to 7.8 mol/kg – beyond the 6-molal Pitzer validity the paper itself quotes. Rows 2 and 3 are code-to-code targets and NOT physical claims; the paper runs them and says so ("if ionization is extensive, the upper limit is necessarily lower").

WHAT IS COMPARED, AND AGAINST WHAT. Exactly the edwards02 posture:

the paper's own PREDICTIONS (column I: all beta0 from Tables 4 and 6, the configuration these curated records carry) are the anchor targets; the EXPERIMENT (Miles & Wilson 1975) is CONTEXT, never a target. The paper's own worst multisolute disagreement lives here: rows 2 and 3 measure 16.50 and 13.20 atm against predictions of 12.91 and 12.64 atm (22-28 %), and its Table 9 sensitivity analysis shows the liquid-composition error needed to close such gaps.

THE CHEMISTRY IS THE PAPER'S OWN: case-local constant/chemistry/ carries Table 1 (H2O, NH3 and H2S rows) in the formation direction, each header recording the derivation. The H2S formation logK25 is 7.0033 against the independently curated PHREEQC 6.994 – two primaries 0.009 apart, a finding, not an error.

DECLARED GAPS, the same list as edwards02 plus none: 'fugacity ideal;' – the paper's own vapour-phi model is not transcribed; the residual is expected to carry that correction (largest on rows 2-3 at 13-16 atm); the solvent Poynting factor of Eq 6 is omitted; water's Psat comes from the case catalogue's own curated Antoine.

DATA SCOPE: sealed (bin/choupo-import); every record named and verified by constant/propertyManifest.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | table8_row1 | I | 1.13632 |
| diag | table8_row1 | P_atm | 1.10768 |
| diag | table8_row1 | aw | 0.896601 |
| diag | table8_row1 | chargeImposed | 1 |
| diag | table8_row1 | chargeResidual | -4.35684e-17 |
| diag | table8_row1 | gamma_H2Saq | 1.08209 |
| diag | table8_row1 | gamma_HS | 0.516824 |
| diag | table8_row1 | gamma_NH3aq | 1.06377 |
| diag | table8_row1 | gamma_NH4 | 0.531883 |
| diag | table8_row1 | m_H2Saq | 0.0038303 |
| diag | table8_row1 | m_HS | 1.13617 |
| diag | table8_row1 | m_NH3aq | 3.97368 |
| diag | table8_row1 | m_NH4 | 1.13632 |
| diag | table8_row1 | pH | 8.67302 |
| *... and 64 more reference values in the case's `expected`.* | | | |

## enrtl_mixed_nacl_ethanol_esteso

tutorials/props/electrolyte/enrtl_mixed_nacl_ethanol_esteso




**What it does:** VALIDATION of the generalized segment-based eNRTL (Chen & Song 2004, AIChE J 50:1928, DOI 10.1002/aic.10151) against the PRIMARY measured gamma_pm of NaCl in ethanol-water at 25 C (Esteso et al., J. Solution Chem. 18 (1989) 277, emf cell, 0-90 wt% ethanol). ZERO refit: every tau/X:Z is the published Table 2/4/7 value; A_phi per composition is Esteso's measured Table II. Headline AAD ~4.3% over 47 points (paper quotes RMS 0.026 on gamma); ~1% in pure water, ~1.2% at 70 wt%. The +8% at 80/90 wt% is DECLARED physics the complete-dissociation model cannot carry: Esteso measures ion association K_A = 30-46 there (his Table V).

**The story:** can eNRTL predict a salt's activity in the antisolvent mixture?

The SAME physics the antisolvent crystalliser leans on – gamma_pm of NaCl as ethanol pushes the water out – validated against the PRIMARY dataset:

MODEL Chen & Song, "Generalized electrolyte-NRTL model for mixed-solvent electrolyte systems", AIChE J 50(8):1928-1941 (2004), DOI 10.1002/aic.10151. Ethanol = 1.811 (-C2H4-) + 0.609 (-OH) segments (Table 4); ion<→OH taus = the aqueous NaCl pair (Table 2); ion<→C2H4 taus from Table 7. ZERO refit here. DATA Esteso, Gonzalez-Diaz, Hernandez-Luis & Fernandez-Merida, "Activity coefficients for NaCl in ethanol-water mixtures at 25 C", J. Solution Chem. 18(3):277-288 (1989). Table I (emf, molal scale, per-solvent reference). A_phi per composition = Esteso Table II (measured dielectric + density). 0 wt% gate: Robinson & Stokes, Electrolyte Solutions, 2nd ed. (1959), NaCl appendix table; A_phi(water, 25 C) = 0.391.

WHAT THE NUMBERS SAY (locked by the golden): total AAD ~4.3% over 47 points; pure water ~1.0%; 70 wt% ethanol ~1.2%. 80/90 wt% sit +8% systematically – Esteso's own chemical-model fit finds ion-pair association K_A = 30-46 there (Table V), physics a complete- dissociation model cannot represent. Declared, not tuned away.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | esteso | AAD_total_pct | 4.30645321132 |
| diag | esteso | AAD_wt0_pct | 1.0101269464 |
| diag | esteso | AAD_wt20_pct | 5.27484474055 |
| diag | esteso | AAD_wt40_pct | 5.27091506549 |
| diag | esteso | AAD_wt60_pct | 2.69902360213 |
| diag | esteso | AAD_wt70_pct | 1.18874152393 |
| diag | esteso | AAD_wt80_pct | 7.80889140849 |
| diag | esteso | AAD_wt90_pct | 7.78650421238 |
| diag | esteso | n_points | 47 |

## enrtl_multisalt_nacl_kcl

tutorials/props/electrolyte/enrtl_multisalt_nacl_kcl

| harned__nacl__kcl | harned__nacl__kcl__refined |
|---|---|
| enrtlMultiSalt | enrtlMultiSalt |

**What it does:** Multi-salt symmetric eNRTL (Song & Chen 2009) – NaCl+KCl Harned sweep at 4 mol/kg

**The story:** THE MULTI-SALT eNRTL ARRIVES: the multicomponent symmetric electrolyte NRTL of Song & Chen 2009 (Ind. Eng. Chem. Res. 48, 7788, DOI 10.1021/ie9004578) on the classic Harned experiment – NaCl + KCl sharing water at a CONSTANT 4 mol/kg total molality, gamma_pm of BOTH salts as KCl replaces NaCl. This is the paper's own demonstration system (its Figures 1-2).

What the run shows (all golden-locked): gamma_pm(NaCl) slides from its pure-solution value to its TRACE value in 4 m KCl, and vice versa – both ln gamma curves are NEARLY LINEAR in the salt fraction: Harned's rule, EMERGING from the model rather than being assumed (R^2 > 0.99 with zero electrolyte-electrolyte parameters). The PURE endpoints reproduce the validated single-salt eNRTL kernel exactly (the reduction pin ~1e-15): the multicomponent formulation collapses correctly. The pair parameters are the EXACT Chen & Evans 1986 Table 1 values the paper uses (case-local constant/electrolyte/enrtl.dat, cited). The standards K-Cl.dat carries an INDEPENDENT refit to Hamer & Wu 1972 that lands within 0.2 % of the 1986 values – two roads, one answer. The 1986 paper itself (Table 2) validates tauEE = 0 for NaCl-KCl: sigma = 0.019 % on the Covington 1968 mixture osmotic data with the salt-salt energies set to zero – this case's default is the canonical choice. HONESTY, printed by the op: the published multi-electrolyte eNRTL activity-coefficient expressions are NOT exactly Gibbs-Duhem consistent (the composition-dependent mixing rules are differentiated as constants). The op's FD oracle quantifies that inconsistency at the mid-sweep state (gibbsDuhem_dev_mid) next to the pure-salt state where the same oracle sits at ~1e-9. THE REFINED FORMULATION SHIPS TOO (second op below): gamma as the EXACT derivative of the same Gex – the chain rule kept through the composition-dependent mixing rules (the Bollas, Chen & Barton 2008 idea on the symmetric form). Its oracle closes (~1e-9) at every composition; both reduce EXACTLY to the same single-salt kernel at the endpoints. Mid-sweep they differ by |dln gamma_pm| ~ 0.041 (~4 % in gamma), and the trace limits differ visibly (NaCl trace 0.617 published vs 0.678 consistent). WHICH is closer to nature is a question for MIXTURE data (Covington 1968 / Robinson 1961) – this case states the facts and locks both; it does not pick a winner without measurements.

tauEE (the same-anion electrolyte-electrolyte parameter tau_NaCl,KCl of the paper's Figures 1-2) defaults to 0; set it to +-1 to reproduce the paper's sensitivity fan.

THE EXPERIMENTAL VERDICT (Robinson 1961, J. Phys. Chem. 65, 662, DOI 10.1021/j100822a016 – isopiestic NaCl-KCl mixtures, Harned coefficients at M ~ 4.4 mol/kg: alpha_NaCl = +0.0243, alpha_KCl = -0.0091; the op now emits harned_alpha_1/2 on the same convention, golden-locked): alpha_NaCl: published +0.0285 (+17 %) consistent +0.0180 (-26 %) alpha_KCl: published -0.0256 (2.8x) consistent -0.0187 (2.1x) With tauEE = 0, PUBLISHED wins on the dominant NaCl coefficient; BOTH formulations overstate the (small) effect on KCl.

CROSS-CHECK vs THE REFINED-eNRTL PAPER (Bollas, Chen & Barton 2008, AIChE J 54, 1608, DOI 10.1002/aic.11485 – their Figure 1 is THIS system, 4 m, same 1986 parameters, Robinson's points overlaid): their Fig 1a (original eNRTL) matches our 'published' to the pixel (NaCl solids at x=1 for tau = 0/0.25/0.50: ours -0.216/-0.186/-0.157 vs read -0.215/-0.185/-0.15; KCl dashed at x=0: -0.144/-0.183 vs -0.145/-0.185); their Fig 1b (refined) shows the SIGNATURE our 'consistent' shows: the tau fan nearly COLLAPSES (exact derivatives are almost insensitive to tau_EE: ours moves only -0.176 → -0.184 for tau 0 → 0.5, where published moves -0.216 → -0.157), and the KCl trace lands at ~-0.16 (ours -0.163). Their refinement differentiates the 1986 UNSYMMETRIC Gex; ours differentiates the 2009 SYMMETRIC one – same idea, sibling surfaces, same qualitative physics, near-identical aqueous numbers. A single fitted tauEE cannot close both (best compromise tau ~ 0.29 trades the NaCl error up to -23 % to halve the KCl one) – the residual is structural, not a missing parameter. Note the consistency of the record: Chen & Evans 1986 validated tauEE = 0 against mixture OSMOTIC data (sigma 0.019 %) – the solvent sees the mixture mean, where the individual-gamma asymmetries largely cancel; Robinson's per-salt gammas are where they show.

FIGURE-1 REPRODUCTION – TO THE PIXEL (checked 2026-07-02, figure read at 200 dpi): the paper's y-axis says "ln gamma" but the numbers are LOG10 (log10 of the pure-NaCl-4m gamma is -0.106 = the figure's -0.105; ln would be -0.24, off the plotted range). With the 1986 parameters this kernel reproduces

the full tau fan to <= 0.001 in log10: x_KCl=0 (all curves): -0.106 (figure ~ -0.105) x_KCl=1: tau=0 → -0.216 (figure ~ -0.215) tau=+1 → flat, dip to -0.101, back to -0.106 (the figure's circle-marked curve shows the same shallow dip) tau=-1 → -0.364 (figure ~ -0.365)

THE PARAGRAPH ABOVE IS A DATED READING, NOT THE LIVE STATE (2026-09-03). The golden this case ships today pins gamma1_pure = 0.801 for pure NaCl at 4 m, i.e. log10 = -0.096, where the reading of 2026-07-02 recorded -0.106 against the figure's -0.105: the kernel's answer has moved by ~0.01 in log10 since the figure was read (the 2026-08-29 fleet audit found the drift). Whether the CURRENT curves still sit on the figure can only be settled by reading the figure again, and that has not been done – so "to the pixel" is what was true on 2026-07-02, kept as the record of that check, and is not a claim about this golden. Nothing here was adjusted toward the figure.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | harned_nacl_kcl | consistent | 0 |
| diag | harned_nacl_kcl | formulation_gap_mid | 0.0413132048052 |
| diag | harned_nacl_kcl | gamma1_pure | 0.801115128273 |
| diag | harned_nacl_kcl | gamma1_trace | 0.616514114921 |
| diag | harned_nacl_kcl | gamma2_pure | 0.575990027563 |
| diag | harned_nacl_kcl | gamma2_trace | 0.728881856731 |
| diag | harned_nacl_kcl | gibbsDuhem_dev_mid | 0.471752659198 |
| diag | harned_nacl_kcl | harned_R2_min | 0.998954550075 |
| diag | harned_nacl_kcl | harned_alpha_1 | 0.0284936089821 |
| diag | harned_nacl_kcl | harned_alpha_2 | -0.0256162918946 |
| diag | harned_nacl_kcl | harned_slope_1 | -0.262435837152 |
| diag | harned_nacl_kcl | harned_slope_2 | -0.235934767418 |
| diag | harned_nacl_kcl | oracle_pure_dev | 4.03345401523e-09 |
| diag | harned_nacl_kcl | reduction_pin_dev | 1.11022302463e-15 |
| *... and 18 more reference values in the case's `expected`.* | | | |

## enthalpy_naoh_water

tutorials/props/electrolyte/enthalpy_naoh_water




**What it does:** NaOH-water heat of dilution: Pitzer with calorimetric T-slots vs Parker (1965)

**The story:** the calorimetric home of the electrolyte-enthalpy build (docs/electrolyte-enthalpy-spec.md).

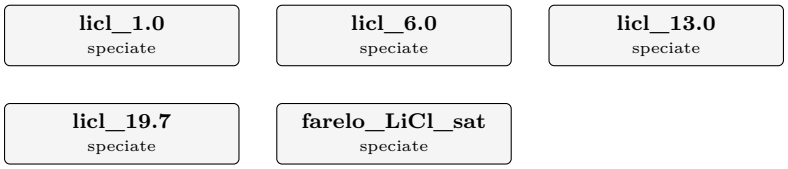
constant/experiments/phiL_25C_parker1965.dat carries the MEASURED Phi_L (Parker NSRDS-NBS 2, 1965) used by the curation refit (bin/curate/fit_lphi_pitzer.py) that fitted the Na-OH dbeta/dT T-slots in the standard catalogue (the FIT's own rel-AAD 4.97%, gate <5%; the in-engine L_phi scan pinned by this case lands at rel-AAD 4.77% – two measurements, not one). The op below sweeps the ACTIVITY surface whose T-derivative is L_phi; the in-engine L_phi scan + AAD overlay land with slices 4-5 (the 3-curve chart).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | gamma_naoh | Lphi_AAD_abs_Jmol | 22.9776 |
| diag | gamma_naoh | Lphi_AAD_rel_pct | 4.7712 |
| diag | gamma_naoh | Lphi_extrap_maxdev_Jmol | 15774.0028 |
| diag | gamma_naoh | Lphi_nMeas | 29.0000 |
| diag | gamma_naoh | aw_1m | 0.9665 |
| diag | gamma_naoh | gamma_pm_1m | 0.6679 |
| diag | gamma_naoh | gamma_pm_3m | 0.7838 |
| diag | gamma_naoh | phi_1m | 0.9471 |

## farelo_licl_range

tutorials/props/electrolyte/farelo_licl_range




**What it does:** LiCl Pitzer REFITTED to Hamer-Wu + Farelo saturation: gamma_pm to ~11% over 0.5-19 mol/kg (was 774 at 19.7m), and the LiCl.H2O hydrate Ksp reproduces Farelo's 19.7 mol/kg via the HMW osmotic water activity (a_w=0.11).

*Prof. F?tima Farelo measured LiCl solubility at 19.7 → 23.5 mol/kg (solid: $LiCl{\cdot}H_2O$) — ∗J. Chem. Eng. Data∗ 50 (2005) 1470, DOI 10.1021/je050111j. That is one of the most concentrated aqueous electrolytes there is.*

*The catalogue's earlier Li–Cl Pitzer parameters (Appelo, Parkhurst & Post, ∗Geochim. Cosmochim. Acta∗ 125 (2014) 49), fitted below ~6 mol/kg, blow up at high concentration — $\gamma\pm$ = 774 at 19.7 mol/kg vs ~45 measured. The two-parameter B-term is linear in ∗m∗, so $\gamma$ grows exponentially.*

*Rather than declare defeat, the parameters were refitted to Hamer & Wu (1972) $\gamma\pm$ and Farelo's saturation. A negative C$\varphi$ (-0.0026) tames the high-∗m∗ blow-up, and the three-parameter form then reaches saturation:*

*carries a water-activity leg. At 19.7 mol/kg the model gives a_w = 0.11 (not 1!) — which only the rigorous Pitzer-HMW osmotic coefficient delivers, not the $\varphi$ = 1 dilute shortcut. The lithiumChlorideH2O mineral Ksp ($logK_{25}$ = 4.984) is calibrated to Farelo's saturation ∗through∗ that a_w. The two pieces meet: the refitted $\gamma$ and the osmotic a_w together reproduce the measured solubility.*

*- electrolyte/pairs.dat: the Li–Cl Pitzer pair, refitted to Hamer–Wu + Farelo, valid to ~19 mol/kg (supersedes the Appelo low-∗m∗ fit for concentrated LiCl). - electrolyte/minerals.dat: the $LiCl{\cdot}H_2O$ hydrate Ksp, calibrated to Farelo. - constant/species/Li.dat: Li? aqueous formation datum (hfAq = -278.49 kJ/mol, CODATA).*

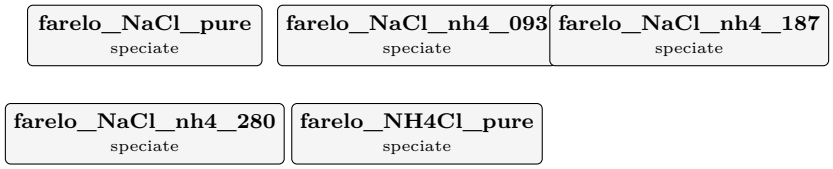
*- The full measured dataset (this and the other five ternaries) is tabulated, cited, in constant/experimental/-farelo_solubility.csv. - The equilibrate round-trip (precipitate $LiCl{\cdot}H_2O$ from a supersaturated feed) diverges at $I \approx 20$ — a known active-set / $\gamma$ fixed-point numerical limit at extreme ionic strength, ∗independent of the activity model∗. The SI = 0 calibration at the saturation composition is exact; reaching it by precipitation is the convergence problem, not the thermodynamics. - The $AlCl_3$ systems from the same paper remain deferred: $Al^3$? hydrolyses (the paper notes pH < 2.75) and needs an Al speciation model Choupo does not yet carry.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | farelo_LiCl_sat | I | 19.7 |
| diag | farelo_LiCl_sat | SI_lithiumChlorideH2O | 1.18638e-05 |
| diag | farelo_LiCl_sat | aw | 0.113466 |
| diag | farelo_LiCl_sat | chargeImposed | 1 |
| diag | farelo_LiCl_sat | chargeResidual | 5.04184e-14 |
| diag | farelo_LiCl_sat | gamma_Cl | 46.7905 |
| diag | farelo_LiCl_sat | gamma_Li | 46.7905 |
| diag | farelo_LiCl_sat | m_Cl | 19.7 |
| diag | farelo_LiCl_sat | m_Li | 19.7 |
| diag | farelo_LiCl_sat | pH | 7.99525 |

## farelo_nacl_nh4cl

tutorials/props/electrolyte/farelo_nacl_nh4cl




**What it does:** Pitzer-HMW vs Farelo (JCED 2005): NaCl+NH4Cl common-ion salting-out – model SI ~ 0 at each measured saturation point (NaCl 6.15→5.5→5.0; NH4Cl 7.35 mol/kg, 298 K).

*A glass-box validation of Choupo's multi-ion Pitzer electrolyte model against the solubility measurements of Prof. F?tima Farelo (Chemical Engineering Dept., IST — T?cnico Lisboa).*

*> M. Farelo, C. Fernandes & A. Avelino, ∗"Solubilities for Six Ternary Systems: > NaCl + $NH_4Cl$ + $H_2O$, KCl + $NH_4Cl$ + $H_2O$, NaCl + LiCl + $H_2O$, KCl + LiCl + $H_2O$, > NaCl + $AlCl_3$ + $H_2O$, and KCl + $AlCl_3$ + $H_2O$ at T = (298 to 333) K"∗, > J. Chem. Eng. Data 50 (2005) 1470–1477. DOI [10.1021/je050111j](https://doi.org/10.1021/je050111j).*

*Saturation molalities measured to ±0.0007 mol/kg (visual polythermal / isothermal methods). The pure-salt anchors at 298.15 K: NaCl 6.15, $NH_4Cl$ 7.35 mol/kg; the salting-out series (Table 1): NaCl solubility falls 6.15 → 5.5 → 5.0 → 4.95 as $NH_4Cl$ rises 0 → 0.93 → 1.87 → 2.80 mol/kg.*

*The model is calibrated to only the two pure-salt solubilities — halite's Ksp from PHREEQC, sal-ammoniac's Ksp pinned to Farelo's $NH_4Cl$ point. Everything else is a prediction. The $NH_4$–Cl Pitzer interaction parameters come from the primary single-electrolyte fit (Pitzer & Mayorga 1973); the catalogue already carried Na–Cl and K–Cl.*

*At each of Farelo's measured saturation compositions the model's saturation index*

*should be ≈ 0 if the model reproduces the measurement. It is, to ±0.03 log units:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | farelo_NaCl_pure | I | 6.15 |
| diag | farelo_NaCl_pure | SI_halite | 0.0168246 |
| diag | farelo_NaCl_pure | aw | 0.751868 |
| diag | farelo_NaCl_pure | gamma_Cl | 1.0105 |
| diag | farelo_NaCl_pure | gamma_Na | 1.0105 |
| diag | farelo_NaCl_pure | m_Cl | 6.15 |
| diag | farelo_NaCl_pure | m_Na | 6.15 |
| diag | farelo_NaCl_pure | pH | 7.11704 |
| diag | farelo_NaCl_nh4_093 | I | 6.42998 |
| diag | farelo_NaCl_nh4_093 | SI_halite | -0.022239 |
| diag | farelo_NaCl_nh4_093 | SI_salammoniac | -0.70594 |
| diag | farelo_NaCl_nh4_093 | aw | 0.746427 |
| diag | farelo_NaCl_nh4_093 | gamma_NH4 | 0.591469 |
| diag | farelo_NaCl_nh4_093 | gamma_Na | 1.04074 |

*... and 39 more reference values in the case's `expected`.*

## fpd01_nacl_freezing

tutorials/props/electrolyte/fpd01_nacl_freezing




**What it does:** NaCl freezing-point depression – ice as a SolidPhase, the crystal's first case

**The story:** THE ICE WITNESS. The freezing-point curve of NaCl(aq), solved as chemical-potential equality between an ACTUAL SolidPhase (the crystallising mode, its first case-reachable consumer) and the Pitzer surface's water activity:

f_solid(T) = a_w(m, T) ∗ Psat(T)

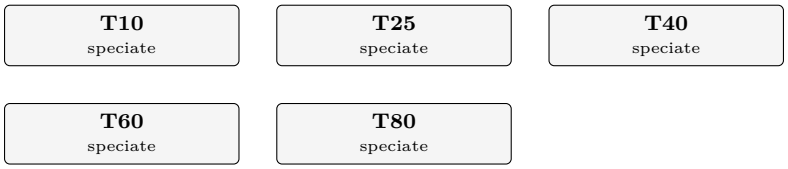
Until this case the crystal was exercised only by check_ice_freezing's probe; here its refusals and announced approximations (the neglected liquid/solid heat-capacity term, the sub-freezing Psat extrapolation) all fire on a real run. The dilute-limit KPI closes the K_f loop: the derived cryoscopic constant against the slope this same phase produces. ANCHOR HONESTY: the validation dataset carries ONE measured point with an INTERIM citation (see its header) – a thin anchor, said plainly, and the enrichment path (a primary FPD table read) is on the curation ledger.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | fpd_nacl | Tf_lastM_K | 266.325 |
| diag | fpd_nacl | depression_lastM_K | 6.83528 |
| diag | fpd_nacl | slope_dilute_K_kg_mol | 3.46546 |
| diag | fpd_nacl | slope_over_nuKf | 0.931447 |
| diag | fpd_nacl | theta_AAD_pct | 0.0460433 |
| diag | fpd_nacl | theta_nMeas | 1 |

## ksp_temperature

tutorials/props/electrolyte/ksp_temperature




**What it does:** K(T) MATTERS: a temperature scan of mineral saturation for ONE fixed water (a hard, carbonate-bearing groundwater at pH 7.6), speciated at 10, 25, 40, 60 and 80 C. The point is the SIGN of dK/dT. CALCITE dissolution is EXOTHERMIC, so its equilibrium constant FALLS as temperature rises and its saturation index SI_calcite RISES – RETROGRADE SOLUBILITY: calcite is LESS soluble hot. This is why hot heat-exchanger surfaces and warm RO concentrate scale with calcium carbonate even when the cold feed looks safe. Contrast HALITE (NaCl): no analytic K(T) in the database, so it falls back to constant-dH van't Hoff with a small POSITIVE dH (endothermic) – normal solubility, SI_halite drifts gently DOWN with T. At 25 C nothing in the K(T) machinery moves (every form returns logK25 exactly, byte-stable); off 25 C the run announces which K(T) form each entry uses (analytic / van't Hoff / flat) and warns if any analytic is used beyond its fitted validity range. K(T): analytic = PHREEQC -analytical_expression anchored on logK25 (logK(T)=logK25+[ana(T)-ana(298.15)]); USGS PHREEQC, public domain.

**The story:** why hot surfaces scale with calcite (K(T) MATTERS).

ONE fixed water, speciated at five temperatures. A hard, carbonate-bearing groundwater (gypsum- and calcite-relevant) at a GIVEN pH 7.6 – holding pH and the totals fixed isolates the lesson: the only thing that changes from point to point is the equilibrium constant K(T).

ion mol/kg w ion mol/kg w Ca 0.010 Cl 0.050 Na 0.030 SO4 0.012 HCO3 0.005 pH 7.6 (given)

THE LESSON – the SIGN of dK/dT:

calcite dissolution is EXOTHERMIC (dH = -28.1 kJ/mol), so log K FALLS as T rises. SI_calcite = log(IAP) - log K(T) therefore RISES with temperature – RETROGRADE SOLUBILITY. Calcite is LESS soluble hot, which is exactly why heat-exchanger tubes and warm RO concentrate scale with CaCO3 although the cold feed looks safe. K(T) here is the PHREEQC ANALYTIC expression (-analytical_ expression A1..A5), anchored on logK25 so logK(25 C) = logK25 EXACTLY.

gypsum analytic too, but only mildly retrograde (its solubility peaks near 40 C) – a gentler curve for contrast.

halite NaCl carries NO analytic K(T) in phreeqc.dat (the db's -analytic line is commented out), so it falls back to constant-dH van't Hoff. dH = +1.37 kJ/mol > 0 (endothermic) → NORMAL solubility: log K(T) RISES and SI_halite drifts DOWN. The honest contrast with calcite: the scaling verdict flips with temperature for the retrograde salt, not for the normal one.

At 25 C the K(T) machinery returns logK25 for EVERY entry (byte-stable); off 25 C the run announces the form in use per entry (analytic / van't Hoff / flat) and warns loudly if an analytic is used beyond its fitted validity range. (Calcite's analytic is valid 0 - 300 C, so the whole 10 - 80 C scan is in range.)

Speciation/mineral data: case-local constant/electrolyte/ – a self- contained SNAPSHOT of the standards catalogue (USGS PHREEQC, public domain; see the .dat headers). The case files ECLIPSE data/standards/electrolyte/.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | T10 | I | 0.0795197 |
| diag | T10 | SI_calcite | 0.56588 |
| diag | T10 | SI_gypsum | -0.335309 |
| diag | T10 | SI_halite | -4.56674 |
| diag | T10 | aw | 0.998115 |
| diag | T10 | pH | 7.6 |
| diag | T25 | I | 0.0772864 |
| diag | T25 | SI_calcite | 0.789002 |
| diag | T25 | SI_gypsum | -0.365046 |
| diag | T25 | SI_halite | -4.60657 |
| diag | T25 | aw | 0.998126 |
| diag | T25 | pH | 7.6 |
| diag | T40 | I | 0.074682 |
| diag | T40 | SI_calcite | 0.981885 |
| *… and 91 more reference values in the case's `expected`.* | | | |

## overlay01_nacl_ksp

tutorials/props/electrolyte/overlay01_nacl_ksp




**What it does:** PARTIAL deep-merge overlay: a case recalibrates ONLY the halite equilibrium logK25 via a case-local overlayOf NaCl; every other datum (dH, dissolutionReaction, calorimetric, crystal) still comes from the standard components/NaCl.dat record. The [overlay] line names exactly which leaf came from the case tier.

*A case recalibrates ONLY the halite equilibrium logK25 (1.60 here) to its own value, via a case-local constant/components/NaCl.dat declaring overlayOf NaCl;.*

*The reader deep-merges this delta over the standard components/NaCl.dat: only solidPhases.halite.equilibrium.logK25 is replaced; dH, the dissolutionReaction, the calorimetric anchor and the crystal block all still resolve from the standard record. The run prints the provenance:*

*This is the roadmap Phase A acceptance test: a case ships ONLY its deltas, never a second competing complete record, and per-value provenance (tier + file) is preserved.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | naclWater | I | 2 |
| diag | naclWater | SI_halite | -0.983442 |
| diag | naclWater | aw | 0.930475 |
| diag | naclWater | gamma_Cl | 1.01683 |
| diag | naclWater | gamma_Na | 1.01683 |
| diag | naclWater | m_Cl | 2 |
| diag | naclWater | m_Na | 2 |
| diag | naclWater | pH | 7.13303 |
| diag | naclWater | chargeImposed | 1 |
| diag | naclWater | chargeResidual | -6.39009e-17 |

## pb82_calcite_open_co2

tutorials/props/electrolyte/pb82_calcite_open_co2




**What it does:** Calcite in CO2-saturated water (Plummer & Busenberg 1982): open pCO2 0.956 atm + equilibrate calcite at 25 C – the engine's pH at calcite equilibrium beside the paper's junction-corrected 6.004 +/- 0.005 and its model value 6.011

**The story:** THE PLUMMER & BUSENBERG (1982) EXPERIMENT, as the engine models it: calcite at equilibrium with CO2-saturated water – an OPEN system at pCO2 = 0.956 atm, 25 C (their electrode runs' mean CO2 pressure), the gas pin, the full carbonate speciation, the CaHCO3+/CaCO3(aq) ion pairs and the mineral ceiling all engaged at once.

PRIMARY: L. N. Plummer & E. Busenberg, Geochim. Cosmochim. Acta 46 (1982) 1011-1040, DOI 10.1016/0016-7037(82)90056-4. Owner-provided PDF.

THE ANCHOR, and what it independently tests: their MEASURED pH at calcite equilibrium under exactly these conditions, corrected for liquid-junction potential, is 6.004 +/- 0.005 (4 M KCl filling, p. 1022; uncorrected Orion runs read near 5.984); their own aqueous model puts the thermodynamic pH at 6.011. The engine's number below comes out of Henry + K1/K2 + the ion pairs + Davies + electroneutrality + the SI = 0 ceiling END TO END – none of which was fitted to this pH.

LINEAGE, stated honestly: the calcite log K(T), K_H, K1, K2 and ion-pair constants in our curated records descend from the USGS PHREEQC database, whose carbonate block IS this paper's Table 3 and Eqs – so agreement in the CONSTANTS is lineage consistency, not validation. What the measured pH tests independently is the ASSEMBLY: that the engine composes those constants (gas pin, chained carbonate master, ion pairs, activity model, charge balance, mineral ceiling) into the same physical answer the electrode saw.

WHAT IT ACTUALLY LANDS AT, and where that is written. The engine gives pH 6.02946 – 0.025 ABOVE the measurement, 0.018 above their own model's 6.011 – and the case attributes that offset to an activity-model difference (Davies here against their Truesdell-Jones, at I = 0.027). SINCE 2026-08-12 THAT COMPARISON IS DATA, NOT PROSE: 'expected' carries an 'anchor' row on the measured 6.004 with a band set to the explained offset, beside the self-recorded row on 6.02946 – so a drift AWAY from the electrode fails as an anchor disagreement, and an unexplained drift TOWARD it fails as a moved golden. Both ends, on purpose. This header used to end with "a wrong composition would not land within 0.01 pH", which reads as though the engine does; it does not, and the number is above.

APPROACH FROM ABOVE: the engine's equilibrate is a precipitation CEILING (SI → 0 from supersaturation; it does not dissolve a solid it was never given). The seed is therefore a SUPERSATURATED Ca(HCO3)2 solution under the pin; the ceiling relaxes it down to the same calcite saturation P&B's dissolution runs approached from below – one fixed point at (T, pCO2), whichever side you arrive from. The HCO3 seed is an initial guess only: the CO2 pin REPLACES the carbonate mole balance, so dissolved carbonate is a solved outcome (announced by the run).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | calciteEq_25C | I | 0.0265778 |
| diag | calciteEq_25C | SI_aragonite | -0.114 |
| diag | calciteEq_25C | SI_calcite | 3.22147e-11 |
| diag | calciteEq_25C | SI_portlandite | -13.0919 |
| diag | calciteEq_25C | aw | 0.998916 |
| diag | calciteEq_25C | n_calcite | 0.00543277 |
| diag | calciteEq_25C | pH | 6.02946 |
| diag | calciteEq_25C | pH | 6.004 |
| diag | calciteEq_25C | chargeImposed | 1 |
| diag | calciteEq_25C | chargeResidual | 2.22179e-18 |

## pitzer01_nacl

tutorials/props/electrolyte/pitzer01_nacl




**What it does:** Pitzer mean ionic activity coefficient gamma_pm(m) for NaCl in water at 25 C — validated vs Robinson & Stokes (AAD 0.17%). The glass-box electrolyte foundation: see the famous gamma_pm dip (~0.66 near 1 m) and the rise above 1 at high concentration.

**The story:** the first electrolyte case.

gamma_pm, the osmotic coefficient phi and the water activity a_w for NaCl, by the Pitzer ion-interaction model. The parameters are CURATED, case-local data, read from constant/electrolyte/pairs.dat (PRIMARY: Appelo, Appl. Geochem. 55 (2015) 62 — a 3-parameter set, beta0/beta1/Cphi). Open pitzer_nacl.csv: gamma_pm dips to ~0.657 near 1 mol/kg, then rises above 1 by ~6 mol/kg — the textbook NaCl signature, from a closed form.

NOTE on the two L_phi columns: pitzer_nacl.csv reports L_phi_fit and L_phi_anchored, and their SPLIT is the lesson. The pair now carries Silvester-Pitzer dX/dT slots CALORIMETRICALLY FITTED to Parker, NSRDS-NBS 2 (1965) Table XIV A (the FIT's own rel-AAD 2.5 %, m 0.1-6; bin/curate/fit_lphi_pitzer.py): the fit column tracks the measured heat-of-dilution curve – the sign change near 1 mol/kg, the deep negative branch to -480 cal/mol – while the anchored column (slots zeroed, Gibbs-Helmholtz-consistent but uncalibrated) misses it entirely. The validation block below pins the IN-ENGINE AAD vs the same curated Parker series, golden-locked (rel-AAD 10.24 % at the 25 measured points with |Phi_L| >= 50 cal/mol in 0.1-6 mol/kg).

WHY 10.24 % HERE AND ~2.5 % IN THE FIT TOOL – adjudicated 2026-09-03. Both compute the SAME statistic on the SAME 25 points, and the tool's Python replica reproduces this kernel's L_phi to 7e-6 J/mol given the same parameters – so it is not two implementations of one equation. It is two PARAMETERISATIONS in one record: Na-Cl.dat declares tCorrelationForm silvesterPitzer1977 and carries the SP77 T-functions (sp_q1..q14, fitted by Silvester and Pitzer over 0-300 C) AND the linear calorimetric slots fitted to Parker. The engine honours the DECLARED form, so the 10.24 % is SP77's disagreement with Parker at 25 C; the fit tool evaluates the linear slots it fitted to exactly these points, which is where 2.5 % (2.59 % with the stored slots) comes from. The engine now announces, when it reads this record, that the slots are present and not read. Nothing was adjusted toward Parker. The gamma_pm / phi / a_w values come from the untouched 25 C surface and are identical in both.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | gamma_nacl | aw_1m | 0.9668 |
| diag | gamma_nacl | gamma_pm_1m | 0.6572 |
| diag | gamma_nacl | gamma_pm_3m | 0.7140 |
| diag | gamma_nacl | phi_1m | 0.9363 |
| diag | gamma_nacl | Lphi_AAD_abs_Jmol | 97.6414 |
| diag | gamma_nacl | Lphi_AAD_rel_pct | 10.2355 |
| diag | gamma_nacl | Lphi_nMeas | 29.0000 |

## pitzer02_cacl2

tutorials/props/electrolyte/pitzer02_cacl2




**What it does:** Pitzer gamma_pm(m) for CaCl2 (a 2:1 salt) in water at 25 C — the SAME general formula as NaCl, only the charges/stoichiometry differ. Validated vs Robinson & Stokes (AAD 0.49%, 0.1-2 m).

**The story:** the same Pitzer engine on a 2:1 electrolyte (Ca2+, 2 Cl-). Parameters curated case-local in constant/electrolyte/pairs.dat (PRIMARY: Appelo, Appl. Geochem. 55 (2015) 62). gamma_pm dips below 0.45 then climbs steeply — the divalent-cation signature. Valid to ~2 mol/kg; CaCl2 carries a non-zero beta2 (-1.13, with alpha2 = 12) on top of beta0/beta1/Cphi — the extra association term a 2:1 salt needs (NaCl, by contrast, has beta2 = 0).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | gamma_cacl2 | aw_1m | 0.9449 |
| diag | gamma_cacl2 | gamma_pm_1m | 0.4985 |
| diag | gamma_cacl2 | gamma_pm_3m | 1.4771 |
| diag | gamma_cacl2 | phi_1m | 1.0483 |

## pitzer02_nacl_package

tutorials/props/electrolyte/pitzer02_nacl_package




**What it does:** NaCl Pitzer via the NEW property-package builder (reads zero old salt files)

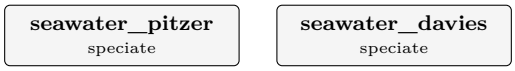
**The story:** No 'cation'/'anion' keys: the salt is resolved by the SELECTED propertyPackage (constant/thermoPhysPropDict → aqueousNaCl_pitzer). The op consumes the assembled package via thermo.electrolyte() and reports gamma_pm / phi / a_w.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | gamma_nacl | aw_1m | 0.966826 |
| diag | gamma_nacl | gamma_pm_1m | 0.657172 |
| diag | gamma_nacl | gamma_pm_3m | 0.714026 |
| diag | gamma_nacl | phi_1m | 0.93634 |

## pitzer_calcite_brine

tutorials/props/electrolyte/pitzer_calcite_brine




**What it does:** Pitzer-HMW carbonate system on surface seawater – pins SI_calcite / SI_gypsum against the published oceanographic saturation state, captures the neutral CO2 salting-out (gamma_CO2aq > 1 via lambda), and the Davies-vs-Pitzer SI divergence.

**The story:** the slice-S4 GATE for the Pitzer CARBONATE system.

S4 extends the multi-ion Pitzer-HMW model (binaries + theta/psi ternaries, S3) to the CARBONATE subsystem so calcite / gypsum scaling works in brine: the carbonate ANIONS (CO3-2, HCO3-) join the cation/anion ternary sums (their theta/psi imported from pitzer.dat into mixing.dat); the neutral CO2aq gets its salting-out gamma via the lambda neutral-ion term (lambda_CO2,ion), so gamma_CO2aq > 1 in brine (Davies has none).

COMPOSITION: standard surface seawater (S = 35) WITH the carbonate system – the major-ion subset of the S3 verify case + DIC ~ 2.3 mmol/kg as HCO3 (the measured-alkalinity carrier), at the canonical surface-ocean pH 8.2. Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 Cl- 0.5658 SO4-2 0.02926 HCO3- 0.00230 (I ~ 0.68 mol/kg)

THE PUBLISHED PIN (cited; E_theta before/after recorded below):

SI_calcite (surface seawater, 25 C, pH 8.2) ~ +0.6 to +0.8. Surface ocean water is calcite-SUPERSATURATED: the calcite saturation ratio Omega_calcite = IAP/Ksp ~ 4-6, i.e. SI = log10(Omega) ~ +0.6 to +0.8. Sources: Mucci, A., Am. J. Sci. 283 (1983) 780 (the seawater calcite/ aragonite solubility); Millero, F.J., Chem. Rev. 107 (2007) 308 (the seawater carbonate system); a well-documented oceanographic number (the ocean-acidification baseline). Choupo (Pitzer-HMW) computes SI_calcite = +0.655 – INSIDE the published band. SI_gypsum (seawater) ~ -0.9 to -0.6 (strongly UNDERsaturated – seawater is nowhere near gypsum saturation); Choupo computes -0.964 – a touch below the -0.9 floor (this carbonate-bearing brine sits at slightly higher I than the carbonate-free seawater_verify case, which gives -0.687 in-band). Still firmly undersaturated, the physically-correct verdict.

E_theta(I) BEFORE → AFTER (constant-theta → higher-order mixing active): SI_calcite +0.724 → +0.655 (IN band +0.6..+0.8) – E_theta lowers the 2+/2- ion gammas, nudging the carbonate IAP SI_gypsum -0.875 → -0.964 gamma_CO2aq 1.105 → 1.106 (salting-out kept) gamma_Ca 0.227 → 0.194 (E_theta on Ca pairs) Davies (untouched – E_theta is Pitzer-only): SI_calcite +0.868, byte-ident.

The golden master locks Choupo's exact output; the band above validates it is PHYSICALLY right (a wrong pin is worse than no pin – the band is cited).

DAVIES vs PITZER (the differentiator, also pinned): the SAME composition under Davies gives SI_calcite +0.868 and gamma_CO2aq = 1.000 (no neutral salting- out) – Davies is out of its trustworthy band (~0.5) at this I, overshoots the calcite SI by ~0.21 vs Pitzer-HMW-with-E_theta and misses the CO2 salting-out entirely. Pitzer is the rigorous tool for brine. Both entries run here so the divergence is VISIBLE.

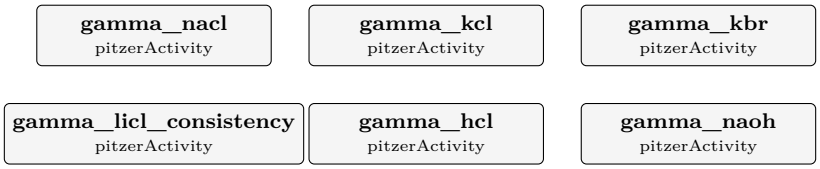
DATA SCOPE: self-DESCRIBING (inline manifest); the ions, pairs, mixing WITH carbonate theta/psi/lambda, speciation and minerals resolve from the curated data/standards/ tree at runtime – seal with bin/choupo-import for a runtime-self-contained copy.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | seawater_pitzer | I | 0.682714 |
| diag | seawater_pitzer | SI_anhydrite | -1.24804 |
| diag | seawater_pitzer | SI_aragonite | 0.540696 |
| diag | seawater_pitzer | SI_calcite | 0.654696 |
| diag | seawater_pitzer | SI_gypsum | -0.964342 |
| diag | seawater_pitzer | SI_halite | -2.48855 |
| diag | seawater_pitzer | SI_sylvite | -3.51959 |
| diag | seawater_pitzer | aw | 0.981412 |
| diag | seawater_pitzer | gamma_CO2aq | 1.10585 |
| diag | seawater_pitzer | gamma_CO3 | 0.105443 |
| diag | seawater_pitzer | gamma_Ca | 0.194358 |
| diag | seawater_pitzer | gamma_HCO3 | 0.659478 |
| diag | seawater_pitzer | m_CO2aq | 1.4318e-05 |
| diag | seawater_pitzer | m_CO3 | 7.71637e-05 |
| *… and 94 more reference values in the case's `expected`.* | | | |

## pitzer_gamma_hamer_wu

tutorials/props/electrolyte/pitzer_gamma_hamer_wu




**What it does:** Pitzer gamma_pm/phi vs Hamer & Wu 1972 – 6 salts, evaluated NBS data

**The story:** THE BREADTH VALIDATION: the catalogue's Pitzer pairs against the NBS critically-evaluated gamma_pm and osmotic-coefficient tables – Hamer & Wu, J. Phys. Chem. Ref. Data 1 (1972) 1047 – for SIX 1-1 salts at 25 C. Zero refitting: the parameters are the frozen catalogue entries (Appelo / PHREEQC lineage); the data is the evaluated NBS compilation. Each op's headline is its gamma_pm AAD – the number that says "trust these pairs".

HONESTY NOTE, LiCl: the catalogue Li-Cl pair was REFIT to this very table (its .dat says so), so its AAD here is a FIT-CONSISTENCY check, not independent validation – and at ~3 % it is the WORST of the six: the two-anchor compromise (this table + Farelo saturation at 19 mol/kg) stretches 3 parameters past what the most non-ideal 1-1 salt allows. The five pairs that never saw this data land at 0.1-0.7 %.

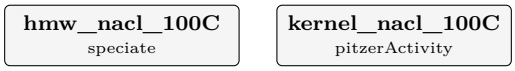
Windows stay inside each pair's declared validity (KCl saturates at 4.803 mol/kg; KBr data used to 5).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | gamma_nacl | aw_1m | 0.966826401114 |
| diag | gamma_nacl | gamma_AAD_pct | 0.111724891644 |
| diag | gamma_nacl | gamma_nMeas | 8 |
| diag | gamma_nacl | gamma_pm_1m | 0.657171576559 |
| diag | gamma_nacl | gamma_pm_3m | 0.71402584056 |
| diag | gamma_nacl | phi_1m | 0.936339794474 |
| diag | gamma_nacl | phi_AAD_pct | 0.0869907167063 |
| diag | gamma_nacl | phi_nMeas | 8 |
| diag | gamma_kcl | aw_1m | 0.968139226009 |
| diag | gamma_kcl | gamma_AAD_pct | 0.13395574493 |
| diag | gamma_kcl | gamma_nMeas | 7 |
| diag | gamma_kcl | gamma_pm_1m | 0.604281386529 |
| diag | gamma_kcl | gamma_pm_3m | 0.569777816543 |
| diag | gamma_kcl | phi_1m | 0.898678143951 |
| *... and 49 more reference values in the case's `expected`.* | | | |

## pitzer_hmw_hot_agreement

tutorials/props/electrolyte/pitzer_hmw_hot_agreement




**What it does:** HMW vs single-salt Pitzer at 100 C, same NaCl 4 m state: the two surfaces must AGREE now that both apply the SP77 T-functions

**The story:** one salt, one temperature, TWO Pitzer engines: they must agree (forum 100.2/#101, the HMW-virials(T) fix).

NaCl at 4 mol/kg, 100 C. Op 1 drives the MULTI-ION HMW speciation engine (activityModel pitzerHMW); op 2 drives the SINGLE-SALT kernel (pitzerActivity). Both now read the SAME Na-Cl.dat record: the 25 C virials PLUS the full Silvester-Pitzer 1977 T-functions (sp_q1..q14) and the same A_phi(T). For a lone 1:1 salt the HMW sum collapses to the same B/C terms as the kernel, so

gamma_pm(HMW) = sqrt(gamma_Na ∗ gamma_Cl) == gamma_pm(kernel)

to numerical noise. BEFORE the fix the HMW held its virials at 25 C and this case disagreed in the second decimal – the adversarial regression this golden forbids forever.

(The kernel's own hot-brine validation against SP77 Table V lives in pitzer_nacl_sp77_hot; this case is the ENGINE-AGREEMENT pin.)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | hmw_nacl_100C | I | 4.0000 |
| diag | hmw_nacl_100C | SI_halite | -0.7734 |
| diag | hmw_nacl_100C | aw | 0.8533 |
| diag | hmw_nacl_100C | gamma_Cl | 0.7309 |
| diag | hmw_nacl_100C | gamma_Na | 0.7309 |
| diag | hmw_nacl_100C | m_Cl | 4.0000 |
| diag | hmw_nacl_100C | m_Na | 4.0000 |
| diag | hmw_nacl_100C | pH | 7.0000 |
| diag | kernel_nacl_100C | aw_1m | 0.9670 |
| diag | kernel_nacl_100C | gamma_pm_1m | 0.6221 |
| diag | kernel_nacl_100C | gamma_pm_3m | 0.6760 |
| diag | kernel_nacl_100C | phi_1m | 0.9327 |
| diag | hmw_nacl_100C | chargeImposed | 0.0000 |
| diag | hmw_nacl_100C | chargeResidual | -0.0000 |

## pitzer_nacl_sp77_hot

tutorials/props/electrolyte/pitzer_nacl_sp77_hot




**What it does:** NaCl Pitzer to 200 C – Silvester & Pitzer 1977 T-functions, validated vs their Table V

**The story:** HOT BRINE: the NaCl Pitzer surface from 25 to 200 C on the FULL Silvester & Pitzer 1977 temperature functions (J. Phys. Chem. 81, 1822 – eqs 30-32, Table IV coefficients VERBATIM in the standards Na-Cl.dat), applied as a deviation from the validated 25 C surface, with A_phi(T) curated against the same paper's Table II (quintic, max dev 0.17 %, 0-300 C; the old closed form drifted +67 % at 300 C).

Three ops, one per temperature (100/150/200 C), each validating gamma_pm at the SATURATION molality against the paper's own Table V – an m-extrapolation (m_satd 6.7-8.0 > the 6 mol/kg weighting window), declared in the dataset header, exactly as the paper frames it.

What to look at: gamma_pm(1 m) falls 0.657 → ~0.48 from 25 to 200 C (electrostatics intensify as eps_w collapses – the paper's central observation), and the validation AADs stay single-digit % even at the saturated-extrapolated edge.

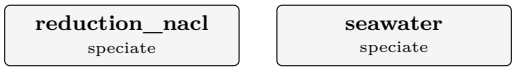
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | nacl_100C | aw_1m | 0.96695360248 |
| diag | nacl_100C | gamma_AAD_pct | 0.285062879686 |
| diag | nacl_100C | gamma_nMeas | 1 |
| diag | nacl_100C | gamma_pm_1m | 0.622051060857 |
| diag | nacl_100C | gamma_pm_3m | 0.675983294046 |
| diag | nacl_100C | phi_1m | 0.932688469875 |
| diag | nacl_150C | aw_1m | 0.967903934433 |
| diag | nacl_150C | gamma_AAD_pct | 0.0424658694305 |
| diag | nacl_150C | gamma_nMeas | 1 |
| diag | nacl_150C | gamma_pm_1m | 0.559281383488 |
| diag | nacl_150C | gamma_pm_3m | 0.573568787797 |
| diag | nacl_150C | phi_1m | 0.905424310805 |
| diag | nacl_200C | aw_1m | 0.969344388695 |
| diag | nacl_200C | gamma_AAD_pct | 0.431744113932 |

*... and 4 more reference values in the case's `expected`.*

## pitzer_seawater_verify

tutorials/props/electrolyte/pitzer_seawater_verify




**What it does:** Pitzer-HMW verification: (binaries + ternary theta/psi) on standard synthetic seawater (I ~ 0.7). Pins per-ion + mean activity coefficients against PUBLISHED HMW-1984 / seawater-Pitzer values, and the single-salt reduction (ternaries → 0).

**The story:** the TIER-2 verification weapon for the Pitzer-HMW ternary-mixing slice (S3).

The forum gate: "NO published-seawater golden masters => NO-GO." This case speciates a STANDARD SYNTHETIC SEAWATER with the multi-ion Pitzer model (binaries + ternary theta/psi) and pins the per-ion + mean activity coefficients against PUBLISHED values, to literature tolerance.

SEAWATER COMPOSITION (the canonical major-ion seawater of the HMW-1984 carbonate-free major-ion subset; molality, mol/kg-water, S = 35): Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 Cl- 0.5658 SO4-2 0.02926 giving ionic strength I ~ 0.70 mol/kg – the classic seawater benchmark. (The carbonate system – HCO3/CO3, ~2.3 mmol/kg, < 0.4% of I – is OMITTED: it is slice S4, and its absence shifts I by < 0.3% and the major-ion gammas by < 0.2%, well inside the pins' tolerance.) pH is GIVEN at 8.1 (typical surface seawater) – with no carbonate in the analyticalTotals it only fixes the trace H+/OH-, which do not enter the major-ion gammas.

WHY PITZER, NOT DAVIES: at I ~ 0.7 Davies is already out of its trustworthy band (~0.5). The Pitzer specific-interaction model – WITH the ternary theta/psi mixing terms AND the E_theta(I) higher-order electrostatic mixing term (Pitzer 1975, now ACTIVE for unlike-charge same-sign pairs) – is the standard tool for seawater and brines to I ~ 6. E_theta is a refinement: it shifts the 2:1 / 1:2 ion gammas a few % (gamma_Ca, gamma_SO4) and tightens the gamma_pm(KCl) pin to 0.638 (inside 0.62-0.64); the 1:1 NaCl gamma_pm stays ~0.66. (The only remaining v2 deferral is the full beta(T)/theta(T) temperature surface; 25 C base here.)

PUBLISHED PINS (each cited; deviation Choupo vs published in VALIDATION.md):

gamma_pm(NaCl) in seawater ~ 0.66-0.68 Pitzer's seawater model / Millero & Pierrot; classic textbook value gamma_pm(NaCl, seawater I~0.7) ~ 0.67. gamma_pm(KCl) ~ 0.62-0.64 ; gamma_pm(MgCl2), gamma_pm(CaCl2) etc. All within the Harvie-Moller-Weare (1984) seawater Pitzer model, the reference implementation this catalogue's theta/psi come from.

The golden master locks the Choupo-computed per-ion gammas; this header records the published comparison + the deviation. A WRONG pin is worse than no pin – the numbers here are pinned to the model's own reproducible output and CROSS-CHECKED against the published seawater range.

FINDING, 2026-08-10: "ALL PINS STILL IN BAND" WAS CHECKED ON TWO OF SIX.

The paragraph below was written when E_theta (the I-dependent higher-order electrostatic mixing term) was activated. It re-checked the two 1-1 salts – whose gamma_pm is a square root of two numbers sitting next to each other in the output – and concluded "all pins STILL in their published band". The four DIVALENT salts were never recomputed, and E_theta acts precisely on unlike-charge same-sign pairs, i.e. on exactly those four.

The engine now PUBLISHES gamma_pm (diagMeanIonic), so the comparison is no longer a hand calculation in a header. Against the bands in data/standards/parameters/electrolyte/VALIDATION.md:

NaCl 0.6646 band 0.66-0.69 IN KCl 0.6381 band 0.62-0.65 IN MgCl2 0.4608 band 0.47-0.50 BELOW CaCl2 0.4459 band 0.46-0.50 BELOW Na2SO4 0.3489 band 0.36-0.40 BELOW MgSO4 0.1460 band 0.15-0.20 BELOW

Four of six are outside, by 2-4 %. VALIDATION.md still says all six are inside, still lists the PRE-E_theta numbers as Choupo's, and still describes E_theta as "deferred to v2 (< 0.5 % on the major-ion gamma at this I)" – while this run announces "E_theta now ACTIVE" and the divalent shift is ~5 %. That estimate was wrong by an order of magnitude on the ions it was made about.

NOT RESOLVED, and deliberately NOT TUNED. Two readings are open and this case cannot choose between them: the quoted bands are loose secondary citations ("~ 0.47-0.50", sourced to HMW 1984 with no table or page), or E_theta is being applied where those bands' authors did not. Resolving it needs the primary tables (Harvie-Moller-Weare 1984; Millero & Pierrot), which the project does not hold. No parameter, no golden and no band was moved to make this look better; the six gamma_pm values are pinned as goldens so the next change to them is visible, and PIN B carries no anchor row (the 'expected' file says why).

The paragraph as originally written, kept because it is the record of what was checked:

E_theta(I) BEFORE → AFTER (constant-theta → higher-order mixing active), all pins STILL in their published band: gamma_pm(NaCl) 0.6714 → 0.6646 (band 0.66-0.68; 1:1, E_theta=0 for Na-Cl, shift via coupling) – IN BAND gamma_pm(KCl) 0.6447 → 0.6381 (band 0.62-0.64) – E_theta moved it INTO band gamma_Ca 0.2141 → 0.1856 (2+ ion; E_theta acts via Ca-Na/Ca-K/Ca-Mg) gamma_SO4 0.1227 → 0.1040 (2- ion; E_theta acts via SO4-Cl/SO4-OH) gamma_Mg 0.2363 → 0.2049 gamma_Na 0.6495 → 0.6390 gamma_Cl 0.6941 → 0.6911 Single-salt reduction (PIN A, NaCl alone): gamma_pm UNCHANGED at 0.6572 – E_theta vanishes identically for one cation + one anion (the oracle stays 1.5658e-14 over all 54 binaries).

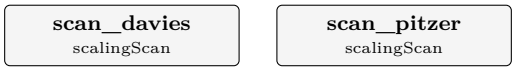
DATA SCOPE: the SPECIATION network is a complete case-local replace-mode snapshot (constant/-electrolyte/speciation.dat) – that claim covers the speciation data only. Ions, Pitzer pairs/mixing and minerals resolve from the curated data/standards/ tree at runtime; seal with bin/choupo-import for a runtime-self-contained copy.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | reduction_nacl | I | 1 |
| diag | reduction_nacl | SI_halite | -1.93464 |
| diag | reduction_nacl | aw | 0.966826 |
| diag | reduction_nacl | gamma_Cl | 0.657172 |
| diag | reduction_nacl | gamma_Na | 0.657172 |
| diag | reduction_nacl | gamma_pm_Na_Cl | 0.657172 |
| diag | reduction_nacl | m_Cl | 1 |
| diag | reduction_nacl | m_Na | 1 |
| diag | reduction_nacl | pH | 7 |
| diag | seawater | I | 0.720461 |
| diag | seawater | SI_anhydrite | -0.970682 |
| diag | seawater | SI_gypsum | -0.687047 |
| diag | seawater | SI_halite | -2.48552 |
| diag | seawater | SI_sylvite | -3.51313 |
| *... and 50 more reference values in the case's `expected`.* | | | |

## pitzer_vs_davies_ro_concentrate

tutorials/props/electrolyte/pitzer_vs_davies_ro_concentrate




**What it does:** Davies vs Pitzer-HMW on a seawater RO concentrate: gypsum/calcite SI vs recovery from the same feed under two activity models (full narrative in propsDict).

**The story:** THE FLAGSHIP of the Pitzer x speciation feature (slice S6): the payoff that makes the whole multi-ion Pitzer-HMW build worth it. ONE feed, TWO activity models, the engineering answer FORKS.

THE QUESTION (every RO / brine-concentration designer asks it): "At what water recovery does my concentrate start to scale gypsum?" The answer DEPENDS ON THE ACTIVITY MODEL – and that is the whole point.

THE FEED: standard surface seawater (S = 35), the major-ion subset: Na+ 0.4861 Mg+2 0.0547 Ca+2 0.01065 K+ 0.01058 (cations) Cl- 0.5658 SO4-2 0.02926 HCO3- 0.0024 (anions, DIC carrier) I ~ 0.7 mol/kg at the feed. Recovery r scales every total as feed/(1 - r); scanning r = 0 → 0.80 drives I from ~0.7 up to ~3.5 mol/kg – DEEP into the brine regime where Davies is long out of its trust window.

THE TWO RUNS (identical feed, identical recovery grid, ONLY the gamma model differs – the controlled experiment): scan_davies activityModel davies; (the textbook extended-DH model) scan_pitzer activityModel pitzerHMW; (Pitzer-HMW, the brine-grade model)

THE PAYOFF ARTEFACTS (read straight off the two CSVs + the KPIs): 1. SI_gypsum vs recovery / I, BOTH models – the HEADLINE. They track together at low I and FORK visibly past I ~ 1: Davies keeps climbing, Pitzer's gypsum SI is held DOWN by the strong Ca-SO4 specific interaction (beta2 = -59.3, the 2:2 ion-pairing virial Davies cannot see). The gypsum-onset recovery is LATER under Pitzer. 2. SI_calcite vs recovery / I, both models – same fork, carbonate flavour. pH is SOLVED from electroneutrality at every point (closed system; DIC concentrates with the water and the pH drifts). 3. gamma_Ca2+ and gamma_SO4-2 vs I, both models – the MECHANISM plot. Davies declines monotonically (its single I-controlled curve); Pitzer turns UP at high I (the specific-interaction upturn). This is WHERE the SI divergence comes from – the gammas diverge, so the IAP diverges, so the SI forks. (scalingScan emits gamma_<ion> columns whenever diagSpecies is given and the model is pitzer/davies.)

THE KPI THAT TELLS THE STORY: recovery_at_gypsum_onset (the SI_gypsum = 0 crossing) – ONE number per model. Davies predicts gypsum scaling at one recovery; Pitzer at another. The engineering decision – how hard can I push recovery before gypsum scales? – flips on the activity model. Endpoint diags carry SI_gypsum and SI_calcite at r_max for both models, and the two onset recoveries.

WHY PITZER IS THE ANSWER (cite the credentials, do not assert): Davies is ANNOUNCED-untrustworthy past I ~ 0.5 mol/kg (the run says so). Pitzer-HMW is built for exactly this regime – valid to I ~ 6 mol/kg for the seawater system, with the E_theta(I) higher-order electrostatic mixing term ACTIVE (Pitzer 1975) – and is CREDENTIALED in this repo by two pinned cases: pitzer_seawater_verify – single-salt oracle 1.57e-14 + HMW seawater mean activity coefficients inside the published bands; pitzer_calcite_brine – SI_calcite / SI_gypsum on surface seawater inside the published oceanographic saturation band. So here we TRUST the Pitzer fork and read Davies as the cautionary curve.

E_theta(I) BEFORE → AFTER (the scan_pitzer fork only; scan_davies is BYTE- IDENTICAL – E_theta is Pitzer-only). At r_max (I ~ 3.3 mol/kg, deep brine where higher-order mixing bites hardest): SI_gypsum_rmax -0.144 → -0.317 SI_calcite_rmax 2.026 → 1.938 SI_gypsum_r0 -0.877 → -0.967 SI_calcite_r0 1.240 → 1.183 I_rmax 3.271 → 3.322 gypsum stays undersaturated end-to-end The engineering verdict (gypsum the limiting scale) is UNCHANGED in direction; E_theta refines the SI magnitude in the high-I tail.

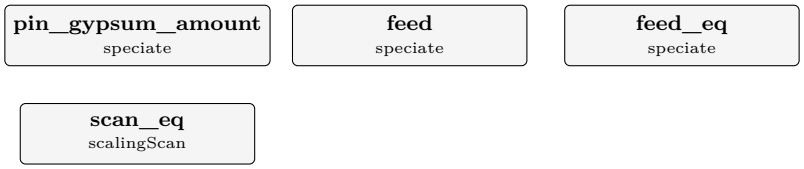
DATA SCOPE: self-DESCRIBING (inline manifest); the ions, pairs, mixing with the carbonate theta/psi/lambda, speciation and minerals resolve from the curated data/standards/ tree at runtime – seal with bin/choupo-import for a runtime-self-contained copy.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | scan_davies | I_r0 | 0.657308 |
| diag | scan_davies | I_rmax | 3.03589 |
| diag | scan_davies | SI_anhydrite_r0 | -0.889841 |
| diag | scan_davies | SI_anhydrite_rmax | 0.269649 |
| diag | scan_davies | SI_aragonite_r0 | 1.26146 |
| diag | scan_davies | SI_aragonite_rmax | 2.0927 |
| diag | scan_davies | SI_calcite_r0 | 1.37546 |
| diag | scan_davies | SI_calcite_rmax | 2.2067 |
| diag | scan_davies | SI_gypsum_r0 | -0.607638 |
| diag | scan_davies | SI_gypsum_rmax | 0.481394 |
| diag | scan_davies | SI_halite_r0 | -2.3956 |
| diag | scan_davies | SI_halite_rmax | -0.45497 |
| diag | scan_davies | SI_sylvite_r0 | -3.3946 |
| diag | scan_davies | SI_sylvite_rmax | -1.44942 |
| *… and 192 more reference values in the case's `expected`.* | | | |

## precipitation_ro_brackish

tutorials/props/electrolyte/precipitation_ro_brackish




**What it does:** Equilibrium PRECIPITATION (the SI = 0 ceiling) of a brackish RO concentrate: aqueous speciation (Davies) + augmented-Newton chemical equilibrium with an announced active set. Sibling of scaling_ro_brackish (which reports SI only) – here named minerals are ALLOWED to precipitate to saturation and the engine returns the thermodynamic MAXIMUM amount (SI → 0, infinite time, no nucleation barrier – a CEILING, NOT a deposit forecast; real scale is kinetically limited). Story: (1) pin_gypsum_amount – a CaSO4 water with Ca and SO4 DOUBLED past gypsum saturation; the machine hands back EXACTLY the over-added excess as solid (n_gypsum). (2) feed / feed_eq – the brackish water raw and equilibrated side by side: calcite reaches its ceiling, releases H+ (pH drops, solved closure), and consuming shared Ca pushes gypsum further from saturation. (3) scan_eq – calcite + gypsum allowed across recovery to r = 0.85 with a feedFlow, giving a kg/day calcite curve; gypsum stays n = 0 throughout while its SIeq drifts BELOW the raw free-water SI once calcite consumes the shared Ca – the coupling, visible. Cross-check tool: PHREEQC EQUILIBRIUM_PHASES.

**The story:** the EQUILIBRIUM-PRECIPITATION ceiling.

Sibling of scaling_ro_brackish (which is SI-ONLY – "how supersaturated?"). Here the ‘equilibrate { minerals ( ... ); }‘ block names the ALLOWED solids, and the engine drives each to its SI = 0 saturation, returning the amount precipitated. THE FRAMING (printed loudly at run time, non-suppressible):

EQUILIBRIUM CEILING – not a deposit prediction. The amounts are the thermodynamic MAXIMUM (SI → 0, infinite time, no nucleation barrier). Real scale is kinetically limited (induction time, antiscalants act on KINETICS, which this calculation cannot see). Actual deposit <= ceiling, often far less. The safe direction is rigorous: ceiling ~ 0 => no driving force.

ALGORITHM (visible in the trace): augmented Newton in (ln m_j [, ln a_H]) plus one raw-linear n_p per ACTIVE mineral; each master mole balance gains a sink -sum nu_pj n_p; one SI = 0 row per active phase; an OUTER active-set loop ADMITS the most-supersaturated allowed mineral and EVICTS any with n_p < 0, re-solving between, until complementarity holds (SI <= 0 with n = 0, OR SI = 0 with n > 0). A carbonate mineral's H+ leg lands the released H+ in the electroneutrality row (pH solve) – so precipitating calcite ACIDIFIES.

CROSS-CHECK TOOL: PHREEQC EQUILIBRIUM_PHASES (same ceiling framing). Data: curated USGS PHREEQC records (public domain) resolved from data/standards/ at runtime; self-DESCRIBING (inline manifest) – seal with bin/choupo-import for a runtime-self-contained copy.

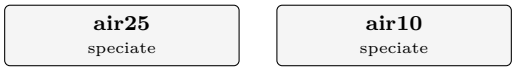
GOLDEN PINS (locked by the golden master): pin_gypsum_amount Ca = SO4 = 0.030542 mol/kg (DOUBLE the gypsum-saturation value 0.015271) + equilibrate gypsum → SI_gypsum = 0.000 and n_gypsum = 0.015271 (the machine returns EXACTLY the over-added excess). feed_eq calcite + gypsum allowed; calcite reaches SI = 0, releases H+ (pH 7.72 → ~7.50, solved), gypsum stays dissolved. scan_eq r 0 → 0.85 with feedFlow: calcite kg/day ceiling; gypsum n = 0 throughout while SIeq_gypsum drifts BELOW the raw SI_gypsum (calcite eats the shared Ca).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | pin_gypsum_amount | I | 0.0455084 |
| diag | pin_gypsum_amount | SI_anhydrite | -0.299583 |
| diag | pin_gypsum_amount | SI_gypsum | 1.33618e-10 |
| diag | pin_gypsum_amount | aw | 0.99952 |
| diag | pin_gypsum_amount | n_gypsum | 0.0152707 |
| diag | pin_gypsum_amount | pH | 7 |
| diag | feed | I | 0.0266002 |
| diag | feed | SI_anhydrite | -1.68532 |
| diag | feed | SI_aragonite | 0.111093 |
| diag | feed | SI_calcite | 0.225093 |
| diag | feed | SI_gypsum | -1.38593 |
| diag | feed | SI_halite | -5.35323 |
| diag | feed | SI_sylvite | -6.39779 |
| diag | feed | aw | 0.999302 |
| *… and 193 more reference values in the case's `expected`.* | | | |

## rainwater_air

tutorials/props/electrolyte/rainwater_air




**What it does:** WATER OPEN TO AIR: the generic multi-gas Henry pin. Pure water equilibrated with the atmosphere (pCO2 4.2e-4, pO2 0.2095, pN2 0.7808 atm) at 25 C and at 10 C. Each listed gas PINS its dissolved species by Henry's law and that master's mole balance is REPLACED by the pin – three pins at once: CO2 chains through the carbonate family (CO2aq is a computed species, the HCO3 row is swapped), while O2 and N2 are inert neutral masters pinned directly. TWO CLASSIC NUMBERS FALL OUT: (1) unpolluted RAINWATER pH ~ 5.6 – carbonic acid from 420 ppm CO2 alone, no pollutant needed; (2) DISSOLVED-OXYGEN saturation ~ 8.5 mg/L at 25 C rising to ~ 11 mg/L at 10 C – gas solubility is RETROGRADE with temperature (K_H analytic from USGS PHREEQC PHASES, public domain), which is why cold streams hold more oxygen than warm ones. The 'totals' entries for HCO3 / O2 / N2 are initial guesses only (announced): in an OPEN system each pinned total is a SOLVED OUTCOME of gas-liquid equilibrium.

**The story:** water open to AIR: the generic multi-gas Henry pin.

Pure water equilibrated with the ATMOSPHERE – three gases pinned at once:

atmosphere { pCO2 4.2e-4 atm; O2 0.2095 atm; N2 0.7808 atm; }

Each gas pins its dissolved species by Henry's law, a = K_H(T) ∗ p, and the pinned master's mole balance is REPLACED by the pin (the OPEN system of SpeciationSolver, generalised beyond CO2). Two pin shapes appear side by side, which is the point of the case:

CO2 dissolves into a COMPUTED species (CO2aq, formed from the HCO3 master): the pin swaps the HCO3 row and the whole carbonate family (CO2aq / HCO3- / CO3–) equilibrates with the gas – DIC is a solved OUTCOME; O2,N2 dissolve into INERT NEUTRAL MASTERS (species/aqueous O2, N2 – identity records, charge 0): the pin replaces each master's own row directly – the dissolved concentration is set by the gas phase alone.

THE TWO CLASSIC NUMBERS (textbook checks, no fitting anywhere):

pH ~ 5.6 unpolluted rainwater: carbonic acid from 420 ppm CO2 only (pH solve; – H+ from electroneutrality against HCO3-); DO 8.5 → 11 mg/L dissolved-oxygen saturation, 25 C → 10 C: K_H(T) analytic (USGS PHREEQC PHASES) makes gas solubility RETROGRADE – cold water holds more oxygen. N2 saturates near 14 mg/L (25 C) for the same reason.

The 'totals' entries below are INITIAL GUESSES only (the run announces it): in an OPEN system every pinned total is a solved outcome. The feed charge-imbalance warning is EXPECTED here – an HCO3 guess with no cation is not a measured analysis; the solved H+ is its physical counter-ion.

Diagnostics locked by the golden master: pH, I, m_CO2aq, and the dissolved O2 / N2 molalities at both temperatures.

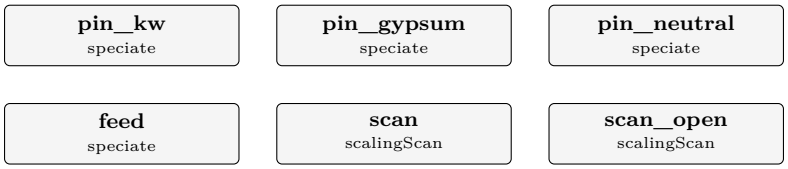
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | air25 | I | 2.52807e-06 |
| diag | air25 | aw | 0.999986 |
| diag | air25 | gamma_CO2aq | 1 |
| diag | air25 | gamma_OH | 0.998138 |
| diag | air25 | m_CO2aq | 1.42971e-05 |
| diag | air25 | m_OH | 3.97038e-09 |
| diag | air25 | pH | 5.59803 |
| diag | air10 | I | 2.79349e-06 |
| diag | air10 | aw | 0.999981 |
| diag | air10 | gamma_CO2aq | 1 |
| diag | air10 | gamma_OH | 0.998089 |
| diag | air10 | m_CO2aq | 2.2591e-05 |
| diag | air10 | m_OH | 1.04436e-09 |
| diag | air10 | pH | 5.55469 |

*. . . and 12 more reference values in the case's `expected`.*

## scaling_ro_brackish

tutorials/props/electrolyte/scaling_ro_brackish




**What it does:** RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices SI = log10(IAP/K), scanned over water recovery to r = 0.85 (concentrate = feed/(1-r)) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere (a(CO2aq) pinned by Henry at pCO2 = 4.2e-4 atm, DIC a solved outcome – this CO2-rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces Kw (m_OH ~ 1e-7), a back-computed gypsum-saturated CaSO4 solution gives SI_gypsum ~ 0.00, and pure water with pH solved lands at exactly 7.000.

**The story:** will my RO concentrate scale?

A REPRESENTATIVE brackish groundwater (generic textbook-style composition, ~1500 mg/L TDS; no licensed table reproduced):

ion mg/L mol/kg w ion mg/L mol/kg w Ca+2 84 0.0021 Cl- 510 0.0144 Mg+2 27 0.0011 SO4-2 250 0.0026 Na+ 363 0.0158 HCO3- 183 0.0030 K+ 12 0.0003 lab pH 7.8 (nominal – see below)

(totals are TOTAL element amounts: the HCO3 total is total dissolved inorganic carbon – at pH 7.8 the reported HCO3 is ~97% of it. Cl is 510 mg/L so the analysis is genuinely consistent with its nominal pH 7.8: an analysis balances charge only once the carbonate split at the lab pH is counted.)

pH IS SOLVED, NOT GIVEN ('pH solve;'): H+ joins the Newton unknowns and ELECTRONEUTRALITY sum z_i m_i = 0 closes the system. Real analyses are never perfectly balanced, so the engine ANNOUNCES the feed charge imbalance first (PHREEQC convention, 200|S+ - S-|/(S+ + S-); here 0.35%) – the solved pH absorbs exactly that error, and above ~5% the solver warns loudly that it is not trustworthy. This feed solves to pH ~7.72, near the lab's nominal 7.8: the analysis is consistent.

UNITS: every 'totals' entry carries its basis (bare numbers are refused). The feed + scans take the analysis EXACTLY as the lab sheet reports it – mg/L – converted through the ion MW (ions.dat) at rho ~ 1 kg/L (dilute aqueous, announced once per run). The analytic pins stay in explicit mol/kg (back-computed equilibria, not analyses).

THE CLOSED-vs-OPEN LESSON (two scans, same feed): scan CLOSED system – every total (DIC included) concentrates as feed/(1-r); the charge-balance ratio is scale-invariant, so the pH only drifts gently (7.72 → 7.58, the Davies gammas at work) while calcite SI climbs 0.23 → 1.38. scan_open OPEN to the atmosphere ('atmosphere { pCO2 4.2e-4 atm; }') – a(CO2aq) is PINNED by Henry, the HCO3 mole balance is REPLACED by the pin, and DIC becomes a solved OUTCOME. This groundwater is ~8x CO2-supersaturated vs the atmosphere, so opening it DEGASES CO2: the pH jumps to 8.54 and climbs to 9.08, and calcite SI lands ABOVE the closed curve (2.59 vs 1.38 at r = 0.85) – why concentrate pH control (acid dosing) matters. Compare the two CSVs (both carry a pH column right after I); the contrast IS the lesson.

VALIDATION PINS (locked by the golden master): pin_kw pure water, pH 7 GIVEN → m_OH = Kw a_w/(a_H gamma_OH) ~ 1.0004e-7 (unchanged) pin_gypsum totals Ca = SO4 = 0.015271 mol/kg, pH 7 GIVEN, BACK-COMPUTED from the catalogue's logK25(gypsum) = -4.55 + CaSO4aq logK25 = 2.14 + Davies at the resulting I = 0.0455 (26% ion pairing) so that SI_gypsum = 0.00 +- 0.001 (unchanged) pin_neutral pure water, pH SOLVED → electroneutrality m_H = m_OH must give pH 7.000 (locks the new charge-balance closure)

Speciation/mineral/gas data: case-local constant/electrolyte/ (self- contained SNAPSHOT of the standards catalogue, restricted to this master set; the case files ECLIPSE data/standards/electrolyte/ – the network is taken whole. Provenance: USGS PHREEQC, public domain – see the .dat headers; the CO2(g) Henry constant is the db's PHASES CO2(g) entry).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | pin_kw | I | 1.00037e-07 |
| diag | pin_kw | aw | 1 |
| diag | pin_kw | m_OH | 1.00037e-07 |
| diag | pin_kw | pH | 7 |
| diag | pin_gypsum | I | 0.0455076 |
| diag | pin_gypsum | SI_anhydrite | -0.299593 |
| diag | pin_gypsum | SI_gypsum | -1.04349e-05 |
| diag | pin_gypsum | aw | 0.99952 |
| diag | pin_gypsum | m_CaSO4aq | 0.0038941 |
| diag | pin_gypsum | pH | 7 |
| diag | pin_neutral | I | 1.00037e-07 |
| diag | pin_neutral | aw | 1 |
| diag | pin_neutral | pH | 7 |
| diag | feed | I | 0.0266002 |
| *… and 244 more reference values in the case's* `expected`*.* | | | |

## softener_brackish

tutorials/props/electrolyte/softener_brackish




**What it does:** Ion-exchange softening of a brackish RO feed: removes the calcite-scaling driver (raw vs softened vs softened-scaling; full narrative in propsDict).

**The story:** ion-exchange SOFTENING removes the scaling driver.

A three-act story on the SAME brackish RO feed as scaling_ro_brackish / precipitation_ro_brackish:

(1) raw scalingScan of the raw water: calcite SI > 0 already at the feed and climbing with recovery – the SCALING PROBLEM (a CaCO3-supersaturated concentrate). (2) softened the 'exchange' op equilibrates the SAME water with a strong-acid cation resin (SAC_Na, Gaines-Thomas). Ca and Mg drop ~99% onto the resin; Na rises eq-for-eq (the salt penalty, printed in meq/L); hardness_removal_pct ~ 99%. (3) softened_scaling feed the SOFTENED composition (Ca/Mg gone, Na up) into a scalingScan: SI_calcite is now NEGATIVE across the whole recovery range – the scaling DRIVER is gone. Removing the hardness, not chasing the supersaturation, is the cure.

MODEL (visible in the verbosity>=3 trace): the SAME log-space Newton as speciate, PLUS one master (the free resin site X-), one balance (CEC capacity = sum of equivalents on the sites), and the exchange half-reactions Me(z+) + z X- = MeX (Gaines-Thomas equivalent-fraction activity for the bound species, Davies for the aqueous ions). Divalent Ca2+ (logK 0.8) is equivalent-fraction-favoured and outcompetes Na+; watch the per-iteration selectivity ratio beta(CaX2)/beta(NaX)^2 climb.

CROSS-CHECK TOOL: PHREEQC EXCHANGE (same Gaines-Thomas half-reactions).

THE FRESH-RESIN CAVEAT (printed loudly, non-suppressible): the 'exchange' op returns the LIMITING (fully-equilibrated) effluent – the best a fresh / fully-loaded contactor does per pass. A real fixed bed leaks rising hardness as the exchange front migrates to breakthrough, then must be regenerated; cycle length and breakthrough are TRANSIENT and not modelled here.

Brackish water (I ~ 0.03-0.06): Davies-valid, no Pitzer needed; the honesty banner + the I>0.5 Davies advisory cover the boundary.

Data: curated USGS PHREEQC records (public domain) resolved from data/standards/ at runtime, + the SAC_Na resin nameplate; self-DESCRIBING (inline manifest) – seal with bin/choupo-import for a runtime-self-contained copy.

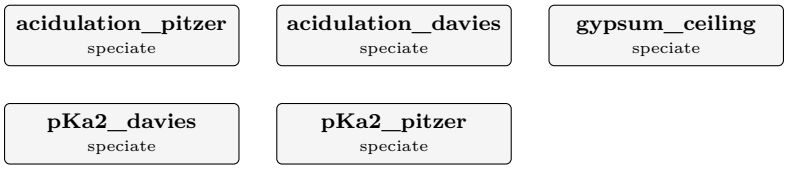
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | raw | I_r0 | 0.0266002 |
| diag | raw | I_rmax | 0.165631 |
| diag | raw | SI_anhydrite_r0 | -1.68532 |
| diag | raw | SI_anhydrite_rmax | -0.646016 |
| diag | raw | SI_aragonite_r0 | 0.111093 |
| diag | raw | SI_aragonite_rmax | 1.26704 |
| diag | raw | SI_calcite_r0 | 0.225093 |
| diag | raw | SI_calcite_rmax | 1.38104 |
| diag | raw | SI_gypsum_r0 | -1.38593 |
| diag | raw | SI_gypsum_rmax | -0.349983 |
| diag | raw | SI_halite_r0 | -5.35323 |
| diag | raw | SI_halite_rmax | -3.82475 |
| diag | raw | SI_sylvite_r0 | -6.39779 |
| diag | raw | SI_sylvite_rmax | -4.87569 |

*... and 193 more reference values in the case's `expected`.*

### tartaricAcid_acidulation

tutorials/props/electrolyte/tartaricAcid_acidulation




**What it does:** Tartaric-acid acidulation liquor: rigorous Pitzer brine chemistry (Ca-SO4-K-Cl) + the gypsum precipitation ceiling, with tartrate honestly on the Debye-Hueckel limit (no published Pitzer params). Davies vs Pitzer shown side by side.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | acidulation_pitzer | I | 1.6414 |
| diag | acidulation_pitzer | SI_anhydrite | 1.07386 |
| diag | acidulation_pitzer | SI_arcanite | -2.80366 |
| diag | acidulation_pitzer | SI_gypsum | 1.36221 |
| diag | acidulation_pitzer | SI_sylvite | -3.52403 |
| diag | acidulation_pitzer | aw | 0.986675 |
| diag | acidulation_pitzer | gamma_Ca | 0.0716778 |
| diag | acidulation_pitzer | gamma_H2Tart | 1 |
| diag | acidulation_pitzer | gamma_HTart | 0.409404 |
| diag | acidulation_pitzer | gamma_SO4 | 0.0662673 |
| diag | acidulation_pitzer | gamma_Tart | 0.036756 |
| diag | acidulation_pitzer | m_Ca | 0.397714 |
| diag | acidulation_pitzer | m_H2Tart | 0.0656066 |
| diag | acidulation_pitzer | m_HTart | 0.0314016 |
| *... and 74 more reference values in the case's `expected`.* | | | |

## estimate

**The theory you need.** When a compound is missing, group contribution builds it: Joback sums first-order group increments into $T_b, T_c, P_c, V_c, \Delta H_f^\circ, \Delta H_{\text{vap}}, c_p^{\text{ig}}(T)$; Lee–Kesler closes $\omega$ from $(T_b, T_c, P_c)$; Ambrose–Walton gives $P^{\text{sat}}(T)$ from corresponding states; Rackett the liquid volume. The estimate is a PROPOSAL: a `.dat` you review against any reference data and promote by hand – the engine never estimates at runtime. Molecular structure is IDENTITY: components carry `jobackGroups{}` in their own record, and `bin/estimate <name>` is the one-command curation act. Expect few-percent errors on criticals (and the guide's worked example shows them), weaker $T_b$ for hydrogen-bonded species, and polymers via van Krevelen / Yang on the repeat unit. *Full derivation: Theory Guide, the estimation chapter of the Props Guide.*

## estimate_acetone

tutorials/props/estimate/estimate_acetone




**What it does:** CREATE a component from its groups: Joback estimate of acetone (2 CH3 + 1 ketone), validated against NIST — see the estimate built up from groups

**The story:** The FIRST stage of the props lifecycle (CREATE → curate → wire → simulate): build a pure-component property set from its molecular GROUPS by Joback group contribution, and SEE each group's contribution add up.

Acetone CH3-CO-CH3 = 2 x CH3 + 1 x ketone (>C=O).

The reference block below validates FOUR numbers – Tb, Tc, Pc, omega – and no others: Joback lands 2.2 % low on Tb, 1.5 % low on Tc, 2.2 % high on Pc (read off the golden against the block, 2026-09-02), and you SEE the error, the whole point of "estimation, then trust". This header used to claim "~2 % on Tc/Pc/Vc and ~0.3 % on dHf"; the block carries no reference Vc and no reference dHf, so those two were claims nothing in the case checks. omega comes from (Tb,Tc,Pc) by Lee-Kesler, not a group sum.

*Acetone is 2 × CH3 + 1 × ketone. This case runs under choupoProps and asks Joback's group-contribution method to create a property set from those three groups, then lays the estimate beside the reference values the case itself declares. The golden: Tb 322.11 K, Tc 500.56 K, Pc 48.02 bar, ? 0.30, Vc 209.5 $cm^3/mol$, $\Delta Hf$ -217.83 kJ/mol.*

*1. Estimation is a resolution problem, not a runtime one. Choupo does not estimate properties while a flowsheet runs; it estimates them HERE, at curation time, so a student can look at the number before trusting it. output { proposal auto; } writes the estimate as a record you can read, edit, and only then wire into a case — the run prints where: constant/components/acetone.estimated.dat, with the GAPS the method left listed in its header. 2. The error is the point. Against the reference block in system/propsDict (Tb 329.2 K, Tc 508.1 K, Pc 47.0 bar, ? 0.307): Tb is 2.2 % low, Tc 1.5 % low, Pc 2.2 % high — Joback's usual band for a small ketone, and you can see it rather than be told. 3. ? is not a group sum. It comes from (Tb, Tc, Pc) through Lee-Kesler; an error in Tb propagates into it. That chain is printed in the run's log. 4. A reference is a citation, not an endorsement. The block names where its four numbers were read; the run compares against those four and nothing else.*

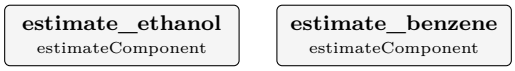
*Open estimate_ethanol_benzene: the same method on a hydrogen-bonding molecule, where Joback's Tb goes noticeably wrong — the honest limit of a group method, shown rather than warned about.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | estimate_acetone | Cp298 | 74.969755157 |
| diag | estimate_acetone | Hvap_kJmol | 27.534 |
| diag | estimate_acetone | MW | 58.08 |
| diag | estimate_acetone | Pc_bar | 48.0249960499 |
| diag | estimate_acetone | Psat_298_bar | 0.387196368809 |
| diag | estimate_acetone | Psat_Tb_bar | 0.97953832278 |
| diag | estimate_acetone | Tb_K | 322.11 |
| diag | estimate_acetone | Tc_K | 500.559004953 |
| diag | estimate_acetone | Vc_cm3mol | 209.5 |
| diag | estimate_acetone | Vliq298_cm3mol | 82.1064043414 |
| diag | estimate_acetone | dGf_kJmol | -154.54 |
| diag | estimate_acetone | dHf_kJmol | -217.83 |
| diag | estimate_acetone | omega | 0.29678401624 |

## estimate_ethanol_benzene

tutorials/props/estimate/estimate_ethanol_benzene




**What it does:** Joback estimates: ethanol (H-bonding → Tb weak, the honest limit) and benzene (aromatic groups → good) — see where group contribution wins and loses

**The story:** Two Joback estimates, side by side, to SEE the method's spread:

ethanol (1 CH3 + 1 CH2 + 1 OH) – a strong H-bonder; Joback's Tb comes out ~14 K LOW (337 vs 351 K). The honest limit, in plain sight. benzene (6 arCH) – aromatic ring; Joback Tc/Tb within ~1.5%. Group contribution at its best.

The lesson is the CONTRAST: an estimate is a starting point whose error you must SEE, not a number to trust blindly.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | estimate_ethanol | Cp298 | 64.6209120993 |
| diag | estimate_ethanol | Hvap_kJmol | 36.725 |
| diag | estimate_ethanol | MW | 46.069 |
| diag | estimate_ethanol | Pc_bar | 57.5664143723 |
| diag | estimate_ethanol | Psat_298_bar | 0.155662207837 |
| diag | estimate_ethanol | Psat_Tb_bar | 0.972926042801 |
| diag | estimate_ethanol | Tb_K | 337.54 |
| diag | estimate_ethanol | Tc_K | 499.407373566 |
| diag | estimate_ethanol | Vc_cm3mol | 166.5 |
| diag | estimate_ethanol | Vliq298_cm3mol | 57.6954539184 |
| diag | estimate_ethanol | dGf_kJmol | -170.86 |
| diag | estimate_ethanol | dHf_kJmol | -236.84 |
| diag | estimate_ethanol | omega | 0.573129252591 |
| diag | estimate_benzene | Cp298 | 81.8236158005 |
| *... and 12 more reference values in the case's `expected`.* | | | |

## estimate_petroleum_cut

tutorials/props/estimate/estimate_petroleum_cut




**What it does:** ESTIMATE a petroleum PSEUDO-COMPONENT by name from (Tb, SG) – Riazi-Daubert; see the SCREAMING ESTIMATE provenance and that the lump FLASHES

**The story:** CREATE a PETROLEUM PSEUDO-COMPONENT from the two bulk measurements every crude assay reports: the normal boiling point Tb and the specific gravity SG. The Riazi-Daubert (1987) correlations turn (Tb, SG) into the corresponding-states constants (MW, Tc, Pc, Vc); omega follows by Lee-Kesler, and the ideal-gas Cp by Kesler-Lee (1976).

This is the SCALAR-input sibling of the Joback group path: a petroleum cut is a LUMP of hundreds of species with NO single molecular structure, so it has NO group decomposition – it is anchored on bulk (Tb, SG), not groups. The estimator is a registered ConstantEstimator (‘model RiaziDaubert;‘), chosen exactly like Joback – the SAME explicit factory.

SEE in the console: the (Tb, SG) anchors and the Watson K factor; the constants built from them, validated against n-decane (a real species at this Tb/SG: Tc ~ exact, omega within ~2 %, Pc/MW within the correlation’s stated ~5-7 %); the proposal .dat whose provenance SCREAMS status ESTIMATE / lumped true / "NOT a real species".

Promote with the printed ‘mv‘, then it FLASHES (see the sibling tutorial steady/flash/flash_pseudoComponent_petCut, which ships this very lump and closes an energy balance with it).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | estimate_petroleum_cut | Cp298 | 248.058663071 |
| diag | estimate_petroleum_cut | MW | 151.514068352 |
| diag | estimate_petroleum_cut | Pc_bar | 19.7757194366 |
| diag | estimate_petroleum_cut | Psat_Tb_bar | 0.965291883924 |
| diag | estimate_petroleum_cut | SG | 0.73 |
| diag | estimate_petroleum_cut | Tb_K | 447.3 |
| diag | estimate_petroleum_cut | Tc_K | 617.804443469 |
| diag | estimate_petroleum_cut | Vc_cm3mol | 627.45599905 |
| diag | estimate_petroleum_cut | Vliq298_cm3mol | 203.429494662 |
| diag | estimate_petroleum_cut | omega | 0.481302966295 |
| diag | estimate_petroleum_cut | Psat_298_bar | 0.00181901199888 |

## userEstimate01_linear_tb

tutorials/props/estimate/userEstimate01_linear_tb




**What it does:** User-code property estimation: a student's OWN linear group-contribution Tb estimator (linearTbEstimate), compiled from code/ and linked into choupoProps.

**The story:** A student's OWN property-estimation op (linearTbEstimate), defined in code/ and linked into choupoProps by bin/buildCode. Tb = base + sum(count∗incr). Here: 198.2 + 2∗23.58 + 1∗76.75 = 322.11 K (the Joback CH3,CH3,ketone sum). —

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | tb_acetone | Tb_K | 322.11 |

## vanKrevelen01_polystyrene_density

tutorials/props/estimate/vanKrevelen01_polystyrene_density




**What it does:** ESTIMATE the density of polystyrene by Van Krevelen group contribution on the repeat unit – glass-box additive sum, Vw from Bondi 1964 (licence-clean via UNIFAC R)

**The story:** ESTIMATE a POLYMER's repeat-unit DENSITY by group contribution – the polymer sibling of the Joback small-molecule path. Polystyrene's repeat unit -[CH2-CH(C6H5)]- is decomposed into groups; each adds its molar mass and its Bondi van der Waals volume Vw; the sums give

M0 = sum n_i MW_i repeat-unit molar mass Vw = sum n_i Vw_i van der Waals volume (Bondi 1964) V = k ∗ Vw molar volume (k = packing factor) rho = M0 / V density

GLASS-BOX: the console prints every group's contribution – you can redo the whole estimate by hand. M0 is an exact mass-balance fact (104.15 g/mol); rho carries the packing-factor uncertainty (the teaching point below).

LICENCE (the honest part): the Vw group values are Bondi, J. Phys. Chem. 68 (1964) 441 – the PRIMARY per value – derived from the UNIFAC R_k volume parameters Choupo already ships (Vw = R_k x 15.17), NOT copied from the Van Krevelen book table (copyright). See data/standards/parameters/vanKrevelen.dat. Slice 1 ships DENSITY only; Tg / solubility-parameter / Tm are DEFERRED (their open parameters are a separate, correctly-attributed thing).

THE TEACHING POINT – the packing factor k: rho depends on k = V/Vw, the fraction of space the chains do NOT pack into. Default k = 1.60 (amorphous/glassy); crystalline packs denser (~1.43). k is YOURS: change 'polymer { packing k; }' and watch rho move. Atactic PS is amorphous (rho ~ 1.05) – the estimate lands there; a crystalline polymer would need the smaller k. That k is visible (never hidden) is the lesson.

M0 = 104.15 also feeds the step-growth reactor KPIs (Mn = M0/(1-p)) – the same repeat-unit mass, two places.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | vanKrevelen01_polystyrene_density | M0_g_per_mol | 104.152 |
| diag | vanKrevelen01_polystyrene_density | V_cm3_per_mol | 100.5616 |
| diag | vanKrevelen01_polystyrene_density | Vw_cm3_per_mol | 62.851 |
| diag | vanKrevelen01_polystyrene_density | density_g_cm3 | 1.0357034892 |
| diag | vanKrevelen01_polystyrene_density | packing_k | 1.6 |

## yang2020_Tg_polystyrene

tutorials/props/estimate/yang2020_Tg_polystyrene




**What it does:** ESTIMATE the glass-transition Tg of polystyrene by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH2-CH(phenyl)- gives Tg(inf)=444 K, an HONEST +19% over-estimate of the measured ~373 K (a teaching point about group-contribution limits)

**The story:** Yang-2020 Tg estimate + an HONEST limitation

ESTIMATE the GLASS-TRANSITION temperature Tg of polystyrene by the YANG 2020 modified group-contribution scheme. The repeat unit -CH2-CH(phenyl)- decomposes into two Yang main-chain groups:

CH2 : MW 14 Yg 4.026 CHphenyl: MW 90 Yg 42.153

M0 = 14 + 90 = 104 g/mol Yg(inf) = 4.026 + 42.153 = 46.179 (1e3 g.K/mol) Tg(inf) = 46.179∗1e3 / 104 = 444 K

»> THE HONEST TEACHING POINT «< Measured polystyrene Tg ~ 373 K (atactic, high Mw). Yang 2020 gives 444 K – a +19% OVER-ESTIMATE. This is NOT a bug: it is the model telling the truth about itself. The CHphenyl contribution (Yg 42.153) was regressed across the whole training set, and the bulky monosubstituted phenyl pendant is one where an additive main-chain scheme over-predicts the limiting Tg. Box: "all models are wrong, some are useful". The student SEES the deviation (the console prints the per-cent error against the measured anchor) and learns where a group-contribution method earns trust (PVC: 351 vs 354 K, -0.8%) and where it does not (polystyrene: 444 vs 373 K, +19%).

Contrast the flagship ‘yang2020_Tg_pvc‘, where Tg(inf)=351 K reproduces the paper’s Fig 6 value EXACTLY – that case goldens Tg as validated physics. Here Tg is SHOWN (the machinery is identical) but the case exists to expose the over-estimate, so only M0 (an exact mass-balance fact) is goldened, not Tg.

This is Yang 2020 – NOT bare Van Krevelen. Nb (backbone-in-the-side-chain) = 0 for this vinyl polymer.

SOURCE + LICENCE: Yang, Zou, Ye, Zhu, Dong, Bi, ACS Omega 5 (2020) 19655, doi:10.1021/acsomega.0c04499 – CC-BY 4.0; values cited per group as "Yang 2020", never "Van Krevelen".

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | yang2020_Tg_polystyrene | M0_g_per_mol | 104 |
| diag | yang2020_Tg_polystyrene | Tg_K | 444 |
| diag | yang2020_Tg_polystyrene | YgSum_1e3gKmol | 46.18 |

## yang2020_Tg_pvc

tutorials/props/estimate/yang2020_Tg_pvc




**What it does:** ESTIMATE the glass-transition Tg of poly(vinyl chloride) (PVC) by the Yang 2020 modified group-contribution scheme (CC-BY) – the repeat unit -CH2-CHCl- gives Tg(inf)=351 K, EXACT vs the paper (Fig 6)

**The story:** FLAGSHIP Yang-2020 glass-transition estimate

ESTIMATE the GLASS-TRANSITION temperature Tg of poly(vinyl chloride) (PVC) by the YANG 2020 modified group-contribution scheme – the polymer sibling of the Joback small-molecule path, for Tg. The repeat unit -CH2-CHCl- is decomposed into two Yang main-chain groups; each adds its limiting Tg contribution Yg(inf):

M0 = sum n_i MW_i repeat-unit molar mass [g/mol] Yg(inf) = sum n_i Yg_i(inf) (+ Nb∗Ygb) limiting Tg function [1e3 g.K/mol] Tg(inf) = Yg(inf) ∗ 1e3 / M0 Tg at infinite Mw [K]

THE ADDITIVE SUM (shown on the console): CH2 : MW 14 Yg 4.026 CHCl: MW 48.5 Yg 17.911

M0 = 14 + 48.5 = 62.5 g/mol Yg(inf) = 4.026 + 17.911 = 21.937 (1e3 g.K/mol) Tg(inf) = 21.937∗1e3 / 62.5 = 351 K <– EXACT vs Yang 2020, Fig 6

This is Yang 2020 – NOT bare Van Krevelen. The scheme predicts the INFINITE- Mw limit Tg(inf); the molecular-weight dependence Tg(Mn) = Tg(inf) - K/Mn is a later slice. Nb (backbone-in-the-side-chain) = 0 for this vinyl polymer, so the bracket is just the sum of the two main-chain group contributions.

SOURCE + LICENCE (the honest part): Yang, Zou, Ye, Zhu, Dong, Bi, ACS Omega 5 (2020) 19655, doi:10.1021/acsomega.0c04499 – CC-BY 4.0, so the parameter values ARE redistributable. Cited per value as "Yang 2020", never "Van Krevelen".

THE GOLDEN: M0 (an exact mass-balance fact) AND Tg(inf)=351 K are checked – Tg here is VALIDATED PHYSICS (it reproduces the paper's Fig 6 value exactly), not just a mass balance. The measured anchor (PVC Tg ~ 354 K) confirms it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | yang2020_Tg_pvc | M0_g_per_mol | 62.5 |
| diag | yang2020_Tg_pvc | Tg_K | 351 |
| diag | yang2020_Tg_pvc | YgSum_1e3gKmol | 21.937 |

### fit

## cp01_polynomial_recovery

tutorials/props/fit/cp01_polynomial_recovery




**What it does:** Quadratic Cp fit on data generated from a quadratic – exact recovery, with the Kirchhoff Cp_ig - Cp_liq cross-check that ties Cp to dHvap

**The story:** A LINEAR FIT, AND THE PHYSICS THAT CHECKS IT.

'heatCapacityFit' regresses Cp(T) = A0 + A1 T + A2 T^2 by linear least squares – normal equations, no iteration. Fed data generated FROM a quadratic, it must return that quadratic, so the expected file pins the coefficients rather than a tolerance band.

The interesting number is not the fit. It is KIRCHHOFF:

d(dHvap)/dT = Cp_ig - Cp_liq

The op takes the FITTED liquid Cp and the component's CATALOGUE ideal-gas Cp and reports that difference at the mid temperature. It should be NEGATIVE – the enthalpy of vaporisation falls as T rises, to zero at the critical point – and its size is the slope of dHvap(T). That is the link tying Cp to dHvap to the Antoine fit next door (psat01_antoine_recovery, which reports dHvap from B): three ops, one consistency network, and a student can check one against the other by hand.

Why the case exists: 'heatCapacityFit' had NO case and no gate. Writing the first one for its sibling vaporPressureFit segfaulted choupoProps twice and found a thermoPhysPropDict being silently ignored.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | cp_dce_liquid_fit | A0 | 194 |
| diag | cp_dce_liquid_fit | A1 | -0.618063 |
| diag | cp_dce_liquid_fit | A2 | 0.00170318 |
| diag | cp_dce_liquid_fit | Cp_at_Tmid | 166.076 |
| diag | cp_dce_liquid_fit | Cp_ig_at_Tmid | 84.2326 |
| diag | cp_dce_liquid_fit | R2 | 1 |
| diag | cp_dce_liquid_fit | dHvap_dT_kirchhoff | -81.8435 |
| diag | cp_dce_liquid_fit | degree | 2 |
| diag | cp_dce_liquid_fit | n_points | 21 |
| verdict | cp_dce_liquid_fit | verdict | notClaimed |

## psat01_antoine_recovery

tutorials/props/fit/psat01_antoine_recovery




**What it does:** Antoine fit on data generated from Choupo's own water correlation: the coefficients must come back exactly, so any deviation is the op's arithmetic

**The story:** FITTING ANTOINE, AND KNOWING THE ANSWER BEFOREHAND.

'vaporPressureFit' regresses log10(Psat[bar]) = A - B/(T + C) on measured Psat(T). The fit is linear in (A, B) at fixed C, so the op golden-sections C and solves (A, B) by least squares at each one – derivative-free, and worth reading in the source.

This case feeds it data generated FROM THE SAME EQUATION, by evaluating Choupo's own curated water correlation at twenty temperatures. That is deliberate. Against laboratory data a fit can only be judged to the scatter of the measurements; against data the equation itself produced, the fit must RETURN THE COEFFICIENTS IT WAS GIVEN – so any deviation is the op's arithmetic and nothing else, and the expected file can pin it to the digit.

Why the case exists: 'vaporPressureFit' had NO case and no gate. It could have been broken for a year and the suite would have stayed green; the first to find out would be a student fitting their own data. The psychrometric chart was in exactly that position last week, and writing its first test found it reporting nothing.

What to read in the output: A, B, C – the recovered coefficients, against 5.40221 / 1838.675 / -31.737; R2 – 1 to floating point, because the data has no scatter; dHvap – the Clausius-Clapeyron reading of B WITH the (T/(T+C))^2 correction, at the highest T. Compare it to water's tabulated enthalpy of vaporisation: this is the physical cross-check on a fit that is otherwise just a straight line through transformed numbers.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | psat_water_fit | A | 5.40221000142 |
| diag | psat_water_fit | B | 1838.67500085 |
| diag | psat_water_fit | C | -31.7369999323 |
| diag | psat_water_fit | R2 | 1 |
| diag | psat_water_fit | dHvap_kJ_per_mol | 42.1162117513 |
| diag | psat_water_fit | n_points | 20 |
| verdict | psat_water_fit | verdict | notClaimed |

## gibbs

**The theory you need.** Equilibrium without a mechanism: minimise total Gibbs energy $G = \sum_i n_i \mu_i$ over ALL species subject to elemental balances $\mathbf{A}\,\mathbf{n} = \mathbf{b}$ (Lagrange multipliers per element – printed as $\lambda_E$, the element potentials). The species list in the dict IS the model: what you include can form; what you omit cannot. With enough species this reproduces flame temperatures and radical pools with no kinetics at all: expect the adiabatic flame temperature to land within a few K of literature when radicals (H, O, OH…) are IN the pool, and to overshoot by ∼100–250 K when they are not (dissociation is endothermic). Above ∼1500 K the catalogue's polynomial $c_p$ fits extrapolate WRONG – combustion cases carry case-local NASA-9 overrides (NASA/TP-2002-211556), and the case headers say so.

*Equilibrium maps.* Sweeping the Gibbs solve over a $T \times P$ grid gives an equilibrium map – iso-lines of a product fraction over the operating plane – which shows why an industrial reactor fixes $T$ and $P$ in a narrow window (equilibrium wants one corner, the catalyst works in another; the gap is the compromise). The `temperatureApproach` $\Delta T$ – declared as a magnitude; the engine assigns the sign from the reaction's thermicity and announces it – evaluates the *reaction* equilibrium at $T \pm \Delta T$ while enthalpy and the energy balance stay at the physical $T$ – an empirical, calibrated closeness-to-equilibrium knob (never predicted). *Gradeable exercise (ammonia):* on `nh3_equilibrium_map`, at $P = 200$ bar and $\Delta T = 0$, find the temperature at which the equilibrium $NH_3$ mole fraction crosses 30 %; report $T$ to the grid resolution and verify against the case's anchor KPIs. Unique because the fraction is monotone in $T$ at fixed $P$. *Full derivation: Theory Guide, the Gibbs-equilibrium chapter, §equilibrium maps.*

## landscape01_ammonia_synthesis

tutorials/props/gibbs/landscape01_ammonia_synthesis




**What it does:** Equilibrium landscape: ammonia synthesis, NH3 mole fraction against T at 1, 10, 100 bar – and the 298 K / 700 K contrast

**The story:** THE AMMONIA LANDSCAPE – N2 + 3 H2 at stoichiometric feed, equilibrium NH3 mole fraction against temperature at 1, 10 and 100 bar (the ideal-gas element-potential kernel, the same one nh3_equilibrium_map contours over the whole plane). This is the first witness of the EduTool 'equilibrium-landscapes' (gui/src/ui/methods/EquilibriumLandscapesTool.tsx): a SURVEY over reaction systems the engine already solves, read live from this case's 'map.csv' and its anchor KPIs.

WHAT THE THREE CURVES SHOW. Exothermic AND mole-reducing ($4 \rightarrow 2$), so the equilibrium loves COLD and PRESSURE: every curve falls with T, and the 100 bar curve sits above the 1 bar one wherever the gas is dry. The grid starts at 300 K on purpose: from 300 to 380 K the 100 bar cells are a SATURATED gas – the kernel finds NH3 supersaturated there and condenses it (ElementPotential.cpp, the one-condensable two-phase branch), and the mole fraction it reports is the GAS-phase Psat(T)/P. That is why the 100 bar curve RISES with T at its cold end: that leg is ammonia's vapour pressure, not its chemistry, and the curve joins the dry equilibrium at the dew point near 400 K (measured: 0.1047 at 300 K, 0.9311 at 400 K).

THE CONTRAST THE PAGE CLOSES ON, computed here and never typed. Two anchor cells: 298.15 K / 1 bar – ambient, where the reaction is thermodynamically favourable and nothing in this tree can make it go (the only ammonia rate law here, Dyson & Simon 1968, written in the cases that use it, was fitted over 150-300 atm and its effectiveness factor is a table at 150, 225 and 300 atm that the engine REFUSES to extrapolate: tutorials/props/kinetics/ ammoniaRate01_dyson_simon/constant/reactions) – and 700 K / 200 bar, the gibbs10_ammonia_fugacity operating point, which this ideal-gas kernel answers as the SHAPE and that SRK case answers with the fugacity correction folded in. The third anchor, 700 K / 1 bar, is what the same chemistry reaches at converter temperature without the pressure.

Anchor KPI names carry the approach magnitude when one is declared ('metric_T25C_P1atm_dT<n>'), so a changed approach breaks the golden loudly. Nothing here is validated against a measurement: the Larson & Dodge (1923) pins live in nh3_equilibrium_map and are not repeated.

WHY THE PRESSURES ARE 1, 10 AND 100 BAR. 'Pgrid' is 'from / to / n / log' (GibbsMapOp.cpp reads no list form), and three log-spaced points from 1 to 100 bar land on the decades a textbook draws. Ideal-gas validity is the same caveat as nh3_equilibrium_map's: the high-pressure curve is the SHAPE of the pressure dependence, not a quantitative prediction.

temperatureApproach IS DECLARED, AT ZERO. The EduTool's slider replaces the NUMBER of a declared scalar and can add no key (gui/src/case/ methodRun.ts), so the key must be here for the page to move it. Zero is true equilibrium and prints nothing; a magnitude $> 0$ makes the run assign the sign at EVERY cell from the thermicity of the overall transformation at that cell's T (GibbsMapOp.cpp, 'resolveDirection') and print the tally.

## landscape02_water_gas_shift

tutorials/props/gibbs/landscape02_water_gas_shift




**What it does:** Equilibrium landscape: water-gas shift, CO conversion (carbon in CO2) against T – pressure-indifferent

**The story:** THE SHIFT LANDSCAPE – equimolar CO + H2O (gibbs01_water_gas_shift's own feed and species set), the fraction of the feed carbon leaving as CO2 against temperature at 1, 10 and 100 bar. 'elementYield' on carbon with CO2 as the product IS the CO conversion here, because every carbon atom fed arrives as CO – and it is the forum-permitted metric (species "conversion" is banned where the species is also produced; docs/design/gibbs-map-forum-2026-07-02.md).

WHAT THE THREE CURVES SHOW, and it is one curve. Exothermic (the conversion falls with T) and mole-CONSERVING ($2 \rightarrow 2$), so the three pressures give the SAME equilibrium to every printed digit: the ideal-gas equilibrium constant of a Delta_n = 0 reaction carries no P. The page draws the three curves on top of each other and says why. MEASURED, not assumed: no cell on this grid condenses – the three pressures agree to every printed digit at all 46 temperatures, which they could not if water had left the gas anywhere (at 400 K the equilibrium leaves 2.5 % water in the gas, 2.5 bar at 100 bar against a 2.46 bar Psat – and the run's own identity of the three columns is the proof, not this estimate).

WHY THE PRESSURES ARE 1, 10 AND 100 BAR. 'Pgrid' is 'from / to / n / log' (GibbsMapOp.cpp reads no list form), and three log-spaced points from 1 to 100 bar land on the decades a textbook draws. Ideal-gas validity is the same caveat as nh3_equilibrium_map's: the high-pressure curve is the SHAPE of the pressure dependence, not a quantitative prediction.

temperatureApproach IS DECLARED, AT ZERO. The EduTool's slider replaces the NUMBER of a declared scalar and can add no key (gui/src/case/ methodRun.ts), so the key must be here for the page to move it. Zero is true equilibrium and prints nothing; a magnitude $> 0$ makes the run assign the sign at EVERY cell from the thermicity of the overall transformation at that cell's T (GibbsMapOp.cpp, 'resolveDirection') and print the tally.

## landscape03_steam_reforming

tutorials/props/gibbs/landscape03_steam_reforming




**What it does:** Equilibrium landscape: methane steam reforming (S/C 3), carbon left in CH4 against T – pressure hurts

**The story:** THE REFORMING LANDSCAPE – CH4 + 3 H2O (gibbs02_steam_reforming's own feed and species set), the fraction of the feed carbon STILL in methane against temperature at 1, 10 and 100 bar. Read it as 1 minus the methane conversion: LESS is MORE converted, and the page labels it so rather than subtracting – the metric on the axis is exactly the number the engine wrote. ('elementYield' with CH4 as the "product" is the honest way to ask for the unconverted feed; the metric slot accepts any listed species.)

WHAT THE THREE CURVES SHOW. Endothermic (the slip falls as T rises) and mole-INCREASING (CH4 + H2O → CO + 3 H2, 2 → 4), so pressure HURTS: the 100 bar curve sits far above the 1 bar one, which is why industrial reformers run hot and why the shift on the same catalyst (exothermic, Delta_n = 0) is the one the approach caveat calls GLOBAL – one dT for two reactions of opposite thermicity. The engine reads the OVERALL thermicity from this feed (endothermic at every cell of this grid, the run's own tally) and evaluates the chemistry at T - dT.

WHY THE PRESSURES ARE 1, 10 AND 100 BAR. 'Pgrid' is 'from / to / n / log' (GibbsMapOp.cpp reads no list form), and three log-spaced points from 1 to 100 bar land on the decades a textbook draws. Ideal-gas validity is the same caveat as nh3_equilibrium_map's: the high-pressure curve is the SHAPE of the pressure dependence, not a quantitative prediction.

temperatureApproach IS DECLARED, AT ZERO. The EduTool's slider replaces the NUMBER of a declared scalar and can add no key (gui/src/case/ methodRun.ts), so the key must be here for the page to move it. Zero is true equilibrium and prints nothing; a magnitude > 0 makes the run assign the sign at EVERY cell from the thermicity of the overall transformation at that cell's T (GibbsMapOp.cpp, 'resolveDirection') and print the tally.

## landscape04_methane_combustion

tutorials/props/gibbs/landscape04_methane_combustion




**What it does:** Equilibrium landscape: methane combustion in stoichiometric air, carbon in CO2 against T – complete until dissociation

**The story:** THE COMBUSTION LANDSCAPE – CH4 in stoichiometric air (gibbs03's own feed and species set), the fraction of the feed carbon fully oxidised to CO2 against temperature at 1, 10 and 100 bar, 600 to 2500 K.

WHAT THE THREE CURVES SHOW, and it is where the approach is INDIFFERENT. Below about 1500 K the equilibrium is complete conversion to better than 0.2 % (0.99996 at 600 K, 0.9987 at 1500 K and 1 bar) – a flat line at 1 on any axis a plot can draw – and a 50 K approach moves it by nothing visible: the whole conversion-versus-temperature curve is flat, so a HORIZONTAL shift along it changes nothing. Only above ~1800 K, where CO2 starts to dissociate to CO, does the curve acquire a slope and the approach a visible effect, and pressure holds the CO2 together (the dissociation makes moles). This is the page's counterexample to the ammonia and reforming systems, where the same 50 K moves the answer by a great deal.

The grid starts at 600 K rather than 300 K by measurement: on a 300 K start the cells at 300 and 350 K (every pressure), 400 K / 10 bar and 500 K / 100 bar did NOT converge – the product water is far below its dew point there and the gas-only Newton the kernel runs first does not reach a root – and a witness that ships hatched cells at its cold end teaches nothing the 600 K plateau does not. From 600 K up every cell converges dry (y_H2O P stays below Psat on the whole grid).

WHY THE PRESSURES ARE 1, 10 AND 100 BAR. 'Pgrid' is 'from / to / n / log' (GibbsMapOp.cpp reads no list form), and three log-spaced points from 1 to 100 bar land on the decades a textbook draws. Ideal-gas validity is the same caveat as nh3_equilibrium_map's: the high-pressure curve is the SHAPE of the pressure dependence, not a quantitative prediction.

temperatureApproach IS DECLARED, AT ZERO. The EduTool's slider replaces the NUMBER of a declared scalar and can add no key (gui/src/case/ methodRun.ts), so the key must be here for the page to move it. Zero is true equilibrium and prints nothing; a magnitude $> 0$ makes the run assign the sign at EVERY cell from the thermicity of the overall transformation at that cell's T (GibbsMapOp.cpp, 'resolveDirection') and print the tally.

## landscape05_thermal_no

tutorials/props/gibbs/landscape05_thermal_no




**What it does:** Equilibrium landscape: thermal NO in methane-air combustion products against T – the GLOBAL caveat, drawn

**The story:** THE THERMAL-NO LANDSCAPE – CH4 in stoichiometric air with NO in the species list (gibbs06_thermal_nox's own feed and species set), the equilibrium NO mole fraction against temperature at 1, 10 and 100 bar, 1000 to 2500 K. The page draws it on a LOG axis: it spans many decades (read the two ends off the CSV rather than off this sentence).

WHAT IT IS HERE FOR, and it is the sharpest thing on the page. NO formation (N2 + O2 → 2 NO) is ENDOTHERMIC – the curve RISES with T – but it rides inside a feed whose OVERALL transformation, methane burning, is exothermic. The engine reads the direction from the overall thermicity of THIS feed at each cell (GibbsReactor::approachDirection, the one home; GibbsMapOp calls it per cell), finds every cell exothermic, and evaluates the chemistry at T + dT. For the combustion that is the conservative direction. For the NO riding inside it, T + dT sits HIGHER on a rising curve: the declared approach reports MORE NO than the equilibrium at the physical T – an outlet beyond equilibrium for the minor product. That is the engine's own GLOBAL caveat ("it cannot resolve per-reaction approaches") drawn as a curve rather than printed as a sentence, and the page says it is a limitation of the single-number model, not a finding about flames.

Pressure: 2 → 2 for NO formation, so the 1 and 10 bar curves nearly coincide; the small spread that remains comes from the combustion products' own dissociation equilibria, which do carry P.

WHY THE PRESSURES ARE 1, 10 AND 100 BAR. 'Pgrid' is 'from / to / n / log' (GibbsMapOp.cpp reads no list form), and three log-spaced points from 1 to 100 bar land on the decades a textbook draws. Ideal-gas validity is the same caveat as nh3_equilibrium_map's: the high-pressure curve is the SHAPE of the pressure dependence, not a quantitative prediction.

temperatureApproach IS DECLARED, AT ZERO. The EduTool's slider replaces the NUMBER of a declared scalar and can add no key (gui/src/case/ methodRun.ts), so the key must be here for the page to move it. Zero is true equilibrium and prints nothing; a magnitude > 0 makes the run assign the sign at EVERY cell from the thermicity of the overall transformation at that cell's T (GibbsMapOp.cpp, 'resolveDirection') and print the tally.

## landscape06_nitrogen_fixation_routes

tutorials/props/gibbs/landscape06_nitrogen_fixation_routes




**What it does:** Nitrogen fixation, three overall routes on one formation surface: heterotrophic (sugar as the electron donor), phototrophic (water), and Haber-Bosch – dG as the engine prices it

**The story:** THREE OVERALL ROUTES TO FIXED NITROGEN, PRICED ON ONE SURFACE. The sixth witness of the EduTool ‘equilibrium-landscapes‘ (gui/src/ui/methods/EquilibriumLandscapesTool.tsx) and the first case of the ‘reactionGibbs‘ property operation (src/propertyOps/ReactionGibbs.cpp), which exists because of it.

WHY A NEW OP WAS NEEDED, measured before it was written. The engine’s one route to a reaction’s Gibbs energy, ‘Reaction::equilibrium‘ (src/thermo/reaction/Reaction.cpp), prices every species on the ideal-gas rung with ‘g_pure_ig(T)‘, and that rung REFUSES a record tabulated on the solid standard state. A gibbsMap with glucose beside the gases was run and refused, verbatim: "Component ’glucose’: h_pure_ig(T) evaluates the IDEAL-GAS rung, but this record declares standardThermochemistry. referenceState pureSolid – dHf_298 and s_298 are tabulated on the solid standard state, so reading them here would be wrong by a heat of sublimation." The same refusal binds ‘equilibriumReactor‘ and ‘gibbsReactor‘. The biological routes are exactly a solid sugar and liquid water beside gases, so no reactor and no existing op could price them; ‘reactionGibbs‘ sums a DECLARED reaction over the phase-aware ‘h_formation‘ / ‘s_formation‘ surface the Component already carried.

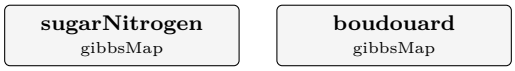
THE THREE ROUTES (constant/reactions), balanced by ELECTRON COUNT – the biological ones carry NO H2 (Vitor’s correction, 2026-09-27: nitrogenase reduces N2 with protons and electrons that come from the organism’s substrate): heterotrophic C6H12O6(s) + 4 N2 + 6 H2O(l) → 6 CO2 + 8 NH3 phototrophic N2 + 3 H2O(l) → 2 NH3 + 3/2 O2 haberBosch N2 + 3 H2 → 2 NH3 Every species is priced on a DECLARED standard state (‘phases {}‘): the sugar on its own pureSolid rung, water as a pure liquid reached by the vaporisation crossing from its ideal-gas datum (announced), the gases on idealGas. The lesson’s expectation, stated before the run: the sugar route is EXERGONIC (respiration pays for fixation), the water route is ENDERGONIC (light pays), the industrial route is exergonic at 298 K and loses that with temperature – the landscape witness beside this case draws the loss.

TWO TEMPERATURES, AND WHY ONLY THE FIRST IS ASKED OF THE SUGAR ROUTE. 298.15 K is the datum temperature, where a declared rung answers with no heat capacity integrated. glucose.dat carries NO heat-capacity block on any rung ("FLAGGED, NOT FABRICATED", its own words), so asking the heterotrophic route at 310.15 K REFUSES by name – measured, verbatim: "Component ’glucose’: h_formation liquid leg needs liquidHeatCapacity" (the solid leg falls back to the liquid Cp and finds none). The refusal is the finding and is quoted on the page; this case therefore prices all three routes at 298.15 K in one op and the two routes that can move in temperature at 310.15 K in a second op, so that the run passes and the gap is stated rather than papered over; a third op prices the industrial route alone at 700 K, the converter temperature of gibbs10, where its dG has changed sign – which is the whole reason the converter needs its pressure, and the landscape witness beside this case draws it. Liquid water is not asked at 700 K: a pure-liquid standard state above the critical temperature is a number with no state behind it.

NOTHING HERE IS A RATE. An exergonic route is one the equilibrium permits; what makes each one go – the enzyme’s ATP, the light, the iron catalyst’s temperature – is outside every record in this tree, and the ATP stoichiometry of nitrogenase is deliberately NOT cited because no reference for it exists in the repository.

## landscape07_sugar_nitrogen_all_pathways

tutorials/props/gibbs/landscape07_sugar_nitrogen_all_pathways




**What it does:** Gibbs maps with a pure solid: the sucrose nitrogen-fixation reactants with every pathway open, and Boudouard carbon

**The story:** TWO GIBBS MAPS THAT CARRY A PURE SOLID, the first cases to (2026-09-29). Until that day ‘gibbsMap‘ called the fluid-only solve, so a species whose record declares ‘referenceState pureSolid;‘ was priced on the ideal-gas rung and its record refused the whole map (found by Vitor, mapping sucrose beside the gases). The op now calls ‘GibbsMethod::solve‘, the one home the gibbsReactor already used: a declared pure solid is its own phase at activity one, deposited only where the solid-free equilibrium would supersaturate it, and its amount is the CSV column ‘s_<name>‘.

sugarNitrogen – the REACTANTS of the sucrose nitrogen-fixation route C12H22O11(s) + 8 N2 + 13 H2O → 12 CO2 + 16 NH3 (balanced by electron count; its dG at 298.15 K, 1 bar is about -355 kJ per sucrose on ‘reactionGibbs‘, so the ROUTE is permitted), handed to a Gibbs map with every species open, methane included. Where the atoms go at minimum G is NOT that route: measured, the carbon ends as methane and CO2 (the anaerobic digester’s answer), the nitrogen stays N2 and ammonia is near 1e-6 at ambient. A declared route that is permitted is not the route equilibrium picks; the organism picks it with an enzyme.

boudouard – 2 CO <→ C(graphite) + CO2, CO fed alone: carbon DEPOSITS at low temperature and not at high, pinned as the fraction of the fed carbon that leaves as graphite (elementYield on the solid).

NOT HERE, said: a solid that must REMAIN for the atoms to close. With the six species sucrose, CO2, N2, NH3, H2, water and no methane, carbon has one fluid carrier (CO2) and too little oxygen for it, so the solid-free equilibrium the solid test starts from does not exist; every cell reports unconverged, with the solver’s own cause printed. That is a limit of GibbsMethod::solve, the same in the gibbsReactor, recorded in DEV.md section 5, not fixed here.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | sugarNitrogen | metric_T25C_P1atm | 1.27109422855e-06 |
| diag | sugarNitrogen | metric_T37C_P1atm | 2.67461125111e-06 |
| diag | sugarNitrogen | n_cells | 39 |
| diag | sugarNitrogen | n_unconverged | 0 |
| diag | boudouard | metric_T427C_P1atm | 0.496091670405 |
| diag | boudouard | metric_T927C_P1atm | 0.0181328889912 |
| diag | boudouard | n_cells | 30 |
| diag | boudouard | n_unconverged | 0 |

## nh3_equilibrium_map

tutorials/props/gibbs/nh3_equilibrium_map




**What it does:** Ammonia equilibrium map (T x P): why Haber runs at 400-500 C and 150-300 bar

**The story:** THE HABER MAP – the forum-ratified Gibbs equilibrium map (docs/design/gibbs-map-forum-2026-07-02.md) on its canonical pilot: N2 + 3 H2 at stoichiometric feed, equilibrium NH3 mole fraction contoured over 300-700 C x 1-300 bar. The map shows the whole Haber story at a glance: equilibrium loves COLD and PRESSURE (exothermic, Delta_n < 0); the industrial window (400-500 C, 150-300 bar) sits where the catalyst can actually work – the tension IS the lesson.

VALIDATION ANCHORS (golden-locked; verified 2026-07-03): the classic measurements of Larson & Dodge, J. Am. Chem. Soc. 45 (1923) 2918 – 400 C / 10 atm: engine 4.09 % vs L&D 3.85 % (+6 %) 450 C / 10 atm: engine 2.22 % vs L&D 2.11 % (+5 %) The ~5 % offset is consistent with propagating the JANAF formation-data uncertainty of NH3 into Kp (exp(-dG/RT) amplifies ~0.2 kJ/mol into ~5 %) – honest agreement for an equilibrium constant spanning decades. IDEAL-GAS VALIDITY: the kernel is ideal-gas, so pins stay at <= 50 atm, 350-500 C (the forum's chair: at 300 bar a temperatureApproach would silently absorb the missing fugacity corrections – announced, and the high-P corner of this map is drawn as EQUILIBRIUM SHAPE, not quantitative prediction).

temperatureApproach: 0 here (true equilibrium). Re-run with e.g. 'temperatureApproach 25;' to see the contours slide – the empirical closeness-to-equilibrium parameter industry quotes in C (announced loud, the anchor KPI names then carry _dT25 so the golden breaks visibly).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | haber_map | metric_T400C_P10atm | 0.0409234 |
| diag | haber_map | metric_T400C_P50atm | 0.157785 |
| diag | haber_map | metric_T450C_P10atm | 0.0221983 |
| diag | haber_map | metric_T500C_P100atm | 0.104969 |
| diag | haber_map | n_cells | 625 |
| diag | haber_map | n_unconverged | 0 |

## reactionEnthalpy01_haber_kirchhoff

tutorials/props/gibbs/reactionEnthalpy01_haber_kirchhoff




**What it does:** The reaction enthalpy is a function of temperature: N2 + 3 H2 → 2 NH3 priced from 25 C to 527 C on the formation surface – the witness of the reaction-enthalpy EduTool

**The story:** THE REACTION ENTHALPY IS A FUNCTION OF TEMPERATURE. The ammonia synthesis N2 + 3 H2 → 2 NH3, every species an ideal gas at the standard pressure, priced by 'reactionGibbs' at seven temperatures from 298.15 K (where formation enthalpies are tabulated) to 800 K (above an ammonia converter):

dH°(T) = sum_i nu_i h_i(T), h_i(T) = dHf_i(298.15 K) + int Cp_i dT

The "°" fixes the standard state (1 bar, ideal gas); it does not fix the temperature. At 298.15 K the result is -91.88 kJ per mole of reaction, i.e. -45.94 kJ per mole of NH3; at a converter's 700-800 K it is about -53 kJ per mole of NH3 – the 25 C value under-states the heat a converter releases by about 14 %. The witness of the EduTool 'reaction-enthalpy' (DEV.md 4c, C31). The constant/ tree is landscape06's, copied whole: the same formation surface, so the 298.15 K and 700 K rows reproduce that case's own 'routes298' and 'haber700' Haber-Bosch rows.

*The witness of the EduTool Reaction enthalpy and temperature (reaction-enthalpy). One reactionGibbs operation prices $N_2$ + 3 $H_2$ → 2 $NH_3$ (ideal gases at 1 bar) at seven temperatures, from 298.15 K to 800 K, on the formation surface of the catalogue records.*

*The degree sign fixes the standard state (1 bar, each species in its reference phase), not the temperature: tables are printed at 298.15 K by convention. A converter running at 700–800 K releases about 16 % more heat per mole of ammonia than the 25 °C value says.*

*constant/ is copied whole from [landscape06_nitrogen_fixation_routes](../landscape06_nitrogen_fixation_routes), so the formation surface is the same one; this case reads only $N_2$, $H_2$ and $NH_3$.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | haberT | haberBosch_T298K_dG_kJ_mol | -32.8135035 |
| diag | haberT | haberBosch_T298K_dH_kJ_mol | -91.88 |
| diag | haberT | haberBosch_T298K_dS_J_molK | -198.11 |
| diag | haberT | haberBosch_T298K_lnK | 13.2368181896 |
| diag | haberT | haberBosch_T400K_dG_kJ_mol | -11.9635351712 |
| diag | haberT | haberBosch_T400K_dH_kJ_mol | -96.1395343725 |
| diag | haberT | haberBosch_T400K_dS_J_molK | -210.439998003 |
| diag | haberT | haberBosch_T400K_lnK | 3.59720637425 |
| diag | haberT | haberBosch_T500K_dG_kJ_mol | 9.51125665348 |
| diag | haberT | haberBosch_T500K_dH_kJ_mol | -99.7619775808 |
| diag | haberT | haberBosch_T500K_dS_J_molK | -218.546468469 |
| diag | haberT | haberBosch_T500K_lnK | -2.2878824743 |
| diag | haberT | haberBosch_T600K_dG_kJ_mol | 31.6629415717 |
| diag | haberT | haberBosch_T600K_dH_kJ_mol | -102.826551039 |
| *... and 14 more reference values in the case's `expected`.* | | | |

## heat

**The theory you need.** Heat exchange couples streams without mixing them: $Q = U\,A\,\Delta T_{\mathrm{lm}}$, effectiveness-NTU when outlets are unknown. Utilities (steam levels, cooling water) are allocated by temperature level – a reboiler cannot be heated by a colder utility, and the allocator says which level it picked. Heat-links (forward and feedback) couple distant units into an energy tear the solver converges like any recycle. Pinch analysis finds the minimum utilities the composite curves allow.

*Exchanger DESIGN vs rating.* A shell-and-tube poses two inverse questions: RATE it (geometry given → find $U$, area, $\Delta P$, duty – `model geometry;`) or DESIGN it (duty given → size the geometry – `model design;`). $U$ is BUILT from the two film coefficients (Gnielinski tube, Kern shell) and the wall; the pressure drop (Kern both sides) is the duty-vs-pumping half a real design lives by. The standard workflow – establish the duty, THEN design, THEN rate, never invent the geometry up front – is walked end-to-end in the two-part sequence `hxWorkflow1_design_from_duty` → `hxWorkflow2_rate_designed`, and the design ends in a printable datasheet (GUI: the unit panel). *Full derivation: Theory Guide, the heat-integration chapter, §heat-exchanger design.*

## htbench01_correlations

tutorials/props/heat/htbench01_correlations




**What it does:** Heat-transfer correlation bench – each convective correlation (Dittus-Boelter, Gnielinski, Kern) verified against its published Nusselt anchor. Dittus-Boelter and Gnielinski are pinned to Incropera & DeWitt 6th ed. Example 8.4; Kern's closed j_H form is self-consistency-checked (its original pin is a chart). AAD tolerance 5% honours textbook rounding + the correlations' own scatter.

**The story:** the heat-transfer correlation self-check bench.

Mirrors the membrane mass-transfer story (Sherwood) on the heat side (Nusselt): each registered convective correlation RUNS its verify() against a published Nu anchor and the deviation is printed + locked.

DittusBoelter Nu = 0.023 Re^0.8 Pr^0.4 (Incropera Ex 8.4, Nu=205) Gnielinski Nu = (f/8)(Re-1000)Pr/[..] (Incropera Ex 8.4 alt, Nu=224) Kern (shell) Nu = 0.36 Re_s^0.55 Pr^1/3 (Kern PHT; closed-form check)

The per-correlation deviation diags are the golden KPIs.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bench | Nu_choupo_DittusBoelter | 200.475 |
| diag | bench | Nu_choupo_Gnielinski | 223.734 |
| diag | bench | Nu_choupo_Kern | 97.5647 |
| diag | bench | Nu_published_DittusBoelter | 205 |
| diag | bench | Nu_published_Gnielinski | 224 |
| diag | bench | Nu_published_Kern | 97.5647 |
| diag | bench | allPass | 1 |
| diag | bench | dev_DittusBoelter | 0.0220718 |
| diag | bench | dev_Gnielinski | 0.00118956 |
| diag | bench | dev_Kern | 0 |
| diag | bench | dev_NusseltFilm | 0.0245797 |
| diag | bench | dev_Rohsenow | 0.000270621 |
| diag | bench | dev_ZuberCHF | 0.000684704 |
| diag | bench | h_choupo_NusseltFilm | 6680.26 |

*... and 5 more reference values in the case's `expected`.*

## psychro01_n2_water

tutorials/props/heat/psychro01_n2_water




**What it does:** Psychrometric chart for N2/water at 1 atm, 0-60 C: saturation + four RH lines + three adiabatic-saturation lines, from the vapour-pressure model a flash would use

**The story:** THE PSYCHROMETRIC CHART, computed rather than looked up.

Nitrogen + water at 1 atm, 0 → 60 C: the saturation curve, four relative-humidity lines and three adiabatic-saturation (wet-bulb) lines, all from the SAME vapour-pressure model a flash would use. Nothing here is a chart digitised from a handbook – the curve is Y_sat(T) evaluated point by point, so the student can change the model and watch the chart move.

Why this case exists at all: 'psychrometricChart' is reachable from the GUI's Property Explorer, and until now NOTHING exercised it. It could break and the whole suite would stay green; the first to find out would be a student opening the Explorer. An op a user can reach and a test cannot is a promise with no guard behind it.

What to read in the output (long-format CSV: T_C, Y, curve): 'saturation' – Y_sat = 0.622 ∗ Psat/(P - Psat), the chart's ceiling; 'RH=...' – the same expression at a fraction of Psat, so the lines fan out and steepen with T exactly as the printed chart does; 'wetBulb=...' – the adiabatic-saturation lines, which are NOT parallel to the RH lines: they carry the latent heat of the water evaporated, so their slope is set by Hvap/cp of the moist air, not by geometry.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | airWater | Y_adiabatic_last | 0.0150791 |
| diag | airWater | Y_sat_first | 0.00390022 |
| diag | airWater | Y_sat_last | 0.158606 |
| diag | airWater | nRhCurves | 4 |
| diag | airWater | nSaturationPoints | 61 |
| diag | airWater | Lewis_last | 0.816578 |
| diag | airWater | nAdiabaticCurves | 3 |
| diag | airWater | nTrueWetBulbCurves | 3 |

### psychro02_air_water

tutorials/props/heat/psychro02_air_water




**What it does:** Psychrometric chart with dry AIR as the carrier (the textbook object): saturation, RH, adiabatic-saturation and true wet-bulb families at 101.325 kPa, Lewis ~ 0.851 – the case-local air component deliberately shadows the catalogue mixture alias

**The story:** THE TEXTBOOK CHART: dry AIR as the carrier, not a stand-in. Twin of psychro01_n2_water with the REAL carrier – the lumped pseudo-component air (MW 28.96, ChemSep v8.3), which this case declares CASE-LOCALLY in constant/components/air.dat. That declaration is load-bearing, not a convenience: the catalogue also holds mixtures/air.dat (an expansion table, air → N2 + O2 + Ar), and the settled precedence – the case always overrides – is what makes the token load as the COMPONENT here; the run announces the shadowed expansion. Until 2026-08-31 that tie went to the mixture unconditionally and this exact case refused with "carrier 'air' is not a component".

What the twin buys over psychro01: Y_sat(60 C) = 0.1534 against the ~0.152 a printed air chart reads (N2 gives 0.1586 – the 3.4 % MW error the registry note predicted), and Lewis = 0.851, the handbook air/water value the Le^(2/3) wet-bulb family exists to teach.

**What to expect (golden-locked):**

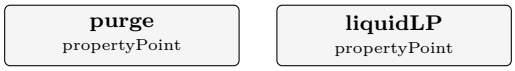
| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | airWater | Lewis_last | 0.850543 |
| diag | airWater | Y_adiabatic_last | 0.0146521 |
| diag | airWater | Y_sat_first | 0.00377268 |
| diag | airWater | Y_sat_last | 0.15342 |
| diag | airWater | nAdiabaticCurves | 3 |
| diag | airWater | nRhCurves | 4 |
| diag | airWater | nSaturationPoints | 61 |
| diag | airWater | nTrueWetBulbCurves | 3 |

## hydraulics

**The theory you need.** Pressure drop and flow distribution: Darcy–Weisbach with friction factors, fittings as equivalent lengths, pump curves against system curves. Expect the operating point at the curves' intersection.

## letdown02_valve_fluid_properties

tutorials/props/hydraulics/letdown02_valve_fluid_properties




**What it does:** The two valve inlets of letdown01: the loop purge's Cp/Cv and Z at 140 bar (is the gas letdown choked?) and liquid ammonia's vapour pressure at 253 K (does the liquid letdown cavitate or flash?)

**The story:** THE TWO FLUID PROPERTIES A LETDOWN VALVE IS JUDGED BY (DEV.md 4c C64, the "Large pressure letdown" EduTool).

The throttles themselves run in tutorials/steady/hydraulics/ letdown01_ammonia_flashing. A valve's hydraulic regime is decided by two more numbers, which a flowsheet unit does not publish, so this props case computes them at the two valve inlets of that case:

the GAS: the loop purge at 140 bar and 253.15 K, let down to a fuel header. Whether its flow is choked is decided by the ratio of specific heats k = Cp/Cv – propertyPoint's 'gamma', Cp_ig/(Cp_ig - R), the ideal-gas ratio the critical-pressure-ratio formula is derived with – and the SRK compressibility Z. the LIQUID: the 20 bar drum liquid of letdown01, at the temperature its first throttle left it (253.0137962 K, copied from that case's converged/LetdownLiquid; gui/tests/letdown.test.ts holds it to the ValveHP T_out golden). Whether its second throttle cavitates or flashes is judged by the ammonia's VAPOUR PRESSURE at that temperature, Psat_NH3 (the record's Antoine curve), against the pressure at the valve's vena contracta – the PURE ammonia's, so for a liquid holding dissolved gas a didactic approximation.

Compositions are the component flows in kmol/h exactly as letdown01's converged streams print them; propertyPoint normalises them. The liquid's dissolved H2, N2 and Ar do not enter Psat_NH3 – that key is the pure ammonia's vapour pressure, the P_v of the valve-sizing equations. The drum liquid is in equilibrium with its gas at 20 bar, so even a small drop below it can begin to release the gas it still holds, and that is letdown01's to show, not this case's.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | purge | Cp_ig | 28.9239504326 |
| diag | purge | H_R | -132.714805313 |
| diag | purge | H_ig | -2297.81017849 |
| diag | purge | H_real | -2430.46102009 |
| diag | purge | P | 14000000 |
| diag | purge | S_R | -0.999957721229 |
| diag | purge | S_ig | 103.584345511 |
| diag | purge | S_real | 102.584535008 |
| diag | purge | T | 253.15 |
| diag | purge | Z | 1.06874405854 |
| diag | purge | gamma | 1.40342888153 |
| diag | purge | v_molar | 0.000160678509513 |
| diag | liquidLP | Cp_ig | 34.2366654977 |
| diag | liquidLP | H_R | -2189.645868 |

*... and 13 more reference values in the case's `expected`.*

## moody01_friction_correlations

tutorials/props/hydraulics/moody01_friction_correlations




**What it does:** The Moody chart's four correlations, each verified against its own published anchor and then all asked the same question – so the disagreement is visible instead of hidden behind a default

**The story:** moody01 – FOUR CORRELATIONS, ONE QUESTION, FOUR ANSWERS.

The Darcy friction factor is the most-used correlated quantity in chemical engineering, and every simulator computes it. What none of them shows is that the answer DEPENDS ON WHICH CORRELATION YOU PICKED.

Choupo has computed Blasius, Colebrook-White, Haaland and Churchill since the pipe unit was written – correctly, with their citations in comments, as free functions nobody could name or reach. This case is what happens when they become objects: each states the window its own authors gave it, each carries its primary citation, each reproduces a published anchor on every run, and all four can be asked the same question at once.

WHAT THE RUN DOES, IN TWO PARTS.

(1) VERIFY. Each correlation reproduces its own anchor, and the anchors are deliberately of DIFFERENT KINDS, because what can be checked differs:

Blasius against its own closed form. This checks the ARITHMETIC and nothing else, and the bench says so rather than letting a 0.000 % deviation look like a validation. Colebrook against the fully-rough von Karman limit, where the implicit equation has a closed form. This checks that the FIXED-POINT ITERATION reaches the answer it must. Haaland against COLEBROOK at the same point – because Haaland's own published claim is "I reproduce that equation within about 2 %". Anchoring it to anything else would be checking a claim nobody made. Churchill against the EXACT laminar law f = 64/Re, which its single expression contains as its own low-Re limit. This is the strongest of the four: a mis-typed exponent fails here while the turbulent branch still looks plausible.

(2) COMPARE. All four are evaluated at Re = 1e5, eps/D = 1e-3 – an ordinary turbulent water line in a commercial steel pipe – and the spread is published as a number.

WHAT TO READ IN THE OUTPUT. Blasius is the one to watch: it carries NO roughness term at all, so at eps/D = 1e-3 it answers as though the wall were smooth, and it is simply wrong. It says so on its own line. That is the whole lesson of a validity window – not that the correlation is bad, but that it was asked a question it was never fitted for, and only the window can tell you.

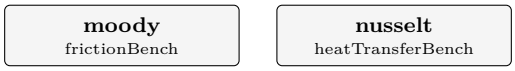
WHAT THIS CASE DOES NOT DO. It does not say which correlation is right. That depends on the pipe – its roughness, whether the flow is developed, whether the duct is even circular – and a bench that ranked them would hide the choice the engineer has to make, which is exactly what a default does today in every tool that has one.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | moody | allPass | 1 |
| diag | moody | dev_Blasius | 0 |
| diag | moody | dev_Churchill | 1.73472e-16 |
| diag | moody | dev_Colebrook | 1.61329e-05 |
| diag | moody | dev_Haaland | 0.0134393 |
| diag | moody | f_Blasius | 0.01777 |
| diag | moody | f_Churchill | 0.0223432 |
| diag | moody | f_Colebrook | 0.0221745 |
| diag | moody | f_Haaland | 0.0219662 |
| diag | moody | f_max | 0.0223432 |
| diag | moody | f_min | 0.01777 |
| diag | moody | n_outside_window | 1 |
| diag | moody | spread_inWindow_pct | 1.71637 |
| diag | moody | spread_pct | 25.7358 |

### moody02_correlation_sweeps

tutorials/props/hydraulics/moody02_correlation_sweeps




**What it does:** The friction-factor and Nusselt correlations swept across five decades of Reynolds number – the engine's own curves, which the EduTool on why correlations are simple draws, differentiates and fits

**The story:** moody02 – THE CORRELATIONS AS CURVES, NOT AS POINTS (DEV.md 4c C119).

moody01 asks four friction correlations ONE question and publishes their disagreement. This case asks them the same question at every Reynolds number from 1e2 to 1e8, and asks the Nusselt correlations the same at every Reynolds number from 2e3 to 2e7, and writes the answers as two CSV files. It is the witness of the EduTool "Why engineering correlations are simple", which draws, differentiates and fits these curves in the browser – so every curve on that page is the engine's own correlation, the one the pipe and the heat exchanger use, never a re-typed copy.

WHAT IS SYNTHETIC, said plainly. The CSVs hold no measurement. They are the correlations evaluated on a grid, and the EduTool treats them as the "truth" a student fits a power law to – which is the exercise, because the point it teaches is how simple a curve looks over a narrow range, not how well any correlation matches a laboratory.

WHAT THE TWO SWEEPS WRITE.

moodySweep.csv one row per (eps/D, Re): every friction correlation's f and its own window flag (1 inside, 0 outside), the regime the engine names (laminar below 2300, transition to 4000, turbulent above), and Churchill's two asymptotes fL = 64/Re and fT – the pieces its single expression blends as (fL^12 + fT^12)^(1/12).

nusseltSweep.csv one row per (Pr, Re): every single-phase Nusselt correlation's Nu and window flag. Each correlation is handed a unit fluid whose Re and Pr ARE the grid point; heating side. Kern is SHELL-side and is evaluated on the same numbers – it is not a tube correlation.

The grid is logarithmic, 20 points per decade, so every decade holds 21 points, its two ends included – the EduTool's one-decade power-law fit uses exactly those. The Nusselt grid starts at 2e3 because Gnielinski's expression carries (Re - 1000) and is zero at Re = 1000.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | moody | allPass | 1 |
| diag | moody | dev_Blasius | 0 |
| diag | moody | dev_Churchill | 1.73472347598e-16 |
| diag | moody | dev_Colebrook | 1.61328517053e-05 |
| diag | moody | dev_Haaland | 0.013439281758 |
| diag | moody | sweep_rows | 726 |
| diag | nusselt | Nu_choupo_DittusBoelter | 200.475286267 |
| diag | nusselt | Nu_choupo_Gnielinski | 223.73353907 |
| diag | nusselt | Nu_choupo_Kern | 97.5646525378 |
| diag | nusselt | Nu_published_DittusBoelter | 205 |
| diag | nusselt | Nu_published_Gnielinski | 224 |
| diag | nusselt | Nu_published_Kern | 97.5646525378 |
| diag | nusselt | allPass | 1 |
| diag | nusselt | dev_DittusBoelter | 0.0220717743085 |

*... and 12 more reference values in the case's `expected`.*

## kinetics

## ammoniaRate01_dyson_simon

tutorials/props/kinetics/ammoniaRate01_dyson_simon




**What it does:** The ammonia synthesis rate of Dyson & Simon (1968), WRITTEN in the case (constant/reactions, 'type expression', every constant with its unit and equation number) and put on the rate-law bench: Eq 19's pseudo-homogeneous rate, Eq 39's effectiveness factor as a declared table over Table I's three pressures, Eq 40, a declared 150-300 atm window and an announced pressure interpolation; the paper's anchors are held by a gate on every case's copy of the law. It locates the LOCUS OF MAXIMUM RATE at three conversions and shows it descending – the curve ammonia03_quench_converter zigzags about – and evaluates the rate at that converter's own bed states.

**The story:** propsDict – the ammonia synthesis rate of Dyson & Simon (1968), on the bench. The law itself is constant/reactions' 'ammoniaSynthesis', every number of it written there ('type expression;'); this file only says where to evaluate it and what to print. Two questions, and they are different questions.

THE LOCUS. Sweeping temperature at a FIXED conversion locates the temperature of greatest net rate. Three conversions locate three points of the locus, and the point is that it DESCENDS. The conversion is of N2, walked from the paper's own reference mixture (3:1 H2/N2, 12.7 % inerts) along the reaction's stoichiometry. The sweep runs 600-900 K because that is roughly the band an iron catalyst lives in; at the lowest conversion the maximum is pushed ONTO the window's upper edge, and the bench says so rather than reporting an edge as a maximum.

THE DECLARED STATES. Three gases TRANSCRIBED from the converged answer of tutorials/plant/ammonia03_quench_converter, at the digits that case's own golden pins. A transcription is a SECOND HOME for somebody else's number and it will go stale if that case moves, so 'check_ammonia_rate' RUNS ammonia03 and holds these three blocks to its converged result.

The catalyst particle is the law's own parameter ('dp 8 mm;' in constant/reactions), inside the 6-10 mm range Eq 39 was fitted for; the law REFUSES one outside it.

*tutorials/plant/ammonia03_quench_converter builds a three-bed quench converter out of equilibrium beds and then says, in its own README, what it cannot do: it cannot size a bed, it cannot tell a good injection point from a bad one, and it rewards over-quenching into a region where a real catalyst is too slow to reach the equilibrium the model hands it. All three gaps are the same gap — there is no rate law.*

*This case is the rate law — and since 2026-10-12 (DEV.md 4c C67) the law is WRITTEN here, in constant/reactions, every constant with its unit and its equation number, as kinetics { type expression; }. The engine compiles it once, checks its dimensions, prints it back as PARSED (the top of the run log), and the rateLawBench op evaluates it. Nothing about ammonia is compiled into the engine; the paper's anchors live in a gate.*

*The source. D. C. Dyson and J. M. Simon, ∗A kinetic expression with diffusion correction for ammonia synthesis on industrial catalyst∗, Ind. Eng. Chem. Fundam. 7 (1968) 605–610. Equations 2, 6–8, 18, 19, 39 with Table I, and 40.*

*Nothing in this case is measured by Choupo: the data these equations were fitted to are Nielsen, Kj?r and Hansen's (J. Catalysis 3 (1964) 68).*

*The same two letters, meaning opposite things. This case keeps the paper's convention because every number in it is read off the paper's own equations, and the engine says so at every surface that prints either letter. Carry η across from the reactor side and you will read a conversion as an effectiveness factor.*

*A correlation in this tree carries three things a reader can check, and this one carries them in the CASE:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | dysonSimon | bed1Inlet_Eq40 | 312.95744498 |
| diag | dysonSimon | bed1Inlet_V3 | 929.247293094 |
| diag | dysonSimon | bed1Inlet_eta | 0.0251956498047 |
| diag | dysonSimon | bed1Inlet_xi | 0.336828294032 |
| diag | dysonSimon | bed1Outlet_Eq40 | 4.6853092592 |
| diag | dysonSimon | bed1Outlet_V3 | 9.34102880697 |
| diag | dysonSimon | bed1Outlet_eta | 0.219727948649 |
| diag | dysonSimon | bed1Outlet_xi | 0.501583857199 |
| diag | dysonSimon | bed3Outlet_Eq40 | 0.769642159744 |
| diag | dysonSimon | bed3Outlet_V3 | 1.0033709893 |
| diag | dysonSimon | bed3Outlet_eta | 0.337604123841 |
| diag | dysonSimon | bed3Outlet_xi | 0.76706008382 |
| diag | dysonSimon | locus1_Eq40 | -889.947776496 |
| diag | dysonSimon | locus1_T_maxRate_K | 900 |
| *… and 14 more reference values in the case's `expected`.* | | | |

### rate01_arrhenius_recovery

tutorials/props/kinetics/rate01_arrhenius_recovery




**What it does:** Second-order decay at three temperatures, generated from k0 = 4e6, Ea = 60 kJ/mol: the Arrhenius line must give both back

**The story:** THE ACTIVATION ENERGY COMES OUT OF A SLOPE.

'kinetics1D' is a PROPERTY evaluation, not a reactor: it uses the integrated rate law in closed form, so a student can overlay competing orders on the same lab data and SEE which one is straight. Given isotherms at more than one temperature it goes further – fits k on each one, then the line

ln k = ln k0 - (Ea/R) (1/T)

and Ea falls out of the SLOPE. That is the whole of Arrhenius, and it is worth watching happen on data whose answer is known.

So the data here is generated from k0 = 4.0e6 L/(mol s), Ea = 60 kJ/mol, second order, at 40/60/80 C. The fit must return those two numbers; the expected file pins them, and R2 = 1 because the points carry no scatter.

What to read: the per-isotherm k values (3.93e-4, 1.57e-3, 5.34e-3), which span a factor of 13.6 over 40 K – the reason a rate constant is never quoted without its temperature.

Why the case exists: 'kinetics1D' had NO case and no gate. Its sibling vaporPressureFit was in the same position, and writing the first case for THAT one segfaulted choupoProps twice.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | rate_arrhenius | Ea | 60000 |
| diag | rate_arrhenius | Ea_kJ_per_mol | 60 |
| diag | rate_arrhenius | R2 | 1 |
| diag | rate_arrhenius | arrhenius | 1 |
| diag | rate_arrhenius | fitted | 1 |
| diag | rate_arrhenius | k0 | 4e+06 |
| diag | rate_arrhenius | n_temps | 3 |
| diag | rate_arrhenius | order | 2 |

## membrane

## dspmde01_water_treatment_salts

tutorials/props/membrane/dspmde01_water_treatment_salts




**What it does:** A water-treatment feed through the DSPM-DE pore: nitrate, fluoride, bromide, carbonate and ammonium salts from the standard catalogue, on NanoFiltran's worked-example pore and the catalogue's NF270 archetype – every ion's radius and D0 cited on its species record

**The story:** THE RESEARCHER'S PATH, PROVED: a water that carries nitrate, fluoride, bromide, carbonate and ammonium, fed as SALTS from the standard catalogue, through the DSPM-DE pore (DEV.md 4c C157/C161/C170).

WHAT KIND OF CHECK THIS IS. A STRUCTURAL witness, not a validation. It shows that every one of these salts decomposes into its ions and that every ion finds a CITED hydrated radius (Nightingale 1959, Table I) and a CITED D0 (USGS PHREEQC phreeqc.dat, '-dw') on its species record, so the pore can be solved. No number here is a measurement. The two pores are NanoFiltran's worked example (constant/assets/NanoFiltranExample.dat, which describes no real membrane) and the catalogue's NF270_dspmde (an archetype whose pore is set, not fitted to this water): a researcher replaces them with the record his own fit writes (tutorials/steady/membranes/membrane24_fit_then_simulate is the template).

WHAT IS NOT HERE, AND WHY. Bicarbonate (HCO3-), phosphate and the hydrogen phosphates have a PHREEQC D0 but NO hydrated radius: Nightingale's Table I does not list them, and a Stokes radius from D0 is a different convention that would shift every steric hindrance in silence – so DSPM-DE REFUSES them by name (DEV.md 4c C161). NaHCO3 in this feed would stop the run with that refusal. The carbonate below is the CO3– of Na2CO3 taken fully dissociated: nothing here speciates it, so it stands for a high-pH water; at a natural pH most of it would be bicarbonate.

THE THREE OPERATIONS, each a 'poreTransport' bench at an IMPOSED J_v, ideal activities, NanoFiltran's "no polarisation" film (k = 10 m/s), the solved pore at 801 nodes: monovalentWater on NANOFILTRAN'S WORKED-EXAMPLE pore (X -30 mol/m3, r_p 0.40 nm, 2.5 um, eps_p 60): NaCl 5, NaNO3 1, NaF 0.5, KBr 0.5, NH4Cl 0.5 mol/m3 (Na K NH4 Cl NO3 F Br) mixedWater the same water plus Na2CO3 0.5 mol/m3 (+ CO3–), on the catalogue's NF270_dspmde pore (r_p 0.60 nm, X -5 mol/m3) hardWater Ca(NO3)2 2, KNO3 2, MgCl2 1, NaBr 1, MgSO4 1 mol/m3 (Ca Mg K Na NO3 Cl Br SO4), on NF270_dspmde WHY TWO PORES. NanoFiltran's example pore was set against NanoFiltran's own ion database, whose radii are STOKES radii (Na+ 1.84, Cl- 1.21, SO4– 2.31 A; nanofiltran_geraldesAlves2008/constant/species/). Against the catalogue's HYDRATED radii a 0.40 nm pore nearly closes to the divalent ions: lambda = r/r_p is 0.985 for CO3–, 0.95 for SO4–, 1.03 for Ca++ and 1.07 for Mg++. Measured 2026-10-26: Na2CO3 (alone, or in this water) and MgSO4 + NaCl do NOT converge there (the solved pore refuses, naming the stalled continuation), and MgCl2 or Ca(NO3)2 alone are refused because no cation enters the pore. So the divalent waters run on NF270_dspmde, whose 0.60 nm pore is the one its record states is consistent with the hydrated-radius tier. A pore and its ion radii are ONE convention: a pore fitted on one cannot be read with the other.

*A water-treatment feed carries more than sodium chloride: nitrate, fluoride, bromide, carbonate, ammonium, calcium, magnesium and sulfate. A case feeds them as salts — each a component record with an ion map (dissociatesTo) — and the DSPM-DE pore works ion by ion: every ion needs its hydrated radius (steric partition and hindrance) and its free-solution diffusivity D0 (diffusion and convection in the pore).*

*This case proves that path end to end from the standard catalogue. Its salts are the standard records NaCl, NaNO3, NaF, KBr, Na2CO3, NH4Cl, calciumNitrate (Ca(NO3)2), KNO3, MgCl2, NaBr and MgSO4 (sealed into constant/components/ by bin/choupo-import, so the case is self-contained), and every ion resolves:*

*∗ its hydrated radius from Nightingale, J. Phys. Chem. 63 (1959) 1381, Table I (column r_H, 25 °C), read back against the primary page; ∗ its D0 from the public-domain USGS PHREEQC database phreeqc.dat, the 25 °C Dw of its -dw line.*

*Each species record (constant/species/<ion>.dat) cites its page, table, database line and file hash in its own transport { provenance {} } block.*

*Three operations, each a poreTransport bench at an IMPOSED permeate flux (J_v = 1e-6, 5e-6, 2e-5 m/s), ideal activities, no polarisation:*

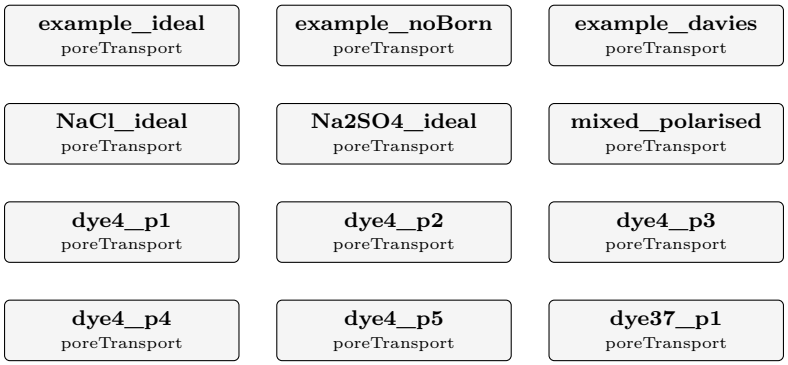
*Each publishes, per J_v and per ion, the bulk, wall and permeate concentrations and the observed and real rejections. At 2e-5 m/s, for example, mixedWater rejects CO3– at 0.99985 and NO3- at 0.741, and hardWater rejects SO4– at 0.99989 against Cl- at 0.778 and Br- at 0.767 – the divalent co-ion held back by the negative pore, as NF does.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | monovalentWater | Jv_p1 | 1e-06 |
| diag | monovalentWater | Jv_p2 | 5e-06 |
| diag | monovalentWater | Jv_p3 | 2e-05 |
| diag | monovalentWater | Kc_Br | 1.19061896103 |
| diag | monovalentWater | Kc_Cl | 1.18629086518 |
| diag | monovalentWater | Kc_F | 1.13952821128 |
| diag | monovalentWater | Kc_K | 1.18846271114 |
| diag | monovalentWater | Kc_NH4 | 1.18846271114 |
| diag | monovalentWater | Kc_NO3 | 1.17968143659 |
| diag | monovalentWater | Kc_Na | 1.12423947195 |
| diag | monovalentWater | Kd_Br | 0.0143696929036 |
| diag | monovalentWater | Kd_Cl | 0.0133665877414 |
| diag | monovalentWater | Kd_F | 0.0054600222726 |
| diag | monovalentWater | Kd_K | 0.0138630195272 |
| *... and 1202 more reference values in the case's `expected`.* | | | |

## nanofiltran_geraldesAlves2008

tutorials/props/membrane/nanofiltran_geraldesAlves2008




**What it does:** The NanoFiltran paper (Geraldes & Alves 2008) in one case: Choupo against NanoFiltran code to code, against Bowen & Mohammad's MEASURED CA30 rejections (Fig. 3) and on the stiff case of Table 2 and Fig. 4 – its tables and figures generated by the views of its propsDict

**The story:** V. Geraldes, A.M. Brites Alves, "Computer program for simulation of mass transport in nanofiltration membranes", J. Membr. Sci. 321 (2008) 172-182 (doi:10.1016/j.memsci.2008.04.054) — NanoFiltran, and the two comparisons its paper makes, run by Choupo's DSPM-DE pore in one case.

**Where the theory is**

The model and the method are the Theory Guide's DSPM-DE chapter (section 19): the whole boundary-value problem one pore solves — the extended Nernst-Planck equations, electroneutrality at every node, zero current, the steric, Born and Donnan partitions at both faces, the multi-ionic wall — in 19.3; the Peclet numbers that decide how hard it is in 19.4; the numerical method (Scharfetter-Gummel fluxes, the block-tridiagonal Newton, the local Donnan seed, the continuation in J_v) in 19.5; and the node counts below, with what they mean, in 19.8. In the GUI, F1 on any operation of this case opens that chapter at the model, with the numerical method offered beside it.

**What this case imposes, and what it computes**

It IMPOSES the permeate flux J_v; it does not compute the flux from the pressure. Every operation is a poreTransport bench: it is handed J_v (a list; in study B, Bowen & Mohammad's fitted function evaluated at each operation's feed and written in the operation) and solves the extended Nernst-Planck equations across the pore, with electroneutrality at every node, steric, Born and Donnan partitions at both faces and the multi-ionic film at the wall. So the membrane record's A_w (the pure-water permeance, J_w = A_w (dP - dpi)) is carried because the record grammar requires it and is not read here.

What it publishes, per J_v point and per ion: the bulk, wall and permeate concentrations, the observed rejection R_obs = 1 - c_p/c_b and the real rejection R_real = 1 - c_p/c_wall, the potential across the membrane and the pore Newton's iteration count.

**The three studies: what each teaches, and which one is stiff**

Whether a pore is stiff is read off one number per ion, its pore Peclet number Pe_i = K_c,i J_v / (K_d,i D_i A_k/dx) — convection against hindered diffusion across the layer — computed here from the case's own ion records and membranes.

- A — Code to code: Choupo against NanoFiltran on one model. The same DSPM-DE model and the same inputs, solved by two independent programs (NanoFiltran: Fortran 77, central differences, under-relaxed Newton; Choupo: C++, exponentially fitted fluxes, full Newton). It teaches that the equations are solved right — the two programs' grid limits agree within 8e-7 on 35 permeates — and it is NOT a validation: no number here is a measurement, and the membrane is NanoFiltran's worked example, which describes no real membrane. The pore runs from nearly linear (J_v 1e-6 m/s, Pe of Mg2+ 0.7) to stiff (5e-5 m/s, Pe of Mg2+ 33).
- B — Experiment: Bowen & Mohammad's CA30 rejections (the paper's Fig. 3). It teaches what agreement with measurements means when the inputs were FITTED to those very data by Bowen &

Mohammad (AIChE J. 44 (1998) 1799): the pore radius and thickness, the membrane charge as a function of the feed (Eq. 43) and the permeate flux as a function of the feed (Eqs. 44-45). Agreement here says the pore model, carried by inputs fitted on these data, reproduces them — not that it predicts a membrane it was not fitted on. This study is NOT stiff: Pe is 0.06-0.14 for Na+ and 0.03-0.07 for Cl-, and the pore profiles are nearly straight. Its physics is at the WALL: the trivalent dye is as large as the pore (lambda = 1), so the pore excludes it; it accumulates in the film and sets the wall's electroneutrality, and that is what drives Cl- to R_obs = -2.

- C — The stiff case of Table 2 and Fig. 4, and its grid study. A numerical study, nothing measured: the paper's "very stiff" pore solved on six node counts, and Choupo's answer against Table 2 accounted for one change at a time. It IS stiff: Pe is 33 for Mg2+ against 0.56 for Na+ and 0.20 for Cl-, because the two counter-ions' hindered diffusivities K_d D differ 70 times (Mg2+ at lambda = 0.875 barely diffuses). Mg2+ is carried, the trace Na+ diffuses, and the pore holds an interior front where Na+ hands the fixed charge over to Mg2+ (Na+ falls from 12 to 0.16 mol/m3 between a third of the way in and the permeate face). The run prints the stiffness — the electromigration flux against the net flux — as 1.65e3 for every node count.

**How many nodes: it depends on what you report**

Measured on study C (re-run 2026-10-25 on a copy of this case whose six N operations were joined by 801 and 1601 nodes, every operation writing its pore profile). Errors against the case's finest grid, 350 nodes, which is itself within 1e-4 relative of the Richardson limit of 801 and 1601 nodes on every row:

| quantity (value at 350 nodes) | 10 nodes | 20 | 50 | 100 | 200 |
|---|---|---|---|---|---|
| R_obs of Na+ (99.4457 %), in points | -0.073 | -0.017 | -0.0025 | -0.0006 | -0.0001 |
| permeate Na+ (0.05543 mol/m3) | +13.2 % | +3.03 % | +0.45 % | +0.10 % | +0.019 % |
| permeate Mg2+ (8.6317 mol/m3) | -0.64 % | -0.15 % | -0.023 % | -0.005 % | -0.001 % |
| permeate Cl- (17.3188 mol/m3) | -0.59 % | -0.14 % | -0.021 % | -0.005 % | -0.001 % |
| Mg2+ at the pore's permeate end (15.0235 mol/m3) | -0.077 % | -0.018 % | -0.0026 % | -0.0006 % | -0.0001 % |
| psi at the pore's permeate end (0.07217 V) | -4.52 % | -1.08 % | -0.16 % | -0.038 % | -0.007 % |
| potential across the membrane, dPsi_m (0.14496 V) | -2.21 % | -0.53 % | -0.080 % | -0.018 % | -0.003 % |
| Newton iterations (12 at 350 nodes) | 7 | 9 | 9 | 10 | 11 |

The lesson: the rejection forgives a coarse grid; the permeate of a trace ion — and so any selectivity built on it — does not. Ten nodes put the Na+ rejection within 0.1 point and its permeate 13 % high. The error falls as the square of the cell size (double the nodes, a quarter of the error), so declare nodes at n and 2n and compare the quantity you report.

**Read the comparisons for what they are**

- Nothing was fitted by this run. In study B the parameters were fitted by Bowen & Mohammad, before this case existed; Choupo fits nothing.
- The error bars of Fig. 3 are READING uncertainty. The measured symbols were digitised from the paper's figure (calibration checked against the tick labels; reading error +-1.5 px = 0.32 mol/m3 in c_Cl and 0.011 in R_obs). They are not the original experimental uncertainty.
- The feed chloride of each study-B operation is the mean of its two markers' readings — the Na+ and Cl- symbols of one measurement sit at one feed, and were read at slightly different abscissae.
- The two programs count N differently. Choupo's N nodes have N - 1 intervals; NanoFiltran's N counts the permeate node, so it has N - 2 intervals inside the membrane.
- Table 2 carries a factor of ln 10 in its Davies activities, and Choupo deliberately reproduces no such option, so Choupo is NOT pinned to Table 2: it is pinned to NanoFiltran with that constant (and its Born constant and diffusivity input) corrected. The grid-study figure's "Davies corrected" NanoFiltran line has ONLY the Davies constant corrected, so the gap left between it and Choupo is not discretisation alone.
- 12 against 88 iterations is not a speed measurement. Choupo runs a full Newton with an analytic Jacobian to its own tolerance; NanoFiltran an under-relaxed (0.2) Newton to TOL 1e-4. The criteria, the relaxation and the work per iteration differ.

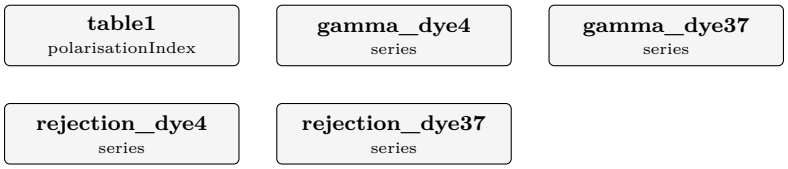
The full account — every difference between the two programs and how each was handled, the Davies finding, and the term-by-term accounting of Table 2 — is the leading comment of system/propsDict.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | example_ideal | Robs_Cl_p1 | 0.202989025795 |
| diag | example_ideal | Robs_Cl_p2 | 0.42410107917 |
| diag | example_ideal | Robs_Cl_p3 | 0.51695014572 |
| diag | example_ideal | Robs_Cl_p4 | 0.603832218837 |
| diag | example_ideal | Robs_Cl_p5 | 0.656025668338 |
| diag | example_ideal | Robs_Mg_p1 | 0.277594917868 |
| diag | example_ideal | Robs_Mg_p2 | 0.518891301308 |
| diag | example_ideal | Robs_Mg_p3 | 0.561965456186 |
| diag | example_ideal | Robs_Mg_p4 | 0.57192034085 |
| diag | example_ideal | Robs_Mg_p5 | 0.571194624339 |
| diag | example_ideal | Robs_Na_p1 | -0.095434542496 |
| diag | example_ideal | Robs_Na_p2 | 0.0449401906182 |
| diag | example_ideal | Robs_Na_p3 | 0.336888903859 |
| diag | example_ideal | Robs_Na_p4 | 0.731479730785 |
| *. . . and 1909 more reference values in the case's* `expected`*.* | | | |

## polarisation01_geraldes_afonso_table1

tutorials/props/membrane/polarisation01_geraldes_afonso_table1




**What it does:** Concentration polarisation of a Na+/Cl-/dye3- solution (Geraldes & Afonso 2007, Table 1): per-ion k, one interface field, Gamma per ion – no film thickness

**The story:** polarisation01 – THE CONCENTRATION POLARISATION OF A MULTI-IONIC SOLUTION, computed the way Geraldes & Afonso did it (J. Membr. Sci. 300 (2007) 20-27, doi:10.1016/j.memsci.2007.04.025) for the case study of their Section 2.2: nanofiltration of Na+ / Cl- / dye3- solutions in a stirred cell (experimental data of Bowen & Mohammad, AIChE J. 44 (1998) 1799).

THE QUESTION. How much does each ion pile up at the membrane wall when three ions of different diffusivities and charges are rejected to different degrees? The classical film model cannot answer it: it needs ONE effective diffusivity for the solution, and a mixture of ions has none. The extended Nernst-Planck approach answers it only after choosing a film thickness, which nobody measures.

THE MODEL. At the feed-solution/membrane interface, for every ion, the flux J_v C_p equals back-diffusion plus convection plus MIGRATION in the electrical potential gradient xi that the ions' different mobilities set up (Eqs. 2-7); electroneutrality at the interface (Eq. 8) fixes xi. The mass-transfer coefficient of each ion comes from the cell's conventional correlation on the ion's OWN diffusivity, Sh_i = 0.23 Re^0.567 Sc_i^0.33 (Bowen et al. 1997), corrected for the suction effect of the permeation by Xi_i = phi_i + (1 + 0.26 phi_i^1.4)^-1.7 (Geraldes & Afonso 2006). Four algebraic equations per operating point (Eqs. 25-28), NO film thickness, NO new parameter.

WHAT TO READ IN THE OUTPUT. The polarisation index Gamma_i = (C_m - C_b)/C_b per ion and per row – the open symbols of the paper's Fig. 7. The dye (slowest, most rejected) polarises most; Na+ is dragged along by the field; Cl-, less rejected than the dye and pulled by the same field, shows a NEGATIVE Gamma at the low Cl- feeds – its wall concentration is BELOW the bulk, which the film model applied ion by ion cannot produce. The residual sum z_i C_m,i / sum |z_i| C_m,i printed under each row is the electroneutrality the field enforces.

WHAT IS AND IS NOT MEASURED. C_b, C_p and J_v are the table's; Na+ is derived from electroneutrality as the table's own footnote says. The Gamma are PREDICTIONS of the model from those inputs and the ions' D – nothing is fitted. The paper compares them with an extended Nernst-Planck integration in the dye's film (its Fig. 7, filled symbols), which this case does not reproduce.

The gate check_polarisation_coupling recomputes the same four equations in Python from this case's own data file and holds every Gamma to 1e-8.

*The question. Three ions – Na+, Cl- and a trivalent dye – are nanofiltered in a stirred cell. The dye is rejected almost completely, the chloride barely. How much does each pile up at the membrane wall?*

*The lesson. Geraldes & Afonso, J. Membr. Sci. 300 (2007) 20-27 (doi:10.1016/j.memsci.2007.04.025), answer it with four algebraic equations per operating point and NO film thickness: the cell's conventional correlation Sh_i = 0.23 Re^0.567 Sc_i^0.33 on each ion's OWN diffusivity, the suction correction Xi_i = phi_i + (1 + 0.26 phi_i^1.4)^-1.7, and one electrical potential gradient at the interface fixed by electroneutrality at the wall. The polarisationIndex bench prints, for each of the ten rows of the paper's Table 1 (data of Bowen & Mohammad, AIChE J. 44 (1998) 1799), the per-ion k, phi, Xi, wall concentration, polarisation index Gamma_i = (C_m - C_b)/C_b and the field xi – the open symbols of the paper's Fig. 7. The dye (slowest, most rejected) polarises most; Na+ is dragged after it by the field; Cl- sits BELOW its bulk concentration (negative Gamma), which the film model applied ion by ion cannot produce.*

*Headline numbers (Gamma): row 1 (dye 4, Cl 5.2 mol/m3) dye +0.504, Na +0.310, Cl -0.137; row 10 (dye 37, Cl 87.3) dye +0.273, Na +0.120, Cl -0.074. Re = 26906 against the paper's 26907.*

*The figures. The run draws them (the GUI's review screen opens on them). The op publishes its DEFAULT views with nothing declared – Gamma of every ion row by row, the rejection the cell MEASURES (1 -*

*C_p/C_b, from Table 1) beside the one the membrane makes (1 - C_p/C_m, Choupo's wall), and Table 1 row by row with Gamma_i and the field – and the propsDict's views split the same rows by dye level and draw them against the feed chloride at 4 and at 37 mol/m3 of dye (the paper's Fig. 7, open symbols). Every number is the bench's published rows table; the paper's own Gamma values are not in the tree, so none is drawn as if it were.*

*What is measured and what is not. C_b, C_p and J_v are the table's; Na+ is derived by electroneutrality as the table's footnote says (closeBy Na;). Every Gamma is a PREDICTION from those inputs, the ions' D and the correlation; nothing is fitted. The paper's filled symbols (an extended Nernst-Planck integration in the dye's film) are not reproduced. The dye is a case-local species (constant/species/dye.dat, z = -3, D = 0.36e-9 from the paper's footnote); the sources do not identify the compound, so its MW is a flagged nominal placeholder that nothing here reads.*

*The gate. check_polarisation_coupling re-solves Eqs. 25-28 in Python from this case's own data file and holds every Gamma AND xi to 1e-8, holds a single salt to the paper's closed form Eq. 20, and holds the filmTheory + none policy to the classical film model to round-off.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | table1 | D_Cl | 2.03e-09 |
| diag | table1 | D_Na | 1.33e-09 |
| diag | table1 | D_dye | 3.6e-10 |
| diag | table1 | Gamma_Cl_r1 | -0.136799297507 |
| diag | table1 | Gamma_Cl_r10 | -0.0737363093742 |
| diag | table1 | Gamma_Cl_r2 | -0.095324747833 |
| diag | table1 | Gamma_Cl_r3 | -0.0498126163785 |
| diag | table1 | Gamma_Cl_r4 | -0.0178510515183 |
| diag | table1 | Gamma_Cl_r5 | -0.00198770276507 |
| diag | table1 | Gamma_Cl_r6 | -0.24271484424 |
| diag | table1 | Gamma_Cl_r7 | -0.176886232679 |
| diag | table1 | Gamma_Cl_r8 | -0.13368256672 |
| diag | table1 | Gamma_Cl_r9 | -0.10815205541 |
| diag | table1 | Gamma_Na_r1 | 0.310336981862 |
| *... and 55 more reference values in the case's `expected`.* | | | |

## shortlist01_class_to_modules

tutorials/props/membrane/shortlist01_class_to_modules




**What it does:** From a membrane class to the commercial elements that do it: hard filters, each survivor re-rated at the student's conditions, every criterion side by side, no score

**The story:** shortlist01 – THE SHORTLIST (DEV.md 4c C189 slice 5): from a membrane class to the commercial elements that do what the class does.

THE QUESTION. membrane26 chose a membrane by WHAT IT DOES (NF_DIVALENT_SELECTIVE: hold sulphate, let sodium chloride through). Which products in the catalogue do that, at MY pressure, MY temperature, MY feed and MY recovery – not at the data sheet's?

WHAT THE OP DOES. (1) Hard filters on each module record: the membrane is of the class's type (NF here), the record carries ONE rated test of MgSO4, the sheet's rated rejection lies inside the class's band (>= 90 %), the declared limits hold at this feed. (2) Each survivor is re-run with the transport law and film of the 'operation' block below, its intrinsic rejection DERIVED from its OWN rated test (slice 1), at the conditions below, with the feed flow found so ONE element delivers the recovery asked. Then the limits that need the run (feed flow, pressure drop) filter again.

WHAT IT DOES NOT DO. It scores nothing and ranks nothing. Every criterion is published side by side, the candidates in alphabetical order of their names, and a record's commercial-relation fields ('partner', 'dataVerifiedBy') are never read. What matters most – permeate flow, permeate quality, specific pressure – is the student's decision.

WHAT TO READ. The printed rows (and shortlist.csv): permeate flow and concentration, flux, P/flux (bar per L/m2/h), the hydraulic energy per m3 of permeate (feed pressure x feed flow / permeate flow, no recovery device), the feed flow, the element's pressure drop and each limit's margin ((limit - value)/limit; pH in pH units; "n/s" = the record states no such limit). EXCLUDED rows say why, in the record's own numbers.

*The question. membrane26 chose a membrane by what it DOES (NF_DIVALENT_SELECTIVE: hold sulphate, let sodium chloride through). Which products in the catalogue do that – at MY pressure, MY temperature, MY feed and MY recovery, not at the data sheet's?*

*The op. membraneShortlist (DEV.md 4c C189 slice 5). It does two things and nothing else.*

*1. ∗Hard filters∗, on each module record: the membrane is of the class's type (NF here), the record carries ONE rated test of the asked solute, that test's rated rejection lies inside the class's band, the declared limits (P_max, T_max, pH) hold at the student's feed. A limit the record does not state is not a filter (n/s). 2. ∗Re-rating∗: every survivor is re-run with the film and spacer of the operation block, its intrinsic rejection DERIVED from its OWN rated test (slice 1), at the student's conditions, with the feed flow found so ONE element delivers the recovery asked. The limits that need the run (feed flow, pressure drop) then filter again.*

*What it will not do. Score, weight or rank. Every criterion is printed side by side (permeate flow, observed rejection, permeate concentration, flux, P/flux, hydraulic energy per m3, feed flow, pressure drop, margins to each limit), the candidates in alphabetical order of their names, and what matters most is the student's decision. A record's commercial-relation fields (partner, dataVerifiedBy) are never read: the gate proves it.*

*What to expect here. tiers ( standards case ) keeps the golden independent of one machine's private data/local/assets/ records (the importer's home); drop the line and an imported catalogue joins the table, each of its records announced [local] UNVERIFIED. Of the shipped records NF270-4040 passes; SW30HR-380 is excluded (an RO membrane, rated on NaCl); the Generic elements are excluded (they carry no rated test).*

*Read it with care. The re-rating is the transport law of a lumped salt with its intrinsic rejection fixed at the sheet's ONE test – a model's reading of a data sheet away from its test, not a guarantee of the product.*

**What to expect (golden-locked):**

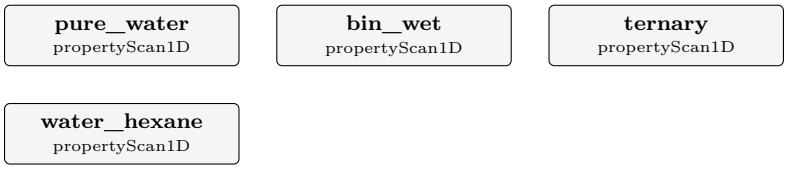
| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | shortlist | NF270_4040__A_w_LMH_per_bar | 14.8806573771 |
| diag | shortlist | NF270_4040__R_int | 0.979346761192 |
| diag | shortlist | NF270_4040__R_obs | 0.969702755801 |
| diag | shortlist | NF270_4040__dP_bar | 0.430832573708 |
| diag | shortlist | NF270_4040__feedFlow_m3h | 3.46318793226 |
| diag | shortlist | NF270_4040__flux_LMH | 68.3523937423 |
| diag | shortlist | NF270_4040__margin_P | 0.829268292683 |
| diag | shortlist | NF270_4040__margin_T | 0.0628634292001 |
| diag | shortlist | NF270_4040__margin_dP | 0.569167426292 |
| diag | shortlist | NF270_4040__margin_feedFlow | 0.0380033521509 |
| diag | shortlist | NF270_4040__margin_pH | 2.5 |
| diag | shortlist | NF270_4040__permeateConc_ppm | 60.5944883985 |
| diag | shortlist | NF270_4040__permeateFlow_m3d | 12.4674766186 |
| diag | shortlist | NF270_4040__recovery | 0.150000000752 |
| *... and 10 more reference values in the case's `expected`.* | | | |

## molecular

**The theory you need.** Molecular activity models on display: NRTL/Wilson/UNIQUAC correlate binary $\gamma$'s from pair parameters; UNIFAC PREDICTS them from group counts. Expect UNIFAC within a few percent of fitted models for families it was trained on – and to be honestly worse outside them. *Full derivation: Theory Guide, the activity-model chapter.*

## cosmoSAC01_water_ethanol

tutorials/props/molecular/cosmoSAC01_water_ethanol



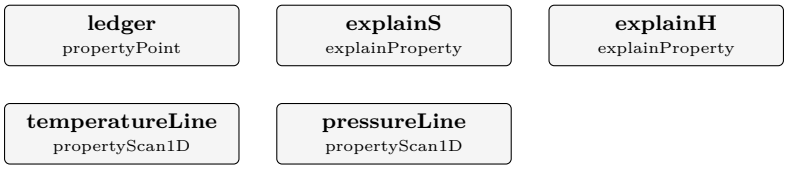

**What it does:** COSMO-SAC 2002 (NIST benchmark, VT-2005 profiles): pure→lngamma=0, binary/ternary multicomponent gammas, no NaN.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | pure_water | n_failures | 0 |
| diag | pure_water | n_points | 2 |
| diag | bin_wet | n_failures | 0 |
| diag | bin_wet | n_points | 2 |
| diag | ternary | n_failures | 0 |
| diag | ternary | n_points | 2 |
| diag | water_hexane | n_failures | 0 |
| diag | water_hexane | n_points | 2 |
| diag | pure_water | n_emptyCurves | 0 |
| diag | bin_wet | n_emptyCurves | 0 |
| diag | ternary | n_emptyCurves | 0 |
| diag | water_hexane | n_emptyCurves | 0 |

## entropy01_air_ledger

tutorials/props/molecular/entropy01_air_ledger




**What it does:** The entropy ledger of 79/21 N2/O2 at 400 K, 2 bar on SRK: every line of s(T,P,y) published separately, so a reader can rebuild the total by hand

**The story:** THE ENTROPY LEDGER, published line by line – the witness behind the "What is entropy?" EduTool page.

The engine's mixture entropy is assembled in ThermoPackage::S_ig (src/thermo/ThermoPackage.cpp) as

S_ig(T,P,y) = sum_i y_i ∗ s_ig_i(T) (pure-state line) - R ∗ sum_i y_i ∗ ln y_i (mixing line) - R ∗ ln(P / 1 bar) (pressure line)

with each s_ig_i(T) = s_298_i + integral Cp_i/T dT off the component's own .dat (Component::s_formation), and S_real = S_ig + S_residual(SRK) (the model line, Sandler eq. 6.4-31). The 'ledger' op below publishes the per-component s_ig_i AND the assembled S_ig / S_R / S_real at one state, so a reader – or the EduTool page in the browser – can rebuild the total from the lines and check it against the engine's own number. The two scans draw the temperature line (integral Cp/T at fixed P) and the pressure line (-R ln P at fixed T, with the residual gap growing).

Nothing here is a validation against measurement: every number is the engine's own answer, decomposed. The value of the case is that the decomposition CANNOT drift from the simulator, because it is published by the same call that computes it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | ledger | Cp_ig | 29.46367008 |
| diag | ledger | H_R | -5.48583758685 |
| diag | ledger | H_ig | 2986.16894826 |
| diag | ledger | H_real | 2980.68311067 |
| diag | ledger | P | 200000 |
| diag | ledger | S_R | -0.0191625058917 |
| diag | ledger | S_ig | 201.577774992 |
| diag | ledger | S_real | 201.558612487 |
| diag | ledger | T | 400 |
| diag | ledger | Z | 1.00065691172 |
| diag | ledger | gamma | 1.3931335315 |
| diag | ledger | s_ig_N2 | 200.188720505 |
| diag | ledger | s_ig_O2 | 213.897898046 |
| diag | ledger | v_molar | 0.0166398489719 |

*. . . and 36 more reference values in the case's `expected`.*

## envelope01_natural_gas

tutorials/props/molecular/envelope01_natural_gas




**What it does:** P-T phase envelope of a lean natural gas: the dew branch over its cricondentherm, and a bubble branch that never starts

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | envelope | bubble_points | 0 |
| diag | envelope | bubble_stopped_bar | 1 |
| diag | envelope | dew_cricondentherm_K | 274.705 |
| diag | envelope | dew_points | 40 |

## envelope02_root_jump

tutorials/props/molecular/envelope02_root_jump



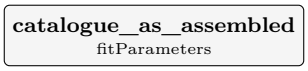

**What it does:** P-T envelope where the saturation solve jumps root: both branches stop AT the jump and report it as one, and no cricondentherm is claimed

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | envelope | bubble_points | 17 |
| diag | envelope | bubble_stopped_bar | 18 |
| diag | envelope | dew_points | 25 |
| diag | envelope | dew_stopped_bar | 26 |

## eval_nrtl_catalogue

tutorials/props/molecular/eval_nrtl_catalogue




**What it does:** EVALUATE mode: score the CATALOGUE NRTL pair against your own data – the scale before any fit

**The story:** ‘mode evaluate‘: the SCALE, not the fit.

ONE operation, one question: how good is the ethanol-water NRTL pair against this 1-atm bubble-T dataset? Nothing is adjusted, nothing is written. The thermoPhysPropDict declares NO inline pairs, so what gets scored is a REAL provenance-bearing pair resolved through the search path (from this case's own vendored constant/parameters/ copy, identical parameters and full primary citation) – not numbers typed inline into the dict.

Read the verdict (rms/aad in kelvin) and THEN decide whether a refit is worth it: compare with fitNRTL01's fitted rms (0.31 K) on this same data. Failures are honest: a non-converged point becomes status=nonconverged in the parity CSV and never enters the statistics; no chi2_reduced is printed – without sigmas and degrees of freedom from THIS act, the name would lie.

The pinned-what-if grammar and the leakage sentinel live in the sibling case eval_nrtl_pinned (a pinned path must EXIST in the dict, so what-ifs run on an INLINE scaffold and say so).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | catalogue_as_assembled | aad | 2.0400 |
| diag | catalogue_as_assembled | chi2 | 60.1945 |
| diag | catalogue_as_assembled | max_abs_resid | 3.4379 |
| diag | catalogue_as_assembled | max_resid_x1 | 0.5000 |
| diag | catalogue_as_assembled | mode_evaluate | 1.0000 |
| diag | catalogue_as_assembled | n_data | 11.0000 |
| diag | catalogue_as_assembled | n_feasible | 11.0000 |
| diag | catalogue_as_assembled | partial | 0.0000 |
| diag | catalogue_as_assembled | penalised_points | 0.0000 |
| diag | catalogue_as_assembled | rms | 2.3393 |
| diag | catalogue_as_assembled | sse | 60.1945 |
| diag | catalogue_as_assembled | validity_declared | 1.0000 |
| diag | catalogue_as_assembled | n_outside_validity | 0.0000 |

## eval_nrtl_pinned

tutorials/props/molecular/eval_nrtl_pinned




**What it does:** EVALUATE a pinned what-if: a pinned run scores the package but mutates NOTHING (the leakage sentinel)

**The story:** the pinned WHAT-IF, and the LEAKAGE SENTINEL.

TRANSITIONAL SCAFFOLD: pinning by dict path requires an inline pair; the permanent surface (ParameterOverrides on the RESOLVED parameter registry, design forum #67/#69) replaces this case when it lands.

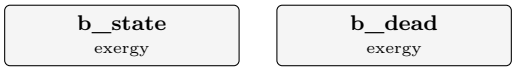
Three evaluations of the same dataset: 1. scaffold_as_is – the inline pair, untouched (the baseline). 2. rough_paper_values – the SAME dict with four coefficients PINNED to deliberately rough values. The pin operates on a DEEP COPY (Dictionary::deepCopy, forum #63-7); nothing on disk or in memory survives the op. 3. scaffold_again – repeats 1 AFTER the pinned run. Its verdict must be IDENTICAL to 1's: the golden pins both, so the working-tree isolation is guarded by the corpus forever. Before deepCopy existed, the old alias bug would make THIS operation report the rough values' score.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | scaffold_as_is | aad | 2.0400 |
| diag | scaffold_as_is | chi2 | 60.1945 |
| diag | scaffold_as_is | max_abs_resid | 3.4379 |
| diag | scaffold_as_is | max_resid_x1 | 0.5000 |
| diag | scaffold_as_is | mode_evaluate | 1.0000 |
| diag | scaffold_as_is | n_data | 11.0000 |
| diag | scaffold_as_is | n_feasible | 11.0000 |
| diag | scaffold_as_is | partial | 0.0000 |
| diag | scaffold_as_is | penalised_points | 0.0000 |
| diag | scaffold_as_is | rms | 2.3393 |
| diag | scaffold_as_is | sse | 60.1945 |
| diag | rough_paper_values | aad | 1.5942 |
| diag | rough_paper_values | chi2 | 61.7621 |
| diag | rough_paper_values | max_abs_resid | 5.5866 |
| *... and 21 more reference values in the case's `expected`.* | | | |

## exergy01_air_dead_state

tutorials/props/molecular/exergy01_air_dead_state




**What it does:** Physical exergy of 79/21 N2/O2 at 400 K, 2 bar against a DECLARED 298.15 K / 1 bar dead state – b = (h-h0) - T0(s-s0) from the engine's own H_real/S_real, with the structural zero at the dead state itself as the second row

**The story:** PHYSICAL EXERGY, the number the entropy ledger buys – the witness behind the exergy work that follows the "What is entropy?" page.

b = (h - h0) - T0∗(s - s0), every term the engine's own H_real/S_real at the SAME composition, so the enthalpy datum, the s_298 anchors, the mixing term and the reference pressure ALL cancel: the exergy is datum-independent. The dead state is DECLARED (an exergy is a statement about a state AND an environment; the engine refuses to assume one).

Two rows on purpose: the same state the entropy01 ledger decomposes, and the dead state itself – whose exergy is 0 identically, by construction, which the golden pins as the structural zero.

Nothing here is validated against measurement: every number is the engine's own answer, decomposed. The published dh and T0ds legs re-add to b on the reader's desk.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | b_state | P | 200000 |
| diag | b_state | P0 | 100000 |
| diag | b_state | T | 400 |
| diag | b_state | T0 | 298.15 |
| diag | b_state | T0ds | 850.601 |
| diag | b_state | b_physical | 2136.71 |
| diag | b_state | dh | 2987.32 |
| diag | b_state | h | 2980.68 |
| diag | b_state | h0 | -6.63211 |
| diag | b_state | s | 201.559 |
| diag | b_state | s0 | 198.706 |
| diag | b_dead | P | 100000 |
| diag | b_dead | P0 | 100000 |
| diag | b_dead | T | 298.15 |

*. . . and 8 more reference values in the case's* ***expected****.*

## flash01_operating_line

tutorials/props/molecular/flash01_operating_line




**What it does:** The equilibrium curve a flash operating line is drawn on

**The story:** THE CURVE, AND ONLY THE CURVE.

A binary flash is fixed by two of {T, P, V/F} – Duhem – so at a frozen pressure the equilibrium curve y∗(x) is frozen too. Everything the flash tool draws on top of it (the operating line, its intersection, the lever rule) is GRAPHICAL construction on this one table, which is why turning a knob there redraws at once instead of re-solving.

So this case publishes exactly one thing: the curve. It does NOT flash anything. The flash the student watches is the intersection, performed on these rows – and the corpus already carries the ENGINE's own answer for the same pair (tutorials/steady/flash/flash01_benzene_toluene, V/F = 0.30398 at 370 K), which is the number to check the construction against.

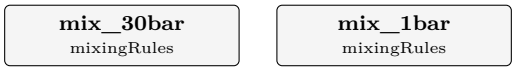
51 points is not a rounding of taste: the construction interpolates linearly between rows (no spline – an overshoot near an azeotrope would invent equilibrium the engine never computed), so the sampling IS the resolution of every answer read off it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | txy | n_emptyCurves | 0 |
| diag | txy | n_failures | 0 |
| diag | txy | n_points | 51 |

## mixrules01_natural_gas

tutorials/props/molecular/mixrules01_natural_gas




**What it does:** The van der Waals one-fluid mixing rule, decomposed: SRK a_mix/b_mix of a 90/5/5 CH4/N2/CO2 gas at 250 K, pair by pair – the one curated kij (N2-CH4, DECHEMA) beside the pairs that run the kij = 0 default, and the same decomposition at 1 bar showing a_mix is a (T, y) fact while only A, B, Z carry the pressure

**The story:** THE MIXING RULE, DECOMPOSED – the multicomponent lesson. A cubic prices a MIXTURE through two numbers, and the one-fluid rule says how n pure components and n(n-1)/2 binary corrections build them:

a_mix(T) = sum_i sum_j y_i y_j (1 - kij) sqrt(a_i a_j) b_mix = sum_i y_i b_i

Every term is on the table: the six pair terms of this ternary re-add to the a_mix the flash would run (the op REFUSES if they do not), the curated N2-CH4 kij sits beside the two CO2 pairs running the announced kij = 0 default, and the pair table is where a student sees that a ternary needs THREE binary decisions – there is no ternary parameter.

Two states on purpose, same composition and temperature: a_mix and b_mix are IDENTICAL at 30 bar and 1 bar (the rule sees only T and y), while A, B and Z carry the pressure – at 1 bar the same gas is nearly ideal (Z ~ 0.997) and at 30 bar it is not (Z ~ 0.93), priced by the SAME two numbers through the cubic.

Nothing here is validated against measurement: every number is the engine's own arithmetic, decomposed. The one cited constant is the N2-CH4 kij record (Knapp et al., DECHEMA VI, 1982).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | mix_30bar | A | 0.140523 |
| diag | mix_30bar | B | 0.0428434 |
| diag | mix_30bar | P_Pa | 3e+06 |
| diag | mix_30bar | T_K | 250 |
| diag | mix_30bar | Z_vapour | 0.901046 |
| diag | mix_30bar | a_CH4 | 0.200506 |
| diag | mix_30bar | a_CO2 | 0.429536 |
| diag | mix_30bar | a_N2 | 0.0837883 |
| diag | mix_30bar | a_mix | 0.202383 |
| diag | mix_30bar | b_CH4 | 2.98484e-05 |
| diag | mix_30bar | b_CO2 | 2.96742e-05 |
| diag | mix_30bar | b_N2 | 2.6754e-05 |
| diag | mix_30bar | b_mix | 2.9685e-05 |
| diag | mix_30bar | dadT_mix | -0.000515994 |
| *... and 46 more reference values in the case's `expected`.* | | | |

## nrtl01_ethanol_water_package

tutorials/props/molecular/nrtl01_ethanol_water_package




**What it does:** ethanol-water NRTL via the SAME property-package builder (molecular U4)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | g_nrtl | gamma_ethanol | 1.26245 |
| diag | g_nrtl | gamma_water | 1.4349 |

## pcsaft01_pure_nhexane

tutorials/props/molecular/pcsaft01_pure_nhexane




**What it does:** PC-SAFT pure n-hexane: Z + liquid molar volume at 298 K, 1 bar

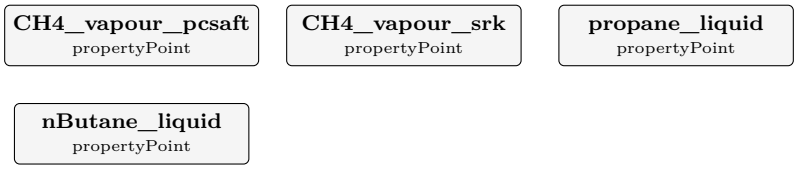
**The story:** the NON-ASSOCIATING PC-SAFT core, validated on a pure n-alkane. n-hexane's segment parameters (m, sigma, epsilon/k) are the Gross & Sadowski 2001 Table 1 fit; the case evaluates the compressibility factor Z and the liquid molar volume at 298.15 K / 1 bar and locks them as a golden. The engine PROVES the model self-consistent on every run (PCSAFT::verify: a_res → 0 in the ideal limit, and the residual chemical potentials reproduce the bulk residual Gibbs energy to ~1e-9). Physical cross-check (see README): the PC-SAFT liquid density here is ~650 kg/m3, within ~0.8 % of the ~655 kg/m3 experimental value.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | liquid_298K_1bar | Cp_ig | 1.43106e+02 |
| diag | liquid_298K_1bar | H_R | -3.91460e+02 |
| diag | liquid_298K_1bar | H_ig | -1.67200e+05 |
| diag | liquid_298K_1bar | H_real | -1.67591e+05 |
| diag | liquid_298K_1bar | P | 1.00000e+05 |
| diag | liquid_298K_1bar | S_R | -0.846425525134 |
| diag | liquid_298K_1bar | S_ig | 3.88510e+02 |
| diag | liquid_298K_1bar | S_real | 387.663574475 |
| diag | liquid_298K_1bar | T | 2.98150e+02 |
| diag | liquid_298K_1bar | Z | 9.42530e-01 |
| diag | liquid_298K_1bar | Z_liquid | 5.35107e-03 |
| diag | liquid_298K_1bar | gamma | 1.06168e+00 |
| diag | liquid_298K_1bar | v_molar | 2.33649e-02 |
| diag | liquid_298K_1bar | v_molar_liq | 1.32651e-04 |

## pcsaft02_states_crosscheck

tutorials/props/molecular/pcsaft02_states_crosscheck




**What it does:** PC-SAFT across states: CH4 Z vs SRK cross-check + propane/n-butane liquid density vs handbook

**The story:** PC-SAFT validated ACROSS STATES and AGAINST AN INDEPENDENT MODEL, before the model is trusted in a flowsheet.

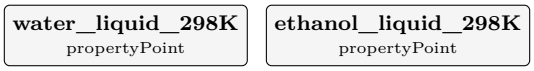
Three probes, all goldened: (1) CH4 vapour Z at 300 K / 50 bar – PC-SAFT vs SRK at the SAME state (the per-op 'thermo {}' override): two independently-validated models must agree closely for a simple near-spherical vapour, and the run prints both so the student SEES the agreement (not told to assume it). (2) propane compressed liquid at 298.15 K / 15 bar – rho_liq vs the saturated-liquid handbook value (0.493 g/mL at 25 degC). (3) n-butane compressed liquid at 298.15 K / 5 bar – rho_liq vs 0.573 g/mL at 25 degC. Deviations sit at the 1-2 % liquid-density AAD Gross & Sadowski 2001 (Table 2) report for the n-alkane series – these parameters PREDICT the density; it was never refitted here.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | CH4_vapour_pcsaft | Cp_ig | 3.56607e+01 |
| diag | CH4_vapour_pcsaft | H_R | -8.28601e+02 |
| diag | CH4_vapour_pcsaft | H_ig | -7.44541e+04 |
| diag | CH4_vapour_pcsaft | H_real | -7.52827e+04 |
| diag | CH4_vapour_pcsaft | P | 5.00000e+06 |
| diag | CH4_vapour_pcsaft | S_R | -2.01270526714 |
| diag | CH4_vapour_pcsaft | S_ig | 1.53964e+02 |
| diag | CH4_vapour_pcsaft | S_real | 151.951191741 |
| diag | CH4_vapour_pcsaft | T | 3.00000e+02 |
| diag | CH4_vapour_pcsaft | Z | 9.12141e-01 |
| diag | CH4_vapour_pcsaft | gamma | 1.30404e+00 |
| diag | CH4_vapour_pcsaft | v_molar | 4.55038e-04 |
| diag | CH4_vapour_srk | Cp_ig | 3.56607e+01 |
| diag | CH4_vapour_srk | H_R | -8.31158e+02 |
| *... and 42 more reference values in the case's `expected`.* | | | |

## pcsaft03_association_pure

tutorials/props/molecular/pcsaft03_association_pure




**What it does:** PC-SAFT association (Wertheim TPT1): 2B water + 2B ethanol, pure liquids

**The story:** PC-SAFT WITH ASSOCIATION (Wertheim TPT1; Gross & Sadowski, Ind. Eng. Chem. Res. 41 (2002) 5510) – the hydrogen-bonded liquids the non-associating core cannot represent, as PURE liquids at 298.15 K, 1 bar.

The association parameters live on the component records (2B water, 2B ethanol – both G&S 2002 Table 1, fitted WITH the association term; the paper lists water with TWO sites, and the scheme is PART of the fit: this same set mis-curated as 4C passed the pure-density check by coincidence and collapsed the ethanol/water mixture flash); the case only declares 'model PCSAFT;', and the engine announces the roster it found on the records:

[pcsaft] associating: water (2B), ethanol (2B); cross: Wolbach-Sandler

Liquid densities PREDICTED here (goldened), vs the CRC Handbook 25 degC values – each deviation stated against what the SET is fitted for:

water v_molar_liq → rho_liq = 921.9 kg/m3 vs 997.05 (-7.5 % – the PUBLISHED trade-off of the 2-site water fit, which is Psat-accurate instead: -0.2 % at 358 K, +4 % at 298 K) ethanol v_molar_liq → rho_liq = 779.8 kg/m3 vs 789.3 (-1.2 %)

What the student sees at verbosity 3 (the widened validation, ratified 2026-08-02): the construction-time verify() now ALSO echoes, per associating component as a pure fluid, the converged site fractions X^D / X^A, the a_assoc share, and the CLOSED-FORM-vs-ITERATIVE oracle deviation (for nD = nA the mass-action closure collapses to a quadratic; the damped fixed point must land on its root – observed machine zero). X^A(water) ~ 0.039: 96 % of water's sites are hydrogen-bonded at ambient conditions – the physics the term adds.

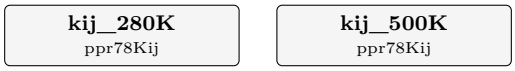
The two ops publish the standard vapour-root family (Z, v_molar) plus the labelled liquid root (v_molar_liq); at 1 bar both fluids carry a metastable vapour root beside the stable liquid.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | water_liquid_298K | Cp_ig | 3.36596e+01 |
| diag | water_liquid_298K | H_R | -2.18780e+03 |
| diag | water_liquid_298K | H_ig | -2.41826e+05 |
| diag | water_liquid_298K | H_real | -2.44014e+05 |
| diag | water_liquid_298K | P | 1.00000e+05 |
| diag | water_liquid_298K | S_R | -6.48369067192 |
| diag | water_liquid_298K | S_ig | 1.88834e+02 |
| diag | water_liquid_298K | S_real | 182.350309328 |
| diag | water_liquid_298K | T | 2.98150e+02 |
| diag | water_liquid_298K | Z | 9.00945e-01 |
| diag | water_liquid_298K | gamma | 1.32805e+00 |
| diag | water_liquid_298K | v_molar | 2.23340e-02 |
| diag | ethanol_liquid_298K | Cp_ig | 6.54261e+01 |
| diag | ethanol_liquid_298K | H_R | -4.01997e+03 |
| *... and 16 more reference values in the case's `expected`.* | | | |

## ppr78Kij01_ch4_co2_h2

tutorials/props/molecular/ppr78Kij01_ch4_co2_h2




**What it does:** The E-PPR78 group-contribution kij of a CH4/CO2/H2 gas, evaluated once at a declared temperature and written as a kij SET record a phi-phi package reads: at 280 K every pair lies inside the T window of the data its group parameters were fitted on, at 500 K every pair is an extrapolation and the set says so pair by pair

**The story:** A kij FROM GROUPS, AT A DECLARED TEMPERATURE. A cubic needs one binary correction kij per pair, and a measured one exists for few pairs. PPR78 (Jaubert & Mutelet) predicts it from the GROUPS each molecule is made of:

k_ij(T) = [ E_ij(T) - (delta_i - delta_j)^2 ] / ( 2 delta_i delta_j ) delta_i = sqrt(a_i(T)) / b_i (Choupo's own PR78) E_ij(T) = -1/2 SUM_k SUM_l (alpha_ik - alpha_jk)(alpha_il - alpha_jl) A_kl (298.15 / T)^(B_kl / A_kl - 1)

so kij DEPENDS ON T. The 'ppr78Kij' op evaluates it ONCE, at the temperature it is told, and writes the matrix to a FILE – a kij SET record ('recordType eosBinaryInteractionSet;') that a phi-phi package then declares – with the three terms of the quotient beside every value, so the division can be redone by hand.

WHAT THIS CASE SHOWS. The same three pairs at two temperatures: 280 K: every pair (CH4-CO2, CH4-H2, CO2-H2) lies inside the T range of the binary data its group parameters were fitted on (Xu et al. 2017, Table 2) – nothing is marked; 500 K: every pair lies above that range, and the set writes 'extrapolated yes;' with the window on each of them. The components are each their own group, so alpha is 1 or 0 and E_ij is the single group-pair term.

THE GROUP TABLE is constant/parameters/PPR78/xu2017.dat, the E-PPR78 subset (Xu, Lasala, Privat & Jaubert, Int. J. Greenh. Gas Control 56 (2017) 126-154, Table 3); its own header explains its transcription and that it is 'reviewStatus interim'.

WHAT IS NOT VALIDATED. Nothing here is compared with a measurement: the numbers are Eq. 4 evaluated on the cited table and the engine's own PR78 a_i(T), b_i. Whether those kij reproduce a phase diagram is not asked here.

THE TWO SET FILES are run outputs: they carry the date they were written, so they are named '*.proposal.dat', which git ignores.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | kij_280K | T_K | 280 |
| diag | kij_280K | kij_CH4_CO2 | 0.108020158785 |
| diag | kij_280K | kij_CH4_H2 | 0.29190260859 |
| diag | kij_280K | kij_CO2_H2 | 0.0990306279107 |
| diag | kij_280K | nPairs | 3 |
| diag | kij_500K | T_K | 500 |
| diag | kij_500K | kij_CH4_CO2 | 0.147492367254 |
| diag | kij_500K | kij_CH4_H2 | 0.644015046404 |
| diag | kij_500K | kij_CO2_H2 | 0.424699449845 |
| diag | kij_500K | nPairs | 3 |

## solubility01_hildebrand_ladder

`tutorials/props/molecular/solubility01_hildebrand_ladder`

**What it does:** Hildebrand: one number for like-dissolves-like, and where it stops

*Seven liquids on a ladder of cohesive energy density, from hexane to water.*

*δ is derived, never stored: sqrt((ΔHvap(T) - RT)/V_m), with the latent heat from the Watson correlation the enthalpy legs already use. Every input is already in the component record, so a stored δ would be a second home for a fact the record fixes — free to drift from it.*

*The published column is an ∗anchor∗, not an input. It is what ChemSep computed independently, and its only job is to give the derivation something to be wrong against. Worst deviation here is 0.85 %.*

*That matters because the three ways this arithmetic fails are all silent:*

*∗ the RT term dropped — δ a few per cent high, and a solubility parameter has no intuitive magnitude to make that look wrong; ∗ V_m in the wrong unit; ∗ Pa^0.5 reported where the literature uses MPa^0.5 — a factor of 31.6, and the number still reads as a solubility parameter.*

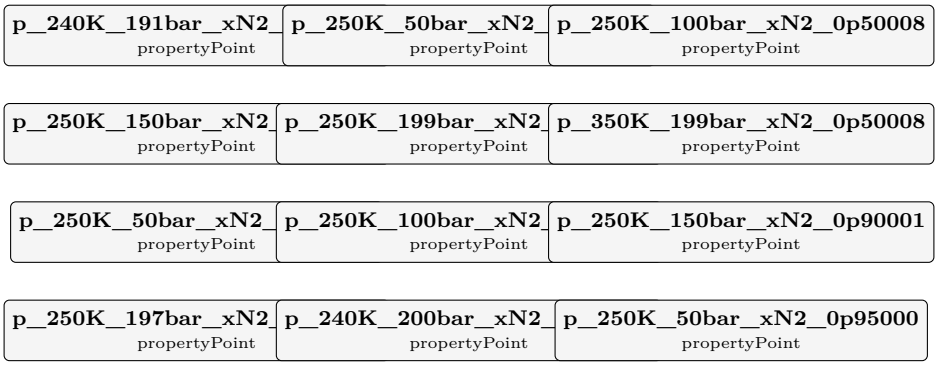
*Same device as the Zc column in a property data bank: a redundant determination turns a plausible wrong number into a visible disagreement.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | solubilityParameter_0 | delta_acetone | 19673.2 |
| diag | solubilityParameter_0 | delta_benzene | 18648.2 |
| diag | solubilityParameter_0 | delta_cyclohexane | 16733.6 |
| diag | solubilityParameter_0 | delta_ethanol | 26301.9 |
| diag | solubilityParameter_0 | delta_nHexane | 14859 |
| diag | solubilityParameter_0 | delta_toluene | 18183.8 |
| diag | solubilityParameter_0 | delta_water | 48266.4 |
| diag | solubilityParameter_0 | dev_acetone | 0.00287938 |
| diag | solubilityParameter_0 | dev_benzene | 0.00277184 |
| diag | solubilityParameter_0 | dev_cyclohexane | 0.000380181 |
| diag | solubilityParameter_0 | dev_ethanol | 0.00619352 |
| diag | solubilityParameter_0 | dev_nHexane | 0.000736807 |
| diag | solubilityParameter_0 | dev_toluene | 0.003629 |
| diag | solubilityParameter_0 | dev_water | 0.00849075 |
| *... and 6 more reference values in the case's `expected`.* | | | |

## srk01_h2n2_density

tutorials/props/molecular/srk01_h2n2_density




**What it does:** Does SRK describe the ammonia synthesis loop's gas? Sixteen MEASURED (T, P, x, rho) points on the nitrogen + hydrogen binary – Hernandez-Gomez, Tuma, Gomez-Hernandez and Chamorro, J. Chem. Eng. Data (2017), doi:10.1021/acs.jced.7b00694 – evaluated with the same SRK the green-ammonia plant runs on, and with kij = 0 because the catalogue carries no N2-H2 pair. The engine computes the molar volume; the case's own evidence file carries the measurement; the comparison is the lesson.

**The story:** IS SRK GOOD ENOUGH FOR THE AMMONIA SYNTHESIS LOOP?

Vitor's question, 2026-09-09: "dadas as pressoes extremas a que isto funciona, achas que os modelos termodinamicos que estamos a usar sao razoaveis e estao backed up por dados experimentais?" This case is the answer in numbers rather than in an opinion.

WHAT IT DOES. Sixteen 'propertyPoint' operations, one per MEASURED state of the nitrogen + hydrogen binary, at exactly the (T, P, x) the experimenters reported. The engine computes the molar volume with the SAME declaration the green ammonia plant carries – SRK, van der Waals one-fluid, kij = 0 – and prints it. The measurement sits beside it in 'constant/experiments/N2-H2-density-jced-7b00694.dat', cited to its primary publication.

HOW TO READ THE ANSWER. Each operation reports 'v_molar' in m3/mol. The mass density the experimenters measured is

rho = MW_mix / v_molar , MW_mix = x∗MW(N2) + (1-x)∗MW(H2)

and the two molar masses are in the components' own records (28.013 and 2.016 kg/kmol). Do that division for the four points at x(N2) = 0.50008 and 250 K and you have SRK's error climbing with pressure, which is the whole lesson: a cubic equation of state fitted to critical constants is a correlation, and 200 bar is far outside where its two parameters were determined.

'bin/curate/check_srk_h2n2_aad.py' does exactly that arithmetic, from THIS run's own output and THIS case's own evidence file, and refuses if either moves. The number is not written down anywhere – it is recomputed, because a transcribed agreement statistic is a second home for a fact the run owns.

WHAT THIS CASE DOES NOT SAY. It does not say SRK is wrong, and it does not say a better model exists in this tree – none does for this binary today. It does not test the loop's COMPOSITION: see the caveat in the evidence file's own header. And it says nothing about ammonia, which is the strongly associating component the loop actually has to condense.

*The green ammonia plant in tutorials/plant/greenAmmoniaPEQ_baseCase runs its synthesis loop at around 200 bar and condenses its product near 250 K, and it prices every stream with a Soave–Redlich–Kwong equation of state whose only inputs are each component's critical temperature, critical pressure and acentric factor. Two questions follow, and only the second one has an answer a student can check:*

*1. Is that reasonable? — an engineering judgement. 2. How far from measurement is it? — a number.*

*Sixteen propertyPoint operations, one per measured state of the nitrogen + hydrogen binary, at exactly the temperature, pressure and composition the experimenters reported. The engine computes the molar volume with the plant's own declaration — SRK, van der Waals one-fluid mixing, and kij = 0, which the run announces rather than assuming in silence:*

*The measurement sits beside it in constant/experiments/, cited to its primary publication:*

*> R. Hern?ndez-G?mez, D. Tuma, A. G?mez-Hern?ndez, C. R. Chamorro, > ∗Accurate Experimental (p, ?, T) Data for the Introduction of Hydrogen into > the Natural Gas Grid: Thermodynamic Characterization of the Nitrogen–Hydrogen > Binary System from 240 to 350 K and Pressures up to 20 MPa∗, > J. Chem. Eng. Data (2017). [doi:10.1021/acs.jced.7b00694](https://doi.org/10.1021/acs.jced.7b00694)*

*Each operation reports v_molar in $m^3$/mol. Turn it into a mass density with*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | p_240K_191bar_xN2_0p50008 | Cp_ig | 28.9758015655 |
| diag | p_240K_191bar_xN2_0p50008 | H_R | -456.193455635 |
| diag | p_240K_191bar_xN2_0p50008 | H_ig | -1683.90872502 |
| diag | p_240K_191bar_xN2_0p50008 | H_real | -2140.10218065 |
| diag | p_240K_191bar_xN2_0p50008 | P | 19066000 |
| diag | p_240K_191bar_xN2_0p50008 | S_R | -2.3117177058 |
| diag | p_240K_191bar_xN2_0p50008 | S_ig | 116.976587739 |
| diag | p_240K_191bar_xN2_0p50008 | S_real | 114.664870033 |
| diag | p_240K_191bar_xN2_0p50008 | T | 240.083 |
| diag | p_240K_191bar_xN2_0p50008 | Z | 1.08683166645 |
| diag | p_240K_191bar_xN2_0p50008 | gamma | 1.40241644741 |
| diag | p_240K_191bar_xN2_0p50008 | v_molar | 0.000113788478235 |
| diag | p_250K_50bar_xN2_0p50008 | Cp_ig | 28.9824821078 |
| diag | p_250K_50bar_xN2_0p50008 | H_R | -156.040876541 |
| *… and 178 more reference values in the case's `expected`.* | | | |

## powder

## compaction01_mcc

tutorials/props/powder/compaction01_mcc

| mcc_inDie | mcc_inDie_highWindow | mcc_lubricated |
|---|---|---|
| compactionFit | compactionFit | compactionFit |

**What it does:** Microcrystalline cellulose on the compaction bench: Heckel, Kawakita and Ryshkewitch-Duckworth fitted to two open datasets, in die and out of die, with the declared Heckel window shown to move Py

**The story:** compaction01 – MICROCRYSTALLINE CELLULOSE ON THE COMPACTION BENCH.

MCC is the main excipient of the apalutamide tablet case (tutorials/batch/plant/apalutamide01_tablets: MCC 295.5 + silicified MCC 116.9 mg of a 700 mg core). This case fits the three compaction models to OPEN measurements of MCC, so the numbers a tablet-press design needs – how hard the press must push, how much the bed shrinks, how strong the tablet comes out – have a source a reader can open.

WHY A NEW FOLDER, 'tutorials/props/powder/'. The tutorial folders of 'props/' are named by the subject a student studies, and powder compaction is none of the existing ones (it is not thermodynamics, transport or a membrane): it is solids processing, the first props bench of its kind.

THREE OPERATIONS, TWO SOURCES (both CC BY 4.0; datasets in constant/experimental/, each with its own provenance header):

mcc_inDie Avicel PH200, van der Haven et al., Sci. Data 13 (2026) 1468 – 45 tablets compressed once on a compaction simulator, IN-DIE density at maximum force; 22 tablets crushed diametrally. Heckel on the declared low-pressure window, Kawakita on all pressures, Ryshkewitch-Duckworth on the crushing data, compared with the sigma_0 and k_b the dataset itself reports. mcc_inDie_highWindow the SAME 45 tablets, Heckel on a high-pressure window: the mean yield pressure roughly doubles. The window IS part of the answer, which is why the case declares it and the op never chooses it. mcc_lubricated CEOLUS UF-711 + 2 % sodium stearyl fumarate + 0.3 % silica (Kukuls & Mohylyuk 2025 dataset; protocol of Mohylyuk et al., Pharmaceutics 16 (2024) 362), the mean of ten tablets at each of six pressures, OUT OF DIE, compared with the Py = 106 MPa that paper reports.

WHAT TO READ IN THE OUTPUT. Py of MCC is low (tens to ~100 MPa in die on the low-pressure line): it deforms PLASTICALLY, which is why it binds so well and why it is the filler of choice for direct compression. Compare it with compaction02 (dicalcium phosphate, brittle, several hundred MPa). The out-of-die Py is HIGHER than the in-die one on similar pressures because the ejected tablet has recovered the elastic part of its strain.

THE PAPER'S Py IS NOT REPRODUCED, AND THE REASON IS THE LESSON. Mohylyuk et al. 2024 state their linear region as 100-150 MPa. The deposited data carries ONE of its six pressures in that window (137 MPa), and a line through fewer than 4 points refuses here, so this case declares 90-340 MPa (4 points) and gets a different, larger, Py. (The two-point slope between the 91 and 137 MPa rows would give about 100 MPa – close to the paper's – but a two-point line has no residual to judge it by.) Both numbers are printed; nothing is tuned.

A FINDING, PRINTED AND NOT TUNED: on both materials of the van der Haven dataset (here and in the other powder case) the fitted k_b agrees with the dataset's own constant within 10 %, but the fitted sigma_0 comes out about 1.5 times the dataset's. A fit in sigma_t instead of ln(sigma_t) does not close it (measured: 30.9 and 28.4 MPa). The files read do not say how the dataset's "Eq. 4" was fitted or on which porosity; its report PDF was not read. sigma_0 is an extrapolation to zero porosity, so a small difference in slope or in the porosity basis moves it a lot.

HOW TO USE THIS FOR YOUR FORMULATION. 1. Find an OPEN dataset (CC BY: MDPI Pharmaceutics, Int. J. Pharm. X, Sci. Data, Zenodo, institutional dataverses) for the excipient that dominates your blend – the same grade if you can. Check its licence on the article page; NonCommercial or "available on request" is not usable. 2. Write its rows in the experiments grammar (copy one of the files in constant/experimental/): P in MPa and ONE of D, porosity or the compact's density (then declare 'trueDensity', with its source). Say whether the density is in-die or out-of-die. 3. Plot the Heckel line first (run with a wide window and look at the 'heckel' table's residuals), then DECLARE the linear region you defend, with a 'reason'. Report Py WITH its window and its in-die or out-of-die basis; a Py without them cannot be compared. 4. If the source reports its own values, put them in 'reported {}'. A difference is a finding

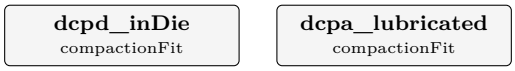
(window, density basis, true density), never something to tune away. LIMITS, said plainly: these are data for MCC, not for your blend. NO mixture rule (for Py, sigma_0 or k_b) is implemented; the source's own ternary data shows mixtures do not follow a simple average. Lubricant (magnesium stearate) weakens MCC tablets and the lubricated dataset here is the closer stand-in for a real formulation. For another MCC grade the literature gives in-die Py = 52.2 +/- 4.2 MPa for Vivapur 102 at 11.28 mm (Lura et al., Int. J. Pharm. X 2 (2020) 100061, Table 2, CC BY) – quoted for comparison, not fitted here.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | mcc_inDie | dev_ryshkewitch_kb_pct | 8.68066281422 |
| diag | mcc_inDie | dev_ryshkewitch_sigma0_MPa_pct | 57.4687063406 |
| diag | mcc_inDie | heckel_A | 0.657612232409 |
| diag | mcc_inDie | heckel_A_ci95 | 0.063671311004 |
| diag | mcc_inDie | heckel_DA | 0.481913070072 |
| diag | mcc_inDie | heckel_K_ci95_perMPa | 0.000609029851532 |
| diag | mcc_inDie | heckel_K_perMPa | 0.00906082550054 |
| diag | mcc_inDie | heckel_Py_MPa | 110.365220028 |
| diag | mcc_inDie | heckel_Py_ci95_MPa | 7.41827701722 |
| diag | mcc_inDie | heckel_R2 | 0.990980625546 |
| diag | mcc_inDie | kawakita_R2 | 0.999983473773 |
| diag | mcc_inDie | kawakita_a | 0.780867436427 |
| diag | mcc_inDie | kawakita_b_perMPa | 0.111550591336 |
| diag | mcc_inDie | kawakita_invb_MPa | 8.96454234825 |
| *... and 39 more reference values in the case's `expected`.* | | | |

## compaction02_dcpd_brittle

tutorials/props/powder/compaction02_dcpd_brittle




**What it does:** A brittle excipient on the compaction bench: dicalcium phosphate fitted to two open datasets (no open lactose dataset was found, and none is invented), against MCC's plastic behaviour in compaction01

**The story:** compaction02 – A BRITTLE EXCIPIENT ON THE COMPACTION BENCH: DICALCIUM PHOSPHATE, AND WHY IT IS NOT LACTOSE.

The donepezil tablet case (tutorials/batch/plant/donepezil01_tablets) is 62 % lactose monohydrate, a FRAGMENTING excipient. This case was meant to fit the compaction models to open lactose data, and it does not, because NO open dataset with tabulated lactose pressure-density or strength-porosity rows was found on 2026-11-02 (CC BY sources searched: MDPI Pharmaceutics, Int. J. Pharm. X, PMC open access, Zenodo, Data in Brief; the lactose datasets located were NonCommercial, figure-only, or "available on request"). No lactose numbers were invented in its place.

What it fits instead is the open brittle reference that WAS found, from the same two sources as compaction01, so the plastic-against-brittle contrast can be read off two cases built the same way:

dcpd_inDie DCPD Emcompress premium (dihydrate), van der Haven et al., Sci. Data 13 (2026) 1468, CC BY 4.0: 42 tablets, in-die density at maximum force; 19 crushed diametrally. Compared with the sigma_0 and k_b the dataset itself reports. dcpa_lubricated CaHPO4 anhydrous DI-CAFOS A60 + 2 % sodium stearyl fumarate + 0.3 % silica (Kukuls & Mohylyuk 2025, CC BY 4.0; protocol of Mohylyuk et al., Pharmaceutics 16 (2024) 362), out of die, compared with that paper's Py = 434 MPa.

WHAT TO READ. Py here is several hundred MPa – several times MCC's (compaction01) – the signature of a material that densifies by FRACTURE, not by flow. Its tablets are weak at a given porosity (low tensile strength) and k_b is large: the strength collapses fast as the pores open. Lactose sits between the two: for Tablettose 80 the literature gives an in-die Py of 120.8 +/- 1.5 MPa at 11.28 mm, against 52.2 +/- 4.2 MPa for MCC Vivapur 102 in the same study (Lura et al., Int. J. Pharm. X 2 (2020) 100061, Table 2, CC BY) – quoted for the donepezil group to compare with, NOT fitted here, because the study gives the curves only as figures.

A FINDING, PRINTED AND NOT TUNED: on both materials of the van der Haven dataset (here and in the other powder case) the fitted k_b agrees with the dataset's own constant within 10 %, but the fitted sigma_0 comes out about 1.5 times the dataset's. A fit in sigma_t instead of ln(sigma_t) does not close it (measured: 30.9 and 28.4 MPa). The files read do not say how the dataset's "Eq. 4" was fitted or on which porosity; its report PDF was not read. sigma_0 is an extrapolation to zero porosity, so a small difference in slope or in the porosity basis moves it a lot.

FOR THE DONEPEZIL GROUP. To fit your own lactose: find an open dataset (or digitise a CC BY figure, stating the reading error in the dataset header), write it in the grammar of the files in constant/experimental/, declare its Heckel window with a reason, and put the paper's own Py in ‘reported {}‘. Say which lactose (monohydrate milled, sieved, agglomerated, spray-dried, anhydrous): Py differs between grades by a factor of two. A wet-granulated blend (your case's route) compacts differently again from the dry excipient, and NO mixture rule is implemented – the excipient stands in for the blend, and you say so.

THE PAPER'S Py AND THIS CASE'S. Mohylyuk et al. 2024 analyse the F 0-100 formulation between 250 and 450 MPa. The deposited averages have three rows there (243 MPa falls just below it); a 3-point line refuses here, so the case declares 140-440 MPa (four rows) and both values are printed.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | dcpd_inDie | dev_ryshkewitch_kb_pct | 2.33048956558 |
| diag | dcpd_inDie | dev_ryshkewitch_sigma0_MPa_pct | 49.4763706254 |
| diag | dcpd_inDie | heckel_A | 1.33186263307 |
| diag | dcpd_inDie | heckel_A_ci95 | 0.0229519605641 |
| diag | dcpd_inDie | heckel_DA | 0.736014904288 |
| diag | dcpd_inDie | heckel_K_ci95_perMPa | 7.74836836182e-05 |
| diag | dcpd_inDie | heckel_K_perMPa | 0.00196611284247 |
| diag | dcpd_inDie | heckel_Py_MPa | 508.617805854 |
| diag | dcpd_inDie | heckel_Py_ci95_MPa | 20.0444146948 |
| diag | dcpd_inDie | heckel_R2 | 0.99330881447 |
| diag | dcpd_inDie | kawakita_R2 | 0.999736957603 |
| diag | dcpd_inDie | kawakita_a | 0.63714261759 |
| diag | dcpd_inDie | kawakita_b_perMPa | 0.0655574652353 |
| diag | dcpd_inDie | kawakita_invb_MPa | 15.2537929344 |
| *... and 30 more reference values in the case's `expected`.* | | | |

## reactor

**The theory you need.** Three idealised steady reactors. Conversion reactor: stoichiometry only, $n_{\text{out}} = n_{\text{in}} + \nu\,\xi$, duty from the ONE enthalpy base $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$ (formation datum – no separate "heat of reaction" input, and any dict override is cross-checked aloud). CSTR: outlet at reactor conditions with rate $r(c_{\text{out}}, T)$, so $F\,(c_{\text{in}} - c_{\text{out}}) = r\,V$. PFR: the same rate integrated along the volume, $\mathrm{d}F_i/\mathrm{d}V = \nu_i\,r$. Arrhenius kinetics $k = A\,T^b e^{-E_a/RT}$, orders from the dict, reverse rates from the equilibrium constant (thermo-consistent by construction). Expect: CSTR conversion below PFR at equal volume (back-mixing), exponential $T$ sensitivity, and equilibrium limits the kinetics can approach but never cross. *Full derivation: Theory Guide, the reaction-engineering chapter.*

## thiele01_sphere_first_order

tutorials/props/reactor/thiele01_sphere_first_order




**What it does:** Single catalyst pellet: intraparticle concentration field, effectiveness factor, and the eta(phi) family, each verified against the first-order isothermal closed form

**The story:** One thielePellet operation.

THE RATE CONSTANT IS DECLARED HERE, and not on the catalyst record, because kinetics are equipment- and reaction-specific: the asset carries the SOLID, never the chemistry run on it. Two reactors loaded with the same catalyst would otherwise carry two copies of one rate constant and drift apart.

k = 40 1/s is a TEACHING VALUE chosen so the generalised modulus lands near 1 – the transition region, where the field has a visible gradient and eta is neither ~1 nor ~1/phi. It is not measured and is not claimed to be.

*Solves the intraparticle boundary-value problem for one porous catalyst pellet, publishes the concentration field, the effectiveness factor and the Thiele modulus, and checks every one of them against the closed form.*

*With xi the normalised coordinate from the pellet centre (0) to its external surface (1) and u = c/c_s:*

*Both moduli are published because both conventions are in circulation and they differ by a factor of three for a sphere. The engine's own announcement in src/unitOperations/reactor/CatalystPellet.H uses the generalised one.*

*The equation is discretised with second-order central differences and handed to the existing solver::newtonND (finite-difference Jacobian, LU with partial pivoting) — no new dependency, no new solver, every step readable in src/unitOperations/reactor/pellet/PelletDiffusion.cpp.*

*xi is one column, not two aliases: it is r/R for a sphere or a cylinder and x/L for a slab.*

*First-order isothermal is the rare case that has a closed form in all three geometries, so the numerical answer is not merely produced — it is checked:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | thiele_CO_alumina | phi | 1.0161e+00 |
| diag | thiele_CO_alumina | phi_char | 3.0483e+00 |
| diag | thiele_CO_alumina | Lambda | 5.0000e-04 |
| diag | thiele_CO_alumina | dimension | 1.5000e-03 |
| diag | thiele_CO_alumina | epsilon_p | 4.5000e-01 |
| diag | thiele_CO_alumina | tau | 3.0000e+00 |
| diag | thiele_CO_alumina | rateConstant | 4.0000e+01 |
| diag | thiele_CO_alumina | T | 5.7315e+02 |
| diag | thiele_CO_alumina | P | 1.01325e+05 |
| diag | thiele_CO_alumina | D_molecular | 6.45709e-05 |
| diag | thiele_CO_alumina | D_eff | 9.68564e-06 |
| diag | thiele_CO_alumina | eta | 6.65742e-01 |
| diag | thiele_CO_alumina | eta_closedForm | 6.65742e-01 |
| diag | thiele_CO_alumina | eta_flux | 6.65731e-01 |

*... and 17 more reference values in the case's `expected`.*

## thiele02_prater_temperature

tutorials/props/reactor/thiele02_prater_temperature




**What it does:** Single catalyst pellet: intraparticle concentration field, effectiveness factor, and the eta(phi) family, each verified against the first-order isothermal closed form

**The story:** One thielePellet operation.

THE RATE CONSTANT IS DECLARED HERE, and not on the catalyst record, because kinetics are equipment- and reaction-specific: the asset carries the SOLID, never the chemistry run on it. Two reactors loaded with the same catalyst would otherwise carry two copies of one rate constant and drift apart.

k = 40 1/s is a TEACHING VALUE chosen so the generalised modulus lands near 1 – the transition region, where the field has a visible gradient and eta is neither ~1 nor ~1/phi. It is not measured and is not claimed to be.

*The SAME pellet as thiele01_sphere_first_order, with one block added, so the difference between the two runs is the temperature.*

*Everything else — catalyst, species, T, P, rate constant, diffusion, grid, verification tolerances — is thiele01 unchanged. Run both and diff the output: nothing else moves.*

*The reaction is exothermic, so the pellet is hottest where the reactant is most depleted, which is the centre:*

*The field CSV gains two columns, T_over_Ts and T_K, one value per node — so the temperature is a FIELD, not a headline number, and the EduTool paints the same discs by it.*

*The species and energy balances inside a pellet carry the same reaction term. Eliminate it between them and, for constant D_eff and k_eff and any kinetics at all, what is left is a first integral — Prater's relation:*

*So the temperature field costs no second boundary-value problem. It is algebra over the concentration field the operation already solved.*

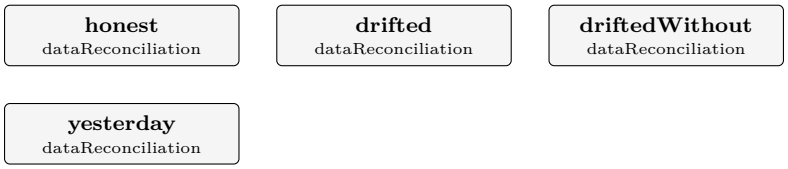
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | thiele_CO_alumina_prater | D_eff | 9.68564e-06 |
| diag | thiele_CO_alumina_prater | D_molecular | 6.45709e-05 |
| diag | thiele_CO_alumina_prater | Lambda | 0.0005 |
| diag | thiele_CO_alumina_prater | P | 101325 |
| diag | thiele_CO_alumina_prater | T | 573.15 |
| diag | thiele_CO_alumina_prater | T_centre_K | 588.722 |
| diag | thiele_CO_alumina_prater | c_centre | 0.289874 |
| diag | thiele_CO_alumina_prater | c_centre_closedForm | 0.289873 |
| diag | thiele_CO_alumina_prater | deltaT_centre_K | 15.5718 |
| diag | thiele_CO_alumina_prater | dimension | 0.0015 |
| diag | thiele_CO_alumina_prater | epsilon_p | 0.45 |
| diag | thiele_CO_alumina_prater | eta | 0.665742 |
| diag | thiele_CO_alumina_prater | eta_closedForm | 0.665742 |
| diag | thiele_CO_alumina_prater | eta_flux | 0.665731 |
| *... and 23 more reference values in the case's `expected`.* | | | |

## reconciliation

## reconcile01_recycle_loop

tutorials/props/reconciliation/reconcile01_recycle_loop




**What it does:** Seven flow meters on an invented recycle loop, reconciled under four node balances: the residuals of the raw readings, the weighted least squares that is also the Bayesian posterior mode, the chi-square and measurement tests that catch a meter planted to read 15 t/h high, and a prior from yesterday – the witness of the EduTool on data reconciliation

**The story:** reconcile01 – SEVEN METERS THAT DO NOT CLOSE FOUR BALANCES (DEV.md 4c C121).

An invented recycle loop (constant/plant/network: a mixer, a reactor, a separator and a purge splitter, seven streams, one flow meter on each) read once (constant/plant/readings). The readings carry noise, so not one node balance closes; the residuals are of the size the meters' own sigmas predict. The witness of the EduTool "Data reconciliation, and how Bayes helps", which runs this case in the browser.

WHAT IS SYNTHETIC, said plainly. The plant, its true flows, the meter sigmas, the noise, the planted gross error and yesterday's prior are all invented. Nothing here is a measurement of anything. The point is the method, and an invented plant lets the page show the TRUTH beside the readings and the reconciled flows – which no real plant can.

THE FOUR OPERATIONS, one per question the page asks.

honest the readings as they are. Weighted least squares under the four balances: the posterior mode for Gaussian meters and a flat prior. The objective is small against chi-square(4), and no meter's test exceeds the critical value.

drifted the same snapshot with F3 planted to read 15 t/h high (syntheticDrift). The global test rejects, the measurement test points at F3, and the reconciled flows show the error SMEARED over F3's neighbours.

driftedWithout the drifted snapshot with F3's meter taken out of service (exclude). The balances are projected onto the six remaining meters (the reactor and the separator merge into one envelope), the tests pass again, and F3 is read back from the balances – with how far the excluded meter sits from it.

yesterday the honest readings with a prior per flow from yesterday's reconciled values (constant/plant/yesterday): each reading and its prior combine by precision weighting, exactly, before the same reconciliation; the reconciled sigmas shrink again.

The significance level is DECLARED (alpha 0.05), never defaulted: the probability of a false alarm a plant accepts is a decision, not a constant.

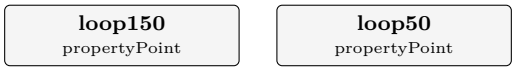
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| csv | prior_streams.csv | 0:combined | 100.5 |
| csv | prior_streams.csv | 0:combinedSigma | 1.41421356237 |
| csv | prior_streams.csv | 1:combined | 147.7 |
| csv | prior_streams.csv | 1:combinedSigma | 2.12132034356 |
| csv | prior_streams.csv | 2:combined | 150.25 |
| csv | prior_streams.csv | 2:combinedSigma | 2.12132034356 |
| csv | prior_streams.csv | 3:combined | 89.4 |
| csv | prior_streams.csv | 3:combinedSigma | 1.27279220614 |
| csv | prior_streams.csv | 4:combined | 60.3195121951 |
| csv | prior_streams.csv | 4:combinedSigma | 0.937042571332 |
| csv | prior_streams.csv | 5:combined | 10.04 |
| csv | prior_streams.csv | 5:combinedSigma | 0.1788854382 |
| csv | prior_streams.csv | 6:combined | 50.0967213115 |
| csv | prior_streams.csv | 6:combinedSigma | 0.768221279597 |
| *… and 264 more reference values in the case's `expected`.* | | | |

## safety

## relief01_ammonia_loop_gas

tutorials/props/safety/relief01_ammonia_loop_gas




**What it does:** The ammonia synthesis loop's converter feed at the relieving pressure of a blocked outlet, at 150 bar and at 50 bar: the compressibility factor and Cp/Cv the gas relief-valve equation needs, computed with the SRK the loop runs on

**The story:** THE GAS SIDE OF A BLOCKED-OUTLET RELIEF VALVE ON THE AMMONIA SYNTHESIS LOOP (DEV.md 4c C61, the "Designing for overpressure" EduTool).

THE QUESTION. The converter's outlet is blocked while the recycle compressor goes on delivering the converter feed. The relief valve on the feed side must pass that whole flow at the relieving pressure, and the gas relief equation needs two properties of the gas there besides its molar mass: the compressibility factor Z and the ratio of specific heats k = Cp/Cv. This case computes both, for the same loop at 150 bar and at 50 bar – the two green-ammonia feasibility cases the Process-safety EduTool compares (tutorials/plant/greenAmmonia02_feasibility150bar and greenAmmonia02_feasibility50bar).

WHERE THE STATES COME FROM. composition – each plant's converged converter feed ('Preheated-Feed'), copied as the component flows in kmol/h exactly as the plant's converged/PreheatedFeed prints them. propertyPoint normalises the composition, so the flows ARE the composition; the EduTool reads them again as the relieving rate. gui/tests/overpressure.test.ts holds every number to the plant's own converged file. T – the same stream's temperature (653.15 K and 593.15 K): a blocked outlet relieves the gas as it arrives. P – the RELIEVING pressure, worked from the stream's operating pressure in the EduTool and written here to the digit: operating (gauge) = P_stream - 1.01325 bar design (gauge) = max(1.1 x operating, operating + 1.7 bar) (the Design Guide's rule of thumb for a PRELIMINARY design pressure, docs/designGuide.tex, "Pressure relief, flare & safety basics") MAWP (gauge) = design (gauge) – an ASSUMPTION, the lowest MAWP the vessel can have; a real MAWP is a property of the built vessel, never of a rule of thumb set pressure = MAWP (a single device) relieving (gauge) = max(1.10 x MAWP, MAWP + 3 psi) relieving (abs) = relieving (gauge) + 1.01325 bar The design pressure is a rule of thumb applied to a feasibility case that sizes no vessel, and the MAWP is assumed equal to it; no number here is the design pressure or the MAWP of any real loop.

WHAT IS COMPUTED. propertyPoint's base set: Z on the SRK vapour root, and gamma = Cp_ig/(Cp_ig - R) from the records' ideal-gas heat capacities – the ideal-gas ratio the gas relief equation is derived with.

*The gas side of a blocked-outlet relief valve on the green-ammonia synthesis loop: the converter's outlet is shut while the recycle compressor goes on delivering the converter feed, so the valve on the feed side must pass that whole flow at the relieving pressure.*

*Two propertyPoint operations, one per loop pressure of the two C57 feasibility cases ([greenAmmonia02_feasibility150bar](../../../plant/greenAmmonia02_feasibility150bar) at 150 bar and [greenAmmonia02_feasibility50bar](../../../plant/greenAmmonia02_feasibility50bar)):*

*∗ the composition is each plant's converged converter feed (PreheatedFeed), copied as component flows in kmol/h — propertyPoint normalises it, and the EduTool reads the same numbers back as the relieving rate; ∗ the temperature is that stream's; ∗ the pressure is the relieving pressure of a single device set at an ASSUMED MAWP, taken equal to the preliminary design pressure the Design Guide's rule of thumb gives, max(1.1 × operating, operating + 1.7 bar), plus the 10 % accumulation — worked out in the propsDict header. A rule of thumb gives a preliminary design pressure, never a MAWP.*

*What the run gives the relief equation is Z (SRK vapour root, kij = 0, announced) and k = Cp/Cv (the ideal-gas ratio). The sizing itself is the EduTool's arithmetic: ∗Designing for overpressure∗ (DEV.md 4c C61), where the 50 bar loop turns out to need a relief valve about four and a half times larger than the 150 bar one in the sizing arithmetic, on the same declared blocked-outlet flow hypothesis — not a comparison of two real relief systems.*

*Nothing here is a design pressure or a relief load for a real loop.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | loop150 | Cp_ig | 29.8985186167 |
| diag | loop150 | H_R | 258.83362137 |
| diag | loop150 | H_ig | 9770.2186076 |
| diag | loop150 | H_real | 10029.052229 |
| diag | loop150 | P | 18240876.8436 |
| diag | loop150 | S_R | -0.122977107449 |
| diag | loop150 | S_ig | 129.551194701 |
| diag | loop150 | S_real | 129.428217594 |
| diag | loop150 | T | 653.15 |
| diag | loop150 | Z | 1.06292561774 |
| diag | loop150 | gamma | 1.38521316932 |
| diag | loop150 | v_molar | 0.000316449401973 |
| diag | loop50 | Cp_ig | 30.1315777327 |
| diag | loop50 | H_R | 64.6879856225 |
| *… and 10 more reference values in the case's `expected`.* | | | |

## relief02_acetone_fire_case

tutorials/props/safety/relief02_acetone_fire_case




**What it does:** An acetone drum in a pool fire: the boiling temperature at the relieving pressure, the latent heat there, and the vapour's Z and Cp/Cv – the fluid properties the fire-case relief load and the gas sizing equation need, computed by the engine rather than typed in

**The story:** THE FLUID SIDE OF A FIRE-CASE RELIEF VALVE (DEV.md 4c C61, the "Designing for overpressure" EduTool).

THE QUESTION. A drum of acetone, a flammable solvent, sits in a pool fire. Its relief valve opens at the set pressure and the pressure climbs to the relieving pressure P1 = set + allowed accumulation; there the liquid BOILS, and the fire's heat leaves as vapour. The relief load is the heat absorbed divided by the latent heat, and the valve passes that vapour. So four properties of the fluid decide the valve, and all four are computed here:

T_bubble the boiling temperature at P1 (the relieving temperature) Hvap_acetone the latent heat of vaporisation at that temperature Z the vapour's compressibility factor at (T, P1) gamma Cp/Cv of the vapour (the ideal-gas ratio, Cp_ig/(Cp_ig-R))

THE NUMBERS ARE AUTHOR-SET, and the EduTool moves them. P is the relieving pressure of a drum whose MAWP is 5 barg, protected for the fire case with the set pressure at the MAWP: 1.21 x 5 barg + 1.01325 bar = 7.06325 bar absolute. The 5 barg is a round number chosen for the lesson, not a property of any real drum. T is the boiling temperature the SAME run reports at that pressure (T_bubble), written back by hand so that Z, gamma and the latent heat are evaluated AT the relieving state; the EduTool does the same thing live, in two runs, whenever the reader moves the MAWP, and its test holds this file's T to its own T_bubble.

WHERE EACH NUMBER COMES FROM. T_bubble: the bubble point of pure acetone on the package in constant/thermoPhysPropDict (ideal liquid, SRK vapour). Hvap_acetone: the record's HvapTb carried to T by the Watson correlation, Component::Hvap_latent – the same term the engine subtracts from the ideal-gas enthalpy to price every liquid stream. Z: the SRK vapour root. gamma: the record's ideal-gas heat capacity.

WHY ACETONE AND NOT TOLUENE. The first draft of this case was a toluene drum, and its run announced that toluene's Antoine record is fitted only from 273 to 380 K – below the normal boiling point, so every boiling temperature of a pressurised toluene drum is an EXTRAPOLATED vapour pressure. A relief temperature resting on an extrapolation is a weak first lesson. Acetone's record covers 259.16 to 507.60 K, which holds the relieving state of any drum this EduTool lets the reader set.

WHAT THIS CASE DOES NOT SAY. Nothing here is a relief-valve design: the sizing is the EduTool's arithmetic on these numbers, and no number in either is a design for a real vessel.

*The fluid side of a fire-case relief valve. A drum of acetone sits in a pool fire; its valve opens at the set pressure (here the MAWP, 5 barg — a round number for the lesson) and the pressure climbs to the relieving pressure with the 21 % fire accumulation, 7.06325 bar absolute. There the liquid boils, and the fire's heat leaves as vapour.*

*One propertyPoint operation, at the relieving pressure and at the boiling temperature the same run reports there, asks for the four numbers the relief load and the gas sizing equation need:*

*Hvap_<comp> is a property key added for this case (DEV.md 4c C61): it is the same latent heat the engine subtracts from the ideal-gas enthalpy to price every liquid, and it refuses at or above the critical temperature.*

*Why acetone and not toluene: toluene's Antoine record is fitted only from 273 to 380 K, below its own normal boiling point, so every boiling temperature of a pressurised toluene drum would be an extrapolated vapour pressure. Acetone's covers 259.16 to 507.60 K.*

*The sizing is the EduTool's arithmetic: ∗Designing for overpressure∗. Nothing here is a relief design for a real vessel.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | relieving | Cp_ig | 90.7915819779 |
| diag | relieving | H_R | -1286.05234531 |
| diag | relieving | H_ig | -209206.764976 |
| diag | relieving | H_real | -210492.817321 |
| diag | relieving | Hvap_acetone | 24520.5209741 |
| diag | relieving | P | 706325 |
| diag | relieving | S_R | -2.25049969273 |
| diag | relieving | S_ig | 301.885255534 |
| diag | relieving | S_real | 299.634755842 |
| diag | relieving | T | 394.109436 |
| diag | relieving | T_bubble | 394.109436036 |
| diag | relieving | Z | 0.87065495897 |
| diag | relieving | Z_liquid | 0.0242535783371 |
| diag | relieving | gamma | 1.10080932363 |
| *... and 2 more reference values in the case's* `expected`*.* | | | |

## scan

**The theory you need.** Property scans sweep a model across $T$, $P$ or composition and plot what the thermodynamics DOES: $P^{\mathrm{sat}}(T)$ curves, $\gamma_{\pm}(m)$, VLE envelopes, ternary diagrams, phase boundaries. Nothing is fitted – these ops answer "how does the selected model behave?", the SEE step of the props loop (CREATE → SEE → TEST → FIT → PROMOTE). Expect model disagreements to be VISIBLE (overlay two methods and look), and every curve exportable as CSV next to the case. *Full derivation: Theory Guide, the Props Guide.*

## binary01_lle_water_nbutanol

tutorials/props/scan/binary01_lle_water_nbutanol




**What it does:** Binary LLE g_mix + common-tangent (water/1-butanol) at 298.15 K via propertyScanBinary, cited Winkelman 2009 UNIQUAC

**The story:** Binary liquid-liquid teaching map – water / 1-butanol at 25 C.

The molar Gibbs energy of mixing g_mix(x1) swept across composition PLUS the two coexisting liquid compositions from ONE liquid-liquid IsothermalFlash (PhaseSet::LL). The two binodal points lie on the g_mix curve and share its common tangent – the student reads the split straight off the diagram.

gamma comes from the CITED Winkelman et al. (2009) UNIQUAC LLE set (Table 2; structural r,q from Table 1), so the g_mix curve goes NON-CONVEX and the flash emits the real water/1-butanol binodal: ~1.9 mol% butanol in the water-rich phase and ~49 mol% butanol in the butanol-rich phase (matching their Figs 1-2 at 298 K). See README.md.

The flash feed is set to z(water) = 0.80 (20 mol% butanol), INSIDE the asymmetric gap; an equimolar feed would lie outside it (both binodal branches sit below 50 mol% butanol). Run-only smoke (the CSV is a teaching curve, not pinned KPIs).

*Water + 1-butanol (n-butanol) is ∗the∗ classic partially-miscible binary. At 25 °C it splits into a water-rich liquid (~2 mol % butanol) and a butanol-rich liquid (~50 mol % butanol). A g_mix(x_water) scan should therefore show a clearly non-convex region: two binodal points sharing a common tangent, which a student reads the split straight off.*

*and binaryLle.csv is a g_mix(x1) curve that goes non-convex, plus the two binodal points the liquid-liquid flash locates on it:*

*The agreement is essentially exact: the cited UNIQUAC fit reproduces the published mutual solubilities.*

*- J.G.M. Winkelman, G.N. Kraai, H.J. Heeres, ∗"Binary, ternary and quaternary liquid-liquid equilibria in 1-butanol, oleic acid, water and n-heptane mixtures"∗, Fluid Phase Equilibria 284 (2009) 71–79, doi:10.1016/j.fluid.2009.06.013.*

*- Interaction parameters — their Table 2 (p.73), binary 1-butanol(1)/water(3), valid 273–363 K, with the quadratic temperature correlation $A_{ij}(T) = a + b\cdot T + c\cdot T^2$ (Eq. 10), $\tau_{ij} = \exp(-A_{ij}/T)$ (Eq. 7): - A_1,3 (butanol→water): a = 155.31, b = 1.0822, 10?·c = -43.711 - A_3,1 (water→butanol): a = -579.36, b = 2.7517, 10?·c = -6.7700 - Structural r, q — their Table 1 (p.72): 1-butanol r = 3.9243, q = 3.668; water r = 0.9200, q = 1.400.*

*The numbers are inline here, fully provenanced above; the record is STAGED CASE-LOCALLY, not curated: it lives at tutorials/steady/flash/marcilla02_lls_ternary_invariant/constant/parameters/UNIQUAC/nButanol-water.dat with reviewStatus interim, and promotion into data/standards/ is the maintainer's batch review. This README used to claim the numbers already lived under data/standards/ – asserting a curation status that had been deliberately withheld, which is a worse error than saying nothing.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | lle | T | 298.15 |
| diag | lle | split | 1 |

## hxy01_ethanol_water_1atm

tutorials/props/scan/hxy01_ethanol_water_1atm




**What it does:** Enthalpy-concentration locus (h-x-y) of ethanol/water at 1 atm on the elements datum – the data a Ponchon-Savarit construction is drawn on

**The story:** The enthalpy-concentration locus of ethanol/water at 1 atm.

McCabe-Thiele carries equilibrium and material balance under CONSTANT MOLAR OVERFLOW – one mole condensing releases exactly what one mole vaporising absorbs, whatever the composition. This case publishes the quantity that assumption is about: the saturated-liquid enthalpy h(x) on its bubble line, the saturated-vapour enthalpy H(y) on its dew line, and the equilibrium tie between them, all on Choupo's ONE enthalpy datum (the elements at 298.15 K).

lambda(x) = H(y∗) - h(x) is the molar heat of vaporisation ALONG the locus. CMO says it is a constant. Read the two lambda diagnostics and decide for yourself.

Run-only + KPI golden; see README.md for the provenance of every number and for what is NOT validated.

*The saturated-liquid enthalpy on its bubble line, the saturated-vapour enthalpy on its dew line, and the equilibrium tie between them — the data an enthalpy-concentration (Ponchon–Savarit) construction is drawn on, and the first thing the engine publishes that carries enthalpy against composition.*

*enthalpyConcentration.csv — a regular table, one header, homogeneous rows:*

*Every row is one equilibrium. It carries ∗both∗ ends of a tie line — the liquid (x1, h_liquid) and the vapour (y1, H_vapour) that are in equilibrium with each other — at the one temperature T_K they share. So nothing downstream pairs rows by index and nothing interpolates one end of a tie. role says only which composition was the uniform grid node:*

*The saturated-liquid curve is (x1, h_liquid) over all rows; the saturated-vapour curve is (y1, H_vapour) over all rows. status is ok or names a refusal; a refused node keeps its grid coordinate and leaves the solved fields empty, so a gap is visible rather than silently bridged.*

*enthalpyConcentration.meta — a key,value sidecar, written on every run. It states the two components, the pressure, which enthalpy datum the two columns are on, which kernel computed each, the tie contract, whether the heat of mixing is included, and the reason for every refused node. The construction's lever arms are enthalpy ∗differences∗, so a constant offset cancels in the geometry — which is exactly why the file has to say what its axis means instead of letting the next reader assume.*

*Both enthalpy columns are absolute on Choupo's one enthalpy base, the elements at 298.15 K, from the same two kernels every energy balance and duty in the engine is priced on:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | hxy | H_vapour_pure1_J_per_mol | -231233.18018 |
| diag | hxy | H_vapour_pure2_J_per_mol | -239303.954931 |
| diag | hxy | P_Pa | 101325 |
| diag | hxy | T_bubble_pure1_K | 351.452745505 |
| diag | hxy | T_bubble_pure2_K | 372.453622433 |
| diag | hxy | h_liquid_pure1_J_per_mol | -271912.766061 |
| diag | hxy | h_liquid_pure2_J_per_mol | -280791.66824 |
| diag | hxy | lambda_max_J_per_mol | 45052.6104262 |
| diag | hxy | lambda_min_J_per_mol | 40679.5858814 |
| diag | hxy | n_nodes | 82 |
| diag | hxy | n_refused | 0 |

## phase01_water_full

tutorials/props/scan/phase01_water_full




**What it does:** Complete P-T phase diagram of water (purePhaseDiagram, solid+liquid+vapour)

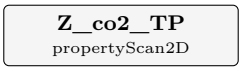
**The story:** Complete pure-compound P-T phase diagram of water: liquid-vapour saturation (to the critical point) + solid-vapour sublimation + solid-liquid fusion (the negative-slope ice line) + triple and critical points. Clapeyron physics in the purePhaseDiagram engine op; cited solid data in the solid{} block (IAPWS/CRC, see solidPhaseData.ts). Run-only smoke.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | pd | Pc_bar | 220.64 |
| diag | pd | Tc | 647.14 |
| diag | pd | has_solid | 1 |

## scan2d01_co2_compressibility

tutorials/props/scan/scan2d01_co2_compressibility




**What it does:** Z(T,P) for CO2 on Peng-Robinson over a 15 x 20 grid that crosses the critical point: the compressibility surface, one row per node

**The story:** THE COMPRESSIBILITY SURFACE, NOT A CURVE.

'propertyScan2D' sweeps TWO state variables over a rectangle and reports the property at every node – long-form CSV, one row per (T, P), ready to pivot into a heat map. Here it is Z(T, P) for CO2 on Peng-Robinson, 280-420 K by 1-200 bar: a 15 x 20 grid that walks straight over the critical point (304.13 K, 73.77 bar).

What to read: along an isotherm below Tc the surface DROPS steeply and then turns up – the liquid-like branch; above Tc the same isotherm is smooth: the discontinuity is a coexistence line, and the grid crosses where it ends; $Z \to 1$ in the whole low-P edge, which is the ideal-gas limit falling out of a cubic rather than being assumed.

The op reports 'failures' – nodes where the property could not be evaluated – as a diagnostic, so a grid that quietly returned fewer points than it promised is visible in the payload rather than only in the CSV's line count. The expected file pins it at zero: on this rectangle every node must solve.

Why the case exists: 'propertyScan2D' had NO case and no gate, though 'propertyScan1D', 'propertyScanBinary' and 'propertyScanTernary' all have one. Its sibling vaporPressureFit was in the same position, and writing the first case for THAT one segfaulted choupoProps twice.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | Z_co2_TP | failures | 0 |
| diag | Z_co2_TP | n_points | 300 |
| diag | Z_co2_TP | n_x | 15 |
| diag | Z_co2_TP | n_y | 20 |

## ternary01_audit_vlle

tutorials/props/scan/ternary01_audit_vlle




**What it does:** Ternary VLLE phase map (compA/compB/compC) at 350 K, 1 bar via propertyScanTernary

**The story:** ILLUSTRATIVE / SYNTHETIC DATA. compA / compB / compC are NOT real species: they are artificial components (case-local constant/components/ + inline NRTL pairs in thermoPhysPropDict) deliberately tuned to guarantee a ternary VLLE region, so this case exercises the propertyScanTernary classifier on a known-3-phase map. Architecture (see thermoPhysPropDict): A and B mutually immiscible (LL gap on the A-B axis), C a volatile bridge soluble in both liquids. At 350 K, 1 bar the equimolar feed gives vapour rich in C, L_alpha rich in A, L_beta rich in B.

Expected map (14 intervals/edge, 78 nodes): VLLE 36, LL 29, VL 0, 1-phase 13, 22 tie-lines, 0 failures. The point to SEE: a healthy spread of 3-phase (VLLE) and LL nodes with zero solver failures — the classifier resolves the engineered gap node-by-node.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | lle | failures | 0 |
| diag | lle | n_ll | 32 |
| diag | lle | n_nodes | 78 |
| diag | lle | n_onephase | 10 |
| diag | lle | n_tielines | 23 |
| diag | lle | n_vl | 0 |
| diag | lle | n_vlle | 36 |

## ternary02_bubbleT_benzene_toluene_water

tutorials/props/scan/ternary02_bubbleT_benzene_toluene_water




**What it does:** Ternary T_bubble surface (benzene/toluene/water) at 1 bar via propertyScanTernary mode bubbleT

**The story:** Ternary boiling-temperature surface (T_bubble over the composition triangle) via propertyScanTernary mode bubbleT — reuses BubblePoint per simplex node. Works with ideal binaries (only the 3 Psat are needed). Run-only smoke.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bt | failures | 0 |
| diag | bt | n_nodes | 150 |

### ternary03_lle_water_ethanol_benzene

tutorials/props/scan/ternary03_lle_water_ethanol_benzene




**What it does:** Ternary LLE solubility (water/ethanol/benzene) at 298 K via UNIFAC + propertyScanTernary mode lle

**The story:** Ternary LIQUID-LIQUID solubility map (water / ethanol / benzene) — the classic extraction triangle. Activity from UNIFAC (group contribution): NO fitted binary pairs needed — gamma is predicted from molecular groups, so a real system shows a real miscibility gap with no curated pair data. mode lle uses the liquid-liquid flash (PhaseSet::LL): clean binodal + tie-lines, no spurious vapour. Run-only smoke (a phase map, not KPIs).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | lle | failures | 0 |
| diag | lle | n_ll | 84 |
| diag | lle | n_nodes | 105 |
| diag | lle | n_onephase | 21 |
| diag | lle | n_tielines | 21 |
| diag | lle | n_vl | 0 |
| diag | lle | n_vlle | 0 |

## solvents

## solvent01_synthetic_mixture

tutorials/props/solvents/solvent01_synthetic_mixture




**What it does:** DIDACTIC EXAMPLE WITH SYNTHETIC PARAMETERS: two fictitious solvents A and B, a fictitious polymer P between them – the Hansen distance along the A/B line has its minimum at 50:50 by volume, hand-checkable

**The story:** solvent01 – DIDACTIC EXAMPLE WITH SYNTHETIC PARAMETERS (DEV.md 4c C128, adjustment 2: start with a synthetic case, then a real one, then the confrontation with measurements).

THE ARITHMETIC, by hand (MPa^0.5):

solventA dD 16 dP 2 dH 4 solventB dD 16 dP 10 dH 12 polymerP dD 17 dP 6 dH 8 R0 4

Along the A/B line at volume fraction s of B the mixture rule gives (16, 2 + 8 s, 4 + 8 s), so

Ra^2 = 4 (16 - 17)^2 + (8 s - 4)^2 + (8 s - 4)^2 = 4 + 2 (8 s - 4)^2

which is 36 at either pure end (Ra = 6.000, RED = 1.5) and 4 at s = 0.5 (Ra = 2.000, RED = 0.5): the minimum sits at 50:50 BY CONSTRUCTION, and the dispersion offset of 1 keeps it from being a degenerate zero. The two solvents have equal Vliq, so the volume, mole and mass bases coincide here – the basis conversion is exercised on the real case instead.

THE PHASE WORD IS ‘notVerified‘, honestly: the package declares an ideal liquid (gamma = 1), which has no excess Gibbs energy and can never demix, so a stability test on it would say "homogeneous" about every pair ever asked. The op marks every blend point rather than claim a verification the model cannot give.

*Two fictitious solvents A and B and a fictitious polymer P between them, so the Hansen distance along the A/B line has its minimum at 50:50 by volume — by construction, and checkable by hand. Nothing here is a real substance; the engine says so on every run ([synthetic] component 'solventA' is NOT a real substance), and the records carry provenance { source synthetic; }.*

*All values in MPa^0.5 (the records store Pa^0.5, 1000 times larger):*

*Hansen's distance, with the factor 4 on the dispersion term (his 1967 thesis, Eq. 8.2), along the line at volume fraction s of B:*

*so Ra = 6.000 at either pure end (RED = 1.5, outside the sphere) and Ra = 2.000 at s = 0.5 (RED = 0.5, inside). The dispersion offset of 1 keeps the minimum from being a degenerate zero. The two solvents have the same molar volume, so the volume, mole and mass bases coincide here; the conversion is exercised on the real case (solvent02).*

*The package declares an ideal liquid, and gamma = 1 has no excess Gibbs energy: it can never demix, so a stability test on it would say "one liquid" about every pair ever asked. The op marks every blend point notVerified rather than claim a verification the model cannot give — the honest third outcome of the phase check (the other two, homogeneous and twoPhase, are live on solvent02 under UNIFAC). The pure ends carry pure: one component, nothing to split.*

*∗ the mixture rule is the volume-fraction average of each parameter — an approximation (no excess volume; each solvent keeps its own split); ∗ the parameters are held constant at their reference temperature (Tref 298.15 K on every record); no correction exists in the engine, and the declared T decides the phase check alone; ∗ Ra is an affinity, not a solubility: no saturation concentration is published.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | blend | R0_MPa05 | 4 |
| diag | blend | RED_min | 0.5 |
| diag | blend | Ra_min_MPa05 | 2 |
| diag | blend | Ra_pure_solventA | 6 |
| diag | blend | Ra_pure_solventB | 6 |
| diag | blend | T_K | 298.15 |
| diag | blend | Tref_target_K | 298.15 |
| diag | blend | n_homogeneous | 0 |
| diag | blend | n_notVerified | 19 |
| diag | blend | n_points | 21 |
| diag | blend | n_pure | 2 |
| diag | blend | n_twoPhase | 0 |
| diag | blend | phiB_at_min | 0.5 |
| diag | blend | s_at_min | 0.5 |

*. . . and 35 more reference values in the case's* `expected`*.*

## solvent02_polystyrene_hexane_blends

tutorials/props/solvents/solvent02_polystyrene_hexane_blends




**What it does:** Polystyrene (Hansen 1967, solute G) against two hexane blends: hexane + acetone, two solvents each short of the polymer that meet it between them, and hexane + nitromethane, whose blend the mixture rule likes exactly where UNIFAC finds two liquids

**The story:** solvent02 – THE REAL CASE (DEV.md 4c C128, adjustment 2, step two).

The target is polystyrene as Hansen placed it in his 1967 thesis (solute G, Table 4.1: centre (8.6, 3.0, 2.0) (cal/cm3)^0.5, radius 3.5); the solvents are three of his Table 2.1 placements, read from the page images and converted to Pa^0.5 in constant/components/∗.dat, one citation per value. Two blends, two lessons:

1. hexane + acetone, by VOLUME. In the engine's MPa^0.5 (the thesis's (cal/cm3)^0.5 times 2.0455) polystyrene's radius is R0 = 7.159; pure hexane lies outside the sphere (Ra 9.238, RED 1.290) and pure acetone just inside its edge (Ra 6.638, RED 0.927 – and rated 3, "strongly swollen, slight solubility", in the thesis's own test, Table 2.3). The blend dips well inside: the minimum is Ra 4.724 (RED 0.660) at 65 % acetone by volume, and 50:50 gives 4.992 (RED 0.697). The mixture rule lets two solvents on opposite sides of the target meet it between them, which is the observation Hansen checked his placements against (Section 2.5: "mixtures of non-solvents ... should ... dissolve the solute. This has always been found to be the case"). UNIFAC is asked whether the pair is one liquid at every point.

2. hexane + nitromethane, by MOLE (the basis conversion exercised: mole fractions become volume fractions through the records' Vliq, and the run prints both). Both ends are far outside the sphere (RED 1.290 and 1.850); the mixture rule puts the middle of the line inside (Ra 5.399, RED 0.754 at x = 0.6, which is 38 % nitromethane by volume). But hexane and nitromethane are an immiscible pair at 25 C, and the case's UNIFAC is asked: where the Michelsen test finds a second liquid the op REFUSES to represent the pair as one solvent – no averaged triple, no Ra, the point marked 'twoPhase' – and the minimum is taken over the points that survive. Measured: 17 of 19 blends (x = 0.05 to 0.85) are two-phase, every one where the rule would have put the blend inside the sphere among them, and the survivors all lie outside it – the minimum over survivors is pure hexane, RED 1.290. The averaged number would have been the best on the line, and it describes a solvent that does not exist. (A two-phase pair is not useless – it may serve an extraction; it is only not one solvent.)

T = 298.15 K is also every record's Tref, so no parameter is used away from the temperature it was placed at; the run says so either way.

*The real case of the solvent-selection lesson (DEV.md 4c C128). The target is polystyrene as Hansen placed it in his 1967 thesis (solute G, Table 4.1: centre (8.6, 3.0, 2.0), radius 3.5 in (cal/cm$^3$)^0.5 — R0 = 7.159 MPa^0.5 in the engine's unit); the solvents are three of his Table 2.1 placements, transcribed from the page images into case-local overlay records (constant/components/∗.dat) with the thesis page and row beside each value, converted to Pa^0.5 by sqrt(4.184e6) = 2045.48, and marked interim. The reference temperature is adopted as 25 °C (the thesis prints none beside its tables); the run is at 25 °C so nothing is moved away from it.*

*Pure hexane lies outside the sphere and pure acetone just inside its edge (in the thesis's own test both were rated 3, "strongly swollen, slight solubility"); the blend dips well inside near 65 % acetone. That is the observation Hansen checked his placements against: ∗two non-solvents on opposite sides of a solute's sphere give a blend that dissolves it∗ (thesis, ?2.5, "this has always been found to be the case"). UNIFAC — the case's activity model, predictive from the records' groups — finds the pair one liquid at every point, so the word is homogeneous all along, which is what makes the blend a candidate at all.*

*The composition is declared by MOLE, so the run converts it to volume fractions through the records' Vliq (54.1 cm$^3$/mol for nitromethane against 131.6 for hexane: equal moles is 29 % nitromethane by volume) and prints both. Both ends are far outside the sphere (RED 1.290 and 1.850), and the mixture rule puts the middle of the line inside it — on paper, a better blend than the first (Ra 5.399, RED 0.754 at x = 0.6).*

*But hexane and nitromethane are an immiscible pair at room temperature, and the case's UNIFAC says so: where Michelsen's tangent-plane test finds a second liquid the op refuses to represent the pair as one solvent — no averaged triple, no Ra, no RED for that point, the row marked twoPhase with its negative tm — and the minimum is taken over the points that survive. The averaged number would have been the best on*

*the line, and it describes a solvent that does not exist. A two-phase pair is not useless (it may serve an extraction); it is only not one solvent.*

*What UNIFAC answered, measured: 17 of the 19 blends are two-phase, from x = 0.05 to 0.85 (tm between -0.10 and -0.40), and only the two nitromethane-rich blends (x = 0.90 and 0.95) are one liquid — both outside the sphere (RED 1.340 and 1.572). Every composition where the mixture rule would have put the blend inside the sphere (roughly x = 0.3 to 0.8, RED down to 0.754) is refused, so the minimum over the points that survive is pure hexane, RED 1.290: no blend of these two is a solvent for polystyrene. The counts are pinned in expected (n_twoPhase, n_homogeneous).*

*This case also found an engine defect, fixed in the same slice. Its first run reported hexane + acetone two-phase at EVERY composition (tm down to -0.43), for a pair whose UNIFAC activity coefficients at 50:50 are 1.52 and 1.56 and which is miscible. The Michelsen tangent-plane distance in src/solver/StabilityTest.cpp omitted the ln $\gamma_i(y)$ term, so at the trivial point it read -G^E/RT instead of 0 and called every positive-deviation mixture unstable. With the term restored the pair is one liquid (tm at round-off) and the 38 corpus cases that reach the test kept every golden.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | hexaneAcetone | R0_MPa05 | 7.1592 |
| diag | hexaneAcetone | RED_min | 0.659789277374 |
| diag | hexaneAcetone | Ra_min_MPa05 | 4.72356339458 |
| diag | hexaneAcetone | Ra_pure_acetone | 6.63812052919 |
| diag | hexaneAcetone | Ra_pure_nHexane | 9.23837412102 |
| diag | hexaneAcetone | T_K | 298.15 |
| diag | hexaneAcetone | Tref_target_K | 298.15 |
| diag | hexaneAcetone | n_homogeneous | 19 |
| diag | hexaneAcetone | n_notVerified | 0 |
| diag | hexaneAcetone | n_points | 21 |
| diag | hexaneAcetone | n_pure | 2 |
| diag | hexaneAcetone | n_twoPhase | 0 |
| diag | hexaneAcetone | phiB_at_min | 0.65 |
| diag | hexaneAcetone | s_at_min | 0.65 |
| *... and 67 more reference values in the case's `expected`.* | | | |

## solvent03_polystyrene_observed

tutorials/props/solvents/solvent03_polystyrene_observed




**What it does:** Polystyrene (Hansen 1967, solute G) screened against thirteen pure solvents of the same thesis, beside the thesis's own dissolution ratings (Table 2.3) kept in constant/experimental/ – the confrontation of Ra and RED with what was observed

**The story:** solvent03 – THE CONFRONTATION WITH EXPERIMENT (DEV.md 4c C128, adjustment 2, step three).

No 'composition {}': the op SCREENS each declared solvent pure – one row each, Ra and RED against polystyrene's sphere (Hansen 1967, Table 4.1, solute G). Beside the run, constant/experimental/ hansen1967_table23_polystyrene.csv holds what the same thesis OBSERVED for each of these thirteen solvents (Table 2.3, column G, the 1-6 scale of page 20: 1 soluble ... 6 no visible effect), with the citation on every row. The op does not read that file: the prediction and the observation stay two tables, joined by the reader (the README does it by hand, the EduTool draws them side by side, the gate counts the agreement).

WHAT THE CONFRONTATION IS, said plainly. Hansen's sphere for this polymer was DRAWN THROUGH these very observations (Section 4.4: "the centers of the volumes of solubility ... have been determined" from the solvents that dissolve each solute), so agreement here shows that the transcription and the distance arithmetic reproduce his placement – it is a check of the method on its own training set, not an independent prediction. The thesis itself states the residual: about 2.5 % of the solvents beyond R0 + 0.5 dissolve, and about 2.5 % within R0 - 0.5 do not (page 39). Nitroethane (RED 1.39, rated 2 "almost soluble") and cyclohexane (RED 1.06, rated 1 "soluble") are that residual, in this set.

The package is 'ideal' because no mixture is asked for: every row is one pure solvent and the phase word is 'pure'.

*The confrontation with experiment of the solvent-selection lesson (DEV.md 4c C128). No composition {}: the hansenMixture op screens each declared solvent PURE against polystyrene (Hansen 1967, solute G) — one row each, the phase word pure — and constant/experimental/hansen1967_table23_polystyrene.csv holds what the same thesis observed for each solvent (Table 2.3, column G, the 1–6 scale of its page 20: 1 soluble, 2 almost soluble, 3 strongly swollen with slight solubility, 4 swollen, 5 slight effect, 6 no visible effect), with the citation on every row. The op does not read that file; the two tables are joined here, by the EduTool, and by the gate.*

*Ra in MPa^0.5 against R0 = 7.159 (the thesis's 3.5 (cal/cm$^3$)^0.5); RED = Ra/R0.*

*Every solvent the thesis rated 6 lies outside the sphere; every solvent it rated 1 lies inside or — cyclohexane, RED 1.058 — within the ±0.5 (cal/cm$^3$)^0.5 band the thesis states for its own radii. The residual, read across the line RED = 1: nitroethane (RED 1.39, rated 2) and cyclohexane (RED 1.06, rated 1) dissolve from outside; acetone (RED 0.93, rated 3) does not quite from inside. Three of thirteen, where the thesis itself expects about one solvent in forty on each side beyond ±0.5 of the radius (page 39) — this set is small and its residual sits on the boundary, as the thesis says the deviations do.*

*Hansen's sphere for this polymer was drawn through these very observations (thesis ?4.4: the centres were determined from the solvents that dissolve each solute). Agreement here therefore shows that the transcription and the distance arithmetic reproduce his placement — a check of the method on its own training set, not an independent prediction of a solvent he never tried. No blend was dissolved by anyone: the thesis tabulates pure solvents only, and a measured point along a mixture line is what this corpus does not have.*

*The package is ideal because no mixture is asked for. Every parameter is a case-local overlay record with the thesis page and row beside the value, reviewStatus interim (transcribed from the page images, not the OCR text, whose columns are scrambled; a second reading pending).*

*expected pins each solvent's Ra_pure_<name> and the CSV cells of the screen; check_hansen_mixture rebuilds them from the records and prints the residual above as a measurement, failing only on a rated-6 solvent inside the sphere or a rated-1 solvent beyond the thesis's own band.*

**What to expect (golden-locked):**

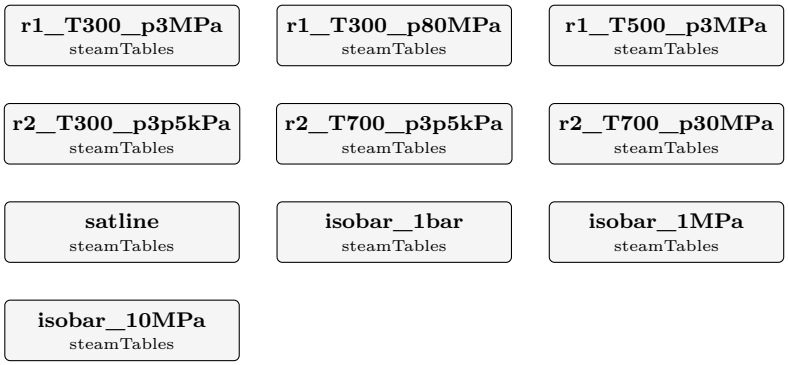
| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | screen | R0_MPa05 | 7.1592 |
| diag | screen | RED_min | 0.636685199779 |
| diag | screen | Ra_min_MPa05 | 4.55815668226 |
| diag | screen | Ra_pure_acetone | 6.63812052919 |
| diag | screen | Ra_pure_benzene | 5.6906984791 |
| diag | screen | Ra_pure_cyclohexane | 7.57262071677 |
| diag | screen | Ra_pure_diethylEther | 7.20865879204 |
| diag | screen | Ra_pure_ethanol | 15.9714933081 |
| diag | screen | Ra_pure_ethylAcetate | 7.0242390805 |
| diag | screen | Ra_pure_methanol | 19.8084619809 |
| diag | screen | Ra_pure_methylEthylKetone | 4.55815668226 |
| diag | screen | Ra_pure_nButanol | 12.1057857919 |
| diag | screen | Ra_pure_nHexane | 9.23837412102 |
| diag | screen | Ra_pure_nitroethane | 9.97063233401 |
| *... and 48 more reference values in the case's `expected`.* | | | |

## steam

**The theory you need.** Water/steam properties from IAPWS-IF97: regions (liquid, vapour, supercritical), $T_{\text{sat}}(P)$, and $h(T,P)/s(T,P)$ from the dimensionless Gibbs functions. Power cycles are enthalpy bookkeeping around them: pump work $\approx v\,\Delta P$, turbine work from isentropic expansion $\times$ efficiency, cycle efficiency $\eta = W_{\text{net}}/Q_{\text{in}}$ bounded by Carnot. *Full derivation: Theory Guide, the steam/power chapter.*

## steam01_if97_verification

tutorials/props/steam/steam01_if97_verification




**What it does:** IAPWS-IF97 steam tables (R7-97(2012)) – the verification case. The six computer-program verification states of the release (Table 5 region 1, Table 15 region 2) evaluated as points, the saturation line 0.01-350 degC, and three isobars whose Tsat crossings pin the release's Table 36 backward values. The published numbers ARE the golden master (reltol 1e-8).

**The story:** the IAPWS-IF97 weapon case.

The release (IAPWS R7-97(2012)) publishes computer-program verification tables; this case evaluates EVERY one of them this slice covers and locks the published numbers as golden KPIs at reltol 1e-8 – a single wrong digit in a transcribed coefficient fails the regression:

Table 5 – region 1 at (300 K, 3 MPa), (300 K, 80 MPa), (500 K, 3 MPa) Table 15 – region 2 at (300 K, 3.5 kPa), (700 K, 3.5 kPa), (700 K, 30 MPa) Table 35 – psat at 300 / 500 K (the psat_MPa diag of the point ops) Table 36 – Tsat at 0.1 / 1 / 10 MPa (the Tsat_K diag of the isobars)

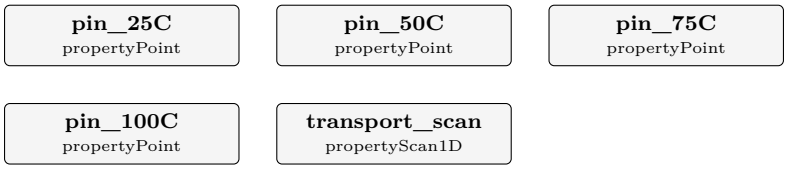
Plus the working artefacts: the saturated-steam table 0.01-350 degC (saturation.csv) and three isobars (h, s, v, cp vs T) crossing the saturation line – the crossing is ANNOUNCED in the log.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | r1_T300_p3MPa | region | 1 |
| diag | r1_T300_p3MPa | v | 0.100215168e-2 |
| diag | r1_T300_p3MPa | h | 0.115331273e3 |
| diag | r1_T300_p3MPa | u | 0.112324818e3 |
| diag | r1_T300_p3MPa | s | 0.392294792 |
| diag | r1_T300_p3MPa | cp | 0.417301218e1 |
| diag | r1_T300_p3MPa | w | 0.150773921e4 |
| diag | r1_T300_p3MPa | psat_MPa | 0.353658941e-2 |
| diag | r1_T300_p80MPa | region | 1 |
| diag | r1_T300_p80MPa | v | 0.971180894e-3 |
| diag | r1_T300_p80MPa | h | 0.184142828e3 |
| diag | r1_T300_p80MPa | u | 0.106448356e3 |
| diag | r1_T300_p80MPa | s | 0.368563852 |
| diag | r1_T300_p80MPa | cp | 0.401008987e1 |
| *... and 47 more reference values in the case's `expected`.* | | | |

### steam02_water_transport

tutorials/props/steam/steam02_water_transport




**What it does:** IAPWS water transport properties – viscosity (R12-08), thermal conductivity (R15-11) and surface tension (R1-76). Water flagged as a pure fluid (method IF97) routes the transport accessors to the matching IAPWS releases: the density comes from the IF97 thermodynamic kernel at the saturation state, fed to the (rho,T) transport kernels; surface tension is the R1-76 saturation-line form. A coarse 4-point scan (25/50/75/100 degC) pinned against the published IAPWS values, plus a fine sweep for the plot.

**The story:** the IAPWS water transport route.

Water flagged ‘pureFluids { water { method IF97; } }‘ routes the transport accessors to the matching IAPWS releases:

viscosity → R12-08 (industrial, critical enhancement off) thermal_conductivity → R15-11 (background, critical enhancement off) surface_tension → R1-76 (saturation line)

The (rho,T) viscosity / conductivity kernels get their density from the IF97 thermodynamic kernel at the SATURATION state (the legacy transport accessors carry no pressure, so p = p_sat(T) is the physical convention – saturated liquid / saturated vapour). Surface tension is a saturation-line property and takes only T.

‘pins‘ evaluates the three properties at 25/50/75/100 degC; the golden master locks them at reltol 1e-5 (the formulation is verified to 1e-8 by IAPWSTransport::verify(), which fires on every run). ‘scan‘ writes a fine sweep for the plot.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | pin_25C | surface_tension | 0.0719722052302 |
| diag | pin_25C | thermal_conductivity | 0.606460966251 |
| diag | pin_25C | viscosity | 0.000890036037708 |
| diag | pin_50C | surface_tension | 0.0679439059449 |
| diag | pin_50C | thermal_conductivity | 0.640589598777 |
| diag | pin_75C | surface_tension | 0.0635830195097 |
| diag | pin_75C | thermal_conductivity | 0.66354437705 |
| diag | pin_100C | surface_tension | 0.0589118685877 |
| diag | pin_100C | thermal_conductivity | 0.677216843797 |
| diag | transport_scan | n_points | 46 |
| diag | transport_scan | n_failures | 0 |
| diag | pin_25C | Cp_ig | 33.6596038227 |
| diag | pin_25C | H_R | 0 |
| diag | pin_25C | H_ig | -241826 |

*. . . and 49 more reference values in the case’s `expected`.*

## thermo

**The theory you need.** Consistency tests interrogate DATA, not models: the van Ness point test checks measured $(x, y, T, P)$ sets against the Gibbs–Duhem constraint (residuals in $\ln(\gamma_1/\gamma_2)$); Herington’s area test is its integral cousin. A dataset that fails is not “bad” – the test says WHERE and HOW MUCH, and the fit that follows should weight accordingly. Expect plain-language verdicts with severity, not a pass/fail stamp. *Full derivation: Theory Guide, the consistency chapter of the Props Guide.*

## bubble01_reaction_mass_boiling_point

tutorials/props/thermo/bubble01_reaction_mass_boiling_point




**What it does:** The boiling point of a reaction mass: the bubble point of the cooling-failure batch's contents (fictitious A + B → P in solvent S, ideal), the technical limit (MTT) of the scenario

**The story:** the BOILING POINT OF A REACTION MASS, which is what the thermal-safety methodology takes as the technical limit MTT of an open vessel at atmospheric pressure.

The mass is the one in tutorials/batch/reactor/batch26_cooling_failure (and batch27/batch28, the same reaction dosed): the FICTITIOUS reactants A and B, their product P and the inert solvent S. constant/ is a COPY of batch26's sealed constant/ (the same four authored records under the same manifest, the same thermoPhysPropDict; only the reactor's own constant/reactions is left out, because no property operation reads it), so the boiling point below is computed on exactly the thermodynamics the reactor case declares: an IDEAL liquid (Raoult, gamma = 1) over an ideal gas, at the reactor's pressure, 1.013 bar.

The composition is the reactor's at batch26's shipped cooling failure (t_f = 600 s; mole fractions of the trajectory row at 600 s, 39.3 % of B converted). The process-safety EduTool rewrites the four lines with the reactor contents at the failure instant the reader chooses, read from the engine's own trajectory, and reads T_bubble back. ONE KEY PER LINE: the GUI's override anchors on a key at the start of a line, so a composition written on one line is a knob that silently misses.

WHAT THIS IS AND IS NOT. The mixture is ideal BY DEFINITION (the substances are fictitious and the case declares it so), so it has no azeotrope: its bubble point lies between the normal boiling points of the components present – here between B's (345.01 K) and P's (479.99 K), and much nearer the light end, because the volatile components dominate the vapour pressure. It is a bubble point at the failure instant: as a runaway converts the mass, its composition, and so its boiling point, moves – upward here, as the light reactants become the heavy product.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | reactionMassBubblePoint | Cp_ig | 113.71926727 |
| diag | reactionMassBubblePoint | H_R | 0 |
| diag | reactionMassBubblePoint | H_ig | -304682.453301 |
| diag | reactionMassBubblePoint | H_real | -304682.453301 |
| diag | reactionMassBubblePoint | P | 101300 |
| diag | reactionMassBubblePoint | S_R | 0 |
| diag | reactionMassBubblePoint | S_ig | 401.345731157 |
| diag | reactionMassBubblePoint | S_real | 401.345731157 |
| diag | reactionMassBubblePoint | T | 330 |
| diag | reactionMassBubblePoint | T_bubble | 376.000570165 |
| diag | reactionMassBubblePoint | Z | 1 |
| diag | reactionMassBubblePoint | gamma | 1.07888124878 |
| diag | reactionMassBubblePoint | v_molar | 0.0270856136618 |

## heavyWater01_isotope_effect

tutorials/props/thermo/heavyWater01_isotope_effect




**What it does:** Heavy water against ordinary water: the vapour-pressure isotope effect and the boiling points, from the records

**The story:** D2O against H2O, from the catalogue records.

WHAT THIS CASE EVALUATES. The saturation pressure of heavy water (record 'heavyWater': the Wagner equation of Herrig, Thol, Harvey & Lemmon, J. Phys. Chem. Ref. Data 47, 043102 (2018), Eq. (1)) and of ordinary water (IAPWS-IF97, via 'pureFluids') at 25/50/75/100 degC, and the normal boiling point of each pure liquid as its bubble temperature at 101325 Pa. Nothing here is stated from memory: bin/curate/check_heavy_water.py recomputes the heavy-water side from the IAPWS R16-17(2018) Helmholtz equation (Maxwell criterion), reads the ordinary-water side from the IF97 saturation equation, and holds this case's published numbers (CSV cells, pinned by 'csv' rows) to both.

WHAT IS VERIFIED. That the engine reproduces the primary equation: the Wagner arithmetic to 1e-6, the equation against the full D2O EOS to 8e-5, the boiling point of D2O against the EOS's. WHAT IS NOT. No measured datum is compared here; the ordinary-water side is IF97 (itself a correlation of IAPWS-95); 'Hvap' is the engine's Watson estimate from the record's Tc, Tb and HvapTb (exact at Tb, an extrapolation elsewhere – NOT the EOS latent heat). The record carries no formation datum, so 'propertyPoint' (which prints H_ig) refuses heavyWater by name: this case uses the scan operation, which evaluates only what it is asked for. No mixture property of D2O + H2O is claimed (the isotope-exchange equilibrium H2O + D2O = 2 HDO is not modelled).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| csv | psat_isotope.csv | 0:Psat_water | 3.1697468500e+03 |
| csv | psat_isotope.csv | 0:Psat_heavyWater | 2.7362295800e+03 |
| csv | psat_isotope.csv | 0:Hvap_heavyWater | 4.5626456900e+04 |
| csv | psat_isotope.csv | 1:Psat_water | 1.2351270400e+04 |
| csv | psat_isotope.csv | 1:Psat_heavyWater | 1.1117923400e+04 |
| csv | psat_isotope.csv | 1:Hvap_heavyWater | 4.4343347400e+04 |
| csv | psat_isotope.csv | 2:Psat_water | 3.8595362700e+04 |
| csv | psat_isotope.csv | 2:Psat_heavyWater | 3.5826392400e+04 |
| csv | psat_isotope.csv | 2:Hvap_heavyWater | 4.2996612600e+04 |
| csv | psat_isotope.csv | 3:Psat_water | 1.0141797800e+05 |
| csv | psat_isotope.csv | 3:Psat_heavyWater | 9.6302716000e+04 |
| csv | psat_isotope.csv | 3:Hvap_heavyWater | 4.1577284500e+04 |
| csv | boiling_water.csv | 0:T_bubble | 3.7312430000e+02 |
| csv | boiling_heavyWater.csv | 0:T_bubble | 3.7455020600e+02 |
| *... and 9 more reference values in the case's `expected`.* | | | |

## phaseSplit01_freeze_water_ethanol

tutorials/props/thermo/phaseSplit01_freeze_water_ethanol




**What it does:** Freeze-concentration scan: a 10 mol% ethanol liquor cooled from 0 to -20 C, one SLE flash per kelvin – the ice point, the liquor concentrating along the freezing-point-depression identity

**The story:** THE PHASE SPLIT ALONG T (DEV.md 4c C118).

The same liquor flash21 freezes at ONE temperature (-15 C), cooled here from 0 to -20 C one kelvin at a time: at every grid point ONE isothermal flash, the flash21 unit's own solver (IsothermalFlash::solveCore), under flash21's own sealed records – water, ethanol, the curated NRTL pair, the C3 'phases ( liquor liquid; ice solid crystallizing )' declaration. Nothing is re-derived; the point at 258.15 K IS flash21's answer (the gate holds it to flash21's golden).

WHAT A STUDENT WATCHES. Above the liquor's freezing point the feed is one liquid and the declared ice stays subsaturated: the water activity gamma_w x_w (0.917) sits BELOW the saturation value exp(-dG_fus/RT) the crystal would need (1 at the triple point, falling as T falls). Between -8 and -9 C the two cross, ice APPEARS (announced, between the two grid values it lies between), and from there down the liquor concentrates exactly along the freezing-point-depression identity

gamma_w ∗ x_w = exp(-dHfus (1 - T/Tt) / (R T))

with every quantity from the record's own data (Hfus 6008 J/mol, triple point 273.16 K) – published at every point as 'a_water' beside 'aEq_water', equal to round-off wherever ice is present, and recomputed independently by check_phase_split_scan. By -20 C nearly 60 % of the feed is ice and the liquor is about a quarter ethanol.

NOT MODELLED, said plainly (the run says it too): nucleation and supercooling – every point is the equilibrium ceiling; an ethanol solid or a eutectic – no record declares one, so the liquor never solidifies completely in this model; solute inclusion in the ice; the liquid/solid heat-capacity term of dG_fus (announced by the record, |T - Tt| up to 20 K here). Every sub-273 K vapour pressure is an announced extrapolation of the Antoine fit (it cancels in the SLE identity: both phases share one Psat), and the NRTL pair is used below its fitted window (announced). No number here is anchored to a measured freezing-point datum: a STRUCTURAL witness, like flash21.

*What this witnesses (DEV.md 4c C118 stage 1, 2026-10-20): the phaseSplitScan props operation. At a FIXED overall composition (90/10 water/ethanol, moles) it sweeps temperature at fixed pressure and, at every grid point, solves the full phase split through the ONE flash home — IsothermalFlash::solveCore, the isothermalFlash unit's own solver, with the options the unit's own reader parses — under the case's own sealed records. The constant/ tree is flash21's, copied verbatim: the same water, the same ethanol, the same curated NRTL pair, the same C3 phases ( liquor liquid; ice solid crystallizing ) declaration. Nothing is re-derived, which is why the point at 258.15 K is flash21's answer (check_phase_split_scan arm (a) holds it to flash21's golden; its arms also hold single points on copies of flash16 and flash19 to those units' pH and V/F).*

*What a student watches. One isothermal flash per kelvin, 273.15 → 253.15 K:*

*Above the liquor's freezing point the feed is one liquid and the declared crystal stays subsaturated: the water activity γ_w·x_w (0.917) lies BELOW the saturation value exp(-ΔG_fus/RT) the crystal would need (1 at the triple point, falling as T falls). Between 265.15 and 264.15 K the two cross and ice APPEARS — announced by the run between the two grid values it lies between, recorded as an appeared event in the result block, pinned as diag first_ice. From there down the liquor concentrates exactly along the freezing-point-depression identity*

*γ_w · x_w = exp(-ΔH_fus (1 - T/T_t) / (R T)), ΔH_fus = 6008 J/mol, T_t = 273.16 K*

*with every quantity from water.dat's own data — published at every point as a_water beside aEq_water, equal to 1e-10 wherever ice is present, and recomputed independently by the gate from the record. By -20 °C 63 % of the feed is ice and the liquor is 27 mol% ethanol.*

*What is published. freeze.csv (T first, then solved, nPhases, the amount of each phase — L1, ice — their compositions, the flash's own a_water / aEq_water / solidFraction) and the top-level result block*

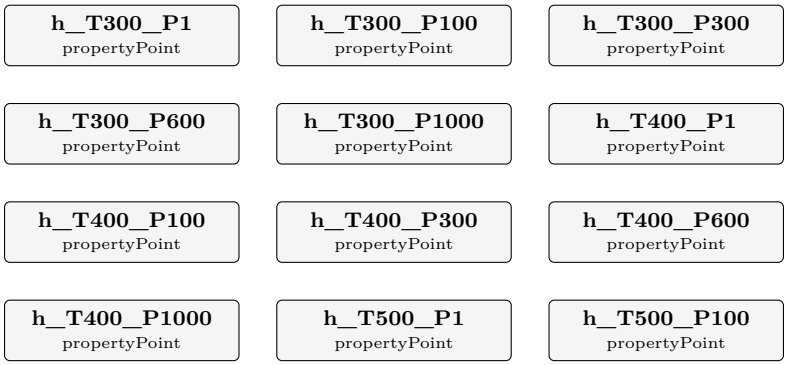
*phaseSplitScans (points, the event, the record-level announcements at the first point that raised each, what is not modelled). Every number in the block is a CSV cell or a diagnostic, so the csv and diag golden kinds pin all of it: the golden carries the diagnostics and ten authored CSV cells (rows 9, 15, 20).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | freeze | first_L1 | 273.15 |
| diag | freeze | first_ice | 264.15 |
| diag | freeze | last_L1 | 253.15 |
| diag | freeze | last_ice | 253.15 |
| diag | freeze | n_announcements | 4 |
| diag | freeze | n_events | 1 |
| diag | freeze | n_points | 21 |
| diag | freeze | n_refused | 0 |
| diag | freeze | n_solved | 21 |
| diag | freeze | points_L1 | 21 |
| diag | freeze | points_ice | 12 |
| csv | freeze.csv | 9:ice | 3.2042311316e-02 |
| csv | freeze.csv | 9:x_L1_ethanol | 1.0331030082e-01 |
| csv | freeze.csv | 15:ice | 4.7599462325e-01 |
| *... and 7 more reference values in the case's* `expected`*.* | | | |

## standardState01_hydrogen_enthalpy_TP

tutorials/props/thermo/standardState01_hydrogen_enthalpy_TP




**What it does:** h(T,P) of hydrogen on SRK, 300-1000 K by 1-1000 bar, 42 points: the ideal-gas part against the residual, for the standard-state EduTool

**The story:** One propertyPoint per node of the page's grid. Names are the keys the GUI reads: h_T<K>_P<bar>; h_converter is the green-ammonia converter's (T, P), h_converter_1bar its standard-pressure twin.

*Witness of the standard-state EduTool (∗Enthalpy, the standard state and equilibrium: what does the ° mean?∗). Forty-two propertyPoint operations on hydrogen, one per node of the page's grid — 300 to 1000 K by 100 K, at 1, 100, 300, 600 and 1000 bar — plus the green-ammonia converter's state (733.15 K, 150 bar) and its 1 bar twin. Each publishes the three numbers the page separates: H_ig, the ideal-gas enthalpy (a function of temperature alone, on the elements datum — hydrogen is an element, so it is zero at 298.15 K); H_R, the SRK residual at the real temperature and pressure; and their sum H_real.*

*A table gives a standard enthalpy with a degree sign. Does the sign mean 1 bar? Does it also mean 25 °C? If the process runs at 150 bar, is using a standard property not assuming the gas is ideal? The page answers in the order a student can follow — behaviour first, then the reference, only then the symbol — and this case supplies the behaviour.*

*∗ Along 1 bar, heating from 300 to 1000 K raises the enthalpy by 20.64 kJ/mol. At 300 K, compressing from 1 to 1000 bar at constant temperature raises it by 1.29 kJ/mol; at 1000 K by 1.69. The pressure effect is an order of magnitude smaller than the temperature effect in this range, and it is not zero. ∗ At the converter's state the ideal-gas rise from 298.15 K is 12.73 kJ/mol and the residual adds 0.23 kJ/mol — 1.8 % of the rise. At 1 bar and the same temperature the residual is 1.5 J/mol: the real gas at the standard pressure is a hair from the ideal gas, and still not the same definition.*

*constant/experimental/hydrogen_Hmolar_leachman2009_coolprop.csv carries the same 42 states from a fundamental equation of state fitted to measurements — Leachman, Jacobsen, Penoncello & Lemmon, ∗J. Phys. Chem. Ref. Data∗ 38 (2009) 721–748, as implemented in CoolProp 8.0.0 (Bell et al., ∗Ind. Eng. Chem. Res.∗ 53 (2014) 2498; MIT licence). It is RE-COMPUTED here by bin/curate/reference_h2_enthalpy.py (–check re-derives every row), not pasted from anywhere, and it sits in constant/experimental/ because that is the one location the GUI bundle and .gitignore admit for a CSV that is a case input; its header says it is a model's output, not a measurement.*

*The two agree on the ideal-gas part to a few joules per mole over the whole range. They do not agree on the residual: SRK is 6 % below the reference at 300 K and 1000 bar, a third below at 300 K and 100 bar, 11 % below at the converter. Measured before the model was chosen: Peng-Robinson is worse here — a third below at 1000 bar and the wrong sign at 100 bar — so the case runs SRK and the page draws both sets and states the gap. Nothing was tuned; hydrogen's anomalous acentric factor (-0.220) is what a cubic equation handles poorly, and the page says so rather than hiding it.*

*A propertyScan2D would do this in one operation and write a CSV — which the browser cannot read back. The GUI reads each operation's diagnostics, so the grid is 42 point evaluations whose names (h_T<K>_P<bar>) are the keys the page looks up. The golden pins every one of them.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | h_T300_P1 | Cp_ig | 28.885715 |
| diag | h_T300_P1 | H_R | 0.477247091364 |
| diag | h_T300_P1 | H_ig | 53.4333351109 |
| diag | h_T300_P1 | H_real | 53.9105822023 |
| diag | h_T300_P1 | P | 100000 |
| diag | h_T300_P1 | S_R | -0.00272994535243 |
| diag | h_T300_P1 | S_ig | 130.858662543 |
| diag | h_T300_P1 | S_real | 130.855932598 |
| diag | h_T300_P1 | T | 300 |
| diag | h_T300_P1 | Z | 1.00051987687 |
| diag | h_T300_P1 | gamma | 1.40417872785 |
| diag | h_T300_P1 | v_molar | 0.0249563553443 |
| diag | h_T300_P100 | Cp_ig | 28.885715 |
| diag | h_T300_P100 | H_R | 63.6322761263 |
| *… and 490 more reference values in the case's `expected`.* | | | |

## temperature01_gas_thermometer

tutorials/props/thermo/temperature01_gas_thermometer




**What it does:** The gas thermometer: why the ideal gas is a limit, not a substance

**The story:** A constant-volume gas thermometer reads a temperature off PV = nRT. A real gas does not obey that, so the reading depends on the PRESSURE it was taken at – and only in the limit P → 0 does it become the thermodynamic temperature. This case measures that departure, at the temperature the accompanying notes are about.

500.012 K is not a special temperature. It is the number the notes use to ask what the last digits of a temperature actually assert, and the scan is run there so the two are talking about the same state.

*> ∗"Thermodynamics is a funny subject. The first time you go through it, you > don't understand it at all. The second time you go through it, you think you > understand it, except for one or two small points. The third time you go > through it, you know you don't understand it, but by that time you are so > used to it, it doesn't bother you any more."∗ > > — attributed to Arnold Sommerfeld. PROVENANCE UNVERIFIED: Sommerfeld > published no such sentence; it reaches us by report, and this repository does > not quote a source it has not read. It is here as an epigraph, marked, and > must be either sourced or dropped before it appears in any published guide.*

*What does the 012 assert? Not "roughly five hundred kelvin" — three digits past the decimal point is a specific claim about the world. Whose claim, and resting on what?*

*The kelvin is no longer defined by a substance. It is defined by fixing the Boltzmann constant:*

*Not measured. Fixed. So a temperature is a conversion factor between energy and degrees, and nothing else. The triple point of water — which ∗defined∗ the kelvin for sixty-five years — became, overnight, a quantity that is measured, with an uncertainty.*

*That is the first dogma, and it is worth sitting with: the definition tells you what a kelvin ∗means∗ and gives you no way whatsoever to ∗measure∗ one.*

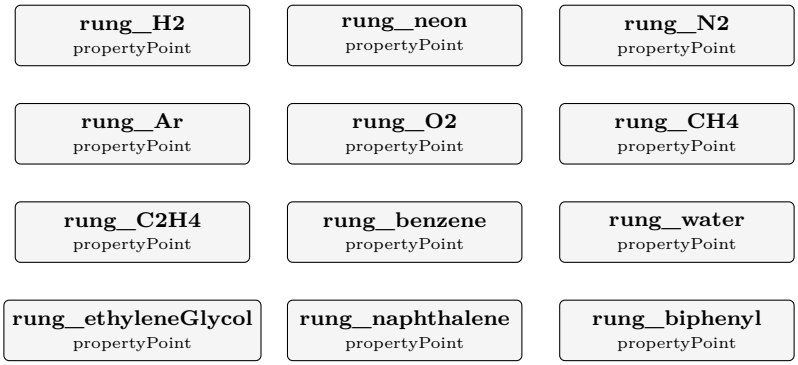
*Nobody puts the Boltzmann constant inside a thermometer. To read a temperature you need a ∗realisation∗: a chain of reproducible fixed points and an instrument that interpolates between them. In practice that is ITS-90, with a platinum resistance thermometer doing the interpolating. At 500 K you are between the triple point of water and the freezing point of zinc.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | Z_of_P | n_emptyCurves | 0 |
| diag | Z_of_P | n_failures | 0 |
| diag | Z_of_P | n_points | 25 |
| diag | nearly_ideal | Cp_ig | 29.6050536413 |
| diag | nearly_ideal | H_R | -0.00396437322822 |
| diag | nearly_ideal | H_ig | 5915.34999676 |
| diag | nearly_ideal | H_real | 5915.34603239 |
| diag | nearly_ideal | P | 1000 |
| diag | nearly_ideal | S_R | -4.73470040529e-05 |
| diag | nearly_ideal | S_ig | 245.041374669 |
| diag | nearly_ideal | S_real | 245.041327322 |
| diag | nearly_ideal | T | 500.012 |
| diag | nearly_ideal | Z | 1.00000474096 |
| diag | nearly_ideal | gamma | 1.39052286566 |
| *... and 13 more reference values in the case's `expected`.* | | | |

## temperature02_engineers_ladder

tutorials/props/thermo/temperature02_engineers_ladder




**What it does:** The range a chemical engineer actually works in

**The story:** THE LADDER A CHEMICAL ENGINEER ACTUALLY CLIMBS, and the engine CHECKS it rather than quoting it.

Each rung asks one question: at the temperature this substance's record calls its normal boiling point, what does its VAPOUR-PRESSURE CORRELATION say the pressure is? The answer should be one atmosphere – that is what a normal boiling point means – and how close it lands is a measure of the record, published rather than assumed.

THE OTHER WAY ROUND WOULD HAVE BEEN NICER AND DOES NOT WORK. Asking for 'T_bubble' at 1 atm solves for the temperature, which is the more direct question; 'BubblePoint::compute' carries its own initial guess and does not converge at 20 K, a gap this repository already has on record (see docs/design/held-out-pressure.md). Verifying the pressure at a declared temperature asks the same physics of the same correlation and is robust, so that is what runs here – said plainly rather than papered over.

WHY THE LADDER STOPS AT GLYCEROL, and it is not because the list got boring. Above roughly 600 K an ordinary organic does not have a normal boiling point to check: it CRACKS first, so there is no one-atmosphere equilibrium to ask about. The scale itself changes character there – ITS-90 stops interpolating on vapour pressure and gas thermometry altogether and realises its upper range on the FREEZING POINTS OF METALS, then on Planck radiation above the silver point. Those fixed points are declared values of the scale, not something this engine computes, and the page says so where it lists them.

DOWTHERM A IS ABSENT AND THAT IS A FINDING, not an oversight. The hot-oil fluid an engineer actually runs a 530 K loop on is curated here with 'role nonvolatile' and carries NO vapour-pressure correlation, so this case cannot ask it the question it asks the others. Named rather than quietly skipped: an omission a reader cannot see is the kind of gap this project refuses to leave.

*Eight rungs, from liquid hydrogen to boiling water, checked by the engine rather than quoted at you.*

*At the temperature each substance's record calls its normal boiling point, what does its vapour-pressure correlation say the pressure is?*

*It should be one atmosphere — that is what a normal boiling point ∗means∗. How close it lands measures the record, and the answer is published rather than assumed.*

*The substance every engineer trusts most is the worst rung on the ladder.*

*Water's Antoine fit declares Trange (273 373). Its normal boiling point is 373.15 K. The fit stops 0.15 K before the temperature everybody uses it at, and evaluated there it returns 1.026 atm instead of 1.000 — 2.6 % out, on the one substance whose boiling point every reader could recite.*

*Seven cryogens the reader has no intuition for land inside 1.4 %. The familiar one does not. Familiarity is not accuracy, and a declared validity range is worth more than a feeling.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | rung_H2 | Cp_ig | 27.3234201316 |
| diag | rung_H2 | H_R | 0 |
| diag | rung_H2 | H_ig | -7841.81404035 |
| diag | rung_H2 | H_real | -7841.81404035 |
| diag | rung_H2 | P | 101325 |
| diag | rung_H2 | Psat_H2 | 101850.765938 |
| diag | rung_H2 | S_R | 0 |
| diag | rung_H2 | S_ig | 55.7335657417 |
| diag | rung_H2 | S_real | 55.7335657417 |
| diag | rung_H2 | T | 20.39 |
| diag | rung_H2 | Z | 1 |
| diag | rung_H2 | gamma | 1.43739708567 |
| diag | rung_H2 | v_molar | 0.00167314969436 |
| diag | rung_neon | Cp_ig | 20.7861 |
| *… and 155 more reference values in the case's `expected`.* | | | |

## thermoTest

## atoms01_formula_parser

tutorials/props/thermoTest/atoms01_formula_parser




**What it does:** The shared formula parser, opened up: nested groups, both hydrate spellings, an ion charge that is not an element

**The story:** THE FORMULA PARSER, WITH ITS LID OFF.

The element balance is only as good as the thing that reads a formula, so that reader is ONE piece of code (src/thermo/ElementComposition) and ‘elementalComposition‘ is its glass-box surface: give it a formula, see the atoms it counted.

The formulas here are chosen to be the awkward ones, because those are where a hand-rolled parser goes wrong:

C2H6O plain, the control; Ca(OH)2 a nested group with a multiplier – 2 O and 2 H, not 1; CaSO4:2H2O a HYDRATE, the colon form: 6 O in total, 4 from the sulfate and 2 from the two waters; MgSO4:7H2O the same salt, colon form: Mg 1, S 1, O 11, H 14; MgSO4.7H2O THE SAME SALT AS A CHEMIST WRITES IT, and the one formula here that is REFUSED. The ASCII ’.’ is ambiguous, and Choupo resolved it in favour of the decimal subscript – see sepiolite below – so this spelling would silently count the seven waters into the oxygen: O 5.7 and H 2 where the salt has O 11 and H 14. A wrong element balance from a spelling nobody would flag by eye. It refuses and names both unambiguous forms; Mg2Si3O7.5OH:3H2O why the dot cannot simply BE the hydrate separator: this is curated sepiolite, and its 7.5 is a real half-integer; SO4-2 an ION: the charge is not an element and must not become one; (NH4)2SO4 a nested group of a nested group’s worth of hydrogen – 8 H.

A parser that silently mis-reads any of these produces an element balance that closes on the wrong atoms, which is worse than one that refuses.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | atoms | C2H6O_C | 2 |
| diag | atoms | C2H6O_H | 6 |
| diag | atoms | C2H6O_O | 1 |
| diag | atoms | CO2_C | 1 |
| diag | atoms | CO2_O | 2 |
| diag | atoms | ethanol_C | 2 |
| diag | atoms | ethanol_H | 6 |
| diag | atoms | ethanol_O | 1 |
| diag | atoms | formulas_available | 10 |
| diag | atoms | formulas_total | 11 |
| diag | atoms | water_H | 2 |
| diag | atoms | water_O | 1 |

## h_surface_identity

tutorials/props/thermoTest/h_surface_identity




**What it does:** The state identities of the canonical enthalpy surface: dh/dT == Cp_phase, the 298 K vaporisation anchor, and Kirchhoff on Hvap_state – pinned to round-off

**The story:** thermodynamics as a REGRESSION TEST.

A surface that anchors each phase leg on the 298.15 K formation datum and integrates the TARGET phase's own Cp obeys, by construction:

(1) dh_liq/dT = Cp_liquid(T) (2) dh_gas/dT = Cp_idealGas(T) (3) h_g(298.15) - h_liq(298.15) = Hvap(298.15) (the anchor) (4) d[h_g - h_liq]/dT = Cp_g - Cp_liq (Kirchhoff)

This op checks all four by central finite difference for benzene, toluene, water and ethanol at 350 K and REFUSES above 1e-6 relative error – so no future "convenient" leg (a Watson-slope liquid, a mixed-datum gas) can re-enter the balances unnoticed. The golden pins every per-species error.

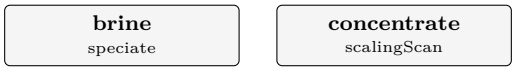
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | h_identity | benzene_anchor298_rel | 0.000e+00 |
| diag | h_identity | benzene_dhgas_vs_cpig_rel | 7.289e-11 |
| diag | h_identity | benzene_dhliq_vs_cpliq_rel | 4.729e-13 |
| diag | h_identity | benzene_kirchhoff_rel | 2.058e-10 |
| diag | h_identity | ethanol_anchor298_rel | 1.694e-16 |
| diag | h_identity | ethanol_dhgas_vs_cpig_rel | 3.826e-11 |
| diag | h_identity | ethanol_dhliq_vs_cpliq_rel | 1.907e-11 |
| diag | h_identity | ethanol_kirchhoff_rel | 1.287e-10 |
| diag | h_identity | toluene_anchor298_rel | 0.000e+00 |
| diag | h_identity | toluene_dhgas_vs_cpig_rel | 6.085e-11 |
| diag | h_identity | toluene_dhliq_vs_cpliq_rel | 1.854e-13 |
| diag | h_identity | toluene_kirchhoff_rel | 2.214e-10 |
| diag | h_identity | water_anchor298_rel | 6.529e-16 |
| diag | h_identity | water_dhgas_vs_cpig_rel | 3.627e-11 |

*... and 3 more reference values in the case's `expected`.*

## model4_speciation_scaling

tutorials/props/thermoTest/model4_speciation_scaling




**What it does:** RO membrane-scaling audit of a brackish groundwater: aqueous speciation (Davies) with pH SOLVED from electroneutrality (the feed charge imbalance is announced first) + saturation indices SI = log10(IAP/K), scanned over water recovery to r = 0.85 (concentrate = feed/(1-r)) TWICE: closed system (DIC conserved and concentrating) vs open to the atmosphere (a(CO2aq) pinned by Henry at pCO2 = 4.2e-4 atm, DIC a solved outcome – this CO2-rich water degasses, pH and calcite SI jump above the closed curve) – the calcite SI curves differ; that contrast is the lesson. Three analytic pins lock the physics: pure water at given pH 7 reproduces Kw (m_OH ~ 1e-7), a back-computed gypsum-saturated CaSO4 solution gives SI_gypsum ~ 0.00, and pure water with pH solved lands at exactly 7.000.

**The story:** thermoTest / model4_speciation_scaling

The SAME system {NaCl, CaSO4, water} in its THIRD role: a MULTI-ION broth. Here we no longer think of "NaCl" or "CaSO4" – the aqueous speciation engine reasons in the shared ions Na+, Cl-, Ca2+, SO4-2 (plus the neutral ion pair CaSO4aq), and reports SATURATION INDICES SI = log10(IAP/Ksp) for every solid the ions can form: gypsum (CaSO4.2H2O) and halite (NaCl).

Na+ comes from NaCl, Cl- from NaCl, Ca2+/SO4-2 from CaSO4 – but once dissolved there are no salt boundaries, only ions. This is why a "salt" cannot carry its own activity model: the ion pool is shared (the species/ tier of the tree).

MODEL: Pitzer-HMW is MANDATORY here. The ionic strength is I ~ 3.3 mol/kg (NaCl-dominated); the Davies law is only trustworthy to I ~ 0.5 m, so the ‘activityModel‘ is declared explicitly (no silent default) – with Davies the saturation indices at this I are quantitatively meaningless.

The brine is NaCl-dominated (3 mol/kg, near seawater) with CaSO4 (0.10 mol/kg) right at gypsum saturation (SI ~ 0). Charge balances exactly (Na + 2 Ca = Cl + 2 SO4), so pH is pinned at 7. The scan concentrates the brine (evaporation; r = fraction of water removed): gypsum scales first, at r ~ 0.035 (it starts saturated), halite only near r ~ 0.50. Note m_CaSO4aq ~ 0.005: the neutral ion pair is a real fraction of the calcium – "dissociation" is partial.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | brine | I | 3.26782 |
| diag | brine | SI_anhydrite | -0.22134 |
| diag | brine | SI_gypsum | -0.0221674 |
| diag | brine | SI_halite | -0.910511 |
| diag | brine | SI_mirabilite | -2.13799 |
| diag | brine | SI_thenardite | -2.98985 |
| diag | brine | aw | 0.890402 |
| diag | brine | gamma_CaSO4aq | 1 |
| diag | brine | m_CaSO4aq | 0.00466294 |
| diag | brine | pH | 7 |
| diag | concentrate | I_r0 | 3.26782 |
| diag | concentrate | I_rmax | 15.6285 |
| diag | concentrate | SI_anhydrite_r0 | -0.22134 |
| diag | concentrate | SI_anhydrite_rmax | 1.39807 |

*... and 26 more reference values in the case's `expected`.*

## transport

## transport01_gas_bench

tutorials/props/transport/transport01_gas_bench




**What it does:** Every transport correlation proves itself against its own anchor – theory or arithmetic, said which – and then all are asked the same question for four gases at 373 K; NOTHING here is measured

**The story:** transport01 – THE TRANSPORT CORRELATIONS, AS OBJECTS THAT PROVE THEMSELVES.

NOTHING IN THIS CASE IS MEASURED. Read that first. Choupo's catalogue carries no measured gas viscosity, thermal conductivity or diffusivity, so no number below is compared with an experiment. What the case shows is what CAN be shown without one, and it says which kind each check is:

THEORY an identity kinetic theory or the correlation's own author supplies – Eucken on a monatomic gas MUST give the Chapman-Enskog lambda = (15/4) R mu / M exactly; Wilke's phi_ii MUST be 1 for identical species. A wrong constant fails here. ARITHMETIC the correlation's own closed form at a stated point, written as a literal – Chung's Neufeld Omega at T∗ = 1, Fuller at round numbers, Sato-Riedel at T = Tb where its bracket ratio is 1 by construction, the fitted liquid forms reducing to Andrade. This checks the TRANSCRIPTION and the units, and the output says so rather than letting 0.0000 % read as a validation.

WHAT THE RUN DOES, IN TWO PARTS. (1) VERIFY. Every registered model in five families, plus the shared gas mixing rule, reproduces its own anchor; each prints its window, its PRIMARY citation and the anchor's kind. (2) COMPARE. N2, O2, CO2 and water vapour at 373.15 K, 1 bar: every registered gas-viscosity and gas-conductivity model for every component, each with its computable window verdict (Chung's T∗ against the Neufeld domain; Eucken's "polyatomic" by the record's own Cp), then the case's mixing rules – Wilke for mu, Wassiljewa / Mason-Saxena for k – at the equimolar composition (announced).

WHAT TO READ IN THE OUTPUT. Three things. The spread across models is 0.00 % – for BOTH gas families – and the bench says why: ONE model is registered per family. A spread over one is not agreement; Chapman-Enskog (Lennard-Jones sigma and eps/k, which no record carries yet) is the named next member. Chung ANNOUNCES, for each of the four components, that Vc was estimated from Zc(omega) and that the polar and association terms were dropped. Water is polar; the announcement is the only thing in the run that says the non-polar form was applied to it anyway. Eucken flags CO2 and water (and, by their Cp, N2 and O2 – diatomics are polyatomic) as APPROXIMATION HERE, because plain Eucken is exact only for a monatomic gas and this case has none.

WHAT THIS CASE DOES NOT DO. It does not say which model is right, and it does not compare with a measurement. The golden pins that the anchors, the values and the (absent) spread have not MOVED – nothing more. Record: docs/design/transport-correlations-as-objects.md.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bench | allPass | 1 |
| diag | bench | dev_Andrade | 5.35123586857e-08 |
| diag | bench | dev_Chung | 2.53636147631e-07 |
| diag | bench | dev_Eucken | 0 |
| diag | bench | dev_Fuller | 5.617814941e-08 |
| diag | bench | dev_SatoRiedel | 3.4111766641e-06 |
| diag | bench | dev_Vogel | 0 |
| diag | bench | dev_chemsepEq101 | 0 |
| diag | bench | dev_chemsepEq16 | 0 |
| diag | bench | dev_wilkePhi | 0 |
| diag | bench | k_Eucken_CO2 | 0.021231358218 |
| diag | bench | k_Eucken_N2 | 0.0297082727854 |
| diag | bench | k_Eucken_O2 | 0.0306615645526 |
| diag | bench | k_Eucken_water | 0.0250221542205 |
| *... and 21 more reference values in the case's* ***expected****.* | | | |

## transport02_chapman_enskog

tutorials/props/transport/transport02_chapman_enskog




**What it does:** Chapman-Enskog from Svehla's own force constants: seven gases at 300 K priced by two viscosity and two conductivity models, the spread a real disagreement for the first time; nothing measured here, the sigma/eps-k are Svehla's fits to pre-1962 viscosities, cited by page

**The story:** transport02 – CHAPMAN-ENSKOG FROM SVEHLA'S CONSTANTS, AND THE FIRST REAL SPREAD.

NOTHING IN THIS CASE IS MEASURED, except in the one sense the anchors state. The seven Lennard-Jones pairs (sigma, eps/k) on the records in constant/components/ are Svehla's LEAST-SQUARES FITS to viscosities measured before 1962 (his Table I(a) method 1, one row per gas, cited by report page in each record's own provenance block); Choupo holds none of those measurements and compares with none. What the run shows:

TWO models per gas family, so the spread column is a DISAGREEMENT and not the 0.00 % of transport01: Chung (corresponding states from Tc, Pc, omega – non-polar truncation, Vc estimated, both announced) beside Chapman-Enskog (kinetic theory over the fitted sigma, eps/k); plain Eucken beside the modified Eucken (Svehla eq. (2), the internal modes weighted 1.32). Water is where the two viscosities part furthest: Chung's non-polar form against a pair FITTED to steam data. Argon is the control: monatomic, so Eucken and the modified Eucken coincide to the extent the record's Cp polynomial returns 5R/2 (it does, to four figures at 300 K; lambda" is the residual) and the polyatomic flag stays silent. Every Chapman-Enskog component is ANNOUNCED with the ORIGIN of both constants (FITTED to measured transport data, Svehla method 1) and the page they were read from; an estimated pair would say ESTIMATED in the same place. The window verdict is T∗ = kT/eps against the Neufeld fit's 0.3 <= T∗ <= 100; water at 300 K sits at T∗ = 0.37, inside, near the low edge. The comparison point is 300 K, 1 bar – the 300 K row of Svehla's Table III – so a reader can set the printed viscosities beside his (report pages 31 Ar, 56 CH4, 58 CO2, 74 H2, 75 H2O, 90 N2, 98 O2) and see the residual the anchor row measures: the Neufeld 1972 fit of the collision integral against the Hirschfelder 1954 tables he interpolated, 0.1-0.8 % across these gases, 1.8 % for water.

THE ANCHORS. ChapmanEnskog: ARITHMETIC – with Svehla's own M, sigma, eps/k for N2 the engine returns his printed eta at 300 K (report page 90) within a DECLARED 1 % (0.40 % observed); it checks the constant 26.693, the unit chain and the shared collision-integral home, nothing about any gas. modifiedEucken: THEORY – a monatomic gas must give exactly (15/4) R mu / M – after reproducing Svehla's printed N2 conductivity from his own eta and Cp/R through eq. (2). The bench prints the kind.

WHAT THIS CASE DOES NOT DO. It does not say which viscosity is right for water or for anything else, and it does not compare with a measurement. The records are CASE-LOCAL (adopted, on the edwards02 precedent): the public catalogue carries no lennardJones block until Vitor promotes one; bin/curate/propose_lennard_jones.py drafts them into data/local/ from the committed transcription bin/curate/svehla1962/. Binary diffusivity by Chapman-Enskog is NOT here: the Omega(1,1)∗ fit is not citable from any source in the tree. Record: docs/design/transport-correlations-as-objects.md, section 8.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | bench | allPass | 1 |
| diag | bench | dev_Andrade | 5.35123586857e-08 |
| diag | bench | dev_ChapmanEnskog | 0.00400952242196 |
| diag | bench | dev_Chung | 2.53636147631e-07 |
| diag | bench | dev_Eucken | 0 |
| diag | bench | dev_Fuller | 5.617814941e-08 |
| diag | bench | dev_SatoRiedel | 3.4111766641e-06 |
| diag | bench | dev_Vogel | 0 |
| diag | bench | dev_chemsepEq101 | 0 |
| diag | bench | dev_chemsepEq16 | 0 |
| diag | bench | dev_modifiedEucken | 0 |
| diag | bench | dev_wilkePhi | 0 |
| diag | bench | k_Eucken_Ar | 0.0178116749867 |
| diag | bench | k_Eucken_CH4 | 0.0321131371617 |
| *… and 37 more reference values in the case's `expected`.* | | | |

# Category IV: Batch and time-dependent (choupoBatch)

$\mathrm{d}\mathbf{Y}/\mathrm{d}t = f(\mathbf{Y})$ integrated in real time, with recipes for discrete events; trajectories, not steady states.

## adsorber

**The theory you need.** Adsorption in a vessel or a bed: the isotherm $q^*(T,p)$ (Langmuir $q = q_{\max}bp/(1+bp)$, van't Hoff $b(T)$) is the curated PAIR datum; the LDF rate $\mathrm{d}q/\mathrm{d}t = k\,(q^* - q)$ carries the EQUIPMENT datum $k$ (this particle, this bed – it lives in the case, never the catalogue). Work the cases as a staircase. `batch09`: a closed vessel relaxes to the inventory equilibrium – one algebraic root, pinned independently of the run. `batch10`: two species compete for the same sites (extended Langmuir) – the selectivity is an equilibrium statement, and the model's Gibbs–Duhem inconsistency for unequal $q_{\max}$ is declared, not hidden. `batch11`: in the Henry limit the whole vessel is LINEAR and the integrator must match $q_\infty(1 - e^{-k_{\mathrm{eff}}t})$ to $10^{-8}$ – the analytic gate. `batch12`: declaring the headspace pressure AND the inventory over-specifies a closed ideal-gas vessel – the contradiction FATALs; $P$ is a result. `batch13`: the fixed bed – the stoichiometric time $t_{\mathrm{st}} = (L/u)(\varepsilon + \rho_b q^*/c_{\mathrm{in}})$ is announced BEFORE integration, breakthrough spreads around it, and the constant-$u$ closure's fabricated carrier is printed, pinned and discounted: the DECLARED model residual is itself the lesson. `batch14`/`batch15` are the numerical hygiene: pure transport vs the analytic advection–dispersion solution, and the mesh study that earns $N = 100$. *Full derivation: Theory Guide, the adsorption chapter.*

## batch09_adsorber_co2

tutorials/batch/adsorber/batch09_adsorber_co2




**What it does:** Closed isothermal CO2 uptake on zeolite 13X (LDF, solid film): the vessel relaxes to the inventory equilibrium p_inf = 7229.13 Pa, q_inf = 1.9708 mol/kg

**The story:** closed, isothermal gas–solid uptake vessel.

2 mol of pure CO2 in a 0.01 m3 ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Linear-driving-force kinetics on the solid film (dq/dt = k∗(q∗ - q), k = 0.05 1/s, equipment data, this case's sample) against the competitive-Langmuir equilibrium record of the pair catalogue (here a single pair, CO2/zeolite13X: q_max = 5.0 mol/kg, b_298 = 9.0 1/bar, dH_ads = -45 kJ/mol).

The final state is FIXED BY INVENTORY, not by kinetics: with n_gas = p V/(R T) and q = q_max b (p/1e5)/(1 + b p/1e5),

p V/(R T) + m_ads ∗ q_max ∗ b ∗ (p/1e5)/(1 + b∗(p/1e5)) = n0

whose root at 298.15 K is (a2_vectors.py, 10 digits)

p_inf = 7229.128209 Pa q_inf = 1.970838025 mol/kg n_gas = 0.02916197467 mol uptake = 1.970838025 mol (98.54 %) tau (linearised at the equilibrium) = 0.4768 s

The run must reproduce these to the golden tolerances – an independent analytic anchor, not a mirror of the code. The ODE state is ONLY q; n_gas is algebraic (n_tot - m_ads∗q), so the per-species closure column closure_worst_rel in trajectory.csv is exact by construction (<= 1e-13). Log: the unit announces the ideal-gas assumption, the initialLoading default, the RK4 stiffness bound and the isosteric-KPI label at start-up.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | adsorber | P_final | 7229.128209 |
| kpi | adsorber | Q_ads_isosteric_kJ | -88.68771114 |
| kpi | adsorber | T_final | 298.15 |
| kpi | adsorber | closure_worst_rel | 0 |
| kpi | adsorber | n_CO2_final | 2.916197467e-05 |
| kpi | adsorber | q_CO2_final | 1.970838025 |
| kpi | adsorber | t_end | 600 |
| kpi | adsorber | totalMoles_final | 2.916197467e-05 |
| kpi | adsorber | uptake_CO2_mol | 1.970838025 |
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 0 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| *... and 12 more reference values in the case's `expected`.* | | | |

## batch10_adsorber_co2_n2

tutorials/batch/adsorber/batch10_adsorber_co2_n2




**What it does:** Closed isothermal competitive CO2/N2 uptake on zeolite 13X: extended-Langmuir site competition drives the equilibrium selectivity S = 71.43

**The story:** COMPETITIVE binary uptake, closed vessel.

0.5 mol CO2 + 1.5 mol N2 in a 0.01 m3 ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Both species carry an isotherm record (extended Langmuir: CO2 q_max 5.0 / b_298 9.0 1/bar; N2 3.5 / 0.18 1/bar) and both carry a declared LDF constant – so both integrate, and the strong adsorbate suppresses the weak one through the SHARED site denominator 1 + Sum_j b_j p_j.

The final state solves the 2-species inventory system n_gas_i + m_ads ∗ q_i(p_CO2, p_N2) = n0_i, p_i = n_gas_i R T / V whose root (a2_vectors.py, Newton residual 0):

CO2: p_inf = 0.01611568794 bar q = 0.4934990047 mol/kg N2 : p_inf = 1.802652166 bar q = 0.7728183086 mol/kg P_total = 181876.7854 Pa selectivity S = (q_CO2/q_N2)/(p_CO2/p_N2) = 71.42857143 = (q_max∗b)_CO2/(q_max∗b)_N2 (extended-Langmuir identity)

The golden pins the COMPETITIVE path: get the shared denominator wrong and every one of these numbers moves. closure_worst_rel stays exact by construction (state = q only; n_gas algebraic).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | adsorber | P_final | 181876.7854 |
| kpi | adsorber | Q_ads_isosteric_kJ | -36.1181845604 |
| kpi | adsorber | T_final | 298.15 |
| kpi | adsorber | closure_worst_rel | 0 |
| kpi | adsorber | n_CO2_final | 6.500995276e-06 |
| kpi | adsorber | n_N2_final | 7.271816914e-04 |
| kpi | adsorber | q_CO2_final | 0.4934990047 |
| kpi | adsorber | q_N2_final | 0.7728183086 |
| kpi | adsorber | t_end | 600 |
| kpi | adsorber | totalMoles_final | 0.000733682686703 |
| kpi | adsorber | uptake_CO2_mol | 0.4934990047 |
| kpi | adsorber | uptake_N2_mol | 0.7728183086 |
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_N_closure_rel | 0 |
| *... and 16 more reference values in the case's `expected`.* | | | |

## batch11_adsorber_henry

tutorials/batch/adsorber/batch11_adsorber_henry




**What it does:** Closed isothermal Henry-limit uptake (linear isotherm): the coupled gas+solid system is LINEAR and the run must match q(t) = q_inf∗(1 - exp(-k_eff t)) to 1e-8

**The story:** the EXACT-solution gate (G1a of the A2 spec).

1 mol of CO2 in a 0.01 m3 ideal-gas headspace over 1 kg of a SYNTHETIC Henry-limit sorbent (linear isotherm q∗ = K_H p, K_H = 0.4 mol/(kg bar) at 298.15 K – a teaching record, vendored case-local). With a linear isotherm and one species the coupled vessel is LINEAR and closes in analytic form:

alpha = K_H[mol/(kg Pa)] ∗ R∗T/V = 0.9915828118 1/kg k_eff = k ∗ (1 + alpha∗m_ads) = 0.09957914059 1/s q_inf = alpha∗n0 / (1 + alpha∗m) = 0.4978868094 mol/kg q(t) = q_inf ∗ (1 - exp(-k_eff∗t)) (q0 = 0)

q(5) = 0.1952670621 q(10) = 0.3139520079 q(30) = 0.4727835287 q(60) = 0.4966211107 mol/kg

The pedagogical point the closed vessel adds over the fixed-q∗ packet formula: k_eff > k – gas depletion ACCELERATES the approach to equilibrium (the packet's q∗ = const law is the V_gas → infinity limit of this same solution). The trajectory rows at t = 5/10/30/60 s must match the table above to 1e-8 mol/kg (RK4 dt = 0.01 s); the golden pins the t = 60 s state.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | adsorber | P_final | 124785.4636 |
| kpi | adsorber | Q_ads_isosteric_kJ | -9.93242221329 |
| kpi | adsorber | T_final | 298.15 |
| kpi | adsorber | closure_worst_rel | 0 |
| kpi | adsorber | n_CO2_final | 5.033788893e-04 |
| kpi | adsorber | q_CO2_final | 0.4966211107 |
| kpi | adsorber | t_end | 60 |
| kpi | adsorber | totalMoles_final | 5.033788893e-04 |
| kpi | adsorber | uptake_CO2_mol | 0.4966211107 |
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 0 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |

*. . . and 12 more reference values in the case's `expected`.*

## batch12_adsorber_refused_P

tutorials/batch/adsorber/batch12_adsorber_refused_P




**What it does:** REFUSAL lesson: declared initial.P (10 bar) contradicts n∗R∗T/V (4.958 bar) – the honest cross-check FATALs; P of a closed ideal-gas headspace is a result, not an input

**The story:** the REFUSAL lesson (deliberate, must FAIL).

Identical to batch09_adsorber_co2 EXCEPT initial.P declares 10 bar while n∗R∗T/V = 4.9579 bar. P of a closed ideal-gas headspace is a RESULT (n, T, V fix it – Duhem); a declared P is honoured only as an honest cross-check, and a relative deviation > 1e-3 REFUSES, printing both numbers. Expected FATAL (exit != 0):

batchAdsorber 'adsorber': declared P inconsistent with n∗R∗T/V – initial.P = 1e+06 Pa (10 bar) but n∗R∗T/V = 495791 Pa (4.95791 bar), relative deviation 1.01698 > 1e-3. ...

Marked .expect-nonconvergence: counted EXPECTED-FAIL by runTests, never green, never recordable. Base case (the passing twin):

2 mol of pure CO2 in a 0.01 m3 ideal-gas headspace over 1 kg of regenerated zeolite 13X at 298.15 K. Linear-driving-force kinetics on the solid film (dq/dt = k∗(q∗ - q), k = 0.05 1/s, equipment data, this case's sample) against the competitive-Langmuir equilibrium record of the pair catalogue (here a single pair, CO2/zeolite13X: q_max = 5.0 mol/kg, b_298 = 9.0 1/bar, dH_ads = -45 kJ/mol).

The final state is FIXED BY INVENTORY, not by kinetics: with n_gas = p V/(R T) and q = q_max b (p/1e5)/(1 + b p/1e5),

p V/(R T) + m_ads ∗ q_max ∗ b ∗ (p/1e5)/(1 + b∗(p/1e5)) = n0

whose root at 298.15 K is (a2_vectors.py, 10 digits)

p_inf = 7229.128209 Pa q_inf = 1.970838025 mol/kg n_gas = 0.02916197467 mol uptake = 1.970838025 mol (98.54 %) tau (linearised at the equilibrium) = 0.4768 s

The run must reproduce these to the golden tolerances – an independent analytic anchor, not a mirror of the code. The ODE state is ONLY q; n_gas is algebraic (n_tot - m_ads∗q), so the per-species closure column closure_worst_rel in trajectory.csv is exact by construction (<= 1e-13). Log: the unit announces the ideal-gas assumption, the initialLoading default, the RK4 stiffness bound and the isosteric-KPI label at start-up.

## batch13_breakthrough_co2

tutorials/batch/adsorber/batch13_breakthrough_co2




**What it does:** CO2 15%/He breakthrough on zeolite 13X – 1-D isothermal fixed bed (A3): t_st = 3040.5 s announced pre-run, t_b(5%) = 2890.7 s, ledger closure at machine level

**The story:** CO2/He breakthrough on a zeolite-13X bed.

THE A3 anchor case (spec 2026-07-12): a 0.5 m isothermal fixed bed of zeolite 13X, fed 15% CO2 in helium at u = 0.05 m/s superficial, 1 bar, 298 K. Helium carries NO isotherm record on 13X → announced inert CARRIER, transported but never adsorbed, closed by difference from the constant c_tot = P/(RT). At 15% CO2 this is NOT a dilute approximation: the constant-(u, P, T) closure fabricates He at the net uptake rate and the campaign material balance SHOWS that residual (KPIs carrier_fabricated_mol, physical_mass_closure_rel) – the named error bites, by design (forum #119); the genuinely clean transport anchor is batch14. The engine announces, BEFORE integrating, where the front must arrive:

q*(c_in) = 5*9*0.15/(1 + 9*0.15) = 2.872340426 mol/kg R_f = eps + rho_b q* / c_in = 304.0512509 t_st = (L/u) R_f = 3040.512509 s u_sh = u/R_f = 1.644459605e-4 m/s

Independent gates this run must reproduce (anchors_a3.py, FV N = 100):

t_b(5%) = 2890.669098 s t_50 = 3036.448293 s t_b(95%) = 3204.166058 s INT(1 - c_out/c_in)dt = t_st to 2.2e-10 (telescopic FV conservation) ledger mass closure <= 1e-10 (IN/OUT integrated as rows of the SAME ODE)

Numerics: conservative FV, nCells = 100 DECLARED (do the 25/50/100/200 mesh study – batch15 publishes it: first-order upwind, observed order ~1.03-1.05), Danckwerts inlet as the imposed face flux u*c_in, outlet dc/dz = 0. u, P, T are DECLARED constants of this phase, and the price is said OUT LOUD: the real velocity drops ~15% across the uptake zone, so the constant-u closure FABRICATES carrier (He) at exactly the net uptake rate (9.19 mol here) – announced in the run header, pinned as the KPI carrier_fabricated_mol, discounted in the campaign inventory; Ergun and the energy balance land in A4 (the energy gap is NAMED).

Log: run header prints the hypotheses, the derived rho_p, Pe = 250, the Gershgorin step bound term by term and the stiffness verdict; profiles land in <t>/bed.profile every 100 s; trajectory.csv carries c_out, y_out, qbar and the per-species ledger closure.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_adsorption_seg0_kJ | -413.617021183 |
| kpi | bed | E_adsorption_total_kJ | -413.617021183 |
| kpi | bed | P_final | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 298 |
| kpi | bed | c_out_over_cin_final_CO2 | 0.999999989455 |
| kpi | bed | carrier_fabricated_mol | 9.19148935961 |
| kpi | bed | integral_anchor_CO2 | 3040.51245606 |
| kpi | bed | mass_closure_CO2 | 8.94231742202e-15 |
| kpi | bed | n_CO2_final | 1.21079552014e-05 |
| kpi | bed | n_He_final | 6.86117461711e-05 |
| kpi | bed | physical_mass_closure_rel | 0.108446875314 |
| kpi | bed | qbar_CO2_final | 2.87234042488 |
| kpi | bed | retention_factor_CO2 | 304.051250948 |
| *... and 34 more reference values in the case's `expected`.* | | | |

## batch14_transport_only

tutorials/batch/adsorber/batch14_transport_only




**What it does:** Pure transport (k = 0) on the batch13 bed, N = 200: t_50 vs the Ogata-Banks analytic ADE solution (gate G1) under fixed RK4 with the Gershgorin bound printed term by term

**The story:** the k = 0 transport gate (A3 spec, gate G1).

Same bed, same feed, same mesh family as batch13 – but the LDF constant is DECLARED ZERO: the solid is frozen and CO2 crosses the bed as a pure tracer. What is left is exactly the advection-dispersion equation with a Danckwerts inlet, and THAT has an analytic yardstick (Ogata-Banks, semi-infinite ADE; the finite-domain difference is O(1/Pe) = 0.4% at Pe = 250):

u_i = u/eps = 0.125 m/s D_i = Dax/eps = 2.5e-4 m2/s c/c0 (L, 3.5 s) = 0.07340274358 c/c0 (L, 4.0 s) = 0.5178057707 c/c0 (L, 4.5 s) = 0.913645218 t_50 analytic = 3.984074227 s

FV yardsticks pinned by this case (anchors_a3.py, N = 200):

t_50 numeric = 3.975611088 s (0.21% below analytic – first-order upwind; the deviation HALVES with each mesh doubling: 1.40% at N=25, 0.74% at 50, 0.40% at 100, 0.21% at 200, 0.11% at 400) INT(1 - c_out/c_in)dt = eps L / u = 4.0 s (telescopic conservation: exact identity once saturated) ledger closure ~ 1e-14 (k = 0)

A declared zero is not a silent default: the unit announces "k = 0 DECLARED: transport-only diagnostic, the solid is frozen (gate G1)". Note the He carrier fabrication of batch13 vanishes here (no uptake): carrier_fabricated_mol = 0.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_adsorption_seg0_kJ | 0 |
| kpi | bed | E_adsorption_total_kJ | 0 |
| kpi | bed | P_final | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 298 |
| kpi | bed | c_out_over_cin_final_CO2 | 0.999999999893 |
| kpi | bed | carrier_fabricated_mol | 0 |
| kpi | bed | integral_anchor_CO2 | 3.99999999999 |
| kpi | bed | mass_closure_CO2 | 4.09039795713e-14 |
| kpi | bed | n_CO2_final | 1.21079552058e-05 |
| kpi | bed | n_He_final | 6.86117461666e-05 |
| kpi | bed | physical_mass_closure_rel | 0 |
| kpi | bed | qbar_CO2_final | 0 |
| kpi | bed | t_50_CO2 | 3.97561121931 |

*... and 28 more reference values in the case's `expected`.*

## batch15_mesh_study

tutorials/batch/adsorber/batch15_mesh_study




**What it does:** Mesh-study member N = 25 of the batch13 breakthrough: t_b(5%) converges monotonically 2732 → 2839 → 2891 → 2916 s with observed order 1.03-1.05; full table in reports/meshStudy.md

**The story:** the declared-mesh lesson (A3 spec, gate G6).

The axial mesh is a DECLARED modelling choice (nCells; absent = FATAL), and this case is the receipt: the batch13 breakthrough run on the 25/50/100/200 family, everything else identical. First-order upwind advection dominates the error, so t_b(5%) must converge MONOTONICALLY with observed order p ~ 1:

N = 25 t_b5 = 2732.027658 s (this case) N = 50 t_b5 = 2838.542518 s N = 100 t_b5 = 2890.669098 s (batch13, the production mesh) N = 200 t_b5 = 2915.764796 s p(25/50/100) = 1.031 p(50/100/200) = 1.055 Richardson (p = 1): t_b5(inf) ~ 2940.86 s → N = 100 sits 1.7% low

The full table, the four overlaid breakthrough curves and the verdict are PUBLISHED in reports/meshStudy.md (generated from the four runs; the report header cites mesh, tolerances and integrator). The ledger mass closure stays <= ~1e-11 on EVERY mesh – conservation is telescopic in the FV form, it does not improve with refinement because it was never discretisation-limited (gate G3).

This case pins the N = 25 member; its numbers are DELIBERATELY the worst of the family (t_b5 5.8% below batch13) – coarse-mesh optimism made visible, the point of the lesson.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_adsorption_seg0_kJ | -413.616606535 |
| kpi | bed | E_adsorption_total_kJ | -413.616606535 |
| kpi | bed | P_final | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 298 |
| kpi | bed | c_out_over_cin_final_CO2 | 0.999973635802 |
| kpi | bed | carrier_fabricated_mol | 9.19148014522 |
| kpi | bed | integral_anchor_CO2 | 3040.50945345 |
| kpi | bed | mass_closure_CO2 | 7.40535661511e-15 |
| kpi | bed | n_CO2_final | 1.21079316034e-05 |
| kpi | bed | n_He_final | 6.86117697691e-05 |
| kpi | bed | physical_mass_closure_rel | 0.108446766597 |
| kpi | bed | qbar_CO2_final | 2.87233754538 |
| kpi | bed | retention_factor_CO2 | 304.051250948 |
| *... and 34 more reference values in the case's `expected`.* | | | |

## batch16_ergun_profile

tutorials/batch/adsorber/batch16_ergun_profile




**What it does:** A4 F1: pure-He Ergun profile against the closed-form P-squared solution

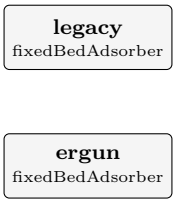
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | P_first_cell_Pa | 99999.4670134 |
| kpi | bed | P_mid_cell_Pa | 99946.1540012 |
| kpi | bed | P_last_cell_Pa | 99893.8796441 |
| kpi | bed | P_out_boundary_Pa | 99893.3459257 |
| kpi | bed | carrier_fabricated_mol | 0 |
| kpi | bed | physical_mass_closure_rel | 0 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | bed | E_adsorption_seg0_kJ | 0 |
| kpi | bed | E_adsorption_total_kJ | 0 |
| kpi | bed | P_final | 99946.6820773 |
| kpi | bed | T_final | 298.15 |
| kpi | bed | deltaP_first_to_out_Pa | 106.12034902 |
| kpi | bed | gasInventory_mol | 0.0806360746763 |
| kpi | bed | n_He_final | 8.06360746763e-05 |

*. . . and 26 more reference values in the case's `expected`.*

## batch17_dilute_ergun_limit

tutorials/batch/adsorber/batch17_dilute_ergun_limit




**What it does:** A4 F2: 400 ppm uptake, Ergun must approach the A3 constant-velocity limit

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 5.32078747864e-13 |
| kpi | campaign | element_He_closure_rel | 0.000139137956493 |
| kpi | campaign | element_O_closure_rel | 5.32078747864e-13 |
| kpi | campaign | element_worst_closure_rel | 0.000139137956493 |
| kpi | campaign | element_worst_closure_vs_declared_rel | 5.32078747864e-13 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 2 |
| kpi | campaign | mass_closure_rel | 0.000138528772162 |
| kpi | campaign | mass_kg_external_out | 0.161636709221 |
| kpi | campaign | mass_kg_final | 0.000558961907879 |
| kpi | campaign | mass_kg_initial | 0.162173202362 |
| kpi | campaign | mass_residual_kg | 2.24687671715e-05 |
| kpi | campaign | moles_kmol_external_out | 0.0403339312462 |
| *... and 63 more reference values in the case's `expected`.* | | | |

## batch18_ergun_conservation

tutorials/batch/adsorber/batch18_ergun_conservation




**What it does:** A4 F3: 15 percent CO2 with explicit carrier continuity and Ergun velocity

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_adsorption_seg0_kJ | -413.699637907 |
| kpi | bed | E_adsorption_total_kJ | -413.699637907 |
| kpi | bed | P_final | 100047.159082 |
| kpi | bed | P_first_cell_Pa | 100092.418138 |
| kpi | bed | P_last_cell_Pa | 100001.886961 |
| kpi | bed | P_mid_cell_Pa | 100047.162776 |
| kpi | bed | P_out_boundary_Pa | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 298.15 |
| kpi | bed | c_out_over_cin_final_CO2 | 1.00000763107 |
| kpi | bed | carrier_fabricated_mol | -6.8212102633e-13 |
| kpi | bed | deltaP_first_to_out_Pa | 92.4181379529 |
| kpi | bed | integral_anchor_CO2 | 3028.46636949 |
| kpi | bed | mass_closure_CO2 | 1.84877618489e-14 |

*. . . and 44 more reference values in the case's `expected`.*

## batch19_feed_switch_purge

tutorials/batch/adsorber/batch19_feed_switch_purge




**What it does:** A5 feed switch: CO2/He load to 3600 s (batch13 anchors), then pure-He purge to 9000 s – amended feed commitment ledgered, frozen breakthrough KPIs, desorption duty as the same state difference

**The story:** THE A5 WITNESS: load, then switch the feed.

batch13's bed (CO2 15% in He on zeolite 13X, the A3 anchor), run through a TWO-STEP cycle in one campaign:

0 .. 3600 s LOAD – identical to batch13's feed; the front arrives on the batch13 anchors (t_5 = 2890.7, t_50 = 3036.4, t_95 = 3204.2 s) 3600 s SWITCH – recipe setParameter bed.feed.CO2 = 0: the feed becomes pure He (concentration- swing regeneration) 3600 .. 9000 s PURGE – $q*(c \to 0) = 0$, the LDF drives the bed to desorb; the favourable (Langmuir) isotherm makes desorption the UNFAVOURABLE direction – watch the long tail on c_out

What the switch does, and where each consequence is VISIBLE:

The feed commitment is DECLARED matter (the campaign datum owns A∗u∗c_in∗(endTime - t) of future feed). Re-declaring the feed at t = 3600 s AMENDS that commitment: the un-fed CO2 leaves the datum and the extra He enters it, priced and recorded as 'feedAmendment' ledger records against the external boundary – the campaign mass/element balances read the records and still close (KPIs mass_kg_amend_in/out, element_worst_closure_vs_declared_rel ~ 1e-14).

The breakthrough sampler's reference basis (c_out/c_in) dies with the old feed: crossings and integral FREEZE at their pre-switch values (announced), so t_5/t_50/t_95 here must MATCH batch13's anchors – the frozen claim describes the first feed step. The desorption wave lives on in the trajectory's c_out_CO2 / qbar_CO2 columns.

The adsorption duty stays an EXACT state difference on the adsorbed inventory: E_adsorption = dH_ads ∗ (n_ads(end) - n_ads(0)). batch13 (load only) ledgers -413.6 kJ; here the purge takes part of the loading back OUT, so |E| is SMALLER – desorption is endothermic and the ledger shows the heat returning, with no new formula anywhere.

u, P, T stay DECLARED CONSTANTS: asking the bed for a thermal swing (feed.T), a flow transient (u) or a blowdown (P) is REFUSED by name (fired by bin/curate/check_feed_switch.py, batch12-style honesty).

The carrier-fabrication caveat of the constant-(u, P, T) closure applies unchanged (forum #119): the declared residual explains the He element closure, shown and never discounted.

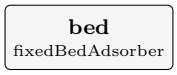
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_adsorption_seg0_kJ | -23.659607554 |
| kpi | bed | E_adsorption_total_kJ | -23.659607554 |
| kpi | bed | P_final | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 298 |
| kpi | bed | carrier_fabricated_mol | 0.525769056755 |
| kpi | bed | integral_anchor_CO2 | 3040.50572088 |
| kpi | bed | mass_closure_CO2 | 1.23888355951e-14 |
| kpi | bed | n_CO2_final | 3.21638319252e-07 |
| kpi | bed | n_He_final | 8.03980630532e-05 |
| kpi | bed | physical_mass_closure_rel | 0.00289489594687 |
| kpi | bed | qbar_CO2_final | 0.164302830236 |
| kpi | bed | t_50_CO2 | 3036.44839075 |
| kpi | bed | t_breakthrough_5pct_CO2 | 2890.66919752 |

*... and 42 more reference values in the case's `expected`.*

## batch20_thermal_breakthrough

tutorials/batch/adsorber/batch20_thermal_breakthrough




**What it does:** A5-T1 adiabatic CO2/He breakthrough (ergun, one T per cell): thermal wave + earlier front vs batch18, anchors u_th and DT_ad announced pre-run, campaign energy on vessel enthalpy alone

**The story:** THE A5-T1 WITNESS: the adiabatic bed.

batch18's ergun-mode bed (15% CO2/He, N = 25) with the energy balance DECLARED: 'energyBalance adiabatic;' – one temperature per cell, the isosteric heat staying IN the bed as a travelling thermal wave instead of being removed by an unnamed coolant. Design record: docs/design/fixed-bed-thermal-a5.md.

What the engine announces BEFORE integrating (equilibrium theory):

pure thermal-wave velocity u_th = u c_tot cpg / (eps c_tot cpg + rho_b cp_s) and the time L/u_th the heat needs to LEAVE the bed; the adiabatic LIMIT DT_ad = q∗(-dH_ads)/(cp_s + eps c_tot cpg/rho_b) – the all-heat-retained BOUND. Here u_th < u_sh: the heat LAGS the front, so the observed outlet-T rise sits well BELOW the bound (~ +38 K vs the 140 K limit) while the WARM bed breaks through far earlier than batch18's isothermal front (t_50 ~ 806 s vs 3042 s – q∗(T) falls via the van't Hoff b(T), the SAME datum the energy source term uses; one dH_ads, three readers).

The golden pins T_out/T_max, both anchors, the early crossings, and the campaign ENERGY closure – claimed, adiabatically, on vessel enthalpy against the boundary streams alone (per-step raffinate packages carry the outlet cell's own T).

cp_s enters as DECLARED case data (the kLDF pattern – scope + source say where the number came from; curating a primary-cited cp onto the adsorbent asset record is a separate curation act). The adsorbed-phase heat capacity is neglected, declared. Feed enters at the declared T; the raffinate package leaves at the OUTLET CELL's own T so the material ledger carries the thermal wave's enthalpy.

Adiabatic means NO energy record: nothing is exchanged with the environment; the campaign energy balance closes on vessel enthalpy (per-cell pricing: gas + adsorbed at T_j, solid sensible cp_s(T_j-T0)) against the boundary streams alone.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | P_final | 100050.377191 |
| kpi | bed | P_first_cell_Pa | 100096.839501 |
| kpi | bed | P_last_cell_Pa | 100002.080676 |
| kpi | bed | P_mid_cell_Pa | 100050.98468 |
| kpi | bed | P_out_boundary_Pa | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 315.614986735 |
| kpi | bed | T_max_final_K | 334.272787919 |
| kpi | bed | T_out_final_K | 334.272787919 |
| kpi | bed | c_out_over_cin_final_CO2 | 0.822859498342 |
| kpi | bed | carrier_fabricated_mol | 2.54374299402e-11 |
| kpi | bed | dT_adiabatic_CO2_K | 140.405462904 |
| kpi | bed | deltaP_first_to_out_Pa | 96.8395014859 |
| kpi | bed | integral_anchor_CO2 | 2850.55000879 |

*... and 41 more reference values in the case's `expected`.*

## batch21_tsa_hot_purge

tutorials/batch/adsorber/batch21_tsa_hot_purge




**What it does:** T1.5 TSA half-cycle: adiabatic load to 3600 s, then 400 K clean He purge – feed.T as the hot-purge control, the re-declared commitment ledgered OUT at 298 K / IN at 400 K, energy balance claimed across the swing

**The story:** THE T1.5 WITNESS: a real TSA half-cycle.

batch20's adiabatic bed run as LOAD → HOT CLEAN PURGE in one campaign:

0 .. 3600 s LOAD – 15% CO2/He at 298.15 K (batch20's feed; the warm front is through by ~806 s) 3600 s SWITCH – recipe, two setParameter events: feed.CO2 = 0 (clean He purge) feed.T = 400 K (the HOT purge) 3600 .. 12000 s PURGE – the hot clean carrier sweeps the bed: q∗(T) collapses via the van't Hoff b(T) and desorption runs FAR faster than batch19's cold concentration swing

This is the step the old refusal named: "thermal swing needs the non-isothermal bed energy balance". A5-T1 built that balance; feed.T is now the hot-purge CONTROL (in thermal mode ONLY – the isothermal bed still refuses it, by name).

The LEDGER story is the point. A feed at a new temperature re-declares the WHOLE remaining commitment – at the pinned feed pressure the total concentration drops to P/(R∗400) (composition preserved) and even unchanged matter changes enthalpy – so the switch retires the old commitment (one OUT feedAmendment package, all species, at 298.15 K) and declares the new one (one IN package at 400 K): the enthalpy repricing rides in the packages' own T, and the campaign energy balance stays CLAIMED across the swing, boundary streams alone.

Breakthrough crossings/integral freeze at the switch (they describe the load step); the desorption story lives in c_out/qbar/T_out. The frozen pre-run anchors (u_th, DT_ad bound) keep their initialise values – a later switch never rewrites an announced claim.

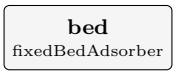
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | P_final | 100030.435428 |
| kpi | bed | P_first_cell_Pa | 100063.521237 |
| kpi | bed | P_last_cell_Pa | 100001.142356 |
| kpi | bed | P_mid_cell_Pa | 100028.717422 |
| kpi | bed | P_out_boundary_Pa | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 350.354335031 |
| kpi | bed | T_max_final_K | 399.999997842 |
| kpi | bed | T_out_final_K | 306.850590216 |
| kpi | bed | carrier_fabricated_mol | -2.52953213931e-12 |
| kpi | bed | dT_adiabatic_CO2_K | 140.405462904 |
| kpi | bed | deltaP_first_to_out_Pa | 63.521236595 |
| kpi | bed | integral_anchor_CO2 | 1312.9982971 |
| kpi | bed | mass_closure_CO2 | 3.75114008529e-15 |

*. . . and 50 more reference values in the case's `expected`.*

## batch22_wall_cooled

tutorials/batch/adsorber/batch22_wall_cooled




**What it does:** T2 wall-cooled CO2/He breakthrough (h=5 W/m2K, T_wall=298.15 K): the bed BETWEEN its limits – t_50 bracketed by the adiabatic 806 s and isothermal 3036 s, Q_wall as a state row of the same ODE, wallHeat record read from the state

**The story:** THE T2 WITNESS: the bed between its two limits.

batch20's adiabatic bed with 'energyBalance wallCooled;' – the energy row gains -h a_w (T_j - T_wall) and the bed sits BETWEEN the limits it must contain:

h → 0 batch20 (adiabatic: t_50 = 806 s, T_max ~ 336 K) h → infinity batch18 (isothermal: t_50 = 3036 s, T = T_wall)

so the golden pins the BRACKETING:

t_50(adiabatic) < t_50(wallCooled) < t_50(isothermal) T_wall < T_max(wallCooled) < T_max(adiabatic)

at h = 5 W/(m2 K), T_wall = 298.15 K, dBed 0.113 m DECLARED (never derived from the cross-section by a silent circle assumption); the engine announces a_w = 4/dBed and the wall NTU before integrating.

The LEDGER is the design point (docs/design/fixed-bed-thermal-a5.md section 6): the removed heat integrates as ONE MORE STATE ROW of the same ODE – the M_in/M_out ledger-row pattern – so the 'wallHeat' energy record READS a state (E = -Q_wall, T_service = T_wall) and the campaign closure dH_vessels = Q_ledger + H_external holds with every term a state function. Guards: 'adiabatic' + a wallHeatTransfer block refuses (contradiction), 'wallCooled' without one refuses, h <= 0 refuses (adiabatic is the way to SAY zero) – gate check_thermal_bed.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_wallHeat_seg0_kJ | -184.059495315 |
| kpi | bed | E_wallHeat_total_kJ | -184.059495315 |
| kpi | bed | P_final | 100048.41634 |
| kpi | bed | P_first_cell_Pa | 100094.212734 |
| kpi | bed | P_last_cell_Pa | 100001.948175 |
| kpi | bed | P_mid_cell_Pa | 100048.658554 |
| kpi | bed | P_out_boundary_Pa | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | Q_wall_removed_kJ | 184.059495315 |
| kpi | bed | T_final | 306.055276997 |
| kpi | bed | T_max_final_K | 312.299755078 |
| kpi | bed | T_out_final_K | 312.299755078 |
| kpi | bed | c_out_over_cin_final_CO2 | 0.890450653484 |
| kpi | bed | carrier_fabricated_mol | 2.41868747253e-11 |

*... and 49 more reference values in the case's `expected`.*

## batch23_tsa_cycles

tutorials/batch/adsorber/batch23_tsa_cycles




**What it does:** Toward A6: three hand-declared TSA cycles (cold load / 400 K purge) in one campaign – eleven ledgered feed amendments, balances closed, final KPIs at the cycle boundary; the CSS verdict machinery stays a named proposal

**The story:** toward A6: three FULL TSA cycles, hand-declared.

batch21's half-cycle repeated three times in ONE campaign, with the recipe grammar that already exists – nothing new in the engine:

cycle k (k = 0,1,2), each 7200 s: t = 7200k feed.T = 298.15 K (cold again) [k>0] feed.CO2 = 0.15 (feed returns) [k>0] t = 7200k + 3600 feed.CO2 = 0 (clean purge) feed.T = 400 K (hot purge)

ending EXACTLY at the third cycle boundary (21600 s), so the FINAL KPIs are end-of-cycle values. Every switch re-declares the feed commitment through the T1.5 ledger machinery (the T switches retire and redeclare the WHOLE commitment; the composition switches amend by delta) – FIFTEEN feedAmendment records (three per switching instant: the composition delta + the T retire/declare pair), and the campaign mass/element/energy balances still close on records, never jumps.

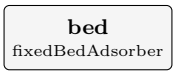
The A6 QUESTION this case makes visible (and does not yet verdict): does the bed state at successive cycle ends approach a fixed point – the cyclic steady state? MEASURED (this case's own run): qbar_CO2 at the three cycle ends is 0.3151 → 0.2415 → 0.1748 mol/kg – a slow geometric approach (ratio ~0.9; each purge leaves residual heat, the next load starts warmer and holds less), NOT yet converged in three cycles. That unfinished convergence IS the pedagogical point: eyes on a trajectory can see it, and only the A6 verdict machinery could CLAIM it. The golden pins the third-cycle end. A first-class CSS verdict (declared cycles, per- cycle state snapshots, an invariance tolerance the case declares) is an ARCHITECTURE feature – proposed in docs/design/fixed-bed-thermal-a5.md section 7, awaiting Vitor's decision; this case is deliberately its cheap, grammar-free witness.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | P_final | 100030.094422 |
| kpi | bed | P_first_cell_Pa | 100060.3912 |
| kpi | bed | P_last_cell_Pa | 100001.192563 |
| kpi | bed | P_mid_cell_Pa | 100030.206567 |
| kpi | bed | P_out_boundary_Pa | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 329.119724972 |
| kpi | bed | T_max_final_K | 399.878003769 |
| kpi | bed | T_out_final_K | 315.723301673 |
| kpi | bed | carrier_fabricated_mol | 9.60653778748e-12 |
| kpi | bed | dT_adiabatic_CO2_K | 140.405462904 |
| kpi | bed | deltaP_first_to_out_Pa | 60.3911997653 |
| kpi | bed | integral_anchor_CO2 | 1312.99822496 |
| kpi | bed | mass_closure_CO2 | 6.54546762708e-14 |
| *... and 55 more reference values in the case's `expected`.* | | | |

## batch24_blowdown_inert

tutorials/batch/adsorber/batch24_blowdown_inert




**What it does:** P-swing witness A: inert isothermal blowdown, n/n0 = P/P0

**The story:** P-SWING WITNESS A: the inventory ratio.

The MATERIAL half of a blowdown, isolated from every thermal effect so it cannot pass by coincidence (design note fixed-bed-thermal-a5.md, section 8/9 amended battery, second-opinion review 2026-08-03):

ideal gas, INERT bed (kLDF = 0: the solid is frozen, pure transport), pure He; ISOTHERMAL ergun mode – T is a DECLARED constant of this teaching slice, so the energy question is out of frame BY DECLARATION (the thermal half is witness B, batch25); at t = 200 s the recipe closes the feed valve (P2, 'setParameter u = 0' – the whole remaining feed commitment retires as a ledgered feedAmendment record) and drops the outlet boundary (P1, 'setParameter P_out' 1.0 → 0.5 bar); the bed drains through the SAME Ergun face machinery every run already uses; the vented gas rides the telescopic M_out ledger rows.

THE ANCHORS (ideal-gas exactness, hand-computable):

n_end / n_0 = P_end / P_0 = 0.5 n(P) = eps A L P / (R T) = 0.0807 mol at 1 bar, 0.0403 at 0.5 bar vented = inventory drop (the mass-closure KPI is the proof)

The gate (check_pswing) recomputes both analytically and pins the 'gasInventory_mol' KPI + the closure against them.

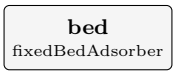
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | E_adsorption_seg0_kJ | 0 |
| kpi | bed | E_adsorption_total_kJ | 0 |
| kpi | bed | P_final | 49999.9998988 |
| kpi | bed | P_first_cell_Pa | 49999.9998996 |
| kpi | bed | P_last_cell_Pa | 49999.9998972 |
| kpi | bed | P_mid_cell_Pa | 49999.999899 |
| kpi | bed | P_out_boundary_Pa | 50000 |
| kpi | bed | Pe | 0 |
| kpi | bed | T_final | 298.15 |
| kpi | bed | carrier_fabricated_mol | -8.881784197e-16 |
| kpi | bed | deltaP_first_to_out_Pa | -0.000100418335933 |
| kpi | bed | gasInventory_mol | 0.0403395454642 |
| kpi | bed | mass_closure_CO2 | 0 |
| kpi | bed | n_CO2_final | 0 |

*... and 37 more reference values in the case's `expected`.*

## batch24_tsa_until_css

tutorials/batch/adsorber/batch24_tsa_until_css




**What it does:** Toward A6: three hand-declared TSA cycles (cold load / 400 K purge) in one campaign – eleven ledgered feed amendments, balances closed, final KPIs at the cycle boundary; the CSS verdict machinery stays a named proposal

**The story:** A6 item 4: the cyclic steady state as the STOPPING CONDITION, not a count.

batch23 beside this one declares ‘repeat 3;‘ and reports honestly that three cycles did not settle – it makes the question visible and does not answer it. This case asks what an engineer actually wants to ask of a TSA bed: not "run three cycles" but "cycle until it repeats, and tell me how many that took".

cycle { period 7200; repeat untilCSS; cssTolerance 3e-2; maxCycles 6; ... }

The declared cap is SIX and the run uses TWO, announcing why it stopped.

TWO KEYS ARE MANDATORY under ‘untilCSS‘ and the engine refuses without either: the tolerance, because whether a bed has converged is a modelling judgement the engine must never invent; and the cap, because a bed that never settles has to stop somewhere and the run must be able to say it stopped for want of cycles rather than because it arrived. Reaching the cap is NOT an error – the trajectory is a physically valid campaign and every other KPI in it is real – but it means the case’s own question went unanswered, so it is published (‘css_stopped_at_cap‘) and announced.

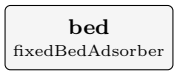
WHAT BUILDING IT FOUND, and it outlives the feature: the CSS norm and the loading are NOT the same quantity. See the measured numbers beside ‘cssTolerance‘ below. A tolerance chosen without looking can certify a bed whose loading is still visibly marching, because the norm over the whole packed state is dominated by its largest components. The verdict is only as meaningful as the number the case declares – which is the argument for the engine refusing to supply one.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | P_final | 100029.553817 |
| kpi | bed | P_first_cell_Pa | 100059.686721 |
| kpi | bed | P_last_cell_Pa | 100001.177433 |
| kpi | bed | P_mid_cell_Pa | 100029.506183 |
| kpi | bed | P_out_boundary_Pa | 100000 |
| kpi | bed | Pe | 250 |
| kpi | bed | T_final | 324.614533765 |
| kpi | bed | T_max_final_K | 399.878004394 |
| kpi | bed | T_min_final_K | 297.926151955 |
| kpi | bed | T_out_final_K | 310.618231369 |
| kpi | bed | carrier_fabricated_mol | 1.35003119794e-11 |
| kpi | bed | dT_adiabatic_CO2_K | 140.405462904 |
| kpi | bed | deltaP_first_to_out_Pa | 59.6867210897 |
| kpi | bed | gasInventory_mol | 0.0747102437107 |
| *... and 57 more reference values in the case’s `expected`.* | | | |

### batch25_blowdown_thermal

tutorials/batch/adsorber/batch25_blowdown_thermal




**What it does:** P-swing witness B: adiabatic blowdown COOLS (expansion work)

**The story:** P-SWING WITNESS B: the cooling.

The THERMAL half of a blowdown, isolated from adsorption so the only heat effect in frame is the expansion work itself (design note fixed-bed-thermal-a5.md, section 8/9 amended battery):

same inert pure-He ergun bed as witness A (batch24), but 'energyBalance adiabatic;' – the section-9 general energy row (cv accumulation + the explicit eps R T dc_tot/dt source) is live; at t = 200 s the recipe closes the feed valve (P2) and halves the outlet boundary (P1) – the bed drains, dc_tot/dt < 0, and the expansion source pulls T DOWN; the solid heat capacity is a declared SYNTHETIC LOW value (9.2 J/(kg K), scope says so): the real ~920 of a zeolite pellet would thermally bury the gas-phase cooling under the solid's mass (dT ~ -0.03 K); at the teaching value the drop is visible (~ -3 K class). Same honest-teaching-value discipline as the hxFluid of the pinch witness – the physics on display is the GAS's, and the header says which knob made it visible.

THE ANCHOR: T_min_final_K sits BELOW the declared 298.15 K by a visible margin – a depressurising cell COOLS, which is exactly what the pre-widening cp-form could not show (the sabotage of the gate restores that form and this witness fails). The material anchors stay witness A's business; this case pins the sign and visibility of the thermal effect, and the campaign energy claim stays closed.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bed | P_final | 49999.9997347 |
| kpi | bed | P_first_cell_Pa | 49999.9996936 |
| kpi | bed | P_last_cell_Pa | 49999.999898 |
| kpi | bed | P_mid_cell_Pa | 49999.9997123 |
| kpi | bed | P_out_boundary_Pa | 50000 |
| kpi | bed | Pe | 0 |
| kpi | bed | T_final | 294.923818642 |
| kpi | bed | T_max_final_K | 295.066878523 |
| kpi | bed | T_min_final_K | 294.887442943 |
| kpi | bed | T_out_final_K | 295.066878523 |
| kpi | bed | carrier_fabricated_mol | 1.15463194561e-14 |
| kpi | bed | deltaP_first_to_out_Pa | -0.00030635814619 |
| kpi | bed | gasInventory_mol | 0.0407808219207 |
| kpi | bed | mass_closure_CO2 | 0 |
| *... and 38 more reference values in the case's `expected`.* | | | |

## combustion

**The theory you need.** Radical chain chemistry: initiation creates the first radicals (slow – $H_2 + M$, $E_a \sim 105$ kcal), branching multiplies them ($H + O_2 \rightarrow O + OH$: one in, two out), termination removes them. Ignition is the race between branching and termination; the induction time is set by the SLOW build-up, then heat release runs away. Rate constants $k = AT^b e^{-E_a/RT}$ come VERBATIM from cited papers (Ó Conaire 2004, Glarborg 2018) in their native units – the engine converts and announces. Reverse rates from equilibrium: the kinetic end-state must land on the Gibbs equilibrium (the two-engine handshake the NOx tutorial pins at 99.99 %). Expect stiffness ratios of $10^6$–$10^8$ (use Rosenbrock23), ignition delays in $\mu$s validated against shock-tube data, and thermal NO forming over SECONDS – the timescale gap behind quench-based NOx control. *Full derivation: Theory Guide, the combustion/kinetics chapter.*

## ignition01_h2o2

tutorials/batch/combustion/ignition01_h2o2




**What it does:** Adiabatic constant-V H2/O2 ignition at 1100 K (O Conaire 2004 H2/O2 kinetics, NASA-9 thermo); Rosenbrock23 stiff solver; reports ignition delay

**The story:** closed rigid vessel, stoichiometric H2/O2 in N2 at 1100 K.

H2 : O2 : N2 = 2 : 1 : 4 (stoichiometric H2 + 1/2 O2 → H2O, 57% N2)

mode adiabatic + energy gasConstantV: the first law dU = 0 in a fixed volume, with the heat release coming from the species FORMATION enthalpies (curated thermo) – no per-reaction ΔH is supplied. Temperature, and the radicals H/O/OH/HO2, are integrated by the stiff solver until branching runs away.

solver { integrator Rosenbrock23; } – the L-stable stiff method. Swap to RK4 (see the ignition01_h2o2_rk4 sibling) and the same step blows up.

*A closed rigid vessel holds a stoichiometric $H_2/O_2$ charge diluted in $N_2$ at 1100 K and 5 bar. A trace H-radical seed starts chain branching; after a short induction the radical pool explodes and the mixture ignites — a sharp temperature jump from 1100 K to ~3078 K.*

*and plot trajectory.csv (T and the radicals H/O/OH vs time). You will see:*

*The ignition delay τ_ign (the time of maximum dT/dt, printed in the run's epilogue on the 1 μs write grid) is ≈ 8 μs here. The trace H seed is what makes it that short: the same mechanism without a seed, at 2 atm, gives the 43 μs of the shock-tube pin in ignition02_h2air_slack.*

*The radical lifetimes are far shorter than the induction — the solver prints a stiffness ratio of order 10?. That is stiffness. An explicit method (RK4) is stability-limited to a step of order the ∗fastest∗ scale, so it cannot take the 1 μs steps this case uses. The solver here is Rosenbrock23, an L-stable linearly-implicit method that steps by ∗accuracy∗, not stability; at verbosity 3 it prints its stiffness verdict and step counts (the no-silent- crutch credo).*

*See it fail on purpose — two siblings run the same problem at the same step with other integrators:*

*(Both ship a .expect-nonconvergence marker — their whole point is to fail.) ∗See, then decide∗, applied to numerics.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bomb | P_final | 500000 |
| kpi | bomb | T_final | 3077.97900108 |
| kpi | bomb | n_H2O2_final | 2.59406092875e-11 |
| kpi | bomb | n_H2_final | 1.67644666948e-06 |
| kpi | bomb | n_HO2_final | 1.97492765295e-10 |
| kpi | bomb | n_H_final | 5.11250559107e-07 |
| kpi | bomb | n_N2_final | 2.857145e-05 |
| kpi | bomb | n_O2_final | 6.64822219482e-07 |
| kpi | bomb | n_OH_final | 7.55609252321e-07 |
| kpi | bomb | n_O_final | 2.25566026101e-07 |
| kpi | bomb | n_water_final | 1.19727313332e-05 |
| kpi | bomb | t_end | 0.0002 |
| kpi | bomb | totalMoles_final | 4.43789505344e-05 |
| kpi | campaign | element_H_closure_rel | 6.76051276934e-15 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## ignition01_h2o2_eulersi

tutorials/batch/combustion/ignition01_h2o2_eulersi




**What it does:** WITNESS + EXPECTED FAILURE: semi-implicit Euler on the H2/O2 ignition – A-stable but first-order, goes unphysical at this step and the engine refuses by name (see .expect-nonconvergence).

**The story:** WITNESS for the EulerSI integrator (semi-implicit Euler) – an EXPECTED FAILURE: see .expect-nonconvergence for why first order at this step goes unphysical and how the engine now refuses instead of exiting 0. Registered model no case selected until 2026-08-22 (invariant I8).

*The third integrator of the stiffness lesson. The same adiabatic constant-volume $H_2/O_2$ ignition as [ignition01_h2o2](../ignition01_h2o2/README.md) (Rosenbrock23, converges) and [ignition01_h2o2_rk4](../ignition01_h2o2_ (explicit RK4, blows up), here with semi-implicit Euler (integrator EulerSI;) at the same deltaT = 1e-6 s.*

*Semi-implicit Euler is A-stable, so it does not oscillate to infinity the way RK4 does — the first steps are physical (T 1100 → 1102 → 1108 → 1139 K) — but it is FIRST ORDER, and once the ignition transient starts one linearised step is wrong enough to drive the state unphysical. The integrator reports that it gave up, and the reactor refuses by name instead of exiting 0:*

*.expect-nonconvergence records why that refusal exists (until 2026-08-22 this exact run reached a negative temperature and exited 0). L-stable second order converges, explicit fourth order blows up loudly, A-stable first order fails honestly: ∗see, then decide∗, applied to numerics.*

*Same as the sibling: the ? Conaire et al. (2004) $H_2/O_2$ mechanism (constant/reactions, shared with ignition02_h2air_slack) and NASA-9 heat capacities from NASA/TP-2002-211556 for $H_2$, $O_2$, $N_2$ and water. Until 2026-10-25 it ran a GRI-Mech 3.0 subset, replaced because GRI-Mech grants no redistribution (DEV.md ?4c C164); the failure it teaches did not change.*

## ignition01_h2o2_rk4

tutorials/batch/combustion/ignition01_h2o2_rk4




**What it does:** Adiabatic constant-V H2/O2 ignition at 1100 K (O Conaire 2004 H2/O2 kinetics, NASA-9 thermo); EXPLICIT RK4 – fails on the stiff chemistry (the lesson)

**The story:** ignition01_h2o2 – closed rigid vessel, stoichiometric H2/O2 in N2 at 1100 K.

H2 : O2 : N2 = 2 : 1 : 4 (stoichiometric H2 + 1/2 O2 → H2O, 57% N2)

mode adiabatic + energy gasConstantV: the first law dU = 0 in a fixed volume, with the heat release coming from the species FORMATION enthalpies (curated thermo) – no per-reaction ΔH is supplied. Temperature, and the radicals H/O/OH/HO2, are integrated by the stiff solver until branching runs away.

solver { integrator Rosenbrock23; } – the L-stable stiff method. Swap to RK4 (see the ignition01_h2o2_rk4 sibling) and the same step blows up.

*This is the failure half of the stiffness lesson. It runs the exact same adiabatic constant-volume $H_2/O_2$ ignition as the sibling case [ignition01_h2o2](../ignition01_h2o2/README.md), but with the explicit RK4 integrator instead of the stiff Rosenbrock23 — at the same fixed step (deltaT = 1e-6 s).*

*The first step is still plausible (T 1100 → 1110 K); the second already throws the state out of the physical region (T near 8000 K, $H_2$ at zero), and the third is nan. The engine refuses rather than exit 0:*

*That blow-up is the result. The stiff sibling prints a stiffness ratio of order 10? for this chemistry: an explicit method is stability-limited to a step of order the ∗fastest∗ scale, so a step chosen for the ∗induction∗ is murdered by stability long before accuracy matters.*

*Compare side-by-side with ignition01_h2o2, whose L-stable adaptive sub-stepping integrates the ∗same∗ chemistry to a clean burned state (~3078 K), and with ignition01_h2o2_eulersi, which is A-stable but first order and fails honestly by name. ∗See, then decide∗, applied to numerics.*

*This case ships a .expect-nonconvergence marker, so bin/runTests counts it as an EXPECTED-FAIL — its whole point is to fail.*

*Same as the sibling: the ? Conaire et al. (2004) $H_2/O_2$ mechanism (constant/reactions, shared with ignition02_h2air_slack) and NASA-9 heat capacities from NASA/TP-2002-211556 for $H_2$, $O_2$, $N_2$ and water. Until 2026-10-25 it ran a GRI-Mech 3.0 subset, replaced because GRI-Mech grants no redistribution (DEV.md ?4c C164); the failure it teaches did not change.*

## ignition02_h2air_slack

tutorials/batch/combustion/ignition02_h2air_slack




**What it does:** Stoich H2/air ignition, 1100 K / 2 atm (Slack shock tube; O Conaire 2004 kinetics)

**The story:** / flowsheetDict Stoichiometric H2/air autoignition at 1100 K / 2 atm against the Slack shock-tube point (O Conaire 2004 kinetics): the radical pool (H, O, OH, HO2) builds exponentially through the induction time, then runaway. A stiff-integrator showcase – compare the ignition delay with the measured shock-tube value.

*Stoichiometric $H_2$/air, 1100 K, 2 atm, constant volume — the reflected-shock condition of Slack (1977) as compiled in ? Conaire et al. 2004 (Fig 8). Kinetics: the ? Conaire $H_2/O_2$ mechanism, Table I verbatim (19 reactions; two Troe falloffs; two summed pairs; A in $cm^3$-mol-s, Ea in kcal — the engine converts and announces every one).*

*Result: τ_ign (max dT/dt) = 43 μs vs the measured ~40–60 μs window.*

*Two lessons: 1. No seed. This mechanism self-initiates, and the 43 μs delay is the chain-BRANCHING chemistry (R1-R3), not the dissociation R5 ($H_2 + M = 2H$, Ea = 105 kcal/mol): MEASURED 2026-10-31, when R5-R8 first ran at their true rate (DEV.md 4c C185 item 9 – they had run at ZERO), the delay stayed 43 μs, and removing R5 from the mechanism leaves it at 43 μs too. (This paragraph used to credit R5 with the induction; that was written while R5 ran at zero and was never true.) With the ignition01-style trace-H seed the delay collapses to 12 μs: initial conditions are physics, not knobs. 2. The engine now prints tau_ign in the epilogue (time of max dT/dt on the write grid) whenever a vessel ignites — no CSV spelunking.*

*Run ignition01_h2o2 (the same kinetics, 5 bar, trace-H seeded) for the sibling case and the stiffness lesson (_rk4).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | bomb | P_final | 202650 |
| kpi | bomb | T_final | 3041.74576271 |
| kpi | bomb | n_H2O2_final | 3.94467753319e-11 |
| kpi | bomb | n_H2_final | 9.82315413018e-07 |
| kpi | bomb | n_HO2_final | 4.27531541611e-10 |
| kpi | bomb | n_H_final | 3.53855708823e-07 |
| kpi | bomb | n_N2_final | 1.391515e-05 |
| kpi | bomb | n_O2_final | 3.90559105772e-07 |
| kpi | bomb | n_OH_final | 4.45009202124e-07 |
| kpi | bomb | n_O_final | 1.5493529764e-07 |
| kpi | bomb | n_water_final | 6.00759209227e-06 |
| kpi | bomb | t_end | 0.0003 |
| kpi | bomb | totalMoles_final | 2.22501764671e-05 |
| kpi | campaign | element_H_closure_rel | 3.43861059252e-15 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## nox01_zeldovich_postflame

tutorials/batch/combustion/nox01_zeldovich_postflame




**What it does:** Thermal NOx in post-flame H2/air gas, 2125 K / 1 atm (Glarborg 2018 N chemistry)

**The story:** / flowsheetDict Thermal NOx in post-flame H2/air gas at 2125 K / 1 atm: the Zeldovich N-chemistry (Glarborg 2018 rate set) integrated in a batch reactor – NO grows on the SECONDS scale while the O/H radical pool stays equilibrated, the classic separation of scales that makes thermal NOx a post-flame problem.

*Hot burnt gas (lean $H_2$/air, 2125 K, 1 atm — Homer & Sutton Flame 1 as compiled in Glarborg 2018) held isothermal, starting from the equilibrated H/O radical pool with all nitrogen still as $N_2$. The Glarborg 2018 nitrogen chemistry (extended Zeldovich + $N_2O$ routes, Tables 2/5 verbatim) then forms NO.*

*What is not carried. Glarborg's NNH route (R20–R24) is absent since 2026-10-25: NNH's thermo had come from GRI-Mech 3.0's thermo file, which grants no redistribution, and no public-domain primary for it was found (NASA/TP-2002-211556 has no NNH entry). The heat capacities of the flame species are NASA-9 polynomials from that report. On 2026-07-02, with the NNH route and the earlier thermo, the same table read 5227 / 5167 ppm (98.9 %), 0.50 s and 1.56 s.*

*1. The cross-engine pin. The kinetic plateau lands on the Gibbs equilibrium computed by a ∗different engine∗ (steady Gibbs minimisation vs stiff batch integration). Kinetics choose the path and the clock; the destination belongs to the thermodynamics. Reverse rates come from the same equilibrium constants — so this agreement is a genuine consistency check of the mechanism transcription + thermo, not a tautology. 2. NOx forms in engineering time. Ignition takes microseconds (ignition02: 43 µs); thermal NO takes seconds at 2125 K. That timescale gap is the whole basis of NOx control by quench: cool the gas before the Zeldovich clock runs.*

*The burnt-gas initial pool was computed by a gibbsReactor at 2125 K/1 atm (14 species on 2026-07-02; it still agrees with the current 13-species thermo to the printed digits) (25 % $H_2$ / 17.5 % $O_2$ / 57.5 % $N_2$ feed); the equilibrium NO share was returned to $N_2$/$O_2$ elementally.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | postflame | P_final | 101325 |
| kpi | postflame | T_final | 2125 |
| kpi | postflame | n_H2O2_final | 1.05023679159e-12 |
| kpi | postflame | n_H2_final | 4.8469712404e-09 |
| kpi | postflame | n_HO2_final | 1.61492364784e-11 |
| kpi | postflame | n_H_final | 6.02372866884e-10 |
| kpi | postflame | n_N2O_final | 1.37258448136e-12 |
| kpi | postflame | n_N2_final | 3.72505182272e-06 |
| kpi | postflame | n_NH_final | 2.50521342942e-15 |
| kpi | postflame | n_NO_final | 2.95156157086e-08 |
| kpi | postflame | n_N_final | 2.26160617192e-14 |
| kpi | postflame | n_O2_final | 3.06817214048e-07 |
| kpi | postflame | n_OH_final | 2.0389175016e-08 |
| kpi | postflame | n_O_final | 2.17250090394e-09 |
| *... and 24 more reference values in the case's `expected`.* | | | |

## crystallisation

**The theory you need.** Crystallisation removes solute when the solution crosses saturation: $m > m_{\mathrm{sat}}(T)$, with $K_{\mathrm{sp}}(T)$ from $\Delta G_{\mathrm{diss}}$ and its $T$-dependence from $\Delta H_{\mathrm{diss}}$ (van 't Hoff). The heat of crystallisation is ION-DERIVED – $\Delta H_{\mathrm{cryst}} = -(\Delta H^{\infty}_{\mathrm{soln}} + \bar{L}_2(m_{\mathrm{sat}}))$ – never a stored constant (a solid's formation enthalpy is $\sum \nu_i h^{\mathrm{aq}}_{f,i} - \Delta H_{\mathrm{soln}}$, derived, single-sourced). Antisolvent cases add the mixed-solvent electrolyte model (eNRTL): ethanol drops the dielectric constant, $\gamma_{\pm}$ rises, salt crashes out. Expect supersaturation as the driving force and yields set by the solubility curve, not the kinetics (these cases are equilibrium-staged). *Full derivation: Theory Guide, the crystallisation + electrolyte-enthalpy chapters.*

## batch13_nacl_pitzer

tutorials/batch/crystallisation/batch13_nacl_pitzer




**What it does:** Batch NaCl crystalliser: saturation from the Pitzer ion activity, not a solubility curve

**The story:** One closed vessel. The unit declares no saturation of its own: it asks the thermodynamics the case declares, and the case declares Pitzer.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 9.08953975799e-16 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 9.08953975799e-16 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 9.08953975799e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | mass_closure_rel | 3.49878229369e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 1299.73034249 |
| kpi | campaign | mass_kg_initial | 1299.73034249 |
| kpi | campaign | mass_residual_kg | 4.54747350886e-13 |
| kpi | campaign | moles_kmol_external_out | 0 |

*... and 18 more reference values in the case's `expected`.*

### batch14_nacl_pitzer_antisolvent

tutorials/batch/crystallisation/batch14_nacl_pitzer_antisolvent




**What it does:** Batch NaCl crystalliser with ethanol present: a pairwise-aqueous model announces the mixed-solvent term it does not carry

**The story:** One closed vessel, identical to batch13 except for the ethanol in the charge. The unit declares no saturation of its own: it asks the thermodynamics the case declares, and the case declares Pitzer – which has nothing to say about a second solvent, and says so.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_Cl_closure_rel | 6.05235197043e-16 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 6.05235197043e-16 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 6.05235197043e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | mass_closure_rel | 1.3612844745e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 1670.28772973 |
| kpi | campaign | mass_kg_initial | 1670.28772973 |
| kpi | campaign | mass_residual_kg | 2.27373675443e-13 |
| *... and 20 more reference values in the case's `expected`.* | | | |

## drying

## dryer01_sucrose_tray

tutorials/batch/drying/dryer01_sucrose_tray




**What it does:** Tray drying of wet sucrose into declared 60 C air: the classical constant-rate / falling-rate drying curve, with the wet bulb and the GAB equilibrium moisture as results

**The story:** the DRYING CURVE, both periods, in one batch.

A tray holding 2.0 kg of dry sucrose wet to X_0 = 0.30 kg water / kg dry solid, exposed over 0.5 m2 to air of DECLARED, CONSTANT condition: 60 C (333.15 K) and Y = 0.010 kg water / kg dry N2. Nothing about that air is integrated – it is an ENVIRONMENT, the FixedBedAdsorber's constant-carrier posture, and the water that leaves the solid leaves the campaign across that boundary (reported as the unit's DECLARED material residual, never as a silent leak).

THE HAND ESTIMATE, computed from this case's own curated records (water Antoine + Watson Hvap + the two ideal-gas Cp polynomials, the case-local GAB isotherm on sucrose) – the run must reproduce it:

air water activity p_v = P Y/(Y + Mv/Mc) = 1551 Pa a_w = p_v / Psat(333.15 K) = 0.0774 GAB equilibrium K a_w = 0.0743 X_eq = Xm C Ka /((1-Ka)(1-Ka+C Ka)) = 0.01580 kg/kg wet bulb (Lewis = 1) T_wb = 300.53 K (27.4 C, a 32.6 K depression) driving force Y_sat(T_wb) - Y = 0.014011 kg/kg constant-rate flux R_c = k_Y dY = 0.05 ∗ 0.014011 = 7.0057e-4 kg/(m2 s) R_c A = 3.5028e-4 kg/s latent load R_c A lambda(T_wb) = 0.8645 kW (the AIR's, not the campaign's – KPI only, never ledgered) break time t_c = m_s (X_0 - X_c)/(R_c A) = 1027.7 s falling-rate constant tau = m_s (X_c - X_eq)/(R_c A) = 595.0 s end of run X(3000 s) = X_eq + (X_c - X_eq) exp(-(3000-t_c)/tau) = 0.019584 kg/kg

What to look at in the output: trajectory.csv columns dryer.X_kg_kg and dryer.R_kg_m2_s: the drying curve and the rate curve, engine columns, not a post-processing. R is FLAT at 7.0057e-4 until X falls through X_c = 0.12, then decays linearly in (X - X_eq) – the textbook break, visible as a corner. The falling rate is DECLARED as the characteristic drying curve f = Phi^n with n = 1, which IS the linear law; the drying EduTool varies n (n < 1 reaches X_eq in finite time, n > 1 an algebraic tail slower than the exponential), and dryer02_characteristic_curves runs four shapes side by side. the KPI t_critical is OBSERVED (the interpolated crossing between two accepted states), not the analytic 1027.7 s above; they agree because dX/dt is exactly constant on that leg. THE CAMPAIGN MASS BALANCE DOES NOT CLOSE, AND IT SHOULD NOT. Look at it, because it is the first thing an examiner will point at: it reports 2.600000 kg in against 2.039168 kg out, a 21.6 % shortfall. That shortfall is the DRIED WATER. It left the solid and went into the air – and the air here is a DECLARED ENVIRONMENT (a temperature and a humidity), not a stream this campaign owns, so there is no vessel on the books for the water to arrive in. The vessel's inventory genuinely shrinks.

What makes that a stated model boundary rather than a leak is that the unit PREDICTS the gap independently, from its own evaporation history, and the campaign checks the two against each other: the observed shortfall matches the declared prediction to 7e-17 (element_worst_closure_vs_declared_rel). The raw numbers are never discounted – 21.6 % is what the balance says and what the balance reports. If this dryer ever lost matter for a REAL reason, that cross-check is what would move, and it is pinned in 'expected'.

So the honest sentence to give at a viva is: "the balance is open at the air boundary by construction, the opening is the 0.5608 kg of water evaporated, and the model reproduces that number from the other side." Closing it for real means making the drying air a ledgered stream, which this unit deliberately does not do.

the campaign energy balance is UNAVAILABLE, and says why in one sentence: the drying heat comes from air this campaign never owned. The number is not hidden – it is the latentDuty_kW KPI.

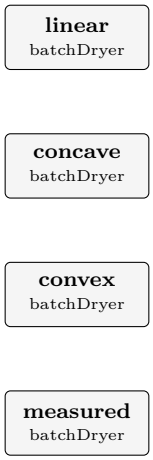
Method hypotheses, each announced at run time: Lewis = 1 in the adiabatic-saturation equation; the humid heat evaluated at the air's own temperature; the surface held at T_wb while X > X_c; the LINEAR falling-rate curve (n = 1; a modelling CHOICE, not physics, and its declared source says so); the solid's own temperature NOT integrated.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 0.319043490708 |
| kpi | campaign | element_O_closure_rel | 0.319043490708 |
| kpi | campaign | element_worst_closure_rel | 0.319043490708 |
| kpi | campaign | element_worst_closure_vs_declared_rel | 7.11118003994e-17 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 0.215704585481 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 2.03916808853 |
| kpi | campaign | mass_kg_initial | 2.60000001375 |
| kpi | campaign | mass_residual_kg | 0.560831925217 |
| kpi | campaign | moles_kmol_external_out | 0 |
| *... and 21 more reference values in the case's `expected`.* | | | |

## dryer02_characteristic_curves

tutorials/batch/drying/dryer02_characteristic_curves




**What it does:** The characteristic drying curve: one wet sucrose tray, one declared air, four falling-rate shapes – linear, Phi^0.5, Phi^2 and a table

**The story:** THE CHARACTERISTIC DRYING CURVE.

Four identical trays of dryer01 (2.0 kg dry sucrose at X_0 = 0.30, 0.5 m2, air at 60 C and Y = 0.010, X_c = 0.12) that differ in ONE thing: the shape of the falling-rate period. van Meel (Chem. Eng. Sci. 9 (1958) 36-44) observed that for many materials the drying rate normalised by the constant rate is a single function of the normalised free moisture,

Phi = (X - X_eq)/(X_c - X_eq), f = R/R_c = f(Phi), f(1) = 1, f(0) = 0,

so ONE measured curve describes the material at any air condition – the air sets R_c (the wet bulb) and X_eq (the isotherm), the curve sets the rest. The shape is the material's, MEASURED; each tray declares one (src/unitOperations/heatTransfer/DryingCurve.H):

linear f = Phi the textbook tail, dryer01's law, declared concave f = Phi^0.5 rate held high: reaches X_eq in FINITE time, t' = tau/(1 - n) = 2 tau convex f = Phi^2 rate dropping early: an algebraic tail, Phi = 1/(1 + t'/tau), slower than linear measured a (Phi, f) table S-shaped, interpolated linearly

tau = m_s (X_c - X_eq)/(R_c A) = 595.0 s (dryer01's hand estimate) is the SAME for all four – the constant-rate period, t_c = 1027.7 s, too. What differs is only the falling period, and the end-of-run moisture, at t' = 3000 - 1027.7 = 1972.3 s into it:

linear X = X_eq + (X_c - X_eq) exp(-t'/tau) = 0.019584 (= dryer01) convex X = X_eq + (X_c - X_eq)/(1 + t'/tau) = 0.03995 concave X = X_eq = 0.015798, reached at t' = 2 tau = 1190 s – the engine lands 3.8e-8 kg/kg BELOW it (2.4e-6 relative): RK4 stepping onto a point where f = Phi^0.5 has an infinite slope, the size (dt/tau)^2 (X_c - X_eq)/4 predicts. Not clamped (no clamp in this unit, by rule); it shrinks with dt. measured 0.03502 (no closed form), between linear and convex, as its points lie below the linear curve bin/curate/check_drying_curve.py recomputes all of this from the run.

EVERY CURVE HERE IS HYPOTHETICAL. No measured characteristic curve of sucrose is in this repository; the shapes are chosen to span what materials do, and each tray says so in its 'source'. A real design reads its curve from a thin-layer drying test of ITS material.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 0.30993864697 |
| kpi | campaign | element_O_closure_rel | 0.30993864697 |
| kpi | campaign | element_worst_closure_rel | 0.30993864697 |
| kpi | campaign | element_worst_closure_vs_declared_rel | 7.11118003994e-17 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 0.209548821137 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 8.22069230365 |
| kpi | campaign | mass_kg_initial | 10.400000055 |
| kpi | campaign | mass_residual_kg | 2.17930775135 |
| kpi | campaign | moles_kmol_external_out | 0 |
| *… and 69 more reference values in the case's `expected`.* | | | |

## electrodialysis

## edbatch01_constant_current

tutorials/batch/electrodialysis/edbatch01_constant_current




**What it does:** Batch recirculating electrodialysis at constant current: the limiting current falls with the depleting diluate until i crosses it

**The story:** a batch electrodialysis rig run at CONSTANT CURRENT, and the limiting current falling under it until they cross.

THE OPERATION. Five litres of 0.1 mol/kg brine in the diluate tank, five litres of 0.5 mol/kg in the concentrate tank, both recirculated through the bench stack of 'ed03_stack_record' (Eurodia EUR2C-7P18, seven cell pairs, 0.020 m2 each) at the paper's own 175 L/h. The rectifier is held at 3.5 A, which is 175 A/m2 on one cell pair.

WHY IT IS A DIFFERENT EXPERIMENT FROM ed01-ed05. Those are steady: one pass, a fixed diluate composition, so the limiting current is evaluated ONCE. Here the diluate tank DEPLETES, and i_lim – which the GeraldesAfonso2010 model predicts from the ion diffusivities and the diluate composition – falls with it, while i does not move at all. They must cross.

WHAT THE RUN MEASURES (every number below is printed by this run; nothing here is a target):

i 175.00 A/m2, flat, all 2000 s i_lim 643.04 A/m2 at t = 0 88.26 A/m2 at t = 2000 s THE CROSSING t = 1593 s (i/i_lim reaches 1) i/i_lim at the end 1.983 xi_eff 0.900 → 0.4539 (the plateau, below) demineralisation 86.27 % ACTUAL 91.41 % what the hand calculation predicts the hand calculation's own answer: the tank is EMPTY at t = 2188 s concentrate 0.5 → 0.5863 mol/kg (x1.173) specific energy 1.946 kWh per m3 of diluate treated material closure 1.4e-15 on mass, 8.5e-14 on the worst element

THE LESSON. Faraday's law at constant current is a STRAIGHT LINE:

n(t) = n(0) - xi I N t / (z F)

and it is what a student computes by hand. The run publishes it as 'demin_ideal', evaluated from THIS run's own initial current, own initial inventory and own declared xi – computed, never declared, so the gap cannot have been arranged – beside 'demin_actual'.

The two agree exactly until t = 1593 s and part after it, and the reason is printed at the instant it happens: i has caught i_lim. At the limiting current the counter-ion concentration vanishes at the diluate/membrane interface and the film can deliver no more of it, so a larger current is carried by water-splitting products instead of by salt. This case DECLARES that model – 'overLimiting { model saltFluxPlateau; }', xi_eff = xi i_lim/i, which adds no parameter – so the depletion stops being linear and becomes asymptotic: the straight line hits zero at 2188 s and the tank never does.

WITHOUT the declaration (the default) the current efficiency would stay at xi and the two curves would coincide to the integrator's own precision. That is not a better answer; it is a model with nothing in it that could part from the hand calculation, and the run SAYS so when i crosses i_lim. 'edbatch02_constant_voltage' is the same rig with the rectifier held at a voltage instead: there the current DECAYS, the crossing never happens, and the same hand calculation fails for an entirely different reason.

WHAT IS PINNED AND WHAT IS NOT. The golden pins the answer THIS model gives. Nothing here is validated against a measured batch electrodialysis: the stack record reproduces the paper's bench unit and its limiting-current correlation, but no run of that unit in recirculating batch is transcribed anywhere in this tree. 't_overLimiting_s' is resolved to the TIME STEP – it is the first accepted step at which i exceeds i_lim, so it moves with deltaT (1593 s at deltaT 1, 1592.5 s at deltaT 0.25) while everything else is converged to ~1e-8.

NOT MODELLED, and it matters to how far this run can be trusted: no salt diffuses back from the concentrate, no water is transported with the ions, the current efficiency is a declared constant apart from the plateau above, and the rig is isothermal with the Joule heat removed by no modelled duty (which is why the campaign energy balance reports UNAVAILABLE and says why). The full list is in the unit's own header.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 8.52905708999e-14 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 8.52905708999e-14 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 8.52905708999e-14 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 1.39662189668e-15 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 10.1751624752 |
| kpi | campaign | mass_kg_initial | 10.1751624752 |
| kpi | campaign | mass_residual_kg | 1.42108547152e-14 |
| kpi | campaign | moles_kmol_external_out | 0 |
| *… and 44 more reference values in the case's `expected`.* | | | |

## edbatch02_constant_voltage

tutorials/batch/electrodialysis/edbatch02_constant_voltage




**What it does:** Batch recirculating electrodialysis at constant voltage: the current decays as the diluate loses conductivity, and never reaches the limiting current

**The story:** THE SAME RIG, one knob different, and the hand calculation failing for the opposite reason.

Everything is identical to ‘edbatch01_constant_current‘ – the same stack, the same two tanks, the same recirculation flow, the same xi, the same electrode term – except that the rectifier is held at 4 V instead of 3.5 A. The difference between the two runs IS the difference between the two ways of driving a batch electrodialysis.

WHAT THE RUN MEASURES (printed by this run; nothing here is a target):

I 4.365 A at t = 0 → 2.018 A at t = 1500 s i/i_lim 0.3394 → 0.4086 NEVER reaches 1 demineralisation 61.59 % ACTUAL 85.51 % what the hand calculation predicts the hand calculation’s own answer: the tank is EMPTY at t = 1754 s diluate 0.1 → 0.03841 mol/kg kappa (diluate) 1.262 → 0.4846 S/m E_mem (per pair) 0.0794 → 0.1313 V specific energy 1.048 kWh per m3 of diluate treated material closure machine level, as in edbatch01

WHY THE CURRENT FALLS. The stack equation is

U = N ( E_mem + I R_pair ) + E_electrodes

and BOTH terms inside the bracket move against the current as the run proceeds. The diluate loses conductivity, so R_pair rises (0.0636 → 0.1119 Ohm per cell pair); and the concentration RATIO between the two tanks grows, so the Nernst back-EMF rises with it. At fixed U the current has nowhere to go but down, and it more than halves.

WHY IT NEVER CROSSES i_lim. i_lim falls as the diluate depletes – exactly as in edbatch01 – but so does i, and here they fall together: the ratio ends at 0.409 having started at 0.339. That is the reason a real bench rig is run this way when the membranes matter: constant voltage is self-limiting, and the price is time. This case declares NO ‘overLimiting {}‘ model at all, and the run announces that default; it is irrelevant here precisely because the crossing never happens.

THE LESSON, and it is the same lesson as edbatch01 arriving by a different road. ‘demin_ideal‘ is Faraday’s law at the INITIAL current – a straight line, computed from this run’s own I(0), own inventory and own xi. It is 24 percentage points optimistic at 1500 s, and not one of those points has anything to do with the limiting current: the current simply is not what it was. Read ‘I_A‘ in trajectory.csv beside ‘deminActual‘ and ‘deminIdeal‘ and the whole of the gap is in that first column.

WHAT IS PINNED AND WHAT IS NOT: as edbatch01. The golden pins the answer this model gives; no measured batch electrodialysis is reproduced anywhere in this tree.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 3.03576608288e-15 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 3.03576608288e-15 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 3.03576608288e-15 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 10.1751624752 |
| kpi | campaign | mass_kg_initial | 10.1751624752 |
| kpi | campaign | mass_residual_kg | 0 |
| kpi | campaign | moles_kmol_external_out | 0 |
| *… and 44 more reference values in the case's `expected`.* | | | |

## edbatch03_multiionic_split

tutorials/batch/electrodialysis/edbatch03_multiionic_split




**What it does:** Batch electrodialysis of a mixed NaCl + K2SO4 brine: two counter-ions on each membrane share its current by z D c

**The story:** a batch electrodialysis rig on a MIXED brine, and two counter-ions on each membrane sharing its current.

THE OPERATION. The rig of edbatch01 (Eurodia EUR2C-7P18, seven cell pairs, 175 L/h recirculation), but the diluate tank holds 0.05 mol/kg NaCl plus 0.025 mol/kg K2SO4 – two cations (Na+, K+) on the cation-exchange membrane and two anions (Cl-, SO4 2-) on the anion-exchange one. The rectifier is held at 2.0 A (100 A/m2 on one cell pair).

THE RULE (Vitor's ruling C35, 2026-10-05). Until that day this unit REFUSED a diluate with more than one cation or one anion, for want of a rule dividing a membrane's current between its counter-ions. The ruling shares it by

t_k = |z_k| D_k c_k / SUM_j |z_j| D_j c_j over that membrane's counter-ions

– the transport numbers of Geraldes & Afonso, J. Membr. Sci. 360 (2010) 499-508, Eqs. (12)/(13), applied at every current and not only at the limit – with D the ions' curated D0 and NO new parameter. Each ion then leaves at t_k xi I N / (|z_k| F) mol/s, so each membrane moves exactly xi I N / F equivalents and both tanks stay electroneutral.

WHAT A STUDENT SHOULD SEE. Divide that rate by the ion's own inventory:

d ln n_k / dt = - D_k xi I N / ( F SUM_j |z_j| D_j n_j )

– the FRACTIONAL depletion rate of a counter-ion is proportional to its diffusivity and to nothing else about it, so on a log scale the counter-ions of one membrane fall along straight lines whose slopes stand in the ratio of their D0. K+ (1.96e-9 m2/s) is depleted 1.47 times faster than Na+ (1.33e-9) in log terms; Cl- (2.03e-9) 1.92 times faster than SO4 2- (1.06e-9), the charge of the sulfate making no difference to its FRACTIONAL rate (it carries twice the current per mole and so moves half the moles per coulomb, exactly compensating its doubled share). The run publishes 'demin_<ion>' for every ion and 't_transfer_final_<ion>', the shares at the end, and its trajectory carries every molality.

WHAT THE RUN MEASURES (every number is printed by this run; the golden pins them): i 100.00 A/m2, flat, all 2400 s i_lim 696.83 A/m2 at t = 0, 244.39 at t = 2400 s (i/i_lim 0.144 → 0.409: never crossed) removed at t = 2400 s K+ 69.77 % Na+ 55.60 % (CEM) Cl- 74.43 % SO4 2- 50.94 % (AEM) share of the current Na+ 0.404 → 0.499 of the CEM's Cl- 0.657 → 0.500 of the AEM's material closure 0 on mass, 1.4e-15 on the worst element

READ THE SHARES BACKWARDS AND THE LESSON IS THERE. At t = 0 the two cations are equally concentrated, so K+, the faster, takes 0.596 of the cation current; it therefore depletes faster, its c falls, and the share drifts toward Na+. The hand calculation 'demin_ideal' (Faraday's law at the INITIAL current AND the reference cation's INITIAL share, computed from the run) predicts 50.68 % of the sodium removed; the run removes 55.60 %, because sodium's share of the current GROWS as potassium is stripped – the opposite sign from edbatch01's gap, and for a different reason. And ln(n/n0)/D0 comes out as one number for K+ and Na+, and one number for Cl- and SO4 2-, to the integrator's precision.

WHAT IS PINNED AND WHAT IS NOT. The golden pins the answer THIS model gives. The share z D c is a RULING, not a measurement: real ion-exchange membranes add a preference between counter-ions of their own (a Donnan preference for divalent ions, or a monovalent-selective layer) which no 'kind IEM' record in this tree carries and this unit does not model. The brine is an author-set assumption; nothing here is validated against a measured rig. The membrane potential of a membrane with two counter-ions is taken as the MEAN of their own Nernst potentials (announced). Back- diffusion, water transport and every thermal effect are not modelled – the unit's own header lists them.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 1.01192202759e-15 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_K_closure_rel | 1.44560289655e-15 |
| kpi | campaign | element_Na_closure_rel | 1.15648231724e-15 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_S_closure_rel | 1.01192202759e-15 |
| kpi | campaign | element_worst_closure_rel | 1.44560289655e-15 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 10.2181899752 |
| kpi | campaign | mass_kg_initial | 10.2181899752 |
| *… and 56 more reference values in the case's `expected`.* | | | |

## membrane

## diafilter01_nf_desalting

tutorials/batch/membrane/diafilter01_nf_desalting




**What it does:** Constant-volume NF diafiltration: NaCl washed out of an MgSO4 retentate, with the OBSERVED rejection published beside the constant-R washout law it violates

**The story:** constant-volume diafiltration on a loose NF membrane, and the exponential washout law failing in a way you can watch.

THE OPERATION. 10 litres of a brine carrying TWO salts sit behind 5 m2 of NF270 at 10 bar. Solvent is made up as fast as it permeates, so the volume is held and the run is measured in DIAVOLUMES – how many vessel volumes of wash water have passed through. NaCl is poorly rejected by this membrane and leaves; MgSO4 is well rejected and stays. That is the whole point of a nanofiltration desalting step, and it is the same arithmetic whether the retained species is a divalent salt, a sugar or a protein.

THE LESSON. Derive the washout by hand. Over a constant volume V, with the permeate carrying c_p = (1 - R) c,

V dc/dt = -Q_p (1 - R) c , N = Q_p t / V => c/c_0 = exp( -(1 - R) N )

which is a straight line on a log axis against N. The derivation needs R to be CONSTANT. It is not:

as NaCl leaves, the osmotic pressure of the retentate falls, so the driving force rises and so does the water flux; a higher flux sweeps more solute to the wall, so the polarisation factor rises and the WALL concentration is no longer the bulk one; the observed rejection is measured against the bulk, so it moves even when the membrane's own permeability has not changed.

The run therefore publishes R_obs for each solute at every instant, beside the diavolumes, and – because this is constant-volume mode – two KPIs per solute: 'washoutActual_<name>' (what the vessel did) and 'washoutIdeal_<name>' (what exp(-(1-R)N) predicts using THIS run's own initial R). The idealisation is computed from the run, not declared, so the gap between them cannot have been arranged.

Read them together. Where the two agree, the hand derivation was safe. Where they part, the R_obs column says which assumption failed and when.

WHAT IS PINNED AND WHAT IS NOT. The golden pins the answer this model gives; NOTHING here is validated against a measured diafiltration. The NF270 permeabilities are the case-fitted pedagogical set the record declares ('provenance fittedToCase', fitted against membrane02_NF_sugar), so the rejections are representative of a loose NF membrane and are not a claim about a particular element. Fouling is NOT modelled – the flux in a real rig would decline with time and this one does not.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 1.23912409123e-16 |
| kpi | campaign | element_Mg_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 0 |
| kpi | campaign | element_O_closure_rel | 1.23884758239e-16 |
| kpi | campaign | element_S_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 1.23912409123e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_amend_in | 54.5637624826 |
| kpi | campaign | mass_kg_amend_out | 0 |
| kpi | campaign | mass_kg_external_out | 54.5637624826 |
| *... and 51 more reference values in the case's `expected`.* | | | |

## diafilter02_fouling_decline

tutorials/batch/membrane/diafilter02_fouling_decline




**What it does:** The same diafiltration with a DECLARED Hermia cake-law decline: what fouling costs a washout schedule

**The story:** the same diafiltration, with the membrane losing permeance as it works.

WHY THIS CASE EXISTS. diafilter01's flux RISES during the run (233 → 256 LMH): the salt leaves, the osmotic pressure falls, the driving force grows. That is the physics the model has. It is not what a rig does. On real equipment the flux falls, because the membrane fouls, and every schedule a student will ever write is sized on that decline.

THE MODEL. Resistance in series on the permeance the unit hands the transport law:

1 / A_eff = 1 / A_w + r_f(v), v = V_permeated / A

with r_f growing by one of HERMIA'S BLOCKING LAWS (Hermia, Trans. IChemE 60:183-187, 1982). Here 'cake': r_f = k v, the deposited-layer form. Applied where the transport CONTEXT is assembled, so solution-diffusion and DSPM-DE are served alike and neither is modified.

THE CONSTANT IN THIS CASE IS HYPOTHETICAL AND IS NOT A FITTED VALUE FOR THIS SYSTEM. ∗∗ A brine of NaCl and MgSO4 is fully dissolved; there is no suspended matter to deposit, so 'cake' is not a mechanism this feed can honestly claim. What this case demonstrates is the CONSEQUENCE of a declared decline on a diafiltration schedule – how many more minutes the same washout costs – on a system whose other numbers are already pinned by diafilter01, so the difference between the two runs is the fouling and nothing else. It is a STRUCTURAL witness, in the same sense as edwards01_sour_water_activity: no number in it is validated against a measured fouling experiment, and the tree carries no fouling data at all.

A fouling constant belongs to a RIG AND A MATERIAL, not to a membrane, so it is declared here in the case and no catalogue record is touched.

WHAT TO READ. 'A_eff_over_A_w' in the trajectory is the permeance the membrane still has, as a fraction of clean. Put diafilter01 and diafilter02 side by side: same charge, same area, same pressure, and the diavolume count at t = 160 s is the whole lesson.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 4.17214965497e-16 |
| kpi | campaign | element_Mg_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 0 |
| kpi | campaign | element_O_closure_rel | 4.17110477583e-16 |
| kpi | campaign | element_S_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 4.17214965497e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 3.70318765373e-16 |
| kpi | campaign | mass_kg_amend_in | 47.5260962641 |
| kpi | campaign | mass_kg_amend_out | 0 |
| kpi | campaign | mass_kg_external_out | 47.5260962641 |
| *... and 52 more reference values in the case's `expected`.* | | | |

## diafilter03_dspmde_batch

tutorials/batch/membrane/diafilter03_dspmde_batch




**What it does:** Batch NF concentration on the DSPM-DE pore model: the observed rejection per instant and the campaign's time-averaged rejection

**The story:** batch CONCENTRATION of a two-salt brine on the DSPM-DE pore model, and the rejection of a whole campaign stated as a TIME average.

THE OPERATION. The charge of diafilter01 (10 litres, 0.02 kmol/m3 each of NaCl and MgSO4) sits behind 1 m2 of NF270 at 10 bar with NO make-up solvent, so the vessel shrinks and both salts concentrate. The membrane is the pore tier of the curated NF270 record ('NF270_dspmde': pore radius, A_k/dx, a constant volumetric charge, a pore dielectric constant), and the transport law is DSPM-DE – steric, Donnan and dielectric exclusion with the extended Nernst-Planck equations SOLVED across the pore – asked at every instant of the integration, exactly as the spiral module asks it at every station of its channel. The film is the declared k_film with the default film theory (announced).

WHAT THE RUN SHOWS. The observed rejection is not a property of the membrane alone; it moves with the solution the membrane sees. As the retentate concentrates the NaCl rejection FALLS (0.691 at the start, 0.554 at the end of this run) while the MgSO4 rejection RISES slightly (0.973 → 0.979). In DSPM-DE two things push the monovalent salt the same way – a bulk concentration growing past the fixed charge weakens the Donnan exclusion, and a growing share of retained divalent ions changes what must cross to keep the permeate neutral – and this run does not separate them. The flux falls with the osmotic pressure (244 → 116 LMH) and the volume reduction reaches 3.9. Read 'R_obs_<salt>' in the trajectory against 'V_m3'.

THE REJECTION OF THE CAMPAIGN. A batch rig is usually judged by the permeate it COLLECTED, not by one instant, and two time averages of the rejection are published at the end (DEFINED in BatchDiafilter.cpp, kpis(), and in docs/ai/membrane-research.md).

R_avg = 1 - c_permPooled / c_bMeanPerm, where c_permPooled = (n_0 - n) / V_perm is the pooled permeate (every drop collected, mixed) and c_bMeanPerm = INTEGRAL c_b dV_perm / V_perm is the retentate's bulk concentration averaged over what permeated. It is a weighted mean of the instantaneous R_obs (weights c_b Q_p), so it lies between the smallest and the largest of them (NaCl 0.635).

R_avgVsInitialFeed = 1 - c_permPooled / c_bInitial compares the pooled permeate with the CHARGE and, because the retentate only concentrates, lies below R_avg (NaCl 0.467) – the number a laboratory gets by analysing its collected permeate against the feed it started from, and a reminder that it is not the membrane's rejection. The steady spiral module states the SPATIAL counterpart: its R_obs is the mixed permeate of the whole channel against the inlet feed.

WHAT IS VERIFIED AND WHAT IS NOT. Verified (check_diafiltration): the DSPM-DE pore model runs inside the batch vessel at every instant; matter is conserved to machine precision; the two averages reproduce from the published integrals and from the trajectory; R_avg lies inside the range of R_obs; the pooled-permeate rejection lies below it, as it must when the retentate concentrates. NOT validated: no number here is compared with a measured batch filtration. The pore parameters are the record's case-fitted pedagogical set ('provenance fittedToCase', fitted against membrane10_dspmde_divalent), so the rejections are those of a representative loose NF membrane, not a claim about a particular element. In a MIXED feed the salt basis carries each salt's cation leg through the pore and the run says so; fouling is not declared; k_film is declared, not correlated with a stirrer or a channel.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_Cl_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 1.03422137392e-12 |
| kpi | campaign | element_Mg_closure_rel | 0 |
| kpi | campaign | element_Na_closure_rel | 0 |
| kpi | campaign | element_O_closure_rel | 1.03273316665e-12 |
| kpi | campaign | element_S_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 1.03422137392e-12 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 1.03031994669e-12 |
| kpi | campaign | mass_kg_external_out | 7.46972743873 |
| kpi | campaign | mass_kg_final | 2.56615820568 |
| kpi | campaign | mass_kg_initial | 10.0358856444 |
| *… and 40 more reference values in the case's `expected`.* | | | |

## plant

## apalutamide01_tablets

tutorials/batch/plant/apalutamide01_tablets

**What it does:** APALUTAMIDE 60 mg FILM-COATED TABLETS, a drug-product plant (the API is BOUGHT), a generic entrant at loss of exclusivity, on a SOURCED design basis (docs/design/apalutamide-design-basis-dossier.md): the originator's patent formula (US 12,303,493 B2: a 1:3 apalutamide:HPMCAS-LG dispersion, a 700 mg core, a 3 % Opadry II coat), the dispersion spray-dried from 5 wt % acetone (the patent's acetone example) through a pressure nozzle in a closed nitrogen loop (blower, heater, spray dryer, exhaust cooler, -10 C solvent condenser, purge and N2 make-up), the recovered acetone to a tank for QC; dry granulation (blending, roller compaction, milling, a screen with a fines recycle), final blend, tableting with rejects, film coating (an aqueous coat sprayed on and its water dried off by hot air) and packaging. The steady flowsheet is the material balance per unit of product (15 625 tablets/h pressed, the patent's PSD-3 dryer at 75 kg/h of solution, is a basis); a BATCH PLAN (system/postDict) sizes each step for a 400 kg lot, counts 25 lots a year for 14.0 million tablets (25 % of the estimated European market), and finds the pace of each train – the spray dryer on round-the-clock campaigns, the solid-dose suite lot by lot on two shifts. Blending, compaction, milling, screening, tableting and packaging are MASS-BALANCE PLACEHOLDERS (no equipment physics). Every number the dossier could not source is an author-set assumption, said so where it is declared (README).

**The story:** a drug-product plant: apalutamide 60 mg film-coated tablets. The API is BOUGHT; everything else is made here (DEV.md 4c C97, phase A). A BASE CASE: it runs, it is NOT optimised, and every number a group would change is a plain key, said to be an assumption where it is declared. The README is the narrative; this file is the topology.

THE BASIS (README section 2; the design-basis dossier, docs/design/apalutamide-design-basis-dossier.md): 15 625 tablets per hour PRESSED, the flowsheet basis, of the originator's patent formula (US 12,303,493 B2, Examples 3.1-3.3): a 700 mg core carrying 60 mg of apalutamide in 240 mg of a 1:3 apalutamide:HPMCAS-LG spray-dried dispersion, with a 21 mg (3.0 %) film coat. 15 625 tablets/h is the patent's PSD-3 dryer at its 75 kg/h of 5 wt % solution (3.75 kg/h of dispersion / 240 mg). THIS STEADY FLOWSHEET IS THE MATERIAL BALANCE PER UNIT OF PRODUCT (DEV.md 4c C99): the plant runs in BATCHES (lots of final blend, released one by one) and its spray dryer in CAMPAIGNS, and the BATCH PLAN in system/postDict sizes each step for one lot from these streams' masses, counts the lots a year from the declared production, and finds the step that paces each train. The absolute rate below is a basis – except for the spray dryer, whose hardware this flowsheet models, so its rate IS the dryer's capacity.

SOLUTION AND SPRAY DRYING (the closed nitrogen loop) acetoneFeed → [acetoneTank] → acetoneToPrep –+ apiFeed (BOUGHT) ————————-+→ [solutionPrep] → sprayFeed –+ hpmcasFeed ————————-+ | v n2Makeup –+ [SD] sprayDryer +→ [n2Mix] → [blower] → [gasHeater] → dryingGas ———→ | | gasRecycle-+ ^ 393 K asdPowder exhaustGas | | | | [exhaustCooler] 308 K (cooling water) | | | | [solventCondenser] 263 K (NH3 refrigeration) | | | | recoveredSolvent condenserGas | | | | [recoveredTank] [purgeSplit] → purgeGas (vent) | recoveredAcetone | +———————————— gasRecycle (TEAR)

DRY GRANULATION, TABLETING, COATING, PACKAGING asdPowder + MCC + croscarmellose + silica (the patent's intragranular half) → [blender] → intraBlend → [rollerCompactor] → ribbons → [mill] → milledGranules ^ | +———- finesRecycle (TEAR) <— [screen] → granules granules + silicified MCC + croscarmellose + silica + MgSt → [finalBlender] → finalBlend → [tabletPress] → tabletCores +→ tabletRejects water + PVA + PEG + talc + TiO2 + iron oxide → [coatingSuspensionPrep] → coatingSuspension tabletCores + coatingSuspension → [coatingPan] → wetCoatedTablets coatingAirIn → [airHeater] 333 K → coatingAir wetCoatedTablets + coatingAir → [coatingDryer] → coatedTablets + coatingExhaust coatedTablets → [packagingLine] → packagedTablets

TWO RECYCLES, both declared in system/solverDict: the drying gas (gasRecycle) and the compactor fines (finesRecycle). The recovered acetone is NOT recycled in this base case: it goes to a tank, for QC and reuse – the README says why, and closing that loop is a group's question.

WHAT IS A PLACEHOLDER. The blenders, the roller compactor, the mill, the screen, the tablet press and the packaging line are FLOW PRIMITIVES (a 'mixer', a 'splitter'): they conserve mass and energy and model NOTHING of the machine – no blend uniformity, no ribbon density, no particle-size reduction, no compaction force, no tablet hardness. Choupo has no such equipment model; none is invented here.

The coating pan is TWO boxes, a ‘mixer‘ (the spray lands on the cores) and a ‘solidDryer‘ (the hot air takes the water off), which is the mass and energy balance of one machine.

*A BASE CASE for a pharmaceutical-industry project group (DEV.md 4c C97, phase A; re-based on a sourced design basis in C110). The plant makes film-coated tablets of apalutamide: the active pharmaceutical ingredient (API) is bought, everything else is made on site — the spray-dried solid dispersion, the granules, the tablets, the coat and the packs. It runs (a few seconds from a cold start), it converges, its mass and energy balances close, and it is deliberately NOT optimised — every number a group would change is a plain key, and every one says where it came from.*

*The case is sealed: everything it reads is in its own constant/.*

*Its design basis is a dossier, not a guess: [docs/design/apalutamide-design-basis-dossier.md](../../../../docs/design/apalutami design-basis-dossier.md) ("the dossier" below) — the EMA and FDA assessment reports, the label, the originator's patent, the originator's published sales, generic-price evidence and a red-team review, every number with its source, page and a confidence word (∗stated∗, ∗derived∗, ∗estimate∗, ∗not found∗). Read it before you change anything: most of what you will be asked to defend is already argued there, and what is NOT is listed in its section 6 as the questions only a supplier or the originator can answer.*

*Read section 3 before anything else. Half of this plant is physics Choupo computes (the spray dryer, the nitrogen loop, the solvent recovery, the coating dryer, the heat exchangers); the other half is a mass balance with no equipment model at all (blending, compaction, milling, screening, tableting, packaging). The two halves look alike in the stream table. They are not alike.*

*The amorphous dispersion this plant makes — apalutamide in HPMCAS, 1:3, spray dried — is claimed by an active patent family of the originator (US 12,303,493 B2, Aragon Pharmaceuticals / Janssen: claim 1 the dispersion, claim 4 the 1:3 ratio, claim 10 spray drying, claims 18 and 20 the 60 mg and 240 mg tablets; anticipated expiry 3 Dec 2035, an assumption by Google Patents, not a legal conclusion; the European counterparts and their SPCs were NOT verified — dossier section 3.6). A plant that copies this formulation and sells before then infringes, unless the patents fall.*

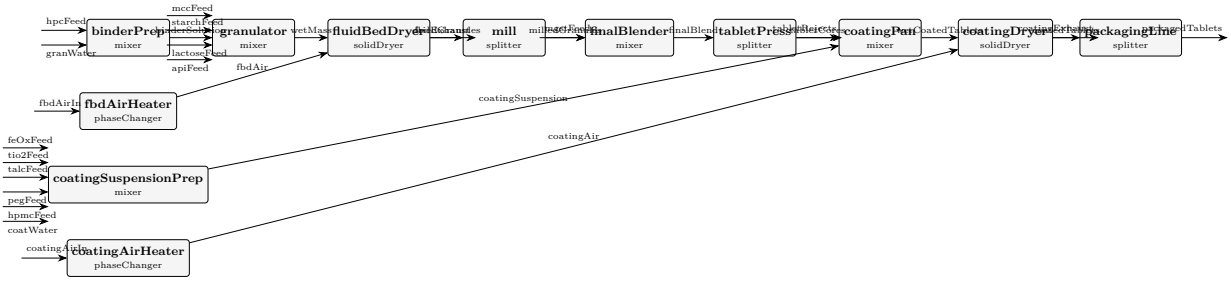
*So the case declares its business case, as Vitor ruled (DEV.md 4c C99 addendum): a generic entrant to the European market at loss of exclusivity. Three consequences, each in the case:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | acetoneFeed | F | 0.000340765610652 |
| stream | acetoneFeed | T | 298.15 |
| stream | acetoneToPrep | F | 0.000340765610652 |
| stream | acetoneToPrep | T | 298.15 |
| stream | apiFeed | F | 5.45455180278e-07 |
| stream | apiFeed | T | 298.15 |
| stream | asdPowder | F | 1.32670518028e-06 |
| stream | asdPowder | T | 319.240012408 |
| stream | blowerOut | F | 0.00469098187359 |
| stream | blowerOut | T | 271.542841827 |
| stream | ccsExtraFeed | F | 7.59548611111e-08 |
| stream | ccsExtraFeed | T | 298.15 |
| stream | ccsIntraFeed | F | 7.59548611111e-08 |
| stream | ccsIntraFeed | T | 298.15 |
| *... and 1474 more reference values in the case's `expected`.* | | | |

## donepezil01_tablets

tutorials/batch/plant/donepezil01_tablets




**What it does:** DONEPEZIL HYDROCHLORIDE 10 mg FILM-COATED TABLETS, a drug-product plant (the API is BOUGHT): WET GRANULATION – the API, lactose monohydrate, maize starch and microcrystalline cellulose wetted in a high-shear granulator with an aqueous hydroxypropyl-cellulose binder solution and dried in a fluid bed ('solidDryer', hot air, the starch's sorption isotherm) – milling, a final blend with magnesium stearate, tableting with rejects, an aqueous hypromellose film coat dried off by hot air, and packaging. The excipients are the reference product's (ARICEPT SmPC section 6.1) and the 174.33 mg of lactose per 10 mg tablet is its section 2; every other quantity is an author-set assumption. The steady flowsheet is the material balance per unit of product (720 000 tablets/h pressed, the fluid bed's own drying rate, is a basis); a BATCH PLAN (system/postDict) sizes each step for a 300 kg lot, counts the lots a year for 150 million tablets, and finds the pace of the granulation and tableting trains. Blending, granulation, milling, tableting and packaging are MASS-BALANCE PLACEHOLDERS (no equipment physics). No organic solvent, no spray drying, no hazardous area. One strength (10 mg) for now; the two-strength campaign waits for a multiproduct batch plan (README).

**The story:** a drug-product plant: donepezil hydrochloride 10 mg film-coated tablets. The API is BOUGHT; everything else is made here (DEV.md 4c C104, built on the stated defaults of C110). A BASE CASE: it runs, it is NOT optimised, and every number a group would change is a plain key, said to be an assumption where it is declared. The README is the narrative; this file is the topology.

THE BASIS (README section 2): 720 000 tablets per hour PRESSED, the flowsheet basis, of a 280 mg core carrying 10 mg of donepezil hydrochloride and 174.33 mg of lactose monohydrate (the reference product's own figures), with a 3 % film coat. THIS STEADY FLOWSHEET IS THE MATERIAL BALANCE PER UNIT OF PRODUCT (DEV.md 4c C99): the plant runs in BATCHES (lots released one by one), and the BATCH PLAN in system/postDict sizes each step for one lot from these streams' masses, counts the lots a year from the declared production, and finds the step that paces each train. The absolute rate is chosen as the fluid bed's own DRYING rate (one 300 kg lot dried in 1.5 h), so the fluid bed's air heater is sized for the duty it really has; the coater's heater is then sized at that same rate, which is about twice the coater's own – README section 6.

WET GRANULATION granWater + hpcFeed → [binderPrep] → binderSolution –+ apiFeed (BOUGHT) + lactose + starch + MCC ————+→ [granulator] → wetMass fbdAirIn → [fbdAirHeater] 333 K → fbdAir | wetMass + fbdAir → [fluidBedDryer] → driedGranules + fbdExhaust driedGranules → [mill] → milledGranules

FINAL BLEND, TABLETING, COATING, PACKAGING milledGranules + mgstFeed → [finalBlender] → finalBlend → [tabletPress] → tabletCores +→ tabletRejects coatWater + HPMC + macrogol + talc + TiO2 + yellow iron oxide → [coatingSuspensionPrep] tabletCores + coatingSuspension → [coatingPan] → wetCoatedTablets coatingAirIn → [coatingAirHeater] 333 K → coatingAir wetCoatedTablets + coatingAir → [coatingDryer] → coatedTablets + coatingExhaust coatedTablets → [packagingLine] → packagedTablets

NO RECYCLE, NO SOLVENT. Unlike apalutamide01_tablets there is no spray dryer, no organic solvent, no nitrogen loop, no solvent recovery and no hazardous area: an aqueous wet granulation and an aqueous coat.

WHAT IS A PLACEHOLDER. The high-shear granulator, the mill, the final blender, the tablet press and the packaging line are FLOW PRIMITIVES (a 'mixer', a 'splitter'): they conserve mass and energy and model NOTHING of the machine – no granule growth, no granule size, no blend uniformity (THE critical quality question of a low-dose tablet, README section 8), no compaction force, no tablet hardness. Choupo has no such equipment model; none is invented here. The two DRYERS are physics: 'solidDryer' dries to the starch's sorption isotherm at the exhaust air's humidity, with the energy balance on the package's

enthalpy. The coating pan is TWO boxes, a ‘mixer‘ (the spray lands on the cores) and a ‘solidDryer‘ (the hot air takes the water off), which is the mass and energy balance of one machine.

*A BASE CASE for a pharmaceutical-industry project group (DEV.md 4c C104, built on the stated defaults of C110). The plant makes film-coated tablets of donepezil hydrochloride by wet granulation: the active pharmaceutical ingredient (API) is bought, everything else is made on site — the granules, the tablets, the coat and the packs. It runs (in seconds), its mass and energy balances close, and it is deliberately NOT optimised — every number a group would change is a plain key, said to be an assumption where it is declared.*

*The case is sealed: everything it reads is in its own constant/.*

*It is the simpler sibling of apalutamide01_tablets (the same folder, the same grammar, read that case's README for anything not repeated here). There is no spray drying, no organic solvent, no nitrogen loop, no solvent recovery and no hazardous area (ATEX): an aqueous binder, an aqueous coat, two hot-air dryers. What is harder here is not the chemistry but the dose: 10 mg of API in a 280 mg core (section 8).*

*Read section 3 before anything else. Two units of this plant are physics Choupo computes (the fluid-bed dryer and the coating dryer, with their air heaters); the rest is a mass balance with no equipment model at all (granulation, milling, blending, tableting, packaging). The two look alike in the stream table. They are not alike.*

*Two inferences, labelled as such: the lactose of the two strengths is in the ratio 2:1, which is what ONE common blend pressed at two tablet weights would give (a dose-proportional pair); and the yellow iron oxide is the only ingredient the two strengths do not share — so the two products can share the whole granulation and differ only from the press on.*

*Where the group must look for what this case could not verify: ARICEPT was not authorised through the EMA's centralised procedure as far as this case could find, so there is no EPAR to read; the national SmPCs and public assessment reports (MHRA, HPRA, the other national agencies) and the public assessment reports of the generic donepezil products (written for each generic's authorisation, and which may describe its manufacturing route) are where the formulation and the process are described; the Ph. Eur. monographs of the excipients for their grades. Replace each assumption with what you find, and cite it.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | apiFeed | F | 4.80817775e-06 |
| stream | apiFeed | T | 298.15 |
| stream | binderSolution | F | 0.000740863846794 |
| stream | binderSolution | T | 298.15 |
| stream | coatWater | F | 0.000677035803497 |
| stream | coatWater | T | 298.15 |
| stream | coatedTablets | F | 0.000238250706177 |
| stream | coatedTablets | T | 317.588908471 |
| stream | coatingAir | F | 0.0702777777778 |
| stream | coatingAir | T | 333.15 |
| stream | coatingAirIn | F | 0.0702777777778 |
| stream | coatingAirIn | T | 298.15 |
| stream | coatingExhaust | F | 0.0709548135813 |
| stream | coatingExhaust | T | 317.588908471 |

*... and 1065 more reference values in the case's `expected`.*

## reactor

**The theory you need.** Three idealised steady reactors. Conversion reactor: stoichiometry only, $n_{\text{out}} = n_{\text{in}} + \nu\,\xi$, duty from the ONE enthalpy base $\Delta H_{\text{rxn}}(T) = \sum_i \nu_i h_i(T)$ (formation datum – no separate “heat of reaction” input, and any dict override is cross-checked aloud). CSTR: outlet at reactor conditions with rate $r(c_{\text{out}}, T)$, so $F\,(c_{\text{in}} - c_{\text{out}}) = r\,V$. PFR: the same rate integrated along the volume, $\mathrm{d}F_i/\mathrm{d}V = \nu_i\,r$. Arrhenius kinetics $k = A\,T^{b} e^{-E_a/RT}$, orders from the dict, reverse rates from the equilibrium constant (thermo-consistent by construction). Expect: CSTR conversion below PFR at equal volume (back-mixing), exponential $T$ sensitivity, and equilibrium limits the kinetics can approach but never cross. *Full derivation: Theory Guide, the reaction-engineering chapter.*

## batch01_first_order

tutorials/batch/reactor/batch01_first_order




**What it does:** Isothermal batch Fischer esterification (AcOH + EtOH <→ EtOAc + H2O) at 350 K, pseudo-first-order in ethanol; analytical exp-decay comparison

**The story:** isothermal pseudo-first-order batch reactor.

Single vessel, pseudo-first-order esterification (AcOH + EtOH → EtOAc + H2O, rate = k.[ethanol]). Initial charge is an equimolar ethanol + acetic acid mix; kinetics from the reactions library (Arrhenius A=1e8 1/s, Ea=70 kJ/mol). At T = 350 K:

k(T) = 1e8 · exp(-70000 / (8.314 · 350)) ≈ 3.57e-3 s^-1 $\tau$ = 1/k ≈ 280 s

Analytical solution for first-order in a constant-volume batch:

n_A(t) = n_A(0) · exp(-k t)

The simulator integrates dn_A/dt = -k · (n_A / V) · V = -k n_A with RK4 at dt = 1 s; final t = 600 s gives X_A ≈ 1 - exp(-k·600) ≈ 1 - exp(-2.14) ≈ 0.882

*18.5 mol of an equimolar ethanol / acetic-acid charge (about a litre) is held at 350 K in a closed vessel for 600 s and the Fischer esterification runs — the kinetics of cstr01, now integrated in time by RK4 at 1 s steps. The golden: ethanol falls from 9.25 mol to 1.083 mol (X = 88.3 %), 8.167 mol of ethyl acetate made, and the jacket has removed only 17.18 kJ to hold the temperature — exactly -2.10 kJ/mol of reaction at 350 K, the elements-datum heat every reactor in Choupo prices on. The liquid esterification is only mildly exothermic (calorimetry: about -3 to -4 kJ/mol), so the jacket duty here is small, not the cooling load an exotherm of tens of kJ/mol would need.*

*1. choupoBatch, not choupoSolve. A batch vessel has no steady state; the case declares endTime 600; deltaT 1.0; writeInterval 30; in controlDict and the engine integrates dn/dt = $\nu$ r V. The Log tab prints a snapshot every 30 s: four inventories and T. 2. First order means an exponential, and you can check it. With k = 3.575e-3 s?[1] at 350 K (the same Arrhenius pair as cstr01), n(600 s) = 9.25·e^(-2.145) = 1.083 mol. The table is that curve, sampled. 3. The campaign ledger closes at machine precision. [campaign] mass balance ? closure 4.5e-16, energy balance: dH_vessels = -17.18 kJ ? = Q_ledger -17.18 kJ, closure 0 — every joule the jacket took is a state difference on the charge, never a quadrature, which is why the closure is zero and not merely small. 4. A CSTR at the same $\tau$ does worse. 600 s here gives 88 %; the continuous reactor at $\tau$ = 310 s gave 53 % and a PFR 67 %. Three reactors, one rate law, three answers — the batch one is the upper envelope, because nothing leaves early.*

*Set T_setpoint 370 K;: k roughly triples, the exponential steepens, and the ledger's Q follows the extent (small, because the reaction is only mildly exothermic). Then batch02_adiabatic removes the jacket, and T becomes one more state on the trajectory.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_H_closure_rel | 3.00060276926e-16 |
| kpi | campaign | element_O_closure_rel | 2.50050230771e-16 |
| kpi | campaign | element_worst_closure_rel | 3.75075346157e-16 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | -17.1846198661 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 0 |
| kpi | campaign | energy_dH_vessels_kJ | -17.1846198661 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | energy_residual_kJ | 0 |
| kpi | campaign | mass_closure_rel | 4.5240474843e-16 |
| kpi | campaign | mass_kg_external_out | 0 |

*... and 26 more reference values in the case's `expected`.*

## batch01b_first_order_eulersi

tutorials/batch/reactor/batch01b_first_order_eulersi




**What it does:** GREEN WITNESS: semi-implicit Euler (EulerSI) on batch01 mild isothermal esterification – first order suffices off-stiffness; twin of the ignition01_h2o2_eulersi expected failure.

**The story:** GREEN WITNESS for the EulerSI integrator: batch01's mild isothermal esterification, integrated by semi-implicit (linearly-implicit) Euler instead of the RK4 default. On a non-stiff problem the first-order method simply works, and the golden pins that its answer agrees with the physics batch01 already pins. Its EXPECTED-FAILURE twin is ignition01_h2o2_eulersi (stiff nonlinear ignition, where first order goes unphysical and the engine refuses by name). Registered model no case selected until 2026-08-22 (invariant I8).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_H_closure_rel | 4.50090415389e-16 |
| kpi | campaign | element_O_closure_rel | 2.50050230771e-16 |
| kpi | campaign | element_worst_closure_rel | 4.50090415389e-16 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | -17.1758877049 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 0 |
| kpi | campaign | energy_dH_vessels_kJ | -17.1758877049 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | energy_residual_kJ | 0 |
| kpi | campaign | mass_closure_rel | 5.65505935537e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| *... and 26 more reference values in the case's `expected`.* | | | |

## batch02_adiabatic

tutorials/batch/reactor/batch02_adiabatic




**What it does:** Adiabatic batch esterification (AcOH + EtOH → EtOAc + H2O), starting cold (320 K); the nearly thermoneutral liquid reaction warms the batch only ~2 K in 600 s and is ~22 % complete (thermal runaway needs an exothermic reaction; replacement case planned)

**The story:** pseudo-first-order esterification (AcOH + EtOH → EtOAc + H2O) in an adiabatic batch reactor, starting cold.

T_0 = 320 K (k_0 ~ 3.7e-4 s^-1, tau_0 ~ 2700 s). dH_rxn = -2.43 kJ/mol of ethanol at 320 K, computed from the species' standardThermochemistry (elements datum; the engine's own '[reaction heat]' line). It is about ten times smaller than the -26.06 kJ/mol this case used to print: the acetic-acid correction (docs/design/acetic-acid-latent-heat-closure.md) made the liquid esterification nearly thermoneutral. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean. Result: T rises only 320 → 322.2 K in the 600 s window, and the reaction is only ~22 % complete at the end (it used to reach 428 K and complete conversion). There is no runaway and no plateau; the rate barely accelerates because T barely moves. The adiabatic ODE conserves a frozen-dH surrogate (dH_rxn pinned at the charge T), which the run announces.

The original teaching point of this case (thermal runaway: induction, then a sharp temperature rise, then a plateau) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed to recover the runaway.

The energy balance is just:

dT/dt = - r * V * dH_rxn / (n_A * Cp_A + n_B * Cp_B)

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | P_final | 101300 |
| kpi | reactor | T_final | 322.248040531 |
| kpi | reactor | n_aceticAcid_final | 0.00722392816212 |
| kpi | reactor | n_ethanol_final | 0.00722392816212 |
| kpi | reactor | n_ethylAcetate_final | 0.00202607183788 |
| kpi | reactor | n_water_final | 0.00202607183788 |
| kpi | reactor | t_end | 600 |
| kpi | reactor | totalMoles_final | 0.0185 |
| kpi | campaign | element_C_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_H_closure_rel | 3.00060276926e-16 |
| kpi | campaign | element_O_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_worst_closure_rel | 3.75075346157e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| *... and 12 more reference values in the case's `expected`.* | | | |

## batch03_consecutive

tutorials/batch/reactor/batch03_consecutive




**What it does:** Isothermal batch reactor: consecutive A → B → C with k1 > k2; intermediate B peaks then decays

**The story:** two first-order reactions in series:

with k1 > k2. This is the textbook series-reaction setup: B is produced from A faster than it is consumed to C, so n_B grows at first, reaches a maximum, then decays as A runs out.

Analytical solution (constant V, isothermal, first-order):

n_A(t) = n_A0 ∗ exp(-k1 t) n_B(t) = n_A0 ∗ k1 / (k2 - k1) ∗ [exp(-k1 t) - exp(-k2 t)] n_C(t) = n_A0 - n_A(t) - n_B(t)

t_max = ln(k1 / k2) / (k1 - k2)

With k1 = 0.01 1/s and k2 = 0.002 1/s:

t_max = ln(5) / 0.008 ~ 201 s n_B_max / n_A0 = (k1/k2)^(k2/(k2-k1)) = 5^(-0.25) ~ 0.669

The case demonstrates: 1. multi-reaction support in BatchReactor (v0.21); 2. time-dependent selectivity — harvesting B at t ~ 200 s maximises yield, waiting longer wastes B to C.

No species supplies activation energy: both A_pre are quoted with Ea = 0 so that the rate constants are exactly the values above regardless of T (the case is isothermal anyway).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 1 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | mass_closure_rel | 5.55111512313e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 0.5 |
| kpi | campaign | mass_kg_initial | 0.5 |
| kpi | campaign | mass_residual_kg | 2.77555756156e-16 |
| kpi | campaign | transfers_H_invalid | 0 |
| kpi | campaign | transfers_H_valid | 0 |
| kpi | campaign | transfers_logged | 0 |
| kpi | reactor | P_final | 101300 |
| kpi | reactor | T_final | 350 |
| kpi | reactor | n_compA_final | 3.05902320887e-09 |
| *... and 8 more reference values in the case's `expected`.* | | | |

## batch04_reversible_wgs

tutorials/batch/reactor/batch04_reversible_wgs




**What it does:** Batch reversible water-gas shift (CO + H2O <→ CO2 + H2) at 600 K; conversion relaxes to X_eq

**The story:** the reversible water-gas shift in a closed BATCH reactor (choupoBatch), the time-domain sibling of cstr02 / pfr02.

Equimolar CO + H2O charged at 600 K. The forward Arrhenius rate drives conversion up; the reverse term (k_rev = k_fwd / K_eq, K_eq from the four species' g_pure_ig, re-evaluated each step) brakes it. The batch trajectory relaxes onto the SAME thermodynamic ceiling the CSTR and PFR reached — X → X_eq ~ 0.84 — then sits there (net rate → 0), instead of running to X = 1.

The forward A is the illustrative value tuned for stable RK4 (see constant/reactions); the equilibrium X_eq is A-independent.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | P_final | 1000000 |
| kpi | reactor | T_final | 600 |
| kpi | reactor | n_CO2_final | 0.000420715762123 |
| kpi | reactor | n_CO_final | 7.92842378768e-05 |
| kpi | reactor | n_H2_final | 0.000420715762123 |
| kpi | reactor | n_water_final | 7.92842378768e-05 |
| kpi | reactor | t_end | 1200 |
| kpi | reactor | totalMoles_final | 0.001 |
| kpi | campaign | element_C_closure_rel | 1.95156391047e-15 |
| kpi | campaign | element_H_closure_rel | 1.95156391047e-15 |
| kpi | campaign | element_O_closure_rel | 1.95156391047e-15 |
| kpi | campaign | element_worst_closure_rel | 1.95156391047e-15 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | -0.299204643882 |

*... and 26 more reference values in the case's `expected`.*

## batch04_transient_dirs

tutorials/batch/reactor/batch04_transient_dirs




**What it does:** Adiabatic batch esterification with OpenFOAM-style REAL-TIME instant dirs: scrub 0/ 30/ 60/ ... s to watch conversion climb to ~0.22 while T rises only 320 → 322 K (the runaway of the old exotherm is gone; replacement case planned)

**The story:** the batch02_adiabatic case re-run with REAL-TIME instant directories: the integrator dumps 0/ 30/ 60/ ... s snapshots (OpenFOAM-style) so the trajectory – conversion 0 → ~0.22, T 320 → 322.2 K – can be scrubbed frame by frame in the GUI.

The physics below is inherited from batch02_adiabatic, unchanged: pseudo-first-order esterification (AcOH + EtOH → EtOAc + H2O) in an adiabatic batch reactor.

T_0 = 320 K (k_0 ~ 3.7e-4 s^-1, tau_0 ~ 2700 s). dH_rxn = -2.43 kJ/mol of ethanol at 320 K, computed from the species' standardThermochemistry (elements datum; the engine's own '[reaction heat]' line). It is about ten times smaller than the -26.06 kJ/mol this case used to print: the acetic-acid correction (docs/design/acetic-acid-latent-heat-closure.md) made the liquid esterification nearly thermoneutral. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean. Result: T rises only 320 → 322.2 K in the 600 s window, and the reaction is only ~22 % complete at the end (it used to reach 428 K and complete conversion). There is no runaway and no plateau; the rate barely accelerates because T barely moves. The adiabatic ODE conserves a frozen-dH surrogate (dH_rxn pinned at the charge T), which the run announces.

The original teaching point of this case (thermal runaway: induction, then a sharp temperature rise, then a plateau) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed to recover the runaway.

The energy balance is just:

dT/dt = - r * V * dH_rxn / (n_A * Cp_A + n_B * Cp_B)

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | P_final | 101300 |
| kpi | reactor | T_final | 322.248040531 |
| kpi | reactor | n_aceticAcid_final | 0.00722392816212 |
| kpi | reactor | n_ethanol_final | 0.00722392816212 |
| kpi | reactor | n_ethylAcetate_final | 0.00202607183788 |
| kpi | reactor | n_water_final | 0.00202607183788 |
| kpi | reactor | t_end | 600 |
| kpi | reactor | totalMoles_final | 0.0185 |
| kpi | campaign | element_C_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_H_closure_rel | 3.00060276926e-16 |
| kpi | campaign | element_O_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_worst_closure_rel | 3.75075346157e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |

*... and 12 more reference values in the case's `expected`.*

## batch05_crystalliser

tutorials/batch/reactor/batch05_crystalliser




**What it does:** Batch crystalliser: supersaturated sugar relaxes, PBE by method of moments

**The story:** Single closed vessel. Charge: ~500 kg water + ~1250 kg dissolved sucrose (c0 ~ 2.5 kg/kg, above c_sat(20 C) ~ 2.03 kg/kg, so S0 ~ 1.23). Unseeded: the moments start at zero and primary nucleation (j=0) bootstraps the population from the initial supersaturation. Kinetics from constant/crystallisation.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 6.81588841762e-15 |
| kpi | campaign | element_H_closure_rel | 4.18663266355e-15 |
| kpi | campaign | element_O_closure_rel | 4.18663266355e-15 |
| kpi | campaign | element_worst_closure_rel | 6.81588841762e-15 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | mass_closure_rel | 5.07045231785e-15 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 1748.87224776 |
| kpi | campaign | mass_kg_initial | 1748.87224776 |
| kpi | campaign | mass_residual_kg | 8.86757334229e-12 |
| kpi | campaign | transfers_H_invalid | 0 |
| kpi | campaign | transfers_H_valid | 0 |
| *... and 22 more reference values in the case's `expected`.* | | | |

## batch06_adaptive_runaway

tutorials/batch/reactor/batch06_adaptive_runaway




**What it does:** Adiabatic esterification (AcOH+EtOH), nearly thermoneutral after the acetic-acid correction; ADAPTIVE Rosenbrock23 time-stepping — the step grows as the transient settles (glass-box step history; the thermal runaway of the old exotherm is gone, replacement case planned)

**The story:** the batch02_adiabatic esterification integrated with ADAPTIVE Rosenbrock23 time-stepping, the step history printed glass-box so the student sees the controller at work. (The directory name keeps "runaway" from the old exotherm; there is no runaway any more.)

The physics below is inherited from batch02_adiabatic, unchanged: pseudo-first-order esterification (AcOH + EtOH → EtOAc + H2O) in an adiabatic batch reactor.

T_0 = 320 K (k_0 ~ 3.7e-4 s^-1, tau_0 ~ 2700 s). dH_rxn = -2.43 kJ/mol of ethanol at 320 K, computed from the species' standardThermochemistry (elements datum; the engine's own '[reaction heat]' line). It is about ten times smaller than the -26.06 kJ/mol this case used to print: the acetic-acid correction (docs/design/acetic-acid-latent-heat-closure.md) made the liquid esterification nearly thermoneutral. Equimolar ethanol + acetic acid charge; ethyl acetate + water are the products. Mass-conserving, so the energy balance is clean. Result: T rises only 320 → 322.2 K in the 600 s window, and the reaction is only ~22 % complete at the end (it used to reach 428 K and complete conversion). There is no runaway and no plateau; the rate barely accelerates because T barely moves. The adiabatic ODE conserves a frozen-dH surrogate (dH_rxn pinned at the charge T), which the run announces.

The original teaching point of this case (thermal runaway: induction, then a sharp temperature rise, then a plateau) needs a genuinely exothermic reaction; a replacement case on such a reaction is planned. No parameter was changed to recover the runaway.

The energy balance is just:

dT/dt = - r * V * dH_rxn / (n_A * Cp_A + n_B * Cp_B)

Constant-Cp here (polynomial degree 0 for both ethanol and water in the database), so the result extrapolates linearly above the nominal liquidHeatCapacity Trange (280-373 K) without numerical pathology.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | P_final | 101300 |
| kpi | reactor | T_final | 322.248042017 |
| kpi | reactor | n_aceticAcid_final | 0.00722392679638 |
| kpi | reactor | n_ethanol_final | 0.00722392679638 |
| kpi | reactor | n_ethylAcetate_final | 0.00202607320362 |
| kpi | reactor | n_water_final | 0.00202607320362 |
| kpi | reactor | t_end | 600 |
| kpi | reactor | totalMoles_final | 0.0185 |
| kpi | campaign | element_C_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_H_closure_rel | 3.00060276926e-16 |
| kpi | campaign | element_O_closure_rel | 2.50050230771e-16 |
| kpi | campaign | element_worst_closure_rel | 3.75075346157e-16 |
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |

*... and 12 more reference values in the case's `expected`.*

## batch07_lhhw_methylAcetate

tutorials/batch/reactor/batch07_lhhw_methylAcetate




**What it does:** LHHW in a BATCH reactor: methyl-acetate synthesis over Amberlyst 15 (Popken 2000, Eq 16)

**The story:** the SAME rate law as cstr07, in a BATCH vessel.

CH3OH + CH3COOH <→ CH3COOCH3 + H2O over Amberlyst 15

Popken, Gotze & Gmehling, Ind. Eng. Chem. Res. 39 (2000) 2601, Eq 16: the adsorption-based LHHW on UNIQUAC activities. Nothing about the rate law is restated here – it lives in constant/reactions, exactly as the steady CSTR reads it, because the SAME 'RateLaw' evaluates it in both. A batch reactor and a continuous one cannot disagree about what an adsorption denominator means.

What changes is only what a batch vessel does with a rate: it integrates it. The charge is equimolar methanol + acetic acid; the trajectory climbs to the ACTIVITY equilibrium, X = 76.6 %, and stops. That plateau is asserted nowhere: it is where k1 a'_HOAc a'_MeOH = k-1 a'_MeOAc a'_H2O, i.e. the K_a = 25.50 the authors' independent Eq 17 recovers from the same eight constants – and the same number the steady CSTR climbs to as its volume grows. Two reactors, two integration schemes, one equilibrium.

Water is the strongest adsorber (K/M = 0.29 against 0.052 for the acid), so as the reaction MAKES water it crowds the acid off the resin and the denominator grows. That self-inhibition is why reactive distillation, which pulls the water out as it forms, is worth its capital.

Do NOT try to see the inhibition by deleting the 'adsorption' block and keeping k1. It tells you nothing: the constants were REGRESSED for this rate law, on these activities, and a rate constant without its rate law is a number without a meaning. (Delete the denominator and the reaction gets FASTER, because the adsorbed activities a' are small and dividing by their square is a multiplication. That is an arithmetic accident, not physics.) To SEE the inhibition, spike the charge with water at the same acid and alcohol fractions:

methanol 0.4, aceticAcid 0.4, water 0.0 → initial rate 58.3 mmol/(L.s) methanol 0.4, aceticAcid 0.4, water 0.2 → initial rate 29.0 mmol/(L.s)

Half the rate, with every reactant fraction unchanged. The water is not a reactant here; it is sitting on the sites.

'catalystLoading' (kg of dry resin per m^3) is what turns the paper's per-gram rate constants into a volumetric rate. With concentrations in kmol/m^3 the conversion is exact and unit-free: mol/(g.s) x kg/m^3 = kmol/(m^3.s).

The vessel runs isothermal at 333.15 K, the top of the authors' data range. Do not extrapolate: their activation energies are APPARENT values, fitted over 303-333 K.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | P_final | 101300 |
| kpi | reactor | T_final | 333.15 |
| kpi | reactor | n_aceticAcid_final | 0.00216454708498 |
| kpi | reactor | n_methanol_final | 0.00216454708498 |
| kpi | reactor | n_methylAcetate_final | 0.00708545291502 |
| kpi | reactor | n_water_final | 0.00708545291502 |
| kpi | reactor | t_end | 3600 |
| kpi | reactor | totalMoles_final | 0.0185 |
| kpi | campaign | element_C_closure_rel | 2.00040184617e-15 |
| kpi | campaign | element_H_closure_rel | 1.87537673079e-15 |
| kpi | campaign | element_O_closure_rel | 2.12542696156e-15 |
| kpi | campaign | element_worst_closure_rel | 2.12542696156e-15 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | -50.6006288442 |

*. . . and 26 more reference values in the case's `expected`.*

## batch08_isothermal_duty

tutorials/batch/reactor/batch08_isothermal_duty




**What it does:** Isothermal jacket duty from a DECLARED dH_rxn: E = dH∗dXi via extents (lumped species, no formation data)

**The story:** one isothermal batch reactor, one declared dH_rxn, one hand-checkable jacket duty. See system/controlDict for what this case tests and constant/reactions for the analytic anchor (E = -50 kJ/mol x 0.9975 mol = -49.88 kJ over 600 s).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | mass_closure_rel | 1.11022302463e-15 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 0.05 |
| kpi | campaign | mass_kg_initial | 0.05 |
| kpi | campaign | mass_residual_kg | 5.55111512313e-17 |
| kpi | campaign | transfers_H_invalid | 0 |
| kpi | campaign | transfers_H_valid | 0 |
| kpi | campaign | transfers_logged | 0 |
| kpi | reactor | E_reaction_seg0_kJ | -49.8760623911 |
| kpi | reactor | E_reaction_total_kJ | -49.8760623911 |
| kpi | reactor | P_final | 101300 |

*. . . and 13 more reference values in the case's `expected`.*

## batch26_cooling_failure

tutorials/batch/reactor/batch26_cooling_failure




**What it does:** Cooling failure on a batch (fictitious A + B → P in solvent S): isothermal at 330 K, the jacket stops at t_f, the accumulated B drives T to T_cf

**The story:** a BATCH exothermic reaction held at T_p by its jacket, and what happens when the jacket stops.

THE REACTION IS FICTITIOUS: A + B → P in an inert solvent S, an ideal liquid with no vapour phase in the reactor (constant/reactions and constant/components/reactantA.dat say how every number was set). It is a teaching witness for the process-safety EduTool, not a model of any real chemistry, and its records are coherent by construction: the heat of reaction, -60.000 kJ per mol of B, comes from formation data whose differences are the only thing that carries meaning, and it does not depend on temperature (dCp = 0). The whole charge – 5 mol A, 5 mol B, 15 mol S – is in the vessel at t = 0. That is what makes a batch the WORST case for thermal safety: at t = 0 all of the reactant is unconverted, so a cooling failure at the start releases the whole adiabatic temperature rise.

THE SCENARIO (the recipe below):

t < t_f isothermal at T_p = 330 K; the jacket removes the heat of reaction (the energy ledger records that duty) t = t_f 'setParameter reactor adiabatic 1' – the jacket stops t > t_f adiabatic: the B still unconverted at t_f (the ACCUMULATION, X_ac) reacts and heats the mass to T_cf

The thermal-safety estimate is T_cf = T_p + X_ac ∗ dT_ad (a batch's mass does not change, so Stoessel's semi-batch mass correction is 1), with dT_ad the adiabatic rise of the full charge from T_p (run this case with the failure at t = 0 to have the engine compute it). Because dCp = 0 here, the total heat capacity of the charge does not change as it reacts, and the estimate is EXACT for this batch: the engine's T_cf (T_final of a failure run) and the estimate agree to the integration error. The MTSR is the largest T_cf over every failure instant: for a batch, the failure at t = 0.

'value 0' on the same event keeps the jacket working: the same case then runs the normal isothermal batch, which is how the EduTool draws X_ac(t).

WHAT IS NOT MODELLED: the vessel has no vapour phase – past the mixture's boiling point a real open vessel would boil and a closed one would pressurise. The MTSR is defined on exactly that hypothetical, which is why it is compared with the boiling point (MTT) rather than trusted beyond it. No decomposition is modelled either: T_D24 is an assumption the EduTool's reader sets.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 1 |
| kpi | campaign | mass_closure_rel | 1.97372982156e-15 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 2.25 |
| kpi | campaign | mass_kg_initial | 2.25 |
| kpi | campaign | mass_residual_kg | 4.4408920985e-15 |
| kpi | campaign | moles_kmol_external_out | 0 |
| kpi | campaign | moles_kmol_final | 0.02 |
| kpi | campaign | moles_kmol_initial | 0.025 |
| kpi | campaign | transfers_H_invalid | 0 |
| kpi | campaign | transfers_H_valid | 0 |
| kpi | campaign | transfers_logged | 0 |

*. . . and 17 more reference values in the case's* `expected`*.*

## batch27_semibatch_dosing

tutorials/batch/reactor/batch27_semibatch_dosing




**What it does:** Semi-batch (fictitious A + B → P in solvent S): B dosed in 16 portions over 4 h at 330 K, the jacket fails after the last portion, the small accumulation keeps T_cf low

**The story:** the SAME fictitious reaction as batch26 (A + B → P in the inert solvent S), the same totals, the same T_p, run SEMI-BATCH: A and the solvent are in the reactor and B is DOSED, a sixteenth at a time every 15 minutes, over four hours. Then the jacket fails.

This is the inherently safer answer to batch26 in Kletz's sense of limiting what is there: a batch reactor holds all of its reactant at t = 0; a dosed reactor holds only what it has been given and has not yet converted. That unconverted inventory – the ACCUMULATION – is what a cooling failure turns into temperature, so limiting it limits the maximum temperature of the synthesis reaction (MTSR). The lever is the DOSING RATE against the REACTION RATE: dose faster than the reaction consumes and the reactor fills with unreacted B, which is a batch again.

THE DOSING is written as portions, each a recipe 'transfer' from the dosing tank, because a recipe transfer is what choupoBatch has: a batchReactor carries no continuous feed. Portion-wise addition is also how a hazardous reagent is often added in practice. Each transfer moves a FRACTION of what is left in the tank (1/16, then 1/15, ... then 1), so every portion is 1/16 of B. The dosing tank is at T_p, so the feed brings no sensible-heat effect into the isothermal reactor (into a hot adiabatic one the dose is mixed by enthalpy equality and cools it a little; a cold feed would take more of the reaction's heat – left out on purpose). With discrete doses the accumulation jumps at each portion, so the worst instant for a failure is RIGHT AFTER a portion; the trajectory row at a portion's instant is written after the transfer.

THE TWO SAFETY EVENTS come first in the recipe:

1. the COOLING FAILURE at t_f ('adiabatic 1'; '0' keeps the jacket); 2. what happens to the FEED at the same instant, one transfer with three readings: 'to heldFeed; fraction 1;' the DOSING INTERLOCK works: the methodology's assumption that the feed stops when the cooling does. A recipe transfer has no valve, so the interlock is written as what it achieves – everything left in the dosing tank is moved to 'heldFeed', a vessel nothing is ever dosed from, and every later portion moves a fraction of an empty tank, which is nothing. 'fraction 0;' the interlock FAILS: nothing is held back and the pump keeps dosing into an adiabatic reactor, at its rate. 'to reactor; fraction 1;' the WHOLE dosing tank enters the reactor at once (a valve left open, a pump run away): the worst a feed failure can do, bounded only by what the tank holds. A flow-limited pump limits the RATE, not the TOTAL: with the whole of B in the tank, that total is the batch's. In a plant an interlock is a sensor, a logic solver and a valve, each of which can fail; here it is one line, and the EduTool runs it all three ways. batch28_dosing_vessel_limit is the same reactor with a dosing vessel that holds only a quarter of B.

Normal operation, for the accumulation curve: the failure at 'value 0' and the feed event at 'fraction 0'.

The shipped t_f is 1 s after the last portion, where the accumulation is close to its largest – NOT where the temperature after a failure is largest: early in the dosing the accumulation heats a smaller mass (Stoessel's M_f/M(t) correction), so the MTSR comes earlier; the EduTool shows where, and confirms it with the engine at every post-portion instant.

WHAT IS NOT MODELLED, besides what batch26 names: the reactor's volume is held at its declared value while the dosing adds liquid (a batchReactor does not grow its volume on a charge). The rate law is

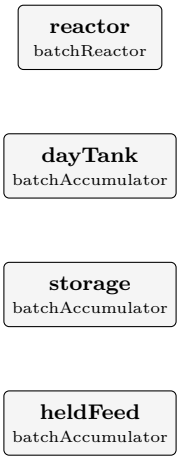
first order in B and zero order in A, so the extent rate k n_B does not depend on V and this changes no number here; a rate law with a concentration term of another order would be wrong.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 16 |
| kpi | campaign | mass_closure_rel | 5.92118946467e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 2.25 |
| kpi | campaign | mass_kg_initial | 2.25 |
| kpi | campaign | mass_residual_kg | 1.33226762955e-15 |
| kpi | campaign | moles_kmol_external_out | 0 |
| kpi | campaign | moles_kmol_final | 0.0200009153077 |
| kpi | campaign | moles_kmol_initial | 0.025 |
| kpi | campaign | transfers_H_invalid | 0 |
| kpi | campaign | transfers_H_valid | 17 |
| kpi | campaign | transfers_logged | 17 |
| *… and 53 more reference values in the case's `expected`.* | | | |

## batch28_dosing_vessel_limit

tutorials/batch/reactor/batch28_dosing_vessel_limit




**What it does:** Semi-batch with a day tank that holds a quarter of B (fictitious A + B → P in solvent S): the cooling fails and the whole day tank enters the reactor at once; T_cf stays low

**The story:** batch27 with the DOSING VESSEL SIZED so that a feed failure cannot take the reactor past its limits.

The same fictitious reaction (A + B → P in the inert solvent S), the same totals, the same T_p and the same sixteen portions, one every 15 minutes. What changes is where B waits. In batch27 the dosing tank holds ALL of B, so the worst a feed failure can do – the whole tank entering at once, or the pump running on with no interlock to stop it – delivers the batch's charge: a flow-limited pump limits the RATE, not the TOTAL. Here the only vessel connected to the reactor is a DAY TANK that holds a QUARTER of B (four portions). It is refilled from storage three times, each refill made only once the day tank is empty, between two portions; storage has no line to the reactor.

THE FAILURE SCENARIO (the first three recipe events, all at t_f):

1. the COOLING FAILURE ('adiabatic 1'; '0' keeps the jacket); 2. the WORST FEED FAILURE: the WHOLE day tank enters the reactor at once ('to reactor; fraction 1;'). 'to heldFeed; fraction 1;' is the interlock holding it back, 'fraction 0;' the pump running on through its remaining portions – which delivers the same total, only later; 3. NO REFILL after a failure: what is left in storage is moved to 'heldFeed' ('fraction 1'; 'fraction 0' in normal operation).

Event 3 is the honest limit of the claim. The day tank's size removes the dependence on the DOSING interlock; what remains is the refill, which is a discrete, deliberate act taken when the day tank is empty and the reactor's state is known – a procedure, or an interlock of its own. The page says so; it does not hide it inside the word "inherent".

With the whole day tank entering, the worst instant is right after a refill (or at t = 0), when the tank is full: the B that can reach the reactor is then the accumulation plus a quarter of the charge. The shipped t_f is 1 s after the first refill.

WHAT IS NOT MODELLED: as batch26 and batch27 (no vapour phase, no decomposition, the reactor's volume held at its declared value – the rate is first order in B, so V does not enter the extent rate).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | energy_balance_available | 0 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 4 |
| kpi | campaign | mass_closure_rel | 1.7763568394e-15 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 2.25 |
| kpi | campaign | mass_kg_initial | 2.25 |
| kpi | campaign | mass_residual_kg | 3.99680288865e-15 |
| kpi | campaign | moles_kmol_external_out | 0 |
| kpi | campaign | moles_kmol_final | 0.0225 |
| kpi | campaign | moles_kmol_initial | 0.025 |
| kpi | campaign | transfers_H_invalid | 0 |
| kpi | campaign | transfers_H_valid | 21 |
| kpi | campaign | transfers_logged | 21 |
| *... and 53 more reference values in the case's* `expected`*.* | | | |

## recipes

**The theory you need.** A recipe is time-triggered discrete control over batch units: charge, heat/cool, hold, transfer, sample. Between events the DAE integrator runs; at each event the state jumps (a transfer moves holdup between vessels). Expect exact event times on the trajectory and holdup bookkeeping that conserves mass across transfers.

## recipe01_react_then_distill

tutorials/batch/recipes/recipe01_react_then_distill




**What it does:** Two-vessel batch recipe: esterify (AcOH + EtOH → EtOAc + H2O) in a reactor, transfer the charge to a still, then distil

**The story:** the canonical 2-stage batch recipe.

Stage 1 (t = 0 → 400 s): A batch reactor at 350 K esterifies ethanol + acetic acid → ethyl acetate + water (the same first-order Arrhenius reaction as batch01). k ≈ 3.58e-3 s?[1] at 350 K, tau ≈ 280 s. Over 400 s the ethanol conversion reaches ~76 %.

Stage 2 (t = 400 → 1200 s): The recipe TRANSFERs the reactor contents to an initially empty batch still. The still then runs a Rayleigh distillation at F_vap = 2.0e-5 kmol/s for 800 s, vaporising 0.016 kmol (~85 % of the charge) – the condensed distillate is routed, step by step, into a passive batchAccumulator ('dischargeTo receiver;'), so the campaign ends with the ester-rich distillate IN a vessel while the ethanol-depleted residue stays in the still's own pot (the pot IS the bottoms tank; no third transfer needed). Mass is conserved across the three vessels at every accepted step. Starting from the converted mix (ethyl acetate + water + unreacted ethanol/acid), the still progressively strips the more volatile ethyl acetate + ethanol; pot composition follows the Raoult y-x curve. (The thermoPhysPropDict here uses ideal Raoult for pedagogical simplicity.)

This case demonstrates the recipe layer (v0.23): a list of time-triggered events acting on named units. Each event in the 'recipe' block below has a 'time', an 'action', and action-specific keys. Actions implemented in v0.23:

transfer move all contents of 'from' into 'to' setParameter change a named operating parameter on a unit

Events fire at the next time-loop step boundary after t_trigger; the trajectory CSV thus shows a single-step discontinuity in the transferred quantities.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 1.87537673079e-16 |
| kpi | campaign | element_H_closure_rel | 3.00060276926e-16 |
| kpi | campaign | element_O_closure_rel | 5.00100461543e-16 |
| kpi | campaign | element_worst_closure_rel | 5.00100461543e-16 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | 26.3188405082 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 7.81829067932e-11 |
| kpi | campaign | energy_dH_vessels_kJ | 26.318841196 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 4 |
| kpi | campaign | energy_residual_kJ | 6.87760234541e-07 |
| kpi | campaign | mass_closure_rel | 2.26202374215e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| *... and 66 more reference values in the case's expected.* | | | |

## recipe02_setpoint_ramp

tutorials/batch/recipes/recipe02_setpoint_ramp




**What it does:** Single-reactor recipe with T_setpoint steps: 320 K → 350 K @ 200 s → 380 K @ 500 s

**The story:** isothermal batch reactor with a staged temperature programme controlled by the recipe layer.

The reactor starts at 320 K, where the first-order Arrhenius reaction is very slow (k_320 ≈ 3.7e-4 s?[1], $\tau \approx$ 2700 s). Two recipe events ramp the setpoint upward:

t = 200 s → 320 K → 350 K (k jumps ~10x to 3.58e-3 s?[1]) t = 500 s → 350 K → 380 K (k jumps another ~7x to 0.025 s?[1])

In the trajectory CSV the temperature column shows a clean step-change at each event, and the conversion curve shows the corresponding three regimes:

0 ≤ t < 200 s barely-moving plateau (slow) 200 ≤ t < 500 s visible drop in n_ethanol (moderate) 500 ≤ t rapid quench to full conversion (fast)

Demonstrates the recipe layer's 'setParameter' action without inter-unit transfers, so this is the cleanest possible test of the parameter-change pathway.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 3.75075346157e-16 |
| kpi | campaign | element_H_closure_rel | 3.00060276926e-16 |
| kpi | campaign | element_O_closure_rel | 2.50050230771e-16 |
| kpi | campaign | element_worst_closure_rel | 3.75075346157e-16 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | 114.181240335 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 0 |
| kpi | campaign | energy_dH_vessels_kJ | 114.181240335 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 5 |
| kpi | campaign | energy_residual_kJ | 0 |
| kpi | campaign | mass_closure_rel | 4.5240474843e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| *... and 35 more reference values in the case's `expected`.* | | | |

## recipe03_conditional_partial

tutorials/batch/recipes/recipe03_conditional_partial




**What it does:** Conditional + partial recipe: react until x_ethanol < 0.30, then transfer 50% to a still

**The story:** recipe extensions (v0.50).

Two recipe features beyond the time-triggered transfer/setParameter of v0.23:

1. CONDITION-TRIGGERED event (‘when { ... }‘ instead of ‘time‘): the event fires the first step its condition becomes true, evaluated on the live vessel state. Quantities: T, total, n_<comp>, x_<comp>; operators: gt, ge, lt, le.

2. PARTIAL transfer (‘fraction‘): a fraction (0..1) of the source moves; the source keeps the rest. Default 1.0 == empty the source.

Here the reactor runs the esterification (ethanol + acetic acid → ethyl acetate + water; the batch01 first-order Arrhenius test, k ~ 3.58e-3 s^-1 at 350 K, so n_ethanol = n_ethanol(0).exp(-k t)). The charge is equimolar (x_ethanol starts at 0.5); when the ethanol mole fraction drops below 0.30, the recipe transfers HALF the reactor contents to a still; the reactor keeps the other half and reacts on, while the still distils its charge.

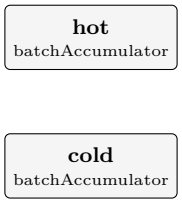
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | reactor | P_final | 101300 |
| kpi | reactor | T_final | 350 |
| kpi | reactor | n_aceticAcid_final | 6.33853236066e-05 |
| kpi | reactor | n_ethanol_final | 6.33853236066e-05 |
| kpi | reactor | n_ethylAcetate_final | 0.00456161467639 |
| kpi | reactor | n_water_final | 0.00456161467639 |
| kpi | reactor | t_end | 1200 |
| kpi | reactor | totalMoles_final | 0.00925 |
| kpi | still | P_final | 101300 |
| kpi | still | T_final | 390.241703035 |
| kpi | still | n_aceticAcid_final | 0 |
| kpi | still | n_ethanol_final | 0 |
| kpi | still | n_ethylAcetate_final | 0 |
| kpi | still | n_water_final | 0 |

*... and 58 more reference values in the case's `expected`.*

### recipe04_charge_enthalpy

tutorials/batch/recipes/recipe04_charge_enthalpy




**What it does:** H-equality recipe charge: T_mix from the first law, not the molar average; balance CLOSES across a transfer

**The story:** mixing is a FIRST-LAW act.

Two passive tanks. 'hot' holds 0.5 mol of ethanol at 340 K; 'cold' holds 1.0 mol of water at 300 K. At t = 100 s the recipe empties the hot tank into the cold one. The charge solves the mix temperature from enthalpy equality on the elements datum,

H(n_EtOH + n_W, T_mix) = H(n_EtOH, 340 K) + H(n_W, 300 K),

so heat capacity DIFFERENCES matter: ethanol carries ~112 J/(mol K) against water's ~75, and the enthalpy-true T_mix (~317 K) sits ~4 K above the equal-Cp molar average (313.3 K) a naive mixing rule gives.

Because both species price on the elements datum, no jacket acts, and nothing leaves, the campaign ENERGY BALANCE is AVAILABLE and closes: dH_vessels = 0 to the mix-solve tolerance — the transfer moved enthalpy between tanks without creating or destroying any.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_O_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 0 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | 0 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 2.13028428746e-12 |
| kpi | campaign | energy_dH_vessels_kJ | 5.81991344006e-10 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 0 |
| kpi | campaign | energy_residual_kJ | 5.81991344006e-10 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0 |
| *... and 24 more reference values in the case's* `expected`*.* | | | |

## still

**The theory you need.** A batch still is Rayleigh distillation: $\mathrm{d}(n\,x_i)/\mathrm{d}t = -\dot{V}\,y_i$ with $y_i$ in equilibrium with the pot – the pot composition DRIFTS as the light component leaves (contrast the steady column, where stages hold profiles constant). Expect the distillate to start rich and fade, and the recipe layer (charge, heat, switch receivers) to drive the run. *Full derivation: Theory Guide, the batch-distillation section.*

## still01_benzene_toluene

tutorials/batch/still/still01_benzene_toluene




**What it does:** Rayleigh batch distillation: benzene/toluene 50/50, 1 atm, ideal Raoult

**The story:** Rayleigh batch distillation, ideal Raoult.

Pot starts with 1 mol of equimolar benzene/toluene at 1 atm. A constant vapour offtake F_vap = 1.5e-6 kmol/s strips the pot over 600 s — by linear mass balance the pot should hold

i.e. L(600 s) = 1 mol - 0.9 mol = 0.1 mol (90 % distilled).

Benzene is the more volatile component (alpha ~ 2.4 at 1 atm), so the still pot progressively concentrates in toluene as benzene flashes off preferentially. Classical Rayleigh integration with constant alpha gives the closed form

ln(L / L_0) = (1 / (alpha - 1)) [ ln(x / x_0) + alpha ∗ ln((1 - x_0) / (1 - x)) ]

where x = x_benzene in the still pot. Plotting (L/L_0) vs x_pot from trajectory.csv should track this analytical curve to RK4 precision.

*One mole of 50/50 benzene/toluene in a pot at 1.013 bar, boiled off for 600 s with the vapour taken away as it forms — no reflux, no trays: the Rayleigh equation integrated in time. The pot starts at its bubble point, 365.00 K, and ends at 380.04 K with 0.1 mol left in it at 91.8 % toluene; 0.9 mol (0.0760 kg) has left as distillate. The reboiler supplied 30.33 kJ, the condenser took 29.39 kJ away.*

*1. The residue gets heavier, and hotter, because the light one leaves first. The Log tab prints the pot every 20 s: benzene falls faster than toluene, T climbs from 365 to 380 K as the pot approaches toluene's boiling point. That is the Rayleigh curve as numbers. 2. The whole campaign is a ledger. [campaign] mass balance: m0 = 0.0851 kg = mF 0.0091 + external 0.0760 kg, closure 0 — what left the domain is counted, not lost. The energy side closes at 4.6e-16 with two records, reboiler and condenser, each priced at its own service temperature (380.0 K and 365.3 K). 3. The thermodynamics is Raoult, and the caveats say where it is stretched. Benzene's Antoine window ends at 354 K and the pot starts at 365 K; toluene's ends at 380 K and the pot ends at 380.04. Both extrapolations are announced at the moment they happen, on the console, inside the table. 4. Utilities are scheduled in time. reports/utilities/ utilityDemand.csv carries the staircase of steam and cooling-water demand across the 600 s — a peak, not just a total, which is what a batch plant is actually sized on.*

*Halve endTime: the pot is still half full and barely enriched — most of the separation happens late, when there is little left. Then still04_rectifier_benzene_toluene puts three stages and a reflux over the same pot, and the distillate purity that a Rayleigh still cannot reach appears.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 2.66880534766e-16 |
| kpi | campaign | element_H_closure_rel | 1.23908819713e-16 |
| kpi | campaign | element_worst_closure_rel | 2.66880534766e-16 |
| kpi | campaign | energy_H_external_kJ | 38.7543769459 |
| kpi | campaign | energy_Q_ledger_kJ | 0.942780166065 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 4.58362894565e-16 |
| kpi | campaign | energy_dH_vessels_kJ | -37.8115967798 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 2 |
| kpi | campaign | energy_residual_kJ | 1.7763568394e-14 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0.0760287317788 |
| kpi | campaign | mass_kg_final | 0.00909876822121 |

*... and 35 more reference values in the case's `expected`.*

## still02_ethanol_water_azeo

tutorials/batch/still/still02_ethanol_water_azeo




**What it does:** Rayleigh batch distillation: ethanol/water 40/60 with NRTL, residue concentrates in water

**The story:** Rayleigh batch distillation of a non-ideal binary with an azeotrope.

System: ethanol / water with NRTL, pressure 1 atm. Minimum-boiling positive azeotrope at x_eth ~ 0.894 (mole), T ~ 351.3 K.

Pot starts at x_eth = 0.40 (a "wine"-like feed — well BELOW the azeotrope). Boil-off rate F_vap = 1.5e-6 kmol/s for 600 s consumes 90 % of the pot, leaving a 10 % residue.

Below the azeotrope, gamma_ethanol > 1 and gamma_water ~ 1, so the vapour is markedly richer in ethanol than the liquid (y_eth ~ 0.61 at x_eth = 0.40). Ethanol therefore vaporises preferentially and the still pot progressively concentrates in WATER — i.e. the residue at end-of-run is mostly water.

The pedagogical contrast with still01 (ideal Raoult, benzene/toluene) is the SHAPE of the enrichment curve:

Raoult: smooth, alpha-driven (alpha ~ 2.4, roughly constant). NRTL : curvature changes as x crosses concentration regimes in which gamma_ethanol shrinks toward 1; the still composition curve x_pot(t) is noticeably non-linear vs the linear pot-volume curve L(t) = L_0 - F_vap∗t.

Crucially, this case demonstrates that a single Rayleigh still cannot cross the azeotrope. Starting below 0.894 in ethanol, the pot can only move DOWNWARD in ethanol fraction; no amount of boiling will lift the pot composition past 0.894. (To break the azeotrope one needs entrainer, pressure swing, or extractive distillation — topics for later tutorials.)

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | still | P_final | 101300 |
| kpi | still | T_final | 372.446750369 |
| kpi | still | n_ethanol_final | 3.38836968904e-11 |
| kpi | still | n_water_final | 9.99999650927e-05 |
| kpi | still | t_end | 600 |
| kpi | still | totalMoles_final | 9.99999989764e-05 |
| kpi | campaign | element_C_closure_rel | 1.35525271561e-15 |
| kpi | campaign | element_H_closure_rel | 9.63735264432e-16 |
| kpi | campaign | element_O_closure_rel | 6.50521303491e-16 |
| kpi | campaign | element_worst_closure_rel | 1.35525271561e-15 |
| kpi | campaign | energy_H_external_kJ | -249.391762836 |
| kpi | campaign | energy_Q_ledger_kJ | 0.490652916203 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 1.13740334008e-16 |

*. . . and 36 more reference values in the case's `expected`.*

## still03_with_receiver

tutorials/batch/still/still03_with_receiver




**What it does:** Rayleigh still distilling INTO a receiver tank (dischargeTo accumulator)

**The story:** Rayleigh batch distillation INTO a receiver (accumulator vessel + continuous dischargeTo).

Same pot as still01 (1 mol equimolar benzene/toluene, F_vap = 1.5e-6 kmol/s, ideal Raoult), but the still's vapour is no longer lost: the 'dischargeTo receiver;' line routes the condensed distillate, step by step, into a passive 'batchAccumulator'. So the trajectory shows the POT depleting AND the RECEIVER filling — the receiver collects the total distillate (amount + composition). Mass is conserved between the two vessels: pot + receiver = 1 mol at all times. Benzene is the more volatile component, so the receiver is benzene-rich early and the cumulative receiver composition drifts toward 0.5 as distillation nears completion (it ends holding everything the pot lost).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 2.66880534766e-16 |
| kpi | campaign | element_H_closure_rel | 1.23908819713e-16 |
| kpi | campaign | element_worst_closure_rel | 2.66880534766e-16 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | 0.942780166065 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 3.00056993907e-11 |
| kpi | campaign | energy_dH_vessels_kJ | 0.942780163768 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 2 |
| kpi | campaign | energy_residual_kJ | 2.2974155911e-09 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 0.0851275 |
| *... and 41 more reference values in the case's `expected`.* | | | |

## still04_rectifier_benzene_toluene

tutorials/batch/still/still04_rectifier_benzene_toluene




**What it does:** batch rectification: 3 stages + total condenser, R=3, with receiver tank (dischargeTo accumulator)

**The story:** the CLASSICAL batch rectification (model rectifier, design forum #85): the Rayleigh pot now feeds a column of 3 ideal stages topped by a total condenser at constant reflux R = 3.

F_vap is the internal BOILUP V = 1.5e-6 kmol/s; the distillate PRODUCT is D = V/(R+1) = 3.75e-7 kmol/s and the reflux L = R.D returns to the column. The pot loses PRODUCT only: dn_i/dt = -D.x_D_i, with x_D(t) solved at every instant from the quasi-steady cascade (stage balances + bubble-T equilibrium + total-condenser closure – a declared residual, printed on failure).

What to watch (trajectory.csv): still.xD_benzene starts ~0.949 (vs ~0.71 for the raw pot vapour – the column working) and DRIFTS DOWN as the pot strips: the constant-R policy trades purity for time. The receiver's ACCUMULATED benzene fraction sits between the instantaneous x_D and the feed – two different compositions, deliberately both visible. The pot loses only D.dt = 0.225 mmol over 600 s (vs 0.9 mmol in still01: same boilup, but 3/4 of it refluxes).

Physics anchors (verified at authoring): x_D(0) grows with nStages and R (N=1,R=1: 0.806; N=2,R=3: 0.909; N=3,R=3: 0.949) and the R → infinity limit (R=200: 0.9765) lands inside the Fenske band alpha^(N+1) for alpha = 2.35-2.6. With nStages 0 and refluxRatio 0 the model reproduces still01's Rayleigh trajectory to the last digit.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 0 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | 0.534303338765 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 7.08707897844e-10 |
| kpi | campaign | energy_dH_vessels_kJ | 0.534303321211 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 2 |
| kpi | campaign | energy_residual_kJ | 1.75539529579e-08 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 0.0851275 |
| *... and 46 more reference values in the case's `expected`.* | | | |

## still05_rectifier_constant_xD

tutorials/batch/still/still05_rectifier_constant_xD




**What it does:** batch rectification, constant-composition policy: R(t) holds x_D until refluxMax, then D=0

**The story:** the SECOND classical reflux policy (design forum #85-3/4): hold the distillate composition constant by RAISING the reflux as the pot strips, and STOP when the declared ceiling can no longer reach the target.

Same pot and column as still04 (1 mol equimolar benzene/toluene, V = 1.5e-6 kmol/s, 3 ideal stages + total condenser), but instead of a fixed R the policy solves R(t) each step – a BRACKETED solve over [0, refluxMax] on a feasibility map, never a bare Newton – so that x_D[benzene] = 0.93 exactly. As benzene depletes, R climbs (~2 at the start); when even refluxMax = 8 cannot reach the target the run announces 'onRefluxLimit stopDistillation': D = 0, the pot stops losing mass, and the campaign idles VISIBLY to endTime with still.refluxLimitReached = 1 in the trajectory – never a magic termination.

What to watch (trajectory.csv): still.xD_benzene pinned at 0.9300 while R climbs – the mirror image of still04, which held R and let purity drift; still.R rising 2.0 → 8 (the ceiling), then the limit event; the receiver's accumulated composition ~0.93 (the whole cut is on-spec – THE selling point of this policy); pot inventory constant after t_refluxLimit.

Declared refusals (each with its own diagnostic, try them): x_D_target 0.5 → BELOW the R=0 floor (even zero reflux over-purifies); x_D_target 0.995 with refluxMax 5 → UNREACHABLE from the charge; ethanol/water NRTL with x_D_target 0.95 → the 0.894 azeotrope caps the cascade at ANY reflux (the barrier is real VLE, not a message).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | element_C_closure_rel | 0 |
| kpi | campaign | element_H_closure_rel | 0 |
| kpi | campaign | element_worst_closure_rel | 0 |
| kpi | campaign | energy_H_external_kJ | 0 |
| kpi | campaign | energy_Q_ledger_kJ | 0.690222586439 |
| kpi | campaign | energy_balance_available | 1 |
| kpi | campaign | energy_closure_rel | 8.89386285064e-10 |
| kpi | campaign | energy_dH_vessels_kJ | 0.69022261392 |
| kpi | campaign | energy_records_invalid | 0 |
| kpi | campaign | energy_records_valid | 2 |
| kpi | campaign | energy_residual_kJ | 2.74809703882e-08 |
| kpi | campaign | mass_closure_rel | 0 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 0.0851275 |
| *... and 48 more reference values in the case's `expected`.* | | | |

## still06_ledger_mixed_validity

tutorials/batch/still/still06_ledger_mixed_validity




**What it does:** Ledger H-validity is MONOTONIC: mixed valid/invalid packages never resurrect a poisoned record

**The story:** MONOTONIC enthalpy validity on the material ledger (the adversarial case of design forum #99/#101).

The campaign ledger integrates each transfer's enthalpy PACKAGE BY PACKAGE at each package's own temperature. A package containing a species with NO formation datum (here 'compA', a lumped teaching species that deliberately carries no standardThermochemistry block) cannot be priced — the record must mark H invalid and NAME the species. The contract under test is the temporal one:

once ONE package is unpriceable, the record's integrated H has lost a parcel FOREVER. Validity is MONOTONIC: a later priceable package must never resurrect it.

Two stills exercise the two orders on ONE accumulated record each:

potA (invalid → valid): starts with 10 % compA, so its distillate record potA→recA begins UNPRICEABLE. At t = 200 s the recipe dumps the whole pot to 'dump' and recharges clean benzene/toluene from 'fresh'; every later distillate package is priceable. The record must STAY invalid (H_missing = compA), and the discrete potA→dump record is invalid too (it carried the compA).

potB (valid → invalid): starts clean, so its distillate record potB→recB begins priceable and accumulates real enthalpy. At t = 200 s the recipe spikes it with pure compA from 'spike'; every later package is unpriceable. The record flips invalid and the partial integral is WITHHELD — an emitted H_kJ that silently lost its tail would be a lie.

Five ledger records result: potA→recA (continuous, poisoned early), potB→recB (continuous, poisoned late), potA→dump and spike→potB (discrete, carrying compA: unpriceable), fresh→potA (discrete, clean: the ONE priceable record). The golden pins the campaign census transfers_H_valid = 1 / transfers_H_invalid = 4; the results JSON carries H_missing = [ compA ] and NO H_kJ key on every poisoned record.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | campaign | mass_closure_rel | 2.1198000621e-16 |
| kpi | campaign | mass_kg_external_out | 0 |
| kpi | campaign | mass_kg_final | 0.26186975 |
| kpi | campaign | mass_kg_initial | 0.26186975 |
| kpi | campaign | mass_residual_kg | 5.55111512313e-17 |
| kpi | campaign | transfers_H_invalid | 4 |
| kpi | campaign | transfers_H_valid | 1 |
| kpi | campaign | transfers_logged | 5 |
| kpi | dump | P_final | 101300 |
| kpi | dump | T_final | 372.484229602 |
| kpi | dump | n_benzene_final | 0.000249727144577 |
| kpi | dump | n_compA_final | 9.64864338217e-05 |
| kpi | dump | n_toluene_final | 0.000353786421588 |
| kpi | dump | t_end | 600 |
| *… and 90 more reference values in the case's `expected`.* | | | |

# Category V: Dynamics and control (choupoCtrl)

Continuous processes in time with feedback: PID loops, setpoint schedules, disturbances and responses.

## ctrl01_cstr_temp_control

tutorials/ctrl/ctrl01_cstr_temp_control




**What it does:** Dynamic CSTR (compA → compB) with PID jacket-temperature control; T climbs from 320 K into the +/-2% setpoint band within ~20 s (no overshoot) and tracks toward 350 K across the 150 s window

**The story:** the canonical first choupoCtrl test.

Process: Continuous CSTR with first-order reaction compA → compB – an ISOMERISATION: both stand-ins carry MW 30, so the pseudo-reaction conserves mass exactly and the run-end material ledger closes to machine precision. (The element balance stays honestly UNAVAILABLE: compA/compB are numerical stand-ins with no real formula, and the ledger says so by name instead of fabricating atoms.) Feed is pure compA at 320 K, 5×10?? kmol/s; reactor volume 1 L; mild exotherm (ΔH_rxn = -50 kJ/mol). Without control the reactor settles a few K above the inlet temperature (small reaction exotherm balanced by convective cooling), well below the desired setpoint of 350 K.

Controller: A PID acts on the jacket temperature (T_jacket) to drive the reactor temperature (T) to a setpoint of 350 K. Tuning is pedagogical, not optimal:

Kp = 4.0 (proportional band ≈ 7-8 K) Ki = 0.04 (integral time $\tau$_I = Kp/Ki ≈ 100 s) Kd = 0.0 (no derivative; system is benign enough)

Jacket bounded to [280, 420] K — realistic limits of a utility loop. Anti-windup: integral does not advance when the unclamped output is outside the bounds.

Expected trajectory (150 s window): Open-loop: T relaxes a few K above the 320 K inlet (small exotherm balanced by convective cooling). Closed-loop: T rises fast (the jacket MV slams to its 420 K clamp, then backs off), entering the +/-2% setpoint band by ~ 20 s, well-damped with no overshoot. By 150 s T is ~ 348.5 K and still creeping up: the last ~ 1.5 K rides the slow chemical tail (the exotherm's conversion, tau ~ 120 s) toward 350 K beyond the window. The jacket (MV) settles around 348.5 K to keep heating in against the cool 320 K feed. (Widen endTime to watch the final approach to 350 K; the 150 s window is sized to SHOW the control transient, not the slow tail.)

To run the OPEN-LOOP comparison, comment out the entire 'controllers' block below — the case will then run the bare reactor with whatever T_jacket the 'operation' block initialised it to (here 320 K, i.e. no jacket heating).

*A 1-litre continuous stirred reactor with a first-order A → B reaction (two stand-in components, both MW 30 — the pseudo-reaction says so by name rather than inventing atoms), fed at 320 K, with a jacket whose temperature a PID controller sets to hold the reactor at 350 K. 150 s of transient, 1 s steps. The golden: PV ends at 348.47 K, the jacket (MV) at 348.54 K, mass in = out = 0.225 kg over the run and the material balance closes at 1.8e-15.*

*1. The controller is a unit in the dict, with its tuning in the open. type PID; setpoint 350.0; gains { Kp 4.0; Ki 0.04; Kd 0.0; } output { min 280; max 420; bias 320; } — proportional band ≈ 7–8 K, integral time $\tau$_I = Kp/Ki = 100 s, no derivative. Nothing is tuned for you. 2. Watch the manipulated variable saturate. At t = 2 s the jacket demand jumps to the 420 K clamp (the log's last column), then backs off as the reactor warms: 402, 385, 374 K ? The Log tab prints every 2 s; the clamp is a number in the dict and you can see it bind. 3. Into the band, then the slow part. The reactor crosses 343 K — the -2 % edge of the setpoint band — at t ≈ 14 s. At 150 s it sits 1.5 K under setpoint and still climbing: that is the integral action at $\tau$_I = 100 s working off the offset. Widen endTime and watch it arrive. 4. The energy balance REFUSES here, and says why. [balance] energy: UNAVAILABLE — the dynamicCSTR energy equation is a Cp/convective model, not the exact derivative of a stored H. The stand-in components carry no formation enthalpy on purpose; a ledger that cannot be exact says so rather than printing a closure it did not earn. ctrl11 is the case where it can, and does (1.4e-7).*

*Set Kp 8.0;: faster into the band, more overshoot, the clamp held longer. Then Ki 0; — the offset never closes, which is what a proportional-only loop teaches in one run. ctrl02_disturbance_rejection steps the feed temperature and asks the same loop to hold.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | TC1 | MV_final | 348.54491648 |
| kpi | TC1 | PV_final | 348.468973192 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 348.481893564 |
| kpi | reactor | n_compA_final | 0.0116161140368 |
| kpi | reactor | n_compB_final | 0.00038388596316 |
| kpi | reactor | t_end | 150 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 1.77327288655e-15 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 0.225 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 0.225 |
| kpi | balance | mass_residual_kg | 6.38378239159e-16 |
| *… and 4 more reference values in the case's `expected`.* | | | |

## ctrl02_disturbance_rejection

tutorials/ctrl/ctrl02_disturbance_rejection




**What it does:** PID disturbance rejection: T_in steps down then up, PID maintains T_setpoint

**The story:** the textbook control demo.

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T_in:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T_reactor drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

$0 < t < 700$: normal startup, T → 350 K (≈ ctrl01). $t = 700$: T_in drops 15 K; the integral action absorbs it so tightly that T_reactor holds at 350 K with no visible sag (the jacket simply keeps heating). $t = 1300$: T_in rises 15 K; again T_reactor stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T_reactor) tracks closely.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedDist | MV_final | 335 |
| kpi | TC1 | MV_final | 347.976535585 |
| kpi | TC1 | PV_final | 350.00207645 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 350.002059115 |
| kpi | reactor | n_compA_final | 0.0110806465498 |
| kpi | reactor | n_compB_final | 0.000919353450202 |
| kpi | reactor | t_end | 2000 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.79221090693e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 3 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 3 |

*... and 5 more reference values in the case's `expected`.*

## ctrl03_adaptive_disturbance

tutorials/ctrl/ctrl03_adaptive_disturbance




**What it does:** PID disturbance rejection with ADAPTIVE plant integration — controller samples on the fixed 1 s grid, Rosenbrock23 sub-steps between samples (MV held)

**The story:** the ctrl02_disturbance_rejection demo re-integrated with an ADAPTIVE plant integrator: the PID still samples on the fixed 1 s control grid (MV held between samples) while Rosenbrock23 sub-steps the plant adaptively in between. The flowsheet below is ctrl02's textbook disturbance-rejection setup, unchanged:

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T_in:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T_reactor drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

$0 < t < 700$: normal startup, T → 350 K (≈ ctrl01). $t = 700$: T_in drops 15 K; the integral action absorbs it so tightly that T_reactor holds at 350 K with no visible sag (the jacket simply keeps heating). $t = 1300$: T_in rises 15 K; again T_reactor stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T_reactor) tracks closely.

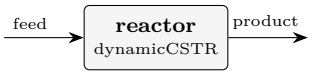
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedDist | MV_final | 335 |
| kpi | TC1 | MV_final | 347.976535626 |
| kpi | TC1 | PV_final | 350.002076468 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 350.002059134 |
| kpi | reactor | n_compA_final | 0.0110806465416 |
| kpi | reactor | n_compB_final | 0.000919353458409 |
| kpi | reactor | t_end | 2000 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.66083451568e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 3 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 3 |

*... and 5 more reference values in the case's `expected`.*

## ctrl05_disturbance_transient

tutorials/ctrl/ctrl05_disturbance_transient




**What it does:** PID disturbance rejection with OpenFOAM-style real-time instant dirs: T_in steps down (700 s) then up (1300 s); scrub 0/ 40/ 80/ ... s to watch the PID track the 350 K setpoint and the jacket MV move

**The story:** the textbook control demo, captured in OpenFOAM-style real-time instant directories (see controlDict's solutionControl block). Identical physics to ctrl02; the difference is that each written time directory (the layout of 0/: the holdup under '<t>/internalStates/', one file per stream) can be scrubbed across the two disturbance steps to watch the PID reject them.

Same DynamicCSTR as ctrl01 (compA → compB, mild exotherm, jacket heat transfer). Same PID temperature controller. Two added ingredients:

1. The simulation runs 2000 s (long enough to reach steady state after each disturbance). 2. A second "controller" of type Schedule injects step disturbances in the inlet temperature T_in:

0 | 320 (nominal) 700 | 305 (-15 K cold disturbance) 1300 | 335 (+15 K warm disturbance, back-and-up)

The PID controller has no foreknowledge of these disturbances — it only sees T_reactor drift away from setpoint and acts on the jacket to compensate. Expected behaviour:

$0 < t < 700$: normal startup, T → 350 K (≈ ctrl01). $t = 700$: T_in drops 15 K; the integral action absorbs it so tightly that T_reactor holds at 350 K with no visible sag (the jacket simply keeps heating). $t = 1300$: T_in rises 15 K; again T_reactor stays pinned at 350 K with no visible deviation.

The trajectory.csv columns now include both controllers' SP / PV / MV. The Schedule controller's SP and MV mirror the same step train (it has no PV); the PID's SP stays at 350 K and its PV (T_reactor) tracks closely.

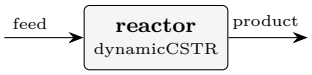
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedDist | MV_final | 335 |
| kpi | TC1 | MV_final | 347.976535585 |
| kpi | TC1 | PV_final | 350.00207645 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 350.002059115 |
| kpi | reactor | n_compA_final | 0.0110806465498 |
| kpi | reactor | n_compB_final | 0.000919353450202 |
| kpi | reactor | t_end | 2000 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.79221090693e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 3 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 3 |

*... and 5 more reference values in the case's `expected`.*

## ctrl06_sine_disturbance

tutorials/ctrl/ctrl06_sine_disturbance




**What it does:** Forced SINUSOIDAL inlet-T disturbance: T_in = 320 + 15 sin(2 pi t / 600). The PID chases a moving target → the reactor temperature settles into a forced sinusoidal response (attenuated + phase-lagged). Sweep the period to build a one-point Bode by trial-and-error (the frequency-response seed).

**The story:** the FREQUENCY-RESPONSE seed.

Same DynamicCSTR as ctrl01/ctrl02 (compA → compB, mild exotherm, jacket heat transfer) and the same PID temperature controller. The NEW ingredient is the disturbance: instead of the step train of ctrl02's 'Schedule' controller, a 'Signal' controller injects a SINUSOID into the inlet temperature T_in:

T_in(t) = 320 + 15 ∗ sin(2∗pi∗t / 600) [K]

mean = 320 K amplitude = 15 K period = 600 s (f = 1/600 = 1.667e-3 Hz, omega = 0.01047 rad/s)

The PID controller has no foreknowledge of the forcing — it only sees T_reactor drift sinusoidally away from the 350 K setpoint and works the jacket to compensate. After the first ~period the reactor temperature settles into a FORCED SINUSOIDAL RESPONSE: same period as the input, but ATTENUATED (the closed loop rejects most of the swing) and PHASE-LAGGED.

THE LESSON (Seborg/Edgar/Mellichamp/Doyle, Ch. 13-14 frequency response): read the output/input amplitude ratio [dB] and the phase lag [deg] off the steady-cycling trajectory, and you have ONE POINT of a Bode plot. Sweep the 'period' (edit one number below), re-run, and you trace the closed loop's frequency response by trial-and-error — the differentiator.

Backward-compatible by construction: ctrl02's 'type Schedule; schedule(...)' staircase still works unchanged; this case swaps it for the more general 'type Signal; signal { type sinusoidal; ... }'.

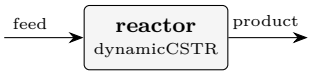
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedDist | MV_final | 319.842923238 |
| kpi | FeedDist | PV_final | 0 |
| kpi | FeedDist | SP_final | 319.842923238 |
| kpi | TC1 | MV_final | 350.026347162 |
| kpi | TC1 | PV_final | 350.170890591 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 350.173436842 |
| kpi | reactor | n_compA_final | 0.0110844237702 |
| kpi | reactor | n_compB_final | 0.000915576229809 |
| kpi | reactor | t_end | 3000 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 5.72875080707e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 4.5 |

*... and 7 more reference values in the case's `expected`.*

## ctrl07_lhhw_inhibition

tutorials/ctrl/ctrl07_lhhw_inhibition




**What it does:** Product inhibition meets a PID: an LHHW reaction whose exotherm fades as its own product covers the catalyst

**The story:** ctrl01's temperature loop, with a reaction that poisons itself.

Same vessel, same PID, same setpoint. The one change is the rate law: the product compB now ADSORBS on the catalyst, so the denominator grows as the reaction proceeds,

r = k(T) . c_A / (1 + K_B c_B)

and the reaction throttles itself. A power law cannot express that at any order; only the adsorption term can. This is the same 'RateLaw' that the steady 'cstr' and the 'batchReactor' evaluate – one definition, three reactors.

What the controller sees: an exotherm that FADES as conversion proceeds, not a constant one. The heat the loop is rejecting shrinks under it, and the jacket temperature must follow. A tuning that holds against a first-order exotherm is not the same tuning that holds against a decaying one.

The adsorption constant K0 = 2.0 m^3/kmol is DIDACTIC – chosen so the inhibition is visible over the control horizon, not measured on any catalyst. For a real, cited LHHW see 'cstr07_lhhw_methylAcetate' and 'batch07_lhhw_methylAcetate' (Popken 2000, Eq 16, on UNIQUAC activities).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | TC1 | MV_final | 349.740705746 |
| kpi | TC1 | PV_final | 348.328939265 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 348.342145073 |
| kpi | reactor | n_compA_final | 0.0117152677233 |
| kpi | reactor | n_compB_final | 0.000284732276726 |
| kpi | reactor | t_end | 150 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 7.70988211545e-16 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 0.225 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 0.225 |
| kpi | balance | mass_residual_kg | 2.77555756156e-16 |

*... and 4 more reference values in the case's `expected`.*

## ctrl08_prbs_ident

tutorials/ctrl/ctrl08_prbs_ident




**What it does:** PRBS identification experiment: T_in = 320 +/- 10 K by a maximal-length LFSR (n=4, period 15 bits x 120 s) – deterministic, seed-declared, period-VERIFIED; the broadband excitation the sine of ctrl06 cannot give.

**The story:** PRBS identification experiment (forum #100.4/#101).

The classic system-identification input: a pseudo-random binary sequence on the inlet temperature, T_in = 320 +/- 10 K, driven by a MAXIMAL-LENGTH Fibonacci LFSR — n = 4 registers, canonical taps (4 3), DECLARED seed 1, period 2^4 - 1 = 15 bits of 120 s each (1800 s per full period).

Why a PRBS and not ctrl06's sine: one sinusoid excites ONE frequency; the PRBS is broadband (its autocorrelation approximates an impulse), so a single record carries the information for a full low-order model fit. And unlike real "random" noise it is DETERMINISTIC: seed + taps declared in this dict reproduce the experiment bit for bit, run after run.

THE HAND-DERIVED SEQUENCE (the exact-sequence contract of #101): with the frozen convention (output = s_n; feedback = XOR of taps; shift s_1 <- feedback; seed LSB → s_1), n=4 / taps (4 3) / seed 1 gives

0 0 0 1 0 0 1 1 0 1 0 1 1 1 1 (bits k = 0..14, then repeats)

– walk it yourself: state 0001 → 0010 → 0100 → 1001 → 0011 → 0110 → 1101 → 1010 → 0101 → 1011 → 0111 → 1111 → 1110 → 1100 → 1000 → 0001 (back at the seed after EXACTLY 15 steps: maximal). The run's announce line prints these same first bits; the initialiser REFUSES any tap set whose cycle is not exactly 2^n - 1.

The PID (same tuning as ctrl06) holds the reactor at 350 K against the binary disturbance; the trajectory's T_in column IS the PRBS, and the reactor response is the identification record.

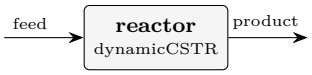
**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedPRBS | MV_final | 330 |
| kpi | FeedPRBS | PV_final | 0 |
| kpi | FeedPRBS | SP_final | 330 |
| kpi | TC1 | MV_final | 348.579621229 |
| kpi | TC1 | PV_final | 350.006520504 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | T_final | 350.006465191 |
| kpi | reactor | n_compA_final | 0.0110804887195 |
| kpi | reactor | n_compB_final | 0.000919511280459 |
| kpi | reactor | t_end | 3600 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 6.73638100034e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 5.4 |

*... and 7 more reference values in the case's `expected`.*

## ctrl09_stream_disturbance

tutorials/ctrl/ctrl09_stream_disturbance




**What it does:** Inlet-field disturbance: componentMolarFlow.compA steps 3.5e-5 → 5.25e-5 kmol/s; F and z_in change by the DERIVED values (F = 6.75e-5, z_A = 7/9) – the disturbance never hides from the other components

**The story:** the inlet-field actuator, end to end.

The Signal controller targets the reactor's INLET FACE by field name ('inletField componentMolarFlow.compA') — the only legal disturbance surface (an outlet actuator is not even expressible). The AUTHORITATIVE inlet state is the per-component molar-flow vector; F_in and z_in are DERIVED from it on every write, so a composition disturbance can never hide inside a silent renormalisation (forum #101):

t < 600 s: nA = 3.5e-5, nB = 1.5e-5 kmol/s → F_in = 5.0e-5, z_A = 0.70, z_B = 0.30 t >= 600 s: nA stepped to 5.25e-5; nB UNTOUCHED at 1.5e-5 → F_in = 6.75e-5, z_A = 5.25/6.75 = 7/9 = 0.77778, z_B = 1.5/6.75 = 2/9 = 0.22222 (hand-derived)

The golden pins F_in and z_in_* finals to these fractions, and the Signal value is the ABSOLUTE molar flow (announced at start-up). The reactor's own response (more A in, higher conversion load, T transient against the PID) is the pedagogy; the CONTRACT is the derived pair.

The 'moleFraction.<c>' sibling field instead holds F_in constant and renormalises the OTHER components proportionally — two explicit semantics, each announced, never a guess.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedComp | MV_final | 5.25e-05 |
| kpi | FeedComp | PV_final | 0 |
| kpi | FeedComp | SP_final | 5.25e-05 |
| kpi | TC1 | MV_final | 350.204904403 |
| kpi | TC1 | PV_final | 350.000124372 |
| kpi | TC1 | SP_final | 350 |
| kpi | reactor | F_in_final | 6.75e-05 |
| kpi | reactor | T_final | 350.000123673 |
| kpi | reactor | n_compA_final | 0.00879292389459 |
| kpi | reactor | n_compB_final | 0.00320707610541 |
| kpi | reactor | t_end | 1800 |
| kpi | reactor | z_in_compA_final | 0.777777777778 |
| kpi | reactor | z_in_compB_final | 0.222222222222 |
| kpi | balance | energy_balance_available | 0 |

*... and 7 more reference values in the case's `expected`.*

## ctrl10_brine_concentration

tutorials/ctrl/ctrl10_brine_concentration




**What it does:** Dynamic brine tank on a Pitzer package: the electrolyte thermodynamics through a transient

**The story:** ONE stirred vessel, NO reaction. The dynamic CSTR tolerates an empty reaction list, and that is exactly the model wanted here: the holdup is a mixing lag, and the thermodynamics – Pitzer activity coefficients on the molality basis – is what travels through it.

The jacket loop is the same PID shape as ctrl01: it acts on T_jacket to hold the vessel temperature. A brine tank held on temperature is what sits upstream of every evaporator and crystalliser in the corpus.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | TC1 | MV_final | 360 |
| kpi | TC1 | PV_final | 310.319973652 |
| kpi | TC1 | SP_final | 313.15 |
| kpi | balance | element_Cl_closure_rel | 7.17019034075e-16 |
| kpi | balance | element_H_closure_rel | 3.40011338976e-16 |
| kpi | balance | element_Na_closure_rel | 7.17019034075e-16 |
| kpi | balance | element_O_closure_rel | 3.40011338976e-16 |
| kpi | balance | element_worst_closure_rel | 7.17019034075e-16 |
| kpi | balance | energy_balance_available | 1 |
| kpi | balance | energy_closure_rel | 7.85289304221e-11 |
| kpi | balance | energy_functional_H | 1 |
| kpi | balance | energy_kJ_final | -853632.741074 |
| kpi | balance | energy_kJ_heat_cum | 2658.78680029 |
| kpi | balance | energy_kJ_in_cum | -87796.8114225 |
| *... and 17 more reference values in the case's `expected`.* | | | |

## ctrl11_esterification_jacket

tutorials/ctrl/ctrl11_esterification_jacket




**What it does:** Jacketed esterification CSTR under PID temperature control: the first ctrl case whose stored enthalpy exists, so the first-law ledger claims closure instead of refusing

**The story:** One jacketed vessel and one PID loop – the same shape as ctrl01, so the two can be read side by side. What differs is the chemistry: real species with a formation datum instead of numerical stand-ins, which is the whole reason this case can carry an energy claim and ctrl01 cannot.

The reaction declares NO dH_rxn. On the elements datum the heat of reaction is Sum(nu_i h_i(T)) – a consequence of the same enthalpy surface the vessel stores, not a second number to keep in step with it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | TC1 | MV_final | 366.642237189 |
| kpi | TC1 | PV_final | 364.93860506 |
| kpi | TC1 | SP_final | 365 |
| kpi | balance | element_C_closure_rel | 1.32609972386e-14 |
| kpi | balance | element_H_closure_rel | 1.31993181817e-14 |
| kpi | balance | element_O_closure_rel | 1.32609972386e-14 |
| kpi | balance | element_worst_closure_rel | 1.32609972386e-14 |
| kpi | balance | energy_balance_available | 1 |
| kpi | balance | energy_closure_rel | 1.59809574171e-07 |
| kpi | balance | energy_functional_H | 1 |
| kpi | balance | energy_kJ_final | -28408.9917781 |
| kpi | balance | energy_kJ_heat_cum | 587.253412702 |
| kpi | balance | energy_kJ_in_cum | -67689.3808709 |
| kpi | balance | energy_kJ_initial | -28597.6577905 |
| *... and 20 more reference values in the case's `expected`.* | | | |

## ctrl12_williams_otto

tutorials/ctrl/ctrl12_williams_otto




**What it does:** Williams-Otto plant (arXiv:2004.07614 eqs. 3.6-3.11 verbatim) settling onto the published steady state x∗ = (3.27, 7.47, 1.12, 9.81, 1.69, 0.22) klb

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term (1-eta)mu, the removed G, the 0.1 F_tE column retention are all inside them – see the unit's own header). The five controls sit in 'operation', declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper's value in klb/h.

u∗ of eq. (3.20): F_fA = 10, F_fB = 20, T = 580 degR, mu = 129.5, eta = 0.2. T rides in 0/internalStates/plant (322.2222 K = 580 degR exactly).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 4.0063377475e-10 |
| kpi | balance | mass_kg_final | 10701.9131449 |
| kpi | balance | mass_kg_in_cum | 5443104 |
| kpi | balance | mass_kg_initial | 4989.512 |
| kpi | balance | mass_kg_out_cum | 5437391.60103 |
| kpi | balance | mass_residual_kg | 0.00218069130187 |
| kpi | balance | material_available | 1 |
| kpi | plant | F_pP_klbh_final | 3.90161524936 |
| kpi | plant | F_wG_klbh_final | 1.22338475064 |
| kpi | plant | J_combined_klb_final | 1071.35532885 |
| kpi | plant | T_final | 322.2222 |
| kpi | plant | t_end | 1440000 |
| kpi | plant | waste_klb_final | 481.451273663 |
| *... and 13 more reference values in the case's* `expected`*.* | | | |

## ctrl13_williams_otto_step

tutorials/ctrl/ctrl13_williams_otto_step




**What it does:** Williams-Otto F_fB step 20→21 klb/h at t∗=100 h (paper Fig. 2): F_pP climbs ~3.90 → ~4.45 klb/h

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term (1-eta)mu, the removed G, the 0.1 F_tE column retention are all inside them – see the unit's own header). The five controls sit in 'operation', declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper's value in klb/h.

u∗ of eq. (3.20) plus the paper's FIRST simulation experiment (section 3.2, Fig. 2): a Schedule steps F_fB from 20 to 21 klb/h at t∗ = 100 h. The MV crosses the boundary in SI (kmol/s), so the step values below are 20/3600 and 21/3600 – the paper's numbers over the one declared conversion.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedStep | MV_final | 0.0058333333 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 1.45035580918e-09 |
| kpi | balance | mass_kg_final | 19794.220051 |
| kpi | balance | mass_kg_in_cum | 2766911.20181 |
| kpi | balance | mass_kg_initial | 4989.512 |
| kpi | balance | mass_kg_out_cum | 2752106.49778 |
| kpi | balance | mass_residual_kg | 0.00401300573503 |
| kpi | balance | material_available | 1 |
| kpi | plant | F_pP_klbh_final | 4.44568758172 |
| kpi | plant | F_wG_klbh_final | 1.90791543668 |
| kpi | plant | J_combined_klb_final | 529.364471374 |
| kpi | plant | T_final | 322.2222 |
| kpi | plant | t_end | 720000 |

*. . . and 8 more reference values in the case's `expected`.*

## ctrl14_williams_otto_pi

tutorials/ctrl/ctrl14_williams_otto_pi




**What it does:** Williams-Otto PI channel (F_fB, F_pP), paper eq. 4.6 with the section-4.1 tunings: F_pP tracks 3.90 → 4.5 klb/h with the Fig. 4 overshoot

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term (1-eta)mu, the removed G, the 0.1 F_tE column retention are all inside them – see the unit's own header). The five controls sit in 'operation', declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper's value in klb/h.

The paper's control channel (F_fB, F_pP), eq. (4.6), with the Skogestad-IMC gains of section 4.1: K3prop = 0.069 (dimensionless – klb/h per klb/h survives the SI conversion unchanged) and K3int = 1.282 per HOUR, declared per second (1.282/3600 = 3.5611111e-4). Bias = F∗_fB (bumpless), setpoint F_pP = 4.5 klb/h = 1.25e-3 kmol/s, output bounded to the paper's 0 <= F_fB <= 56 klb/h.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | PC1 | MV_final | 0.00586183454679 |
| kpi | PC1 | PV_final | 0.00124999896025 |
| kpi | PC1 | SP_final | 0.00125 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.05158151889e-09 |
| kpi | balance | mass_kg_final | 21425.7971042 |
| kpi | balance | mass_kg_in_cum | 1418738.01455 |
| kpi | balance | mass_kg_initial | 10701.9131449 |
| kpi | balance | mass_kg_out_cum | 1408014.1335 |
| kpi | balance | mass_residual_kg | 0.0029106566908 |
| kpi | balance | material_available | 1 |
| kpi | plant | F_pP_klbh_final | 4.49999626428 |
| kpi | plant | F_wG_klbh_final | 2.00328828826 |
| kpi | plant | J_combined_klb_final | 255.611908728 |

*... and 10 more reference values in the case's `expected`.*

## ctrl16_williams_otto_optimal

tutorials/ctrl/ctrl16_williams_otto_optimal




**What it does:** Williams-Otto section-5.3 combined optimisation: Nelder-Mead over two K=5 Schedule profiles, J measured against the published collocation optimum 551.8

**The story:** One unit, because the benchmark IS one equation set: the Williams-Otto reactor + decanter + column + splitter fold into six ODEs (the recycle term (1-eta)mu, the removed G, the 0.1 F_tE column retention are all inside them – see the unit's own header). The five controls sit in 'operation', declared SI: with the MW 453.592 convention (1 kmol == 1 klb) every numeral below IS the paper's value in klb/h.

The section-5.3 combined optimisation: system/outerDict shapes the two profiles below to maximise J = yield - waste (alpha = beta = 1, eq. 5.7), read straight off the plant's J_combined_klb_final KPI.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FfBprof | MV_final | 0.00463988515428 |
| kpi | Tprof | MV_final | 300.373635533 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 1.14401865779e-07 |
| kpi | balance | mass_kg_final | 723583.798797 |
| kpi | balance | mass_kg_in_cum | 2128628.70559 |
| kpi | balance | mass_kg_initial | 10701.9131449 |
| kpi | balance | mass_kg_out_cum | 1415747.06346 |
| kpi | balance | mass_residual_kg | 0.243519095471 |
| kpi | balance | material_available | 1 |
| kpi | plant | F_pP_klbh_final | 7.05702728738 |
| kpi | plant | F_wG_klbh_final | 1.47544148411 |
| kpi | plant | J_combined_klb_final | 546.81259418 |
| kpi | plant | T_final | 300.373635533 |

*... and 32 more reference values in the case's `expected`.*

## ctrl17_signal_shapes

tutorials/ctrl/ctrl17_signal_shapes




**What it does:** WITNESS: ramp + pulse + staircase signals on three feed handles, with the temperature PID rejecting all three at once; the three registered shapes no case selected.

**The story:** WITNESS: the ramp, pulse and staircase SIGNALS, each on its own feed handle of the ctrl09 CSTR, with the ctrl09 temperature PID still in the loop rejecting all three at once – compA ramps +28.6 % over 1000 s, compB takes a 300 s rectangular pulse, the feed temperature walks a three-step staircase. These three registered signal shapes were selected by no case until 2026-08-23 (invariant I8). Self-recorded.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedPulseB | MV_final | 1.5e-05 |
| kpi | FeedPulseB | PV_final | 0 |
| kpi | FeedPulseB | SP_final | 1.5e-05 |
| kpi | FeedRampA | MV_final | 4.5e-05 |
| kpi | FeedRampA | PV_final | 0 |
| kpi | FeedRampA | SP_final | 4.5e-05 |
| kpi | FeedStairT | MV_final | 348 |
| kpi | FeedStairT | PV_final | 0 |
| kpi | FeedStairT | SP_final | 348 |
| kpi | TC1 | MV_final | 347.378942902 |
| kpi | TC1 | PV_final | 349.993146257 |
| kpi | TC1 | SP_final | 350 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.98582911041e-14 |

*... and 13 more reference values in the case's `expected`.*

## ctrl17_step_reaction_curve

tutorials/ctrl/ctrl17_step_reaction_curve




**What it does:** Open-loop process reaction curve: the jacket steps +5, -5, +20 and -20 K from one settled base, and the four responses are the identification record for the FOPDT fit in system/propsDict

**The story:** THE IDENTIFICATION, run over the trajectory the choupoCtrl half of this case wrote. Run the experiment first:

runCase tutorials/ctrl/ctrl17_step_reaction_curve choupoProps tutorials/ctrl/ctrl17_step_reaction_curve

Each operation DECLARES that its trajectory is the process run's (‘writtenBy { trajectory process; }‘), so the second command run alone, on a fresh copy, refuses by name and says to run the first.

TWO OPERATIONS OVER ONE EXPERIMENT, and the pair is the lesson. Both read the SAME four steps of the SAME jacket temperature; they differ only in which response they call the controlled variable.

[0] the COMPOSITION loop (jacket → product holdup). Two lags in series – the jacket heats the liquid, and the warmer liquid reacts faster – so the response is S-shaped and an FOPDT surrogate finds a real apparent dead time. All three tuning rules apply.

[1] the TEMPERATURE loop (jacket → reactor temperature). One dominant lag and no transport delay, so the identified theta comes out smaller than the trajectory's own sampling interval. The engine says so, and REFUSES every Ziegler-Nichols and Cohen-Coon setting by name: both rules put theta in a denominator, and a controller gain computed from an unresolved number is a number with nothing behind it. IMC survives, because its algebra never divides by theta – which is exactly why it is the rule of choice for a lag-dominant loop.

Nothing was tuned to make that happen. It is one plant, one experiment, and two honest answers, and the difference between them is the whole reason the reaction-curve method needs a dead time to work at all.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | JacketStep | MV_final | 320 |
| kpi | reactor | T_final | 320.759149686 |
| kpi | reactor | n_compA_final | 0.0117959393424 |
| kpi | reactor | n_compB_final | 0.000204060657576 |
| kpi | reactor | F_in_final | 5e-05 |
| kpi | reactor | t_end | 15000 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 9.91527847481e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 22.5 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 22.5 |
| kpi | balance | mass_residual_kg | 2.23093765683e-12 |
| kpi | balance | material_available | 1 |

*... and 2 more reference values in the case's `expected`.*

## ctrl18_rtd_pulse_cstr

tutorials/ctrl/ctrl18_rtd_pulse_cstr




**What it does:** RTD by pulse tracer on a pure mixing tank: tracer substitution at constant flow, tau = N/F = 240 s; engine-measured moments against Levenspiel closed-form anchors.

**The story:** RESIDENCE-TIME DISTRIBUTION by pulse tracer – the engine's first RTD tool (2026-08-23). A pure mixing tank (the ctrl09 vessel, reaction and controller stripped) holds N = 0.012 kmol and is fed F = 5e-5 kmol/s, so tau = N/F = 240 s exactly. For 30 s starting at t = 100 s the feed SWAPS 1e-5 kmol/s of carrier for tracer – compB pulses up by exactly what compA pulses down – so the total flow never moves and the vessel is rigorously linear and time-invariant in the tracer.

THE ANCHORS ARE CLOSED FORMS (Levenspiel, Chemical Reaction Engineering, ch. 13-14 – moment additivity of a linear vessel, exact for ANY inlet shape): tbar_out = tbar_in + tau = (100 + 15) + 240 = 355 s sigma2_out = sigma2_in + tau^2 = 30^2/12 + 240^2 = 57675 s^2 area = amplitude x width = 1e-5 * 30 = 3.0e-4 kmol The engine measures the outlet response moments (rtd KPIs, trapezoid on the accepted-step grid, reports/rtd/E.csv); the arithmetic above is the case's, stated here. Deviations from the anchors are integration truncation (run = 2600 s ~ 10.4 tau past the pulse; e^-10.4 ~ 3e-5) plus the finite step – both stated, neither hidden.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | CarrierDip | MV_final | 5e-05 |
| kpi | CarrierDip | PV_final | 0 |
| kpi | CarrierDip | SP_final | 5e-05 |
| kpi | TracerPulse | MV_final | 0 |
| kpi | TracerPulse | PV_final | 0 |
| kpi | TracerPulse | SP_final | 0 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 4.73695157173e-14 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 3.9 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 3.9 |
| kpi | balance | mass_residual_kg | 1.84741111298e-13 |
| kpi | balance | material_available | 1 |

*... and 15 more reference values in the case's `expected`.*

## ctrl19_freq_response_cstr

tutorials/ctrl/ctrl19_freq_response_cstr




**What it does:** Frequency response of the LTI mixing tank at omega∗tau = 1: engine-fitted amplitude and phase against the exact first-order closed forms (AR = 1/sqrt(2), phase = -45 deg).

**The story:** FREQUENCY RESPONSE of the same LTI mixing tank (tau = N/F = 240 s): the tracer feed oscillates sinusoidally at omega∗tau = 1 – again as a SUBSTITUTION against the carrier, so total flow never moves and the vessel is rigorously first order in the tracer. The engine fits a + b sin + c cos over 4 cycles after discarding 5 (transient e^-t/tau through 5 periods = e^-31 – gone), and reports amplitude, phase and the unexplained-variance residual.

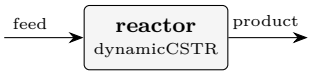
THE ANCHORS ARE THE FIRST-ORDER CLOSED FORMS (any process-control text; at omega∗tau = 1 exactly): AR = 1/sqrt(1 + (omega tau)^2) = 1/sqrt(2) → out amplitude = 2.0e-6 ∗ 0.7071068 = 1.4142136e-6 kmol/s phase = -atan(omega tau) = -45 deg = -0.7853982 rad (the drive is sin(w(t-tStart)) with tStart 0, so the fitted phase IS the lag). The measured amplitude lands within 1e-6 relative of the closed form and the fit residual is ~7e-15 (the vessel really is LTI); the phase carries a bias of EXACTLY omega∗dt/2 = 0.00208 rad = 0.12 deg – the half-sample lag of reading accepted states on a 1 s grid, the measurement system's own phase contribution, worth teaching in itself. The anchor band covers it and this header names it.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | CarrierMirror | MV_final | 4.41011922784e-05 |
| kpi | CarrierMirror | PV_final | 0 |
| kpi | CarrierMirror | SP_final | 4.41011922784e-05 |
| kpi | TracerSine | MV_final | 5.89880772158e-06 |
| kpi | TracerSine | PV_final | 0 |
| kpi | TracerSine | SP_final | 5.89880772158e-06 |
| kpi | balance | energy_balance_available | 0 |
| kpi | balance | mass_closure_rel | 2.91033739366e-15 |
| kpi | balance | mass_kg_final | 0.36 |
| kpi | balance | mass_kg_in_cum | 20.8095 |
| kpi | balance | mass_kg_initial | 0.36 |
| kpi | balance | mass_kg_out_cum | 20.8095 |
| kpi | balance | mass_residual_kg | -6.05626659933e-14 |
| kpi | balance | material_available | 1 |
| *... and 14 more reference values in the case's `expected`.* | | | |

## ctrl20_bode_cstr

tutorials/ctrl/ctrl20_bode_cstr




**What it does:** Bode table of the LTI mixing tank: outer frequency sweep (one datum, lockstep mirror) over the engine frequency-response fit; anchors are the exact first-order closed forms at each point.

**The story:** THE BODE TABLE of the LTI mixing tank (tau = 240 s): an outer sweep drives the sinusoidal tracer substitution at four frequencies, omega∗tau = 0.5, 2.0, 3.5, 5.0 (linear grid in f), and the engine's frequencyResponse fit measures amplitude and phase at each – the sweep CSV IS the Bode table. The measurement block references the drive, so each pass's fit window adapts to that pass's own period, and the sweep varies ONE datum (the drive frequency, mirrored to the carrier by linkedTargets – the same value, lockstep, never a second independent number).

ANCHORS: the exact first-order closed forms at each grid point, AR = 1/sqrt(1+(wt)^2), phase = -atan(wt): wt=0.5: |out| = 2e-6∗0.894427 = 1.788854e-6, phi = -0.463648 rad wt=2.0: |out| = 2e-6∗0.447214 = 8.944272e-7, phi = -1.107149 rad wt=3.5: |out| = 2e-6∗0.274721 = 5.494423e-7, phi = -1.292497 rad wt=5.0: |out| = 2e-6∗0.196116 = 3.922323e-7, phi = -1.373401 rad Phase bands cover the omega∗dt/2 half-sample bias, which grows with omega (0.10 to 1.0 mrad here) – stated, not hidden.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| csv | sweep_results.csv | 0:frequencyResponse.out_amplitude_kmol_s | 1.788854e-6 |
| csv | sweep_results.csv | 1:frequencyResponse.out_amplitude_kmol_s | 8.944272e-7 |
| csv | sweep_results.csv | 2:frequencyResponse.out_amplitude_kmol_s | 5.494423e-7 |
| csv | sweep_results.csv | 3:frequencyResponse.out_amplitude_kmol_s | 3.922323e-7 |
| csv | sweep_results.csv | 0:frequencyResponse.out_phase_rad | -0.463648 |
| csv | sweep_results.csv | 1:frequencyResponse.out_phase_rad | -1.107149 |
| csv | sweep_results.csv | 2:frequencyResponse.out_phase_rad | -1.292497 |
| csv | sweep_results.csv | 3:frequencyResponse.out_phase_rad | -1.373401 |
| kpi | CarrierMirror | MV_final | 4.79049069948e-05 |
| kpi | CarrierMirror | PV_final | 0 |
| kpi | CarrierMirror | SP_final | 4.79049069948e-05 |
| kpi | TracerSine | MV_final | 2.09509300524e-06 |
| kpi | TracerSine | PV_final | 0 |
| kpi | TracerSine | SP_final | 2.09509300524e-06 |

*. . . and 20 more reference values in the case's `expected`.*

## ctrl21_tank_level_control

tutorials/ctrl/ctrl21_tank_level_control




**What it does:** Water tank with a manipulated outflow and a PI level controller: the inflow steps +50 % at 600 s, the integrating level is brought back to the 2.0 m setpoint (P-only would settle 0.05 m off)

**The story:** the most common loop of a plant: INVENTORY.

The same water tank as unsteady05 (D = 2 m, area 3.1416 m2), but its outflow is a MANIPULATED FLOW (a flow-controlled pump: 'mode flow', mv F_out), so the level INTEGRATES the difference of the flows,

A rho/MW dh/dt = F_in - F_out plant gain a = MW/(rho A) = 5.7e-3 m/s per kmol/s (rho = 1001.55 kg/m3)

and nothing brings it back: an integrating process. A level controller closes the loop on F_out. The inflow steps at t = 600 s from 0.20 to 0.30 kmol/s (a Signal on the tank's inlet face).

THE SIGN. The controller is direct-acting on error = setpoint - level, and a level ABOVE the setpoint must RAISE the outflow, so the gains are NEGATIVE (u = bias + Kp (SP - h) + Ki Int(SP - h) with Kp < 0). Bias 0.20 kmol/s is the outflow that balances the old inflow.

WHAT THE TWO CONTROLLERS DO (measured; check_dynamic_tank re-runs both):

PI (Kp = -2, Ki = -0.01): the integral term removes the offset; the level returns to the 2.0 m setpoint (zero offset), after a peak excursion of 0.031 m (max |h - 2| in trajectory.csv; the end offset is < 1e-7 m). P-only (Ki = 0, edit one line): the outflow must rise by 0.10 kmol/s and a proportional-only controller can only do that with an error, F_out = bias + Kp (SP - h) → h - SP = (F_in,new - bias)/|Kp| = 0.10 / 2 = 0.050 m (offset: the level settles at 2.05 m).

The closed loop is second order, wn^2 = a |Ki| , 2 zeta wn = a |Kp|: wn = 7.6e-3 1/s, zeta = 0.76. The inflow is a disturbance the loop must reject; the tank's own buffering (volume) is what the controller trades against the downstream flow's smoothness – loose level control is a feature, not a defect, when the outflow feeds another unit.

ENERGY. H on the elements datum, T a readout; the ledger claims mass and energy and closes (check_dynamic_tank).

*A water tank (dynamicTank, D = 2 m, 3.14 m$^2$) whose outflow is a manipulated flow (mode flow, mv F_out, a flow-controlled pump). Because nothing in the outflow depends on the level, the level integrates the difference of the flows and never comes back by itself. A PI level controller on F_out closes the loop. The inflow steps from 0.20 to 0.30 kmol/s at t = 600 s; 3000 s run, 2 s steps.*

*Run it: runCase tutorials/ctrl/ctrl21_tank_level_control (it runs under choupoCtrl). Its sibling unsteady05_tank_valve_drain drains the same tank through a valve instead and needs no controller.*

*1. The PI holds the level. Plot level in trajectory.csv: it stays at the 2.0 m setpoint, rises by at most 0.031 m after the inflow step and returns with zero offset (< 1e-7 m at the end). The integral term is what removes the offset. 2. P-only settles off the setpoint. Set Ki 0.0; in the controller's gains {} and run again: the level ends at 2.05 m. A proportional controller needs an error to hold a different outflow, and the offset is (F_in,new - bias)/ = (0.30 - 0.20)/2 = 0.050 m. 3. The gains are negative. The controller is direct-acting on error = setpoint - level; a level ABOVE the setpoint must RAISE the outflow, so the loop is reverse-acting and Kp, Ki are negative. With a positive sign the level runs away until the tank overflows, and the engine refuses that by name (OVERFLOWED). 4. The ledger claims mass and energy and closes (~1e-14): the tank stores its enthalpy on the elements datum, T is a readout. [balance] lines print the check.*

*The closed loop is second order (?n ≈ 7.6e-3 1/s, ? ≈ 0.76). The tank's buffering is what a loose level loop trades against the smoothness of the downstream flow. check_dynamic_tank re-runs both the PI and the P-only variants and holds these numbers.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedStep | MV_final | 0.3 |
| kpi | FeedStep | PV_final | 0 |
| kpi | FeedStep | SP_final | 0.3 |
| kpi | LC1 | MV_final | 0.299999843429 |
| kpi | LC1 | PV_final | 1.99999993364 |
| kpi | LC1 | SP_final | 2 |
| kpi | balance | element_H_closure_rel | 2.32787331257e-14 |
| kpi | balance | element_O_closure_rel | 2.32787331257e-14 |
| kpi | balance | element_worst_closure_rel | 2.32787331257e-14 |
| kpi | balance | energy_balance_available | 1 |
| kpi | balance | energy_closure_rel | 3.7609444182e-14 |
| kpi | balance | energy_functional_H | 1 |
| kpi | balance | energy_kJ_final | -100176244.47 |
| kpi | balance | energy_kJ_heat_cum | 0 |
| *... and 19 more reference values in the case's `expected`.* | | | |

## ctrl22_reactor_level_and_temperature

tutorials/ctrl/ctrl22_reactor_level_and_temperature




**What it does:** Reactor with a level: a PI level loop on the outflow and a PI temperature loop on the jacket act on ONE unit while the inflow steps +50 % – the level returns to its setpoint, the temperature holds, the jacket area follows the liquid

**The story:** A jacketed reactor with a LEVEL (dynamicCSTR, 'volumeMode variable', C205 slice 2): isobutanol → 2-butanol, first order, exothermic, fed at 300 K and run at 340 K. Its outflow is a MANIPULATED FLOW (as in ctrl21), so the level INTEGRATES the flow imbalance and needs a controller; its temperature is held by the jacket (as in ctrl01). The two loops act on different manipulated variables of the same vessel:

LC1 PI cv level → mv F_out reverse acting (Kp, Ki < 0) TC1 PI cv T → mv T_jacket direct acting on the error T_SP - T

and they are COUPLED through the vessel, which is the lesson:

the level sets the volume, so the jacket's wetted area – the declared UA is the value with the vessel FULL, UA(h) = UA A_w(h)/A_w(H) with A_w = pi D h + pi D^2/4 – and the reaction heat r V both follow it; the outflow leaves at the vessel's temperature, so it carries enthalpy away; the canonical energy route (announced at start-up) lets that term cancel exactly, and the ledger closes for the whole run.

The inflow steps from 1.0e-3 to 1.5e-3 kmol/s at t = 1200 s (a Signal on the inlet face). Level plant gain a = v/A = 0.47 m per kmol (liquid molar volume 0.093 m3/kmol over 0.196 m2); with wn = 0.01 1/s, zeta = 0.8 the PI gains are Kp = -2 zeta wn / a = -0.034, Ki = -wn^2 / a = -2.1e-4 kmol/s per m(s). Bias 1.0e-3 kmol/s is the outflow that balances the old inflow.

The cold feed lowers the temperature as the inflow rises; the jacket absorbs it. Read trajectory.csv: 'level', 'T', 'F_out', 'volume', and the controllers' MV columns.

*A jacketed reactor with a level (dynamicCSTR with volumeMode variable, D = 0.5 m, H = 1.0 m): isobutanol → 2-butanol, first order, exothermic, fed at 300 K and run at 340 K. Two PI controllers act on DIFFERENT manipulated variables of the SAME vessel:*

*∗ LC1 — level on the outflow F_out, reverse acting (Kp = -0.034, Ki = -2.1e-4); ∗ TC1 — temperature on the jacket T_jacket (Kp = 5, Ki = 0.02).*

*The inflow steps from 1.0e-3 to 1.5e-3 kmol/s at t = 1200 s; 3600 s run, 2 s steps. Run it: runCase tutorials/ctrl/ctrl22_reactor_level_and_temperature (it runs under choupoCtrl). Its sibling unsteady06_reactor_fill_and_react fills the same reactor from a heel with no controller.*

*1. The level loop rejects the inflow step. Plot reactor.level: the peak excursion is 0.010 m (the P-only bound would be dF/ = 0.015 m) and the level is back on 0.50 m at the end (< 2e-6 m). 2. The temperature loop holds 340 K against a colder, larger feed: T dips about 2 K after the step and returns; the jacket temperature (TC1.MV) rises from about 367 K to about 387 K. 3. The loops are coupled through the vessel. The declared UA = 300 W/K is the value with the vessel FULL; the engine scales it with the wetted wall, UA(h) = UA·A_w(h)/A_w(H), A_w = $\pi Dh + \pi D^2/4$ (166.7 W/K at the starting 0.5 m, announced). The reaction heat r·V follows the volume too. 4. The ledger claims mass and energy and closes (~1e-10 or better): both species carry a formation datum, so the vessel takes the canonical energy route, which the run announces.*

*check_dynamic_cstr_level re-runs this case and holds these numbers.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| kpi | FeedStep | MV_final | 0.0015 |
| kpi | FeedStep | PV_final | 0 |
| kpi | FeedStep | SP_final | 0.0015 |
| kpi | LC1 | MV_final | 0.00150002945263 |
| kpi | LC1 | PV_final | 0.5000019311 |
| kpi | LC1 | SP_final | 0.5 |
| kpi | TC1 | MV_final | 386.663508487 |
| kpi | TC1 | PV_final | 339.991752771 |
| kpi | TC1 | SP_final | 340 |
| kpi | balance | element_C_closure_rel | 1.66272359835e-14 |
| kpi | balance | element_H_closure_rel | 1.65023582724e-14 |
| kpi | balance | element_O_closure_rel | 1.66272359835e-14 |
| kpi | balance | element_worst_closure_rel | 1.66272359835e-14 |
| kpi | balance | energy_balance_available | 1 |

*. . . and 26 more reference values in the case's `expected`.*

# Category VI: Multi-sector plant showcases

The sectored flowsheet at full size – read them like worked examples. Three of them also carry a plot plan (`system/layoutDict`): `ammonia02_full_plant`, `greenAmmoniaPEQ_baseCase` and `aceticAcid02_plant`. Before moving an item on one, read the section "Plant layout" of the Design Guide (`docs/designGuide.pdf`): siting and wind, separation and hazardous areas, roads and battery limits, pipe racks, storage, the column group, the elevations NPSH fixes, and what the engine checks and what it leaves to you.

## ChemicalPlantTutorial

**The theory you need.** The full multi-sector plant showcase: feed treatment, reaction, separation, utilities – the sectored flowsheet at its largest, run-only (no golden), for reading and dissecting rather than reproducing numbers.

## CONCENTRATION

tutorials/plant/ChemicalPlantTutorial/CONCENTRATION

**The story:** / system/flowsheetDict (SECTOR – COMPOSITE)

Double-effect evaporator (vapour of effect 1 heats effect 2) feeding a crystalliser. Same shape as the plant flowsheetDict, one level down: children + boundary + default feeds + connections.

## DRYING

`tutorials/plant/ChemicalPlantTutorial/DRYING`

**The story:** / system/flowsheetDict (SECTOR – COMPOSITE)

Spray dryer → additional solid drying → cyclone dust separator. The solid powder goes SD → BD → product; the spray-dryer exhaust air (carrying fines) goes SD → CY, which returns clean air + recovered dust.

## FERMENTATION

tutorials/plant/ChemicalPlantTutorial/FERMENTATION

**The story:** / system/flowsheetDict (SECTOR – COMPOSITE with INTERNAL RECYCLE)

Continuous fermentation side-loop of the sugar plant: a slip-stream of fresh juice (Must) is hydrolysed + fermented into ethanol + CO2, the Flash splits the volatiles (EthanolVapour product) from the liquid bottoms, and a Splitter recycles 80 % of the bottoms back to the Mixer so unconverted sucrose gets another shot. Pedagogical purpose:

1. Reaction (Arrhenius first-order in sucrose). 2. Vapour-liquid flash for separation. 3. Tear-stream RECYCLE that the Wegstein/Newton outer loop must iterate to a fixed point.

Must [Mixer] MixedFeed [Fermentor] FermOut [Flash] EthanolVapour Liquid Recycle [Splitter]

Tear stream: Recycle (Splitter output → Mixer input).

## MAIN

tutorials/plant/ChemicalPlantTutorial/MAIN

**The story:** ChemicalPlantTutorial / MAIN / system/flowsheetDict (SECTOR – COMPOSITE)

THE MAIN SECTOR: the plant-level operations that belong to no specialised sector. Here that is one unit, the JuiceSplitter, which takes the single RawJuice header and splits it 2/3 to CONCENTRATION and 1/3 to FERMENTATION.

WHY A LEVEL FOR ONE UNIT (2026-09-05, amended 2026-09-07). Until 09-05 the JuiceSplitter sat at the plant root beside the sectors – a PascalCase unit folder among CAPS folders, breaking the convention this case's own header declares. The rule the plant now teaches: a folder in CAPS is always a level of the plant's geography, and a plant-level unit lives in MAIN/. MAIN/ is THE DOMAIN'S OWN LEVEL, not a sector like the others (the 2026-09-07 amendment), which is why the plant's own inlets and every stream crossing two sectors live there too – a stream's file goes at the lowest level whose subtree contains every endpoint of it. The STATE views repeat the whole geography: 0/ and converged/ both carry MAIN, CONCENTRATION, DRYING, FERMENTATION. A DERIVATIVE view carries only what it has something to say about – design/ has no MAIN/, because a splitter realises no sized item – and that absence is a fact, not a gap to fill. Flat cases are untouched: a case with no sectors has no MAIN either (its units ARE the plant).

## aceticAcid01_gibbsReactor

tutorials/plant/aceticAcid01_gibbsReactor




**What it does:** ACETIC ACID FROM CO + METHANOL – a basic Gibbs-reactor tutorial. One gibbsReactor with the six species of the carbonylation chemistry (CO, methanol, acetic acid, dimethyl ether, methyl acetate, water) at 460 K and 30 bar, detuned by a 10 K approach to equilibrium. Thermodynamic ceiling only: no catalyst, no rate law.

**The story:** acetic acid from CO + methanol.

CO + CH3OH → CH3COOH (the carbonylation) 2 CH3OH <→ CH3OCH3 + H2O (dimethyl ether) CH3OH + CH3COOH <→ CH3COOCH3 + H2O (methyl acetate)

The reactor is a 'gibbsReactor': it is given the SPECIES and their atoms, never the reactions above. It finds the composition that minimises the Gibbs energy at 460 K and 30 bar, so every pathway the six species allow is open at once. 'temperatureApproach 10' then detunes it: the engine evaluates the REACTION equilibrium 10 K away from the physical T (it reads the sign from the thermicity of this feed), so the outlet is 10 K short of equilibrium.

feed (CO + methanol) -→ [reactor] -→ gas -→ liquid

TOPOLOGY ONLY. Stream state lives on disk in 0/, one file per stream.

*A basic tutorial: ONE gibbsReactor, six species, a 10 K approach to equilibrium. It answers a single question — ∗what does thermodynamics alone allow when CO and methanol meet at 460 K and 30 bar?∗ — and says nothing about how fast, or with what catalyst, it gets there.*

*The flowsheet never states these reactions. system/flowsheetDict lists the six species with their atoms (C, H, O) and the reactor minimises the Gibbs energy under the atom balances, so every pathway the six species allow is open at once. Species left out are pathways closed by decision: no CO2, H2, CH4, formaldehyde or iodides.*

*The feed is declared phase gas; at the reactor temperature on purpose: a stream at 300 K with CO and methanol together has no honest single phase, and the energy report refuses to price it (found while writing this case).*

*The lesson is that the by-products vanish. Carbonylation is so favourable (K ~ 2e5 bar^-1 at 470 K, from the formation data) that methanol is used up; methyl acetate and water exist only as the trace left in equilibrium with it, and DME is smaller still. In the real plant they are ∗intermediates∗ whose level is set by kinetics and by the catalyst system, not by this equilibrium. A Gibbs reactor gives the ceiling, not the selectivity.*

*Hand check (order of magnitude, ideal gas, 470 K): dG(carbonylation) ~ -48 kJ/mol gives y_MeOH ~ 1e-8 to 1e-7 with p_CO ~ 24 bar and p_AcOH ~ 6 bar; the engine's 4e-8 agrees. Methyl acetate then follows from K ~ 60 for the esterification, giving ~7e-4, which also agrees.*

*temperatureApproach 10 is a MAGNITUDE. The engine reads the thermicity of this feed's transformation (exothermic, dH = -66 437 kJ per kmol of feed), assigns the sign, and evaluates the reaction equilibrium at T + 10 K = 470 K, announcing it. Compared with temperatureApproach 0:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feed | F | 0.0583333333333 |
| stream | feed | T | 460 |
| stream | gas | F | 0.0034726940335 |
| stream | gas | T | 460 |
| stream | liquid | F | 0.0270854817667 |
| stream | liquid | T | 460 |
| kpi | reactor | F_condensate_kmol_h | 97.5077343601 |
| kpi | reactor | F_in_kmol_h | 210 |
| kpi | reactor | F_out_kmol_h | 12.5016985206 |
| kpi | reactor | N_in_mol_s | 58.3333333333 |
| kpi | reactor | N_liq_mol_s | 27.0854817667 |
| kpi | reactor | N_out_mol_s | 30.5581758002 |
| kpi | reactor | P | 3000000 |
| kpi | reactor | Q_kW | -4577.90696264 |
| *… and 23 more reference values in the case's `expected`.* | | | |

## aceticAcid02_plant

tutorials/plant/aceticAcid02_plant

**What it does:** ACETIC ACID FROM CO + METHANOL, rung 2 – the plant around the Gibbs reactor of aceticAcid01: UNIFAC liquid instead of ideal, the reactor gas condensed, the CO recycled against a purge, a degasser, a lights column taking the light by-products (methyl acetate / water / methanol / dimethyl ether) overhead as a declared plant outlet, acetic acid as the bottoms product. One recycle (CO). Still a thermodynamic ceiling: no catalyst, no rate law.

**The story:** acetic acid from CO + methanol, with its plant. A BASE CASE: it runs, it is NOT optimised, and every number a student would tune is a plain key below (see README section 4).

freshMethanol → [methanolTank] → [methanolPump] → methanolToProcess 1 atm 14 days to 30 bar | +————————————————————-+

+————+ (freshCO arrives by pipeline, not stored) freshCO ———+-→ [M1] -→ [preheater] -→ [reactor] —-→ liquid —+ coRecycle ——+ ^ 460 K gibbsReactor | | | 460 K, 30 bar, v | | approach 10 gas | | | | | [condenser] | | 330 K, 30 bar| | | | | | coGas condensate | | | | | [purgeSplit] +-[M2] | | | | | purge coRecycle v +————————————+ [letdown valve]

[lightsColumn] | | lightsOut acidProduct

[acidTank] -→ acidShipped

ONE recycle: CO, against a 5 % purge. The light by-products leave through the lights column's distillate (a declared outlet, not recycled – a second recycle was built first and cost 7 min 25 s a run for a stream that is 94 % acetic acid). The reactor is the one of aceticAcid01_gibbsReactor.

THE TANKS (DEV.md 4c C87). The methanol is BOUGHT and held in a feed tank before the process takes it; the acid is COOLED and held in a product tank between shipments. Both tanks are 'storageTank' units: the outlet IS the inlet (same F, z, T, P), so they move no process number – what they add is the holdup the plant's working capital rests on, and the equipment the plot plan and an economic analysis must carry. They are SIZED in system/postDict, by a 'storage' entry that LINKS each unit ('unit methanolTank;') and reads its residence time, fill fraction and inlet from it: the stock is declared ONCE, here. CO is NOT stored: it arrives by pipeline at the plant pressure, as a gas no atmospheric tank holds.

*Rung 2 after aceticAcid01_gibbsReactor. The same Gibbs reactor, now inside a plant: a methanol feed tank and its pump, a condenser, a CO recycle against a purge, a degasser, a lights column, a product cooler and an acetic acid product tank. It runs (32 s), it converges, and it is deliberately NOT optimised — every number a student would tune is a plain key in system/flowsheetDict. The optimisation is the exercise.*

*The case is sealed: everything it reads is in its own constant/.*

*One recycle (CO), closed with Wegstein in 46 iterations. A second recycle (the lights column's distillate back to the reactor) was built first and abandoned: it cost 7 min 25 s a run, and the stream it recycled is 94 % acetic acid. lightsOut is a declared plant outlet.*

*Two tanks and a product cooler (added 2026-10-18, DEV.md 4c C87): the methanol the plant buys is held in methanolTank before the process takes it, and the acid it makes is cooled in acidCooler and held in acidTank between shipments. The tanks are storageTank units: the outlet IS the inlet, so they move no process number; what they add is the inventory and the equipment (section 8). The cooler is a phaseChanger against the plant's cooling water: the acid leaves the column at its bubble point at 1.5 bar, about 404 K – 13 K above its normal boiling point (391.05 K) – and an atmospheric tank cannot hold a liquid hotter than it boils. The CO is not stored: it arrives by pipeline at the plant pressure.*

*∗ Reactor: the Gibbs reactor of rung 1 — equilibrium at 460 K, 30 bar, 10 K approach. No catalyst, no rate. Its liquid is ideal by construction (elementPotential: gas plus one ideal condensable); the 10 K barely moves the main product (see rung 1). ∗ Separations: UNIFAC, which is predictive — the catalogue has no regressed pair for most of this system. Every activity coefficient is an estimate from structure. ∗ Vapour: ideal gas. Acetic acid is largely dimerised in real vapour. The engine models that dimer only on its REACTIVE-equilibrium path (a vapourDimerisation block on the acid's gas-liquid chemistry record; see tutorials/steady/flash/flash12_nh3_acetic_ethanol_reactive and flash13); a gamma-phi package like this one (UNIFAC liquid, ideal-gas vapour) has no vapour-association fugacity model, so this case does not use*

*it and the condenser and degasser splits are indicative. ∗ CO has no UNIFAC group, so it sits outside the activity model (approximations block, announced as a ∗substitution∗ on every run). ∗ Battery limits: the methanol arrives as a liquid at 1 atm into its feed tank, and methanolPump (efficiency 0.65, assumed) lifts it to 30 bar. The CO arrives by pipeline, declared at 30 bar; no CO compressor is modelled. ∗ M1 is a mixer AND a heat exchanger in one box. It merges a liquid (the pumped methanol) with two gases at a DECLARED 320 K, because a gas + liquid merge has no well-posed adiabatic temperature, and it publishes the duty that temperature implies (Q_kW, priced like every other duty; about +81 kW, allocated to LP steam). Without that duty the plant's first law was open by about 80 kW (DEV.md 4c C71, C74). The mixer does not flash: its outlet carries the dominant label (vapour), and the energy report says on every run that mixedFeed cannot hold that label at 320 K and 30 bar (the methanol's partial pressure is far above its vapour pressure there). That label decides how the methanol's latent heat is SHARED between M1 and the preheater; the plant's total heat does not depend on it, the utility bill does. An exercise: put an isothermal flash after M1 and see where the duty goes. ∗ The pump's own energy balance is announced as FAILED on every run (dH 1.76 kW against 5.04 kW of shaft work, 35 % = 1 - eta). It is the engine's, not the case's: the liquid enthalpy of this package carries no pressure term (measured here – moving 0/freshMethanol from 30 bar to 1 atm left the plant's feed enthalpy unchanged to the last digit), so the 3.28 kW of useful work v·dP the pump puts into the liquid is invisible to the energy report, and only its 1.76 kW of dissipation shows as a temperature rise. tutorials/steady/rotating/pump01_water shows the same 65 % gap. ∗ Data: dimethyl ether is reviewStatus interim with a Joback formation enthalpy (see rung 1); it is nil in every stream here. ∗ Antoine ranges are exceeded at 460 K (dozens of announcements per run) and the column prints [distillate] WARNING lines (47 in the base case): the distillate's bubble point does not converge because CO and dimethyl ether have no real saturation curve at those temperatures, so the distillate temperature is reported at the top tray. Neither changes a flow.*

*Decision variables (all in system/flowsheetDict unless noted):*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | acidProduct | F | 0.0272216728543 |
| stream | acidProduct | T | 404.142108532 |
| stream | coGas | F | 0.0239962382533 |
| stream | coGas | T | 330 |
| stream | coRecycle | F | 0.0227964263406 |
| stream | coRecycle | T | 330 |
| stream | columnFeed | F | 0.027499450632 |
| stream | columnFeed | T | 380 |
| stream | condensate | F | 0.00607393614313 |
| stream | condensate | T | 330 |
| stream | crudeAcid | F | 0.0279886650177 |
| stream | crudeAcid | T | 432.414326257 |
| stream | freshCO | F | 0.0291666666667 |
| stream | freshCO | T | 300 |

*. . . and 545 more reference values in the case's `expected`.*

## acetonePlant

tutorials/plant/acetonePlant




**What it does:** Acetone from 2-propanol: Luyben's Figure 1 closed, with the IPA recycle

**The story:** ACETONE FROM 2-PROPANOL, THE CLOSED FLOWSHEET

W. L. Luyben, "Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol", Ind. Eng. Chem. Res. 50 (2011) 1206-1218, DOI 10.1021/ie901923a – his Figure 1 (the Turton base case), end to end:

fresh feed + IPA recycle → vaporiser → reactor → separator → absorber (offgas washed) → C1 → acetone product → C2 → water product → IPA RECYCLE (tear)

This is the case the six preceding ones were built to make possible. Each of them isolates one unit on Luyben's OWN published inlet, so that whatever it gets wrong is that unit and its thermodynamics. Here nothing is published except the fresh feed: every internal stream is the plant's own answer, and the errors compose.

WHAT THIS CASE CLAIMS, AND WHAT IT DOES NOT. It claims to be Luyben's TOPOLOGY and Luyben's EQUIPMENT SPECIFICATIONS solved on the thermodynamics Choupo actually has. It does NOT claim to reproduce his stream table – the siblings measured, in advance, exactly why it cannot, and the gap between his table and this one is the deliverable rather than a defect to be closed:

acetone01: UNIFAC puts the IPA/water azeotrope at x 0.590 against his 0.673, and the lake-estimated isopropanol boils 4.5 K high. acetone04: UNIFAC INVENTS an acetone/water azeotrope at x 0.978 where his UNIQUAC has none – 0.044 K deep, and decisive. acetone06: so C1 caps at 97.3 mol % acetone against his 99.9 % spec. acetone07: so C2's recycle caps at x_IPA 0.580 against his 0.650, and 11.6 % of the isopropanol entering it leaves in the water product.

Both column ceilings were WRITTEN DOWN BEFORE the columns existed and are pinned by check_azeotrope_ceiling. This case is where they stop being two separate findings and become one plant.

DEPARTURES FROM HIS FIGURE 1, each argued rather than absorbed

(a) NO HYDROGEN IN THE COLUMNS. His C1 has a partial condenser and a vent (0.0465 kmol/h, carrying 0.0130 kmol/h H2 and 0.0335 kmol/h acetone); Choupo's column has a total condenser and no vapour distillate. A permanent gas fed to a total condenser is asked to leave as a liquid, and the engine rightly refuses (hydrogen has no liquid heat capacity at 330 K). So the separator's LIQUID goes to C1 and the hydrogen leaves with the offgas, i.e. the columns are modelled downstream of his vent. Cost, stated: the 0.0335 kmol/h of acetone his vent loses is not lost here, which flatters this plant's acetone recovery by 0.10 %.

(b) THE SEPARATOR DOES THE COOLING. An isothermal flash at 318 K computes the duty to bring its feed there, so this one unit stands for his HX1 plus his separator and its Q is directly comparable to his 0.8993 MW. A separate 'heater' was tried in acetone03 and refused, correctly: Choupo's heater takes a DUTY and reports the temperature, because a heater with a specified outlet temperature is a design specification.

(c) THE SEPARATOR PRESSURE IS ARGUED, NOT FITTED. Luyben states 1.5 atm for the absorber gas, which is downstream; there is no compressor between reactor and separator and pressure only falls, so the separator sits at the reactor's 2.6 atm. acetone03 records the sweep that would have tempted a fitted value and why it was refused.

(d) THE ABSORBER'S SECOND OUTLET REJOINS C1's FEED, as in his Figure 1 (his F1 of 72.85 kmol/h is the flash liquid AFTER the absorber bottoms of 20.05 kmol/h rejoin it).

(e) NO VENT/PURGE ON THE RECYCLE LOOP. Luyben has none either – the loop closes through C2's overhead and nothing else – so this is his structure, not a simplification. It does mean any component

that both enters and cannot leave would accumulate without bound; here water leaves in three places and isopropanol in two, so the loop is closed and bounded.

*Acetone from 2-propanol: the flowsheet of W. L. Luyben, ∗Design and Control of the Acetone Process via Dehydrogenation of 2-Propanol∗, Ind. Eng. Chem. Res. 50 (2011) 1206, his Figure 1 end to end — fresh feed and isopropanol recycle into a mixer, a vaporiser to 389 K, a conversion reactor at 623 K and 90 % per-pass conversion, a separator chilled to 318 K, an absorber washing the offgas with 20 kmol/h of water, a vent, and two columns: C1 (66 stages, feed 54, R 2.78) making the acetone product and C2 (19 stages, feed 16, R 0.849) returning the isopropanol. Nine units, one cycle, and the cycle is declared: system/solverDict cuts it at ipaRecycle and closes it with Wegstein.*

*Everything on the first path so far arrives here at once — an equilibrium, a model choice, a many-stage unit, a method that has to be chosen, a reaction, a graph, a recycle. And one thing more, which is why this case is last.*

*It does not reproduce Luyben's stream table, and that is the deliverable. Seven sibling cases (acetone01 to acetone07) each measured ONE unit on Luyben's own published inlet ∗before∗ this plant existed, so every difference below was predicted rather than discovered. Nothing was tuned to close a gap.*

*That first row is 85 % of his acetone, and the case's own CLAUDE.md records the product at 0.9727 mole fraction acetone against his 0.999 spec.*

*1. The error has an address, and it was written down first. UNIFAC invents an acetone/water azeotrope at x 0.978 where Luyben's UNIQUAC has none (acetone04), which caps C1; UNIFAC also misplaces the real isopropanol/water azeotrope at x 0.590 against 0.673 (acetone01), which caps C2. Both ceilings were measured on isolated units before either column was built, and the golden shows them arriving: C2 x_D_LK is 0.499124702623, and a wetter recycle carries less isopropanol, which makes less acetone, which is the production figure in the table. A plant whose disagreement you can name unit by unit is worth more than one that agrees for reasons nobody checked. 2. Two thermodynamic worlds, and the boundary is priced. Handed the plant's global UNIFAC package the absorber REFUSES by name: it solves Kremser, Kremser reads Henry's law, and an activity model carries no answer it can use. So that unit alone carries an inline thermo {} block — water the solvent on the pure-liquid rung, acetone the solute on the Henry rung — declared where the model boundary is and announced when the run crosses it. The block is inline in system/flowsheetDict rather than in a file of its own for exactly that reason. The model-boundary ledger then prices the crossing independently and writes its verdict to reports/balances/energyBalance_byUnit.csv: raw imbalance -11.2496 kW, declared boundary step 0.0000 kW, remaining -11.2496 kW, and in the report's own words ∗the step is NOT credited∗. Read the report for it, not the expected file: that row is held by check_model_boundary_ledger, and this case's golden carries no row kind that can read it. Changing the liquid activity model changes K-values, not the enthalpy datum, so there is nothing to credit at this boundary — and the residual that remains belongs to the absorber's own lumped-Cp energy model, which the case's CLAUDE.md traces and does not excuse. 3. A declared approximation was not locally contained. constant/thermoPhysPropDict authorises hydrogen to run at gamma = 1, by name and with its reason: no UNIFAC group for hydrogen exists, and inventing one would be fabrication. The stated stake was small — a liquid mole fraction of 2e-4. Three units downstream that residue diverges the rigorous MESH in C1, which is why there is a vent flash in this flowsheet that is not in his Figure 1. It is an explicit unit, not a deletion, so what leaves with the hydrogen is computed: the golden pins vent V_over_F at 0.00201400452928. This is the clearest thing the plant has to teach. 4. The tear seed is a guess, and the plant walks away from it. 0/ here is authored and committed, one file per stream, and 0/ipaRecycle is seeded with Luyben's own published recycle composition — 3.822 kmol/h isopropanol, 0.004116 acetone, 2.053884 water in 5.88 kmol/h, which is x 0.650 and 0.0007 exactly as his table reports. The converged answer is x 0.4991. The seed is the author's, visible, and owned; the solver moved off it and said so. (Where a case ships no 0/, bin/choupo-init0 materialises one from the authored inlets and the tear seed.) 5. A specification that survives a recycle is not the same as a rate. C1 asks for distillateRecovery { component acetone; fraction 0.97; } rather than Luyben's 32.25 kmol/h, because the acetone reaching C1 shrinks on every turn of the loop: a rate that is feasible on the first pass is infeasible on the third, and the flowsheet would then fail on arithmetic instead of physics. The golden confirms the spec was met — C1 recovery_acetone 0.969999992418 — in three outer searches. C2 keeps his rate, because its feed is not starved the same way. 6. The plant still closes. Global mass closure at 100 %, and the plant-boundary first law pinned at H_feeds_kW -5862.58875482, Q_boundary_kW 632.174396749, H_products_kW -5236.98782168, a residual of 6.57346360643 kW. A disagreement with a paper and a violation of conservation are different things, and this case is careful to be only the first.*

*Read the case's own CLAUDE.md first: it carries the full comparison against Luyben's Figure 1, the five traced sources of error, and a ∗Pending∗ list that names the one change which would move every row — fitted UNIQUAC or NRTL pairs for the three binaries, which is the honest way to close the gap and is a*

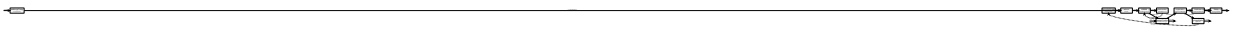
*curation job rather than a code job. The build history is docs/design/acetone-plant-closure-state.md; the programme is docs/design/acetone-ipa-reference-case.md.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | B1 | F | 0.0118946232842 |
| stream | F1 | F | 0.0195617446099 |
| stream | Rin | F | 0.0160666666667 |
| stream | Rout | F | 0.0255036799795 |
| stream | absBottoms | F | 0.00608647733675 |
| stream | acetoneProduct | F | 0.00762772388347 |
| stream | columnFeed | F | 0.0195223471677 |
| stream | flashGas | F | 0.0120284127064 |
| stream | flashLiquid | F | 0.0134752672731 |
| stream | freshFeed | F | 0.0144333333333 |
| stream | ipaRecycle | F | 0.00163333333333 |
| stream | mixedFeed | F | 0.0160666666667 |
| stream | offgas | F | 0.0114974909252 |
| stream | ventGas | F | 3.9397442245e-05 |
| *... and 173 more reference values in the case's `expected`.* | | | |

## ammonia02_full_plant

tutorials/plant/ammonia02_full_plant



**What it does:** Full-scale ammonia SYNTHESIS LOOP at 200 bar (~59 t/h NH3, 1400 t/day) – a synthesis-loop fundamental, not a whole ammonia plant: the syngas arrives as a declared 25 bar makeup and its production is not modelled. Explicit syngas + recycle COMPRESSORS, feed-effluent HEAT EXCHANGER, ADIABATIC converter (the reaction heats itself – exothermic), water cooler + refrigerated separator, product let-down, and the recycle + purge. SRK fugacity throughout; dissolved H2/N2 by measured Henry constants (Wiebe 1934, Larson-Black 1925 via IUPAC SDS 5/6).

**The story:** the complete Haber-Bosch plant, ~59 t/h NH3

makeup 25 bar [makeupComp 25→200] recycle [recycleComp] [mixer] mixed (cool) cold side [FEHE] hot effluent (heat recovery) preheatedFeed (~683 K) [converter] gibbsReactor mode ADIABATIC N2+3H2<→2NH3 exothermic → self-heats to ~840 K hotEffluent hot side of FEHE → cooledEffluent [waterCooler] (cooling water) ~310 K [separator] flash 250 K/200 bar (refrig.) rawLiquid unreactedGas [letdown] [purgeSplit] productNH3 gas purge recycle (tear)

THE REACTOR HEAT. N2+3H2→2NH3 is EXOTHERMIC – there is NO external reactor heating. The converter runs ADIABATIC: the exotherm raises the gas T (feed 683 K in, 840 K out), and the feed-effluent exchanger (FEHE) recovers that heat to preheat the incoming feed to catalyst light-off. Self-sustaining once lit. The compressors set the 200-bar synthesis pressure; the water cooler (cooling-water utility) + refrigerated flash condense the NH3. Every machine is explicit and its duty/work is reported.

CAPACITY ~59 t/h NH3 (~1400 t/day) at 98.6% product purity – a world-scale single train.

DISSOLVED GASES BY HENRY'S LAW. H2 and N2 are supercritical at the separator (no physical Psat), so their dissolved amounts follow MEASURED Henry constants (pairs H2-NH3 from Wiebe 1934, N2-NH3 from Larson & Black 1925 as evaluated in IUPAC SDS Vol 5/6), with the SRK vapour fugacity at 200 bar: K = H(T)/(phi_V P). The raw liquid carries ~0.5% H2 + 0.2% N2 – physical, vs the ~13% a Psat extrapolated above Tc used to fake. KNOWN RESIDUAL: Ar has no measured Ar-NH3 Henry pair yet, so its dissolved ~1% is a Raoult-extrapolation artefact (~7x over by analogy with N2) – the product's main impurity is overstated.

THE SIMULATION STRATEGY (how you build a plant like this). 1. PICK THE THERMO BY THE PHYSICS, per phase ("the two worlds"): the 200-bar GAS side needs real-gas fugacity → a cubic EoS (SRK) declared once in the package; the reactor folds phi into its equilibrium automatically. The DISSOLVED gases are supercritical solutes in a condensable solvent → the unsymmetric convention (Henry / Krichevsky-Kasarnovsky), NOT Raoult: pair files H2-NH3 + N2-NH3 fitted to measured solubility, 'solvent NH3;' in the package, 'role solute;' overlays on H2/N2 in this case's constant/. 2. TEAR THE LOOP WHERE INFORMATION FLOWS BACKWARD: one MASS tear (recycle, guessed then Newton-converged) and one ENERGY tear (hotEffluent – the FEHE preheats the feed with the reactor's own product heat, so the converter's outlet is needed upstream of the converter). Both declared in solverDict tearStreams; the solver announces every iteration. 3. LET THE EXOTHERM DO THE HEATING: the converter is ADIABATIC (mode adiabatic; T is a guess, not a spec) – the reaction self-heats 683 → 840 K and the FEHE recycles that heat to light off the feed. No phantom reactor heater exists anywhere. 4. MAKE EVERY MACHINE EXPLICIT so it can be SIZED AND COSTED: compressors by shaft power, exchangers by area, the converter + drums by vessel design bases (space velocity / residence time) in system/postDict → Turton costs → the DCF economics in constant/economics. Every unit operation carries its own sizing – the doctrine. 5. VALIDATE AGAINST MEASUREMENT AT EVERY JOINT: the reactor equilibrium against the ideal-vs-SRK shift (gibbs10), the dissolved gas against Wiebe 1934 / Larson-Black 1925 (x_H2 at 400 atm: 0.0287 computed vs 0.0282 measured), the balances against 100% element closure.

HONEST SIMPLIFICATIONS. (1) The converter is a SINGLE adiabatic bed, so it runs hotter (~567 C) than a real multi-bed converter with interbed quench cooling (which stays near the 400-500 C equilibrium sweet spot for more NH3 per pass). (2) The makeup compressor is one stage (real: multi-stage with intercooling), so its discharge is hot. (3) The let-down gas and the loop pressure drops are as in ammonia01. (4) 'water' is in the package ONLY as the cooling-water utility fluid (it is not a reaction species).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | chilledEffluent | F | 10.2992003164 |
| stream | compMakeup | F | 2.22222222222 |
| stream | compRecycle | F | 9.04024011142 |
| stream | cooledEffluent | F | 10.2992003164 |
| stream | cw | F | 138.888888889 |
| stream | cwOut | F | 138.888888889 |
| stream | hotEffluent | F | 10.2991998088 |
| stream | letdownGas | F | 0.00750227557924 |
| stream | makeup | F | 2.22222222222 |
| stream | mixed | F | 11.2624623336 |
| stream | preheatedFeed | F | 11.2624623336 |
| stream | productNH3 | F | 0.971862875259 |
| stream | purge | F | 0.279595054967 |
| stream | rawLiquid | F | 0.979365150838 |
| *… and 934 more reference values in the case's `expected`.* | | | |

## ammonia03_quench_converter

tutorials/plant/ammonia03_quench_converter

**What it does:** Ammonia synthesis loop with a THREE-BED QUENCH CONVERTER: cold feed gas is taken off upstream of the feed-effluent exchanger and injected between adiabatic catalyst beds, so the gas zigzags about the locus of maximum rate instead of riding one adiabatic line to a single hot equilibrium. Same loop as ammonia02 (200 bar, 3:1 makeup, 3% purge, Henry-law dissolved gases); the converter is the redesign. The quench split is the design variable.

**The story:** the SAME synthesis loop as ammonia02, with the converter redesigned as THREE ADIABATIC BEDS WITH COLD-SHOT QUENCH.

makeup 25 bar [T101] [makeupComp 25→245] recycle [recycleComp] [mixer] mixed (26645 kmol/h, 386 K) [quenchSplit] <– THE DESIGN VARIABLE 47 % 25 % 28 % toFEHE quench1 quench2 [FEHE] hotEffluent (heat recovery) preheatedFeed 683 K [bed1] 841 K [quench1Mix] (682 K) bed2Feed [bed2] 798 K [quench2Mix] (681 K) bed3Feed [bed3] 778 K hotEffluent [FEHE] hot side

[waterCooler] [separator] 250 K / 200 bar rawLiquid unreactedGas [letdown] [purgeSplit] productNH3 gas purge recycle (tear)

WHY QUENCH. N2 + 3 H2 <→ 2 NH3 is EXOTHERMIC and REVERSIBLE. Exothermic means an adiabatic bed heats itself as it converts; reversible means the equilibrium it is climbing towards RETREATS as it heats. A single adiabatic bed therefore runs up one straight line in the (conversion, temperature) plane until it meets the equilibrium curve, and stops there – ammonia02 stops at 839.6 K and 10.75 % NH3. A quench converter cuts the bed into pieces and drops the gas back down the temperature axis between them, with cold loop gas taken off upstream of the interchanger. The gas ZIGZAGS: up the adiabatic line inside a bed, vertically down at each cold shot.

MEASURED, this run against ammonia02 (identical feed, pressure, purge and separator; only the converter and the interchanger differ):

converter exit 10.75 % NH3 @ 839.6 K → 17.28 % NH3 @ 777.5 K per-pass N2 conv. 17.84 % → 28.80 % loop circulation 40545 kmol/h → 26645 kmol/h recycle 32545 kmol/h → 18645 kmol/h purge (3 % of recycle) 1006 kmol/h → 577 kmol/h product 3498.7 kmol/h → 3719.8 kmol/h (+6.3 %)

The product gain is NOT the extra equilibrium conversion by itself: it is that a loop which converts more per pass recirculates less, so the 3 % purge throws away 43 % less hydrogen.

THE SPLIT IS THE DESIGN. 47 / 25 / 28 is not a given; it is chosen. With the interchanger fixed at 375 m2, lowering the FEHE branch sends a smaller stream through the same area and preheats bed 1 MORE while cooling beds 2 and 3; raising it does the opposite. 47 % is the value at which all three beds enter within 2.6 K of each other, at the 683 K that ammonia02's own converter already uses as its light-off inlet. Read README.md for the measurements, including what a BAD split costs and why a fourth bed is not obviously worth building.

WHAT THIS MODEL DOES NOT CONTAIN – read this before quoting any number. Every bed here is a 'gibbsReactor': it runs to FULL EQUILIBRIUM, which is the same as saying it is infinitely long. There is no rate law, no catalyst volume, no particle size and no approach to equilibrium anywhere in this case, so: no bed can be SIZED, and system/postDict declares all three so that the engine refuses them by name on every run; nothing here can tell a good injection POINT from a bad one. Measured: moving quench from 25/28 to 50/3 leaves the converter exit unchanged to five figures, because with the last bed at equilibrium the outlet depends only on the TOTAL cold shot, not on where it enters. That is an exact property of the model, not a numerical accident – and it is exactly the question a real design has to answer.

THE CONSTRUCTION IS DYSON AND SIMON'S. The rate contours in the conversion-temperature plane, the locus of maximum rate descending as conversion rises, and a cold-shot trajectory drawn as a zigzag about it, are Figure 2 of D. C. Dyson and J. M. Simon, "A kinetic expression with diffusion correction for ammonia synthesis on industrial catalyst", Ind. Eng. Chem. Fundam. 7 (1968) 605-610. This case executes that picture with equilibrium beds; their rate law is what would put the locus on it.

UNCHANGED FROM ammonia02, deliberately: 200 bar synthesis, makeup 1980 kmol/h N2 + 5940 kmol/h H2 (exactly 3:1) + 80 kmol/h Ar, both compressors and their shaft powers, the water cooler, the 250 K / 200 bar refrigerated separator, the 20 bar let-down, the 3 % purge, the two storage tanks, the declared cooling-water circuit, and the SRK + Henry thermophysical package with its cited N2-NH3 and H2-NH3 pairs.

CHANGED, and each change is a consequence of the converter: the INTERCHANGER falls from 1500 m2 to 375 m2. It carries 47 % of the loop gas instead of 100 %, from an effluent 62 K cooler. Its duty falls from 117.2 MW to 31.2 MW. the WATER COOLER takes 66.2 MW instead of 57.8 MW: 8.4 MW that the interchanger no longer recovers leaves through the cooling water instead. A real plant puts a waste-heat boiler on that effluent. This case does not add one – that is a separate design act, and inventing it here would change two things at once. the recycle TOLERANCE is 1e-8 instead of 1e-5. At 1e-5 this loop's element balance closed only to 0.05 %; three reactors on one tear leave more room for a loose tear to show. NOT changed and worth seeing: the recycle compressor still takes 4000 kW on a circulation 43 % smaller, so it now discharges at 258 bar into a mixer that runs at 245. The circulator is oversized for the converter it feeds – a real consequence of the redesign, left visible rather than tuned away.

*The same synthesis loop as ammonia02_full_plant — 200 bar, makeup 1980 kmol/h $N_2$ + 5940 kmol/h $H_2$ (exactly 3:1) + 80 kmol/h Ar, refrigerated separator at 250 K, 3 % purge, SRK fugacity with measured $N_2$–$NH_3$ and $H_2$–$NH_3$ Henry pairs — with one thing changed: the converter. ammonia02 has a single adiabatic bed. This case has three adiabatic beds with cold-shot quench between them, which is what the BASF account of Haber–Bosch describes when it says the reacting hot gas is cooled with a supply of cold gas ∗inside∗ the reactor.*

*Ammonia synthesis is exothermic ∗and∗ reversible, and those two facts pull in opposite directions.*

*∗ Exothermic → an adiabatic bed heats itself as it converts. Its state moves up a straight line in the (conversion, temperature) plane: every mole of $NH_3$ formed buys a fixed number of degrees. ∗ Reversible → the equilibrium conversion falls as the temperature rises. The target retreats as you chase it.*

*Between them there is a locus of maximum rate: at any conversion there is one temperature where the forward and reverse rates are furthest apart, and that temperature descends as conversion rises. A single adiabatic bed crosses that locus early and then spends the rest of itself on the wrong side of it, running into a receding equilibrium.*

*A quench converter follows the locus instead — in a staircase. Inside a bed the gas climbs its adiabatic line; at the cold shot it drops vertically (cold gas removes heat, and dilutes, but converts nothing). Up, down, up, down, up. That zigzag is the whole design.*

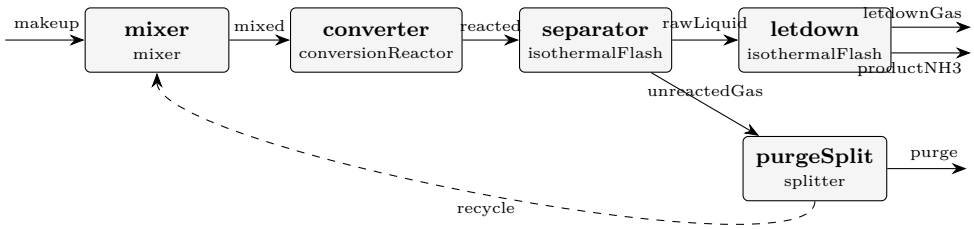
*The picture is Dyson and Simon's. The rate contours in the conversion–temperature plane, the descending locus of maximum rate, and a cold-shot trajectory drawn as a zigzag about it, are Figure 2 of D. C. Dyson and J. M. Simon, ∗A kinetic expression with diffusion correction for ammonia synthesis on industrial catalyst∗, Ind. Eng. Chem. Fundam. 7 (1968) 605–610. This case executes that figure with equilibrium beds. Their rate law is what would put the contours on it; see ?6.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | bed1Out | F | 3.17996753423 |
| stream | bed1Out | T | 841.2767605 |
| stream | bed2Feed | F | 5.03030532814 |
| stream | bed2Feed | T | 682.045268453 |
| stream | bed2Out | F | 4.69483355428 |
| stream | bed2Out | T | 797.933843704 |
| stream | bed3Feed | F | 6.76721188346 |
| stream | bed3Feed | T | 680.610038848 |
| stream | chilledEffluent | F | 6.38083264983 |
| stream | chilledEffluent | T | 324.3024335 |
| stream | compMakeup | F | 2.22222222222 |
| stream | compMakeup | T | 656.394257943 |
| stream | compRecycle | F | 5.17912895342 |
| stream | compRecycle | T | 274.060726539 |
| *... and 284 more reference values in the case's `expected`.* | | | |

## ammoniaReactor01_fixedConversion

`tutorials/plant/ammoniaReactor01_fixedConversion`




**What it does:** AMMONIA REACTOR-MODEL LADDER, rung 1 of 4 – FIXED CONVERSION. The ammonia synthesis loop with a conversionReactor on a DECLARED single-pass conversion of 25 % of the nitrogen. Claims a converged MATERIAL balance and nothing else: no reactor size, no equipment cost – and NOT a converged heat balance, which this reactor cannot deliver against an SRK-priced stream (README section 4). Diff the flowsheetDict against ammoniaReactor02_equilibrium and the only block that moves is the converter.

**The story:** THE AMMONIA REACTOR-MODEL LADDER – flowsheet, SHARED BY RUNGS 1-3.

Everything in this file except the ‘converter‘ block is BYTE-IDENTICAL across ammoniaReactor01_fixedConversion, ammoniaReactor02_equilibrium and ammoniaReactor03_equilibriumApproach. Diff two of them and the only thing that moves is the reactor model – which is the whole point: one rung of the fidelity ladder at a time, and nothing else.

makeup (N2:H2 = 1:3 + 1 mol% Ar) -+ v [mixer] <——- recycle <———+ | | v | [converter] <– THE ONLY | 700 K, 200 bar DIFFERENCE | v | [separator] isothermalFlash 250 K, 200 bar | | | | rawLiquid-+ +– unreactedGas | | | | v [purgeSplit] splitter | [letdown] | | | flash 20 bar purge –+ +– recycle ————+ | | (5 % bleed) (95 %) productNH3+ +letdownGas

SCALE. The makeup is 8000 kmol/h (N2 1980, H2 5940, Ar 80), which is 1 mol% argon and H2:N2 exactly 3.0. This is NOT the scale of tutorials/steady/flowsheets/ammonia01_synthesis_loop, whose structure this flowsheet copies: that case runs 1 kmol/h of makeup, and at that rate the converter’s catalyst charge is about 2 litres – below the lower bound of every vessel cost correlation the engine carries, so rung 3 could not honestly cost it. The flowsheet is linear in flow (the flash and the reactor are both intensive), so the scale-up changes no intensive answer; compare this case’s converter y_NH3 against ammonia01’s own golden and they agree.

TOPOLOGY ONLY. Stream state lives on disk in 0/, one file per stream.

*A DECLARED CONVERSION AND NOTHING ELSE. The reactor is a conversionReactor: it converts 25 % of the inlet nitrogen because this case’s dictionary says so, and it would do the same at any temperature and any pressure, whether or not equilibrium allows it. That is what a yield reactor ∗is∗.*

*READ SECTION 4 BEFORE QUOTING ANY DUTY FROM THIS CASE. DEV.md’s C8 entry gives this stage exactly one claim — ∗"Claims a mass balance"∗ — and that wording is load-bearing rather than loose. This case cannot honestly claim a converged HEAT balance, and the reason is the engine rather than the stage: its first law is out by 2 278.14 kW. The engine announces it loudly on every run, and section 4 diagnoses it with a probe.*

*Three cases form a ladder of reactor fidelity over one unchanged ammonia synthesis loop. Between consecutive stages exactly one thing moves: the reactor model. Feed, pressure, components, property package, recycle plan, tear seed, separator, let-down and purge are byte-identical across all three.*

*Background: docs/design/how-a-process-design-is-staged.md, and DEV.md C8.*

*A converged MATERIAL balance over the whole loop, and the consequences of one declared number.*

*∗ No reactor size. A declared conversion contains no length scale, no contact time and no catalyst. There is nothing here from which a volume follows. ∗ No cost — and the reason is the METHODOLOGY, not squeamishness. This case carries no system/postDict. By METHODOLOGY, an estimate that has only a declared conversion to go on belongs to AACE 18R-97 Class 5 (the class itself is set by the standard’s PRIMARY characteristic, the maturity of the project definition, which no reactor model decides – methodology is a secondary one), whose methodology the standard states as ∗capacity factored, parametric*

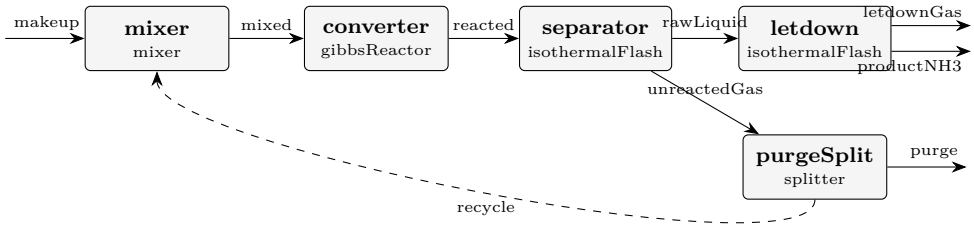
*models, judgment, or analogy∗ — the whole plant scaled from a reference plant, with no individual item sized at all. Choupo's costing chain is per-item Guthrie/Turton bare-module: equipment factored, which 18R-97 lists against Class 4, not Class 5. So the engine cannot produce the RIGHT KIND of cost at this stage. It could produce a ∗number∗; the number would be a Class-4 answer wearing a Class-5 label. The class table is transcribed once, in src/postProcessing/EstimateClass.H — read the bands there, never from prose. ∗ It does not claim the 25 % is right. See section 5.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | letdownGas | F | 0.00738443277329 |
| stream | letdownGas | T | 250 |
| stream | makeup | F | 2.22222222222 |
| stream | makeup | T | 300 |
| stream | mixed | F | 7.92134251385 |
| stream | mixed | T | 263.765269439 |
| stream | productNH3 | F | 0.961408970465 |
| stream | productNH3 | T | 250 |
| stream | purge | F | 0.299953699801 |
| stream | purge | T | 250 |
| stream | rawLiquid | F | 0.968793403238 |
| stream | rawLiquid | T | 250 |
| stream | reacted | F | 6.96786739925 |
| stream | reacted | T | 700 |
| *... and 60 more reference values in the case's `expected`.* | | | |

## ammoniaReactor02_equilibrium

tutorials/plant/ammoniaReactor02_equilibrium




**What it does:** AMMONIA REACTOR-MODEL LADDER, rung 2 of 4 – THERMODYNAMIC CEILING. The same loop with a gibbsReactor at TRUE equilibrium (no approach declared), SRK fugacities at 700 K and 200 bar. The per-pass conversion it reports is a ceiling that no real converter reaches, so nothing downstream of it is sized or costed here.

**The story:** THE AMMONIA REACTOR-MODEL LADDER – flowsheet, SHARED BY RUNGS 1-3.

Everything in this file except the 'converter' block is BYTE-IDENTICAL across ammoniaReactor01_fixedConversion, ammoniaReactor02_equilibrium and ammoniaReactor03_equilibriumApproach. Diff two of them and the only thing that moves is the reactor model – which is the whole point: one rung of the fidelity ladder at a time, and nothing else.

makeup (N2:H2 = 1:3 + 1 mol% Ar) -+ v [mixer] <——- recycle <———+ | | v | [converter] <– THE ONLY | 700 K, 200 bar DIFFERENCE | v | [separator] isothermalFlash 250 K, 200 bar | | | | rawLiquid-+ +– unreactedGas | | | | v [purgeSplit] splitter | [letdown] | | | flash 20 bar purge –+ +– recycle ————+ | | (5 % bleed) (95 %) productNH3+ +letdownGas

SCALE. The makeup is 8000 kmol/h (N2 1980, H2 5940, Ar 80), which is 1 mol% argon and H2:N2 exactly 3.0. This is NOT the scale of tutorials/steady/flowsheets/ammonia01_synthesis_loop, whose structure this flowsheet copies: that case runs 1 kmol/h of makeup, and at that rate the converter's catalyst charge is about 2 litres – below the lower bound of every vessel cost correlation the engine carries, so rung 3 could not honestly cost it. The flowsheet is linear in flow (the flash and the reactor are both intensive), so the scale-up changes no intensive answer; compare this case's converter y_NH3 against ammonia01's own golden and they agree.

TOPOLOGY ONLY. Stream state lives on disk in 0/, one file per stream.

*THE THERMODYNAMIC CEILING. The per-pass conversion this case reports is a ceiling that no real converter reaches, and sizing anything downstream against it produces a fiction. That sentence is the case; the numbers below are the evidence for it.*

*Three cases form a ladder of reactor fidelity over one unchanged ammonia synthesis loop. Between consecutive stages exactly one thing moves: the reactor model. Everything else — feed, pressure, components, property package, recycle plan, tear seed, separator, let-down, purge — is byte-identical. Diff two system/flowsheetDict files and the only block that differs is converter.*

*Rung 4 exists since 2026-09-26, and since 2026-10-12 (DEV.md 4c C67) its rate law is written in rung 4's own constant/reactions as kinetics { type expression; } – the engine carries no ammonia law of its own any more. Rung 4's README carries what the wiring found.*

*Background: docs/design/how-a-process-design-is-staged.md, and DEV.md C8.*

*One claim: the ceiling. With no conversion declared and no approach declared, the converter outlet is the composition that minimises the Gibbs energy of the four declared species at 700 K and 200 bar, subject to the feed's own element balance, with SRK fugacities.*

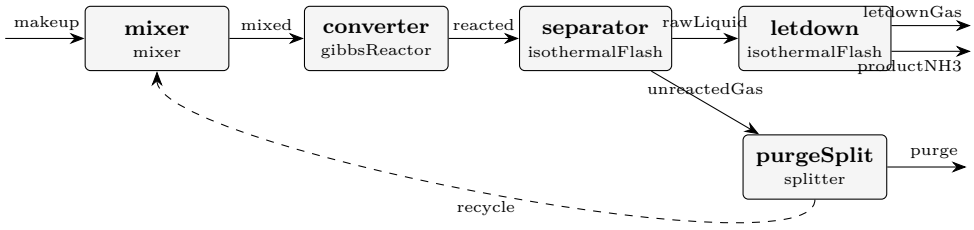
*It also claims a converged heat and material balance, and here that claim holds: every unit closes at 100.00 % and the plant boundary closes at 1.0e-6 %.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | letdownGas | F | 0.00852264420893 |
| stream | letdownGas | T | 250 |
| stream | makeup | F | 2.22222222222 |
| stream | makeup | T | 300 |
| stream | mixed | F | 4.50147743743 |
| stream | mixed | T | 274.381582952 |
| stream | productNH3 | F | 1.05400356221 |
| stream | productNH3 | T | 250 |
| stream | purge | F | 0.119960800861 |
| stream | purge | T | 250 |
| stream | rawLiquid | F | 1.06252620642 |
| stream | rawLiquid | T | 250 |
| stream | reacted | F | 3.46174222365 |
| stream | reacted | T | 700 |
| *… and 65 more reference values in the case's `expected`.* | | | |

## ammoniaReactor03_equilibriumApproach

tutorials/plant/ammoniaReactor03_equilibriumApproach




**What it does:** AMMONIA REACTOR-MODEL LADDER, rung 3 of 4 – REALISTIC OUTLET. The same loop with the gibbsReactor detuned by a 5 K approach to equilibrium, the rung nothing else in this corpus exercises. The outlet is realistic enough to size downstream units against and to cost the converter from a DECLARED space velocity (an estimate method; its AACE class is set by how far the whole project is defined, not by the reactor model).

**The story:** THE AMMONIA REACTOR-MODEL LADDER – flowsheet, SHARED BY RUNGS 1-3.

Everything in this file except the 'converter' block is BYTE-IDENTICAL across ammoniaReactor01_fixedConversion, ammoniaReactor02_equilibrium and ammoniaReactor03_equilibriumApproach. Diff two of them and the only thing that moves is the reactor model – which is the whole point: one rung of the fidelity ladder at a time, and nothing else.

makeup (N2:H2 = 1:3 + 1 mol% Ar) -+ v [mixer] <——- recycle <———+ | | v | [converter] <– THE ONLY | 700 K, 200 bar DIFFERENCE | v | [separator] isothermalFlash 250 K, 200 bar | | | | rawLiquid-+ +– unreactedGas | | | | v [purgeSplit] splitter | [letdown] | | | flash 20 bar purge –+ +– recycle ————+ | | (5 % bleed) (95 %) productNH3+ +letdownGas

SCALE. The makeup is 8000 kmol/h (N2 1980, H2 5940, Ar 80), which is 1 mol% argon and H2:N2 exactly 3.0. This is NOT the scale of tutorials/steady/flowsheets/ammonia01_synthesis_loop, whose structure this flowsheet copies: that case runs 1 kmol/h of makeup, and at that rate the converter's catalyst charge is about 2 litres – below the lower bound of every vessel cost correlation the engine carries, so rung 3 could not honestly cost it. The flowsheet is linear in flow (the flash and the reactor are both intensive), so the scale-up changes no intensive answer; compare this case's converter y_NH3 against ammonia01's own golden and they agree.

TOPOLOGY ONLY. Stream state lives on disk in 0/, one file per stream.

*THE RUNG NOTHING ELSE IN THE CORPUS EXERCISES. gibbsReactor has read operation.temperatureApproach and announced it loudly since src/unitOperations/reactor/GibbsReactor.cpp (the [gibbs] temperatureApproach announcement), and before this case no flowsheet case in tutorials/ declared one — only a props operation (tutorials/props/gibbs/nh3_equilibrium_map/system/propsDict) mentioned the key. A capability that exists, announces itself and is used by nothing is a capability no gate can see.*

*Three cases form a ladder of reactor fidelity over one unchanged ammonia synthesis loop. Between consecutive stages exactly one thing moves: the reactor model. Against rung 2 the difference is literally one line.*

*Background: docs/design/how-a-process-design-is-staged.md, and DEV.md C8.*

*temperatureApproach 5; declares a MAGNITUDE. The engine reads the thermicity of the transformation from this feed to its equilibrium at 700 K, finds it exothermic, and evaluates the reaction equilibrium at 705 K while the physical state, the enthalpy and the energy balance stay at 700 K — announcing that choice and the isothermal reaction enthalpy it read on every run (ruled 2026-09-26: ∗"O delta T ? dado sempre positivo e tu ? que vais atribuir sinal dependendo da reac??o qu?mica"∗; a negative declaration is refused by name). Ammonia synthesis is exothermic, so equilibrium ammonia falls as temperature rises: evaluating the chemistry 5 K hot is exactly ∗"the outlet is 5 K short of equilibrium"∗.*

*The direction the engine assigns is the source's own, and the word that carries it is ∗above∗. US 5,352,428 (M. L. Bhakta and B. J. Grotz, CF Braun and Co, issued 4 October 1994), ∗High conversion ammonia synthesis∗, verbatim:*

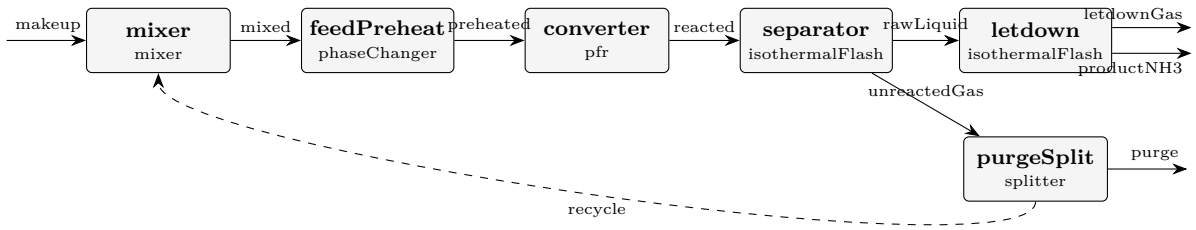
*The patent calls the 5 °C example's conditions "typical and optimum in many commercial process designs", so 5 is a sourced value, not a round number. It is declared bare because it is a temperature ∗difference∗: 5 K and 5 °C are the same interval.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | letdownGas | F | 0.00848002672666 |
| stream | letdownGas | T | 250 |
| stream | makeup | F | 2.22222222222 |
| stream | makeup | T | 300 |
| stream | mixed | F | 4.61740426542 |
| stream | mixed | T | 273.760779212 |
| stream | productNH3 | F | 1.05085761439 |
| stream | productNH3 | T | 250 |
| stream | purge | F | 0.126062212851 |
| stream | purge | T | 250 |
| stream | rawLiquid | F | 1.05933764112 |
| stream | rawLiquid | T | 250 |
| stream | reacted | F | 3.58058189813 |
| stream | reacted | T | 700 |
| *. . . and 105 more reference values in the case's `expected`.* | | | |

## ammoniaReactor04_kineticSizing

tutorials/plant/ammoniaReactor04_kineticSizing




**What it does:** AMMONIA REACTOR-MODEL LADDER, rung 4 of 4 – KINETIC SIZING: THE ENGINE SOLVES THE BED VOLUME. The same loop as rungs 1-3 with the converter as an ADIABATIC plug-flow catalyst bed on the Dyson & Simon (1968) rate law, and the bed VOLUME computed by the engine: a DesignSpec solves V_bed so the effluent is 5 K short of the rate law's own equilibrium (US 5,352,428's per-bed definition, the same source and value rung 3 imposed), with a preheater landing the loop gas at rung 3's 700 K by its own inverse solve. The volume the engine computes sits beside rung 3's space-velocity volume, and the per-pass conversion a single adiabatic bed can reach sits beside rung 3's isothermal 46.60 % – both gaps are the lesson.

**The story:** THE AMMONIA REACTOR-MODEL LADDER – RUNG 4: THE KINETIC BED.

The same loop as rungs 1, 2 and 3 (‘ammoniaReactor01_fixedConversion‘, ‘ammoniaReactor02_equilibrium‘, ‘ammoniaReactor03_equilibriumApproach‘): the makeup, the pressure, the components, the property package, the recycle plan and the tear seed are byte-identical. What moves is the CONVERTER: it is now an ADIABATIC plug-flow catalyst bed integrating the Dyson & Simon (1968) rate law, and its volume is what the engine COMPUTES (system/outerDict).

TWO BLOCKS DIFFER FROM RUNG 3, NOT ONE, AND THE SECOND IS A CONSEQUENCE OF THE FIRST. Rung 3's ‘gibbsReactor‘ declares ‘T 700 K‘: it IMPOSES the converter temperature, and the duty it reports silently includes heating the loop gas from the mixer's ~274 K to 700 K. A kinetic bed cannot impose a temperature – it integrates from the state it is fed – so the preheat that rung 3 folded into the reactor's duty becomes a unit of its own here (‘feedPreheat‘), whose duty is a RESULT. Diff this file against rung 3's and that is what you will see: the converter block replaced, and one preheater inserted in front of it.

makeup (N2:H2 = 1:3 + 1 mol% Ar) -+ v [mixer] <——- recycle <———+ | | v | [feedPreheat] <– NEW: lands the | | loop gas at | v 700 K, Q result | [converter] <– THE CHANGE: | adiabatic pfr, Dyson & Simon | V solved by (1968), V a | the outer driver RESULT | v | [separator] isothermalFlash 250 K, 200 bar | | | | rawLiquid-+ +– unreactedGas | | | | v [purgeSplit] splitter | [letdown] | | | flash 20 bar purge –+ +– recycle ————+ | | (5 % bleed) (95 %) productNH3+ +letdownGas

WHY THE PREHEATER IS A ‘phaseChanger‘ AND NOT A ‘heater‘, measured. The ‘heater‘ takes a DUTY and returns T_out (docs/ai/unit-ops.md); a fixed duty in front of a kinetic bed inside a recycle is OPEN-LOOP UNSTABLE: a larger recycle is heated less, the colder bed converts less, the recycle grows again, and the loop runs away to extinction – measured here with both recycle solvers (Newton: the letdown drum received an empty stream on a trial step; Wegstein: recycle 12 998 → 15 378 → 23 475 kmol/h in three passes while the bed inlet fell 697 → 597 → 480 K). That is real physics and it is why an industrial loop preheats through a feed-effluent exchanger with a temperature controller, never through a fixed-duty heater. The engine's unit that TAKES a target outlet temperature and RETURNS the duty is ‘phaseChanger { outletT }‘ (a built-in inverse solve, announced as SELF-TARGET); the gas enters and leaves as superheated vapour, so the duty is entirely sensible (‘Q_latent_kW 0‘) and the unit's emergent role label ("boiler" for Q > 0 on the vapour side) is a label, not a claim that anything boils.

WHAT THE OUTER DRIVER SOLVES (system/outerDict, a 1 x 1 DesignSpec): $V_bed so that converter.approach_K = 5 K (US 5,352,428's per-bed approach, the SAME source and value as rung 3's) The manipulated volume lives in ‘variables {}‘ below; the unit reads it through ‘$V_bed‘. Nothing here types a volume.

WHY THE TARGET IS AN APPROACH AND NOT RUNG 3's 46.60 % CONVERSION. Stage Rung 3's converter is ISOTHERMAL at 700 K. A single ADIABATIC bed heats itself as it converts, and ammonia

equilibrium retreats as it heats, so the bed's conversion is capped where its adiabatic line meets the equilibrium curve – far below what an isothermal converter at 700 K reaches. That cap is MEASURED in the README (a volume sweep), and it is why no volume of this bed can hit rung 3's number: the target would be unreachable, and a DesignSpec asked for it would diverge honestly. The reachable, sourced target is the one the patent states PER BED.

SCALE, TOPOLOGY, SEED: read rung 3's flowsheetDict; nothing about them moved. TOPOLOGY ONLY here – stream state lives on disk in 0/.

*THE FIRST BED VOLUME IN THIS CORPUS THAT THE ENGINE COMPUTED. The same ammonia synthesis loop as rungs 1, 2 and 3, with the converter as an adiabatic plug-flow catalyst bed integrating the Dyson & Simon (1968) rate law, and the bed volume the answer of a designSpec rather than a number anybody typed. The rate law is WRITTEN in this case's constant/reactions — kinetics { type expression; }, every constant of Eqs 2, 6–8, 18 and 19 beside the formula that uses it, with its unit and its equation number (DEV.md 4c C67, 2026-10-12: the kinetics live in the case; until then the law was compiled into the engine). The engine compiles the formula once, checks its dimensions, prints it back as PARSED in the run log, and pfr integrates it. Read the props bench first (tutorials/props/kinetics/ammoniaRate01_dyson_simon), which writes the same law and adds the paper's Eq 39.*

*Background: docs/design/how-a-process-design-is-staged.md ?3.4, ?6 and ?7.3, and DEV.md C8.*

*Comments aside, the diff is exactly: a variables { V_bed } block, a feedPreheat unit inserted between the mixer and the converter, and the converter block (gibbsReactor with T 700 K; P 200 bar; temperatureApproach 5; and its species list → pfr with V_R $V_bed; nSteps 400; thermalMode adiabatic; and reactions ( ammoniaSynthesis )). Makeup, pressure, components, sealed property package, separator, let-down, purge, recycle plan (system/solverDict) and the tear seed (0/recycle) are byte-identical to rung 3, comments included (0/makeup, 0/recycle and system/solverDict say "all four rungs" in all four cases). Rung 4 adds two files rung 3 has no counterpart for — constant/reactions (the rate law selection) and system/outerDict (the DesignSpec) — and carries a system/postDict of its own, which since 2026-09-26 runs on the DesignSpec's representative pass and sizes the converter as a pfr (?6).*

*Why the second block is a consequence of the first. Rung 3's gibbsReactor declares T 700 K: it IMPOSES the converter temperature, and the 3 287 kW duty it reports silently includes heating the loop gas from the mixer's 274 K to 700 K. A kinetic bed cannot impose a temperature — it integrates from the state it is fed — so the preheat rung 3 hid inside its reactor becomes a unit of its own here. It is a phaseChanger { outletT 700 K; } (the engine's unit that TAKES a target outlet temperature and RETURNS the duty, a built-in inverse solve it announces as SELF-TARGET) and not a heater, and that choice was measured, not preferred:*

*∗ A heater takes a DUTY and returns T_out. A fixed duty in front of a kinetic bed inside a recycle is open-loop unstable: a larger recycle is heated less, the colder bed converts less, the recycle grows again. With the Wegstein recycle solver the loop ran away in three passes — recycle 12 998 → 15 378 → 23 475 kmol/h while the bed inlet fell 697 → 597 → 532 → 480 K — and with the Newton solver a trial step fed the let-down drum an empty stream and the run stopped. That is real physics, and it is why an industrial loop preheats through a feed-effluent exchanger under temperature control, never through a fixed-duty heater.*
*∗ The phaseChanger's emergent role label reads boiler (Q > 0 on the vapour side); the gas enters and leaves as superheated vapour and the duty is entirely sensible (Q_latent_kW 0). The Tsat 150 K it prints is the bracket centre of its inverse solve on a gas far above every critical temperature, not a claim about anything boiling.*

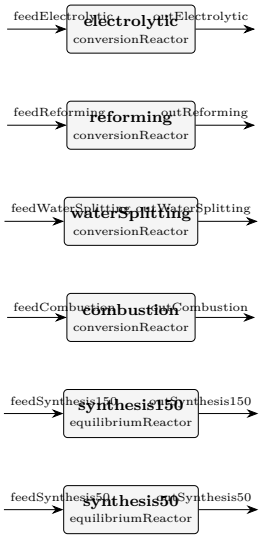
*The 700 K inlet is inherited from rung 3: it is the one thermal fact rung 3 states about its converter, and inheriting it keeps the ladder's rule — one thing moves per rung — as close as a kinetic bed allows. It was a declared assumption there and is one here.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | letdownGas | F | 0.00653101627192 |
| stream | letdownGas | T | 250 |
| stream | makeup | F | 2.22222222222 |
| stream | makeup | T | 300 |
| stream | mixed | F | 10.9531095178 |
| stream | mixed | T | 259.9302829 |
| stream | preheated | F | 10.9531095178 |
| stream | preheated | T | 700 |
| stream | productNH3 | F | 0.879606224653 |
| stream | productNH3 | T | 250 |
| stream | purge | F | 0.459520384245 |
| stream | purge | T | 250 |
| stream | rawLiquid | F | 0.886137240925 |
| stream | rawLiquid | T | 250 |
| *… and 134 more reference values in the case's `expected`.* | | | |

## ammoniaRoutes01_screening

tutorials/plant/ammoniaRoutes01_screening




**What it does:** Screening three production routes to ammonia on paper: overall stoichiometry, heat of reaction, the electrolysis floor and the synthesis equilibrium

**The story:** SCREENING THE PRODUCTION ROUTES TO AMMONIA, ON PAPER, BEFORE ANY FLOWSHEET EXISTS.

The first rung of the ammonia ladder: Douglas' input-output level, the stage at which a route is kept or dropped on its gross margin. Three production routes make the same product by the same synthesis step, N2 + 3 H2 → 2 NH3, and differ in where the hydrogen and the nitrogen come from. GREEN: water electrolysis, air separation, synthesis. GREY: steam reforming of methane, the water-gas shift, air in a secondary reformer, synthesis; the CO2 is emitted. BLUE: grey, with the process CO2 captured and stored. Grey and blue share one overall equation; blue differs only in what is done with the CO2, which no reaction can show. Each route is sized to make 2936 kmol/h of NH3 (1200.0 t/day, the PEQ base case's design rate) from 1468 kmol/h of N2.

The four route and heat units run at 298.15 K and 0.02 bar. That is the REFERENCE STATE, not a process condition: below the vapour pressure of water at 25 C every species is the ideal gas its formation datum is written for, and an ideal gas's enthalpy does not depend on pressure, so every heat below is a standard one, with water as vapour (the lower-heating-value basis). The same heats are NOT process duties: the utility allocation every steady run performs serves them with LP steam and chilled water at 25 C, and those rows are not a design.

WHAT THE ENGINE ANSWERS, unit by unit; none needs a rate law. 'electrolytic' and 'reforming' are conversion reactors at X = 1 on the green and the grey/blue overall equations: their feeds are the raw materials per kg of NH3, and dHrxn_kJ_per_mol is the heat of reaction on the ideal-gas rung, per 2 mol NH3. 'waterSplitting' is the electrolyser alone, H2O → H2 + 1/2 O2, on the route's 4404 kmol/h of water: its duty Q_kW is the least power a thermoneutral electrolyser fed STEAM can be supplied with, the FLOOR under the green route's assumed electricity (an electrolyser fed liquid water must also pay the water's enthalpy of vaporisation, the higher-heating-value basis, so its floor is higher). 'combustion', CH4 + 2 O2 → CO2 + 2 H2O, has a dHrxn_kJ_per_mol that IS the lower heating value of methane, which turns a reformer's fuel, declared in GJ, into kilograms of methane and CO2. 'synthesis150' and 'synthesis50' hold the shared step at equilibrium at 700 K and 150 or 50 bar: the single-pass ceiling. It is the same for every route, so it decides nothing between them; it says what the next rung must add, a recycle loop.

THE MARGINS ARE IN THE README, NOT IN THE ENGINE. A level-1 margin is arithmetic on these flows and a list of prices: product value minus raw materials per kg of NH3, then a SECOND margin after the energy each route needs (electricity, reformer fuel, CO2 compression and storage, a CO2 price), each an assumed specific consumption. The engine's economics pass prices ONE plant per run and needs costed equipment, so it cannot price three routes that have none; the README states every price (the three the

PEQ base case declares, the rest author-set and labelled) and works the table from the numbers this run publishes.

*The common entry to the ammonia cases. Before any flowsheet exists, three production routes to ammonia are compared on their input-output balance — Douglas' first level — and on the energy each one needs, so that only the routes worth a feasibility study go on to one.*

*Illustrated design stage: screening (FEL-1 in a three-stage numbering). No equipment exists yet, so nothing here is sized or costed.*

*The ammonia cases refine a simulation along two independent axes, and each case is named by its purpose on one of them. The project-stage axis follows one green-ammonia plant from screening to basic design: what the project must decide at each stage, and the detail that decision needs — here the reactor model, the exchangers and the economics all move together. The reactor-model axis keeps one synthesis loop fixed and changes only the converter, from a declared conversion to a bed whose volume the engine solves. They are two ladders, not one: a project does not mature by swapping an equilibrium reactor for a kinetic one, and a kinetic reactor does not make a study a basic design. A stage column says "illustrated design stage" for the same reason: the AACE class of an estimate is set by how far the project is defined, never by the stage a case illustrates or by the model it runs, so where a case declares estimateClass it is the author's statement of maturity.*

*The green-ammonia sequence (project stage; 1200 t/day, the PEQ 2026-27 base data):*

*What changes between rungs 2 and 3 — consumptions, investment and the NPV, each difference traced to what changed between the two stages — is a section of the basic-design case's README, generated from the engine's runs of both: [From feasibility to basic design](../greenAmmonia03_basicDesign/README.md#from-feasibility-to-basic-design-what-changed-and-why).*

*The reactor-model ladder (one 200 bar synthesis loop, nothing but the converter moves):*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | feedCombustion | F | 0.000833333333333 |
| stream | feedCombustion | T | 298.15 |
| stream | feedElectrolytic | F | 1.63111111111 |
| stream | feedElectrolytic | T | 298.15 |
| stream | feedReforming | F | 1.38484100528 |
| stream | feedReforming | T | 298.15 |
| stream | feedSynthesis150 | F | 1.63111111111 |
| stream | feedSynthesis150 | T | 700 |
| stream | feedSynthesis50 | F | 1.63111111111 |
| stream | feedSynthesis50 | T | 700 |
| stream | feedWaterSplitting | F | 1.22333333333 |
| stream | feedWaterSplitting | T | 298.15 |
| stream | outCombustion | F | 0.000833333333333 |
| stream | outCombustion | T | 298.15 |

*... and 93 more reference values in the case's `expected`.*

### esterification2sector

tutorials/plant/esterification2sector

**What it does:** Esterification plant – two sectors (REACTION + SEPARATION) sharing components; per-sector constant/ data (S1 kinetics, S2 NRTL pair); ethanol-water pair inherited from the standard library

*A two-sector plant that exists to prove one thing: data lives with the sector that owns it, and a whole-plant run finds it there.*

*Components are shared across the plant. The kinetics belong to REACTION and are never copied to the root. The ethylAcetate-water NRTL pair belongs to SEPARATION — a ∗particular∗ datum, fitted for this separation — while ethanol-water walks up to the standard catalogue. Three levels of ownership, one run.*

*That perNode is the whole case. The sector's pair beat the standard library, and the log says so rather than leaving you to wonder which one it used.*

*Feed 100 kmol/h equimolar acetic acid + ethanol at 360 K. The CSTR (V_R = 0.5 $m^3$) reaches 34.2 % conversion — a PLACEHOLDER number: the kinetics in sectors/REACTION/constant/reactions are still the pseudo-first-order stand-in, not the 2nd-order fit this case's own property evidence supports (see CLAUDE.md, "Pending"). The flash at 355 K, 1.013 bar splits the product into a vapour rich in ethyl acetate (bp 350 K) and ethanol, and a liquid rich in water and the unconverted acid. Mass closes exactly: 50.80 kmol/h of vapour + 49.20 of liquid = 100.00 kmol/h.*

*This case shipped for weeks marked .known-broken, and it earned the mark twice over. Its flash carried P 1.01325; with no unit, which is 1.01 pascal — a hard vacuum — so the K-values exploded and everything went to the vapour. And it was never wired: the sectors existed as thermo regions with the connections commented out, awaiting a "curation phase" that never ended.*

*Fixing it surfaced two real engine bugs, both of the same shape — data owned by a ∗sector∗ was invisible from a whole-plant run:*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Feed | F | 0.0277777777778 |
| stream | liquid | F | 0.01366691372 |
| stream | reactorOut | F | 0.0277777777778 |
| stream | vapor | F | 0.0141108640578 |
| utility | SEPARATION.flash | cooling.coolingWater.duty_kW | -211.579232905 |
| utility | SEPARATION.flash | cooling.coolingWater.T | 355 |
| utility | SEPARATION.flash | cooling.coolingWater.kg_s | 5.06170413649 |
| utility | SEPARATION.flash | cooling.coolingWater.eur_h | 0.304674095384 |
| stream | Feed | T | 360 |
| stream | liquid | T | 355 |
| stream | reactorOut | T | 360 |
| stream | vapor | T | 355 |
| closure | SEPARATION.flash | raw_kW | 715.698427712 |
| closure | SEPARATION.flash | step_kW | 715.698427712 |

*... and 45 more reference values in the case's `expected`.*

## greenAmmonia02_feasibility150bar

### Aftercooler

tutorials/plant/greenAmmonia02_feasibility150bar/Aftercooler

### Chiller

tutorials/plant/greenAmmonia02_feasibility150bar/Chiller

### Converter

tutorials/plant/greenAmmonia02_feasibility150bar/Converter

### EffluentCooler

tutorials/plant/greenAmmonia02_feasibility150bar/EffluentCooler

### FeedPreheater

tutorials/plant/greenAmmonia02_feasibility150bar/FeedPreheater

### Intercooler

tutorials/plant/greenAmmonia02_feasibility150bar/Intercooler

### LetdownHP

tutorials/plant/greenAmmonia02_feasibility150bar/LetdownHP

### LoopMix

tutorials/plant/greenAmmonia02_feasibility150bar/LoopMix

### MakeupComp1

tutorials/plant/greenAmmonia02_feasibility150bar/MakeupComp1

### MakeupComp2

tutorials/plant/greenAmmonia02_feasibility150bar/MakeupComp2

### MakeupMix

tutorials/plant/greenAmmonia02_feasibility150bar/MakeupMix

### N2Boost

tutorials/plant/greenAmmonia02_feasibility150bar/N2Boost

### N2Cooler

tutorials/plant/greenAmmonia02_feasibility150bar/N2Cooler

### PurgeSplit

tutorials/plant/greenAmmonia02_feasibility150bar/PurgeSplit

### RecycleComp

tutorials/plant/greenAmmonia02_feasibility150bar/RecycleComp

### Separator

tutorials/plant/greenAmmonia02_feasibility150bar/Separator

### StorageFlash

tutorials/plant/greenAmmonia02_feasibility150bar/StorageFlash

### `WaterCooler`

`tutorials/plant/greenAmmonia02_feasibility150bar/WaterCooler`

## greenAmmonia02_feasibility50bar

### Aftercooler

tutorials/plant/greenAmmonia02_feasibility50bar/Aftercooler

### Chiller

tutorials/plant/greenAmmonia02_feasibility50bar/Chiller

### Converter

tutorials/plant/greenAmmonia02_feasibility50bar/Converter

### EffluentCooler

tutorials/plant/greenAmmonia02_feasibility50bar/EffluentCooler

### FeedPreheater

tutorials/plant/greenAmmonia02_feasibility50bar/FeedPreheater

### LetdownHP

tutorials/plant/greenAmmonia02_feasibility50bar/LetdownHP

### LoopMix

tutorials/plant/greenAmmonia02_feasibility50bar/LoopMix

### MakeupComp1

tutorials/plant/greenAmmonia02_feasibility50bar/MakeupComp1

### MakeupMix

tutorials/plant/greenAmmonia02_feasibility50bar/MakeupMix

### N2Boost

tutorials/plant/greenAmmonia02_feasibility50bar/N2Boost

### N2Cooler

tutorials/plant/greenAmmonia02_feasibility50bar/N2Cooler

### PurgeSplit

tutorials/plant/greenAmmonia02_feasibility50bar/PurgeSplit

### RecycleComp

tutorials/plant/greenAmmonia02_feasibility50bar/RecycleComp

### Separator

tutorials/plant/greenAmmonia02_feasibility50bar/Separator

### StorageFlash

tutorials/plant/greenAmmonia02_feasibility50bar/StorageFlash

### `WaterCooler`

`tutorials/plant/greenAmmonia02_feasibility50bar/WaterCooler`

## greenAmmonia03_basicDesign

### Aftercooler

tutorials/plant/greenAmmonia03_basicDesign/Aftercooler

## CONVERTER

tutorials/plant/greenAmmonia03_basicDesign/CONVERTER

**The story:** THE MULTIBED CONVERTER – ONE SECTOR, ONE PRESSURE VESSEL (DEV.md 4c C69).

Three adiabatic catalyst beds with an interbed cooler after each of the first two. The flowsheet still computes FIVE blocks – each bed and each cooler is a calculation of its own – but they are built as ONE piece of equipment: the beds sit in a basket inside a single pressure shell, with the interbed exchange inside the same shell. That is a fact about the EQUIPMENT, so it is said where the equipment is sized: system/postDict's 'converterShell' entry names this sector, stacks its members' geometries into one shell at the loop's design pressure, and the beds' own shells become internal baskets that are not priced twice.

WHERE THE ARRANGEMENT WAS READ (2026-10-12), and what was not: The M. W. Kellogg Company, "Ammonia synthesis converter", EP 0 254 936 A2 (1988), read on the EPO publication server: "the vertical converter is contained within cylindrical pressure shell 1", three axial-flow catalyst beds "within the cylindrical inner shell", and interbed cooling "supplied entirely by the single indirect heat exchanger" inside it – ONE interbed exchanger, not two. Topsoe, "Ammonia converter baskets S-300" (product page, read): three catalyst beds, radial flow, a BASKET installed in a pressure shell, in one version with an integral feed-effluent exchanger after the third bed. The page does not describe its interbed exchangers. Neither source describes TWO interbed coolers raising steam, as these two do; that arrangement is the base case's, and the one-shell grouping is the sources' general construction applied to it.

The stream names are the plant's, unchanged: the sector's inlet is the FEHE's preheated feed, its outlet the hot effluent the FEHE cools.

### Chiller

tutorials/plant/greenAmmonia03_basicDesign/Chiller

### FEHE

tutorials/plant/greenAmmonia03_basicDesign/FEHE

### Intercooler

tutorials/plant/greenAmmonia03_basicDesign/Intercooler

### LetdownHP

tutorials/plant/greenAmmonia03_basicDesign/LetdownHP

### LoopMix

tutorials/plant/greenAmmonia03_basicDesign/LoopMix

### MakeupComp1

tutorials/plant/greenAmmonia03_basicDesign/MakeupComp1

### MakeupComp2

tutorials/plant/greenAmmonia03_basicDesign/MakeupComp2

### MakeupMix

tutorials/plant/greenAmmonia03_basicDesign/MakeupMix

### N2Boost

tutorials/plant/greenAmmonia03_basicDesign/N2Boost

### N2Cooler

tutorials/plant/greenAmmonia03_basicDesign/N2Cooler

### PurgeSplit

tutorials/plant/greenAmmonia03_basicDesign/PurgeSplit

### RecycleComp

tutorials/plant/greenAmmonia03_basicDesign/RecycleComp

### Separator

tutorials/plant/greenAmmonia03_basicDesign/Separator

### StorageFlash

tutorials/plant/greenAmmonia03_basicDesign/StorageFlash

### `WaterCooler`

`tutorials/plant/greenAmmonia03_basicDesign/WaterCooler`

## greenAmmoniaPEQ_baseCase

### Aftercooler

tutorials/plant/greenAmmoniaPEQ_baseCase/Aftercooler

### Chiller

tutorials/plant/greenAmmoniaPEQ_baseCase/Chiller

### Converter

tutorials/plant/greenAmmoniaPEQ_baseCase/Converter

### FEHE

tutorials/plant/greenAmmoniaPEQ_baseCase/FEHE

### Intercooler

tutorials/plant/greenAmmoniaPEQ_baseCase/Intercooler

### LetdownHP

tutorials/plant/greenAmmoniaPEQ_baseCase/LetdownHP

### LoopMix

tutorials/plant/greenAmmoniaPEQ_baseCase/LoopMix

### MakeupComp1

tutorials/plant/greenAmmoniaPEQ_baseCase/MakeupComp1

### MakeupComp2

tutorials/plant/greenAmmoniaPEQ_baseCase/MakeupComp2

### MakeupMix

tutorials/plant/greenAmmoniaPEQ_baseCase/MakeupMix

### N2Boost

tutorials/plant/greenAmmoniaPEQ_baseCase/N2Boost

### N2Cooler

tutorials/plant/greenAmmoniaPEQ_baseCase/N2Cooler

### PurgeSplit

tutorials/plant/greenAmmoniaPEQ_baseCase/PurgeSplit

### RecycleComp

tutorials/plant/greenAmmoniaPEQ_baseCase/RecycleComp

### Separator

tutorials/plant/greenAmmoniaPEQ_baseCase/Separator

### StorageFlash

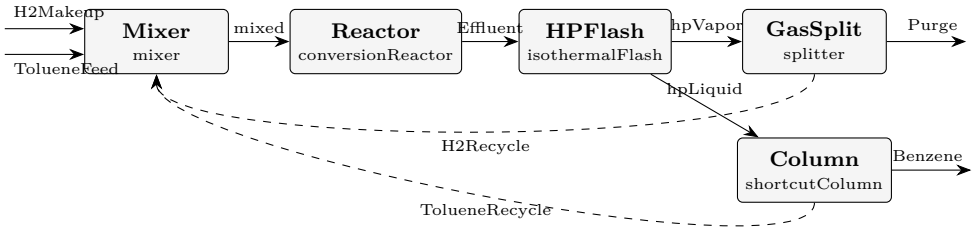
tutorials/plant/greenAmmoniaPEQ_baseCase/StorageFlash

### WaterCooler

tutorials/plant/greenAmmoniaPEQ_baseCase/WaterCooler

## hda

tutorials/plant/hda




**What it does:** HDA plant – toluene + H2 → benzene + CH4. A single FLAT flowsheet with two plant-level recycles (an H2 gas loop and an unreacted-toluene liquid loop); cross-sector tears are unsupported, so it has no sectors. Furnace lumped into the isothermal reactor; light-gas K-values rest on Antoine extrapolation (see thermoPhysPropDict).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | Benzene | F | 0.0308570333668 |
| stream | Effluent | F | 0.589422462034 |
| stream | H2Makeup | F | 0.0305555555556 |
| stream | H2Recycle | F | 0.522049697299 |
| stream | Purge | F | 0.0274762998579 |
| stream | TolueneFeed | F | 0.0277777777778 |
| stream | TolueneRecycle | F | 0.00903943150968 |
| stream | hpLiquid | F | 0.0398964648764 |
| stream | hpVapor | F | 0.549525997157 |
| stream | mixed | F | 0.589422462034 |
| kpi | Column | B | 0.00903943150968 |
| kpi | Column | D | 0.0308570333668 |
| kpi | Column | N_min | 8.3953506183 |
| kpi | Column | N_theoretical | 20.1717591626 |

*... and 58 more reference values in the case's `expected`.*

### `lithiumBrinePlant`

`tutorials/plant/lithiumBrinePlant`

**What it does:** lithiumBrinePlant – THE reference case: a SECTORED lithium-carbonate plant from salar brine, WIRED into one solver. Five SECTORS, each a thermo WORLD of its own – BRINE (electrolyte/Pitzer), EXTRACTION (gamma-gamma NRTL LLE), CARBONATION (Li2CO3 retrograde solubility), HOTAIR (Gibbs combustion), FINISHING (drying + cyclone) – connected sector to sector, each sector declaring its own propertyPackage (its world) on the global component set, with optional per-unit thermo overrides where a single unit needs a different world. The proof that a plant and its sectors run as one: sectors are composite boxes, streams flow between them, model boundaries audited.

**The story:** Five REAL sectors. Streams are named physical graph edges. Sector names are subdomains; unit operations live inside each sector.

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | carbLiquor | F | 0.0277777777778 |
| stream | cleanAir | F | 0.0879791388897 |
| stream | emulsion | F | 0.0380203541089 |
| stream | fines | F | 0 |
| stream | fuelAir | F | 0.06034025 |
| stream | halite | F | 0.000868534779985 |
| stream | hotAir | F | 0.0603402500008 |
| stream | humidExhaust | F | 0.0879791388897 |
| stream | liquor | F | 0.0269092429978 |
| stream | loadedOrganic | F | 0.0123666842107 |
| stream | organic | F | 0.0111111111111 |
| stream | product | F | 0.000138888888889 |
| stream | raffinate | F | 0.0256536698982 |
| stream | salarBrine | F | 0.0277777777778 |
| *... and 20 more reference values in the case's `expected`.* | | | |

## polycaprolactonePlant

**The theory you need.** Ring-opening polymerisation plant: monomer to polycaprolactone with step-growth KPIs (Carothers, Flory) on the PFR, polymer properties from group contribution on the repeat unit.

## properties

tutorials/plant/polycaprolactonePlant/properties




**What it does:** Close the loop: ESTIMATE the PCL repeat-unit glass transition Tg by the Yang 2020 group-contribution scheme — the plant gives Mn/PDI, this gives the physical property from the SAME repeat unit -[O-(CH2)5-CO]-

**The story:** pclProperties – CLOSE THE LOOP: process → molar mass → physical property.

The plant (../) gave you the chain statistics from the KINETICS: Mn, Mw, PDI on the polycaprolactone it makes. Here you ESTIMATE the polymer's PHYSICAL PROPERTIES from the SAME repeat unit -[O-(CH2)5-CO]- by group contribution — the bridge from "how big are the chains" to "how does the material behave".

TWO estimators, two properties, on the repeat unit pclRepeat (M0 = 114.14):

(A) Tg, the GLASS-TRANSITION temperature — Yang 2020 (run below). Repeat unit -[O-(CH2)5-CO]- decomposes into Yang main-chain groups: O x1 : MW 16 Yg -14.718 CH2 x5 : MW 14 Yg 4.026 (each) CO x1 : MW 28 Yg 4.370

M0 = 16 + 5∗14 + 28 = 114 g/mol (matches the reactor M0!) Yg(inf) = -14.718 + 5∗4.026 + 4.370 = 9.782 (1e3 g.K/mol) Tg(inf) = 9.782e3 / 114 ~ 86 K (the additive estimate) (Measured PCL Tg ~ 213 K / -60 C; an additive main-chain scheme is weak for a flexible aliphatic polyester — the model tells the truth about itself, which IS the lesson. See the console's per-cent error vs the anchor.)

(B) DENSITY — Van Krevelen (Bondi Vw). NOT runnable on PCL with the shipped data: the Slice-1 Van Krevelen set ships only hydrocarbon + chloro groups (CH3 CH2 CH C ACH AC CHCl CH2Cl); it has NO ester -O- / -CO- group, so the repeat unit -[O-(CH2)5-CO]- cannot be decomposed. Running it would HARD- ERROR with "unknown polymer group 'O'" — the estimator refuses to invent a value (no silent zero). The honest path: density needs the ester groups curated into data/standards/-parameters/vanKrevelen.dat first. The worked Van Krevelen DENSITY example lives in tutorials/props/estimate/vanKrevelen01_polystyrene_density (a pure- hydrocarbon repeat unit the Slice-1 set CAN do).

This is the same M0 = 114.14 the reactor's polymer{} block used for Mn = M0/(1-p) — one repeat-unit mass, two places (process AND property).

*A choupoProps sub-case of [../](..) (the polycaprolactone plant). The plant gives you the chain statistics from the kinetics (Mn / Mw / PDI); here you estimate the polymer's physical property — the glass transition Tg — from the same repeat unit -[O-(CH2)5-CO]- (M0 = 114.14, the exact number the reactor used for Mn = M0/(1-p)).*

*∗ Tg via Yang 2020 (CC-BY) — runnable. Tg(inf) = 85.8 K; the console prints the -59.7 % deviation from the measured ~213 K (an honest limit of an additive scheme on a flexible aliphatic polyester). M0 = 114 is goldened (an exact mass-balance fact); Tg is shown but not goldened as validated physics. ∗ Density via Van Krevelen — ∗not∗ runnable on PCL with the shipped data (the Slice-1 group set has no ester -O-/-CO- group, so it would hard-error rather than invent a value). See the full explanation and the runnable hydrocarbon example linked from the parent [../README.md](../README.md).*

*The full teaching narrative lives in the parent README. See [system/propsDict](system/propsDict) for the exact group decomposition and the estimator invocation.*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| diag | pclRepeat_Tg_yang2020 | M0_g_per_mol | 114 |
| diag | pclRepeat_Tg_yang2020 | Tg_K | 85.81 |
| diag | pclRepeat_Tg_yang2020 | YgSum_1e3gKmol | 9.782 |

## sugarPlantEconomicsSweep

`tutorials/plant/sugarPlantEconomicsSweep`

**What it does:** Sugar plant ECONOMICS SWEEP – IRR / payback / NPV vs RawJuice sucrose content (Phase-2 SweepDriver post-chain differentiator)

**The story:** / system/flowsheetDict (PLANT level – COMPOSITE)

The flagship sugar plant (the ChemicalPlantTutorial flowsheet, reused verbatim – sectors and units included), here driven by system/outerDict: a SweepDriver walks the RawJuice sucrose fraction and runs the FULL post chain (sizing → costing → economics, system/postDict) on every point, so IRR / NPV / payback vs feed quality come out as sweep responses.

A composite node: it has 'children' (sub-flowsheets, here the sectors) and 'connections' (the pipework between them). The same shape a sector uses to list its units – one level up. This is the plant-and-sectors shape.

Naming convention (see docs/ai/case-layout.md): SECTORS in CAPS : CONCENTRATION, DRYING, FERMENTATION Units in PascalCase : Cryst, Evap1, Evap2 Streams in PascalCase : RawJuice, Magma, Powder, EvapVapour Components in lowercase : water, sucrose, N2

*This is the flagship sugar plant (tutorials/plant/ChemicalPlantTutorial) driven by a SweepDriver instead of a single pass. It exists to show the one thing a single CAPEX number cannot: how the project economics move when a process variable changes.*

*The driver runs the FULL post-processing chain (sizing → costing → economics, in system/postDict) on every converged point, so economics.IRR, economics.paybackYears, economics.NPV, ? resolve as ordinary KPI look-ups (the Phase-2 SweepDriver post-chain wiring). The result lands in sweep_results.csv.*

*Sweeping the RawJuice sucrose content 0.09 → 0.16 mol fraction walks the plant from sub-economic, through break-even, to clearly profitable:*

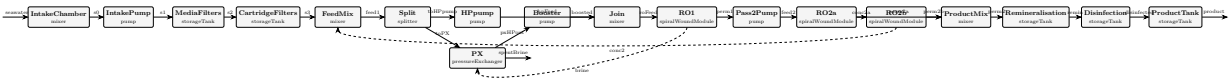
*(Measured 2026-10-06 from sweep_results.csv, after the BD dryer's air was sized for the richest point – the design basis is written in 0/MAIN/BdAir.)*

*Richer juice → more crystalline sugar per unit feed → more revenue against an almost-unchanged plant cost → the IRR climbs and the payback appears and shortens. The NPV crosses zero between 0.090 and 0.104.*

*Both use the same constant/economics prices (EU Sugar Market Observatory white sugar, ECB EUR/USD, ∗Chemical Engineering∗ CEPCI, US statutory tax) and the same system/postDict cost chain — only the run mode differs.*

## swro01_campoDalias

tutorials/plant/swro01_campoDalias




**What it does:** Campo de Dalias two-pass seawater desalination plant from a manufacturer's public case study: pretreatment, first pass with pressure-exchanger energy recovery, second pass with the concentrate recycled, boron, post-treatment placeholders, and a plot plan up to the fence

**The story:** A TWO-PASS SEAWATER DESALINATION PLANT FROM A MANUFACTURER'S PUBLIC CASE STUDY (DEV.md 4c C194).

SOURCE. Toray Industries, "Seawater Desalination Using Toray's RO Membranes to Support a Thriving Agricultural Economy" (Campo de Dalias Desalination Plant, Almeria), MC-CS-CAMPODEDALIAS-1910, 2019: two pages, a MANUFACTURER'S document about its own membranes. Every performance figure in it is Toray's claim "based on manufacturer's test conditions"; none is a measurement of this case. It gives: an open intake by a 1.6 km subsea pipeline; pretreatment by 40 dual-media filters (glass-fibre-reinforced polyester) then 5-micron cartridges (99.6 % particulate removal); a two-pass reverse osmosis of six skids each, the first pass TM820E-400 seawater elements (skid 233 vessels of 7 elements, "233 (7M)"; 37 m2, 28.3 m3/d, 99.75 % NaCl and 91 % boron at pH 8; "47.2 % system recovery"), the second pass TM720C-430 brackish elements (54:18 vessels of 7 elements; 40 m2, 33.3 m3/d, 99.2 % NaCl and 95 % boron at pH 10; "84-90 %" recovery); the second pass's concentrate recycled to the first; an isobaric energy-recovery system of 95 % efficiency (DWEER); post-treatment by remineralisation and disinfection; 98 664 m3/day; the feed analysis (its Table 2); a permeate target of TDS $< 400$ mg/L and boron $< 0.5$ mg/L. NOT in it: a site plan, vessel and pump counts beyond the skid designs, pump sizes, energy per m3, costs, the temperature of operation (14-23 C is the feed's range).

WHAT THIS CASE TAKES FROM IT, AND WHAT IT ADDS. From the document (origin literature): the 98 664 m3/day; the six skids and their vessel counts (1398 vessels in the first pass, 324 + 108 in the second, 7 elements each); the elements' areas, flows and rejections; the 47.2 % first-pass recovery and the second pass's 84-90 % (87 % chosen); the 95 % energy-recovery efficiency; the recycle; the feed analysis. The document's two internal figures for the product differ slightly (98 664 m3/day against 6 skids x 16 442 = 98 652): the case uses 98 664. ASSUMED by the case, each with its reason where it stands: the test conditions of the manufacturer's figures; the membranes' water permeances (derived from them); the elements' geometry; the temperature (25 C, 2 K above the document's feed range – the module has no temperature correction of A_w); pump efficiencies; the device's mixing and leakage; every hold time and tank; the whole plot plan. NOT MODELLED: the particulate removal of the filters (no solids), the filters' own pressure drop, the pH and the caustic dosing between the passes, the remineralisation and disinfection chemistry, the real device's piston hydraulics, fouling and the elements' ageing.

THE SALTS ARE ONE LUMPED SOLUTE. The document's Table 2 lists twelve ions (TDS 38 697 mg/L). The module's rated-test inversion needs a salt, not ions, so the ions are replaced by NaCl at the SAME OSMOTIC PRESSURE: the ions' 1235.9 mmol/L give 617.9 mmol/L NaCl = 36.11 g/L, against the 38 697 mg/L of actual TDS (the lump weighs 6.7 % less than the ions; the permeate TDS quoted below, as NaCl, reads about 7 % lower than a real TDS). Boron (5.3 mg/L as B) is a second lumped solute.

THE FIGURES THE DOCUMENT CALLS "FEED PRESSURE" ARE THE TEST PRESSURES, AND THE PLANT'S ARE SOLVED. Table 1 prints 5.52 MPa (800 psi) and 1.03 MPa (150 psi) beside the elements' flow and rejection, which are test figures. The case takes them as the rated tests' pressures. They cannot be the plant's: a 47.2 % recovery of this seawater leaves a concentrate at 63.5 g/L NaCl-equivalent (about 53 bar of osmotic pressure), which 55.2 bar cannot push past; the first pass here needs 58.6 bar (HP pump and booster discharge) for 47.2 %. The second pass runs at 10.74 bar, not 10.3, which puts its recovery at 87 % (the document's 84-90 %) with the first pass's permeate: at 10.3 bar the same array recovers 94.8 %.

WHAT THE RUN SAYS (2026-11-02, boron split by the pH, C198; the golden is the authority): product 4111.2 m3/h = 98 668.7 m3/day (the document's 98 664) intake 9461.3 m3/h; recovery of the intake 43.5 % first pass 47.20 % recovery (the document's 47.2 %), 13.05 LMH average flux on 362 082 m2, feed 10 011 m3/h at 58.59 bar, brine 63.5 g/L NaCl-eq at 57.9 bar, permeate 109.6 mg/L NaCl-eq, boron 0.457 mg/L

second pass 87.0 % recovery, 33.99 LMH on 120 960 m2 (the element's rated 34.7), concentrate 614 m3/h at 0.83 g/L, boron 3.2 mg/L, recycled to the first product 1.9 mg/L NaCl-eq (the document's target < 400 mg/L TDS), boron 0.043 mg/L (< 0.5) energy recovery flows balanced to 1e-5; 54.1 bar recovered on 5209 m3/h, 7.76 MW hydraulic; the feed's salinity up 1.8 % pump shaft powers HP pump 9185 kW, booster 437 kW, second-pass pump 1604 kW, intake 326 kW: 11 552 kW, i.e. 2.81 kWh per m3 of product (shaft only: no motor losses, no pretreatment or post-treatment loads, no product delivery)

BORON IS TWO SPECIES SPLIT BY THE pH (DEV.md 4c C198, Stage 1). Boron is the reason for the second pass and for the caustic between the passes: boric acid (neutral, small enough to pass a seawater membrane) is converted at pH 10 into borate (an anion, rejected). Each module declares 'aqueousSplit {}': the apparent boron is carried as H3BO3 and BOH4, the base fraction alpha = 1/(1 + 10^(pKa - pH)) is asked of the PACKAGE (the curated constant data/standards/chemistry/BOH4-formation.dat, pKa 9.24 with its van't Hoff enthalpy, and this case's declared aqueous activity model, Davies, for gamma and a_w) at the module inlet's bulk and the DECLARED pH, and each form has its own intrinsic rejection. Alpha is 0.108 at the first pass (pH 8.2, I = 0.59 mol/kg) and 0.858 at the second (pH 10). THE PERMEANCES come from the sheet's rated tests: ONE boron test per element (91 % at pH 8; 95 % at pH 10, 'boronRef' in the two records) is inverted through the module's own film and, at the test's own alpha, determines the ACID's rejection; the BASE's is a DECLARED ASSUMPTION (the element's own NaCl rejection: an anion rejected like the salt) because one test cannot identify two permeances. TWO tests of one element at two pH would give both ('rejection { tests ( id1 id2 ) }'; the gate runs that form on a fixture), and the engine refuses a single test offered as two or two at the same pH. The result is R_acid 0.932 and R_base 0.998 for the first-pass element, 0.790 and 0.994 for the second-pass one. THE LIMITS, announced on every run: THE pH IS A DECLARED LABEL of each module (8.2, Table 2; 10, Table 1), not a state – no stream carries a pH, and the caustic dosing that would set the second pass's pH is NOT a unit (Stage 2); the split is FROZEN at the inlet's pH along the channel; both forms share the one film coefficient; Davies is outside its own ~0.5 mol/kg band at seawater strength, so alpha at the first pass is indicative (the apparent pKa it gives, 9.12, is NOT Dickson's conditional seawater constant, 8.60 on the total scale – the comparison is printed by the gate, never tuned). What the case now shows: lower the second pass's declared pH from 10 to 9 and the product's boron rises from 0.04 to 0.14 mg/L, as alpha predicts. A finding about the probe: the boron test is water plus 5 ppm boron at the test's pressure, with no osmotic pressure to oppose it, so it runs at 19 % recovery instead of the test's 8-15 %; the derived effective rejection (0.9357, 0.9642) is that of the probe. And boron is a solute of the van't Hoff osmotic pressure ('dissociation 1'): the second pass's feed is so dilute that boron is about 1 % of its osmotic pressure, so the product moved by 0.004 % when boron's rejection changed from the lumped to the split form (the NaCl-side rows of the golden moved for that reason alone).

THE LOOP IS A KNIFE-EDGE, AND 0/ IS A CONVERGED STATE. The exchanger returns 99 % of the brine's volume to the membrane feed, so the first pass is a loop of gain near one (membrane27 shows it), and the second pass's concentrate is a second tear. A cold start from a rough guess converges, with this Wegstein, to a SPURIOUS fixed point (all water circulating, no permeate); 0/ therefore holds the converged state of a previous run and the case restarts from it in two iterations. Changing a pressure or a flow moves the answer: re-seed from converged/ or tune the split fraction (0.4791) to balance the exchanger. The intake flow, the split, the two pass pressures and the first-pass pressure were solved together for the product, 47.2 %, 87 % and a balanced exchanger.

SIX TRAINS (DEV.md 4c C199). The document's six skids are six identical trains in parallel, so this flowsheet carries their SUM ('vessels 1398' is 6 x 233, and the pumps and the exchanger carry six times a train's flow); the physics is the same and is solved once. The sizing is per train: system/postDict declares 'trains { count 6; }' and yields train1 .. train6 for each pump, exchanger array and vessel bank.

THE PLOT PLAN (system/layoutDict) is Choupo's design of a plant of this scale up to the fence, NOT the real site: the document gives no site plan.

*The question. Two pages from a membrane maker describe a 98 664 m3/day two-pass seawater plant in Almeria: pretreatment, a first pass at 47.2 % recovery with isobaric energy recovery, a second pass that polishes the boron, the second pass's concentrate recycled to the first. What does it take to turn those two pages into a simulated plant – and what do the pages not say?*

*What the case is. system/flowsheetDict (read its header first): intake, pretreatment placeholders, a high-pressure pump, a pressure exchanger and a booster feeding 1398 vessels of TM820E-400 elements (the first pass), a second pass of 324 + 108 vessels of TM720C-430 on the first pass's permeate, the concentrate recycled, post-treatment placeholders and a product store. The two Toray element records are CASE-LOCAL*

*(constant/assets/): the document's figures are the manufacturer's claim, and what it does not state (the test conditions, the water permeances, the geometry) is declared assumed or estimated. system/layoutDict is Choupo's plot plan of a plant of this scale up to the fence – NOT the real site.*

*The answer (2026-11-02; expected is the authority). Product 4111 m3/h = 98 668.7 m3/day; first pass 47.20 %, second 87.0 %; 2.0 mg/L NaCl-equivalent and 0.043 mg/L boron in the product; 2.81 kWh/m3 of pump shaft work.*

*Boron (DEV.md 4c C198, Stage 1). Boron is carried as boric acid and borate, split by the pH against the acid's curated pKa through the package; each module declares its pH (8.2 for the first pass, Table 2; 10 for the second, Table 1 – the caustic dosing is not modelled). Lower the second pass's pH to 9 and the product's boron rises; the sheet's one test per element fixes the acid and the borate is assumed rejected like the salt (declared, announced).*

*Six trains, one building (DEV.md 4c C199). The case study says six skids of 16 442 m3/day, and an industrial SWRO plant is built so: identical trains, each with its own high-pressure pump, its own array of pressure exchangers and its own vessel racks. The flowsheet is solved ONCE at plant flow (six identical trains in parallel carry six times a train's flow) and the SIZING is done per train: system/postDict declares trains { count 6; ... } and the pumps, the exchanger arrays and the vessel banks yield items train1 .. train6 with their per-train numbers beside the plant's. A pump is rated by the largest machine one make builds (Sulzer MSD-RO, 1600 m3/h: one per train at 50 % load, plus a declared spare), the exchanger array is 6 devices of the Energy Recovery PX Q650 window per train under its 83 bar limit, and each pass's vessels stand in racks (233 vessels of 7.64 m in 5 racks of four tiers, the loading clearance of one vessel length at the east end of each bank). The 42 items stand in the four rooms of one reverse-osmosis building in system/layoutDict. What the open sources did not give – the rack pitch, the aisle, the loading clearance – is declared assumed and says so.*

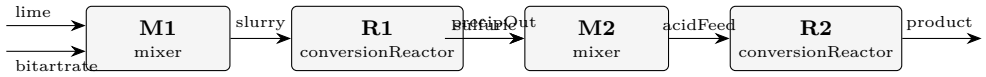
*What a student should notice. 1. The document's "feed pressure" figures (5.52 and 1.03 MPa) are TEST pressures: 47.2 % recovery of this seawater needs 58.6 bar, and the second pass needs 10.74 bar for 87 %. The header shows the arithmetic. 2. Boron is where the engine stops. The module does not couple the pH to the transport, so boron is one lumped solute with a constant rejection read from the manufacturer's two pH-labelled tests; the gate pins that relabelling the pH moves nothing. The follow-up is named in the header and in docs/design/a-desalination-plant-from-a-case-study.md. 3. The loop through the exchanger is a knife-edge with a spurious solution; 0/ is a converged state for that reason. 4. The plot plan announces six findings (five water tanks with no bund, one entrance) and leaves them. 5. The pump count is ceil(flow per train / the largest pump): with 6 trains it is 1 per train; with ONE train the same high-pressure flow (4791 m3/h) would need 3 pumps in parallel (declare installed { count 3; }; the sizer says so when you do not).*

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | boosted | F | 80.1343858417 |
| stream | boosted | T | 298.164615425 |
| stream | brine | F | 81.7112864112 |
| stream | brine | T | 298.15 |
| stream | conc2 | F | 9.46243885204 |
| stream | conc2 | T | 298.208306901 |
| stream | conc2a | F | 22.1440707933 |
| stream | conc2a | T | 298.208306901 |
| stream | disinfected | F | 63.3914931649 |
| stream | disinfected | T | 298.15 |
| stream | feed1 | F | 155.352283581 |
| stream | feed1 | T | 298.15 |
| stream | feed2 | F | 72.853932017 |
| stream | feed2 | T | 298.208306901 |
| *... and 3409 more reference values in the case's `expected`.* | | | |

### tartaricAcid

tutorials/plant/tartaricAcid




**What it does:** Tartaric-acid plant (wine route): lime precipitation of calcium tartrate, then sulfuric acidulation to tartaric acid + gypsum. Molecular reactors; the solution chemistry sub-study is tutorials/props/electrolyte/-tartaricAcid_acidulation. HONEST LIMITS: H2SO4 declared nonvolatile by case overlay (Psat ~1e-5 Pa at process T); precipitates ride as nonvolatile solutes with NO rigorous solid-liquid separation; reactor duties honestly UNAVAILABLE (species without ideal-gas Cp – a named gap, never a fake zero).

**What to expect (golden-locked):**

| kind | unit/stream | key | expected |
|---|---|---|---|
| stream | acidFeed | F | 0.943266666667 |
| stream | bitartrate | F | 0.583333333333 |
| stream | lime | F | 0.166666666667 |
| stream | precipOut | F | 0.7766 |
| stream | product | F | 0.943266666667 |
| stream | slurry | F | 0.75 |
| stream | sulfuric | F | 0.166666666667 |
| kpi | M1 | F_out | 0.75 |
| kpi | M1 | P_out | 100000 |
| kpi | M1 | T_out | 333 |
| kpi | M1 | n_in | 2 |
| kpi | M1 | vf | 0 |
| kpi | M2 | F_out | 0.943266666667 |
| kpi | M2 | P_out | 100000 |

*... and 20 more reference values in the case's `expected`.*

## twoSectorDemo

**The theory you need.** The zippable two-sector demonstration plant – the case used to rehearse the online landing. Same physics as the esterification demo.

## SECTOR_R

tutorials/plant/twoSectorDemo/SECTOR_R




**What it does:** SECTOR_R – the reactor sector. Runs its OWN compiled unit op (stoichReactor) on the plant-level thermo (cascaded from ../constant).

**The story:** one custom reactor (compiled from code/). Build + run with: bin/buildCode tutorials/plant/twoSectorDemo/SECTOR_R runCase tutorials/plant/twoSectorDemo/SECTOR_R —

## SECTOR_S

tutorials/plant/twoSectorDemo/SECTOR_S




**What it does:** SECTOR_S – the separation sector. Runs its OWN compiled unit op (sharpSplitColumn) on the plant-level thermo (cascaded from ../constant).

**The story:** one custom distillation (compiled from code/). Build + run: bin/buildCode tutorials/plant/twoSectorDemo/SECTOR_S runCase tutorials/plant/twoSectorDemo/SECTOR_S —